

“Now! Now!” cried the Queen. “Faster! Faster! . . . Now *here*, you see, it takes all the running you can do, to keep in the same place. If you want to get somewhere else, you must run at least twice as fast as that!”

LEWIS CARROLL, *Alice’s Adventures in Wonderland* (1865)

OUTLINE

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- 14.3 • Kinetics of Enzyme-Catalyzed Reactions
- 14.4 • Enzyme Inhibition
- 14.5 • Kinetics of Enzyme-Catalyzed Reactions Involving Two or More Substrates
- 14.6 • RNA and Antibody Molecules as Enzymes: Ribozymes and Abzymes

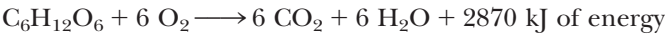
Chapter 14

Enzyme Kinetics



“Alice and the Queen of Hearts,” illustrated by John Tenniel, *The Nursery Alice*. (Mary Evans Picture Library, London)

Living organisms seethe with metabolic activity. Thousands of chemical reactions are proceeding very rapidly at any given instant within all living cells. Virtually all of these transformations are mediated by **enzymes**, proteins (and occasionally RNA) specialized to catalyze metabolic reactions. The substances transformed in these reactions are often organic compounds that show little tendency for reaction outside the cell. An excellent example is glucose, a sugar that can be stored indefinitely on the shelf with no deterioration. Most cells quickly oxidize glucose, producing carbon dioxide and water and releasing lots of energy:



(−2870 kJ/mol is the standard free energy change [ΔG°] for the oxidation of glucose; see Chapter 3). In chemical terms, 2870 kJ is a large amount of energy, and glucose can be viewed as an energy-rich compound even though at ambi-

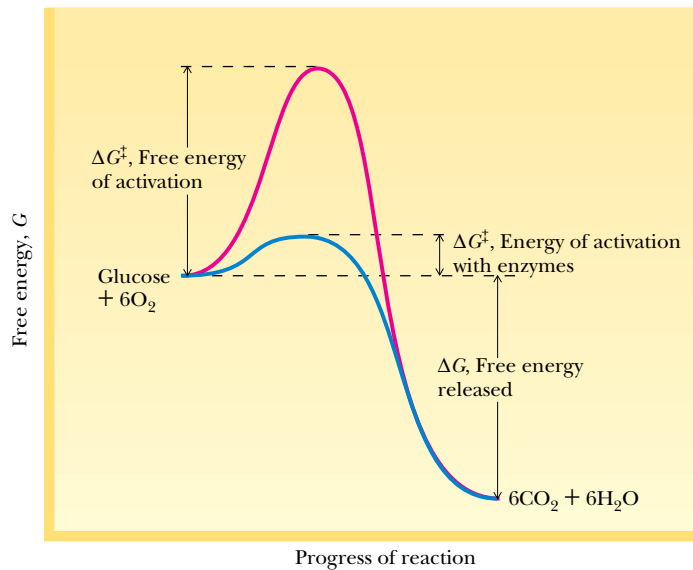


FIGURE 14.1 • Reaction profile showing large $\Delta G^\ddagger$ for glucose oxidation, free energy change of $-2,870$ kJ/mol; catalysts lower $\Delta G^\ddagger$, thereby accelerating rate.

ent temperature it is not readily reactive with oxygen outside of cells. Stated another way, glucose represents **thermodynamic potentiality**: its reaction with oxygen is strongly exergonic, but it just doesn't occur under normal conditions. On the other hand, enzymes can catalyze such thermodynamically favorable reactions so that they proceed at extraordinarily rapid rates (Figure 14.1). In glucose oxidation and countless other instances, enzymes provide cells with the ability to exert *kinetic control over thermodynamic potentiality*. That is, living systems use enzymes to accelerate and control the rates of vitally important biochemical reactions.

Enzymes Are the Agents of Metabolic Function

Acting in sequence, enzymes form metabolic pathways by which nutrient molecules are degraded, energy is released and converted into metabolically useful forms, and precursors are generated and transformed to create the literally thousands of distinctive biomolecules found in any living cell (Figure 14.2). Situated at key junctions of metabolic pathways are specialized **regulatory enzymes** capable of sensing the momentary metabolic needs of the cell and adjusting their catalytic rates accordingly. The responses of these enzymes ensure the harmonious integration of the diverse and often divergent metabolic activities of cells so that the living state is promoted and preserved.

14.1 • Enzymes—Catalytic Power, Specificity, and Regulation

Enzymes are characterized by three distinctive features: **catalytic power**, **specificity**, and **regulation**.

Catalytic Power

Enzymes display enormous catalytic power, accelerating reaction rates as much as 10^{16} over uncatalyzed levels, which is far greater than any synthetic catalysts can achieve, and enzymes accomplish these astounding feats in dilute aqueous

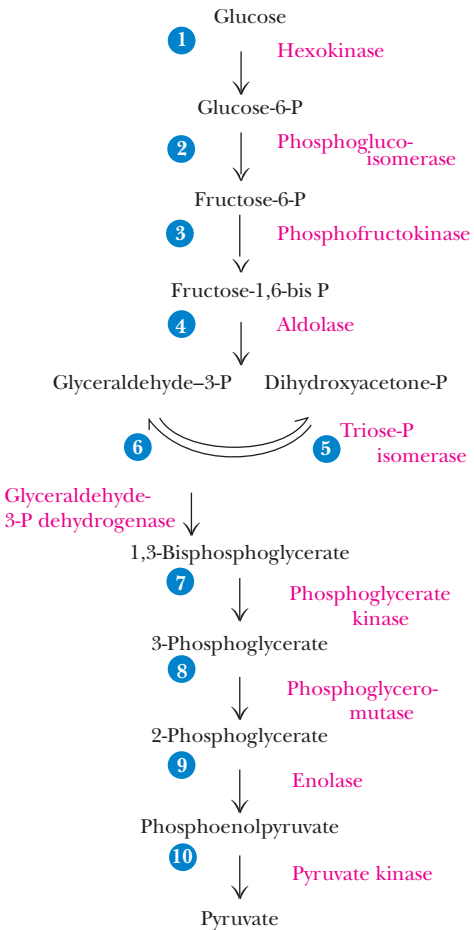
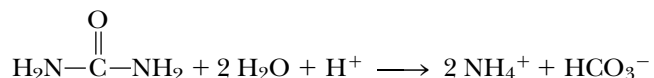


FIGURE 14.2 • The breakdown of glucose by *glycolysis* provides a prime example of a metabolic pathway. Ten enzymes mediate the reactions of glycolysis. Enzyme 4, *fructose 1,6, biphosphate aldolase*, catalyzes the C—C bond-breaking reaction in this pathway.

solutions under mild conditions of temperature and pH. For example, the enzyme jack bean *urease* catalyzes the hydrolysis of urea:



At 20°C, the rate constant for the enzyme-catalyzed reaction is $3 \times 10^4/\text{sec}$; the rate constant for the uncatalyzed hydrolysis of urea is $3 \times 10^{-10}/\text{sec}$. Thus, 10^{14} is the ratio of the catalyzed rate to the uncatalyzed rate of reaction. Such a ratio is defined as the relative **catalytic power** of an enzyme, so the catalytic power of urease is 10^{14} .

Specificity

A given enzyme is very selective, both in the substances with which it interacts and in the reaction that it catalyzes. The substances upon which an enzyme acts are traditionally called **substrates**. In an enzyme-catalyzed reaction, none of the substrate is diverted into nonproductive side-reactions, so no wasteful by-products are produced. It follows then that the products formed by a given enzyme are also very specific. This situation can be contrasted with your own experiences in the organic chemistry laboratory, where yields of 50% or even 30% are viewed as substantial accomplishments (Figure 14.3). The selective qualities of an enzyme are collectively recognized as its **specificity**. Intimate interaction between an enzyme and its substrates occurs through molecular recognition based on structural complementarity; such mutual recognition is the basis of specificity. The specific site on the enzyme where substrate binds and catalysis occurs is called the **active site**.

Regulation

Regulation of enzyme activity is achieved in a variety of ways, ranging from controls over the amount of enzyme protein produced by the cell to more rapid, reversible interactions of the enzyme with metabolic inhibitors and activators. Chapter 15 is devoted to discussions of enzyme regulation. Because most enzymes are proteins, we can anticipate that the functional attributes of enzymes are due to the remarkable versatility found in protein structures.

Enzyme Nomenclature

Traditionally, enzymes often were named by adding the suffix *-ase* to the name of the substrate upon which they acted, as in *urease* for the urea-hydrolyzing enzyme or *phosphatase* for enzymes hydrolyzing phosphoryl groups from organic phosphate compounds. Other enzymes acquired names bearing little resemblance to their activity, such as the peroxide-decomposing enzyme *catalase* or the proteolytic enzymes (*proteases*) of the digestive tract, *trypsin* and *pepsin*. Because of the confusion that arose from these trivial designations, an International Commission on Enzymes was established in 1956 to create a systematic basis for enzyme nomenclature. Although common names for many enzymes remain in use, all enzymes now are classified and formally named according to the reaction they catalyze. Six classes of reactions are recognized (Table 14.1). Within each class are subclasses, and under each subclass are sub-subclasses within which individual enzymes are listed. Classes, subclasses, sub-subclasses, and individual entries are each numbered, so that a series of four numbers serves to specify a particular enzyme. A systematic name, descriptive

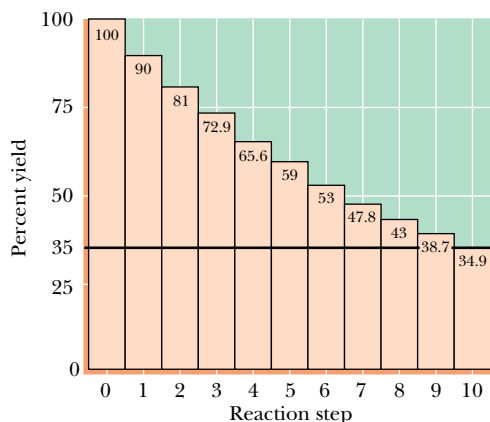


FIGURE 14.3 • A 90% yield over 10 steps, for example, in a metabolic pathway, gives an overall yield of 35%. Therefore, yields in biological reactions *must be substantially greater*; otherwise, unwanted by-products would accumulate to unacceptable levels.

Table 14.1

Systematic Classification of Enzymes According to the Enzyme Commission

E.C. Number	Systematic Name and Subclasses
1	<i>Oxidoreductases</i> (oxidation–reduction reactions)
1.1	Acting on CH—OH group of donors
1.1.1	With NAD or NADP as acceptor
1.1.3	With O ₂ as acceptor
1.2	Acting on the C=O group of donors
1.2.3	With O ₂ as acceptor
1.3	Acting on the CH—CH group of donors
1.3.1	With NAD or NADP as acceptor
2	<i>Transferases</i> (transfer of functional groups)
2.1	Transferring C-1 groups
2.1.1	Methyltransferases
2.1.2	Hydroxymethyltransferases and formyltransferases
2.1.3	Carboxyltransferases and carbamoyltransferases
2.2	Transferring aldehydic or ketonic residues
2.3	Acytransferases
2.4	Glycosyltransferases
2.6	Transferring N-containing groups
2.6.1	Aminotransferases
2.7	Transferring P-containing groups
2.7.1	With an alcohol group as acceptor
3	<i>Hydrolases</i> (hydrolysis reactions)
3.1	Cleaving ester linkage
3.1.1	Carboxylic ester hydrolases
3.1.3	Phosphoric monoester hydrolases
3.1.4	Phosphoric diester hydrolases
4	<i>Lyases</i> (addition to double bonds)
4.1	C=C lyases
4.1.1	Carboxy lyases
4.1.2	Aldehyde lyases
4.2	C=O lyases
4.2.1	Hydrolases
4.3	C=N lyases
4.3.1	Ammonia-lyases
5	<i>Isomerases</i> (isomerization reactions)
5.1	Racemases and epimerases
5.1.3	Acting on carbohydrates
5.2	Cis-trans isomerases
6	<i>Ligases</i> (formation of bonds with ATP cleavage)
6.1	Forming C—O bonds
6.1.1	Amino acid–RNA ligases
6.2	Forming C—S bonds
6.3	Forming C—N bonds
6.4	Forming C—C bonds
6.4.1	Carboxylases

of the reaction, is also assigned to each entry. To illustrate, consider the enzyme that catalyzes this reaction:



A phosphate group is transferred from ATP to the C-6-OH group of glucose, so the enzyme is a *transferase* (Class 2, Table 14.1). Subclass 7 of transferases is

enzymes transferring phosphorus-containing groups, and sub-subclass 1 covers those *phosphotransferases with an alcohol group as an acceptor*. Entry 2 in this sub-subclass is **ATP: D-glucose-6-phosphotransferase**, and its classification number is **2.7.1.2**. In use, this number is written preceded by the letters **E.C.**, denoting the Enzyme Commission. For example, entry 1 in the same sub-subclass is E.C.2.7.1.1, ATP: D-hexose-6-phosphotransferase, an ATP-dependent enzyme that transfers a phosphate to the 6-OH of hexoses (that is, it is nonspecific regarding its hexose acceptor). These designations can be cumbersome, so in everyday usage, trivial names are employed frequently. The glucose-specific enzyme, E.C.2.7.1.2, is called *glucokinase* and the nonspecific E.C.2.7.1.1 is known as *hexokinase*. *Kinase* is a trivial term for enzymes that are ATP-dependent phosphotransferases.

Coenzymes

Many enzymes carry out their catalytic function relying solely on their protein structure. Many others require nonprotein components, called **cofactors** (Table 14.2). Cofactors may be metal ions or organic molecules referred to as **coenzymes**. Cofactors, because they are structurally less complex than proteins, tend to be stable to heat (incubation in a boiling water bath). Typically, proteins are denatured under such conditions. Many coenzymes are vitamins or contain vitamins as part of their structure. Usually coenzymes are actively involved in the catalytic reaction of the enzyme, often serving as intermediate carriers of functional groups in the conversion of substrates to products. In most cases, a coenzyme is firmly associated with its enzyme, perhaps even by covalent bonds, and it is difficult to

Table 14.2

Enzyme Cofactors: Some Metal Ions and Coenzymes and the Enzymes with Which They Are Associated				
Metal Ions and Some Enzymes That Require Them		Coenzymes Serving as Transient Carriers of Specific Atoms or Functional Groups		Representative Enzymes Using Coenzymes
Metal Ion	Enzyme	Coenzyme	Entity Transferred	
Fe ²⁺ or Fe ³⁺	Cytochrome oxidase	Thiamine pyrophosphate (TPP)	Aldehydes	Pyruvate dehydrogenase
	Catalase	Flavin adenine dinucleotide (FAD)	Hydrogen atoms	Succinate dehydrogenase
	Peroxidase	Nicotinamide adenine dinucleotide (NAD)	Hydride ion (H [−])	Alcohol dehydrogenase
Cu ²⁺	Cytochrome oxidase			
Zn ²⁺	DNA polymerase	Coenzyme A (CoA)	Acyl groups	Acetyl-CoA carboxylase
	Carbonic anhydrase	Pyridoxal phosphate (PLP)	Amino groups	Aspartate aminotransferase
	Alcohol dehydrogenase	5'-Deoxyadenosylcobalamin (vitamin B ₁₂)	H atoms and alkyl groups	Methylmalonyl-CoA mutase
Mg ²⁺	Hexokinase			
	Glucose-6-phosphatase	Biotin (biocytin)	CO ₂	Propionyl-CoA carboxylase
Mn ²⁺	Arginase	Tetrahydrofolate (THF)	Other one-carbon groups	Thymidylate synthase
K ⁺	Pyruvate kinase (also requires Mg ²⁺)			
Ni ²⁺	Urease			
Mo	Nitrate reductase			
Se	Glutathione peroxidase			

separate the two. Such tightly bound coenzymes are referred to as **prosthetic groups** of the enzyme. The catalytically active complex of protein and prosthetic group is called the **holoenzyme**. The protein without the prosthetic group is called the **apoenzyme**; it is catalytically inactive.

14.2 • Introduction to Enzyme Kinetics

Kinetics is the branch of science concerned with the rates of chemical reactions. The study of **enzyme kinetics** addresses the biological roles of enzymatic catalysts and how they accomplish their remarkable feats. In enzyme kinetics, we seek to determine the maximum reaction velocity that the enzyme can attain and its binding affinities for substrates and inhibitors. Coupled with studies on the structure and chemistry of the enzyme, analysis of the enzymatic rate under different reaction conditions yields insights regarding the enzyme's mechanism of catalytic action. Such information is essential to an overall understanding of metabolism.

Significantly, this information can be exploited to control and manipulate the course of metabolic events. The science of pharmacology relies on such a strategy. **Pharmaceuticals**, or **drugs**, are often special inhibitors specifically targeted at a particular enzyme in order to overcome infection or to alleviate illness. A detailed knowledge of the enzyme's kinetics is indispensable to rational drug design and successful pharmacological intervention.

Review of Chemical Kinetics

Before beginning a quantitative treatment of enzyme kinetics, it will be fruitful to review briefly some basic principles of chemical kinetics. **Chemical kinetics** is the study of the rates of chemical reactions. Consider a reaction of overall stoichiometry



Although we treat this reaction as a simple, one-step conversion of A to P, it more likely occurs through a sequence of elementary reactions, each of which is a simple molecular process, as in



where I and J represent intermediates in the reaction. Precise description of all of the elementary reactions in a process is necessary to define the overall reaction mechanism for $A \rightarrow P$.

Let us assume that $A \rightarrow P$ is an elementary reaction and that it is spontaneous and essentially irreversible. Irreversibility is easily assumed if the rate of P conversion to A is very slow *or* the concentration of P (expressed as [P]) is negligible under the conditions chosen. The **velocity**, v , or **rate**, of the reaction $A \rightarrow P$ is the amount of P formed or the amount of A consumed per unit time, t . That is,

$$v = \frac{d[P]}{dt} \quad \text{or} \quad v = \frac{-d[A]}{dt} \quad (14.1)$$

The mathematical relationship between reaction rate and concentration of reactant(s) is the **rate law**. For this simple case, the rate law is

$$v = \frac{-d[A]}{dt} = k[A] \quad (14.2)$$

From this expression, it is obvious that the rate is proportional to the concentration of A, and k is the proportionality constant, or **rate constant**. k has the units of $(\text{time})^{-1}$, usually sec^{-1} . v is a function of $[A]$ to the first power, or, in the terminology of kinetics, v is first-order with respect to A. For an elementary reaction, the **order** for any reactant is given by its exponent in the rate equation. The number of molecules that must simultaneously interact is defined as the **molecularity** of the reaction. Thus, the simple elementary reaction of $A \rightarrow P$ is a **first-order reaction**. Figure 14.4 portrays the course of a first-order reaction as a function of time. The rate of decay of a radioactive isotope, like ^{14}C or ^{32}P , is a first-order reaction, as is an intramolecular rearrangement, such as $A \rightarrow P$. Both are **unimolecular reactions** (the molecularity equals 1).

Bimolecular Reactions

Consider the more complex reaction, where two molecules must react to yield products:



Assuming this reaction is an elementary reaction, its molecularity is 2; that is, it is a **bimolecular reaction**. The velocity of this reaction can be determined from the rate of disappearance of either A or B, or the rate of appearance of P or Q:

$$v = \frac{-d[A]}{dt} = \frac{-d[B]}{dt} = \frac{d[P]}{dt} = \frac{d[Q]}{dt} \quad (14.3)$$

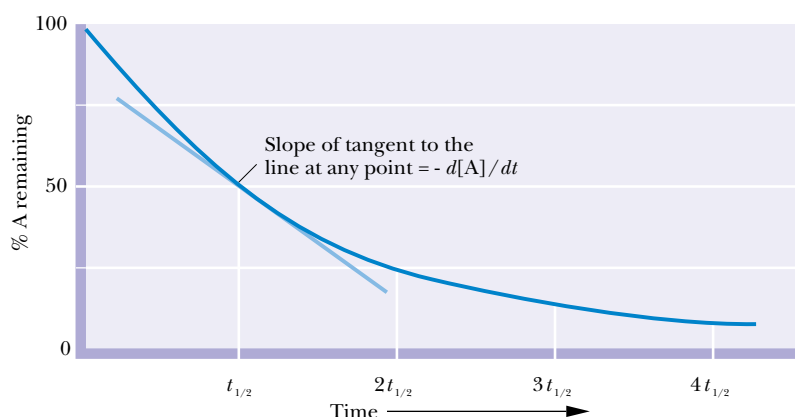
The rate law is

$$v = k[A][B] \quad (14.4)$$

The rate is proportional to the concentrations of both A and B. Because it is proportional to the product of two concentration terms, the reaction is **second-order** overall, first-order with respect to A and first-order with respect to B. (Were the elementary reaction $2A \rightarrow P + Q$, the rate law would be $v = k[A]^2$, second-order overall and second-order with respect to A.) Second-order rate constants have the units of $(\text{concentration})^{-1}(\text{time})^{-1}$, as in $M^{-1}\text{sec}^{-1}$.

Molecularities greater than two are rarely found (and greater than three, never). When the overall stoichiometry of a reaction is greater than two (for example, as in $A + B + C \rightarrow$ or $2A + B \rightarrow$), the reaction almost always proceeds via uni- or bimolecular elementary steps, and the overall rate obeys a simple first- or second-order rate law.

FIGURE 14.4 • Plot of the course of a first-order reaction. The half-time, $t_{1/2}$, is the time for one-half of the starting amount of A to disappear.



At this point, it may be useful to remind ourselves of an important caveat that is the first principle of kinetics: *Kinetics cannot prove a hypothetical mechanism*. Kinetic experiments can only rule out various alternative hypotheses because they don't fit the data. However, through thoughtful kinetic studies, a process of elimination of alternative hypotheses leads ever closer to the reality.

Free Energy of Activation and the Action of Catalysts

In a first-order chemical reaction, the conversion of A to P occurs because, at any given instant, a fraction of the A molecules has the energy necessary to achieve a reactive condition known as the **transition state**. In this state, the probability is very high that the particular rearrangement accompanying the $A \rightarrow P$ transition will occur. This transition state sits at the apex of the energy profile in the energy diagram describing the energetic relationship between A and P (Figure 14.5). The average free energy of A molecules defines the initial state and the average free energy of P molecules is the final state along the reaction coordinate. The rate of any chemical reaction is proportional to the concentration of reactant molecules (A in this case) having this transition-state energy. Obviously, the higher this energy is above the average energy, the smaller the fraction of molecules that will have this energy, and the slower the reaction will proceed. The height of this energy barrier is called the **free energy of activation**, $\Delta G^\ddagger$. Specifically, $\Delta G^\ddagger$ is the energy required to raise the average energy of one mole of reactant (at a given temperature) to the transition-state energy. The relationship between activation energy and the rate constant of the reaction, k , is given by the **Arrhenius equation**:

$$k = Ae^{-\Delta G^\ddagger/RT} \quad (14.5)$$

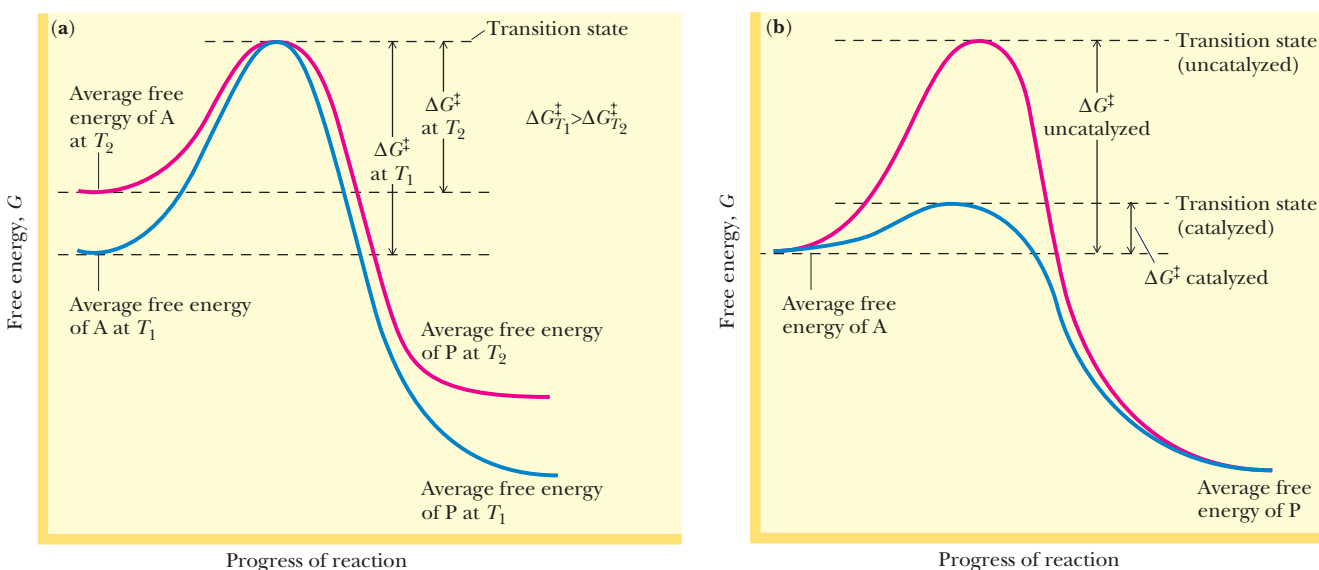


FIGURE 14.5 • Energy diagram for a chemical reaction ($A \rightarrow P$) and the effects of (a) raising the temperature from T_1 to T_2 or (b) adding a catalyst. Raising the temperature raises the average energy of A molecules, which increases the population of A molecules having energies equal to the activation energy for the reaction, thereby increasing the reaction rate. In contrast, the average free energy of A molecules remains the same in uncatalyzed versus catalyzed reactions (conducted at the same temperature). The effect of the catalyst is to lower the free energy of activation for the reaction.

where A is a constant for a particular reaction (not to be confused with the reactant species, A , that we're discussing). Another way of writing this is $1/k = (1/A)e^{\Delta G^\ddagger/RT}$. That is, k is inversely proportional to $e^{\Delta G^\ddagger/RT}$. Therefore, if the energy of activation decreases, the reaction rate increases.

Decreasing $\Delta G^\ddagger$ Increases Reaction Rate

We are familiar with two general ways that rates of chemical reactions may be accelerated. First, the temperature can be raised. This will increase the average energy of reactant molecules, which in effect lowers the energy needed to reach the transition state (Figure 14.5a). The rates of many chemical reactions are doubled by a 10°C rise in temperature. Second, the rates of chemical reactions can also be accelerated by **catalysts**. Catalysts work by lowering the energy of activation rather than by raising the average energy of the reactants (Figure 14.5b). Catalysts accomplish this remarkable feat by combining transiently with the reactants in a way that promotes their entry into the reactive, transition-state condition. Two aspects of catalysts are worth noting: (a) they are regenerated after each reaction cycle ($A \rightarrow P$), and so can be used over and over again; and (b) catalysts have *no* effect on the overall free energy change in the reaction, the free energy difference between A and P (Figure 14.5b).

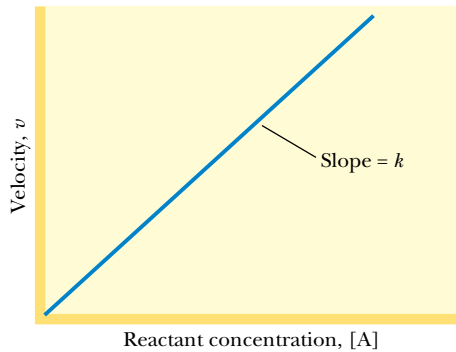


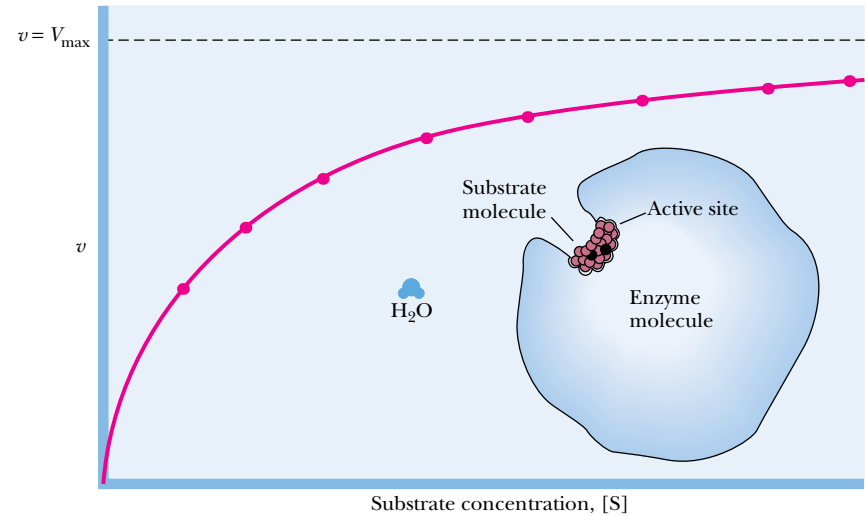
FIGURE 14.6 • A plot of v versus $[A]$ for the unimolecular chemical reaction, $A \rightarrow P$, yields a straight line having a slope equal to k .

14.3 • Kinetics of Enzyme-Catalyzed Reactions

Examination of the change in reaction velocity as the reactant concentration is varied is one of the primary measurements in kinetic analysis. Returning to $A \rightarrow P$, a plot of the reaction rate as a function of the concentration of A yields a straight line whose slope is k (Figure 14.6). The more A that is available, the greater the rate of the reaction, v . Similar analyses of enzyme-catalyzed reactions involving only a single substrate yield remarkably different results (Figure 14.7). At low concentrations of the substrate S , v is proportional to $[S]$, as expected for a first-order reaction. However, v does not increase proportionally as $[S]$ increases, but instead begins to level off. At high $[S]$, v becomes virtually independent of $[S]$ and approaches a maximal limit. The value of v at this limit is written V_{max} . Because rate is no longer dependent on $[S]$ at these high concentrations, the enzyme-catalyzed reaction is now obeying **zero-order**



FIGURE 14.7 • Substrate saturation curve for an enzyme-catalyzed reaction. The amount of enzyme is constant, and the velocity of the reaction is determined at various substrate concentrations. The reaction rate, v , as a function of $[S]$ is described by a rectangular hyperbola. At very high $[S]$, $v = V_{\text{max}}$. That is, the velocity is limited only by conditions (temperature, pH, ionic strength) and by the amount of enzyme present; v becomes independent of $[S]$. Such a condition is termed *zero-order kinetics*. Under zero-order conditions, velocity is directly dependent on $[\text{enzyme}]$. The H_2O molecule provides a rough guide to scale. The substrate is bound at the active site of the enzyme.



kinetics; that is, the rate is independent of the reactant (substrate) concentration. This behavior is a **saturation effect**: when v shows no increase even though $[S]$ is increased, the system is saturated with substrate. Such plots are called **substrate saturation curves**. The physical interpretation is that every enzyme molecule in the reaction mixture has its substrate-binding site occupied by S. Indeed, such curves were the initial clue that an enzyme interacts directly with its substrate by binding it.

The Michaelis–Menten Equation

Lenore Michaelis and Maud L. Menten proposed a general theory of enzyme action in 1913 consistent with observed enzyme kinetics. Their theory was based on the assumption that the enzyme, E, and its substrate, S, associate reversibly to form an enzyme-substrate complex, ES:



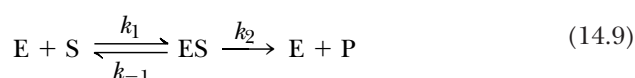
This association/dissociation is assumed to be a rapid equilibrium, and K_s is the *enzyme:substrate dissociation constant*. At equilibrium,

$$k_{-1}[ES] = k_1[E][S] \quad (14.7)$$

and

$$K_s = \frac{[E][S]}{[ES]} = \frac{k_{-1}}{k_1} \quad (14.8)$$

Product, P, is formed in a second step when ES breaks down to yield $E + P$.



E is then free to interact with another molecule of S.

Steady-State Assumption

The interpretations of Michaelis and Menten were refined and extended in 1925 by Briggs and Haldane, by assuming the concentration of the enzyme–substrate complex ES quickly reaches a constant value in such a dynamic system. That is, ES is formed as rapidly from $E + S$ as it disappears by its two possible fates: dissociation to regenerate $E + S$, and reaction to form $E + P$. This assumption is termed the **steady-state assumption** and is expressed as

$$\frac{d[ES]}{dt} = 0 \quad (14.10)$$

That is, the change in concentration of ES with time, t , is 0. Figure 14.8 illustrates the time course for formation of the ES complex and establishment of the steady-state condition.

Initial Velocity Assumption

One other simplification will be advantageous. Because enzymes accelerate the rate of the reverse reaction as well as the forward reaction, it would be helpful to ignore any back reaction by which $E + P$ might form ES. The velocity of this back reaction would be given by $v = k_{-2}[E][P]$. However, if we observe only the *initial velocity* for the reaction immediately after E and S are mixed in the absence of P, the rate of any back reaction is negligible because its rate will

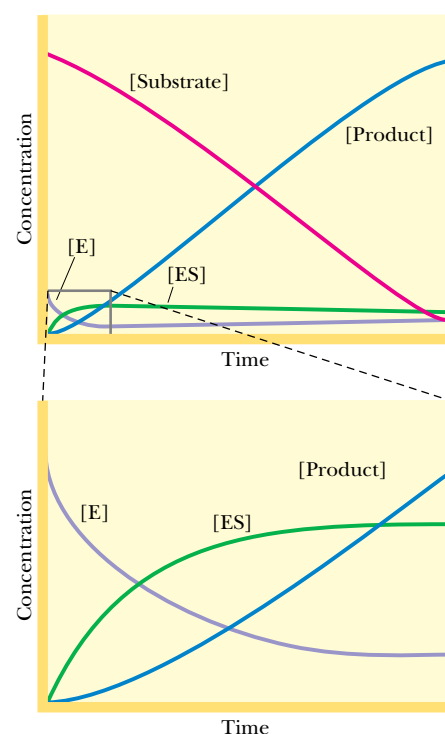


FIGURE 14.8 • Time course for the consumption of substrate, the formation of product, and the establishment of a steady-state level of the enzyme-substrate [ES] complex for a typical enzyme obeying the Michaelis–Menten, Briggs–Haldane models for enzyme kinetics. The early stage of the time course is shown in greater magnification in the bottom graph.

be proportional to $[P]$, and $[P]$ is essentially 0. Given such simplification, we now analyze the system described by Equation (14.9) in order to describe the initial velocity v as a function of $[S]$ and amount of enzyme.

The total amount of enzyme is fixed and is given by the formula

$$\text{Total enzyme, } [E_T] = [E] + [ES] \quad (14.11)$$

where $[E]$ = free enzyme and $[ES]$ = the amount of enzyme in the enzyme–substrate complex. From Equation (14.9), the rate of $[ES]$ formation is

$$v_f = k_1([E_T] - [ES])[S]$$

where

$$[E_T] - [ES] = [E] \quad (14.12)$$

From Equation (14.9), the rate of $[ES]$ disappearance is

$$v_d = k_{-1}[ES] + k_2[ES] = (k_{-1} + k_2)[ES] \quad (14.13)$$

At steady state, $d[ES]/dt = 0$, and therefore, $v_f = v_d$.

So,

$$k_1([E_T] - [ES])[S] = (k_{-1} + k_2)[ES] \quad (14.14)$$

Rearranging gives

$$\frac{([E_T] - [ES])[S]}{[ES]} = \frac{(k_{-1} + k_2)}{k_1} \quad (14.15)$$

The Michaelis Constant, K_m

The ratio of constants $(k_{-1} + k_2)/k_1$ is itself a constant and is defined as the **Michaelis constant, K_m**

$$K_m = \frac{(k_{-1} + k_2)}{k_1} \quad (14.16)$$

Note from (14.15) that K_m is given by the ratio of two concentrations ($([E_T] - [ES])$ and $[S]$) to one ($[ES]$), so K_m has the units of *molarity*. From Equation (14.15), we can write

$$\frac{([E_T] - [ES])[S]}{[ES]} = K_m \quad (14.17)$$

which rearranges to

$$[ES] = \frac{[E_T][S]}{K_m + [S]} \quad (14.18)$$

Now, the most important parameter in the kinetics of any reaction is the **rate of product formation**. This rate is given by

$$v = \frac{d[P]}{dt} \quad (14.19)$$

and for this reaction

$$v = k_2[ES] \quad (14.20)$$

Substituting the expression for $[ES]$ from Equation (14.18) into (14.20) gives

$$v = \frac{k_2[E_T][S]}{K_m + [S]} \quad (14.21)$$

The product $k_2[E_T]$ has special meaning. When $[S]$ is high enough to saturate all of the enzyme, the velocity of the reaction, v , is maximal. At saturation, the

amount of [ES] complex is equal to the total enzyme concentration, E_T , its maximum possible value. From Equation (14.20), the initial velocity v then equals $k_2[E_T] = V_{\max}$. Written symbolically, when $[S] \gg [E_T]$ (and K_m), $[E_T] = [ES]$ and $v = V_{\max}$. Therefore,

$$V_{\max} = k_2[E_T] \quad (14.22)$$

Substituting this relationship into the expression for v gives the **Michaelis–Menten equation**

$$v = \frac{V_{\max}[S]}{K_m + [S]} \quad (14.23)$$

This equation says that the rate of an enzyme-catalyzed reaction, v , at any moment is determined by two constants, K_m and $V_{\max}$, and the concentration of substrate at that moment.

When $[S] = K_m$, $v = V_{\max}/2$

We can provide an operational definition for the constant K_m by rearranging Equation (14.23) to give

$$K_m = [S] \left(\frac{V_{\max}}{v} - 1 \right) \quad (14.24)$$

Then, at $v = V_{\max}/2$, $K_m = [S]$. That is, K_m is defined by the substrate concentration that gives a velocity equal to one-half the maximal velocity. Table 14.3 gives the K_m values of some enzymes for their substrates.

Relationships Between $V_{\max}$, K_m , and Reaction Order

The Michaelis–Menten equation (14.23) describes a curve known from analytical geometry as a *rectangular hyperbola*.¹ In such curves, as $[S]$ is increased, v approaches the limiting value, $V_{\max}$, in an asymptotic fashion. $V_{\max}$ can be approximated experimentally from a substrate saturation curve (Figure 14.7), and K_m can be derived from $V_{\max}/2$, so the two constants of the Michaelis–Menten equation can be obtained from plots of v versus $[S]$. Note, however, that actual estimation of $V_{\max}$, and consequently K_m , is only approximate from such graphs. That is, according to Equation (14.23), in order to get $v = 0.99 V_{\max}$, $[S]$ must equal $99 K_m$, a concentration that may be difficult to achieve in practice.

From Equation (14.23), when $[S] \gg K_m$, then $v = V_{\max}$. That is, v is no longer dependent on $[S]$, so the reaction is obeying zero-order kinetics. Also, when $[S] < K_m$, then $v \approx (V_{\max}/K_m)[S]$. That is, the rate, v , approximately follows a first-order rate equation, $v = k'[A]$, where $k' = V_{\max}/K_m$.

K_m and $V_{\max}$, once known explicitly, define the rate of the enzyme-catalyzed reaction, *provided*:

1. The reaction involves only one substrate, *or* if the reaction is multisubstrate, the concentration of only one substrate is varied while the concentration of all other substrates is held constant.
2. The reaction $ES \rightarrow E + P$ is irreversible, *or* the experiment is limited to observing only initial velocities where $[P] = 0$.
3. $[S]_0 > [E_T]$ and $[E_T]$ is held constant.
4. All other variables that might influence the rate of the reaction (temperature, pH, ionic strength, and so on) are constant.

¹A proof that the Michaelis–Menten equation describes a rectangular hyperbola is given by Naqui, A., 1986. Where are the asymptotes of Michaelis–Menten? *Trends in Biochemical Sciences* **11**:64–65.

Table 14.3

<i>K_m</i> Values for Some Enzymes		
Enzyme	Substrate	<i>K_m</i> (mM)
Carbonic anhydrase	CO ₂	12
Chymotrypsin	<i>N</i> -Benzoyltyrosinamide	2.5
	Acetyl-L-tryptophanamide	5
	<i>N</i> -Formyltyrosinamide	12
	<i>N</i> -Acetyltyrosinamide	32
	Glycyltyrosinamide	122
Hexokinase	Glucose	0.15
	Fructose	1.5
β-Galactosidase	Lactose	4
Glutamate dehydrogenase	NH ₄ ⁺	57
	Glutamate	0.12
	α-Ketoglutarate	2
	NAD ⁺	0.025
	NADH	0.018
Aspartate aminotransferase	Aspartate	0.9
	α-Ketoglutarate	0.1
	Oxaloacetate	0.04
	Glutamate	4
Threonine deaminase	Threonine	5
Arginyl-tRNA synthetase	Arginine	0.003
	tRNA ^{Arg}	0.0004
	ATP	0.3
Pyruvate carboxylase	HCO ₃ [−]	1.0
	Pyruvate	0.4
	ATP	0.06
Penicillinase	Benzylpenicillin	0.05
Lysozyme	Hexa- <i>N</i> -acetylglucosamine	0.006

Enzyme Units

In many situations, the actual molar amount of the enzyme is not known. However, its amount can be expressed in terms of the activity observed. The International Commission on Enzymes defines **One International Unit** of enzyme as *the amount that catalyzes the formation of one micromole of product in one minute*. (Because enzymes are very sensitive to factors such as pH, temperature, and ionic strength, the conditions of assay must be specified.) Another definition for units of enzyme activity is the **katal**. One katal is *that amount of enzyme catalyzing the conversion of one mole of substrate to product in one second*. Thus, one katal equals 6×10^7 international units.

Turnover Number

The **turnover number** of an enzyme, k_{cat} , is a measure of its maximal catalytic activity. k_{cat} is defined as the number of substrate molecules converted into product per enzyme molecule per unit time when the enzyme is saturated with substrate. The turnover number is also referred to as the **molecular activity** of the enzyme. For the simple Michaelis–Menten reaction (14.9) under conditions of initial velocity measurements, $k_2 = k_{\text{cat}}$. Provided the concentration of

enzyme, $[E_T]$, in the reaction mixture is known, k_{cat} can be determined from V_{max} . At saturating $[S]$, $v = V_{\text{max}} = k_2[E_T]$. Thus,

$$k_2 = \frac{V_{\text{max}}}{[E_T]} = k_{\text{cat}} \tag{14.25}$$

The term k_{cat} represents the kinetic efficiency of the enzyme. Table 14.4 lists turnover numbers for some representative enzymes. Catalase has the highest turnover number known; each molecule of this enzyme can degrade 40 million molecules of H_2O_2 in one second! At the other end of the scale, lysozyme requires 2 seconds to cleave a glycosidic bond in its glycan substrate.

k_{cat}/K_m

Under physiological conditions, $[S]$ is seldom saturating, and k_{cat} itself is not particularly informative. That is, the *in vivo* ratio of $[S]/K_m$ usually falls in the range of 0.01 to 1.0, so active sites often are not filled with substrate. Nevertheless, we can derive a meaningful index of the efficiency of Michaelis–Menten-type enzymes under these conditions by employing the following equations. As presented in Equation (14.23), if

$$v = \frac{V_{\text{max}}[S]}{K_m + [S]}$$

and $V_{\text{max}} = k_{\text{cat}}[E_T]$, then

$$v = \frac{k_{\text{cat}}[E_T][S]}{K_m + [S]} \tag{14.26}$$

When $[S] \ll K_m$, the concentration of free enzyme, $[E]$, is approximately equal to $[E_T]$, so that

$$v = \left(\frac{k_{\text{cat}}}{K_m}\right)[E][S] \tag{14.27}$$

That is, k_{cat}/K_m is an *apparent second-order rate constant* for the reaction of E and S to form product. Because K_m is inversely proportional to the affinity of the enzyme for its substrate and k_{cat} is directly proportional to the kinetic efficiency of the enzyme, k_{cat}/K_m provides an index of the catalytic efficiency of an enzyme operating at substrate concentrations substantially below saturation amounts.

An interesting point emerges if we restrict ourselves to the simple case where $k_{\text{cat}} = k_2$. Then

$$\frac{k_{\text{cat}}}{K_m} = \frac{k_1k_2}{(k_{-1} + k_2)} \tag{14.28}$$

But k_1 must always be greater than or equal to $k_1k_2/(k_{-1} + k_2)$. That is, the reaction can go no faster than the rate at which E and S come together. Thus, k_1 sets the upper limit for k_{cat}/K_m . In other words, *the catalytic efficiency of an enzyme cannot exceed the diffusion-controlled rate of combination of E and S to form ES*. In H_2O , the rate constant for such diffusion is approximately $10^9/M \cdot \text{sec}$. Those enzymes that are most efficient in their catalysis have k_{cat}/K_m ratios approaching this value. Their catalytic velocity is limited only by the rate at which they encounter S; enzymes this efficient have achieved so-called *catalytic perfection*. All E and S encounters lead to reaction because such “catalytically perfect” enzymes can channel S to the active site, regardless of where S hits E. Table 14.5 lists the kinetic parameters of several enzymes in this category. Note that k_{cat} and K_m both show a substantial range of variation in this table, even though their ratio falls around $10^8/M \cdot \text{sec}$.

Table 14.4

Values of k_{cat} (Turnover Number) for Some Enzymes

Enzyme	k_{cat} (sec ^{−1})
Catalase	40,000,000
Carbonic anhydrase	1,000,000
Acetylcholinesterase	14,000
Penicillinase	2,000
Lactate dehydrogenase	1,000
Chymotrypsin	100
DNA polymerase I	15
Lysozyme	0.5

Table 14.5

Enzymes Whose k_{cat}/K_m Approaches the Diffusion-Controlled Rate of Association with Substrate

Enzyme	Substrate	k_{cat} (sec ^{−1})	K_m (M)	k_{cat}/K_m (sec ^{−1} M ^{−1})
Acetylcholinesterase	Acetylcholine	1.4×10^4	9×10^{-5}	1.6×10^8
Carbonic anhydrase	CO ₂	1×10^6	0.012	8.3×10^7
	HCO ₃ [−]	4×10^5	0.026	1.5×10^7
Catalase	H ₂ O ₂	4×10^7	1.1	4×10^7
Crotonase	Crotonyl-CoA	5.7×10^3	2×10^{-5}	2.8×10^8
Fumarase	Fumarate	800	5×10^{-6}	1.6×10^8
	Malate	900	2.5×10^{-5}	3.6×10^7
Triosephosphate isomerase	Glyceraldehyde- 3-phosphate*	4.3×10^3	1.8×10^{-5}	2.4×10^8
β-Lactamase	Benzylpenicillin	2×10^3	2×10^{-5}	1×10^8

* K_m for glyceraldehyde-3-phosphate is calculated on the basis that only 3.8% of the substrate in solution is unhydrated and therefore reactive with the enzyme.
Adapted from Fersht, A. 1985. *Enzyme Structure and Mechanism*, 2nd ed. New York: W.H. Freeman & Co.

Linear Plots Can Be Derived from the Michaelis–Menten Equation

Because of the hyperbolic shape of v versus $[S]$ plots, V_{max} can only be determined from an extrapolation of the asymptotic approach of v to some limiting value as $[S]$ increases indefinitely (Figure 14.7); and K_m is derived from that value of $[S]$ giving $v = V_{\text{max}}/2$. However, several rearrangements of the Michaelis–Menten equation transform it into a straight-line equation. The best known of these is the **Lineweaver–Burk double-reciprocal plot**:

Taking the reciprocal of both sides of the Michaelis–Menten equation, Equation (14.23), yields the equality

$$\frac{1}{v} = \left(\frac{K_m}{V_{\text{max}}}\right)\left(\frac{1}{[S]}\right) + \frac{1}{V_{\text{max}}}$$

(14.29)

This conforms to $y = mx + b$ (the equation for a straight line), where $y = 1/v$; m , the slope, is K_m/V_{max} ; $x = 1/[S]$; and $b = 1/V_{\text{max}}$. Plotting $1/v$ versus $1/[S]$ gives a straight line whose x -intercept is $-1/K_m$, whose y -intercept is $1/V_{\text{max}}$, and whose slope is K_m/V_{max} (Figure 14.9).

The **Hanes–Woelf plot** is another rearrangement of the Michaelis–Menten equation that yields a straight line:

Multiplying both sides of Equation (14.29) by $[S]$ gives

$$\frac{[S]}{v} = [S]\left(\frac{K_m}{V_{\text{max}}}\right)\left(\frac{1}{[S]}\right) + \frac{[S]}{V_{\text{max}}} = \frac{K_m}{V_{\text{max}}} + \frac{[S]}{V_{\text{max}}}$$

(14.30)

and

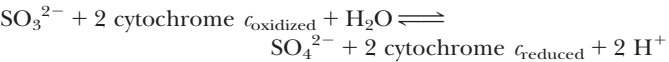
$$\frac{[S]}{v} = \left(\frac{1}{V_{\text{max}}}\right)[S] + \frac{K_m}{V_{\text{max}}}$$

(14.31)

Graphing $[S]/v$ versus $[S]$ yields a straight line where the slope is $1/V_{\text{max}}$, the

An Example of the Effect of Amino Acid Substitutions on K_m and k_{cat} :
Wild-Type and Mutant Forms of Human Sulfite Oxidase

Mammalian sulfite oxidase is the last enzyme in the pathway for degradation of sulfur-containing amino acids. Sulfite oxidase (SO) catalyzes the oxidation of sulfite (SO_3^{2-}) to sulfate (SO_4^{2-}), using the heme-containing protein, cytochrome c , as electron acceptor:



Isolated sulfite oxidase deficiency is a rare and often fatal genetic disorder in humans. The disease is characterized by severe neurological abnormalities, revealed as convulsions shortly after birth. R. M. Garrett and K. V. Rajagopalan at Duke University Medical Center have isolated the human cDNA for sulfite oxidase from the cells of normal (*wild-type*) and SO-deficient individuals. Expression of these SO cDNAs in transformed *Escherichia coli* cells allowed the isolation and kinetic analysis of wild-type and mutant forms of SO, including one (designated R160Q) in which the Arg at position 160 in the polypeptide chain is replaced by Gln. A genetically engineered version of SO (designated R160K) in which Lys replaces Arg¹⁶⁰ was also studied.

Kinetic Constants for Wild-Type and Mutant Sulfite Oxidase

Enzyme	$K_m^{\text{ sulfite}}$ (μM)	$k_{cat}(\text{sec}^{-1})$	k_{cat}/K_m ($10^6 \text{ M}^{-1}\text{sec}^{-1}$)
Wild-type	17	18	1.1
R160Q	1900	3	0.0016
R160K	360	5.5	0.015

Replacing R¹⁶⁰ in sulfite oxidase by Q increases K_m , decreases k_{cat} , and markedly diminishes the catalytic efficiency (k_{cat}/K_m) of the enzyme. The R160K mutant enzyme has properties intermediate between wild-type and the R160Q mutant form. The substrate, SO_3^{2-} , is strongly anionic, and R¹⁶⁰ is one of several Arg residues situated within the SO substrate-binding site. Positively charged side chains in the substrate-binding site facilitate SO_3^{2-} binding and catalysis, with Arg being optimal in this role.

y-intercept is K_m/V_{max} , and the x-intercept is $-K_m$, as shown in Figure 14.10. The common advantage of these plots is that they allow both K_m and V_{max} to be accurately estimated by extrapolation of straight lines rather than asymptotes. Computer fitting of v versus $[S]$ data to the Michaelis–Menten equation is more commonly done than graphical plotting.

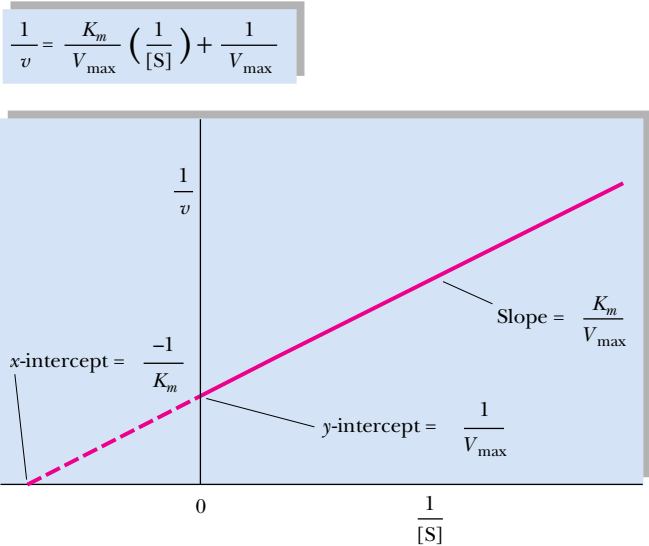


FIGURE 14.9 • The Lineweaver–Burk double-reciprocal plot, depicting extrapolations that allow the determination of the x- and y-intercepts and slope.

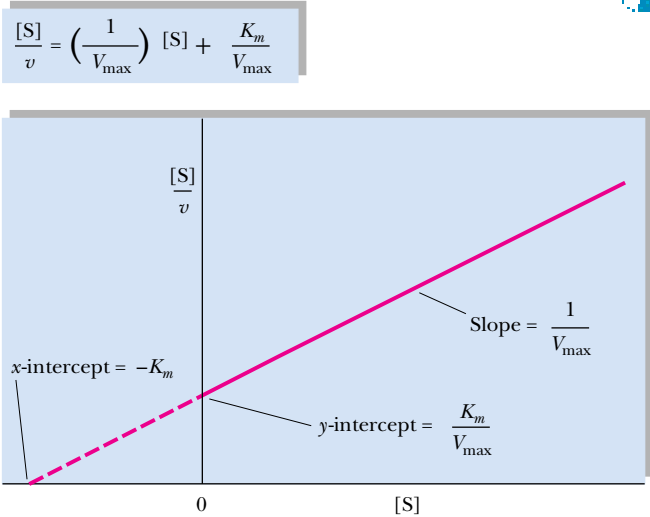


FIGURE 14.10 • A Hanes–Wolff plot of $[S]/v$ versus $[S]$, another straight-line rearrangement of the Michaelis–Menten equation.

Departures from Linearity: A Hint of Regulation?

If the kinetics of the reaction disobey the Michaelis–Menten equation, the violation is revealed by a departure from linearity in these straight-line graphs. We shall see in the next chapter that such deviations from linearity are characteristic of the kinetics of regulatory enzymes known as **allosteric enzymes**. Such regulatory enzymes are very important in the overall control of metabolic pathways.

Effect of pH on Enzymatic Activity

Enzyme-substrate recognition and the catalytic events that ensue are greatly dependent on pH. An enzyme possesses an array of ionizable side chains and prosthetic groups that not only determine its secondary and tertiary structure but may also be intimately involved in its active site. Further, the substrate itself often has ionizing groups, and one or another of the ionic forms may preferentially interact with the enzyme. Enzymes in general are active only over a limited pH range and most have a particular pH at which their catalytic activity is optimal. These effects of pH may be due to effects on K_m or V_{max} or both. Figure 14.11 illustrates the relative activity of four enzymes as a function of pH. Although the pH optimum of an enzyme often reflects the pH of its normal environment, the optimum may not be precisely the same. This difference suggests that the pH-activity response of an enzyme may be a factor in the intracellular regulation of its activity.

Effect of Temperature on Enzymatic Activity

Like most chemical reactions, the rates of enzyme-catalyzed reactions generally increase with increasing temperature. However, at temperatures above 50° to 60°C, enzymes typically show a decline in activity (Figure 14.12). Two effects are operating here: (a) the characteristic increase in reaction rate with temperature, and (b) thermal denaturation of protein structure at higher tem-

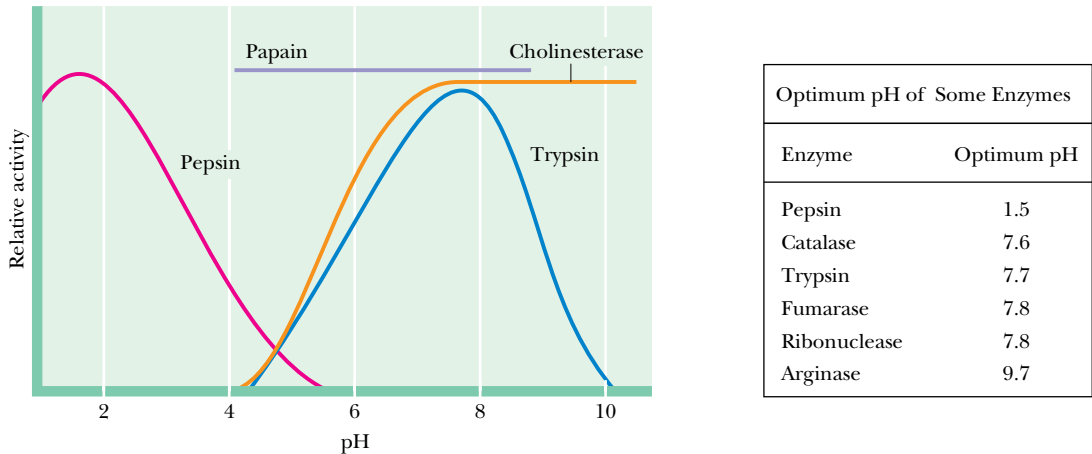


FIGURE 14.11 • The pH activity profiles of four different enzymes. *Trypsin*, an intestinal protease, has a slightly alkaline pH optimum, whereas *pepsin*, a gastric protease, acts in the acidic confines of the stomach and has a pH optimum near 2. *Papain*, a protease found in papaya, is relatively insensitive to pHs between 4 and 8. *Cholinesterase* activity is pH-sensitive below pH 7 but not between pH 7 and 10. The cholinesterase pH activity profile suggests that an ionizable group with a pK' near 6 is essential to its activity. Might it be a histidine residue within the active site?

peratures. Most enzymatic reactions double in rate for every 10°C rise in temperature (that is, $Q_{10} = 2$, where Q_{10} is defined as *the ratio of activities at two temperatures 10° apart*) as long as the enzyme is stable and fully active. Some enzymes, those catalyzing reactions having very high activation energies, show proportionally greater Q_{10} values. The increasing rate with increasing temperature is ultimately offset by the instability of higher orders of protein structure at elevated temperatures, where the enzyme is inactivated. Not all enzymes are quite so thermally labile. For example, the enzymes of thermophilic bacteria (*thermophilic* = “heat-loving”) found in geothermal springs retain full activity at temperatures in excess of 85°C.

14.4 • Enzyme Inhibition

If the velocity of an enzymatic reaction is decreased or **inhibited**, the kinetics of the reaction obviously have been perturbed. Systematic perturbations are a basic tool of experimental scientists; much can be learned about the normal workings of any system by inducing changes in it and then observing the effects of the change. The study of enzyme inhibition has contributed significantly to our understanding of enzymes.

Reversible Versus Irreversible Inhibition

Enzyme inhibitors are classified in several ways. The inhibitor may interact either reversibly or irreversibly with the enzyme. **Reversible inhibitors** interact with the enzyme through noncovalent association/dissociation reactions. In contrast, **irreversible inhibitors** usually cause stable, covalent alterations in the enzyme. That is, the consequence of irreversible inhibition is a decrease in the concentration of active enzyme. The kinetics observed are consistent with this interpretation, as we shall see later.

Reversible Inhibition

Reversible inhibitors fall into two major categories: competitive and noncompetitive (although other more unusual and rare categories are known). **Competitive inhibitors** are characterized by the fact that the substrate and inhibitor compete for the same binding site on the enzyme, the so-called **active site** or **S-binding site**. Thus, increasing the concentration of S favors the likelihood of S binding to the enzyme instead of the inhibitor, I. That is, high [S] can overcome the effects of I. The other major type, noncompetitive inhibition, cannot be overcome by increasing [S]. The two types can be distinguished by the particular patterns obtained when the kinetic data are analyzed in linear plots, such as double-reciprocal (Lineweaver–Burk) plots. A general formulation for common inhibitor interactions in our simple enzyme kinetic model would include



That is, we consider here reversible combinations of the inhibitor with E and/or ES.

Competitive Inhibition

Consider the following system

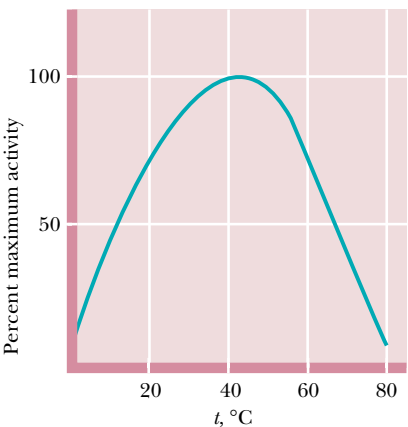
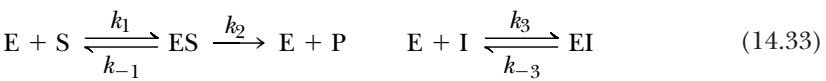


FIGURE 14.12 • The effect of temperature on enzyme activity. The relative activity of an enzymatic reaction as a function of temperature. The decrease in the activity above 50°C is due to thermal denaturation.

Table 14.6
The Effect of Various Types of Inhibitors on the Michaelis–Menten Rate Equation and on Apparent K_m and Apparent $V_{\max}$

Inhibition Type	Rate Equation	Apparent K_m	Apparent $V_{\max}$
None	$v = V_{\max}[S] / (K_m + [S])$	K_m	$V_{\max}$
Competitive	$v = V_{\max}[S] / ([S] + K_m(1 + [I] / K_I))$	$K_m(1 + [I] / K_I)$	$V_{\max}$
Noncompetitive	$v = (V_{\max}[S] / (1 + [I] / K_I)) / (K_m + [S])$	K_m	$V_{\max} / (1 + [I] / K_I)$
Mixed	$v = V_{\max}[S] / ((1 + [I] / K_I) K_m + (1 + [I] / K_I')[S])$	$K_m(1 + [I] / K_I) / (1 + [I] / K_I')$	$V_{\max} / (1 + [I] / K_I')$

K_I is defined as the enzyme:inhibitor dissociation constant $K_I = [E][I] / [EI]$; K_I' is defined as the enzyme substrate complex :inhibitor dissociation constant $K_I' = [ES][I] / [ESI]$

where an inhibitor, I, binds *reversibly* to the enzyme at the same site as S. S-binding and I-binding are mutually exclusive, *competitive* processes. Formation of the ternary complex, EIS, where both S and I are bound, is physically impossible. This condition leads us to anticipate that S and I must share a high degree of structural similarity because they bind at the same site on the enzyme. Also notice that, in our model, EI does not react to give rise to E + P. That is, I is not changed by interaction with E. The rate of the product-forming reaction is $v = k_2[ES]$.

It is revealing to compare the equation for the uninhibited case, Equation (14.23) (the Michaelis–Menten equation) with Equation (14.43) for the rate of the enzymatic reaction in the presence of a fixed concentration of the competitive inhibitor, [I]

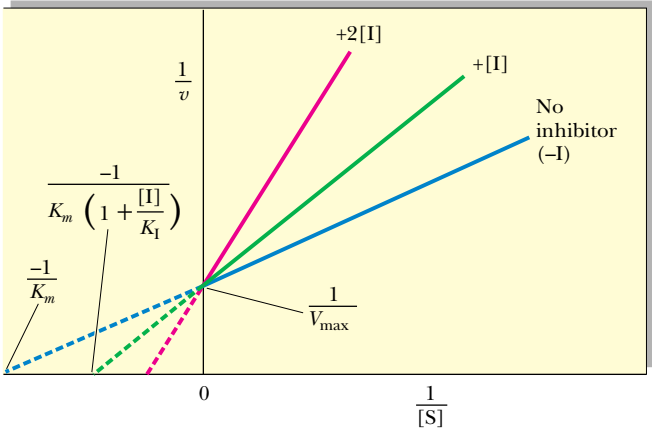
$$v = \frac{V_{\max}[S]}{[S] + K_m}$$

$$v = \frac{V_{\max}[S]}{[S] + K_m\left(1 + \frac{[I]}{K_I}\right)}$$

(see also Table 14.6). The K_m term in the denominator in the inhibited case is increased by the factor $(1 + [I] / K_I)$; thus, v is less in the presence of the inhibitor, as expected. Clearly, in the absence of I, the two equations are identical. Figure 14.13 shows a Lineweaver–Burk plot of competitive inhibition.



FIGURE 14.13 • Lineweaver–Burk plot of competitive inhibition, showing lines for no I, [I], and 2[I]. Note that when [S] is infinitely large ($1/[S] = 0$), $V_{\max}$ is the same, whether I is present or not. In the presence of I, the negative x-intercept $= -1/K_m(1 + [I] / K_I)$.



DEEPER LOOK

The Equations of Competitive Inhibition

Given the relationships between E, S, and I described previously and recalling the steady-state assumption that $d[ES]/dt = 0$, from Equations (14.14) and (14.16) we can write

[ES] = (k1[E][S]) / (k2 + k-1) = [E][S] / Km (14.34)

Assuming that E + I ⇌ EI reaches rapid equilibrium, the rate of EI formation, $v_f' = k_3[E][I]$, and the rate of disappearance of EI, $v_d' = k_{-3}[EI]$, are equal. So,

k3[E][I] = k-3[EI] (14.35)

Therefore,

[EI] = (k3/k-3) [E][I] (14.36)

If we define K_I as k_{-3}/k_3 , an enzyme-inhibitor dissociation constant, then

[EI] = ([E][I]) / KI (14.37)

knowing $[E_T] = [E] + [ES] + [EI]$. Then

[E_T] = [E] + ([E][S]) / Km + ([E][I]) / KI (14.38)

Solving for [E] gives

[E] = (KI Km [E_T]) / (KI Km + KI [S] + Km [I]) (14.39)

Because the rate of product formation is given by $v = k_2[ES]$, from Equation (14.34) we have

v = (k2[E][S]) / Km (14.40)

So,

v = (k2 KI [E_T] [S]) / (KI Km + KI [S] + Km [I]) (14.41)

Because $V_{max} = k_2[E_T]$,

v = (Vmax [S]) / (Km + [S] + (Km [I]) / KI) (14.42)

or

v = (Vmax [S]) / ([S] + Km (1 + [I] / KI)) (14.43)

Several features of competitive inhibition are evident. First, at a given [I], v decreases ($1/v$ increases). When [S] becomes infinite, $v = V_{max}$ and is unaffected by I because all of the enzyme is in the ES form. Note that the value of the $-x$ -intercept decreases as [I] increases. This $-x$ -intercept is often termed the *apparent* K_m (or K_{mapp}) because it is the K_m apparent under these conditions. The diagnostic criterion for competitive inhibition is that V_{max} is unaffected by I; that is, all lines share a common y -intercept. This criterion is also the best experimental indication of binding at the same site by two substances. Competitive inhibitors resemble S structurally.

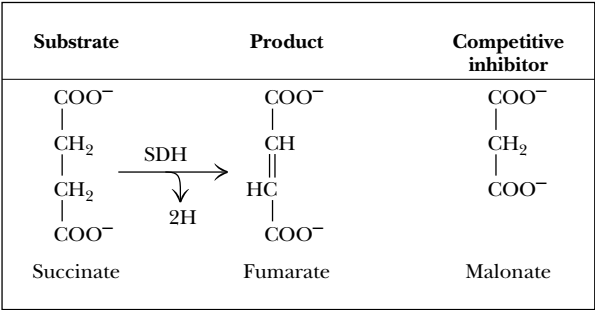
Succinate Dehydrogenase—A Classic Example of Competitive Inhibition

The enzyme *succinate dehydrogenase* (SDH) is competitively inhibited by malonate. Figure 14.14 shows the structures of succinate and malonate. The structural similarity between them is obvious and is the basis of malonate’s ability to mimic succinate and bind at the active site of SDH. However, unlike succinate, which is oxidized by SDH to form fumarate, malonate cannot lose two hydrogens; consequently, it is unreactive.

Noncompetitive Inhibition

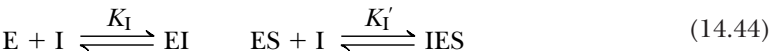
Noncompetitive inhibitors interact with both E and ES (or with S and ES, but this is a rare and specialized case). Obviously, then, the inhibitor is not binding to the same site as S, and the inhibition cannot be overcome by raising [S]. There are two types of noncompetitive inhibition: pure and mixed.

FIGURE 14.14 • Structures of succinate, the substrate of succinate dehydrogenase (SDH), and malonate, the competitive inhibitor. Fumarate (the product of SDH action on succinate) is also shown.



Pure Noncompetitive Inhibition

In this situation, the binding of I by E has no effect on the binding of S by E. That is, S and I bind at different sites on E, and binding of I does not affect binding of S. Consider the system

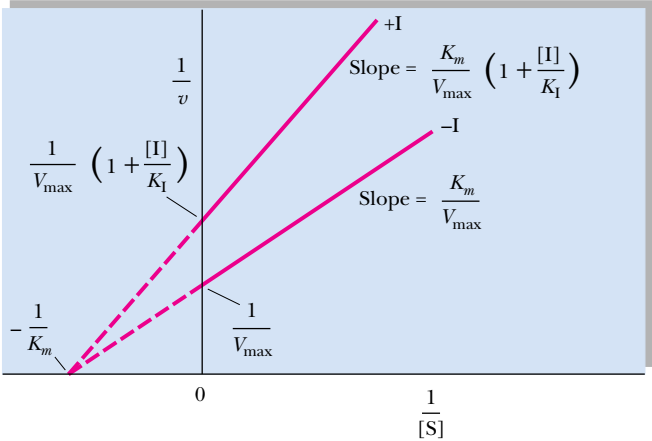


Pure noncompetitive inhibition occurs if $K_I = K_I'$. This situation is relatively uncommon; the Lineweaver–Burk plot for such an instance is given in Figure 14.15. Note that K_m is unchanged by I (the x -intercept remains the same, with or without I). Note also that V_{max} decreases. A similar pattern is seen if the amount of enzyme in the experiment is decreased. Thus, it is as if I lowered $[E]$.

Mixed Noncompetitive Inhibition

In this situation, the binding of I by E influences the binding of S by E. Either the binding sites for I and S are near one another or conformational changes in E caused by I affect S binding. In this case, K_I and K_I' , as defined previously, are not equal. Both K_m and V_{max} are altered by the presence of I, and K_m/V_{max} is not constant (Figure 14.16). This inhibitory pattern is commonly encountered. A reasonable explanation is that the inhibitor is binding at a site distinct from the active site, yet is influencing the binding of S at the active site.

FIGURE 14.15 • Lineweaver–Burk plot of pure noncompetitive inhibition. Note that I does not alter K_m but that it decreases V_{max} . In the presence of I, the y -intercept is equal to $(1/V_{\text{max}})(1 + I/K_I)$.



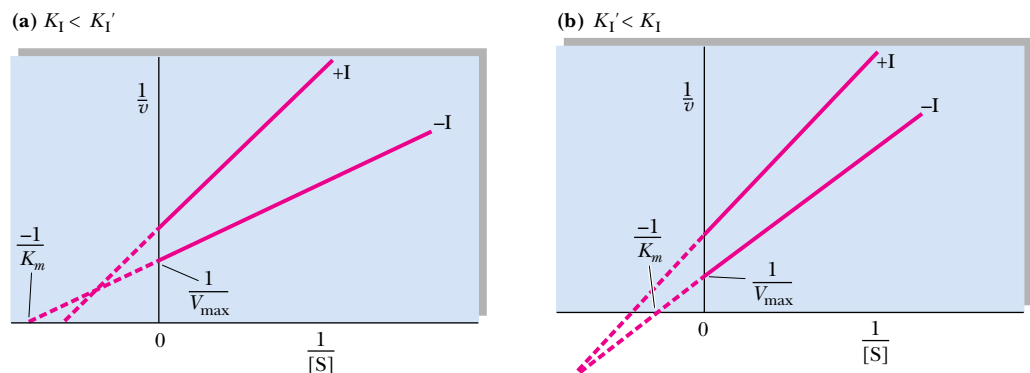


FIGURE 14.16 • Lineweaver–Burk plot of mixed noncompetitive inhibition. Note that both intercepts and the slope change in the presence of I. (a) When K_I is less than K_I' ; (b) when K_I is greater than K_I' .

Presumably, these effects are transmitted via alterations in the protein’s conformation. Table 14.6 includes the rate equations and apparent K_m and $V_{\max}$ values for both types of noncompetitive inhibition.

Irreversible Inhibition

If the inhibitor combines irreversibly with the enzyme—for example, by covalent attachment—the kinetic pattern seen is like that of noncompetitive inhibition, because the net effect is a loss of active enzyme. Usually, this type of inhibition can be distinguished from the noncompetitive, reversible inhibition case since the reaction of I with E (and/or ES) is not instantaneous. Instead, there is a *time-dependent decrease in enzymatic activity* as $E + I \rightarrow EI$ proceeds, and the rate of this inactivation can be followed. Also, unlike reversible inhibitions, dilution or dialysis of the enzyme:inhibitor solution does not dissociate the EI complex and restore enzyme activity.

Suicide Substrates —Mechanism-Based Enzyme Inactivators

Suicide substrates are inhibitory substrate analogs designed so that, via normal catalytic action of the enzyme, a very reactive group is generated. This reactive group then forms a covalent bond with a nearby functional group within the active site of the enzyme, thereby causing irreversible inhibition. Suicide substrates, also called *Trojan horse substrates*, are a type of **affinity label**. As substrate analogs, they bind with specificity and high affinity to the enzyme active site; in their reactive form, they become covalently bound to the enzyme. This covalent link effectively labels a particular functional group within the active site, identifying the group as a key player in the enzyme’s catalytic cycle.

Penicillin—A Suicide Substrate

Several drugs in current medical use are mechanism-based enzyme inactivators. For example, the antibiotic **penicillin** exerts its effects by covalently reacting with an essential serine residue in the active site of *glycoprotein peptidase*, an enzyme that acts to cross-link the peptidoglycan chains during synthesis of bacterial cell walls (Figure 14.17). Once cell wall synthesis is blocked, the bacterial cells are very susceptible to rupture by osmotic lysis, and bacterial growth is halted.

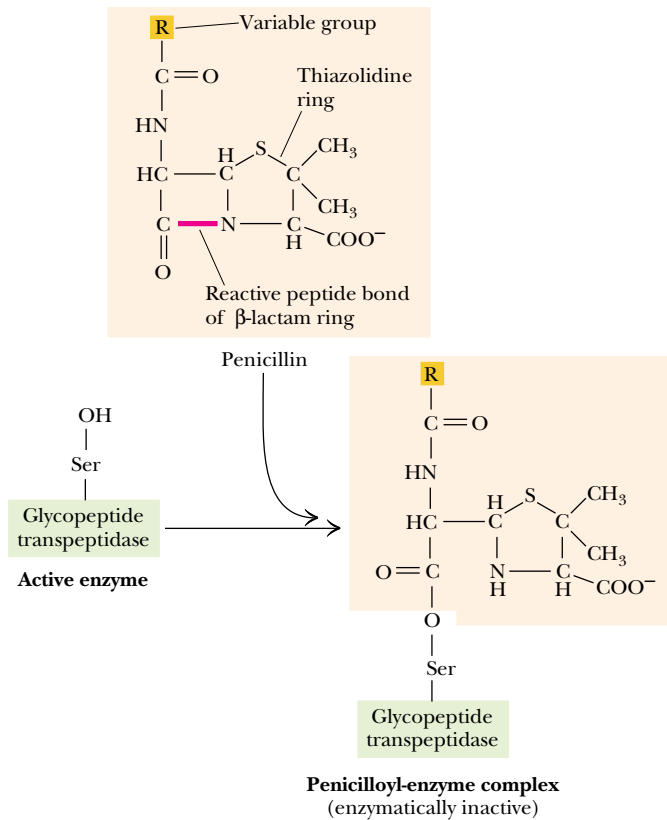


FIGURE 14.17 • Penicillin is an irreversible inhibitor of the enzyme *glycoprotein peptidase*, which catalyzes an essential step in bacterial cell wall synthesis. Penicillin consists of a thiazolidine ring fused to a β -lactam ring to which a variable group R is attached. A reactive peptide bond in the β -lactam ring covalently attaches to a serine residue in the active site of the glycopeptide transpeptidase. (The conformation of penicillin around its reactive peptide bond resembles the transition state of the normal glycoprotein peptidase substrate.) The penicilloyl–enzyme complex is catalytically inactive. The bond between the enzyme and penicillin is indefinitely stable; that is, penicillin binding is irreversible.

14.5 • Kinetics of Enzyme-Catalyzed Reactions Involving Two or More Substrates

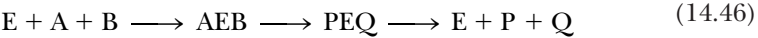
Thus far, we have considered only the simple case of enzymes that act upon a single substrate, S . This situation is not common. Usually, enzymes catalyze reactions in which two (or even more) substrates take part.

Consider the case of an enzyme catalyzing a reaction involving two substrates, A and B , and yielding the products P and Q :



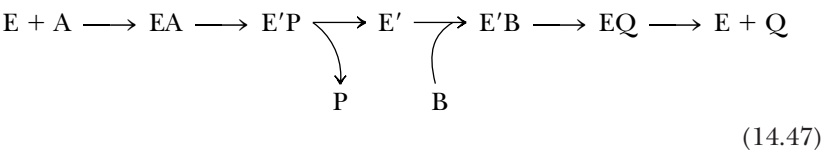
Such a reaction is termed a **bisubstrate reaction**. In general, bisubstrate reactions proceed by one of two possible routes:

- 1. Both A and B are bound to the enzyme and then reaction occurs to give $P + Q$:



Reactions of this type are defined as **sequential** or **single-displacement reactions**. They can be either of two distinct classes:

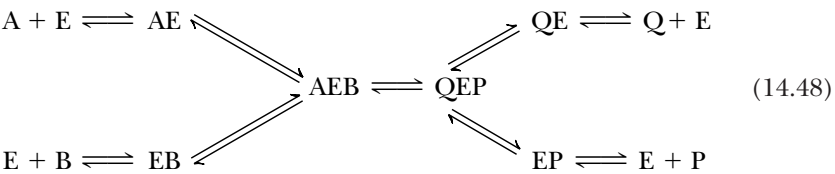
- a. **random**, where either A or B may bind to the enzyme first, followed by the other substrate, or
 - b. **ordered**, where A, designated the *leading substrate*, must bind to E first before B can be bound.
- Both classes of single-displacement reactions are characterized by lines that intersect to the left of the $1/v$ axis in Lineweaver–Burk double-reciprocal plots (Figure 14.18).
2. The other general possibility is that one substrate, A, binds to the enzyme and reacts with it to yield a chemically modified form of the enzyme (E') plus the product, P. The second substrate, B, then reacts with E' , regenerating E and forming the other product, Q.



Reactions that fit this model are called **ping-pong** or **double-displacement reactions**. Two distinctive features of this mechanism are the obligatory formation of a modified enzyme intermediate, E' , and the pattern of parallel lines obtained in double-reciprocal plots (Figure 14.19).

Random, Single-Displacement Reactions

In this type of sequential reaction, all possible binary enzyme:substrate complexes (AE , EB , QE , EP) are formed rapidly and reversibly when the enzyme is added to a reaction mixture containing A, B, P, and Q:



The rate-limiting step is the reaction $AEB \rightarrow QEP$. It doesn't matter whether A or B binds first to E, or whether Q or P is released first from QEP . Sometimes,

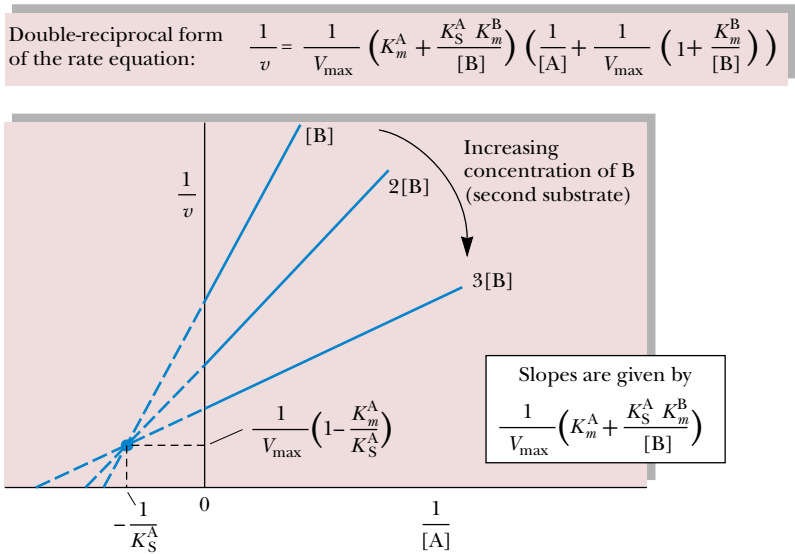
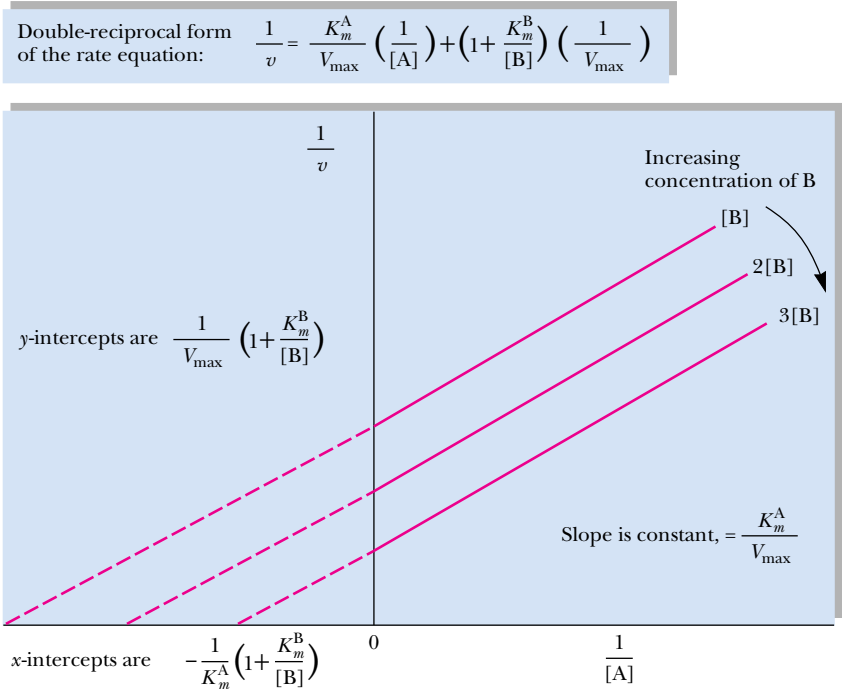


FIGURE 14.18 • Single-displacement bisubstrate mechanism. Double-reciprocal plots of the rates observed with different fixed concentrations of one substrate (B here) are graphed versus a series of concentrations of A. Note that, in these Lineweaver–Burk plots for single-displacement bisubstrate mechanisms, the lines intersect to the left of the $1/v$ axis.

FIGURE 14.19 • Double-displacement (ping-pong) bisubstrate mechanisms are characterized by Lineweaver–Burk plots of parallel lines when double-reciprocal plots of the rates observed with different fixed concentrations of the second substrate, B, are graphed versus a series of concentrations of A.



reactions that follow this random order of addition of substrates to E can be distinguished mechanistically from reactions obeying an ordered, single-displacement mechanism, *if* A has *no* influence on the binding constant for B (and vice versa); that is, the mechanism is purely random. Then, the lines in a Lineweaver–Burk plot intersect at the $1/[A]$ axis (Figure 14.20).

Creatine Kinase Acts by a Random, Single-Displacement Mechanism

An example of a random, single-displacement mechanism is seen in the enzyme *creatine kinase*, a phosphoryl-transfer enzyme that uses ATP as a phosphoryl

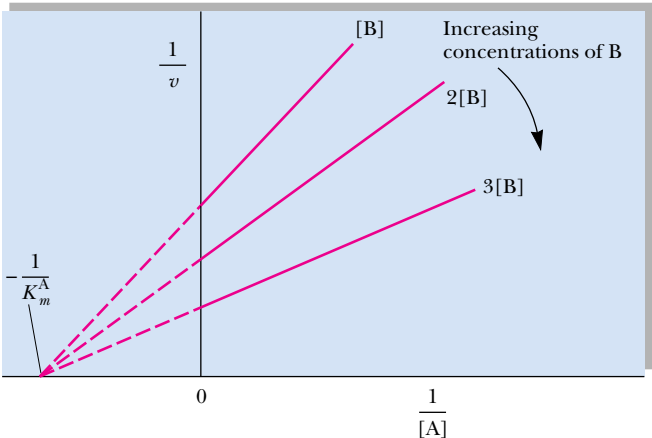
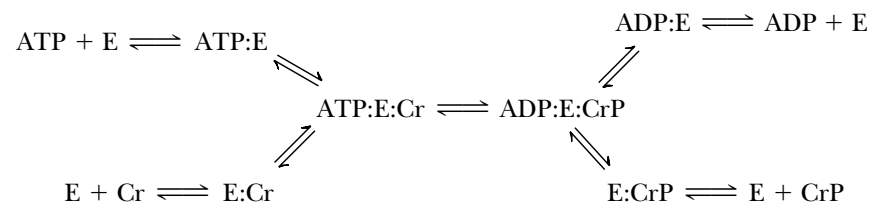


FIGURE 14.20 • Random, single-displacement bisubstrate mechanism where A does not affect B binding, and vice versa. Note that the lines intersect at the $1/[A]$ axis. (If $[B]$ were varied in an experiment with several fixed concentrations of A, the lines would intersect at the $1/[B]$ axis in a $1/v$ versus $1/[B]$ plot.)

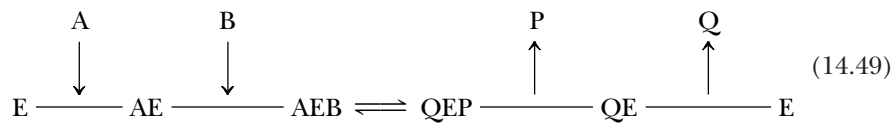
donor to form creatine phosphate (CrP) from creatine (Cr). Creatine-P is an important reservoir of phosphate-bond energy in muscle cells (Figure 14.21).



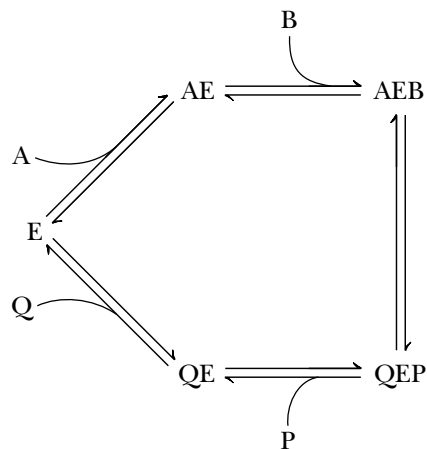
The overall direction of the reaction will be determined by the relative concentrations of ATP, ADP, Cr, and CrP and the equilibrium constant for the reaction. The enzyme can be considered to have two sites for substrate (or product) binding: an adenine nucleotide site, where ATP or ADP binds, and a creatine site, where Cr or CrP is bound. In such a mechanism, ATP and ADP compete for binding at their unique site, while Cr and CrP compete at the specific Cr-, CrP-binding site. Note that no modified enzyme form (E'), such as an E-PO₄ intermediate, appears here. The reaction is characterized by rapid and reversible binary ES complex formation, followed by addition of the remaining substrate, and the rate-determining reaction taking place within the ternary complex.

Ordered, Single-Displacement Reactions

In this case, the **leading substrate**, A (also called the **obligatory** or **compulsory substrate**), must bind first. Then the second substrate, B, binds. Strictly speaking, B cannot bind to free enzyme in the absence of A. Reaction between A and B occurs in the ternary complex, and is usually followed by an ordered release of the products of the reaction, P and Q. In the schemes below, Q is the product of A and is released last. One representation, suggested by W. W. Cleland, follows:



Another way of portraying this mechanism is as follows:



Note that A and Q are competitive for binding to the free enzyme, E, but not A and B (or Q and B).

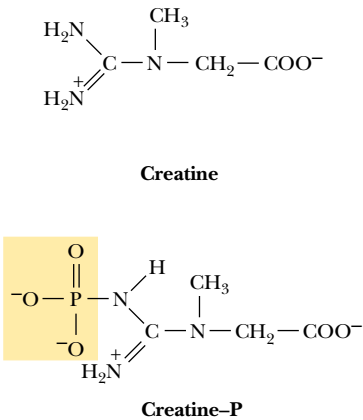
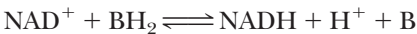


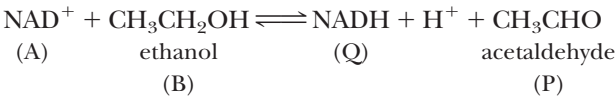
FIGURE 14.21 • The structures of creatine and creatine phosphate, guanidinium compounds that are important in muscle energy metabolism.

NAD⁺-Dependent Dehydrogenases Show Ordered Single-Displacement Mechanisms

Nicotinamide adenine dinucleotide (NAD⁺)-dependent dehydrogenases are enzymes that typically behave according to the kinetic pattern just described. A general reaction of these dehydrogenases is



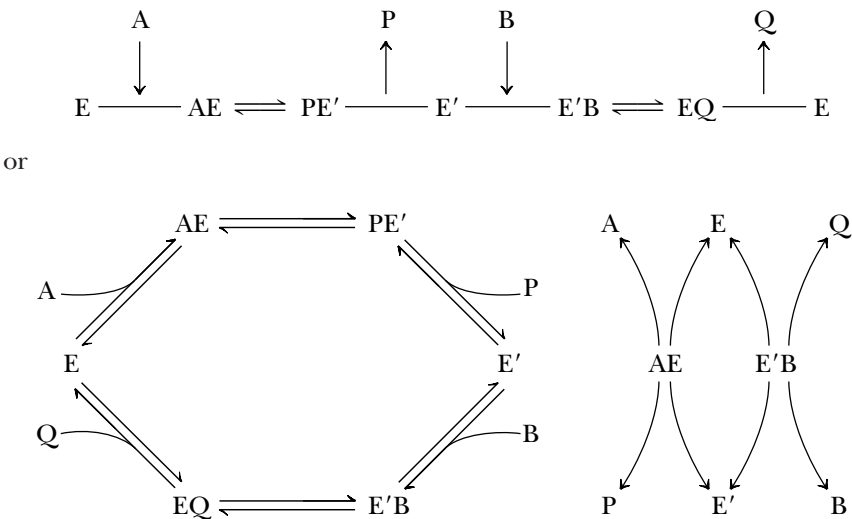
The leading substrate (A) is nicotinamide adenine dinucleotide (NAD⁺), and NAD⁺ and NADH (product Q) compete for a common site on E. A specific example is offered by *alcohol dehydrogenase (ADH)*:



We can verify that this ordered mechanism is not random by demonstrating that no B (ethanol) is bound to E in the absence of A (NAD⁺).

Double-Displacement (Ping-Pong) Reactions

Reactions conforming to this kinetic pattern are characterized by the fact that the product of the enzyme’s reaction with A (called P in the following schemes) is released *prior* to reaction of the enzyme with the second substrate, B. As a result of this process, the enzyme, E, is converted to a modified form, E’, which then reacts with B to give the second product, Q, and regenerate the unmodified enzyme form, E:



Note that these schemes predict that A and Q compete for the free enzyme form, E, while B and P compete for the modified enzyme form, E’. A and Q do not bind to E’, nor do B and P combine with E.

Aminotransferases Show Double-Displacement Catalytic Mechanisms

One class of enzymes that follow a ping-pong-type mechanism are *aminotransferases* (previously known as transaminases). These enzymes catalyze the transfer of an amino group from an amino acid to an α-keto acid. The products are a new amino acid and the keto acid corresponding to the carbon skeleton of the amino donor:



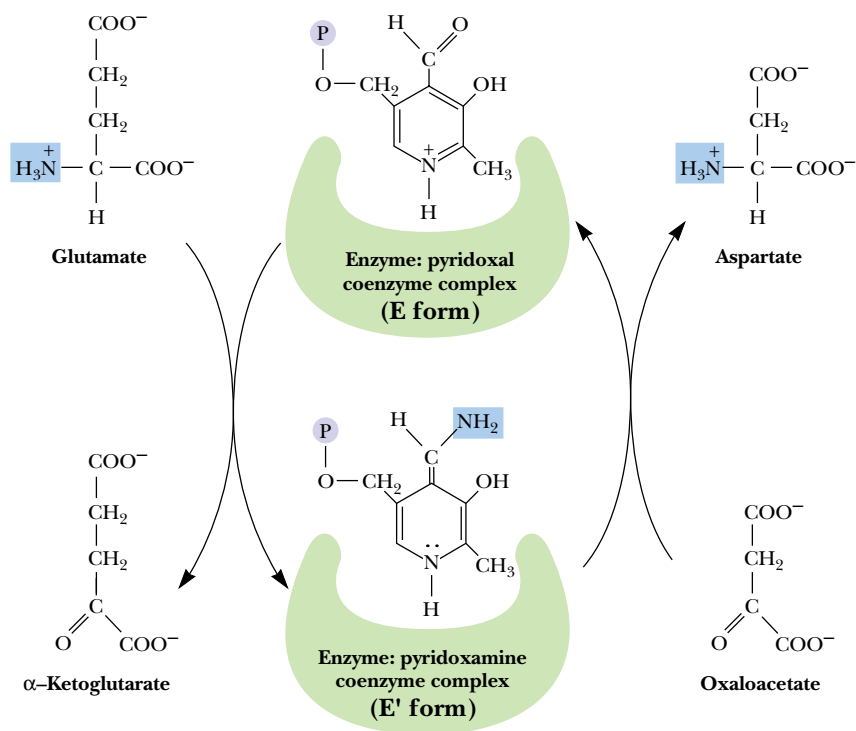
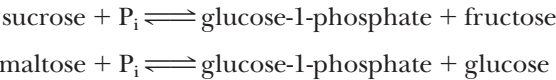


FIGURE 14.22 • *Glutamate:aspartate aminotransferase*, an enzyme conforming to a double-displacement bisubstrate mechanism. Glutamate:aspartate aminotransferase is a pyridoxal phosphate–dependent enzyme. The pyridoxal serves as the —NH_2 acceptor from glutamate to form pyridoxamine. Pyridoxamine is then the amino donor to oxaloacetate to form aspartate and regenerate the pyridoxal coenzyme form. (The pyridoxamine:enzyme is the E' form.)

A specific example would be *glutamate:aspartate aminotransferase*. Figure 14.22 depicts the scheme for this mechanism. Note that glutamate and aspartate are competitive for E, and that oxaloacetate and α -ketoglutarate compete for E'. In glutamate:aspartate aminotransferase, an enzyme-bound coenzyme, *pyridoxal phosphate* (a vitamin B₆ derivative), serves as the amino group acceptor/donor in the enzymatic reaction. The unmodified enzyme form, E, has the coenzyme in the aldehydic pyridoxal form, whereas the modified enzyme form, E', is actually pyridoxamine phosphate (Figure 14.22). Not all enzymes displaying ping-pong–type mechanisms require coenzymes as carriers for the chemical substituent transferred in the reaction.

Diagnosis of Bisubstrate Mechanisms

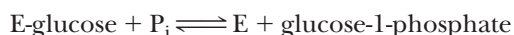
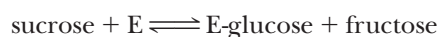
Kineticists rely on a number of diagnostic tests for the assignment of a reaction mechanism to a specific enzyme. One is the graphic analysis of the kinetic patterns observed. It is usually easy to distinguish between single- and double-displacement reactions in this manner, and examining competitive effects between substrates aids in assigning reactions to random versus ordered patterns of S-binding. A second diagnostic test is to determine whether the enzyme catalyzes an exchange reaction. Consider as an example the two enzymes *sucrose phosphorylase* and *maltose phosphorylase*. Both catalyze the phosphorolysis of a disaccharide and both yield glucose-1-phosphate and a free hexose:



Interestingly, in the absence of sucrose and fructose, sucrose phosphorylase will catalyze the exchange of inorganic phosphate, P_i , into glucose-1-phosphate. This reaction can be followed by using $^{32}P_i$ as a radioactive tracer and observing the appearance of ^{32}P into glucose-1-phosphate:



Maltose phosphorylase cannot carry out a similar reaction. The ^{32}P exchange reaction of sucrose phosphorylase is accounted for by a double-displacement mechanism where $E' = E\text{-glucose}$:



Thus, in the presence of just $^{32}P_i$ and glucose-1-phosphate, sucrose phosphorylase still catalyzes the second reaction and radioactive P_i is incorporated into glucose-1-phosphate over time.

Maltose phosphorylase proceeds via a single-displacement reaction that necessarily requires the formation of a ternary maltose : $E : P_i$ (or glucose : E : glucose-1-phosphate) complex for any reaction to occur. Exchange reactions are a characteristic of enzymes that obey double-displacement mechanisms at some point in their catalysis.

Multisubstrate Reactions

Thus far, we have considered enzyme-catalyzed reactions involving one or two substrates. How are the kinetics described in those cases in which more than two substrates participate in the reaction? An example might be the glycolytic enzyme *glyceraldehyde-3-phosphate dehydrogenase* (Chapter 19):



Many other multisubstrate examples abound in metabolism. In effect, these situations are managed by realizing that the interaction of the enzyme with its many substrates can be treated as a series of uni- or bisubstrate steps in a multi-step reaction pathway. Thus, the complex mechanism of a multisubstrate reaction is resolved into a sequence of steps, each of which obeys the single- and double-displacement patterns just discussed.

14.6 • RNA and Antibody Molecules as Enzymes: Ribozymes and Abzymes

Catalytic RNA Molecules: Ribozymes

It was long assumed that all enzymes are proteins. However, in recent years, more and more instances of biological catalysis by RNA molecules have been discovered. These catalytic RNAs, or **ribozymes**, satisfy several enzymatic criteria: They are substrate-specific, they enhance the reaction rate, and they emerge from the reaction unchanged. For example, RNase P, an enzyme responsible for the formation of mature tRNA molecules from tRNA precursors, requires an RNA component as well as a protein subunit for its activity in the cell. *In vitro*, the protein alone is incapable of catalyzing the maturation reaction, but the RNA component by itself can carry out the reaction under appropriate conditions. In another case, in the ciliated protozoan *Tetrahymena*, formation of mature ribosomal RNA from a pre-rRNA precursor involves the removal of an internal RNA segment and the joining of the two ends in a process known as **splicing out**. The excision of this intervening internal sequence of RNA and

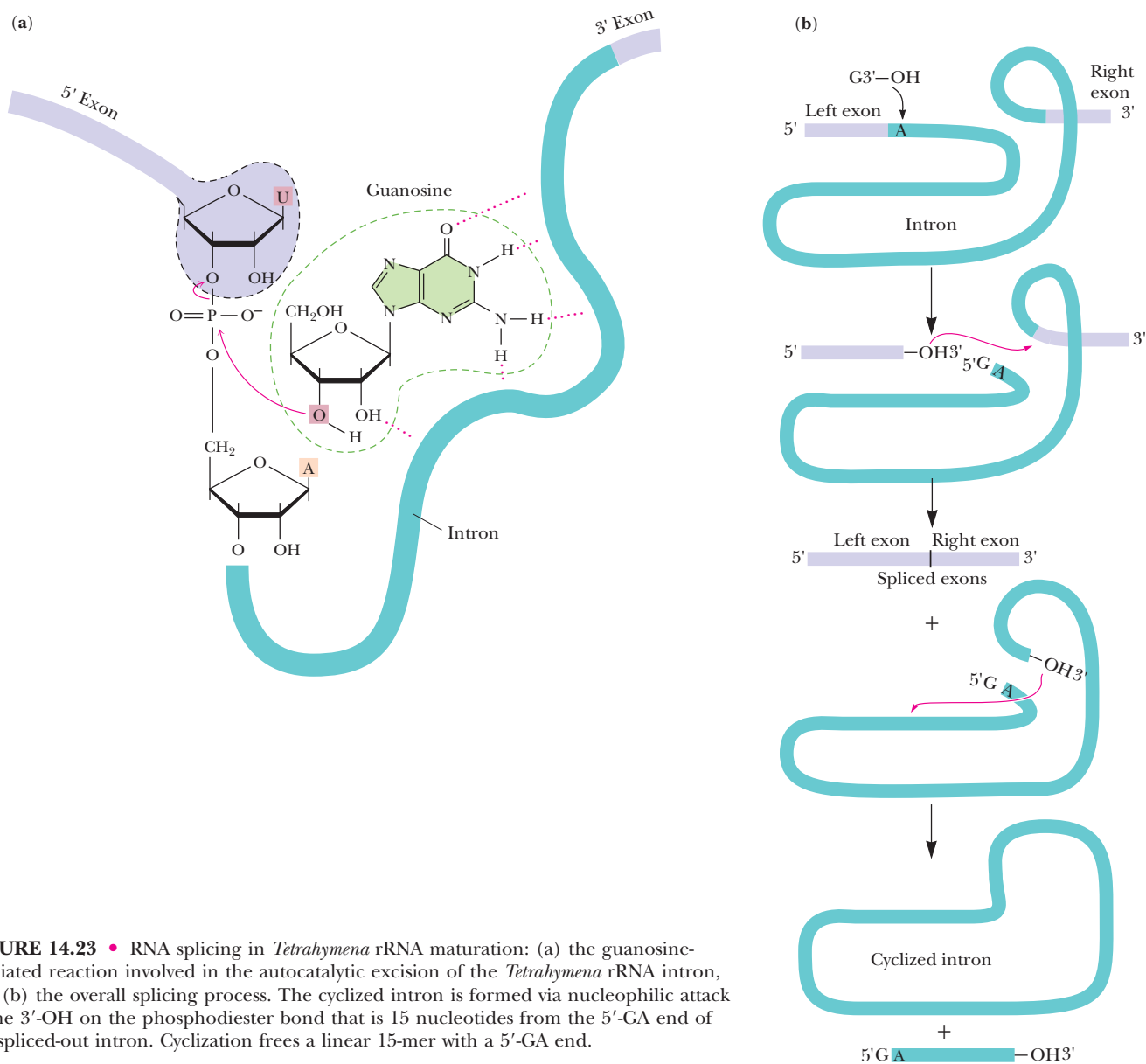


FIGURE 14.23 • RNA splicing in *Tetrahymena* rRNA maturation: (a) the guanosine-mediated reaction involved in the autocatalytic excision of the *Tetrahymena* rRNA intron, and (b) the overall splicing process. The cyclized intron is formed via nucleophilic attack of the 3'-OH on the phosphodiester bond that is 15 nucleotides from the 5'-GA end of the spliced-out intron. Cyclization frees a linear 15-mer with a 5'-GA end.

ligation of the ends is, remarkably, catalyzed by the intervening sequence of RNA itself, in the presence of Mg^{2+} and a free molecule of guanosine nucleoside or nucleotide (Figure 14.23). *In vivo*, the intervening sequence RNA probably acts only in splicing itself out; *in vitro*, however, it can act many times, turning over like a true enzyme.

Protein-Free 50S Ribosomal Subunits Catalyze Peptide Bond Formation In Vitro

Perhaps the most significant case of catalysis by RNA occurs in protein synthesis. Harry F. Noller and his colleagues have found that the **peptidyl transferase reaction**, which is the reaction of peptide bond formation during protein synthesis (Figure 14.24), can be catalyzed by 50S ribosomal subunits (see Chapter 12) from which virtually all of the protein has been removed. These

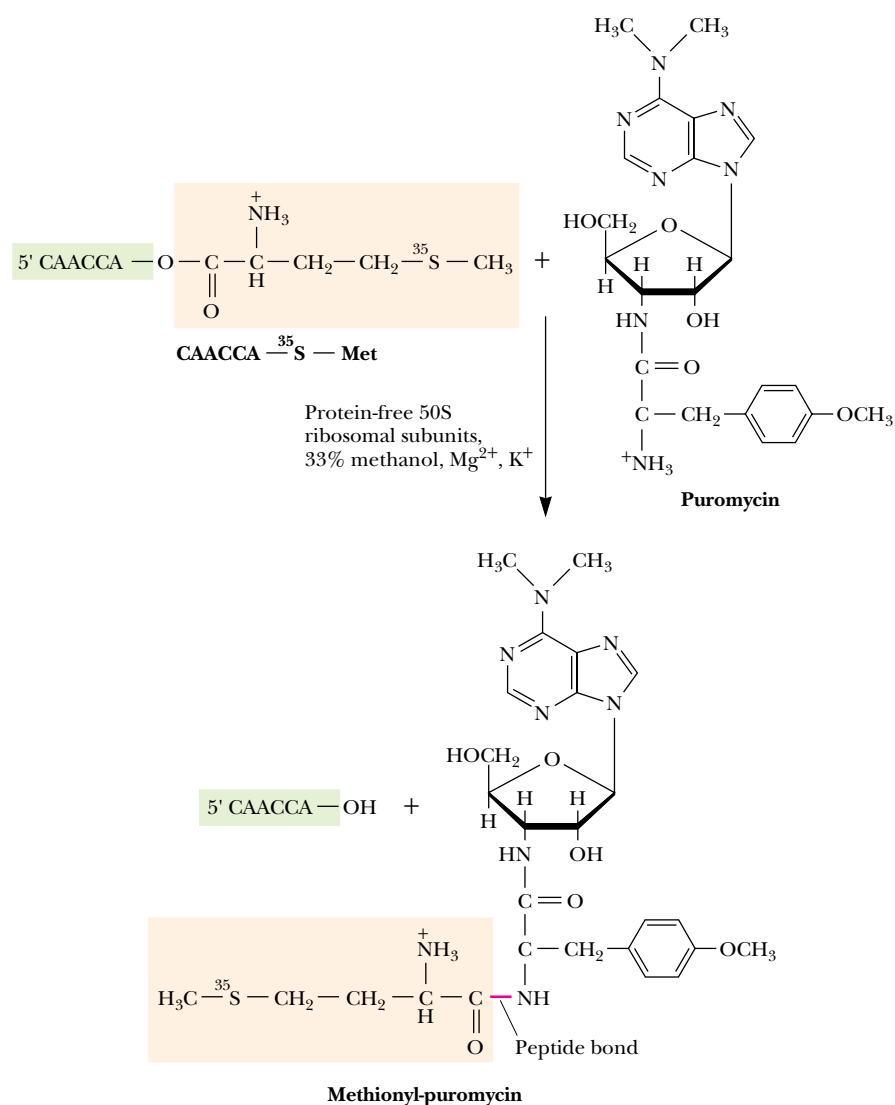


FIGURE 14.24 • Protein-free 50S ribosomal subunits have peptidyl transferase activity. Peptidyl transferase is the name of the enzymatic function that catalyzes peptide bond formation. The presence of this activity in protein-free 50S ribosomal subunits was demonstrated using a model assay for peptide bond formation in which an aminoacyl-tRNA analog (a short RNA oligonucleotide of sequence CAACCA carrying ³⁵S-labeled methionine attached at its 3'-OH end) served as the peptidyl donor and puromycin (another aminoacyl-tRNA analog) served as the peptidyl acceptor. Activity was measured by monitoring the formation of ³⁵S-labeled methioninyl-puromycin.

experiments imply that just the 23S rRNA by itself is capable of catalyzing peptide bond formation. Also, the laboratory of Thomas Cech has created a synthetic 196-nucleotide-long ribozyme capable of performing the peptidyl transferase reaction.

Several features of these “RNA enzymes,” or **ribozymes**, lead to the realization that their biological efficiency does not challenge that achieved by proteins. First, RNA enzymes often do not fulfill the criterion of catalysis *in vivo* because they act only once in intramolecular events such as self-splicing. Second, the catalytic rates achieved by RNA enzymes *in vivo* and *in vitro* are

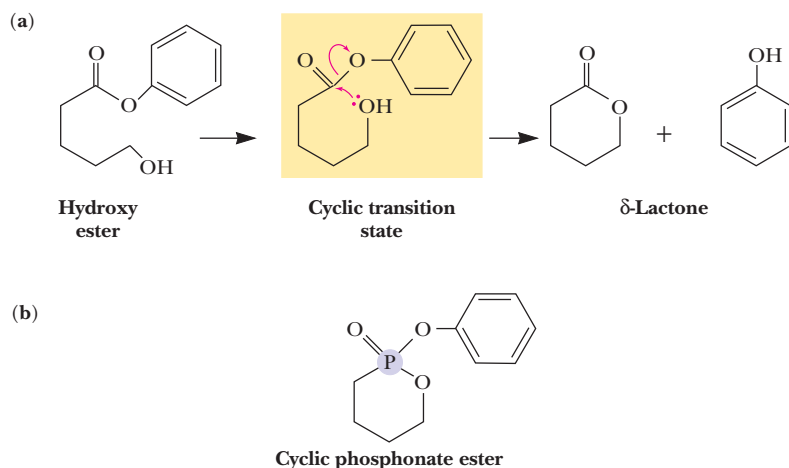


FIGURE 14.25 • Catalytic antibodies are designed to specifically bind the transition-state intermediate in a chemical reaction. (a) The intramolecular hydrolysis of a hydroxy ester to yield as products a δ -lactone and the alcohol phenol. Note the cyclic transition state. (b) The cyclic phosphonate ester analog of the cyclic transition state. Antibodies raised against this phosphonate ester act as *enzymes*: they are catalysts that markedly accelerate the rate of ester hydrolysis.

significantly enhanced by the participation of protein subunits. Nevertheless, the fact that RNA can catalyze certain reactions is experimental support for the idea that a primordial world dominated by RNA molecules existed before the evolution of DNA and proteins.

Catalytic Antibodies: Abzymes

Antibodies are *immunoglobulins*, which, of course, are proteins. Like other antibodies, **catalytic antibodies**, so-called **abzymes**, are elicited in an organism in response to immunological challenge by a foreign molecule called an **antigen** (see Chapter 29 for discussions on the molecular basis of immunology). In this case, however, the antigen is purposefully engineered to be *an analog of the transition-state intermediate in a reaction*. The rationale is that a protein specific for binding the transition-state intermediate of a reaction will promote entry of the normal reactant into the reactive, transition-state conformation. Thus, a catalytic antibody facilitates, or catalyzes, a reaction by forcing the conformation of its substrate in the direction of its transition state. (A prominent explanation for the remarkable catalytic power of conventional enzymes is their great affinity for the transition-state intermediates in the reactions they catalyze; see Chapter 16.)

One strategy has been to prepare ester analogs by substituting a phosphorus atom for the carbon in the ester group (Figure 14.25). The phospho-compound mimics the natural transition state of ester hydrolysis, and antibodies elicited against these analogs act like enzymes in accelerating the rate of ester hydrolysis as much as 1000-fold. Abzymes have been developed for a number of other classes of reactions, including C—C bond formation via aldol condensation (the reverse of the aldolase reaction [see Figure 14.2, reaction 4 and Chapter 19]) and the pyridoxal 5'-P-dependent aminotransferase reaction shown in Figure 14.22. In this latter instance, N^{α} -(5'-phosphopyridoxyl)-lysine (Figure 14.26a) coupled to a carrier protein served as the antigen. An antibody raised against this antigen catalyzed the conversion of D-alanine and pyridoxal 5'-P to pyruvate and pyridoxamine 5'-P (Figure 14.26b). This biotechnology offers the real possibility of creating “**designer enzymes**,” specially tailored enzymes designed to carry out specific catalytic processes.

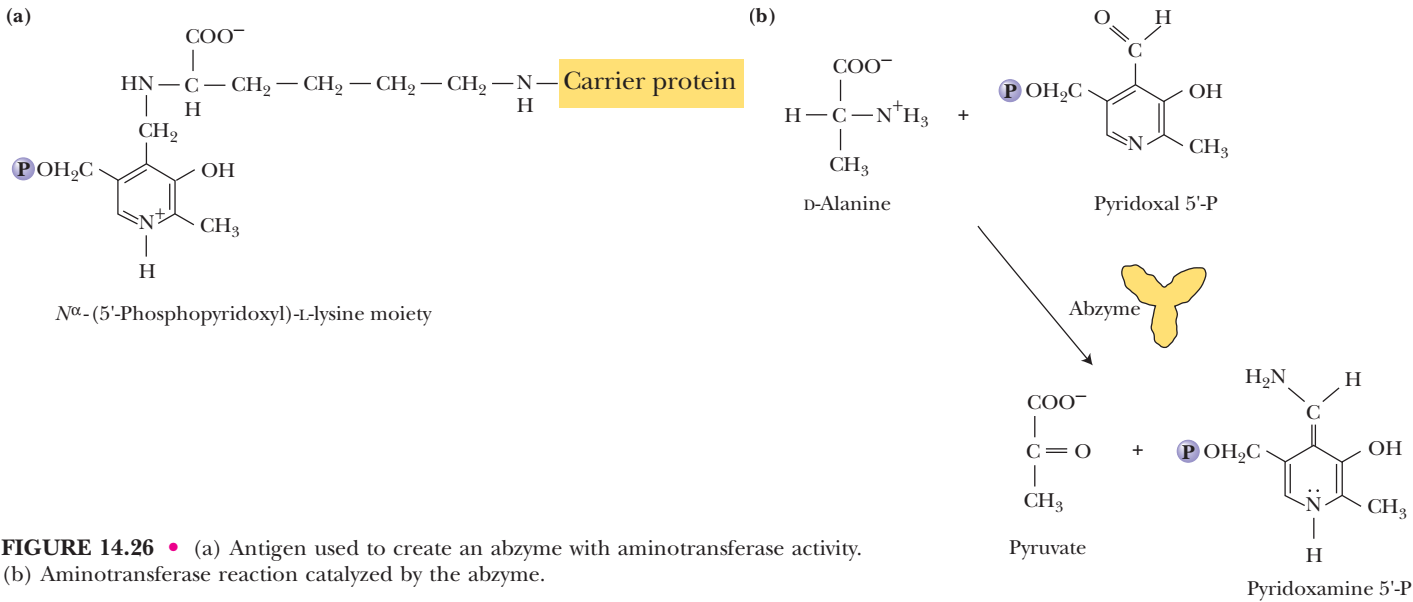


FIGURE 14.26 • (a) Antigen used to create an abzyme with aminotransferase activity. (b) Aminotransferase reaction catalyzed by the abzyme.

PROBLEMS

1. According to the Michaelis–Menten equation, what is the $v/V_{\max}$ ratio when $[S] = 4 K_m$?
2. If $V_{\max} = 100 \mu\text{mol/mL sec}$ and $K_m = 2 \text{ mM}$, what is the velocity of the reaction when $[S] = 20 \text{ mM}$?
3. For a Michaelis–Menten reaction, $k_1 = 7 \times 10^7 / \text{M} \cdot \text{sec}$, $k_{-1} = 1 \times 10^3 / \text{sec}$, and $k_2 = 2 \times 10^4 / \text{sec}$. What are the values of K_S and K_m ? Does substrate binding approach equilibrium or does it behave more like a steady-state system?
4. The following kinetic data were obtained for an enzyme in the absence of any inhibitor (1), and in the presence of two different inhibitors (2) and (3) at 5 mM concentration. Assume $[E_T]$ is the same in each experiment.

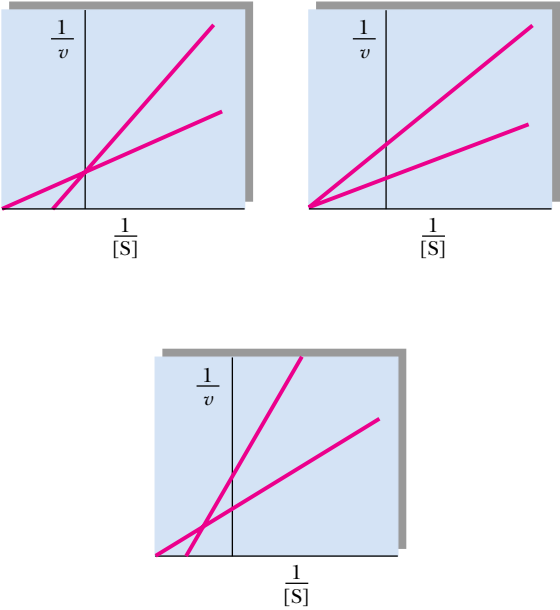
[S] (mM)	(1) $v(\mu\text{mol/mL sec})$	(2) $v(\mu\text{mol/mL sec})$	(3) $v(\mu\text{mol/mL sec})$
1	12	4.3	5.5
2	20	8	9
4	29	14	13
8	35	21	16
12	40	26	18

- a. Determine $V_{\max}$ and K_m for the enzyme.
 - b. Determine the type of inhibition and the K_i for each inhibitor.
5. The general rate equation for an ordered, single-displacement reaction where A is the leading substrate is

$$v = \frac{V_{\max}[A][B]}{(K_S^A K_m^B + K_m^A [B] + K_m^B [A] + [A][B])}$$

Write the Lineweaver–Burk (double-reciprocal) equivalent of this equation, and from it calculate algebraic expressions for (a) the slope; (b) the y -intercepts; and (c) the horizontal and vertical coor-

- dinate of the point of intersection when $1/v$ is plotted versus $1/[B]$ at various *fixed* concentrations of A.
6. The following graphical patterns obtained from kinetic experiments have several possible interpretations depending on the nature of the experiment and the variables being plotted. Give at least two possibilities for each.



7. Liver alcohol dehydrogenase (ADH) is relatively nonspecific and will oxidize ethanol or other alcohols, including methanol. Methanol oxidation yields formaldehyde, which is quite toxic, causing, among other things, blindness. Mistaking it for the cheap

wine he usually prefers, my dog Clancy ingested about 50 mL of windshield washer fluid (a solution 50% in methanol.) Knowing that methanol would be excreted eventually by Clancy’s kidneys if its oxidation could be blocked, and realizing that, in terms of methanol oxidation by ADH, ethanol would act as a competitive inhibitor, I decided to offer Clancy some wine. How much of Clancy’s favorite vintage (12% ethanol) must he consume in order to lower the activity of his ADH on methanol to 5% of its normal

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value if the K_m values of canine ADH for ethanol and methanol are 1 millimolar and 10 millimolar, respectively? (The K_1 for ethanol in its role as competitive inhibitor of methanol oxidation by ADH is the same as its K_m .) Both the methanol and ethanol will quickly distribute throughout Clancy’s body fluids, which amount to about 15 L. Assume the densities of 50% methanol and the wine are both 0.9 g/mL.

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Allostery is the key chemical process that makes possible intracellular and intercellular regulation: “. . . the molecular interactions which ensure the transmission and interpretation of (regulatory) signals rest upon (allosteric) proteins endowed with discriminatory stereospecific recognition properties.”

JACQUES MONOD in *Chance and Necessity*

OUTLINE

- 15.1 • Specificity Is the Result of Molecular Recognition
 - 15.2 • Controls Over Enzymatic Activity—General Considerations
 - 15.3 • The Allosteric Regulation of Enzyme Activity
 - 15.4 • Models for the Allosteric Behavior of Proteins
 - 15.5 • Glycogen Phosphorylase: Allosteric Regulation and Covalent Modification
- Special Focus: Hemoglobin and Myoglobin—Paradigms of Protein Structure and Function

Chapter 15

Enzyme Specificity and Regulation



Metabolic regulation is achieved through an exquisitely balanced interplay among enzymes and small molecules, a process symbolized by the delicate balance of forces in this mobile. (Sea Scape by Alexander Calder/Whitney Museum of American Art, NY)

The extraordinary ability of an enzyme to catalyze only one particular reaction is a quality known as **specificity** (Chapter 14). Specificity means an enzyme acts only on a specific substance, its substrate, invariably transforming it into a specific product. That is, an enzyme binds only certain compounds, and then, only a specific reaction ensues. Some enzymes show absolute specificity, catalyzing the transformation of only one specific substrate to yield a unique product. Other enzymes carry out a particular reaction but act on a class of compounds. For example, *hexokinase* (ATP:hexose-6-phosphotransferase) will carry out the ATP-dependent phosphorylation of a number of hexoses at the 6-position, including glucose.

15.1 • Specificity Is the Result of Molecular Recognition

An enzyme molecule is typically orders of magnitude larger than its substrate. Its active site comprises only a small portion of the overall enzyme structure. The active site is part of the conformation of the enzyme molecule arranged to create a special pocket or cleft whose three-dimensional structure is complementary to the structure of the substrate. The enzyme and the substrate molecules “recognize” each other through this structural complementarity. The substrate binds to the enzyme through relatively weak forces—H bonds, ionic bonds (salt bridges), and van der Waals interactions between sterically complementary clusters of atoms. Specificity studies on enzymes entail an examination of the rates of the enzymatic reaction obtained with various **structural analogs** of the substrate. By determining which functional and structural groups within the substrate affect binding or catalysis, enzymologists can map the properties of the active site, analyzing questions such as: Can it accommodate sterically bulky groups? Are ionic interactions between E and S important? Are H bonds formed?

The “Lock and Key” Hypothesis

Pioneering enzyme specificity studies at the turn of the century by the great organic chemist Emil Fischer led to the notion of an enzyme resembling a “**lock**” and its particular substrate the “**key**.” This analogy captures the essence of the specificity that exists between an enzyme and its substrate, but enzymes are not rigid templates like locks.

The “Induced Fit” Hypothesis

Enzymes are highly flexible, conformationally dynamic molecules, and many of their remarkable properties, including substrate binding and catalysis, are due to their structural pliancy. Realization of the conformational flexibility of proteins led Daniel Koshland to hypothesize that the binding of a substrate (S) by an enzyme is an interactive process. That is, the shape of the enzyme’s active site is actually modified upon binding S, in a process of dynamic recognition between enzyme and substrate aptly called **induced fit**. In essence, substrate binding alters the conformation of the protein, so that the protein and the substrate “fit” each other more precisely. The process is truly interactive in that the conformation of the substrate also changes as it adapts to the conformation of the enzyme.

This idea also helps to explain some of the mystery surrounding the enormous catalytic power of enzymes: In enzyme catalysis, precise orientation of catalytic residues comprising the active site is necessary for the reaction to occur; substrate binding induces this precise orientation by the changes it causes in the protein’s conformation.

“Induced Fit” and the Transition-State Intermediate

The catalytically active enzyme:substrate complex is an interactive structure in which the enzyme causes the substrate to adopt a form that mimics the transition-state intermediate of the reaction. Thus, a poor substrate would be one that was less effective in directing the formation of an optimally active enzyme:transition-state intermediate conformation. This active conformation of the enzyme molecule is thought to be relatively unstable in the absence of substrate, and free enzyme thus reverts to a conformationally different state.

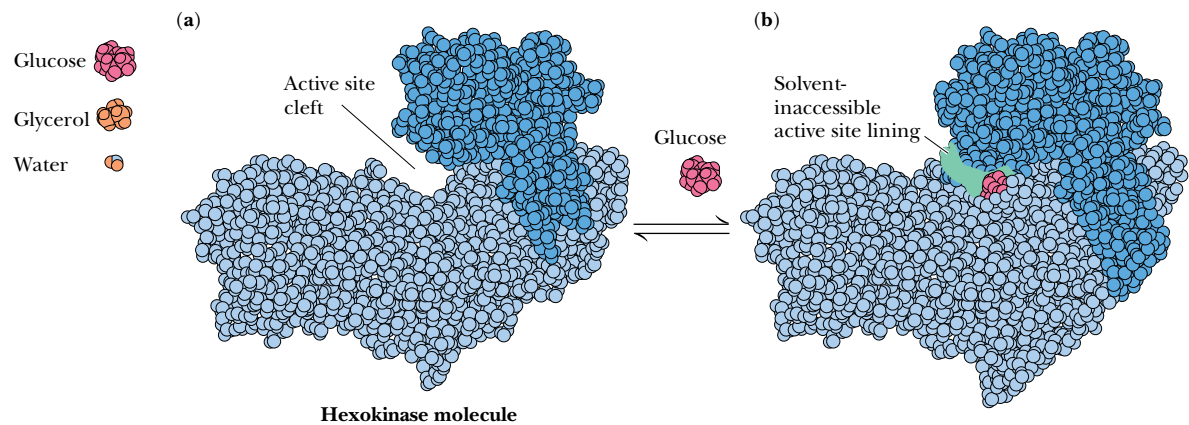


FIGURE 15.1 • A drawing, roughly to scale, of H₂O, glycerol, glucose, and an idealized hexokinase molecule. Note the two domains of structure in hexokinase, (a), between which the active site is located. Binding of glucose induces a conformational change in hexokinase. The two domains close together, creating the catalytic site (b). The shaded area in (b) represents solvent-inaccessible surface area in the active site cleft that results when the enzyme binds substrate.

Specificity and Reactivity

Consider, for example, why hexokinase catalyzes the ATP-dependent phosphorylation of hexoses but not smaller phosphoryl-group acceptors such as glycerol, ethanol, or even water. Surely these smaller compounds are not sterically forbidden from approaching the active site of hexokinase (Figure 15.1). Indeed, water should penetrate the active site easily and serve as a highly effective phosphoryl-group acceptor. Accordingly, hexokinase should display high ATPase activity. It does not. Only the binding of hexoses induces hexokinase to assume its fully active conformation.

In Chapter 16, we explore in greater detail the factors that contribute to the remarkable catalytic power of enzymes and examine specific examples of enzyme reaction mechanisms. Here we focus on another essential feature of enzymes: *the regulation of their activity*.

15.2 • Controls Over Enzymatic Activity—
General Considerations

The activity displayed by enzymes is affected by a variety of factors, some of which are essential to the harmony of metabolism.

1. The enzymatic rate, $v = d[P]/dt$, “slows down” as product accumulates and equilibrium is approached. The apparent decrease in rate is due to the conversion of P to S by the reverse reaction as [P] rises. Once $[P]/[S] = K_{eq}$, no further reaction is apparent. K_{eq} defines thermodynamic equilibrium. Enzymes have no influence on the thermodynamics of a reaction. Also, product inhibition can be a kinetically valid phenomenon: Some enzymes are actually inhibited by the products of their action.
2. The availability of substrates and cofactors will determine the enzymatic reaction rate. In general, enzymes have evolved such that their K_m values approximate the prevailing *in vivo* concentration of their substrates. (It is also true that the concentration of some enzymes in cells is within an order of magnitude or so of the concentrations of their substrates.)

3. There are genetic controls over the amounts of enzyme synthesized (or degraded) by cells. If the gene encoding a particular enzyme protein is turned on or off, changes in the amount of enzyme activity soon follow. **Induction**, which is the activation of enzyme synthesis, and **repression**, which is the shutdown of enzyme synthesis, are important mechanisms for the regulation of metabolism. By controlling the amount of an enzyme that is present at any moment, cells can either activate or terminate various metabolic routes. Genetic controls over enzyme levels have a response time ranging from minutes in rapidly dividing bacteria to hours (or longer) in higher eukaryotes.
4. Enzymes can be regulated by **covalent modification**, the reversible covalent attachment of a chemical group. For example, a fully active enzyme can be converted into an inactive form simply by the covalent attachment of a functional group, such as a phosphoryl moiety (Figure 15.2). Alternatively, some enzymes exist in an inactive state unless specifically converted into the active form through covalent addition of a functional group. Covalent modification reactions are catalyzed by special **converter enzymes**, which are themselves subject to metabolic regulation. Although covalent modification represents a stable alteration of the enzyme, a different converter enzyme operates to remove the modification, so that when the conditions that favored modification of the enzyme are no longer present, the process can be reversed, restoring the enzyme to its unmodified state. Many examples of covalent modification at important metabolic junctions will be encountered in our discussions of metabolic pathways. Because covalent modification events are enzyme-catalyzed, they occur very quickly, with response times of seconds or even less for significant changes in metabolic activity. The 1992 Nobel Prize in physiology or medicine was awarded to Edmond Fischer and Edwin Krebs for their pioneering studies of reversible protein phosphorylation as an important means of cellular regulation.
5. Enzymatic activity can also be activated or inhibited through noncovalent interaction of the enzyme with small molecules (metabolites) other than the substrate. This form of control is termed **allosteric regulation**, because the activator or inhibitor binds to the enzyme at a site *other* than (*allo* means “other”) the active site. Further, such allosteric regulators, or **effector molecules**, are often quite different sterically from the substrate. Because this form of regulation results simply from reversible binding of regulatory ligands to the enzyme, the cellular response time can be virtually instantaneous.

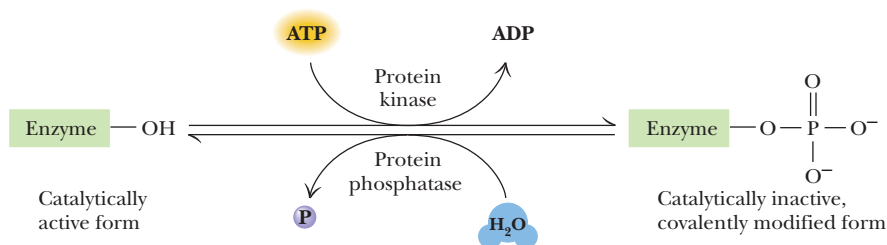


FIGURE 15.2 • Enzymes regulated by covalent modification are called **interconvertible enzymes**. The enzymes (*protein kinase* and *protein phosphatase*, in the example shown here) catalyzing the conversion of the interconvertible enzyme between its two forms are called **converter enzymes**. In this example, the free enzyme form is catalytically active, whereas the phosphoryl-enzyme form represents an inactive state. The —OH on the interconvertible enzyme represents an —OH group on a specific amino acid side chain in the protein (for example, a particular Ser residue) capable of accepting the phosphoryl group.

6. Specialized controls: Enzyme regulation is an important matter to cells, and evolution has provided a variety of additional options, including zymogens, isozymes, and modulator proteins.

Zymogens

Most proteins become fully active as their synthesis is completed and they spontaneously fold into their native, three-dimensional conformations. Some proteins, however, are synthesized as inactive precursors, called **zymogens** or **proenzymes**, that only acquire full activity upon specific proteolytic cleavage of one or several of their peptide bonds. Unlike allosteric regulation or covalent modification, zymogen activation by specific proteolysis is an irreversible process. Activation of enzymes and other physiologically important proteins by specific proteolysis is a strategy frequently exploited by biological systems to switch on processes at the appropriate time and place, as the following examples illustrate.

INSULIN. Some protein hormones are synthesized in the form of inactive precursor molecules, from which the active hormone is derived by proteolysis. For instance, **insulin**, an important metabolic regulator, is generated by proteolytic excision of a specific peptide from **proinsulin** (Figure 15.3).

PROTEOLYTIC ENZYMES OF THE DIGESTIVE TRACT. Enzymes of the digestive tract that serve to hydrolyze dietary proteins are synthesized in the stomach and pancreas as zymogens (Table 15.1). Only upon proteolytic activation are these enzymes able to form a catalytically active substrate-binding site. The activation of chymotrypsinogen is an interesting example (Figure 15.4). **Chymotrypsinogen** is a 245-residue polypeptide chain cross-linked by five disulfide bonds. Chymotrypsinogen is converted to an enzymatically active form called π -chymotrypsin when trypsin cleaves the peptide bond joining Arg¹⁵ and Ile¹⁶. The enzymatically active π -chymotrypsin acts upon other π -chymotrypsin molecules, excising two dipeptides, Ser¹⁴-Arg¹⁵ and Thr¹⁴⁷-Asn¹⁴⁸. The end product of this processing pathway is the mature protease **α -chymotrypsin**, in which the three peptide chains, A (residues 1 through 13), B (residues 16 through 146), and C (residues 149 through 245), remain together because they are linked by two disulfide bonds, one from A to B, and one from B to C. It is interesting to note that the transformation of inactive chymotrypsinogen to active π -chymotrypsin requires the cleavage of just one particular peptide bond.

BLOOD CLOTTING. The formation of blood clots is the result of a series of zymogen activations (Figure 15.5). The amplification achieved by this cascade of enzymatic activations allows blood clotting to occur rapidly in response to injury. Seven of the clotting factors in their active form are serine proteases:

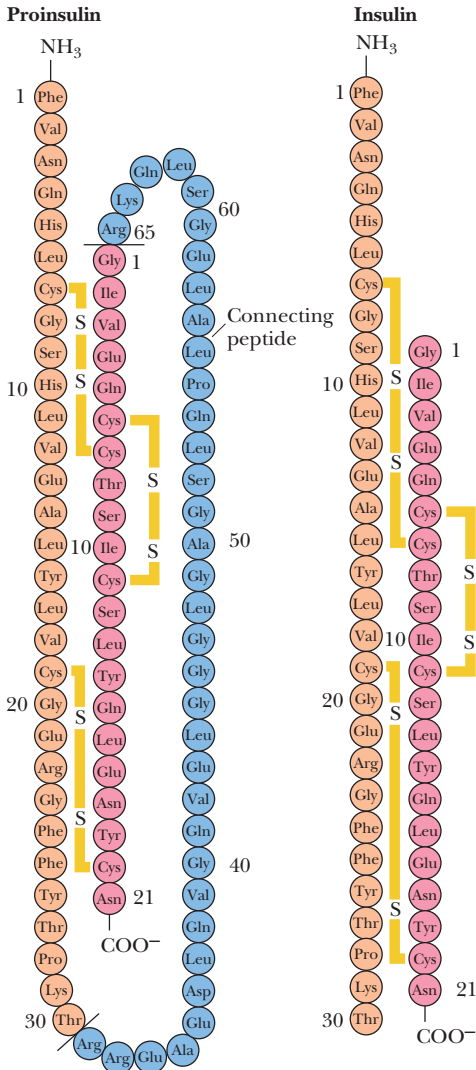


FIGURE 15.3 • Proinsulin is an 86-residue precursor to insulin (the sequence shown here is human proinsulin). Proteolytic removal of residues 31 to 65 yields insulin. Residues 1 through 30 (the B chain) remain linked to residues 66 through 87 (the A chain) by a pair of interchain disulfide bridges.

Table 15.1 Pancreatic and Gastric Zymogens		
Origin	Zymogen	Active Protease
Pancreas	Trypsinogen	Trypsin
Pancreas	Chymotrypsinogen	Chymotrypsin
Pancreas	Procarboxypeptidase	Carboxypeptidase
Pancreas	Proelastase	Elastase
Stomach	Pepsinogen	Pepsin

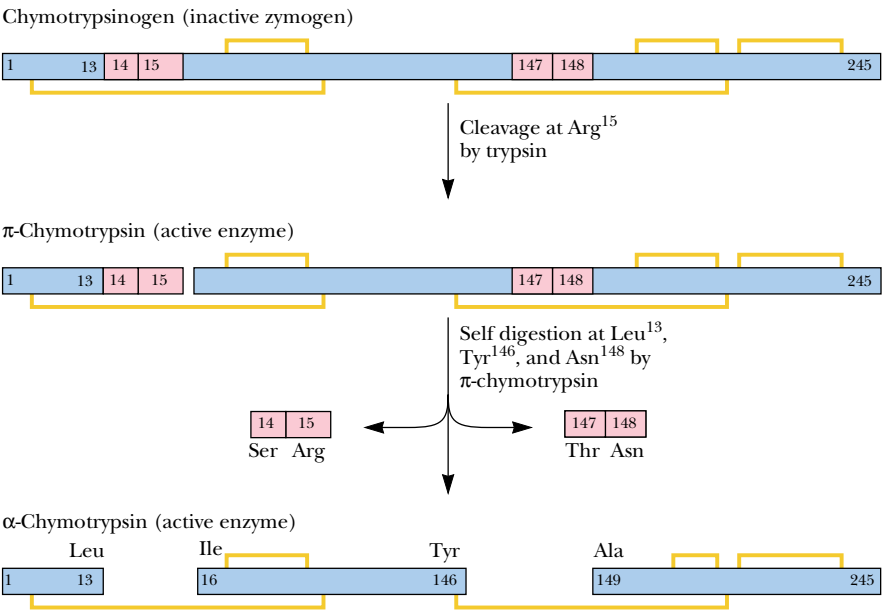


FIGURE 15.4 • The proteolytic activation of chymotrypsinogen.

kallikrein, XII_a, XI_a, IX_a, VII_a, X_a, and thrombin. Two routes to blood clot formation exist. The **intrinsic pathway** is instigated when the blood comes into physical contact with abnormal surfaces caused by injury; the **extrinsic pathway** is initiated by factors released from injured tissues. The pathways merge at Factor X and culminate in clot formation. Thrombin excises peptides rich

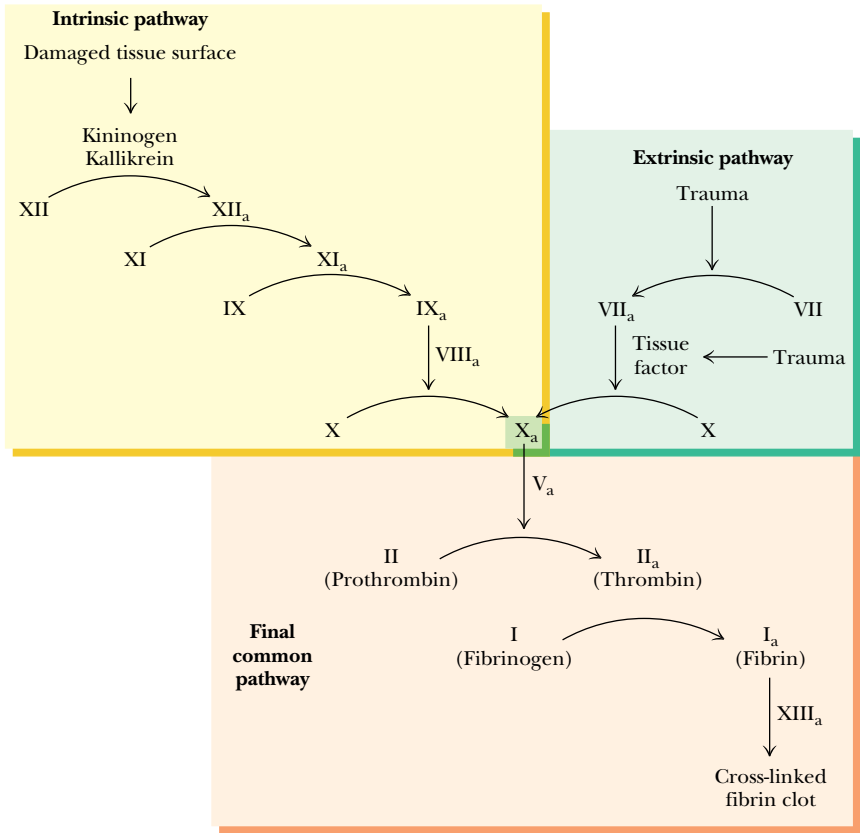


FIGURE 15.5 • The cascade of activation steps leading to blood clotting. The intrinsic and extrinsic pathways converge at Factor X, and the final common pathway involves the activation of thrombin and its conversion of fibrinogen into fibrin, which aggregates into ordered filamentous arrays that become cross-linked to form the clot.

DEEPER LOOK

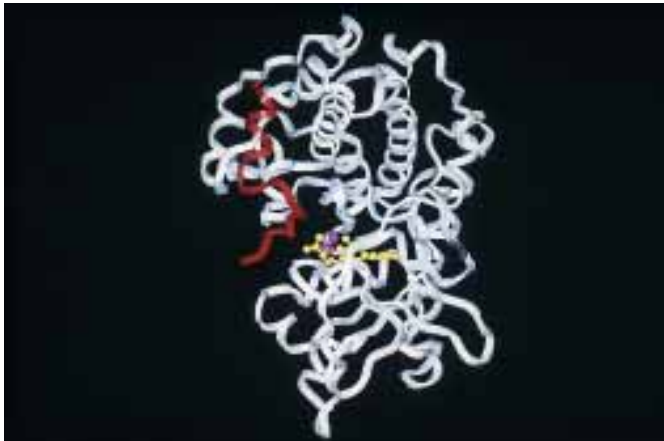
Protein Kinases: Target Recognition and Intrasteric Control

Protein kinases are converter enzymes that catalyze the ATP-dependent phosphorylation of serine, threonine, and/or tyrosine hydroxyl groups in target proteins (table). Phosphorylation introduces a bulky group bearing two negative charges, causing conformational changes that alter the target protein’s function. (Unlike a phosphoryl group, no amino acid side chain can provide *two* negative charges.) Protein kinases represent a protein superfamily whose members are widely diverse in terms of size, subunit structure, and subcellular localization. Nevertheless, all share a common catalytic mechanism based on a conserved catalytic core/kinase domain of approximately 260 amino acid residues (see figure). Protein kinases are classified as Ser/Thr- and/or Tyr-specific and are subclassified in terms of the allosteric activators they require *and* the consensus amino acid sequence within the target protein that is recognized by the kinase. For example, cAMP-dependent protein kinase (**PKA**) phosphorylates proteins having Ser or Thr residues within an R(R/K)X(S*/T*) target consensus sequence (* denotes the residue that becomes phosphorylated). That is, PKA phosphorylates Ser or Thr residues that occur in an Arg-(Arg or Lys)-(any amino acid)-(Ser or Thr) sequence segment (table).

Targeting of protein kinases to particular consensus sequence elements within proteins creates a means to regulate these kinases by **intrasteric control**. Intrasteric control occurs when a regulatory subunit (or protein domain) has a **pseudo-substrate sequence** that mimics the target sequence but lacks a OH-bearing side chain at the right place. For example, the cAMP-binding regulatory subunits of PKA (R subunits in Figure 15.7) possess the pseudosubstrate sequence RRGGA*I and this sequence binds to the active site of PKA catalytic subunits, blocking their activity. This pseudosubstrate sequence has an alanine residue where serine occurs in the PKA target sequence; Ala is sterically similar to serine but lacks a phosphorylatable OH-group. When

these PKA regulatory subunits bind cAMP, they undergo a conformational change and dissociate from the catalytic (C) subunits, and the active site of PKA is free to bind and phosphorylate its targets.

The abundance of many protein kinases in cells is an indication of the great importance of protein phosphorylation in cellular regulation. Exactly 113 protein kinase genes have been recognized in yeast, and it is estimated that the human genome encodes more than 1000 different protein kinases. **Tyrosine kinases** (protein kinases that phosphorylate Tyr residues) occur only in multicellular organisms (yeast has no tyrosine kinases). Tyrosine kinases are components of signaling pathways involved in cell–cell communication (see Chapter 34).



Cyclic AMP-dependent protein kinase is shown complexed with a pseudosubstrate peptide (red). This complex also includes ATP (yellow) and two Mn²⁺ ions (violet) bound at the active site.

in negative charge from **fibrinogen**, converting it to **fibrin**, a molecule with a different surface charge distribution. Fibrin readily aggregates into ordered fibrous arrays that are subsequently stabilized by covalent cross-links. Thrombin specifically cleaves Arg-Gly peptide bonds and is homologous to trypsin, which is also a serine protease (recall that trypsin acts only at Arg and Lys residues).

Isozymes

A number of enzymes exist in more than one quaternary form, differing in their relative proportions of structurally equivalent but catalytically distinct polypeptide subunits. A classic example is mammalian **lactate dehydrogenase (LDH)**, which exists as five different isozymes, depending on the tetrameric association of two different subunits, A and B: A₄, A₃B, A₂B₂, AB₃, and B₄ (Figure 15.6). The kinetic properties of the various LDH isozymes differ in terms of their relative affinities for the various substrates and their sensitivity to inhibition by product. Different tissues express different isozyme forms, as appropriate to their particular metabolic needs. By regulating the relative

Classification of Protein Kinases		
Protein Kinase Class	Target Sequence*	Activators
I. Ser/Thr protein kinases		
A. Cyclic nucleotide-dependent		
cAMP-dependent (PKA)	—R(R/K)X(S*/T*)—	cAMP
cGMP-dependent	—(R/K)KKX(S*/T*)—	cGMP
B. Ca ²⁺ -calmodulin (CaM)-dependent		
Phosphorylase kinase (PhK)	—KRKQIS*VRGL—	phosphorylation by PKA
Myosin light-chain kinase (MLCK)	—KKRPQRATS*NV—	Ca ²⁺ CaM
C. Protein kinase C (PKC)		Ca ²⁺ , diacylglycerol
D. Mitogen-activated protein kinases (MAP kinases)	—PXX(S*/T*)P—	phosphorylation by MAPK kinase
E. G protein-coupled receptors		
β-Adrenergic receptor kinase (BARK)		
Rhodopsin kinase		
II. Ser/Thr/Tyr protein kinases		
MAP kinase kinase (MAPK kinase)	—TEY—	phosphorylation by Raf (a protein kinase)
III. Tyr protein kinases		
A. Cytosolic tyrosine kinases (<i>src</i> , <i>fgr</i> , <i>abl</i> , etc.)		
B. Receptor tyrosine kinases (RTKs)		
Plasma membrane receptors for hormones such as <i>epidermal growth factor</i> (EGF) or <i>platelet-derived growth factor</i> (PDGF)		

*X denotes any amino acid.

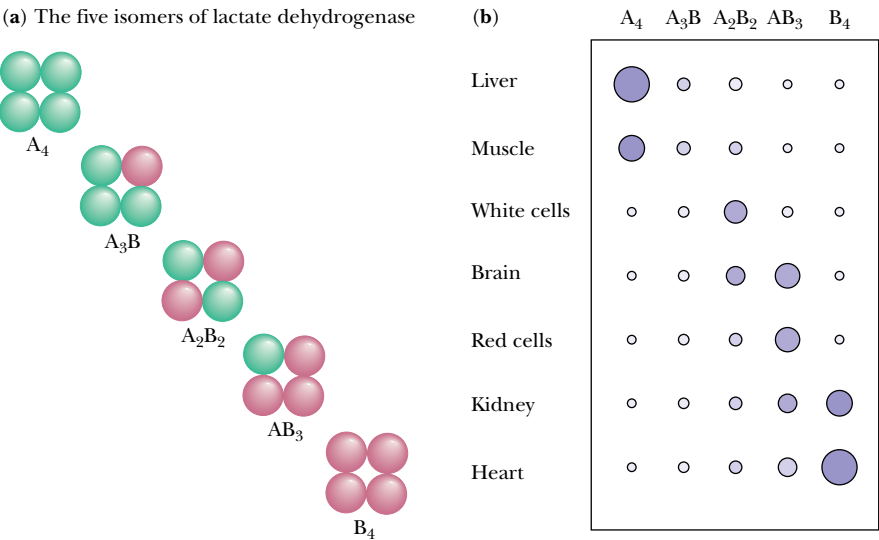


FIGURE 15.6 • The isozymes of lactate dehydrogenase (LDH). Active muscle tissue becomes anaerobic and produces pyruvate from glucose via glycolysis (Chapter 19). It needs LDH to regenerate NAD⁺ from NADH so glycolysis can continue. The lactate produced is released into the blood. The muscle LDH isozyme (A₄) works best in the NAD⁺-regenerating direction. Heart tissue is aerobic and uses lactate as a fuel, converting it to pyruvate via LDH and using the pyruvate to fuel the citric acid cycle to obtain energy. The heart LDH isozyme (B₄) is inhibited by excess pyruvate so the fuel won't be wasted.

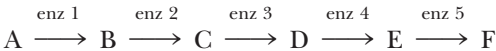
amounts of A and B subunits they synthesize, the cells of various tissues control which isozymic forms are likely to assemble, and, thus, which kinetic parameters prevail.

Modulator Proteins

Modulator proteins are yet another way that cells mediate metabolic activity. **Modulator proteins** are proteins that bind to enzymes, and by binding, influence the activity of the enzyme. For example, some enzymes, such as *cAMP-dependent protein kinase* (Chapter 23), exist as dimers of catalytic subunits and regulatory subunits. These regulatory subunits are *modulator proteins* that suppress the activity of the catalytic subunits. Dissociation of the regulatory subunits (modulator proteins) activates the catalytic subunits; reassociation once again suppresses activity (Figure 15.7). **Phosphoprotein phosphatase inhibitor-1 (PPI-1)** is another example of a modulator protein. When PPI-1 is phosphorylated on one of its serine residues, it binds to *phosphoprotein phosphatase* (Figure 15.2), inhibiting its phosphatase activity. The result is an increased phosphorylation of the interconvertible enzyme targeted by the protein kinase/phosphoprotein phosphatase cycle (Figure 15.2). We will meet other important representatives of this class as the processes of metabolism unfold in subsequent chapters. For now, let us focus our attention on the fascinating kinetics of allosteric enzymes.

15.3 • The Allosteric Regulation of Enzyme Activity

Allosteric regulation acts to modulate enzymes situated at key steps in metabolic pathways. Consider as an illustration the following pathway, where A is the precursor for formation of an end product, F, in a sequence of five enzyme-catalyzed reactions:



In this scheme, F symbolizes an essential metabolite, such as an amino acid or a nucleotide. In such systems, F, the essential end product, inhibits *enzyme 1*, the *first step* in the pathway. Therefore, when sufficient F is synthesized, it blocks further synthesis of itself. This phenomenon is called **feedback inhibition** or **feedback regulation**.

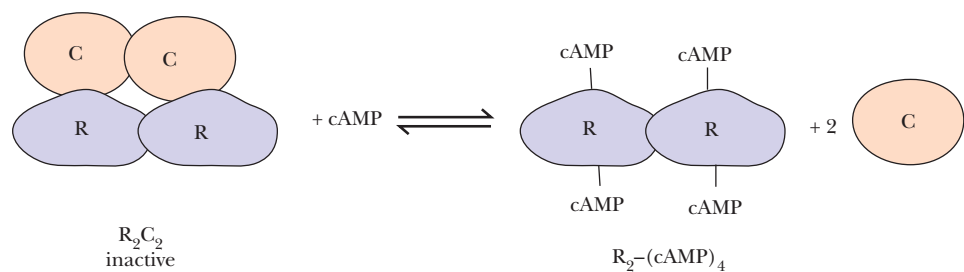


FIGURE 15.7 • Cyclic AMP–dependent protein kinase (also known as *PKA*) is a 150- to 170-kD R_2C_2 tetramer in mammalian cells. The two R (regulatory) subunits bind cAMP ($K_b = 3 \times 10^{-8} M$); cAMP binding releases the R subunits from the C (catalytic) subunits. C subunits are enzymatically active as monomers.

General Properties of Regulatory Enzymes

Enzymes such as enzyme 1, which are subject to feedback regulation, represent a distinct class of enzymes, the **regulatory enzymes**. As a class, these enzymes have certain exceptional properties:

1. Their kinetics do not obey the Michaelis–Menten equation. Their v versus $[S]$ plots yield **sigmoid-** or **S-shaped** curves rather than rectangular hyperbolas (Figure 15.8). Such curves suggest a second-order (or higher) relationship between v and $[S]$; that is, v is proportional to $[S]^n$, where $n > 1$. A qualitative description of the mechanism responsible for the S-shaped curves is that binding of one S to a protein molecule makes it easier for additional substrate molecules to bind to the same protein molecule. In the jargon of allostery, substrate binding is **cooperative**.
2. Inhibition of a regulatory enzyme by a feedback inhibitor does not conform to any normal inhibition pattern, and the feedback inhibitor F bears little structural similarity to A, the substrate for the regulatory enzyme. F apparently acts at a binding site distinct from the substrate-binding site. The term *allosteric* is apt, because F is sterically dissimilar and, moreover, acts at a site other than the site for S. Its effect is called **allosteric inhibition**.
3. Regulatory or allosteric enzymes like enzyme 1 are, in some instances, regulated by activation. That is, whereas some effector molecules such as F exert negative effects on enzyme activity, other effectors show stimulatory, or positive, influences on activity.
4. Allosteric enzymes have an oligomeric organization. They are composed of more than one polypeptide chain (subunit) and have more than one S-binding site per enzyme molecule.
5. The working hypothesis is that, by some means, interaction of an allosteric enzyme with effectors alters the distribution of conformational possibilities or subunit interactions available to the enzyme. That is, the regulatory effects exerted on the enzyme's activity are achieved by conformational changes occurring in the protein when effector metabolites bind.

In addition to enzymes, noncatalytic proteins may exhibit many of these properties; hemoglobin is the classic example. The allosteric properties of hemoglobin are the subject of a Special Focus beginning on page 480.

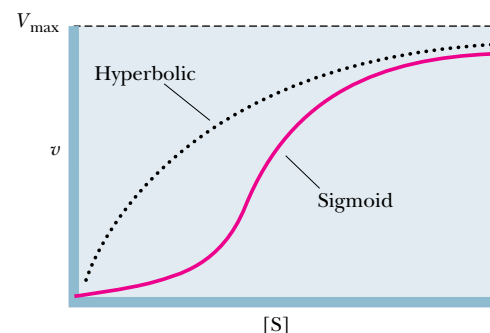


FIGURE 15.8 • Sigmoid v versus $[S]$ plot. The dotted line represents the hyperbolic plot characteristic of normal Michaelis–Menten-type enzyme kinetics.

15.4 • Models for the Allosteric Behavior of Proteins

The Symmetry Model of Monod, Wyman, and Changeux

In 1965, Jacques Monod, Jeffries Wyman, and Jean-Pierre Changeux proposed a theoretical model of allosteric transitions based on the observation that allosteric proteins are oligomers. They suggested that allosteric proteins can exist in (at least) two conformational states, designated **R**, signifying “relaxed,” and **T**, or “taut,” and that, in each protein molecule, all of the subunits have the same conformation (either R or T). That is, molecular symmetry is conserved. Molecules of mixed conformation (having subunits of both R and T states) are not allowed by this model.

In the absence of ligand, the two states of the allosteric protein are in equilibrium:



(Note that the subscript “0” signifies “in the absence of ligand.”) The equilibrium constant is termed L : $L = T_0/R_0$. L is assumed to be large; that is, the

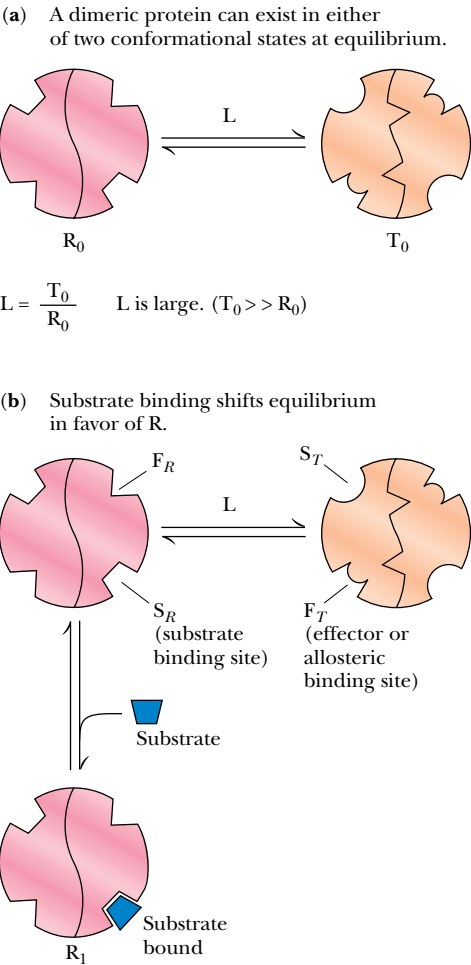


FIGURE 15.9 • Monod–Wyman–Changeux (MWC) model for allosteric transitions. Consider a dimeric protein that can exist in either of two conformational states, R or T. Each subunit in the dimer has a binding site for substrate S and an allosteric effector site, F. The promoters are symmetrically related to one another in the protein, and symmetry is conserved regardless of the conformational state of the protein. The different states of the protein, with or without bound ligand, are linked to one another through the various equilibria. Thus, the relative population of protein molecules in the R or T state is a function of these equilibria and the concentration of the various ligands, substrate (S), and effectors (which bind at F_R or F_T). As [S] is increased, the T/R equilibrium shifts in favor of an increased proportion of R-conformers in the total population (that is, more protein molecules in the R conformational state).

amount of the protein in the T conformational state is much greater than the amount in the R conformation. Let us suppose that $L = 10^4$.

The affinities of the two states for substrate, S, are characterized by the respective dissociation constants, K_R and K_T . The model supposes that $K_T \gg K_R$. That is, the affinity of R_0 for S is much greater than the affinity of T_0 for S. Let us choose the extreme where $K_R/K_T = 0$ (that is, K_T is infinitely greater than K_R). In effect, we are picking conditions where S binds only to R. (If K_T is infinite, T does not bind S.)

Given these parameters, consider what happens when S is added to a solution of the allosteric protein at conformational equilibrium (Figure 15.9). Although the relative $[R_0]$ concentration is small, S will bind “only” to R_0 , forming R_1 . This depletes the concentration of R_0 , perturbing the T_0/R_0 equilibrium. To restore equilibrium, molecules in the T_0 conformation undergo a transition to R_0 . This shift renders more R_0 available to bind S, yielding R_1 , diminishing $[R_0]$, perturbing the T_0/R_0 equilibrium, and so on. Thus, these linked equilibria (Figure 15.9) are such that S-binding by the R_0 state of the allosteric protein perturbs the T_0/R_0 equilibrium with the result that S-binding drives the conformational transition, $T_0 \rightarrow R_0$.

In just this simple system, *cooperativity* is achieved because each subunit has a binding site for S, and thus, *each protein molecule has more than one binding site for S*. Therefore, the increase in the population of R conformers gives a progressive increase in the number of sites available for S. The extent of cooperativity depends on the relative T_0/R_0 ratio and the relative affinities of R and T for S. If L is large (that is, the equilibrium lies strongly in favor of T_0) and if $K_T \gg K_R$, as in the example we have chosen, cooperativity is great (Figure 15.10). Ligands such as S here that bind in a cooperative manner, so that binding of one equivalent enhances the binding of additional equivalents of S to the same protein molecule, are termed **positive homotropic effectors**. (The prefix “homo” indicates that the ligand influences the binding of like molecules.)

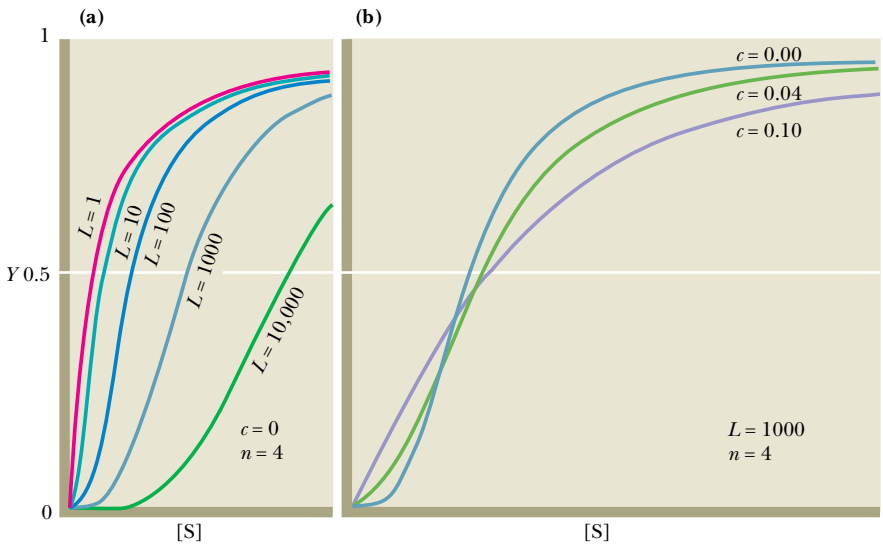


FIGURE 15.10 • The Monod–Wyman–Changeux model. Graphs of allosteric effects for a tetramer ($n = 4$) in terms of Y , the saturation function, versus $[S]$. Y is defined as [ligand-binding sites that are occupied by ligand]/[total ligand-binding sites]. (a) A plot of Y as a function of $[S]$, at various L values. (b) Y as a function of $[S]$, at different c , where $c = K_R/K_T$. (When $c = 0$, K_T is infinite.) (Adapted from Monod, J., Wyman, J., and Changeux, J.-P., 1965. *On the nature of allosteric transitions: A plausible model*. Journal of Molecular Biology 12:92.)

Heterotropic Effectors

This simple system also provides an explanation for the more complex substrate-binding responses to positive and negative effectors. Effectors that influence the binding of something other than themselves are termed **heterotropic effectors**. For example, effectors that promote S binding are termed **positive heterotropic effectors** or **allosteric activators**. Effectors that diminish S binding are **negative heterotropic effectors** or **allosteric inhibitors**. Feedback inhibitors fit this class. Consider a protein composed of two subunits, each of which has two binding sites: one for the substrate, S, and one to which allosteric effectors bind, the *allosteric site*. Assume that S binds preferentially (“only”) to the R conformer; further assume that the *positive heterotropic effector*, A, binds to the allosteric site only when the protein is in the R conformation, and the *negative allosteric effector*, I, binds at the allosteric site only if the protein is in the T conformation. Thus, with respect to binding at the allosteric site, A and I are competitive with each other.

Positive Effectors

If A binds to R₀, forming the new species R_{1(A)}, the relative concentration of R₀ is decreased and the T₀/R₀ equilibrium is perturbed (Figure 15.11). As a consequence, a relative T₀ → R₀ shift occurs in order to restore equilibrium. The net effect is an increase in the number of R conformers in the presence of A, meaning that more binding sites for S are available. For this reason, A

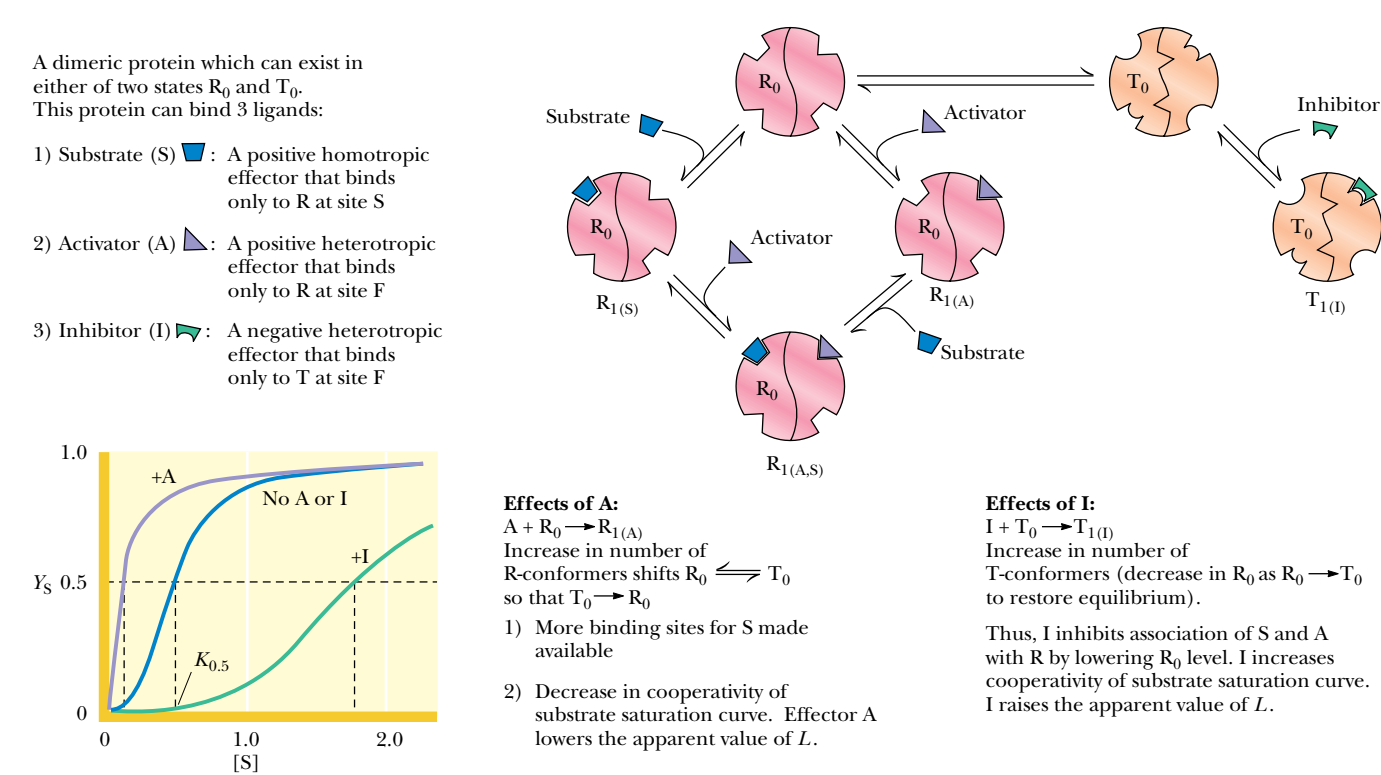
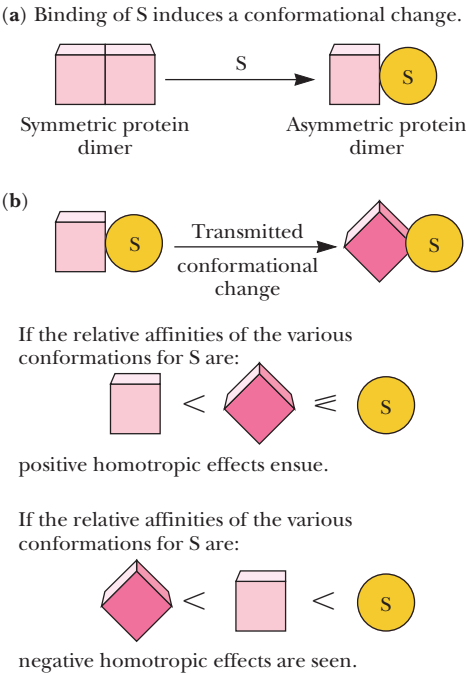


FIGURE 15.11 • Heterotropic allosteric effects: A and I binding to R and T, respectively. The linked equilibria lead to changes in the relative amounts of R and T and, therefore, shifts in the substrate saturation curve. This behavior, depicted by the graph, defines an allosteric “K” system. The parameters of such a system are: (1) S and A (or I) have different affinities for R and T and (2) A (or I) modifies the apparent $K_{0.5}$ for S by shifting the relative R versus T population.

A D E E P E R L O O K

An Alternative Allosteric Model: The Sequential Allosteric Model of Koshland, Nemethy, and Filmer

Daniel Koshland has championed the idea that proteins are inherently flexible molecules whose conformations are altered when ligands bind. This notion serves as the fundamental tenet of the “induced-fit hypothesis” discussed earlier. Because this is so, ligand binding can potentially cause conformational changes in the protein. Depending on the nature of these conformational changes, virtually any sort of allosteric interaction is possible. That is, the binding of one ligand could result in conformational transitions in the protein that make it easier or harder for other ligands (of the same or different kinds) to bind. In 1966, Koshland and his colleagues proposed an allosteric model in which ligand-induced conformational changes caused transition to a conformational state with altered affinities. Because ligand binding and conformational transitions were distinct steps in a sequential pathway, the Koshland, Nemethy, Filmer (or KNF) model is dubbed the **sequential model** for allosteric transitions. The figure depicts the essential features of this model in a hypothetical dimeric protein. Binding of the ligand S induces a conformational change in the subunit to which it binds. Note that there is no requirement for conservation of symmetry here; the two subunits can assume different conformations (represented as a square and a circle). If the *subunit interactions are tightly coupled*, then binding of S to one subunit could cause the other subunit(s) to assume a conformation having a greater, or a lesser, affinity for S (or some other ligand). The underlying mechanism rests on the fact that the ligand-induced conformational change in one subunit can transmit its effects to neighboring subunits by changing the interactions and alignments of amino acid residues at the interface between subunits. Depending on the relative ligand affinity of the conformation adopted by the neighboring subunit, the overall effect for further ligand binding may be positive, negative, or neutral (figure).



The Koshland–Nemethy–Filmer sequential model for allosteric behavior. (a) S-binding can, by induced fit, cause a conformational change in the subunit to which it binds. (b) If subunit interactions are tightly coupled, binding of S to one subunit may cause the other subunit to assume a conformation having a greater (positive homotropic) or lesser (negative homotropic) affinity for S. That is, the ligand-induced conformational change in one subunit can affect the adjoining subunit. Such effects could be transmitted between neighboring peptide domains by changing alignments of nonbonded amino acid residues.

leads to a decrease in the cooperativity of the substrate saturation curve, as seen by a shift of this curve to the left (Figure 15.11). Effectively, the presence of A lowers the apparent value of *L*.

Negative Effectors

The converse situation applies in the presence of I, which binds “only” to T. I-binding will lead to an increase in the population of T conformers, at the expense of *R*₀ (Figure 15.11). The decline in [*R*₀] means that it is less likely for S (or A) to bind. Consequently, the presence of I increases the cooperativity (that is, the sigmoidicity) of the substrate saturation curve, as evidenced by the shift of this curve to the right (Figure 15.11). The presence of I raises the apparent value of *L*.

K Systems and V Systems

The allosteric model just presented is called a **K system** because the concentration of substrate giving half-maximal velocity, defined as $K_{0.5}$, changes in response to effectors (Figure 15.11). Note that $V_{\max}$ is constant in this system.

An allosteric situation where $K_{0.5}$ is constant but the apparent $V_{\max}$ changes in response to effectors is termed a **V system**. In a V system, all v versus S plots are hyperbolic rather than sigmoid (Figure 15.12). The positive heterotropic effector A activates by raising $V_{\max}$, whereas I, the negative heterotropic effector, decreases it. Note that neither A nor I affects $K_{0.5}$. This situation arises if R and T have the *same* affinity for the substrate, S, but differ in their catalytic ability and their affinities for A and I. A and I thus can shift the relative T/R distribution. Acetyl-coenzyme A carboxylase, the enzyme catalyzing the committed step in the fatty acid biosynthetic pathway, behaves as a V system in response to its allosteric activator, citrate (see Chapter 25).

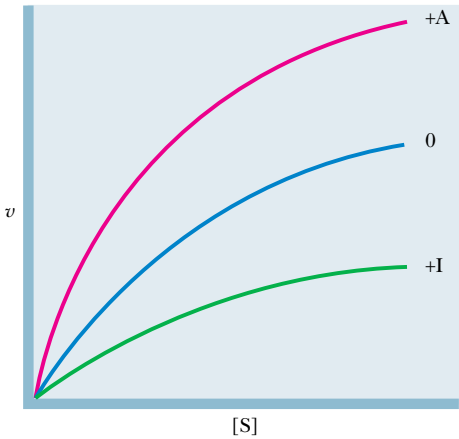


FIGURE 15.12 • v versus $[S]$ curves for an allosteric “V” system. The V system fits the model of Monod, Wyman, and Changeux, given the following conditions: (1) R and T have the *same* affinity for the substrate, S. (2) The effectors A and I have different affinities for R and T and thus can shift the relative T/R distribution. (That is, A and I change the apparent value of L .) Assume as before that A binds “only” to the R state and I binds “only” to the T state. (3) R and T differ in their catalytic ability. Assume that R is the enzymatically active form, whereas T is inactive. Because A perturbs the T/R equilibrium in favor of more R, A increases the apparent $V_{\max}$. I favors transition to the inactive T state.

15.5 • Glycogen Phosphorylase: Allosteric Regulation and Covalent Modification

The Glycogen Phosphorylase Reaction

The cleavage of glucose units from the nonreducing ends of glycogen molecules is catalyzed by **glycogen phosphorylase**, an allosteric enzyme. The enzymatic reaction involves phosphorolysis of the bond between C-1 of the departing glucose unit and the glycosidic oxygen, to yield *glucose-1-phosphate* and a glycogen molecule that is shortened by one residue (Figure 15.13). (Because

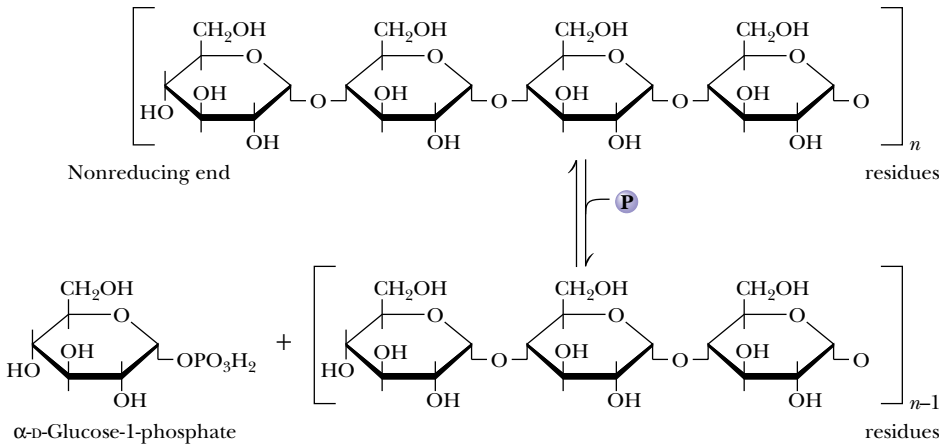
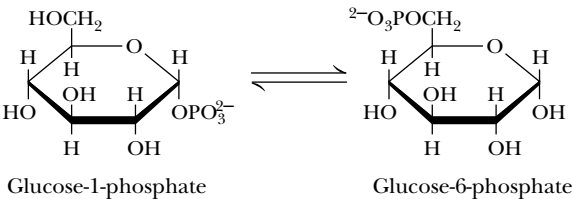


FIGURE 15.13 • The glycogen phosphorylase reaction.

FIGURE 15.14 • The phosphoglucomutase reaction.



the reaction involves attack by phosphate instead of H₂O, it is referred to as a **phosphorolysis** rather than a hydrolysis.) Phosphorolysis produces a phosphorylated sugar product, glucose-1-P, which is converted to the glycolytic substrate, glucose-6-P, by *phosphoglucomutase* (Figure 15.14). In muscle, glucose-6-P proceeds into glycolysis, providing needed energy for muscle contraction. In liver, hydrolysis of glucose-6-P yields glucose, which is exported to other tissues via the circulatory system.

The Structure of Glycogen Phosphorylase

Muscle glycogen phosphorylase is a dimer of two identical subunits (842 residues, 97.44 kD). Each subunit contains a pyridoxal phosphate cofactor, covalently linked as a Schiff base to Lys⁶⁸⁰. Each subunit contains an active site (at the center of the subunit) and an allosteric effector site near the subunit interface (Figure 15.15). In addition, a regulatory phosphorylation site is located at Ser¹⁴ on each subunit. A glycogen-binding site on each subunit facilitates prior association of glycogen phosphorylase with its substrate and also exerts regulatory control on the enzymatic reaction.

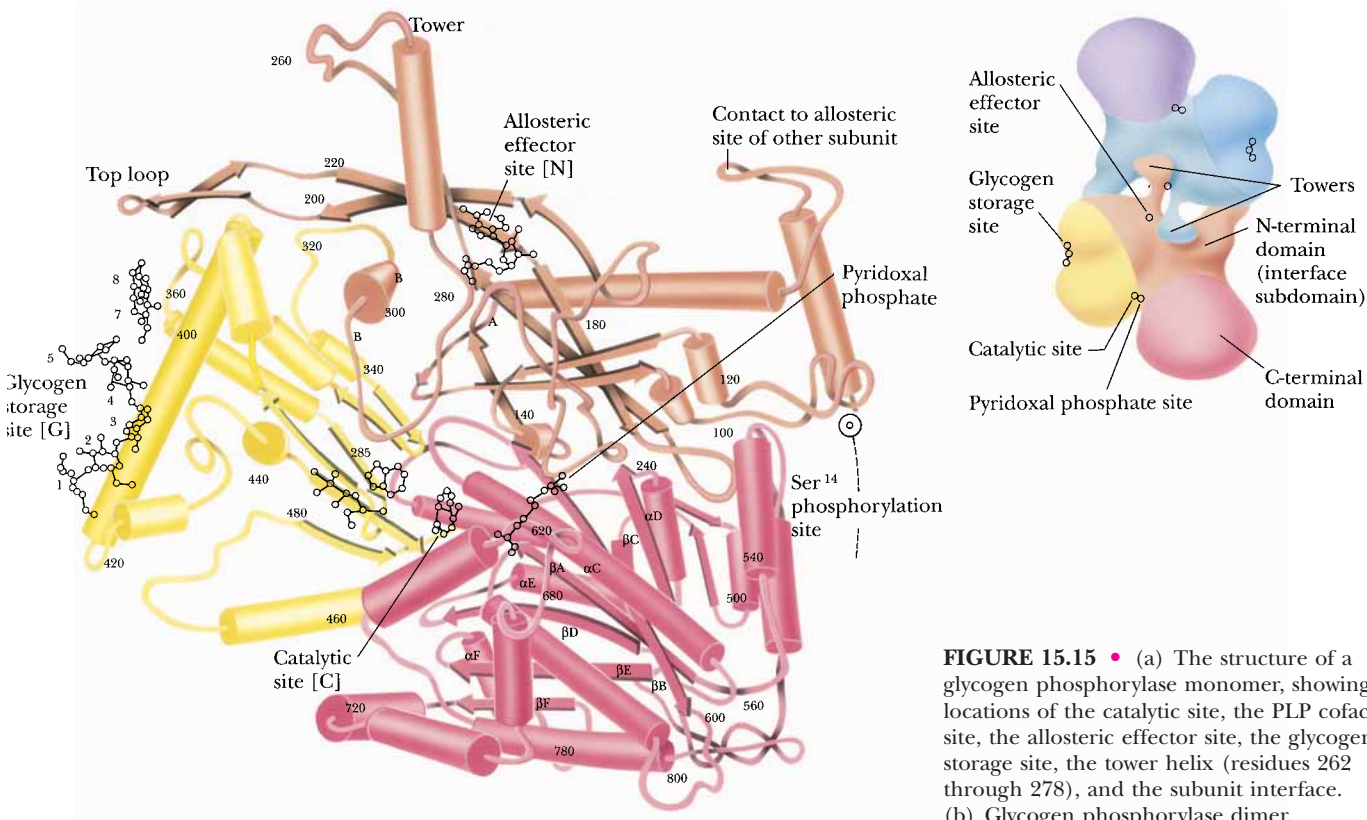


FIGURE 15.15 • (a) The structure of a glycogen phosphorylase monomer, showing the locations of the catalytic site, the PLP cofactor site, the allosteric effector site, the glycogen storage site, the tower helix (residues 262 through 278), and the subunit interface. (b) Glycogen phosphorylase dimer.

Each subunit contributes a tower helix (residues 262 to 278) to the subunit–subunit contact interface in glycogen phosphorylase. In the phosphorylase dimer, the tower helices extend from their respective subunits and pack against each other in an antiparallel manner.

Regulation of Glycogen Phosphorylase by Allosteric Effectors

Muscle Glycogen Phosphorylase Shows Cooperativity in Substrate Binding

The binding of the substrate *inorganic phosphate* (P_i) to muscle glycogen phosphorylase is highly cooperative (Figure 15.16a), which allows the enzyme activity to increase markedly over a rather narrow range of substrate concentration. P_i is a *positive homotropic effector* with regard to its interaction with glycogen phosphorylase.

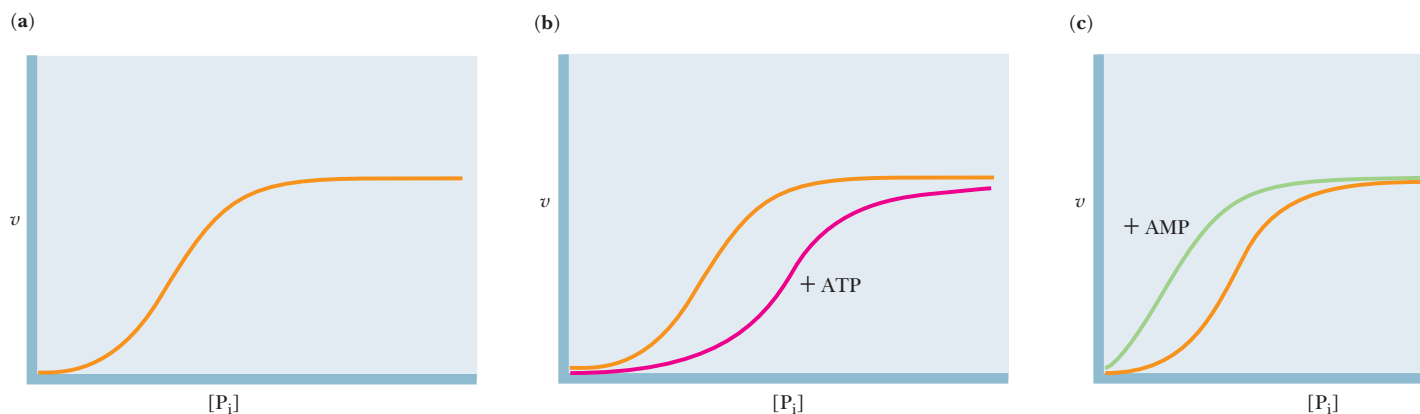
ATP and Glucose-6-P Are Allosteric Inhibitors of Glycogen Phosphorylase

ATP can be viewed as the “end product” of glycogen phosphorylase action, in that the glucose-1-P liberated by glycogen phosphorylase is degraded in muscle via metabolic pathways whose purpose is energy (ATP) production. Glucose-1-P is readily converted into glucose-6-P to feed such pathways. (In the liver, glucose-1-P from glycogen is converted to glucose and released into the bloodstream to raise blood glucose levels.) Thus, feedback inhibition of glycogen phosphorylase by ATP and glucose-6-P provides a very effective way to regulate glycogen breakdown. Both ATP and glucose-6-P act by decreasing the affinity of glycogen phosphorylase for its substrate P_i (Figure 15.16b). Because the binding of ATP or glucose-6-P has a negative effect on substrate binding, these substances act as *negative heterotropic effectors*. Note in Figure 15.16b that the substrate saturation curve is displaced to the right in the presence of ATP or glucose-6-P, and a higher substrate concentration is needed to achieve half-maximal velocity ($V_{\max}/2$). When concentrations of ATP or glucose-6-P accumulate to high levels, glycogen phosphorylase is inhibited; when [ATP] and [glucose-6-P] are low, the activity of glycogen phosphorylase is regulated by availability of its substrate, P_i .

AMP Is an Allosteric Activator of Glycogen Phosphorylase

AMP also provides a regulatory signal to glycogen phosphorylase. It binds to the same site as ATP, but it stimulates glycogen phosphorylase rather than inhibiting it (Figure 15.16c). AMP acts as a *positive heterotropic effector*, meaning that it enhances the binding of substrate to glycogen phosphorylase. Significant levels of AMP indicate that the energy status of the cell is low and that more

FIGURE 15.16 • v versus S curves for glycogen phosphorylase. (a) The sigmoid response of glycogen phosphorylase to the concentration of the substrate phosphate (P_i) shows strong positive cooperativity. (b) ATP is a feedback inhibitor that affects the affinity of glycogen phosphorylase for its substrates but does not affect $V_{\max}$. (Glucose-6-P shows similar effects on glycogen phosphorylase.) (c) AMP is a positive heterotropic effector for glycogen phosphorylase. It binds at the same site as ATP. AMP and ATP are competitive. Like ATP, AMP affects the affinity of glycogen phosphorylase for its substrates, but does not affect $V_{\max}$.



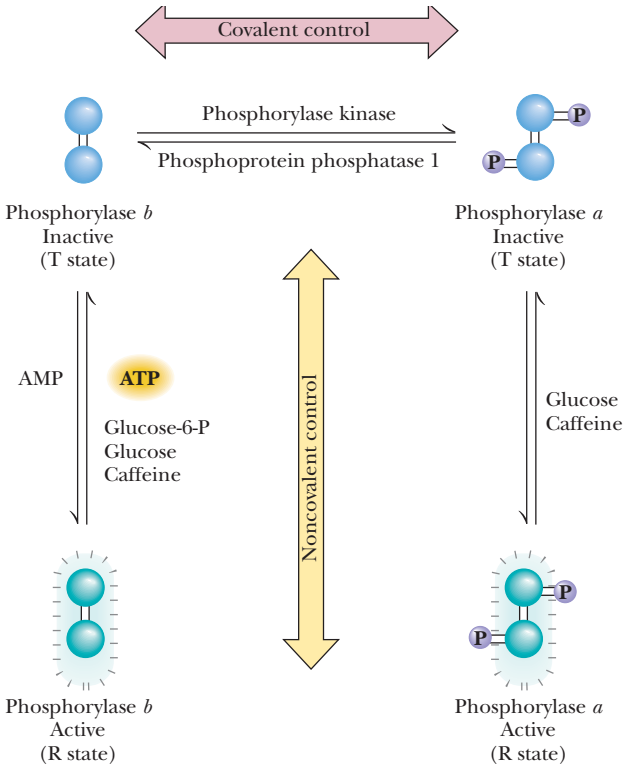
energy (ATP) should be produced. Reciprocal changes in the cellular concentrations of ATP and AMP and their competition for binding to the same site (the *allosteric site*) on glycogen phosphorylase, with opposite effects, allow these two nucleotides to exert *rapid and reversible control* over glycogen phosphorylase activity. Such reciprocal regulation ensures that the production of energy (ATP) is commensurate with cellular needs.

To summarize, muscle glycogen phosphorylase is allosterically activated by AMP and inhibited by ATP and glucose-6-P; caffeine can also act as an allosteric inhibitor (Figure 15.17). When ATP and glucose-6-P are abundant, glycogen breakdown is inhibited. When cellular energy reserves are low (i.e., high [AMP] and low [ATP] and [G-6-P]), glycogen catabolism is stimulated.

Glycogen phosphorylase conforms to the Monod–Wyman–Changeux model of allosteric transitions, with the active form of the enzyme designated the **R state** and the inactive form denoted as the **T state** (Figure 15.17). Thus, AMP promotes the conversion to the active R state, whereas ATP, glucose-6-P, and caffeine favor conversion to the inactive T state.

X-ray diffraction studies of glycogen phosphorylase in the presence of allosteric effectors have revealed the molecular basis for the $T \rightleftharpoons R$ conversion. Although the structure of the central core of the phosphorylase subunits is identical in the T and R states, a significant change occurs at the subunit interface between the T and R states. This conformation change at the subunit interface is linked to a structural change at the active site that is important for catalysis. In the T state, the negatively charged carboxyl group of Asp²⁸³ faces the active site, so that binding of the anionic substrate phosphate is unfavorable. In the conversion to the R state, Asp²⁸³ is displaced from the active site and replaced by Arg⁵⁶⁹. The exchange of negatively charged aspartate for positively charged arginine at the active site provides a favorable binding site for phosphate. These allosteric controls serve as a mechanism for adjusting the activity of glycogen phosphorylase to meet normal metabolic demands. However, in crisis situations in which abundant energy (ATP) is needed imme-

FIGURE 15.17 • The mechanism of covalent modification and allosteric regulation of glycogen phosphorylase. The T states are blue and the R states blue-green.



diately, these controls can be overridden by covalent modification of glycogen phosphorylase. Covalent modification through phosphorylation of Ser¹⁴ in glycogen phosphorylase converts the enzyme from a less active, allosterically regulated form (the *b* form) to a more active, allosterically unresponsive form (the *a* form). Covalent modification is like a “permanent” allosteric transition that is independent of [allosteric effector], such as AMP.

Regulation of Glycogen Phosphorylase by Covalent Modification

As early as 1938, it was known that glycogen phosphorylase existed in two forms: the less active **phosphorylase *b*** and the more active **phosphorylase *a***. In 1956, Edwin Krebs and Edmond Fischer reported that a “converting enzyme” could convert phosphorylase *b* to phosphorylase *a*. Three years later, Krebs and Fischer demonstrated that the conversion of phosphorylase *b* to phosphorylase *a* involved covalent phosphorylation, as in Figure 15.17.

Phosphorylation of Ser¹⁴ causes a dramatic conformational change in phosphorylase. Upon phosphorylation, the amino-terminal end of the protein (including residues 10 through 22) swings through an arc of 120°, moving into the subunit interface (Figure 15.18). This conformational change moves Ser¹⁴ by more than 3.6 nm.

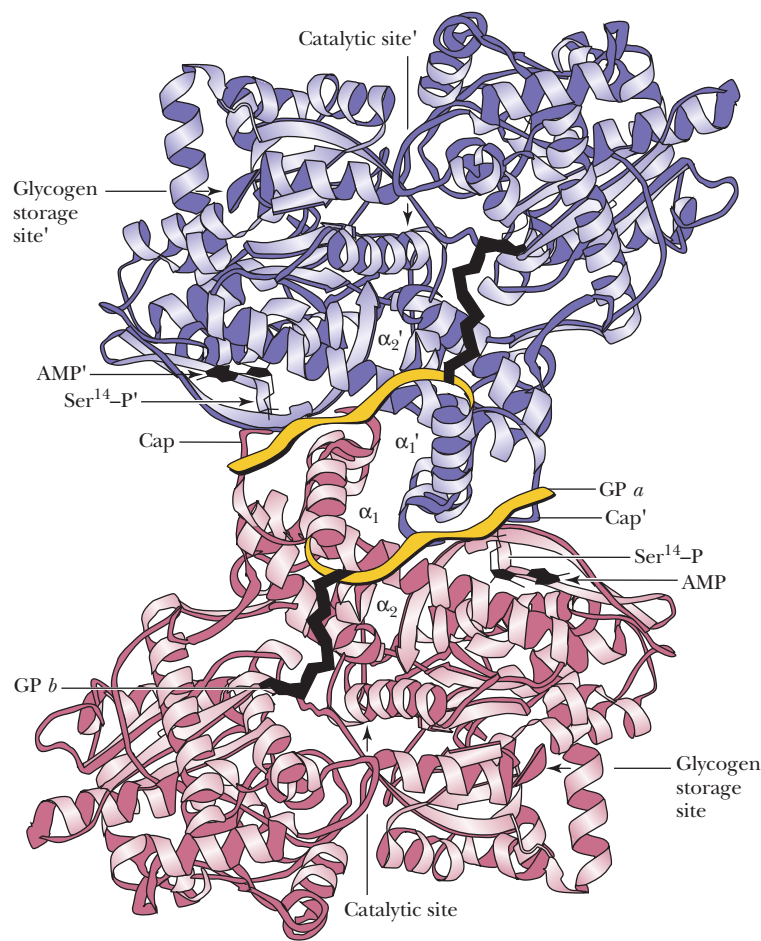


FIGURE 15.18 • In this diagram of the glycogen phosphorylase dimer, the phosphorylation site (Ser¹⁴) and the allosteric (AMP) site face the viewer. Access to the catalytic site is from the opposite side of the protein. The diagram shows the major conformational change that occurs in the N-terminal residues upon phosphorylation of Ser¹⁴. The solid black line shows the conformation of residues 10 to 23 in the *b*, or unphosphorylated, form of glycogen phosphorylase. The conformational change in the location of residues 10 to 23 upon phosphorylation of Ser¹⁴ to give the *a* (phosphorylated) form of glycogen phosphorylase is shown in yellow. Note that these residues move from intrasubunit contacts into intersubunit contacts at the subunit interface. (Sites on the two respective subunits are denoted, with those of the upper subunit designated by primes (').) (Adapted from Johnson, L. N., and Barford, D., 1993. The effects of phosphorylation on the structure and function of proteins. Annual Review of Biophysics and Biomolecular Structure 22:199–232.)

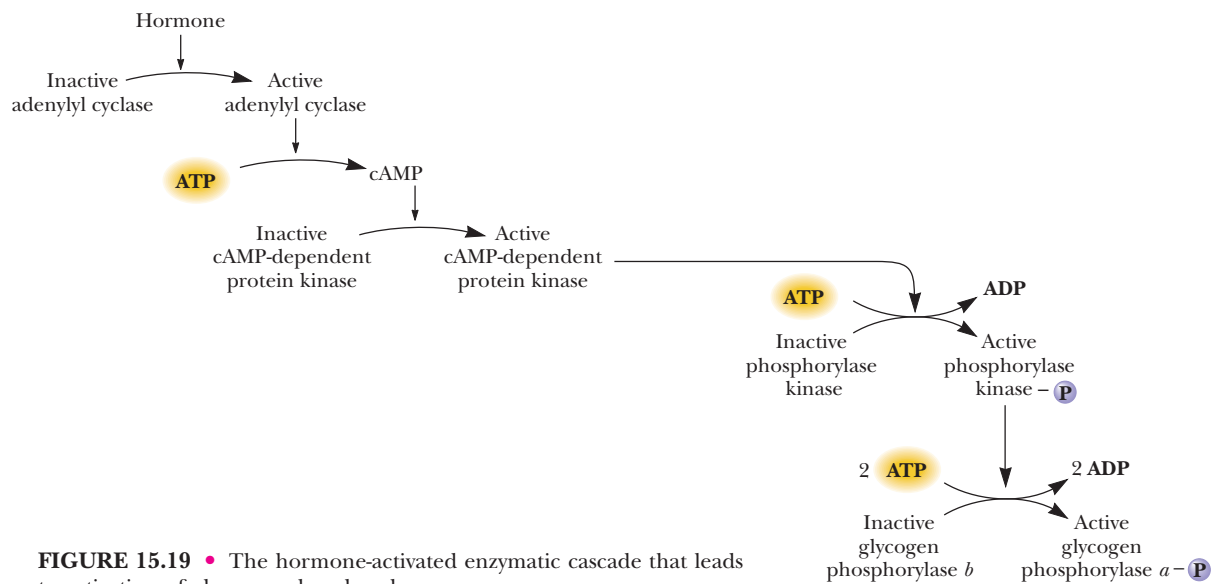


FIGURE 15.19 • The hormone-activated enzymatic cascade that leads to activation of glycogen phosphorylase.

Dephosphorylation of glycogen phosphorylase is carried out by **phosphoprotein phosphatase 1**. The action of phosphoprotein phosphatase 1 inactivates glycogen phosphorylase.

Enzyme Cascades Regulate Glycogen Phosphorylase

The phosphorylation reaction that activates glycogen phosphorylase is mediated by an **enzyme cascade** (Figure 15.19). The first part of the cascade leads to hormonal stimulation (described in the next section) of **adenylyl cyclase**, a membrane-bound enzyme that converts ATP to *adenosine-3',5'-cyclic monophosphate*, denoted as *cyclic AMP* or simply *cAMP* (Figure 15.20). This regulatory molecule is found in all eukaryotic cells and acts as an intracellular messenger molecule, controlling a wide variety of processes. Cyclic AMP is known as a **second messenger** because it is the intracellular agent of a hormone (the “first messenger”). (The myriad cellular roles of cyclic AMP are described in detail in Chapter 34.)

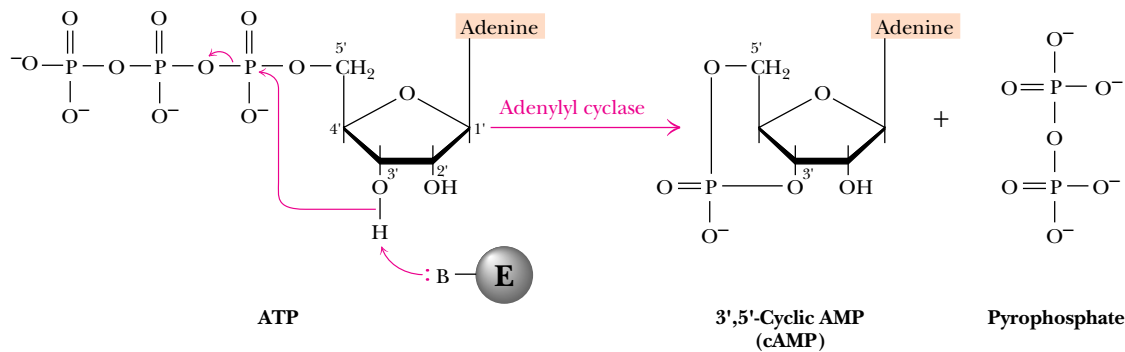


FIGURE 15.20 • The adenylyl cyclase reaction yields 3',5'-cyclic AMP and pyrophosphate. The reaction is driven forward by subsequent hydrolysis of pyrophosphate by the enzyme inorganic pyrophosphatase.

The hormonal stimulation of adenylyl cyclase is effected by a transmembrane signaling pathway consisting of three components, all membrane-associated. Binding of hormone to the external surface of a hormone receptor causes a conformational change in this transmembrane protein, which in turn stimulates a **GTP-binding protein** (abbreviated **G protein**). G proteins are heterotrimeric proteins consisting of α - (45–47 kD), β - (35 kD), and γ - (7–9 kD) subunits. The α -subunit binds GDP or GTP and has an intrinsic, slow GTPase activity. In the inactive state, the $G_{\alpha\beta\gamma}$ complex has GDP at the nucleotide site. When a G protein is stimulated by a hormone-receptor complex, GDP dissociates and GTP binds to G_{α} , causing it to dissociate from $G_{\beta\gamma}$ and to associate with adenylyl cyclase (Figure 15.21). *Binding of $G_{\alpha}(GTP)$ activates adenylyl cyclase to form cAMP from ATP.* However, the intrinsic GTPase activity of G_{α} eventually hydrolyzes GTP to GDP, leading to dissociation of $G_{\alpha}(GDP)$ from adenylyl cyclase and reassociation with $G_{\beta\gamma}$ to form the inactive $G_{\alpha\beta\gamma}$ complex. This cascade amplifies the hormonal signal because a single hormone-receptor complex can activate many G proteins before the hormone dissociates from the receptor, and because the G_{α} -activated adenylyl cyclase can synthesize many cAMP molecules before bound GTP is hydrolyzed by G_{α} . More than 100 different G protein-coupled receptors and at least 21 distinct G_{α} proteins are known (Chapter 34).

Cyclic AMP is an essential activator of *cAMP-dependent protein kinase (PKA)*. This enzyme is normally inactive because its two catalytic subunits (C) are strongly associated with a pair of regulatory subunits (R), which serve to block activity. Binding of cyclic AMP to the regulatory subunits induces a conformation change that causes the dissociation of the C monomers from the R dimer (Figure 15.7). The free C subunits are active and can phosphorylate other proteins. One of the many proteins phosphorylated by PKA is *phosphorylase kinase* (Figure 15.19). Phosphorylase kinase is inactive in the unphosphorylated state and active in the phosphorylated form. As its name implies, phosphorylase kinase functions to phosphorylate (and activate) glycogen phosphorylase. Thus, stimulation of adenylyl cyclase leads to activation of glycogen breakdown.

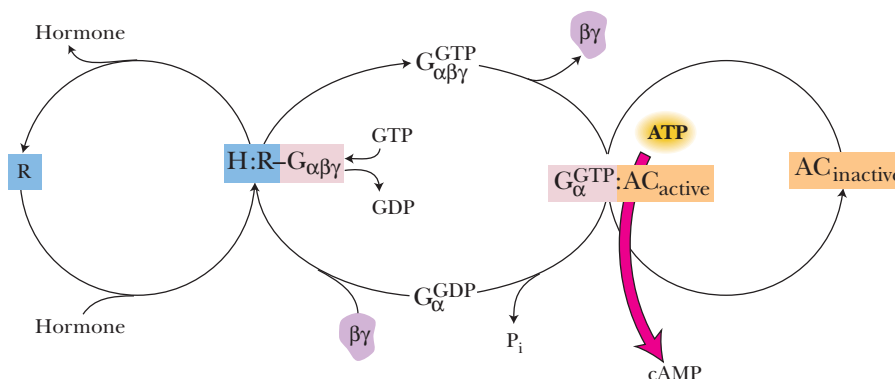


FIGURE 15.21 • Hormone (H) binding to its receptor (R) creates a hormone:receptor complex (H:R) that catalyzes GDP-GTP exchange on the α -subunit of the heterotrimer G protein ($G_{\alpha\beta\gamma}$), replacing GDP with GTP. The G_{α} -subunit with GTP bound dissociates from the $\beta\gamma$ -subunits and binds to adenylyl cyclase (AC). AC becomes active upon association with $G_{\alpha}:GTP$ and catalyzes the formation of cAMP from ATP. With time, the intrinsic GTPase activity of the G_{α} -subunit hydrolyzes the bound GTP, forming GDP; this leads to dissociation of $G_{\alpha}:GDP$ from AC, reassociation of G_{α} with the $\beta\gamma$ subunits, and cessation of AC activity. AC and the hormone receptor H are integral plasma membrane proteins; G_{α} and $G_{\beta\gamma}$ are membrane-anchored proteins.

SPECIAL FOCUS:
Hemoglobin and Myoglobin—Paradigms of Protein Structure and Function

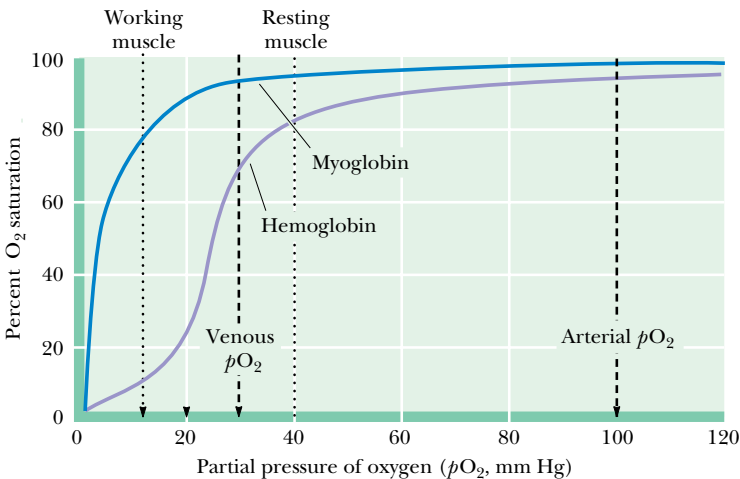
Ancient life forms evolved in the absence of oxygen and were capable only of anaerobic metabolism. As the earth’s atmosphere changed over time, so too did living things. Indeed, the production of O₂ by photosynthesis was a major factor in altering the atmosphere. Evolution to an oxygen-based metabolism was highly beneficial. Aerobic metabolism of sugars, for example, yields far more energy than corresponding anaerobic processes. Two important oxygen-binding proteins appeared in the course of evolution so that aerobic metabolic processes were no longer limited by the solubility of O₂ in water. These proteins are represented in animals as **hemoglobin (Hb)** in blood and **myoglobin (Mb)** in muscle. Because hemoglobin and myoglobin are two of the most-studied proteins in nature, they have become paradigms of protein structure and function. Moreover, hemoglobin is a model for protein quaternary structure and allosteric function. The binding of O₂ by hemoglobin, and its modulation by effectors such as protons, CO₂, and 2,3-bisphosphoglycerate, depend on interactions between subunits in the Hb tetramer. Subunit–subunit interactions in Hb reveal much about the functional significance of quaternary associations and allosteric regulation.

The Comparative Biochemistry of Myoglobin and Hemoglobin

A comparison of the properties of hemoglobin and myoglobin offers insights into allosteric phenomena, even though these proteins are *not* enzymes. Hemoglobin displays sigmoid-shaped O₂-binding curves (Figure 15.22). The unusual shape of these curves was once a great enigma in biochemistry. Such curves closely resemble allosteric enzyme:substrate saturation graphs (see Figure 15.8). In contrast, myoglobin’s interaction with oxygen obeys classical Michaelis–Menten-type substrate saturation behavior.

Before examining myoglobin and hemoglobin in detail, let us first encapsulate the lesson: Myoglobin is a compact globular protein composed of a single polypeptide chain 153 amino acids in length; its molecular mass is 17.2 kD (Figure 15.23). It contains **heme**, a porphyrin ring system complexing an iron ion, as its prosthetic group (see Figure 5.15). Oxygen binds to Mb via its heme. Hemoglobin (Hb) is also a compact globular protein, but Hb is a tetramer. It consists of four polypeptide chains, each of which is very similar structurally

FIGURE 15.22 • O₂-binding curves for hemoglobin and myoglobin.



to the myoglobin polypeptide chain, and each bears a heme group. Thus, a hemoglobin molecule can bind four O₂ molecules. In adult human Hb, there are two identical chains of 141 amino acids, the α -chains, and two identical β -chains, each of 146 residues. The human Hb molecule is an $\alpha_2\beta_2$ -type tetramer of molecular mass 64.45 kD. The tetrameric nature of Hb is crucial to its biological function: *When a molecule of O₂ binds to a heme in Hb, the heme Fe ion is drawn into the plane of the porphyrin ring. This slight movement sets off a chain of conformational events that are transmitted to adjacent subunits, dramatically enhancing the affinity of their heme groups for O₂.* That is, the binding of O₂ to one heme of Hb makes it easier for the Hb molecule to bind additional equivalents of O₂. Hemoglobin is a marvelously constructed molecular machine. Let us dissect its mechanism, beginning with its monomeric counterpart, the myoglobin molecule.

Myoglobin

Myoglobin is the oxygen-storage protein of muscle. The muscles of diving mammals such as seals and whales are especially rich in this protein, which serves as a store for O₂ during the animal’s prolonged periods underwater. Myoglobin is abundant in skeletal and cardiac muscle of nondiving animals as well. Myoglobin is the cause of the characteristic red color of muscle.

The Mb Polypeptide Cradles the Heme Group

The myoglobin polypeptide chain is folded to form a cradle (4.4 × 4.4 × 2.5 nm) that nestles the heme prosthetic group (Figure 15.24). O₂ binding depends on the heme’s oxidation state. The iron ion in the heme of myoglobin is in the +2 oxidation state, that is, the *ferrous* form. This is the form that binds O₂. Oxidation of the ferrous form to the +3 *ferric* form yields **metmyoglobin**, which will not bind O₂. It is interesting to note that free heme in solution will readily interact with O₂ also, but the oxygen quickly oxidizes the iron atom to the ferric state. Fe³⁺:protoporphyrin IX is referred to as **hematin**. Thus, the

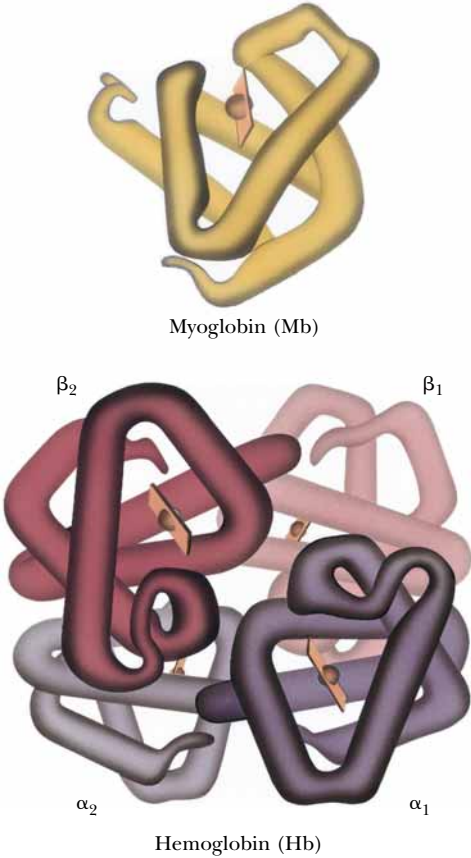


FIGURE 15.23 • The myoglobin and hemoglobin molecules. *Myoglobin* (sperm whale): one polypeptide chain of 153 aa residues (mass = 17.2 kD) has one heme (mass = 652 D) and binds one O₂. *Hemoglobin* (human): four polypeptide chains, two of 141 aa residues (α) and two of 146 residues (β); mass = 64.45 kD. Each polypeptide has a heme; the Hb tetramer binds four O₂. (Irving Geis)

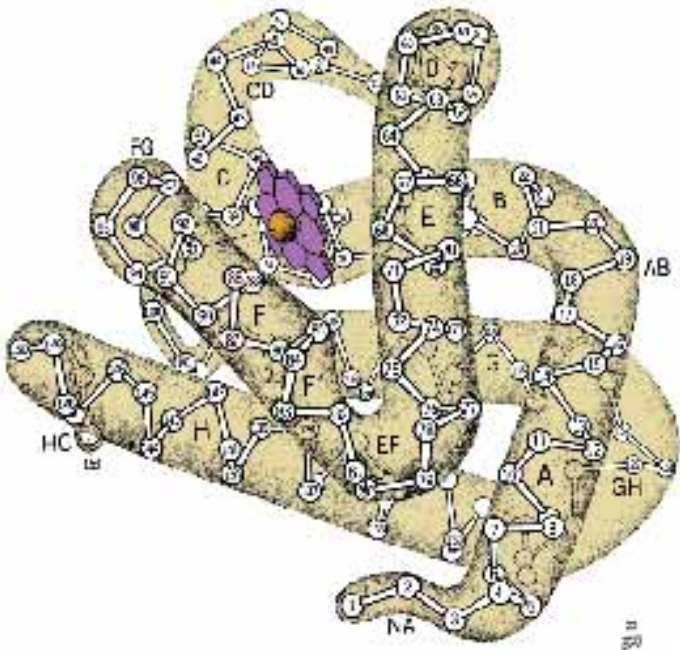


FIGURE 15.24 • Detailed structure of the myoglobin molecule. The myoglobin polypeptide chain consists of eight helical segments, designated by the letters A through H, counting from the N-terminus. These helices, ranging in length from 7 to 26 residues, are linked by short, unordered regions that are named for the helices they connect, as in the AB region or the EF region. The individual amino acids in the polypeptide are indicated according to their position within the various segments, as in His F8, the eighth residue in helix F, or Phe CD1, the first amino acid in the interhelical CD region. Occasionally, amino acids are specified in the conventional way, that is, by the relative position in the chain, as in Gly¹⁵³. The heme group is cradled within the folded polypeptide chain. (Irving Geis)

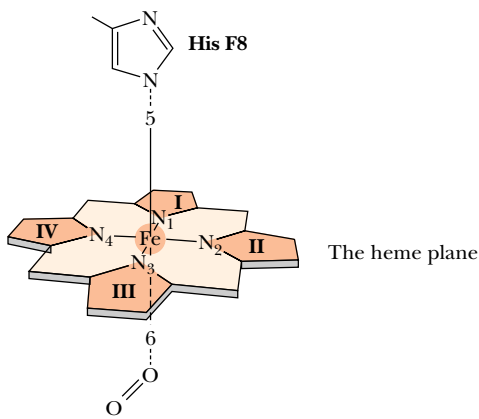


FIGURE 15.25 • The six liganding positions of an iron ion. Four ligands lie in the same plane; the remaining two are, respectively, above and below this plane. In myoglobin, His F8 is the fifth ligand; in oxymyoglobin, O₂ becomes the sixth.

polypeptide of myoglobin may be viewed as serving three critical functions: it cradles the heme group, it protects the heme iron atom from oxidation, and it provides a pocket into which the O₂ can fit.

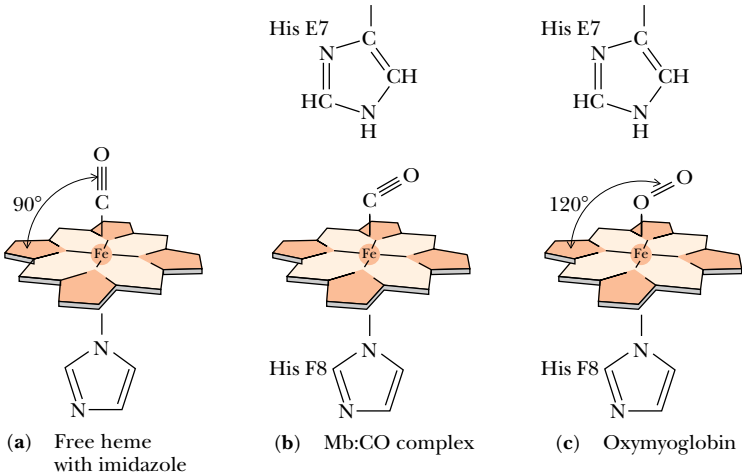
O₂ Binding to Mb

Iron ions, whether ferrous or ferric, prefer to interact with six ligands, four of which share a common plane. The fifth and sixth ligands lie above and below this plane (see Figure 15.25). In heme, four of the ligands are provided by the nitrogen atoms of the four pyrroles. A fifth ligand is donated by the imidazole side chain of amino acid residue His F8. When myoglobin binds O₂ to become **oxymyoglobin**, the O₂ molecule adds to the heme iron ion as the sixth ligand (Figure 15.25). O₂ adds end on to the heme iron, but it is not oriented perpendicular to the plane of the heme. Rather, it is tilted about 60° with respect to the perpendicular. In **deoxymyoglobin**, the sixth ligand position is vacant, and in metmyoglobin, a water molecule fills the O₂ site and becomes the sixth ligand for the ferric atom. On the oxygen-binding side of the heme lies another histidine residue, His E7. While its imidazole function lies too far away to interact with the Fe atom, it is close enough to contact the O₂. Therefore, the O₂-binding site is a sterically hindered region. Biologically important properties stem from this hindrance. For example, the affinity of *free heme* in solution for carbon monoxide (CO) is 25,000 times greater than its affinity for O₂. But CO only binds 250 times more tightly than O₂ to the heme of myoglobin, because His E7 forces the CO molecule to tilt away from a preferred perpendicular alignment with the plane of the heme (Figure 15.26). This diminished affinity of myoglobin for CO guards against the possibility that traces of CO produced during metabolism might occupy all of the heme sites, effectively preventing O₂ from binding. Nevertheless, CO is a potent poison and can cause death by asphyxiation.

O₂ Binding Alters Mb Conformation

What happens when the heme group of myoglobin binds oxygen? X-ray crystallography has revealed that a crucial change occurs in the position of the iron atom relative to the plane of the heme. In deoxymyoglobin, the ferrous ion has but five ligands, and it lies 0.055 nm above the plane of the heme, in the direction of His F8. The iron:porphyrin complex is therefore dome-shaped.

FIGURE 15.26 • Oxygen and carbon monoxide binding to the heme group of myoglobin.



When O₂ binds, the iron atom is pulled back toward the porphyrin plane and is now displaced from it by only 0.026 nm (Figure 15.27). The consequences of this small motion are trivial as far as the biological role of myoglobin is concerned. However, as we shall soon see, this slight movement profoundly affects the properties of hemoglobin. Its action on His F8 is magnified through changes in polypeptide conformation that alter subunit interactions in the Hb tetramer. These changes in subunit relationships are the fundamental cause of the allosteric properties of hemoglobin.

The Physiological Significance of Cooperative Binding of Oxygen by Hemoglobin

The oxygen-binding equations for myoglobin and hemoglobin are described in detail in the Appendix at the end of this chapter. The relative oxygen affinities of hemoglobin and myoglobin reflect their respective physiological roles (see Figure 15.22). Myoglobin, as an oxygen storage protein, has a greater affinity for O₂ than hemoglobin at all oxygen pressures. Hemoglobin, as the oxygen carrier, becomes saturated with O₂ in the lungs, where the partial pressure of O₂ (*p*O₂) is about 100 torr.¹ In the capillaries of tissues, *p*O₂ is typically 40 torr, and oxygen is released from Hb. In muscle, some of it can be bound by myoglobin, to be stored for use in times of severe oxygen deprivation, such as during strenuous exercise.

The Structure of the Hemoglobin Molecule

As noted, hemoglobin is an α₂β₂ tetramer. Each of the four subunits has a conformation virtually identical to that of myoglobin. Two different types of subunits, α and β, are necessary to achieve cooperative O₂-binding by Hb. The β-chain at 146 amino acid residues is shorter than the myoglobin chain (153 residues), mainly because its final helical segment (the H helix) is shorter. The α-chain (141 residues) also has a shortened H helix and lacks the D helix as well (Figure 15.28). Max Perutz, who has devoted his life to elucidating the atomic structure of Hb, noted very early in his studies that the molecule was

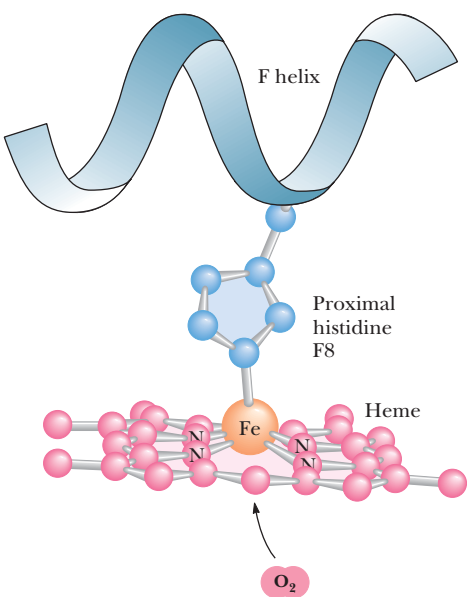
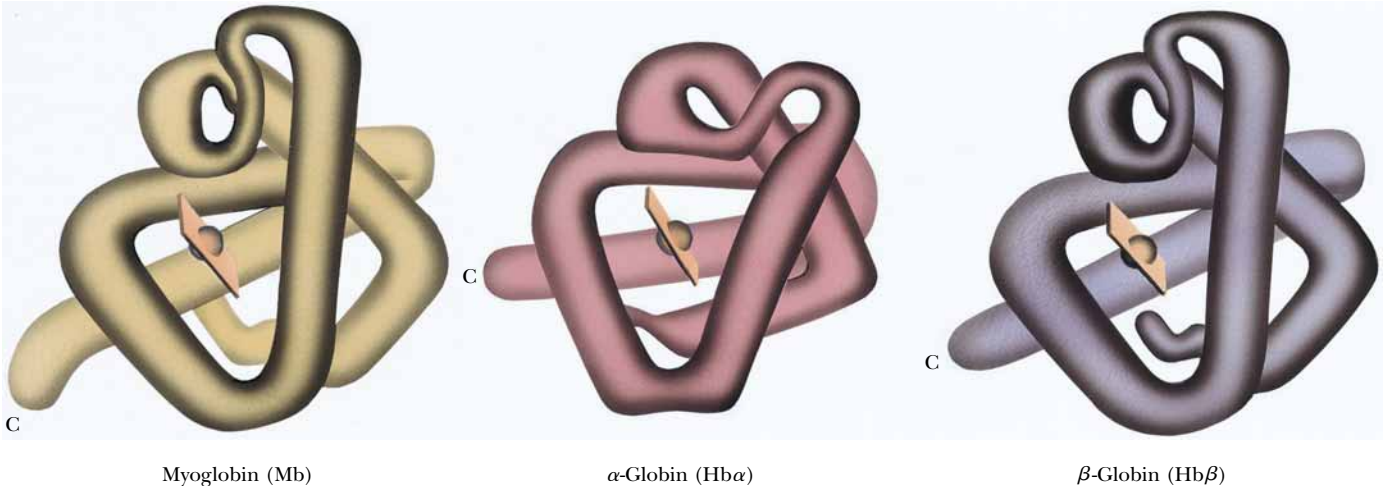


FIGURE 15.27 • The displacement of the Fe ion of the heme of deoxymyoglobin from the plane of the porphyrin ring system by the pull of His F8. In oxymyoglobin, the bound O₂ counteracts this effect.

FIGURE 15.28 • Conformational drawings of the α- and β-chains of Hb and the myoglobin chain. (Irving Geis)



¹The **torr** is a unit of pressure named for Torricelli, inventor of the barometer. 1 torr corresponds to 1 mm Hg (1/760th of an atmosphere).

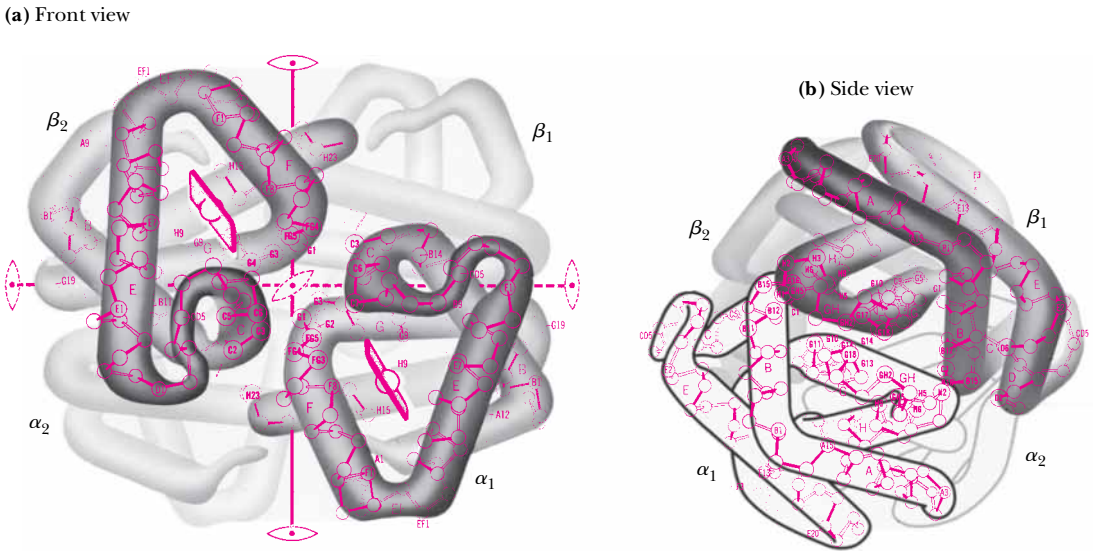


FIGURE 15.29 • The arrangement of subunits in horse methemoglobin, the first hemoglobin whose structure was determined by X-ray diffraction. The iron atoms on metHb are in the oxidized, ferric (Fe^{3+}) state. (Irving Geis)

highly symmetrical. The actual arrangement of the four subunits with respect to one another is shown in Figure 15.29 for horse methemoglobin. All vertebrate hemoglobins show a three-dimensional structure essentially the same as this. The subunits pack in a tetrahedral array, creating a roughly spherical molecule $6.4 \times 5.5 \times 5.0$ nm. The four heme groups, nestled within the easily recognizable cleft formed between the E and F helices of each polypeptide, are exposed at the surface of the molecule. The heme groups are quite far apart; 2.5 nm separates the closest iron ions, those of hemes α_1 and β_2 , and those of hemes α_2 and β_1 . The subunit interactions are mostly between dissimilar chains: each of the α -chains is in contact with both β -chains, but there are few α – α or β – β interactions.

A
DEEPER LOOK

The Physiological Significance of the Hb:O₂ Interaction

We can determine quantitatively the physiological significance of the sigmoid nature of the hemoglobin oxygen-binding curve, or, in other words, the biological importance of cooperativity. The equation

$$\frac{Y}{(1 - Y)} = \left(\frac{p\text{O}_2}{P_{50}} \right)^n$$

describes the relationship between $p\text{O}_2$, the affinity of hemoglobin for O₂ (defined as P_{50} , the partial pressure of O₂ giving half-maximal saturation of Hb with O₂), and the fraction of hemoglobin with O₂ bound, Y , versus the fraction of Hb with no O₂ bound, $(1 - Y)$ (see Appendix Equation [A15.16]). The coefficient n is the *Hill coefficient*, an index of the cooperativity (sigmoidicity) of the hemoglobin oxygen-binding curve (see

Appendix for details). Taking $p\text{O}_2$ in the lungs as 100 torr, P_{50} as 26 torr, and n as 2.8, Y , the fractional saturation of the hemoglobin heme groups with O₂, is 0.98. If $p\text{O}_2$ were to fall to 10 torr within the capillaries of an exercising muscle, Y would drop to 0.06. The oxygen delivered under these conditions would be proportional to the difference, $Y_{\text{lungs}} - Y_{\text{muscle}}$, which is 0.92. That is, virtually all the oxygen carried by Hb would be released. Suppose instead that hemoglobin binding of O₂ were not cooperative; in that case, the hemoglobin oxygen-binding curve would be hyperbolic, and $n = 1.0$. Then Y in the lungs would be 0.79 and Y in the capillaries, 0.28, and the difference in Y values would be 0.51. Thus, under these conditions, the cooperativity of oxygen binding by Hb means that 0.92/0.51 or 1.8 times as much O₂ can be delivered.

Oxygenation Markedly Alters the Quaternary Structure of Hb

Crystals of deoxyhemoglobin shatter when exposed to O₂. Further, X-ray crystallographic analysis reveals that oxy- and deoxyhemoglobin differ markedly in quaternary structure. In particular, specific $\alpha\beta$ -subunit interactions change. The $\alpha\beta$ contacts are of two kinds. The $\alpha_1\beta_1$ and $\alpha_2\beta_2$ contacts involve helices B, G, and H and the GH corner. These contacts are extensive and important to subunit packing; they remain unchanged when hemoglobin goes from its deoxy to its oxy form. The $\alpha_1\beta_2$ and $\alpha_2\beta_1$ contacts are called **sliding contacts**. They principally involve helices C and G and the FG corner (Figure 15.30). When hemoglobin undergoes a conformational change as a result of ligand binding to the heme, these contacts are altered (Figure 15.31). Hemoglobin, as a conformationally dynamic molecule, consists of two dimeric halves, an $\alpha_1\beta_1$ -subunit pair and an $\alpha_2\beta_2$ -subunit pair. Each $\alpha\beta$ dimer moves as a rigid body, and the two halves of the molecule slide past each other upon oxygenation of the heme. The two halves rotate some 15° about an imaginary pivot passing through the $\alpha\beta$ -subunits; some atoms at the interface between $\alpha\beta$ dimers are relocated by as much as 0.6 nm.

Movement of the Heme Iron by Less Than 0.04 nm Induces the Conformational Change in Hemoglobin

In deoxyhemoglobin, histidine F8 is liganded to the heme iron ion, but steric constraints force the Fe²⁺:His-N bond to be tilted about 8° from the perpendicular to the plane of the heme. Steric repulsion between histidine F8 and the nitrogen atoms of the porphyrin ring system, combined with electrostatic repulsions between the electrons of Fe²⁺ and the porphyrin π -electrons, forces the iron atom to lie out of the porphyrin plane by about 0.06 nm. Changes in electronic and steric factors upon heme oxygenation allow the Fe²⁺ atom to move about 0.039 nm closer to the plane of the porphyrin, so now it is displaced only 0.021 nm above the plane. It is as if the O₂ were drawing the heme Fe²⁺ into the porphyrin plane (Figure 15.32). This modest displacement of 0.039 nm seems a trivial distance, but its biological consequences are far-reaching. As the iron atom moves, it drags histidine F8 along with it, causing helix

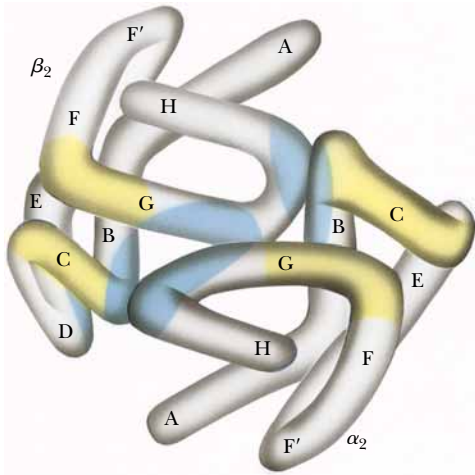


FIGURE 15.30 • Side view of one of the two $\alpha\beta$ dimers in Hb, with packing contacts indicated in blue. The sliding contacts made with the other dimer are shown in yellow. The changes in these sliding contacts are shown in Figure 15.31. (Irving Geis)

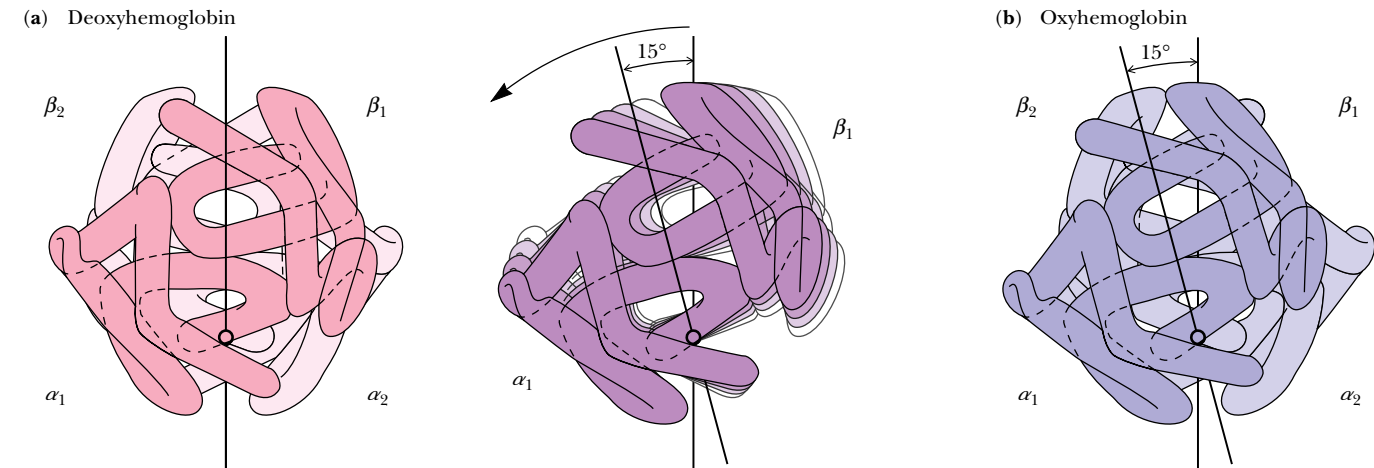


FIGURE 15.31 • Subunit motion in hemoglobin when the molecule goes from the (a) deoxy to the (b) oxy form. (Irving Geis)

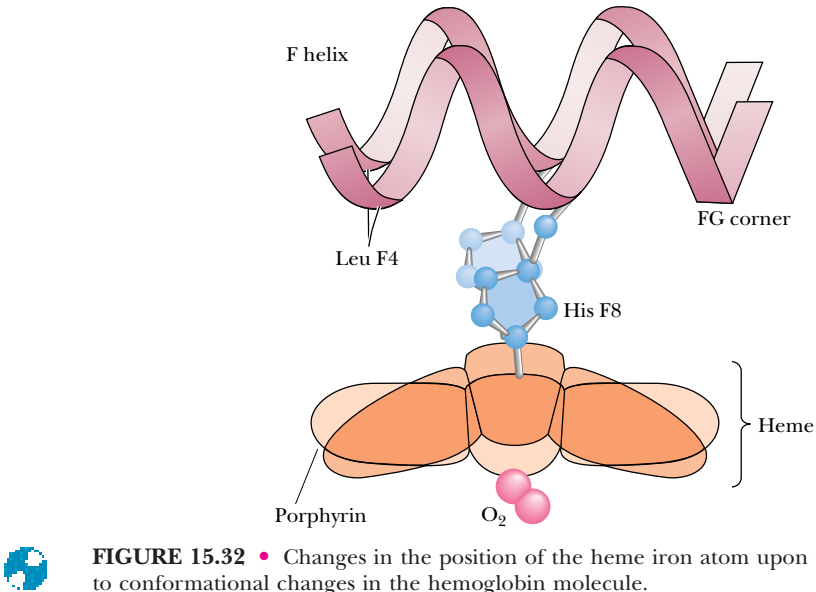


FIGURE 15.32 • Changes in the position of the heme iron atom upon oxygenation lead to conformational changes in the hemoglobin molecule.

F, the EF corner, and the FG corner to follow. These shifts are transmitted to the subunit interfaces, where they trigger conformational readjustments that lead to the rupture of interchain salt links.

The Oxy and Deoxy Forms of Hemoglobin Represent Two Different Conformational States

Hemoglobin resists oxygenation (see Figure 15.22) because the deoxy form is stabilized by specific hydrogen bonds and salt bridges (ion-pair bonds). All of these interactions are broken in oxyhemoglobin, as the molecule stabilizes into a new conformation. A crucial H bond in this transition involves a particular tyrosine residue. Both α - and β -subunits have Tyr as the penultimate C-terminal residue (Tyr α 140 = Tyr HC2; Tyr β 145 = Tyr HC2, respectively²). The phenolic —OH groups of these Tyr residues form intrachain H bonds to the peptide C=O function contributed by Val FG5 in deoxyhemoglobin. (Val FG5 is α 93 and β 98, respectively.) The shift in helix F upon oxygenation leads to rupture of this Tyr HC2:Val FG5 hydrogen bond. Further, eight salt bridges linking the polypeptide chains are broken as hemoglobin goes from the deoxy to the oxy form (Figure 15.33). Six of these salt links are between different subunits. Four of these six involve either carboxyl-terminal or amino-terminal amino acids in the chains; two are between the amino termini and the carboxyl termini of the α -chains, and two join the carboxyl termini of the β -chains to the ϵ -NH₃⁺ groups of the two Lys α 140 residues. The other two interchain electrostatic bonds link Arg and Asp residues in the two α -chains. In addition, ionic interactions between Asp β 94 and His β 146 form an intrachain salt bridge in each β -subunit. In deoxyhemoglobin, with all of these interactions intact, the C-termini of the four subunits are restrained, and this conformational state is termed **T**, the **tense** or **taut form**. In oxyhemoglobin, these C-termini have almost complete freedom of rotation, and the molecule is now in its **R**, or **relaxed, form**.

²C here designates the C-terminus; the H helix is C-terminal in these polypeptides. “C2” symbolizes the next-to-last residue.

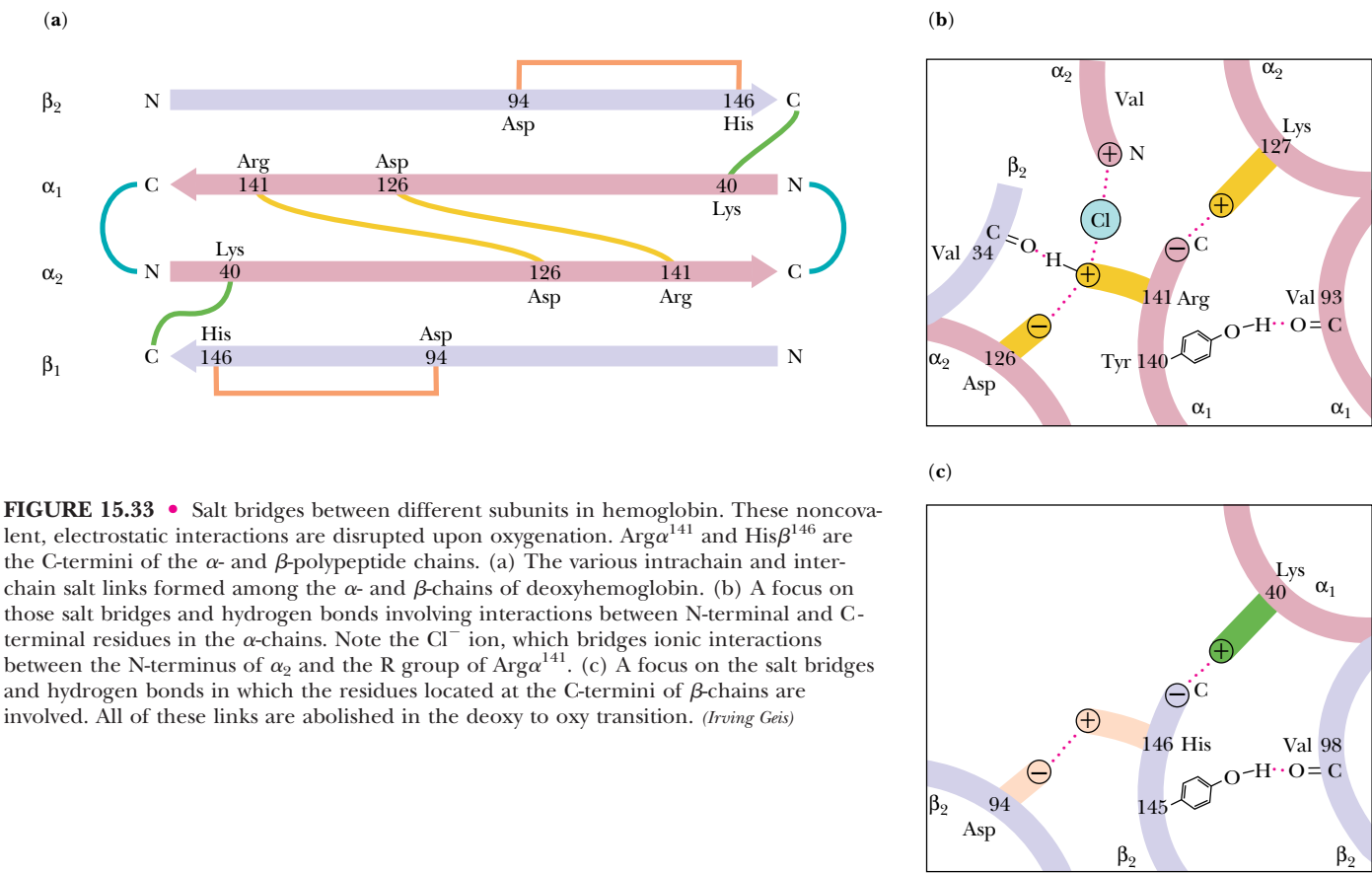


FIGURE 15.33 • Salt bridges between different subunits in hemoglobin. These noncovalent, electrostatic interactions are disrupted upon oxygenation. Arg ^{α 141} and His ^{β 146} are the C-termini of the α - and β -polypeptide chains. (a) The various intrachain and interchain salt links formed among the α - and β -chains of deoxyhemoglobin. (b) A focus on those salt bridges and hydrogen bonds involving interactions between N-terminal and C-terminal residues in the α -chains. Note the Cl^- ion, which bridges ionic interactions between the N-terminus of α_2 and the R group of Arg ^{α 141}. (c) A focus on the salt bridges and hydrogen bonds in which the residues located at the C-termini of β -chains are involved. All of these links are abolished in the deoxy to oxy transition. (Irving Geis)

A Model for the Allosteric Behavior of Hemoglobin

A model for the allosteric behavior of hemoglobin is based on recent observations that oxygen is accessible only to the heme groups of the α -chains when hemoglobin is in the T conformational state. Perutz has pointed out that the heme environment of β -chains in the T state is virtually inaccessible because of steric hindrance by amino acid residues in the E helix. This hindrance dis-

DEEPER LOOK

Changes in the Heme Iron upon O₂ Binding

In deoxyhemoglobin, the six *d* electrons of the heme Fe²⁺ exist as four unpaired electrons and one electron pair, and five ligands can be accommodated: the four N-atoms of the porphyrin ring system and histidine F8. In this electronic configuration, the iron atom is paramagnetic and in the **high-spin state**. When the heme binds O₂ as a sixth ligand, these electrons are rearranged into three *e*⁻ pairs and the iron changes to the **low-spin state** and is diamagnetic. This change in spin state allows the bond between

the Fe²⁺ ion and histidine F8 to become perpendicular to the heme plane and to shorten. In addition, interactions between the porphyrin N-atoms and the iron strengthen. Also, high-spin Fe²⁺ has a greater atomic volume than low-spin Fe²⁺ because its four unpaired *e*⁻ occupy four orbitals rather than two when the electrons are paired in low-spin Fe²⁺. So, low-spin iron is less sterically hindered and able to move nearer to the porphyrin plane.

appears when the hemoglobin molecule undergoes transition to the R conformational state. Binding of O₂ to the β-chains is thus dependent on the T to R conformational shift, and this shift is triggered by the subtle changes that occur when O₂ binds to the α-chain heme groups.

H⁺ Promotes the Dissociation of Oxygen from Hemoglobin

Protons, carbon dioxide, and chloride ions, as well as the metabolite 2,3-bisphosphoglycerate (or *BPG*), all affect the binding of O₂ by hemoglobin. Their effects have interesting ramifications, which we shall see as we discuss them in turn. Deoxyhemoglobin has a higher affinity for protons than oxyhemoglobin. Thus, as the pH decreases, dissociation of O₂ from hemoglobin is enhanced. In simple symbolism, ignoring the stoichiometry of O₂ or H⁺ involved:



Expressed another way, H⁺ is an antagonist of oxygen binding by Hb, and the saturation curve of Hb for O₂ is displaced to the right as acidity increases (Figure 15.34). This phenomenon is called the **Bohr effect**, after its discoverer, the Danish physiologist Christian Bohr (the father of Niels Bohr, the atomic physicist). The effect has important physiological significance because actively metabolizing tissues produce acid, promoting O₂ release where it is most needed. About two protons are taken up by deoxyhemoglobin. The N-termini of the two α-chains and the His β146 residues have been implicated as the major players in the Bohr effect. (The pK_a of a free amino terminus in a protein is about 8.0, but the pK_a of a protein histidine imidazole is around 6.5.) Neighboring carboxylate groups of Asp β94 residues help to stabilize the protonated state of the His β146 imidazoles that occur in deoxyhemoglobin. However, when Hb binds O₂, changes in the conformation of β-chains upon Hb oxygenation move the negative Asp function away, and dissociation of the imidazole protons is favored.

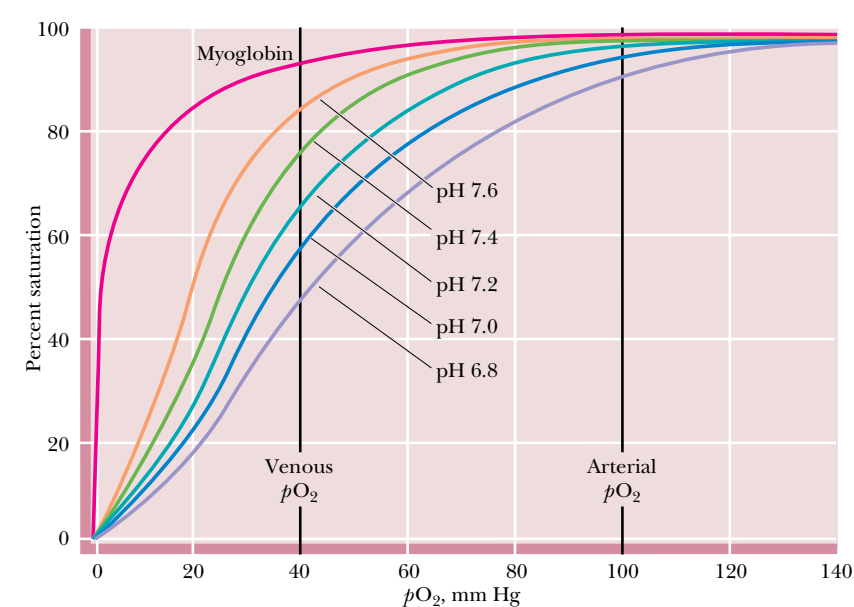
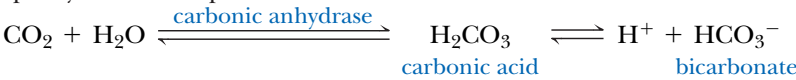


FIGURE 15.34 • The oxygen saturation curves for myoglobin and for hemoglobin at five different pH values: 7.6, 7.4, 7.2, 7.0, and 6.8.

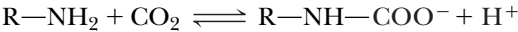
CO₂ Also Promotes the Dissociation of O₂ from Hemoglobin

Carbon dioxide has an effect on O₂ binding by Hb that is similar to that of H⁺, partly because it produces H⁺ when it dissolves in the blood:



The enzyme *carbonic anhydrase* promotes the hydration of CO₂. Many of the protons formed upon ionization of carbonic acid are picked up by Hb as O₂ dissociates. The bicarbonate ions are transported with the blood back to the lungs. When Hb becomes oxygenated again in the lungs, H⁺ is released and reacts with HCO₃[−] to re-form H₂CO₃, from which CO₂ is liberated. The CO₂ is then exhaled as a gas.

In addition, some CO₂ is directly transported by hemoglobin in the form of *carbamate* (−NHCOO[−]). Free α-amino groups of Hb react with CO₂ reversibly:



This reaction is driven to the right in tissues by the high CO₂ concentration; the equilibrium shifts the other way in the lungs where [CO₂] is low. Thus, carbamylation of the N-termini converts them to anionic functions, which then form salt links with the cationic side chains of Arg α141 that stabilize the deoxy or T state of hemoglobin.

In addition to CO₂, Cl[−] and BPG also bind better to deoxyhemoglobin than to oxyhemoglobin, causing a shift in equilibrium in favor of O₂ release. These various effects are demonstrated by the shift in the oxygen saturation curves for Hb in the presence of one or more of these substances (Figure 15.35). Note that the O₂-binding curve for Hb + BPG + CO₂ fits that of whole blood very well.

2,3-Bisphosphoglycerate Is an Important Allosteric Effector for Hemoglobin

The binding of 2,3-bisphosphoglycerate (BPG) to Hb promotes the release of O₂ (Figure 15.35). Erythrocytes (red blood cells) normally contain about 4.5 mM BPG, a concentration equivalent to that of tetrameric hemoglobin molecules. Interestingly, this equivalence is maintained in the Hb:BPG binding stoichiometry because the tetrameric Hb molecule has but one binding site for BPG. This site is situated within the central cavity formed by the association of the four subunits. The strongly negative BPG molecule (Figure 15.36) is electrostatically bound via interactions with the positively charged functional

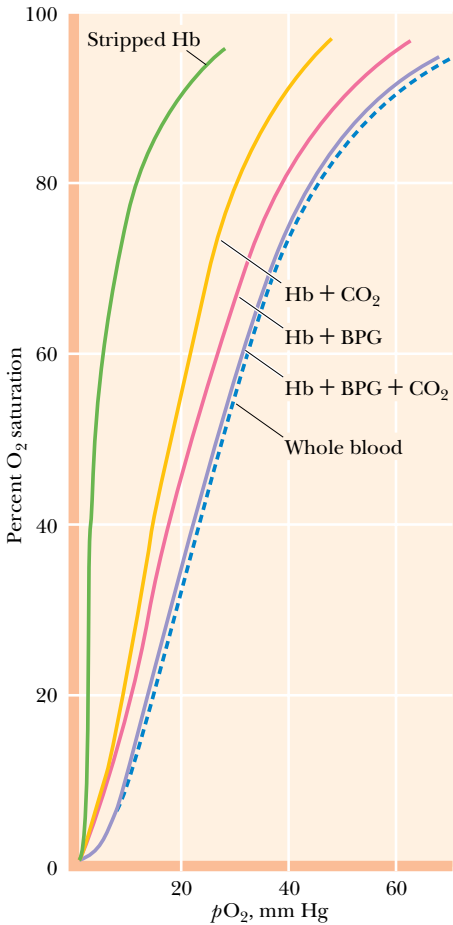


FIGURE 15.35 • Oxygen-binding curves of blood and of hemoglobin in the absence and presence of CO₂ and BPG. From left to right: stripped Hb, Hb + CO₂, Hb + BPG, Hb + BPG + CO₂, and whole blood.

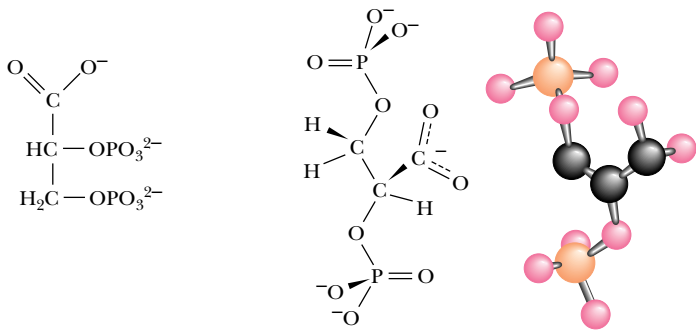


FIGURE 15.36 • The structure, in ionic form, of BPG or 2,3-bisphosphoglycerate, an important allosteric effector for hemoglobin.

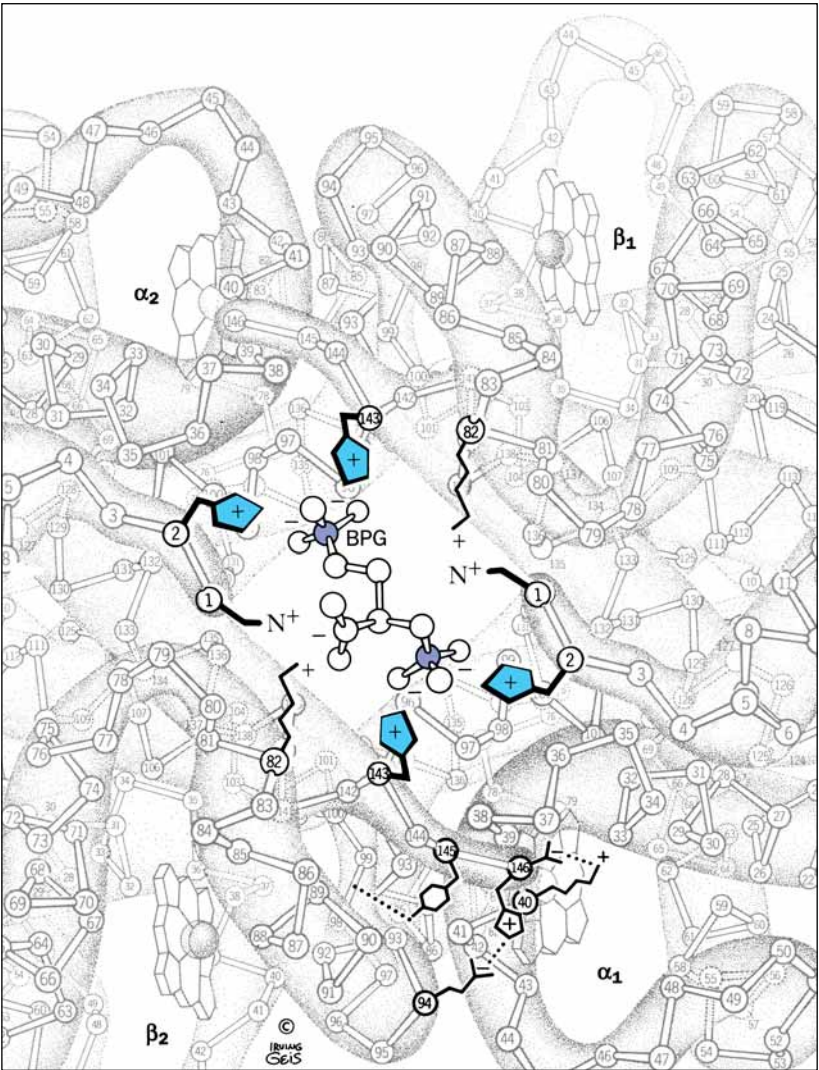


FIGURE 15.37 • The ionic binding of BPG to the two β -subunits of Hb. (*Irving Geis*)

groups of each Lys β 82, His β 2, His β 143, and the NH_3^+ -terminal group of each β -chain. These positively charged residues are arranged to form an electrostatic pocket complementary to the conformation and charge distribution of BPG (Figure 15.37). In effect, BPG cross-links the two β -subunits. The ionic bonds between BPG and the two β -chains aid in stabilizing the conformation of Hb in its deoxy form, thereby favoring the dissociation of oxygen. In oxy-hemoglobin, this central cavity is too small for BPG to fit. Or, to put it another way, the conformational changes in the Hb molecule that accompany O_2 -binding perturb the BPG-binding site so that BPG can no longer be accommodated. Thus, BPG and O_2 are mutually exclusive allosteric effectors for Hb, even though their binding sites are physically distinct.

The Physiological Significance of BPG Binding

The importance of the BPG effect is evident in Figure 15.35. Hemoglobin stripped of BPG is virtually saturated with O_2 at a $p\text{O}_2$ of only 20 torr, and it cannot release its oxygen within tissues, where the $p\text{O}_2$ is typically 40 torr. BPG shifts the oxygen saturation curve of Hb to the right, making the Hb an O_2

delivery system eminently suited to the needs of the organism. BPG serves this vital function in humans, most primates, and a number of other mammals. However, the hemoglobins of cattle, sheep, goats, deer, and other animals have an intrinsically lower affinity for O_2 , and these Hbs are relatively unaffected by BPG. In fish, whose erythrocytes contain mitochondria, the regulatory role of BPG is filled by ATP or GTP. In reptiles and birds, a different organophosphate serves, namely inositol pentaphosphate (IPP) or inositol hexaphosphate (IHP) (Figure 15.38).

Fetal Hemoglobin Has a Higher Affinity for O_2 Because It Has a Lower Affinity for BPG

The fetus depends on its mother for an adequate supply of oxygen, but its circulatory system is entirely independent. Gas exchange takes place across the placenta. Ideally then, fetal Hb should be able to absorb O_2 better than maternal Hb so that an effective transfer of oxygen can occur. Fetal Hb differs from adult Hb in that the β -chains are replaced by very similar, but not identical, 146-residue subunits called γ -chains (gamma chains). Fetal Hb is thus $\alpha_2\gamma_2$. Recall that BPG functions through its interaction with the β -chains. BPG binds less effectively with the γ -chains of fetal Hb (also called Hb F). (Fetal γ -chains have Ser instead of His at position 143, and thus lack two of the positive charges in the central BPG-binding cavity.) Figure 15.39 compares the relative affinities of adult Hb (also known as Hb A) and Hb F for O_2 under similar conditions of pH and [BPG]. Note that Hb F binds O_2 at pO_2 values where most of the oxygen has dissociated from Hb A. Much of the difference can be attributed to the diminished capacity of Hb F to bind BPG (compare Figures 15.35 and 15.39); Hb F thus has an intrinsically greater affinity for O_2 , and oxygen transfer from mother to fetus is ensured.

Sickle-Cell Anemia

In 1904, a Chicago physician treated a 20-year-old black college student complaining of headache, weakness, and dizziness. The blood of this patient revealed serious anemia—only half the normal number of red cells were pre-

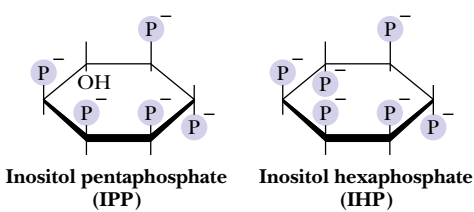


FIGURE 15.38 • The structures of inositol pentaphosphate and inositol hexaphosphate, the functional analogs of BPG in birds and reptiles.

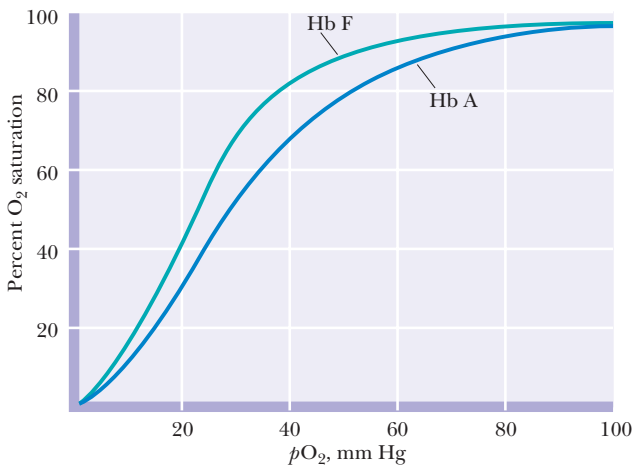


FIGURE 15.39 • Comparison of the oxygen saturation curves of Hb A and Hb F under similar conditions of pH and [BPG].

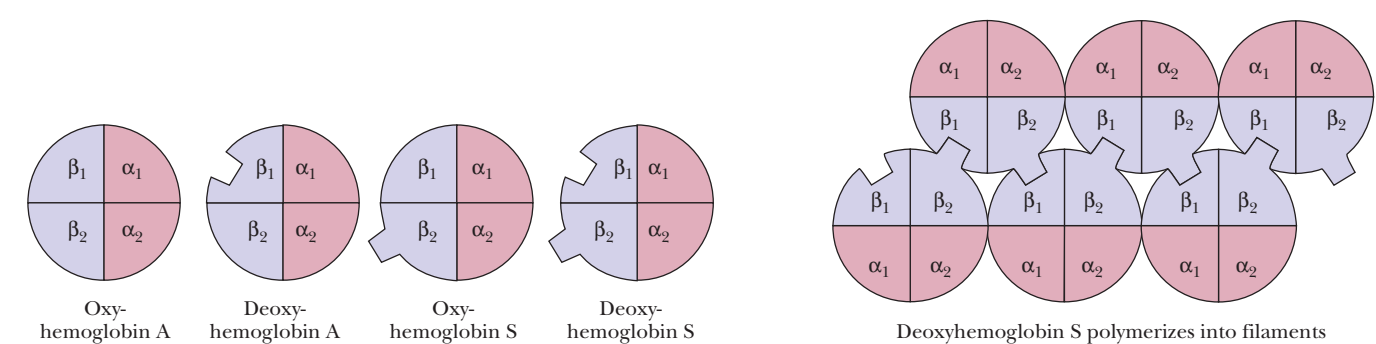


FIGURE 15.40 • The polymerization of Hb S via the interactions between the hydrophobic Val side chains at position $\beta 6$ and the hydrophobic pockets in the EF corners of β -chains in neighboring Hb molecules. The protruding “block” on Oxy S represents the Val hydrophobic protrusion. The complementary hydrophobic pocket in the EF corner of the β -chains is represented by a square-shaped indentation. (This indentation is probably present in Hb A also.) Only the β_2 Val protrusions and the β_1 EF pockets are shown. (The β_1 Val protrusions and the β_2 EF pockets are not involved, although they are present.)

sent. Many of these cells were abnormally shaped; in fact, instead of the characteristic disc shape, these erythrocytes were elongate and crescentlike in form, a feature that eventually gave name to the disease **sickle-cell anemia**. These sickle cells pass less freely through the capillaries, impairing circulation and causing tissue damage. Further, these cells are more fragile and rupture more easily than normal red cells, leading to anemia.

Sickle-Cell Anemia Is a Molecular Disease

A single amino acid substitution in the β -chains of Hb causes sickle-cell anemia. Replacement of the glutamate residue at position 6 in the β -chain by a valine residue marks the only chemical difference between Hb A and sickle-cell hemoglobin, Hb S. The amino acid residues at position $\beta 6$ lie at the surface of the hemoglobin molecule. In Hb A, the ionic R groups of the Glu residues fit this environment. In contrast, the aliphatic side chains of the Val residues in Hb S create hydrophobic protrusions where none existed before. To the detriment of individuals who carry this trait, a hydrophobic pocket forms in the EF corner of each β -chain of Hb when it is in the deoxy state, and this pocket nicely accommodates the Val side chain of a neighboring Hb S molecule (Figure 15.40). This interaction leads to the aggregation of Hb S molecules into long, chainlike polymeric structures. The obvious consequence is that deoxyHb S is less soluble than deoxyHb A. The concentration of hemoglobin in red blood cells is high (about 150 mg/mL), so that even in normal circumstances it is on the verge of crystallization. The formation of insoluble deoxyHb S fibers distorts the red cell into the elongated sickle shape characteristic of the disease.³

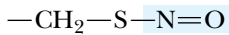
³In certain regions of Africa, the sickle-cell trait is found in 20% of the people. Why does such a deleterious heritable condition persist in the population? For reasons as yet unknown, individuals with this trait are less susceptible to the most virulent form of malaria. The geographic distribution of malaria and the sickle-cell trait are positively correlated.

HUMAN BIOCHEMISTRY

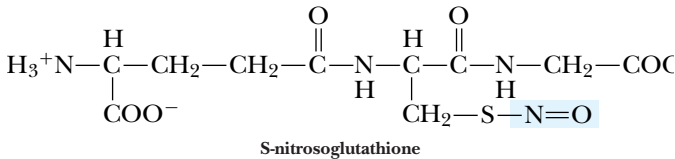
Hemoglobin and Nitric Oxide

Nitric oxide (NO ·) is a simple gaseous molecule whose many remarkable physiological functions are still being discovered. For example, NO · is known to act as a neurotransmitter and as a second messenger in signal transduction (see Chapter 34). Further, **endothelial relaxing factor** (ERF, also known as endothelium-derived relaxing factor, or EDRF), an elusive hormone-like agent that acts to relax the musculature of the walls (**endothelium**) of blood vessels and lower blood pressure, has been identified as NO ·. It has long been known that NO · is a high-affinity ligand for Hb, binding to its heme-Fe²⁺ atom with an affinity 10,000 times greater than that of O₂. An enigma thus arises: Why is NO · not instantaneously bound by Hb within human erythrocytes and prevented from exerting its vasodilation properties?

The reason that Hb doesn't block the action of NO · is due to a unique interaction between Cys93β of Hb and NO · recently described by Li Jia, Celia and Joseph Bonaventura, and Johnathan Stamler at Duke University. Nitric oxide reacts with the sulfhydryl group of Cys93β, forming an S-nitroso derivative:



This S-nitroso group is in equilibrium with other S-nitroso compounds formed by reaction of NO · with small-molecule thiols such as free cysteine or glutathione (an isoglutamylcysteinylglycine tripeptide):



These small-molecule thiols serve to transfer NO · from erythrocytes to endothelial receptors, where it acts to relax vascular tension. NO · itself is a reactive free-radical compound whose biological half-life is very short (1–5 sec). S-nitrosoglutathione has a half-life of several hours.

The reactions between Hb and NO · are complex. NO · forms a ligand with the heme-Fe²⁺ that is quite stable in the absence of O₂. However, in the presence of O₂, NO · is oxidized to NO₃[–] and the heme-Fe²⁺ of Hb is oxidized to Fe³⁺, forming methemoglobin. Fortunately, the interaction of Hb with NO · is controlled by the allosteric transition between R-state Hb (oxyHb) and T-state Hb (deoxyHb). Cys93β is more exposed and reactive in R-state Hb than in T-state Hb, and binding of NO · to Cys93β precludes reaction of NO · with heme iron. Upon release of O₂ from Hb in tissues, Hb shifts conformation from R state to T state, and binding of NO · at Cys93β is no longer favored. Consequently, NO · is released from Cys93β and transferred to small-molecule thiols for delivery to endothelial receptors, causing capillary vasodilation. This mechanism also explains the puzzling observation that free Hb produced by recombinant DNA methodology for use as a whole blood substitute causes a transient rise of 10 to 12 mm Hg in diastolic blood pressure in experimental clinical trials. (Conventional whole blood transfusion has no such effect.) It is now apparent that the “synthetic” Hb, which has no bound NO ·, is binding NO · in the blood and preventing its vasoregulatory function.

In the course of hemoglobin evolution, the only invariant amino acid residues in globin chains are HisF8 (the obligatory heme ligand) and a Phe residue acting to wedge the heme into its pocket. However, in mammals and birds, Cys93β is also invariant, no doubt due to its vital role in NO · delivery.

Adapted from Jia, L., et al., 1996. S-Nitrosohaemoglobin: A dynamic activity of blood involved in vascular control. *Nature* 380:221–226.

PROBLEMS

1. List six general ways in which enzyme activity is controlled.
2. Why do you suppose proteolytic enzymes are often synthesized as inactive zymogens?
3. First draw both Lineweaver–Burk plots and Hanes–Wolf plots for the following: a Monod–Wyman–Changeux allosteric *K* enzyme system, showing separate curves for the kinetic response in (1) the absence of any effectors; (2) the presence of allosteric activator A; and (3) the presence of allosteric inhibitor I. Then draw a similar set of curves for a Monod–Wyman–Changeux allosteric *V* enzyme system.
4. In the Monod–Wyman–Changeux model for allosteric regulation, what values of *L* and relative affinities of R and T for A will lead activator A to exhibit positive homotropic effects? (That is, under what conditions will the binding of A enhance further A-binding, in the same manner that S-binding shows positive coop-

erativity?) Similarly, what values of *L* and relative affinities of R and T for I will lead inhibitor I to exhibit positive homotropic effects? (That is, under what conditions will the binding of I promote further I-binding?)

5. The equation $\frac{Y}{(1 - Y)} = \left(\frac{p\text{O}_2}{P_{50}}\right)^n$ allows the calculation of *Y* (the fractional saturation of hemoglobin with O₂), given *P*₅₀ and *n* (see Box on page 484). Let *P*₅₀ = 26 torr and *n* = 2.8. Calculate *Y* in the lungs where *p*O₂ = 100 torr, and *Y* in the capillaries where *p*O₂ = 40 torr. What is the efficiency of O₂ delivery under these conditions (expressed as *Y*_{lungs} – *Y*_{capillaries})? Repeat the calculations, but for *n* = 1. Compare the values for *Y*_{lungs} – *Y*_{capillaries} for *n* = 2.8 versus *Y*_{lungs} – *Y*_{capillaries} for *n* = 1 to determine the effect of cooperative O₂ binding on oxygen delivery by hemoglobin.

6. The cAMP formed by adenylyl cyclase (Figure 15.20) does not persist because 5'-phosphodiesterase activity prevalent in cells hydrolyzes cAMP to give 5'-AMP. Caffeine inhibits 5'-phosphodiesterase activity. Describe the effects on glycogen phosphorylase activity that arise as a consequence of drinking lots of caffeinated coffee.
7. If no precautions are taken, blood that has been stored for some time becomes depleted in 2,3-BPG. What happens if such blood is used in a transfusion?

FURTHER READING

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Dickerson, R. E., and Geis, I., 1983. *Hemoglobin: Structure, Function, Evolution and Pathology*. Menlo Park, CA: Benjamin/Cummings.

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8. Enzymes have evolved such that their K_m values (or $K_{0.5}$ values) for substrate(s) are roughly equal to the *in vivo* concentration(s) of the substrate(s). Assume that glycogen phosphorylase is assayed at $[P_i] \approx K_{0.5}$ in the absence and presence of AMP or ATP. Estimate from Figure 15.15 the relative glycogen phosphorylase activity when (a) neither AMP or ATP is present, (b) AMP is present, and (c) ATP is present.

Lin, K., et al., 1996. Comparison of the activation triggers in yeast and muscle glycogen phosphorylase. *Science* **273**:1539–1541. Despite structural and regulatory differences between yeast and muscle glycogen phosphorylases, both are activated through changes in their intersubunit interface.

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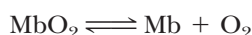
Weiss, J. N., 1997. The Hill equation revisited: Uses and abuses. *The FASEB Journal* **11**:835–841.

Appendix to Chapter 15

The Oxygen-Binding Curves of Myoglobin and Hemoglobin

Myoglobin

The reversible binding of oxygen to myoglobin,



can be characterized by the equilibrium dissociation constant, K .

$$K = \frac{[\text{Mb}][\text{O}_2]}{[\text{MbO}_2]} \quad (\text{A15.1})$$

If Y is defined as the **fractional saturation** of myoglobin with O_2 , that is, the fraction of myoglobin molecules having an oxygen molecule bound, then

$$Y = \frac{[\text{MbO}_2]}{[\text{MbO}_2] + [\text{Mb}]} \quad (\text{A15.2})$$

The value of Y ranges from 0 (no myoglobin molecules carry an O_2) to 1.0 (all myoglobin molecules have an O_2 molecule bound). Substituting from Equation (A15.1), $([\text{Mb}][\text{O}_2])/K$ for $[\text{MbO}_2]$ gives

$$Y = \frac{\left(\frac{[\text{Mb}][\text{O}_2]}{K}\right)}{\left(\frac{[\text{Mb}][\text{O}_2]}{K} + [\text{Mb}]\right)} = \frac{\left(\frac{[\text{O}_2]}{K}\right)}{\left(\frac{[\text{O}_2]}{K} + 1\right)} = \frac{[\text{O}_2]}{[\text{O}_2] + K} \quad (\text{A15.3})$$

and, if the concentration of O_2 is expressed in terms of the partial pressure (in torr) of oxygen gas in equilibrium with the solution of interest, then

$$Y = \frac{p\text{O}_2}{p\text{O}_2 + K} \quad (\text{A15.4})$$

(In this form, K has the units of torr.) The relationship defined by Equation (A15.4) plots as a hyperbola. That is, the MbO_2 saturation curve resembles an enzyme:substrate saturation curve. For myoglobin, a partial pressure of 1 torr for $p\text{O}_2$ is sufficient for half-saturation (Figure A15.1). We can define P_{50} as the partial pressure of O_2 at which 50% of the myoglobin molecules have a molecule of O_2 bound (that is, $Y = 0.5$), then

$$0.5 = \frac{p\text{O}_2}{p\text{O}_2 + P_{50}} \quad (\text{A15.5})$$

(Note from Equation (A15.1) that when $[\text{MbO}_2] = [\text{Mb}]$, $K = [\text{O}_2]$, which is the same as saying when $Y = 0.5$, $K = P_{50}$.) The general equation for O_2 binding to Mb becomes

$$Y = \frac{p\text{O}_2}{p\text{O}_2 + P_{50}} \quad (\text{A15.6})$$

The ratio of the fractional saturation of myoglobin, Y , to free myoglobin, $1 - Y$, depends on $p\text{O}_2$ and K according to the equation

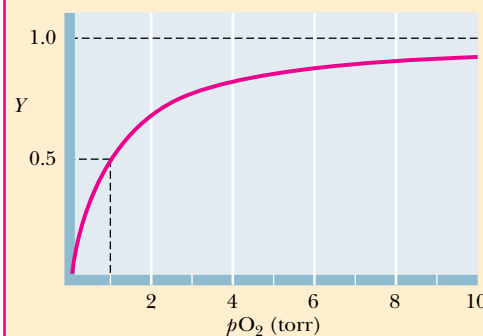


FIGURE A15.1 • Oxygen saturation curve for myoglobin in the form of Y versus $p\text{O}_2$ showing P_{50} is at a $p\text{O}_2$ of 1 torr (1 mm Hg).

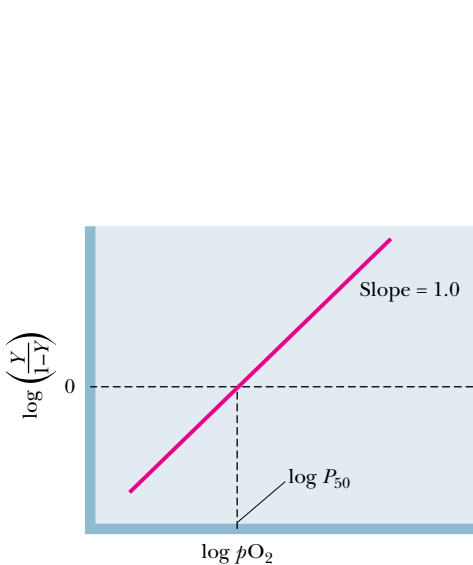


FIGURE A15.2 • Hill plot for the binding of O₂ to myoglobin. The slope of the line is the **Hill coefficient**. For Mb, the Hill coefficient is 1.0. At log(*Y*/(1 − *Y*)) = 0, log *pO*₂ = log *P*₅₀.



$$\frac{Y}{1 - Y} = \frac{pO_2}{K} \tag{A15.7}$$

Taking the logarithm yields

$$\log \left(\frac{Y}{1 - Y} \right) = \log pO_2 - \log K \tag{A15.8}$$

A graph of log (*Y*/(1 − *Y*)) versus log *pO*₂ is known as a **Hill plot** (in honor of Archibald Hill, a pioneer in the study of O₂ binding by hemoglobin). A Hill plot for myoglobin (Figure A15.2) gives a straight line. At half-saturation, defined as *Y* = 0.5, *Y*/(1 − *Y*) = 1, and log (*Y*/(1 − *Y*)) = 0. At this value of log (*Y*/(1 − *Y*)), the value for *pO*₂ = *K* = *P*₅₀. The slope of the Hill plot at the point where log (*Y*/(1 − *Y*)) = 0, the midpoint of binding, is known as the **Hill coefficient**. The Hill coefficient for myoglobin is 1.0. A Hill coefficient of 1.0 means that O₂ molecules bind independently of one another to myoglobin, a conclusion entirely logical because each Mb molecule can bind only one O₂.

Hemoglobin

New properties emerge when four heme-containing polypeptides come together to form a tetramer. The O₂-binding curve of hemoglobin is sigmoid rather than hyperbolic (see Figure 15.21), and Equation (A15.4) does not describe such curves. Of course, each hemoglobin molecule has four hemes and can bind up to four oxygen molecules. Suppose for the moment the O₂ binding to hemoglobin is an “all-or-none” phenomenon, where Hb exists either free of O₂ or with four O₂ molecules bound. This supposition represents the extreme case for cooperative binding of a ligand by a protein with multiple binding sites. In effect, it says that if one ligand binds to the protein molecule, then all other sites are immediately occupied by ligand. Or, to say it another way for the case in hand, suppose that four O₂ molecules bind to Hb simultaneously:



Then the *dissociation* constant, *K*, would be

$$K = \frac{[Hb][O_2]^4}{[Hb(O_2)_4]} \tag{A15.9}$$

By analogy with Equation (A15.4), the equation for fractional saturation of Hb is given by

$$Y = \frac{[pO_2]^4}{[pO_2]^4 + K} \tag{A15.10}$$

A plot of *Y* versus *pO*₂ according to Equation (A15.10) is presented in Figure A15.3. This curve has the characteristic sigmoid shape seen for O₂ binding by Hb. Half-saturation is set to be a *pO*₂ of 26 torr. Note that, when *pO*₂ is low, the fractional saturation, *Y*, changes very little as *pO*₂ increases. The interpretation is that Hb has little affinity for O₂ at these low partial pressures of O₂. However, as *pO*₂ reaches some threshold value and the first O₂ is bound, *Y*, the fractional saturation, increases rapidly. Note that the slope of the curve is steepest in the region where *Y* = 0.5. The sigmoid character of this curve is diagnostic of the fact that the binding of O₂ to one site on Hb strongly enhances binding of additional O₂ molecules to the remaining vacant sites on the same Hb molecule, a phenomenon aptly termed **cooperativity**. (If each O₂ bound independently, exerting no influence on the affinity of Hb for more O₂ binding, this plot would be hyperbolic.)

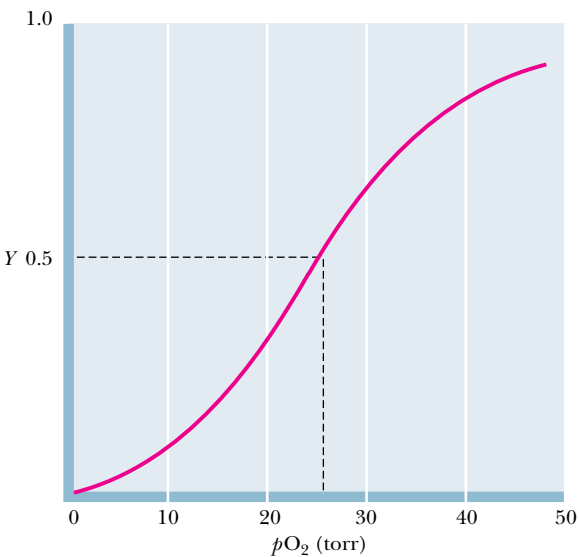


FIGURE A15.3 • Oxygen saturation curve for Hb in the form of Y versus pO_2 , assuming $n = 4$, and $P_{50} = 26$ torr. The graph has the characteristic experimentally observed sigmoid shape.

The experimentally observed oxygen-binding curve for Hb does not fit the graph given in Figure A15.3 exactly. If we generalize Equation (A15.10) by replacing the exponent 4 by n , we can write the equation as

$$Y = \frac{[pO_2]^n}{[pO_2]^n + K} \tag{A15.11}$$

Rearranging yields

$$\frac{Y}{1 - Y} = \frac{[pO_2]^n}{K} \tag{A15.12}$$

This equation states that the ratio of oxygenated heme groups (Y) to O_2 -free heme ($1 - Y$) is equal to the n th power of the pO_2 divided by the apparent dissociation constant, K .

Archibald Hill demonstrated in 1913, well before any knowledge about the molecular organization of Hb existed, that the O_2 -binding behavior of Hb could be described by Equation (A15.12). If a value of 2.8 is taken for n , Equation (A15.12) fits the experimentally observed O_2 -binding curve for Hb very well (Figure A15.4). If the binding of O_2 to Hb were an all-or-none phenomenon,

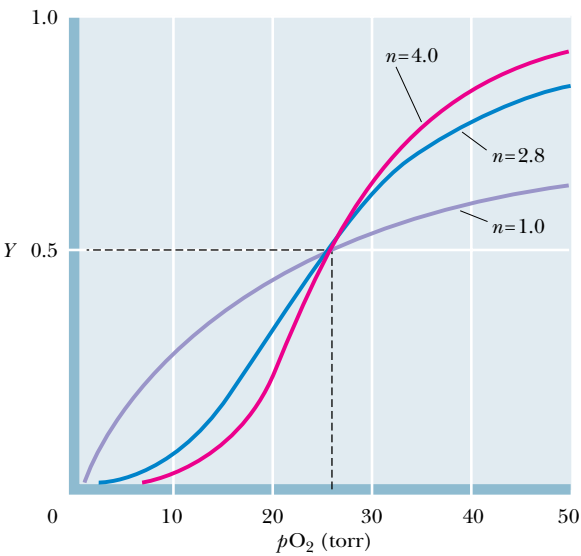


FIGURE A15.4 • A comparison of the experimentally observed O_2 curve for Hb yielding a value for n of 2.8, the hypothetical curve if $n = 4$, and the curve if $n = 1$ (noninteracting O_2 -binding sites).



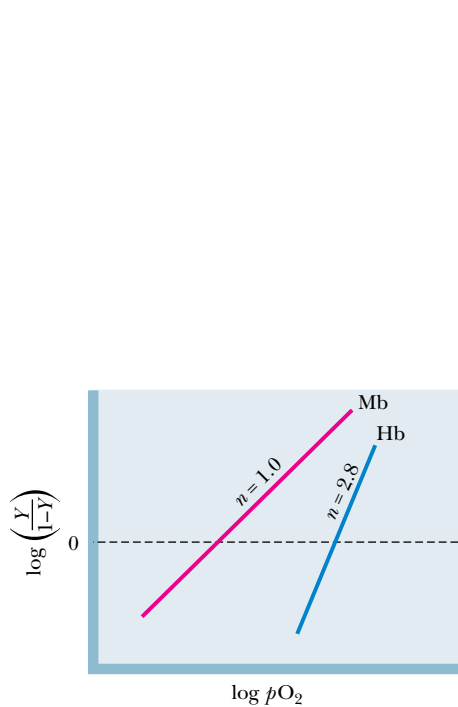


FIGURE A15.5 • Hill plot ($\log(Y/(1 - Y))$ versus $\log pO_2$) for Mb and Hb, showing that at $\log(Y/(1 - Y)) = 0$, that is, $Y = (1 - Y)$, the slope for Mb is 1.0 and for Hb is 2.8. The plot for Hb only approximates a straight line.

n would equal 4, as discussed above. If the O_2 -binding sites on Hb were completely noninteracting, that is, if the binding of one O_2 to Hb had no influence on the binding of additional O_2 molecules to the same Hb, n would equal 1. Figure A15.4 compares these extremes. Obviously, the real situation falls between the extremes of $n = 1$ or 4. The qualitative answer is that O_2 binding by Hb is highly cooperative, and the binding of the first O_2 markedly enhances the binding of subsequent O_2 molecules. However, this binding is not quite an all-or-none phenomenon.

If we take the logarithm of both sides of Equation (A15.12):

$$\log \left(\frac{Y}{1 - Y} \right) = n(\log pO_2) - \log K \tag{A15.13}$$

this expression is, of course, the generalized form of Equation (A15.8), the *Hill equation*, and a plot of $\log(Y/(1 - Y))$ versus $(\log pO_2)$ *approximates* a straight line in the region around $\log(Y/(1 - Y)) = 0$. Figure A15.5 represents a *Hill plot* comparing hemoglobin and myoglobin.

Because the binding of oxygen to hemoglobin is cooperative, the Hill plot is actually sigmoid (Figure A15.6). Cooperativity is a manifestation of the fact that the dissociation constant for the *first* O_2 , K_1 , is very different from the dissociation constant for the *last* O_2 bound, K_4 . The tangent to the lower asymptote of the Hill plot, when extrapolated to the $\log(Y/(1 - Y)) = 0$ axis, gives the dissociation constant, K_1 , for the binding of the first O_2 by Hb. Note that the value of K_1 is quite large ($>10^2$ torr), indicating a low affinity of Hb for this first O_2 (or conversely, a ready dissociation of the Hb (O_2)₁ complex). By a similar process, the tangent to the upper asymptote gives K_4 , the dissociation constant for the last O_2 to bind. K_4 has a value of less than 1 torr. The K_1/K_4 ratio exceeds 100, meaning the affinity of Hb for binding the fourth O_2 is over 100 times greater than for binding the first oxygen.

The value P_{50} has been defined above for myoglobin as the pO_2 that gives 50% saturation of the oxygen-binding protein with oxygen. Noting that at 50% saturation, $Y = (1 - Y)$, then we have from Equation (A15.13).

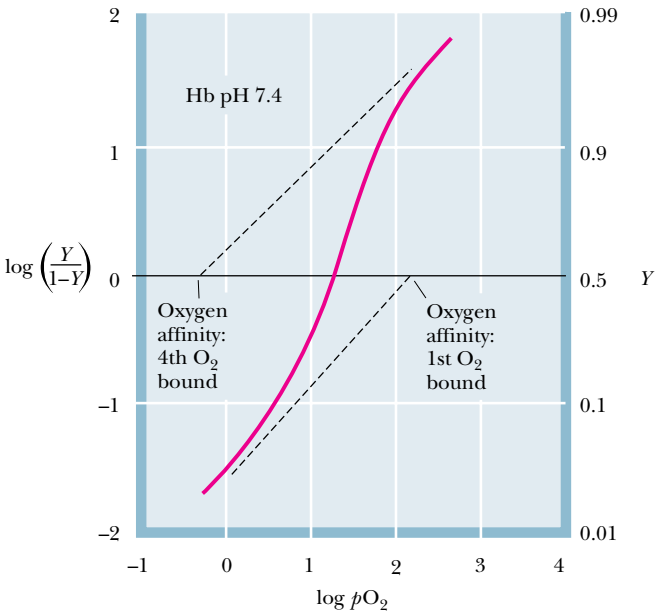


FIGURE A15.6 • Hill plot of Hb showing its nonlinear nature and the fact that its asymptotes can be extrapolated to yield the dissociation constants, K_1 and K_4 , for the first and fourth oxygens.

$$0 = n(\log pO_2) - \log K = n(\log P_{50}) - \log K \tag{A15.14}$$

$$\log K = n(\log P_{50}) \text{ or } K = (P_{50})^n \tag{A15.15}$$

That is, the situations for myoglobin and hemoglobin differ; therefore, P_{50} and K cannot be equated for Hb because of its multiple, interacting, O_2 -binding sites. The relationship between pO_2 and P_{50} for hemoglobin, by use of Equation (A15.12), becomes

$$\frac{Y}{1 - Y} = \left(\frac{pO_2}{P_{50}} \right)^n \tag{A15.16}$$

No single thing abides but all things flow.
Fragment to fragment clings and thus they
grow
Until we know them by name.
Then by degrees they change and are no
more the things we know.

LUCRETIVS (ca. 94 B.C.–50 B.C.)

OUTLINE

- 16.1 • The Basic Principle—Stabilization of the Transition State
- 16.2 • Enzymes Provide Enormous Rate Accelerations
- 16.3 • The Binding Energy of ES Is Crucial to Catalysis
- 16.4 • Entropy Loss and Destabilization of the ES Complex
- 16.5 • Transition-State Analogs Bind Very Tightly to the Active Site
- 16.6 • Covalent Catalysis
- 16.7 • General Acid–Base Catalysis
- 16.8 • Metal Ion Catalysis
- 16.9 • Proximity
- 16.10 • Typical Enzyme Mechanisms
- 16.11 • Serine Proteases
- 16.12 • The Aspartic Proteases
- 16.13 • Lysozyme

Chapter 16

Mechanisms of Enzyme Action

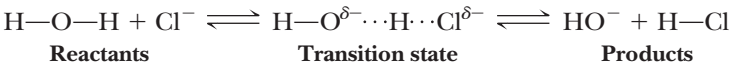


Like the workings of an ancient clock, the details of enzyme mechanisms are at once complex and simple. (David Parker/Science Photo Library/Photo Researchers, Inc.)

Although the catalytic properties of enzymes may seem almost magical, it is simply chemistry—the breaking and making of bonds—that gives enzymes their prowess. This chapter will explore the unique features of this chemistry. The mechanisms of hundreds of enzymes have been studied in at least some detail. In this chapter, it will be possible to examine only a few of these. Nonetheless, the chemical principles that influence the mechanisms of these few enzymes are universal, and many other cases are understandable in light of the knowledge gained from these examples.

16.1 • The Basic Principle—Stabilization of the Transition State

In all chemical reactions, the reacting atoms or molecules pass through a state that is intermediate in structure between the reactant(s) and the product(s). Consider the transfer of a proton from a water molecule to a chloride anion:



In the middle structure, the proton undergoing transfer is shared equally by the hydroxyl and chloride anions. This structure represents, as nearly as possible, the transition between the reactants and products, and it is known as the **transition state**.¹

Chemical reactions in which a substrate (S) is converted to a product (P) can be pictured as involving a transition state (which we henceforth denote as X[‡]), a species intermediate in structure between S and P (Figure 16.1). As seen in Chapter 14, the catalytic role of an enzyme is to reduce the energy barrier between substrate and transition state. This is accomplished through the formation of an **enzyme–substrate complex** (ES). This complex is converted to product by passing through a transition state, EX[‡] (Figure 16.1). As shown, the energy of EX[‡] is clearly lower than X[‡]. One might be tempted to conclude that this decrease in energy explains the rate acceleration achieved by the enzyme, but there is more to the story.

The energy barrier for the uncatalyzed reaction (Figure 16.1) is of course the difference in energies of the S and X[‡] states. Similarly, the energy barrier to be surmounted in the enzyme-catalyzed reaction, assuming that E is saturated with S, is the energy difference between ES and EX[‡]. *Reaction rate acceleration by an enzyme means simply that the energy barrier between ES and EX[‡] is less than the energy barrier between S and X[‡].* In terms of the free energies of activation, ΔG_e[‡] < ΔG_u[‡].

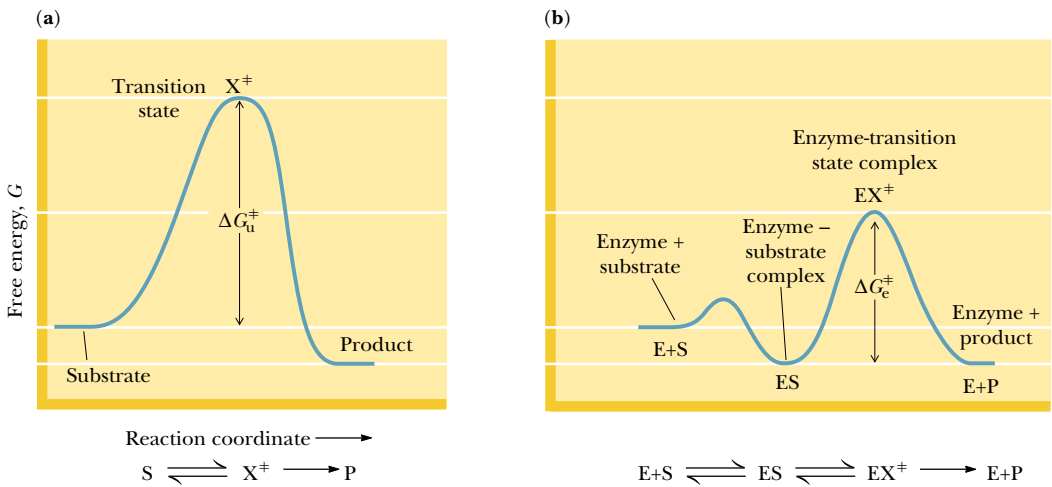


FIGURE 16.1 • Enzymes catalyze reactions by lowering the activation energy. Here the free energy of activation for (a) the uncatalyzed reaction, ΔG_u[‡], is larger than that for (b) the enzyme-catalyzed reaction, ΔG_e[‡].

¹It is important here to distinguish **transition states** from **intermediates**. A transition state is envisioned as an extreme distortion of a bond, and thus the lifetime of a typical transition state is viewed as being on the order of the lifetime of a bond vibration, typically 10^{−13} sec. Intermediates, on the other hand, are longer-lived, with lifetimes in the range of 10^{−13} sec to 10^{−3} sec.

DEEPER LOOK

What Is the Rate Enhancement of an Enzyme?

Enigmas abound in the world of enzyme catalysis. One of these surrounds the discussion of how the rate enhancement by an enzyme can be best expressed. Notice that the uncatalyzed conversion of a substrate S to a product P is usually a simple first-order process, described by a first-order rate constant k_u :

$$v_u = k_u[S]$$

On the other hand, for an enzyme that obeys Michaelis–Menten kinetics, the reaction is viewed as being first-order in S at low S and zero-order in S at high S. (See Chapter 14, where this distinction is discussed.)

$$v_e = \frac{k_{cat}[E_T][S]}{K_m + [S]}$$

If the “rate enhancement” effected by the enzyme is defined as $\text{rate enhancement} = v_e/v_u$

then we can write:

$$\text{rate enhancement} = \frac{k_{cat}}{k_u} \left(\frac{[E_T]}{K_m + [S]} \right)$$

Depending on the relative sizes of K_m and $[S]$, there are two possible results:

Case 1: When $[S]$ is large compared to K_m , the enzyme is saturated with S and the kinetics are zero-order in S.

$$\text{rate enhancement} = \frac{k_{cat}}{k_u} \left(\frac{[E_T]}{[S]} \right)$$

where $[E_T]/[S]$ is the fraction of the total S that is in the ES complex. Note here that defining the rate enhancement in terms of k_{cat}/k_u is equivalent to comparing the quantities $\Delta G_e^\ddagger$ and $\Delta G_u^\ddagger$ in the figure at right.

Case 2: When $[S]$ is small compared to K_m , not all the enzyme molecules have S bound, and the kinetics are first order in S.

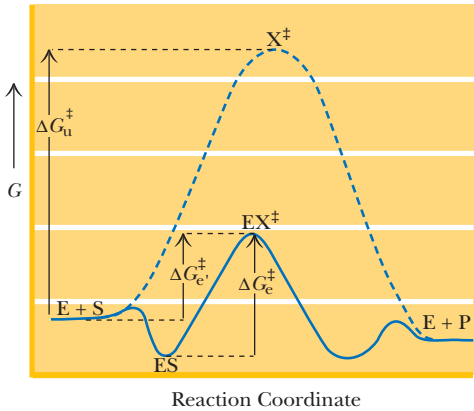
$$\text{rate enhancement} = \frac{k_{cat}}{k_u} \left(\frac{[E_T]}{K_m} \right)$$

Here, defining the rate enhancement in terms of $\frac{k_{cat}}{k_u K_m}$ is equivalent to comparing the quantities $\Delta G_e^\ddagger$ and $\Delta G_u^\ddagger$ in the figure below. Moreover, to the extent that K_m is approximated by K_S (see Equation 16.1), this rate enhancement can be rewritten as

$$\text{rate enhancement} = \frac{[E_T]}{K_T}$$

where K_T is the dissociation constant for the $EX^\ddagger$ complex (see Equation 16.2).

Viewed in this way, the best definition of “rate enhancement” depends upon the relationship between enzyme and substrate concentrations and the enzyme’s kinetic parameters.



There are important consequences for this statement. The enzyme must stabilize the transition-state complex, $EX^\ddagger$, more than it stabilizes the substrate complex, ES. Put another way, enzymes are “designed” by nature to bind the transition-state structure more tightly than the substrate (or the product). The dissociation constant for the enzyme-substrate complex is

$$K_S = \frac{[E][S]}{[ES]} \tag{16.1}$$

and the corresponding dissociation constant for the transition-state complex is

$$K_T = \frac{[E][X^\ddagger]}{[EX^\ddagger]} \tag{16.2}$$

Enzyme catalysis requires that $K_T < K_S$. According to **transition-state theory** (see references at end of chapter), the rate constants for the enzyme-catalyzed (k_e) and uncatalyzed (k_u) reactions can be related to K_S and K_T by:

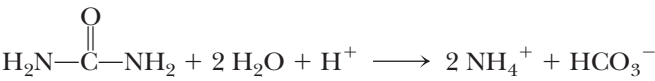
$$k_e/k_u \cong K_S/K_T \tag{16.3}$$

Thus, the enzymatic rate acceleration is approximately equal to the ratio of the dissociation constants of the enzyme–substrate and enzyme–transition-state complexes, at least when E is saturated with S.

16.2 • Enzymes Provide Enormous Rate Accelerations

Enzymes are powerful catalysts. Enzyme-catalyzed reactions are typically 10^7 to 10^{14} times faster than their uncatalyzed counterparts (Table 16.1). (There is even a report of a rate acceleration of $>10^{16}$ for the alkaline phosphatase-catalyzed hydrolysis of methylphosphate!)

These large rate accelerations correspond to substantial changes in the free energy of activation for the reaction in question. The urease reaction, for example,



shows an energy of activation some 84 kJ/mol smaller than the corresponding uncatalyzed reaction. To fully understand any enzyme reaction, it is important to account for the rate acceleration in terms of the structure of the enzyme and its mechanism of action. There are a limited number of catalytic mechanisms or factors that contribute to the remarkable performance of enzymes.

Table 16.1

A Comparison of Enzyme-Catalyzed Reactions and Their Uncatalyzed Counterparts				
Reaction	Enzyme	Uncatalyzed Rate, v_u (sec ⁻¹)	Catalyzed Rate, v_e (sec ⁻¹)	v_e/v_u
$\text{CH}_3-\text{O}-\text{PO}_3^{2-} + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{OH} + \text{HPO}_4^{2-}$	Alkaline phosphatase	1×10^{-15}	14	1.4×10^{16}
$\text{H}_2\text{N}-\overset{\text{O}}{\overset{\parallel}{\text{C}}}-\text{NH}_2 + 2 \text{H}_2\text{O} + \text{H}^+ \longrightarrow 2 \text{NH}_4^+ + \text{HCO}_3^-$	Urease	3×10^{-10}	3×10^4	1×10^{14}
$\text{R}-\overset{\text{O}}{\overset{\parallel}{\text{C}}}-\text{O}-\text{CH}_2\text{CH}_3 + \text{H}_2\text{O} \longrightarrow \text{RCOOH} + \text{HOCH}_2\text{CH}_3$	Chymotrypsin	1×10^{-10}	1×10^2	1×10^{12}
$\text{Glycogen} + \text{P}_i \longrightarrow \text{Glycogen} + \text{Glucose-1-P}$ $(n) \qquad \qquad \qquad (n-1)$	Glycogen phosphorylase	$<5 \times 10^{-15}$	1.6×10^{-3}	$>3.2 \times 10^{11}$
$\text{Glucose} + \text{ATP} \longrightarrow \text{Glucose-6-P} + \text{ADP}$	Hexokinase	$<1 \times 10^{-13}$	1.3×10^{-3}	$>1.3 \times 10^{10}$
$\text{CH}_3\text{CH}_2\text{OH} + \text{NAD}^+ \longrightarrow \text{CH}_3\overset{\text{O}}{\overset{\parallel}{\text{C}}}\text{H} + \text{NADH} + \text{H}^+$	Alcohol dehydrogenase	$<6 \times 10^{-12}$	2.7×10^{-5}	$>4.5 \times 10^6$
$\text{CO}_2 + \text{H}_2\text{O} \longrightarrow \text{HCO}_3^- + \text{H}^+$	Carbonic anhydrase	10^{-2}	10^5	1×10^7
$\text{Creatine} + \text{ATP} \longrightarrow \text{Cr-P} + \text{ADP}$	Creatine kinase	$<3 \times 10^{-9}$	4×10^{-5}	$>1.33 \times 10^4$

Adapted from Koshland, D., 1956. *Journal of Cellular Comparative Physiology*, Supp. 1, 47:217.

These include the following:

- 1. Entropy loss in ES formation
- 2. Destabilization of ES due to strain, desolvation, or electrostatic effects
- 3. Covalent catalysis
- 4. General acid or base catalysis
- 5. Metal ion catalysis
- 6. Proximity and orientation

Any or all of these mechanisms may contribute to the net rate acceleration of an enzyme-catalyzed reaction relative to the uncatalyzed reaction. A thorough understanding of any enzyme would require that the net acceleration be accounted for in terms of contributions from one or (usually) more of these mechanisms. Each of these will be discussed in detail in this chapter, but first it is important to appreciate how the formation of the enzyme-substrate (ES) complex makes all these mechanisms possible.

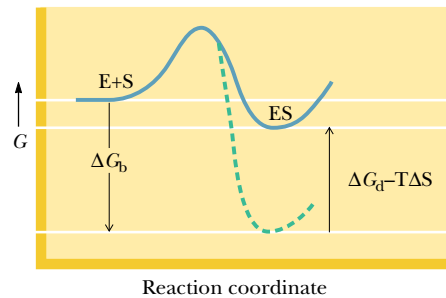


FIGURE 16.2 • The intrinsic binding energy of the enzyme-substrate (ES) complex (ΔG_b) is compensated to some extent by entropy loss due to the binding of E and S ($T\Delta S$) and by destabilization of ES (ΔG_d) by strain, distortion, desolvation, and similar effects. If ΔG_b were not compensated by $T\Delta S$ and ΔG_d , the formation of ES would follow the dashed line.

16.3 • The Binding Energy of ES Is Crucial to Catalysis

How is it that $X^\ddagger$ is stabilized more than S at the enzyme active site? To understand this, we must dissect and analyze the formation of the enzyme-substrate complex, ES. There are a number of important contributions to the free energy difference between the uncomplexed enzyme and substrate ($E + S$) and the ES complex (Figure 16.2). The favorable interactions between the substrate and amino acid residues on the enzyme account for the **intrinsic binding energy**, ΔG_b . The intrinsic binding energy ensures the favorable formation of the ES complex, but, if uncompensated, it makes the activation energy for the enzyme-catalyzed reaction unnecessarily large and wastes some of the catalytic power of the enzyme.

Compare the two cases in Figure 16.3. Because the enzymatic reaction rate is determined by the difference in energies between ES and $EX^\ddagger$, the smaller

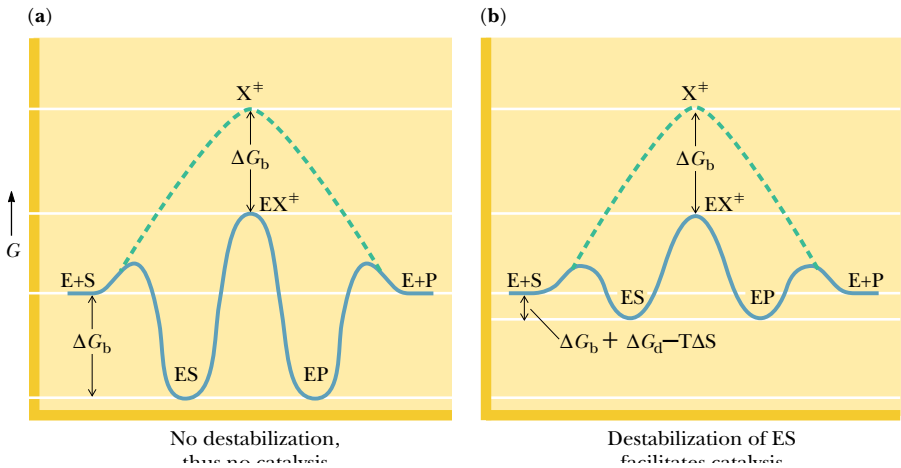


FIGURE 16.3 • (a) Catalysis does not occur if the ES complex and the transition state for the reaction are stabilized to equal extents. (b) Catalysis *will* occur if the transition state is stabilized to a greater extent than the ES complex (*right*). Entropy loss and destabilization of the ES complex ΔG_d ensure that this will be the case.

this difference, the faster the enzyme-catalyzed reaction. Tight binding of the substrate deepens the energy well of the ES complex and actually lowers the rate of the reaction.

16.4 • Entropy Loss and Destabilization of the ES Complex

The message of Figure 16.3 is that raising the energy of ES will increase the enzyme-catalyzed reaction rate. This is accomplished in two ways: (a) **loss of entropy** due to the binding of S to E, and (b) **destabilization of ES** by strain, distortion, desolvation, or other similar effects. The entropy loss arises from the fact that the ES complex (Figure 16.4) is a highly organized (low-entropy) entity compared to E + S in solution (a disordered, high-entropy situation). The entry of the substrate into the active site brings all the reacting groups and coordinating residues of the enzyme together with the substrate in just the proper position for reaction, with a net loss of entropy. The substrate and enzyme both possess **translational entropy**, the freedom to move in three dimensions, as well as **rotational entropy**, the freedom to rotate or tumble about any axis through the molecule. Both types of entropy are lost to some extent when two molecules (E and S) interact to form one molecule (the ES complex). Because ΔS is negative for this process, the term $-T\Delta S$ is a positive quantity, and *the intrinsic binding energy of ES is compensated to some extent by the entropy loss that attends the formation of the complex.*

Destabilization of the ES complex can involve **structural strain**, **desolvation**, or **electrostatic effects**. Destabilization by strain or distortion is usually just a consequence of the fact (noted previously) that *the enzyme is designed to bind the transition state more strongly than the substrate*. When the substrate binds, the imperfect nature of the “fit” results in distortion or strain in the substrate, the enzyme, or both. This means that the amino acid residues that make up the active site are oriented to coordinate the transition-state structure precisely, but will interact with the substrate or product less effectively.

Destabilization may also involve desolvation of charged groups on the substrate upon binding in the active site. Charged groups are highly stabilized in

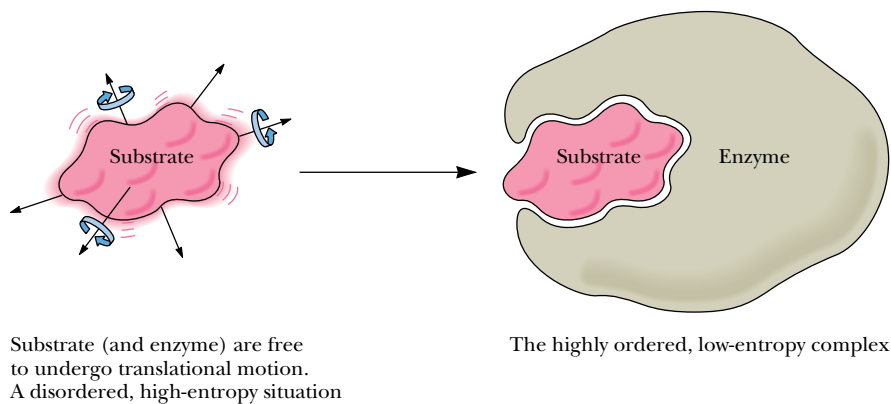


FIGURE 16.4 • Formation of the ES complex results in a loss of entropy. Prior to binding, E and S are free to undergo translational and rotational motion. By comparison, the ES complex is a more highly ordered, low-entropy complex.

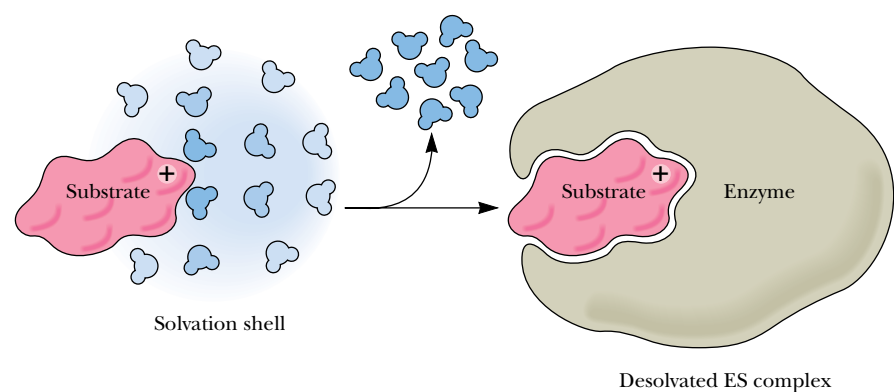


FIGURE 16.5 • Substrates typically lose waters of hydration in the formation of the ES complex. Desolvation raises the energy of the ES complex, making it more reactive.

water. For example, the transfer of Na^+ and Cl^- from the gas phase to aqueous solution is characterized by an **enthalpy of solvation**, ΔH_{solv} , of -775 kJ/mol. (Energy is given off and the ions become more stable.) When charged groups on a substrate move from water into an enzyme active site (Figure 16.5), they are often desolvated to some extent, becoming less stable and therefore more reactive.

When a substrate enters the active site, charged groups may be forced to interact (unfavorably) with charges of like sign, resulting in **electrostatic destabilization** (Figure 16.6). The reaction pathway acts in part to remove this stress. If the charge on the substrate is diminished or lost in the course of reaction, electrostatic destabilization can result in rate acceleration.

Whether by strain, desolvation, or electrostatic effects, destabilization raises the energy of the ES complex, and this increase is summed in the term ΔG_d , the free energy of destabilization. As noted in Figure 16.2, the net energy difference between $\text{E} + \text{S}$ and the ES complex is the sum of the intrinsic binding energy, ΔG_b ; the entropy loss on binding, $-T\Delta S$; and the distortion energy, ΔG_d . ES is destabilized (raised in energy) by the amount $\Delta G_d - T\Delta S$. The transition state is subject to no such destabilization, and the difference between the energies of $\text{X}^\ddagger$ and $\text{EX}^\ddagger$ is essentially ΔG_b , the full intrinsic binding energy.

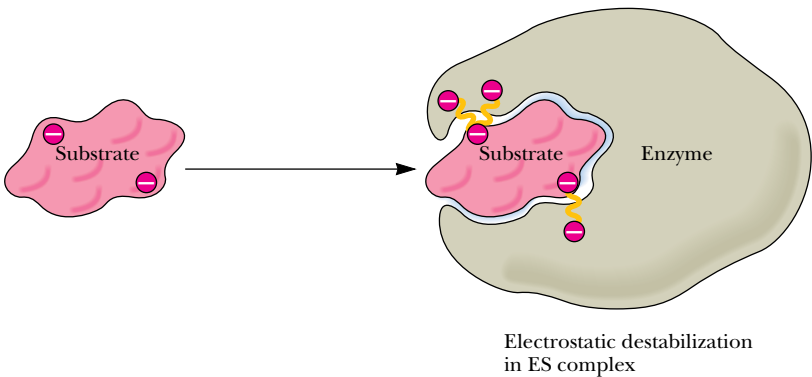


FIGURE 16.6 • Electrostatic destabilization of a substrate may arise from juxtaposition of like charges in the active site. If such charge repulsion is relieved in the course of the reaction, electrostatic destabilization can result in a rate increase.

16.5 • Transition-State Analogs Bind Very Tightly to the Active Site

Although not apparent at first, there are other important implications of Equation 16.3. It is important to consider the magnitudes of K_S and K_T . The ratio k_e/k_u may even exceed 10^{16} , as noted previously. Given a typical ratio of 10^{12} and a typical K_S of 10^{-3} M, the value of K_T should be 10^{-15} M! This is the dissociation constant for the transition-state complex from the enzyme, and this very low value corresponds to very tight binding of the transition state by the enzyme.

It is unlikely that such tight binding in an enzyme transition state will ever be measured experimentally, however, because the transition state itself is a “moving target.” It exists only for about 10^{-14} to 10^{-13} sec, less than the time required for a bond vibration. The nature of the elusive transition state can be explored, on the other hand, using **transition-state analogs**, stable molecules that are chemically and structurally similar to the transition state. Such molecules should bind more strongly than a substrate and more strongly than competitive inhibitors that bear no significant similarity to the transition state. Hundreds of examples of such behavior have been reported. For example, Robert Abeles studied a series of inhibitors of **proline racemase** (Figure 16.7) and found that *pyrrole-2-carboxylate* bound to the enzyme 160 times more tightly than L-proline, the normal substrate. This analog binds so tightly because it is planar and is similar in structure to the planar transition state for the racemization of proline. Two other examples of transition-state analogs are shown in Figure 16.8. *Phosphoglycolohydroxamate* binds 40,000 times more tightly to yeast aldolase than the substrate dihydroxyacetone phosphate. Even more remarkable, the 1,6-hydrate of *purine ribonucleoside* has been estimated to bind to adenosine deaminase with a K_i of 3×10^{-13} M!

It should be noted that transition-state analogs are only approximations of the transition state itself and will never bind as tightly as would be expected for the true transition state. These analogs are, after all, stable molecules and cannot be expected to resemble a true transition state too closely.

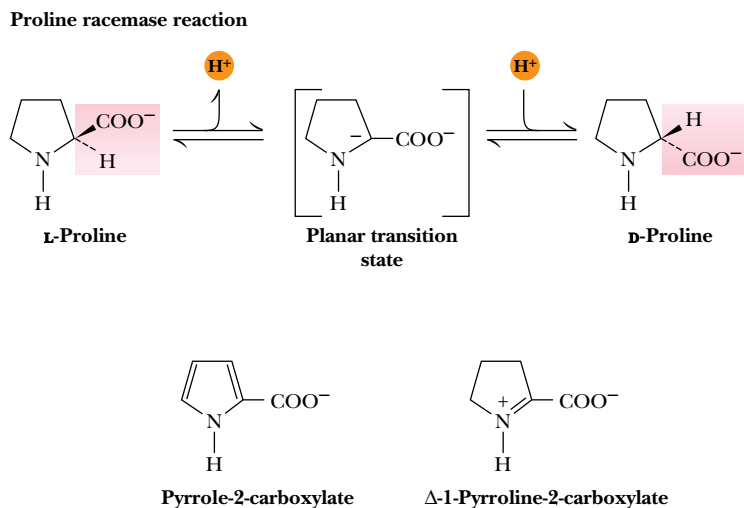
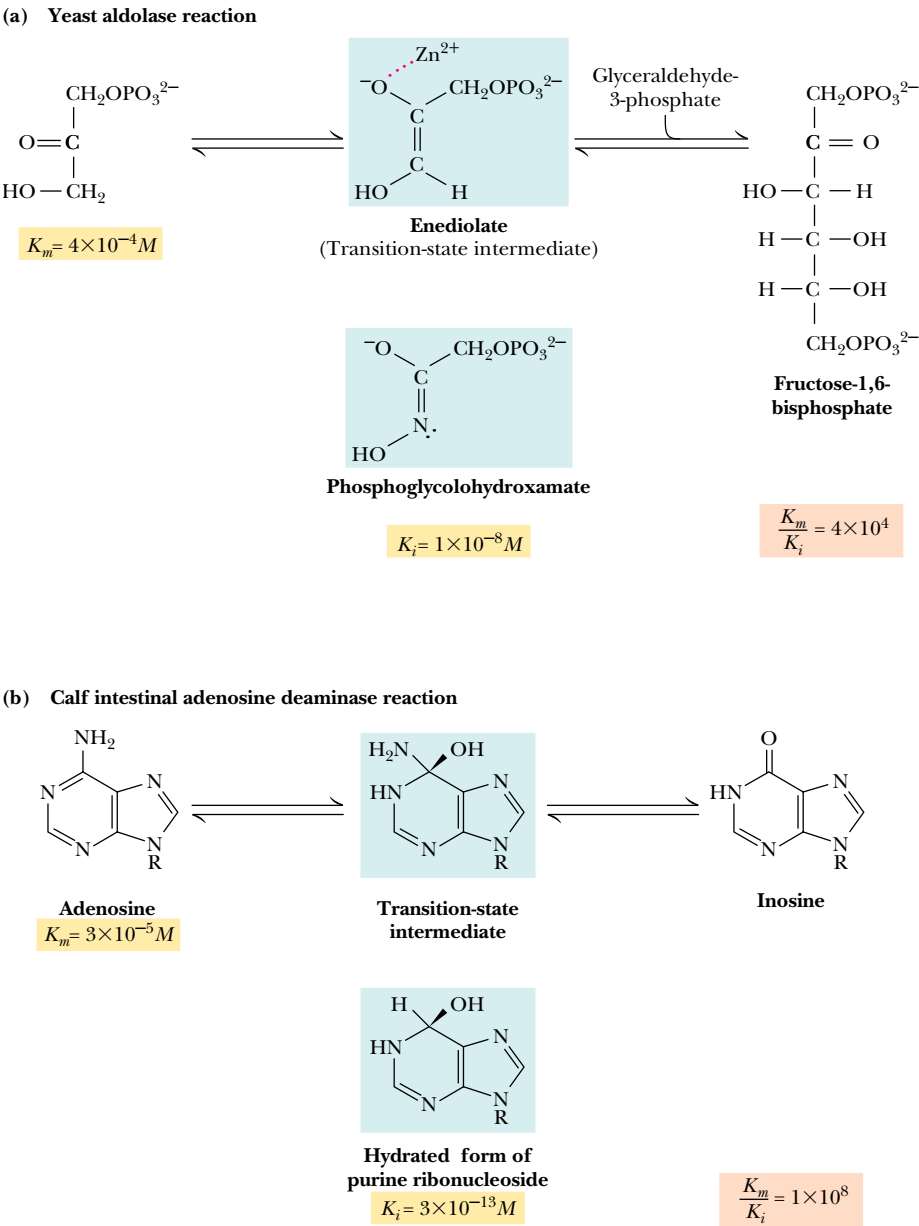
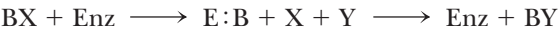


FIGURE 16.7 • The proline racemase reaction. Pyrrole-2-carboxylate and Δ-1-pyrroline-2-carboxylate mimic the planar transition state of the reaction.



and an enzymatic version of this reaction involving formation of a **covalent intermediate**:



If the enzyme-catalyzed reaction is to be faster than the uncatalyzed case, the acceptor group on the enzyme must be a better attacking group than Y and a better leaving group than X. Note that most enzymes that carry out covalent catalysis have ping-pong kinetic mechanisms.

The side chains of amino acids in proteins offer a variety of **nucleophilic** centers for catalysis, including amines, carboxylates, aryl and alkyl hydroxyls, imidazoles, and thiol groups. These groups readily attack electrophilic centers of substrates, forming covalently bonded enzyme-substrate intermediates. Typical electrophilic centers in substrates include phosphoryl groups, acyl groups, and glycosyl groups (Figure 16.9). The covalent intermediates thus formed can be attacked in a subsequent step by a water molecule or a second substrate, giving the desired product. **Covalent electrophilic catalysis** is also observed, but usually involves coenzyme adducts that generate electrophilic centers. Well over 100 enzymes are now known to form covalent intermediates during catalysis. Table 16.2 lists some typical examples, including that of glyceraldehyde-3-phosphate dehydrogenase, which catalyzes the reaction:

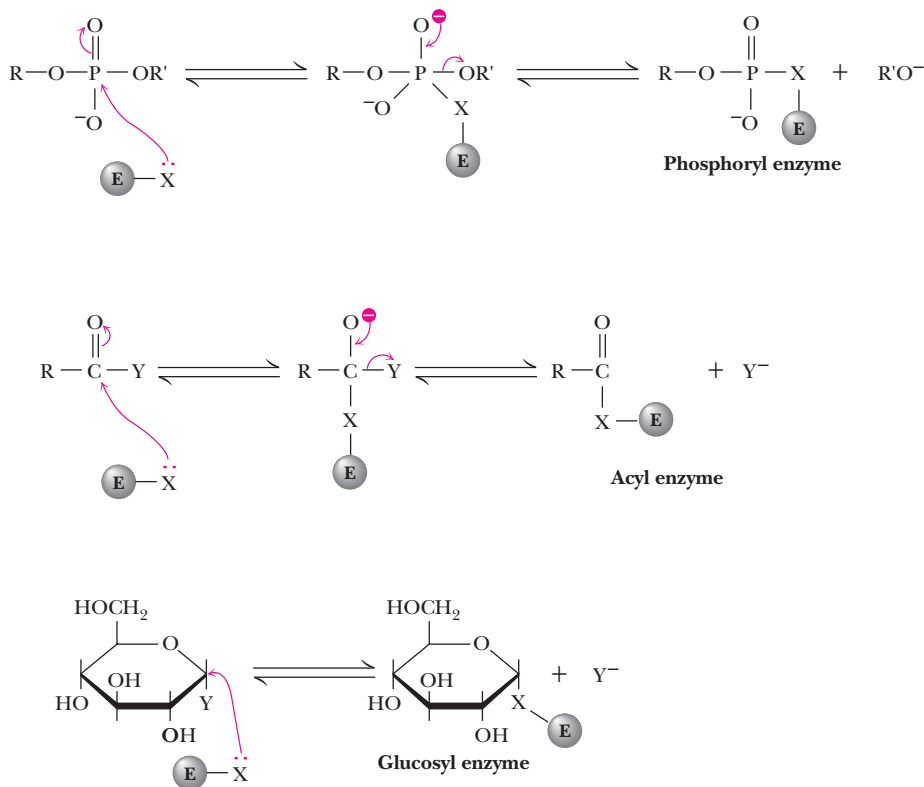
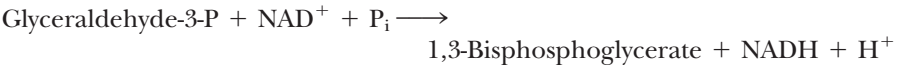


FIGURE 16.9 • Examples of covalent bond formation between enzyme and substrate. In each case, a nucleophilic center (X:) on an enzyme attacks an electrophilic center on a substrate.

Table 16.2

Enzymes That Form Covalent Intermediates

Enzymes	Reacting Group	Covalent Intermediate
1. Chymotrypsin Elastase Esterases Subtilisin Thrombin Trypsin	$\left. \begin{array}{c} \text{CH}-\text{CH}_2\text{OH} \\ \text{(Ser)} \end{array} \right\}$	$\left. \begin{array}{c} \text{CH}-\text{CH}_2\text{O}-\text{C}(=\text{O})-\text{R} \\ \text{(Acyl-Ser)} \end{array} \right\}$
2. Glyceraldehyde-3-phosphate dehydrogenase Papain	$\left. \begin{array}{c} \text{CH}-\text{CH}_2\text{SH} \\ \text{(Cys)} \end{array} \right\}$	$\left. \begin{array}{c} \text{CH}-\text{CH}_2\text{S}-\text{C}(=\text{O})-\text{R} \\ \text{(Acyl-Cys)} \end{array} \right\}$
3. Alkaline phosphatase Phosphoglucomutase	$\left. \begin{array}{c} \text{CH}-\text{CH}_2\text{OH} \\ \text{(Ser)} \end{array} \right\}$	$\left. \begin{array}{c} \text{CH}-\text{CH}_2\text{O}-\text{PO}_3^{2-} \\ \text{(Phosphoserine)} \end{array} \right\}$
4. Phosphoglycerate mutase Succinyl-CoA synthetase	$\left. \begin{array}{c} \text{CH}_2 \\ \text{HN} \diagup \text{C} \diagdown \\ \text{N} \end{array} \right\}$ (His)	$\left. \begin{array}{c} \text{CH}_2 \\ \text{O} \\ \text{O}=\text{P}-\text{N} \diagup \text{C} \diagdown \\ \text{O}^- \end{array} \right\}$ (Phosphohistidine)
5. Aldolase Decarboxylases Pyridoxal phosphate-dependent enzymes	$\left. \begin{array}{c} \text{R}-\text{NH}_3^+ \\ \text{(Amino)} \end{array} \right\}$	$\left. \begin{array}{c} \text{R}-\text{N}=\text{C} \\ \text{(Schiff base)} \end{array} \right\}$

As shown in Figure 16.10, this reaction mechanism involves nucleophilic attack by —SH on the substrate glyceraldehyde-3-P to form a covalent *acylcysteine* (or *hemithioacetal*) intermediate. Hydride transfer to NAD⁺ generates a *thioester* intermediate. Nucleophilic attack by phosphate yields the desired mixed carboxylic–phosphoric anhydride product, 1,3-bisphosphoglycerate. Several examples of covalent catalysis will be discussed in detail in later chapters.

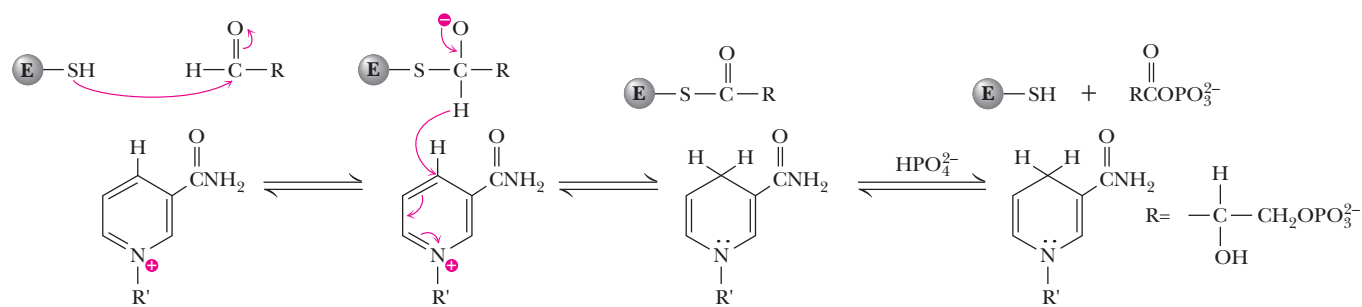


FIGURE 16.10 • Formation of a covalent intermediate in the glyceraldehyde-3-phosphate dehydrogenase reaction. Nucleophilic attack by a cysteine —SH group forms a covalent acylcysteine intermediate. Following hydride transfer to NAD⁺, nucleophilic attack by phosphate yields the product, 1,3-bisphosphoglycerate.

16.7 • General Acid–Base Catalysis

Nearly all enzyme reactions involve some degree of acid or base catalysis. There are two types of acid–base catalysis: (1) **specific acid–base catalysis**, in which H^+ or OH^- accelerates the reaction, and (2) **general acid–base catalysis**, in which an acid or base other than H^+ or OH^- accelerates the reaction. For ordinary solution reactions, these two cases can be distinguished on the basis of simple experiments. As shown in Figure 16.11, in specific acid or base catalysis, the buffer concentration has no effect. In general acid or base catalysis, however, the buffer may donate or accept a proton in the transition state and thus affect the rate. *By definition, general acid–base catalysis is catalysis in which a proton is transferred in the transition state.* Consider the hydrolysis of *p*-nitrophenylacetate with imidazole acting as a general base (Figure 16.12). Proton transfer apparently stabilizes the transition state here. The water has been made more nucleophilic without generation of a high concentration of OH^- or without the formation of unstable, high-energy species. General acid or general base catalysis may increase reaction rates 10- to 100-fold. In an enzyme, ionizable groups on the protein provide the H^+ transferred in the transition state. Clearly, an ionizable group will be most effective as a H^+ transferring agent at or near its $\text{p}K_a$. Because the $\text{p}K_a$ of the histidine side chain is near 7, histidine is often the most effective general acid or base. Descriptions of several cases of general acid–base catalysis in typical enzymes follow.

16.8 • Metal Ion Catalysis

Many enzymes require metal ions for maximal activity. If the enzyme binds the metal very tightly or requires the metal ion to maintain its stable, native state, it is referred to as a **metalloenzyme**. Enzymes that bind metal ions more weakly, perhaps only during the catalytic cycle, are referred to as **metal activated**. One role for metals in metal-activated enzymes and metalloenzymes is to act as electrophilic catalysts, stabilizing the increased electron density or negative charge that can develop during reactions. Among the enzymes that function in this

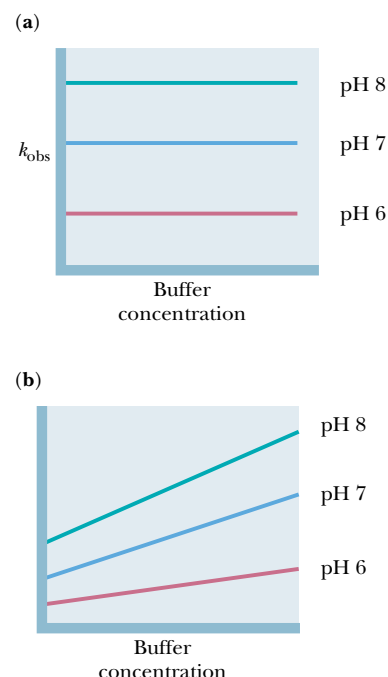
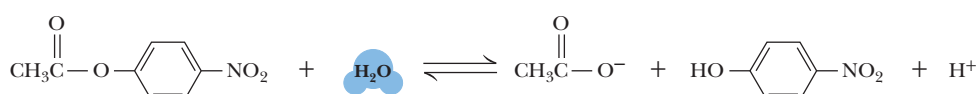


FIGURE 16.11 • Specific and general acid–base catalysis of simple reactions in solution may be distinguished by determining the dependence of observed reaction rate constants (k_{obs}) on pH and buffer concentration. (a) In specific acid–base catalysis, H^+ or OH^- concentration affects the reaction rate, k_{obs} is pH-dependent, but buffers (which accept or donate H^+/OH^-) have no effect. (b) In general acid–base catalysis, in which an ionizable buffer may donate or accept a proton in the transition state, k_{obs} is dependent on buffer concentration.

Reaction



Mechanism

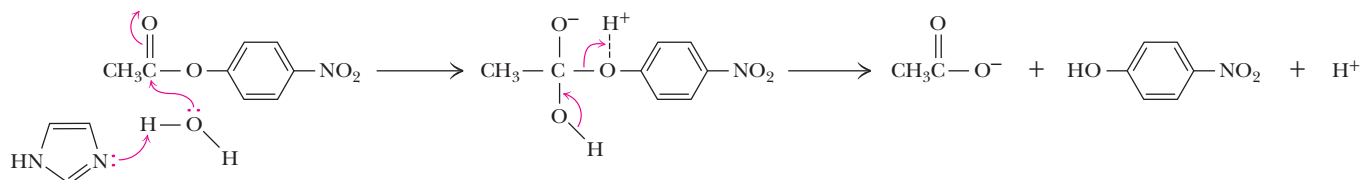


FIGURE 16.12 • Catalysis of *p*-nitrophenylacetate hydrolysis by imidazole—an example of general base catalysis. Proton transfer to imidazole in the transition state facilitates hydroxyl attack on the substrate carbonyl carbon.

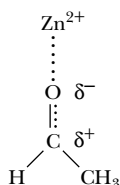


FIGURE 16.13 • Liver alcohol dehydrogenase catalyzes the transfer of a hydride ion (H^-) from NADH to acetaldehyde (CH_3CHO), forming ethanol ($\text{CH}_3\text{CH}_2\text{OH}$). An active-site zinc ion stabilizes negative charge development on the oxygen atom of acetaldehyde, leading to an induced partial positive charge on the carbonyl C atom. Transfer of the negatively charged hydride ion to this carbon forms ethanol.

manner (Figure 16.13) is liver alcohol dehydrogenase. Another potential function of metal ions is to provide a powerful nucleophile at neutral pH. Coordination to a metal ion can increase the acidity of a nucleophile with an ionizable proton:



The reactivity of the coordinated, deprotonated nucleophile is typically intermediate between that of the un-ionized and ionized forms of the nucleophile. Carboxypeptidase (Chapter 5) contains an active site Zn^{2+} , which facilitates deprotonation of a water molecule in this manner.

16.9 • Proximity

Chemical reactions go faster when the reactants are *in proximity*, that is, near each other. In solution or in the gas phase, this means that increasing the concentrations of reacting molecules, which raises the number of collisions, causes higher rates of reaction. Enzymes, which have specific binding sites for particular reacting molecules, essentially take the reactants out of dilute solution and hold them close to each other. This proximity of reactants is said to raise the “effective” concentration over that of the substrates in solution, and leads to an increased reaction rate. In order to measure proximity effects in enzyme reactions, enzymologists have turned to model studies comparing intermolecular reaction rates with corresponding or similar intramolecular reaction rates. A typical case is the imidazole-catalyzed hydrolysis of *p*-nitrophenylacetate (Figure 16.14a). Under certain conditions the rate constant for this bimolecular reaction is $35 \text{ M}^{-1} \text{ min}^{-1}$. By comparison, the first-order rate constant for the analogous but intramolecular reaction shown in Figure 16.14b is 839 min^{-1} . The ratio of these two rate constants

$$(839 \text{ min}^{-1}) / (35 \text{ M}^{-1} \text{ min}^{-1}) = 23.97 \text{ M}$$

has the units of concentration and can be thought of as an effective concentration of imidazole in the intramolecular reaction. Put another way, a concentration of imidazole of 23.9 M would be required in the intermolecular reaction to make it proceed as fast as the intramolecular reaction.

There is more to this story, however. Enzymes not only bring substrates and catalytic groups close together, they orient them in a manner suitable for catalysis as well. Comparison of the rates of reaction of the molecules shown

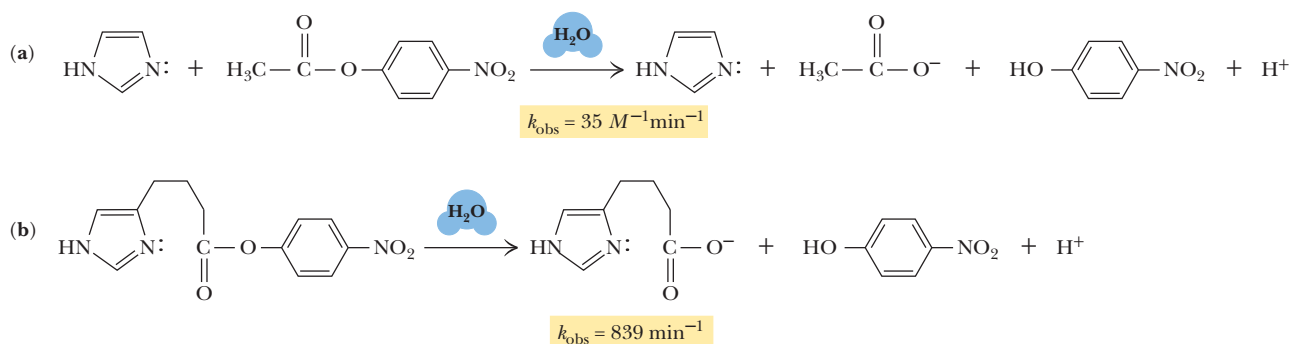


FIGURE 16.14 • An example of proximity effects in catalysis. (a) The imidazole-catalyzed hydrolysis of *p*-nitrophenylacetate is slow, but the corresponding intramolecular reaction is 24-fold faster (assuming $[\text{imidazole}] = 1 \text{ M}$ in [a]).

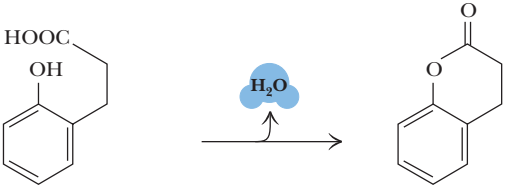
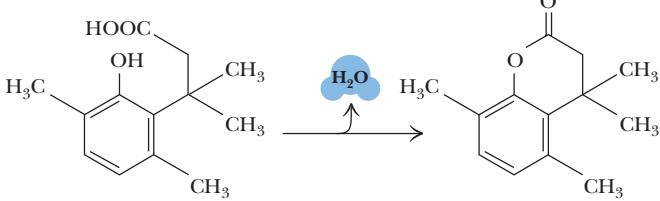
Reaction	Rate const. (M ⁻¹ sec ⁻¹)	Ratio
	5.9x10 ⁻⁶	
	1.5x10 ⁶	2.5x10 ¹¹

FIGURE 16.15 • Orientation effects in intramolecular reactions can be dramatic. Steric crowding by methyl groups provides a rate acceleration of 2.5×10^{11} for the lower reaction compared to the upper reaction. (Adapted from Milstien, S., and Cohen, L. A., 1972. Stereopopulation control I. Rate enhancements in the lactonization of *o*-hydroxyhydrocinnamic acid. Journal of the American Chemical Society 94:9158–9165.)

in Figure 16.15 makes it clear that the bulky methyl groups force an orientation on the alkyl carboxylate and the aromatic hydroxyl groups that makes them approximately 250 billion times more likely to react. Enzymes function similarly by placing catalytically functional groups (from the protein side chains or from another substrate) in the proper position for reaction.

Clearly, proximity and orientation play a role in enzyme catalysis, but there is a problem with each of the above comparisons. In both cases, it is impossible to separate true proximity and orientation effects from the effects of entropy loss when molecules are brought together (described the Section 16.4). The actual rate accelerations afforded by proximity and orientation effects in Figures 16.14 and 16.15, respectively, are much smaller than the values given in these figures. Simple theories based on probability and nearest-neighbor models, for example, predict that proximity effects may actually provide rate increases of only 5- to 10-fold. For any real case of enzymatic catalysis, it is nonetheless important to remember that proximity and orientation effects are significant.

16.10 • Typical Enzyme Mechanisms

The balance of this chapter will be devoted to several classic and representative enzyme mechanisms. These particular cases are well understood, because the three-dimensional structures of the enzymes and the bound substrates are known at atomic resolution, and because great efforts have been devoted to kinetic and mechanistic studies. They are important because they represent reaction types that appear again and again in living systems, and because they demonstrate many of the catalytic principles cited above. Enzymes are the catalytic machines that sustain life, and what follows is an intimate look at the inner workings of the machinery.

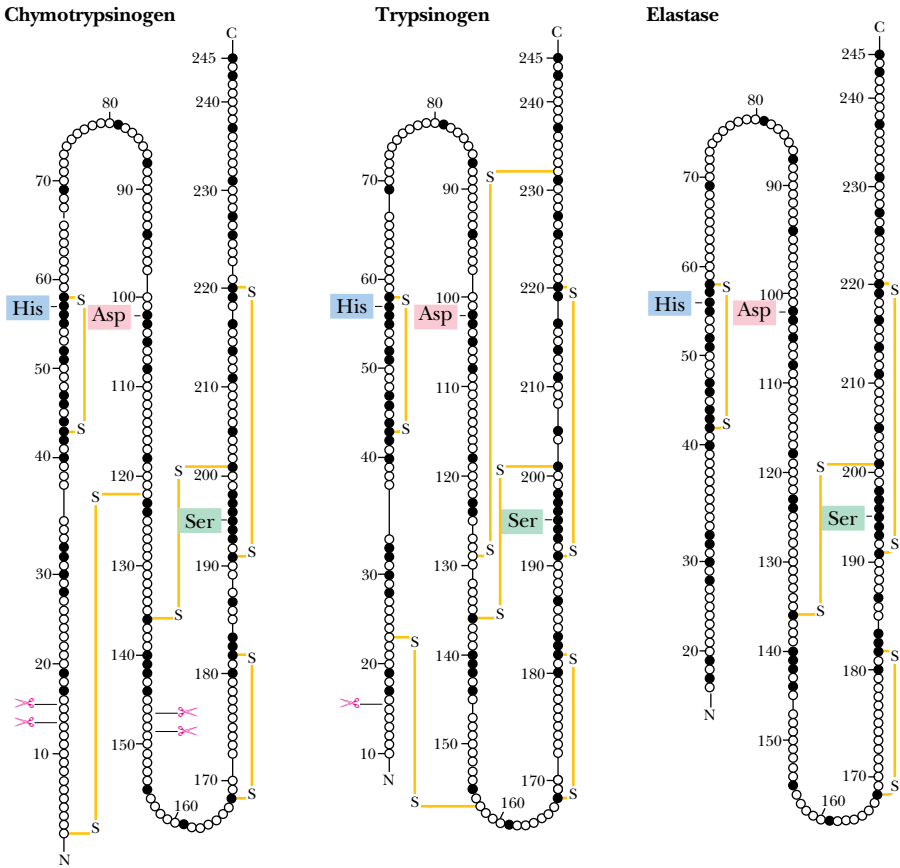
16.11 • Serine Proteases

Serine proteases are a class of proteolytic enzymes whose catalytic mechanism is based on an active-site serine residue. Serine proteases are one of the best-characterized families of enzymes. This family includes *trypsin*, *chymotrypsin*, *elastase*, *thrombin*, *subtilisin*, *plasmin*, *tissue plasminogen activator*, and other related enzymes. The first three of these are digestive enzymes and are synthesized in the pancreas and secreted into the digestive tract as inactive **proenzymes**, or **zymogens**. Within the digestive tract, the zymogen is converted into the active enzyme form by cleaving off a portion of the peptide chain. Thrombin is a crucial enzyme in the blood-clotting cascade, subtilisin is a bacterial protease, and plasmin breaks down the fibrin polymers of blood clots. Tissue plasminogen activator (TPA) specifically cleaves the proenzyme *plasminogen*, yielding plasmin. Owing to its ability to stimulate breakdown of blood clots, TPA can minimize the harmful consequences of a heart attack, if administered to a patient within 30 minutes of onset. Finally, although not itself a protease, *acetylcholinesterase* is a *serine esterase* and is related mechanistically to the serine proteases. It degrades the neurotransmitter acetylcholine in the synaptic cleft between neurons.

The Digestive Serine Proteases

Trypsin, chymotrypsin, and elastase all carry out the same reaction—the cleavage of a peptide chain—and although their structures and mechanisms are quite similar, they display very different specificities. Trypsin cleaves peptides

FIGURE 16.16 • Comparison of the amino acid sequences of chymotrypsinogen, trypsinogen, and elastase. Each circle represents one amino acid. Numbering is based on the sequence of chymotrypsinogen. Filled circles indicate residues that are identical in all three proteins. Disulfide bonds are indicated in yellow. The positions of the three catalytically important active-site residues (His⁵⁷, Asp¹⁰², and Ser¹⁹⁵) are indicated.



on the carbonyl side of the basic amino acids, arginine or lysine (see Table 5.6). Chymotrypsin prefers to cleave on the carbonyl side of aromatic residues, such as phenylalanine and tyrosine. Elastase is not as specific as the other two; it mainly cleaves peptides on the carbonyl side of small, neutral residues. These three enzymes all possess molecular weights in the range of 25,000, and all have similar sequences (Figure 16.16) and three-dimensional structures. The structure of chymotrypsin is typical (Figure 16.17). The molecule is ellipsoidal in shape and contains an α -helix at the C-terminal end (residues 230 to 245) and several β -sheet domains. Most of the aromatic and hydrophobic residues are buried in the interior of the protein, and most of the charged or hydrophilic residues are on the surface. Three polar residues—His⁵⁷, Asp¹⁰², and Ser¹⁹⁵—form what is known as a **catalytic triad** at the active site (Figure 16.18). These three residues are conserved in trypsin and elastase as well. The active site in this case is actually a depression on the surface of the enzyme, with a small pocket that the enzyme uses to identify the residue for which it is specific (Figure 16.19). Chymotrypsin, for example, has a pocket surrounded by hydrophobic residues and large enough to accommodate an aromatic side chain. The pocket in trypsin has a negative charge (Asp¹⁸⁹) at its bottom, facilitating the binding of positively charged arginine and lysine residues. Elastase, on the other hand, has a shallow pocket with bulky threonine and valine residues at the opening. Only small, nonbulky residues can be accommodated in its pocket. The backbone of the peptide substrate is hydrogen-bonded in antiparallel fashion to residues 215 to 219 and bent so that the peptide bond to be cleaved is bound close to His⁵⁷ and Ser¹⁹⁵.

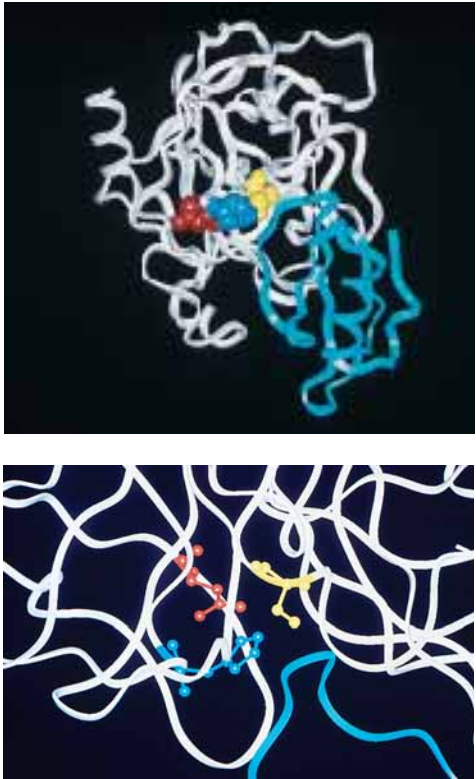


FIGURE 16.17 • Structure of chymotrypsin (white) in a complex with eglin C (blue ribbon structure), a target protein. The residues of the catalytic triad (His⁵⁷, Asp¹⁰², and Ser¹⁹⁵) are highlighted. His⁵⁷ (blue) is flanked above by Asp¹⁰² (red) and on the right by Ser¹⁹⁵ (yellow). The catalytic site is filled by a peptide segment of eglin. Note how close Ser¹⁹⁵ is to the peptide that would be cleaved in a chymotrypsin reaction.

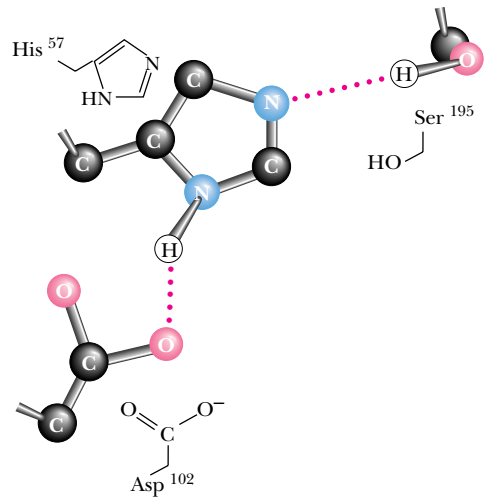


FIGURE 16.18 • The catalytic triad of chymotrypsin.

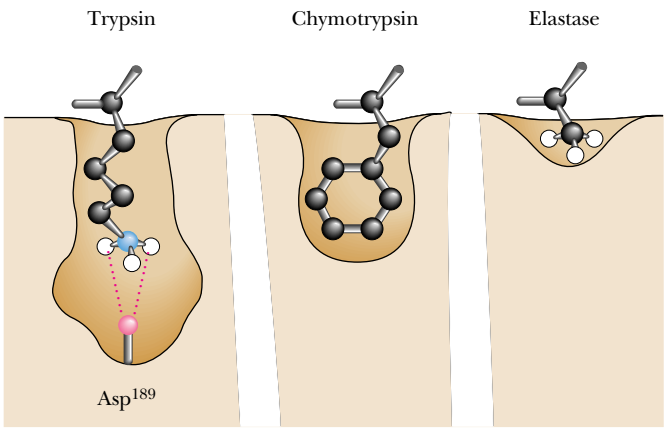


FIGURE 16.19 • The substrate-binding pockets of trypsin, chymotrypsin, and elastase. (Irving Geis)

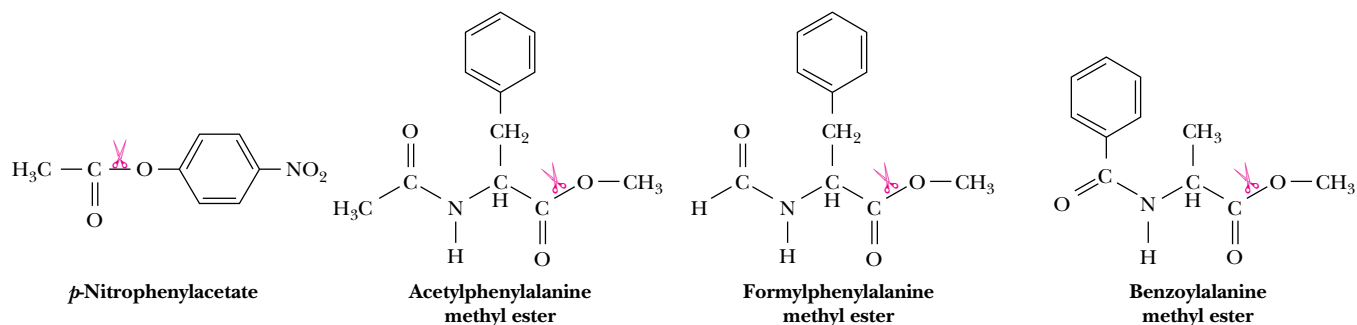


FIGURE 16.20 • Artificial substrates used in studies of the mechanism of chymotrypsin.

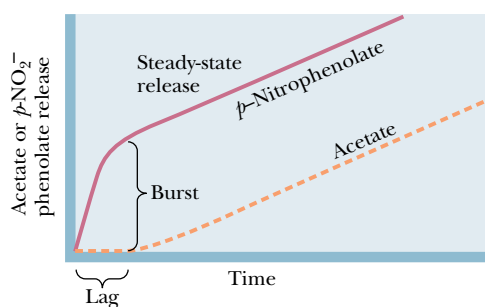


FIGURE 16.21 • Burst kinetics observed in the chymotrypsin reaction. A burst of nitrophenolate production is followed by a slower, steady-state release. After an initial lag period, acetate release is also observed. This kinetic pattern is consistent with rapid formation of an acyl-enzyme intermediate (and the burst of nitrophenolate). The slower, steady-state release of products corresponds to rate-limiting breakdown of the acyl-enzyme intermediate.

formylphenylalanine methyl ester and acetylphenylalanine methyl ester (Figure 16.20). *p*-Nitrophenylacetate is an especially useful model substrate, because the nitrophenolate product is easily observed, owing to its strong absorbance at 400 nm. When large amounts of chymotrypsin are used in kinetic studies with this substrate, a **rapid initial burst** of *p*-nitrophenolate is observed (in an amount approximately equal to the enzyme concentration), followed by a much slower, linear rate of nitrophenolate release (Figure 16.21). Observation of a burst, followed by slower, steady-state product release, is strong evidence for a multistep mechanism, with a fast first step and a slower second step.

In the chymotrypsin mechanism, the nitrophenylacetate combines with the enzyme to form an ES complex. This is followed by a rapid second step in which an **acyl-enzyme intermediate** is formed, with the acetyl group covalently bound to the very reactive Ser¹⁹⁵. The nitrophenyl moiety is released as nitrophenolate (Figure 16.22), accounting for the burst of nitrophenolate product. Attack of a water molecule on the acyl-enzyme intermediate yields acetate as the second product in a subsequent, slower step. The enzyme is now free to bind another molecule of *p*-nitrophenylacetate, and the *p*-nitrophenolate product produced at this point corresponds to the slower, steady-state formation of product in the upper right portion of Figure 16.21. In this mechanism, the release of acetate is the **rate-limiting step**, and accounts for the observation of **burst kinetics**—the pattern shown in Figure 16.21.

Serine proteases like chymotrypsin are susceptible to inhibition by **organic fluorophosphates**, such as *diisopropylfluorophosphate* (*DIFP*, Figure 16.23). DIFP

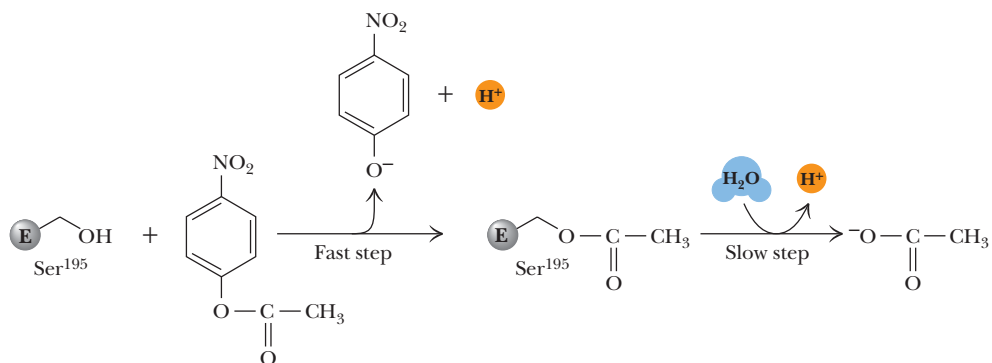


FIGURE 16.22 • Rapid formation of the acyl-enzyme intermediate is followed by slower product release.

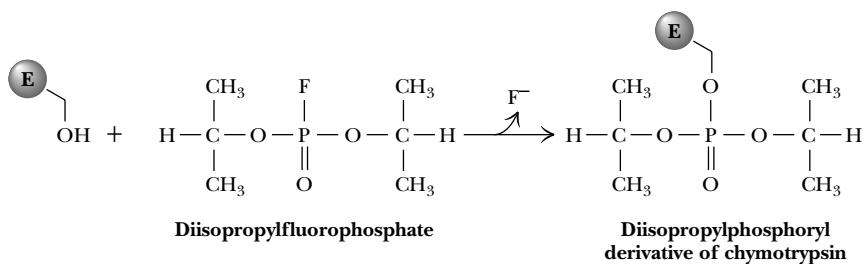


FIGURE 16.23 • Diisopropylfluorophosphate (DIFP) reacts with active-site serine residues of serine proteases (and esterases), causing permanent inactivation.

reacts rapidly with active-site serine residues, such as Ser¹⁹⁵ of chymotrypsin and the other serine proteases (but not with any of the other serines in these proteins), to form a DIP-enzyme. This covalent enzyme-inhibitor complex is extremely stable, and chymotrypsin is thus permanently inactivated by DIFP.

The Serine Protease Mechanism in Detail: Events at the Active Site

A likely mechanism for peptide hydrolysis is shown in Figure 16.24. As the backbone of the substrate peptide binds adjacent to the catalytic triad, the specific side chain fits into its pocket. Asp¹⁰² of the catalytic triad positions His⁵⁷ and immobilizes it through a hydrogen bond as shown. In the first step of the reaction, His⁵⁷ acts as a general base to withdraw a proton from Ser¹⁹⁵, facilitating nucleophilic attack by Ser¹⁹⁵ on the carbonyl carbon of the peptide bond to be cleaved. This is probably a *concerted step*, because proton transfer prior to Ser¹⁹⁵ attack on the acyl carbon would leave a relatively unstable negative charge on the serine oxygen. In the next step, donation of a proton from His⁵⁷ to the peptide's amide nitrogen creates a protonated amine on the covalent, tetrahedral intermediate, facilitating the subsequent bond breaking and dissociation of the amine product. The negative charge on the peptide oxygen is unstable; the tetrahedral intermediate is short-lived and rapidly breaks down to expel the amine product. The acyl-enzyme intermediate that results is reasonably stable; it can even be isolated using substrate analogs for which further reaction cannot occur. With normal peptide substrates, however, subsequent nucleophilic attack at the carbonyl carbon by water generates another transient tetrahedral intermediate (Figure 16.24). His⁵⁷ acts as a general base in this step, accepting a proton from the attacking water molecule. The subsequent collapse of the tetrahedral intermediate is assisted by proton donation from His⁵⁷ to the serine oxygen in a concerted manner. Deprotonation of the carboxyl group and its departure from the active site complete the reaction as shown.

Until recently, the catalytic role of Asp¹⁰² in trypsin and the other serine proteases had been surmised on the basis of its proximity to His⁵⁷ in structures obtained from X-ray diffraction studies, but it had never been demonstrated with certainty in physical or chemical studies. As can be seen in Figure 16.17, Asp¹⁰² is buried at the active site and is normally inaccessible to chemical modifying reagents. In 1987, however, Charles Craik, William Rutter, and their colleagues used site-directed mutagenesis (see Chapter 13) to prepare a mutant trypsin with an asparagine in place of Asp¹⁰². This mutant trypsin possessed a hydrolytic activity with ester substrates only 1/10,000 that of native trypsin, demonstrating that Asp¹⁰² is indeed essential for catalysis and that its ability to immobilize and orient His⁵⁷ is crucial to the function of the catalytic triad.

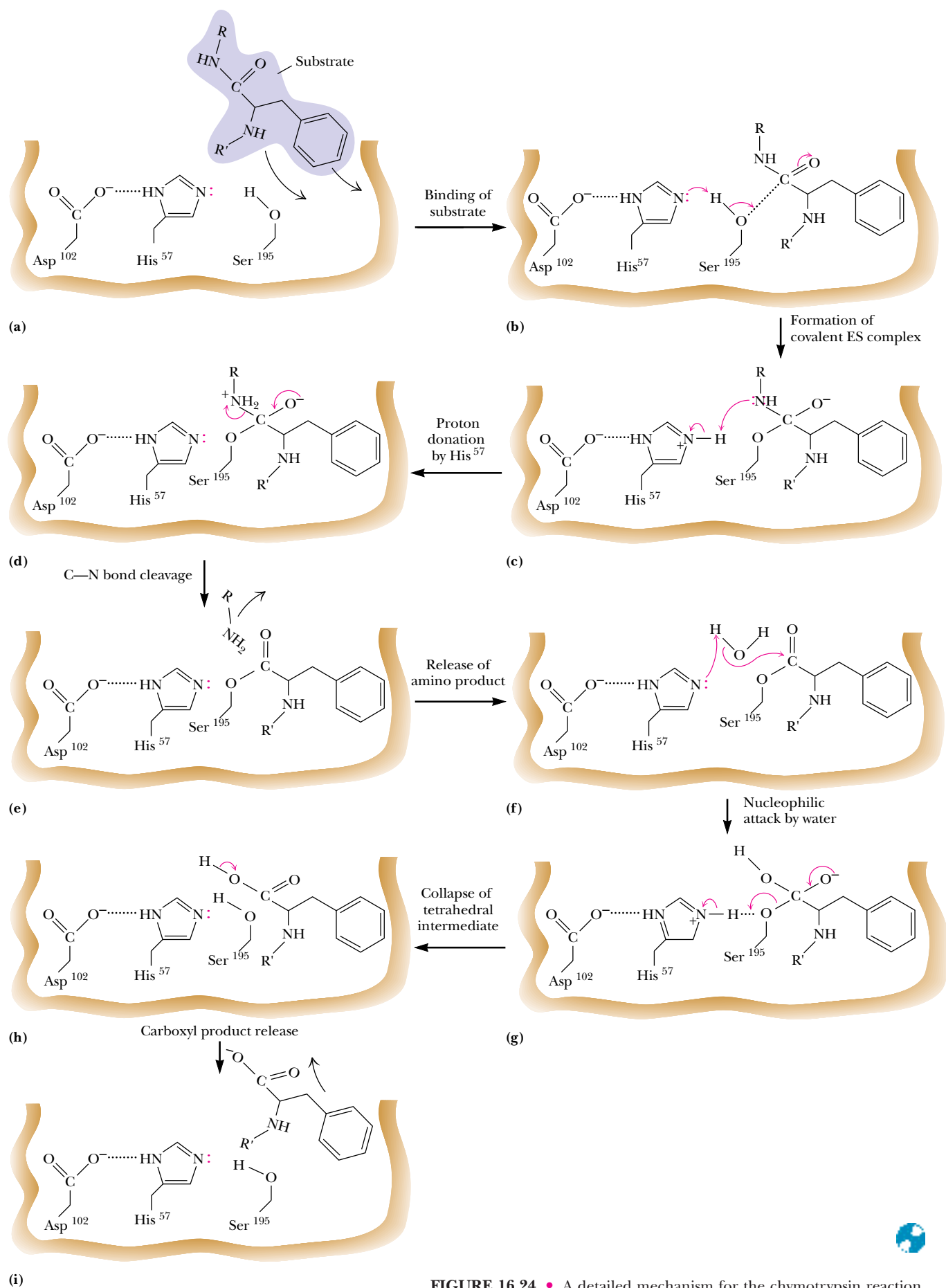


FIGURE 16.24 • A detailed mechanism for the chymotrypsin reaction.

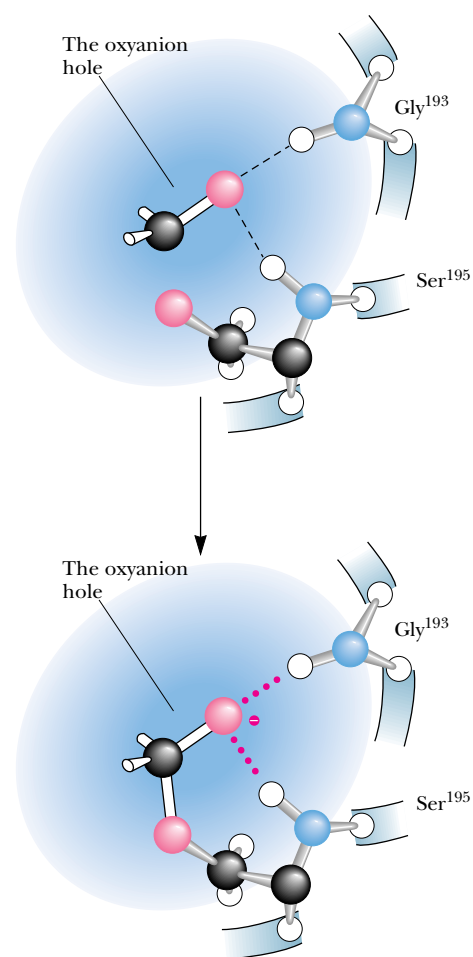
CRITICAL DEVELOPMENTS IN BIOCHEMISTRY

Transition-State Stabilization in the Serine Proteases

X-ray crystallographic studies of serine protease complexes with transition-state analogs have shown how chymotrypsin stabilizes the **tetrahedral oxyanion transition states** (structures (c) and (g) in Figure 16.24) of the protease reaction. The amide nitrogens of Ser¹⁹⁵ and Gly¹⁹³ form an “oxyanion hole” in which the substrate carbonyl oxygen is hydrogen-bonded to the amide N-H groups.

Formation of the tetrahedral transition state increases the interaction of the carbonyl oxygen with the amide N-H groups in two ways. Conversion of the carbonyl double bond to the *longer* tetrahedral single bond brings the oxygen atom closer to the amide hydrogens. Also, the hydrogen bonds between the charged oxygen and the amide hydrogens are significantly stronger than the hydrogen bonds with the uncharged carbonyl oxygen.

Transition-state stabilization in chymotrypsin also involves the side chains of the substrate. The side chain of the departing amine product forms stronger interactions with the enzyme upon formation of the tetrahedral intermediate. When the tetrahedral intermediate breaks down (Figure 16.24d and e), steric repulsion between the product amine group and the carbonyl group of the acyl-enzyme intermediate leads to departure of the amine product.



The “oxyanion hole” of chymotrypsin stabilizes the tetrahedral oxyanion transition states of the mechanism in Figure 16.24.

16.12 • The Aspartic Proteases

Mammals, fungi, and higher plants produce a family of proteolytic enzymes known as **aspartic proteases**. These enzymes are active at acidic (or sometimes neutral) pH, and each possesses two aspartic acid residues at the active site. Aspartic proteases carry out a variety of functions (Table 16.3), including digestion (*pepsin* and *chymosin*), lysosomal protein degradation (*cathepsin D* and *E*), and regulation of blood pressure (*renin* is an aspartic protease involved in the production of *angiotensin*, a hormone that stimulates smooth muscle contraction and reduces excretion of salts and fluid). The aspartic proteases display a variety of substrate specificities, but normally they are most active in the cleavage of peptide bonds between two hydrophobic amino acid residues. The preferred substrates of pepsin, for example, contain aromatic residues on both sides of the peptide bond to be cleaved.

Most aspartic proteases are composed of 323 to 340 amino acid residues, with molecular weights near 35,000. Aspartic protease polypeptides consist of

Table 16.3

Some Representative Aspartic Proteases

Name	Source	Function
Pepsin*	Animal stomach	Digestion of dietary protein
Chymosin [†]	Animal stomach	Digestion of dietary protein
Cathepsin D	Spleen, liver, and many other animal tissues	Lysosomal digestion of proteins
Renin [‡]	Kidney	Conversion of angiotensinogen to angiotensin I; regulation of blood pressure
HIV-protease [§]	AIDS virus	Processing of AIDS virus proteins

*The second enzyme to be crystallized (by John Northrup in 1930). Even more than urease before it, pepsin study by Northrup established that enzyme activity comes from proteins.
[†]Also known as rennin, it is the major pepsinlike enzyme in gastric juice of fetal and newborn animals.
[‡]A drop in blood pressure causes release of renin from the kidneys, which converts more angiotensinogen to angiotensin.
[§]A dimer of identical monomers, homologous to pepsin.

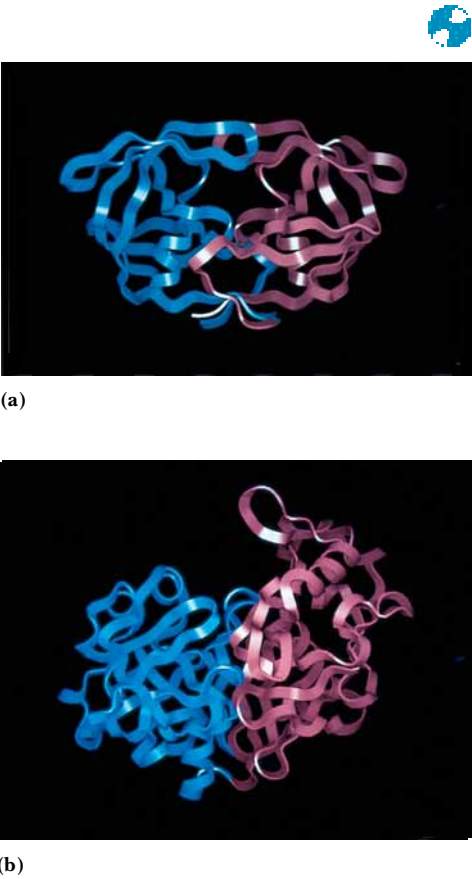


FIGURE 16.25 • Structures of (a) HIV-1 protease, a dimer, and (b) pepsin (a monomer). Pepsin’s N-terminal half is shown in red; C-terminal half is shown in blue.

two homologous domains that fold to produce a tertiary structure composed of two similar lobes, with approximate twofold symmetry (Figure 16.25). Each of these lobes or domains consists of two β -sheets and two short α -helices. The two domains are bridged and connected by a six-stranded, antiparallel β -sheet. The active site is a deep and extended cleft, formed by the two juxtaposed domains and large enough to accommodate about seven amino acid residues. The two catalytic aspartate residues, residues 32 and 215 in porcine pepsin, for example, are located deep in the center of the active site cleft. The N-terminal domain forms a “flap” that extends over the active site, which may help to immobilize the substrate in the active site.

On the basis, in part, of comparisons with chymotrypsin, trypsin, and the other serine proteases, it was hypothesized that aspartic proteases might function by formation of covalent enzyme-substrate intermediates involving the active-site aspartate residues. Two possibilities were proposed: an acyl-enzyme intermediate involving an acid anhydride bond and an amino-enzyme intermediate involving an amide (peptide) bond (Figure 16.26). All attempts to trap or isolate a covalent intermediate failed, and a mechanism (see following paragraph) favoring noncovalent enzyme-substrate intermediates and general acid–general base catalysis is now favored for aspartic proteases.

The Mechanism of Action of Aspartic Proteases

A crucial datum supporting the general acid–general base model is the pH dependence of protease activity (see Critical Developments in Biochemistry: *The pH Dependence of Aspartic Proteases and HIV-1 Protease*, page 525). Enzymologists hypothesize that the aspartate carboxyl groups function alternately as general acid and general base. This model requires that one of the aspartate carboxyls be protonated and one be deprotonated when substrate binds. X-ray diffraction data on aspartic proteases show that the active-site structure in the vicinity of the two aspartates is highly symmetric. The two aspartates appear to act as a “catalytic dyad” (analogous to the catalytic triad of the serine proteases). The dyad proton may thus be covalently bound to either of

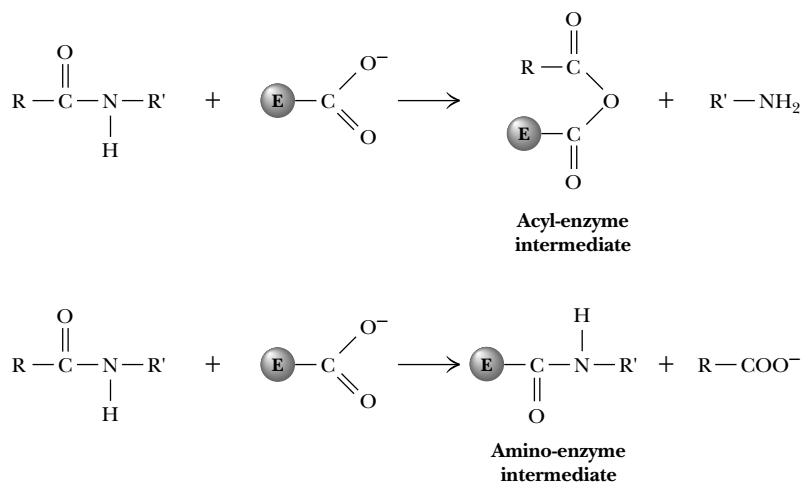


FIGURE 16.26 • Acyl-enzyme and amino-enzyme intermediates originally proposed for aspartic proteases were modeled after the acyl-enzyme intermediate of the serine proteases.

the aspartate groups in the free enzyme or in the enzyme-substrate complex. Thus, in pepsin, for example, Asp³² may be deprotonated while Asp²¹⁵ is protonated, or vice versa.

In the most widely accepted mechanism (Figure 16.27), substrate binding is followed by a step in which two concerted proton transfers facilitate nucleophilic attack on the carbonyl carbon of the substrate by water. In the mechanism shown, Asp³² acts as a general base, accepting a proton from an active-site water molecule, whereas Asp²¹⁵ acts a general acid, donating a proton to the oxygen of the peptide carbonyl group. *By virtue of these two proton transfers, nucleophilic attack occurs without explicit formation of hydroxide ion at the active site.* The resulting intermediate is termed an **amide dihydrate**. Note that the protonation states of the two aspartate residues are now opposite to those in the free enzyme (Figure 16.27).

Breakdown of the amide dihydrate occurs by a mechanism similar to its formation. The ionized aspartate carboxyl (Asp³² in Figure 16.27) acts as a general base to accept a proton from one of the hydroxyl groups of the amide dihydrate, while the protonated carboxyl of the other aspartate (Asp²¹⁵ in this case) simultaneously acts as a general acid to donate a proton to the nitrogen of one of the departing peptide products.

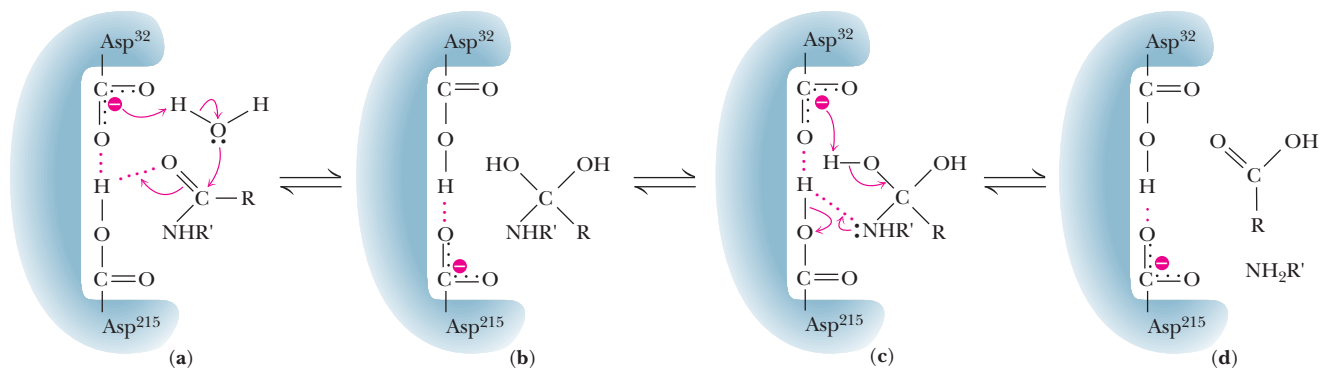


FIGURE 16.27 • A mechanism for the aspartic proteases. In the first step, two concerted proton transfers facilitate nucleophilic attack of water on the substrate carbonyl carbon. In the third step, one aspartate residue (Asp³² in pepsin) accepts a proton from one of the hydroxyl groups of the amine dihydrate, and the other aspartate (Asp²¹⁵) donates a proton to the nitrogen of the departing amine.

The AIDS Virus HIV-1 Protease Is an Aspartic Protease

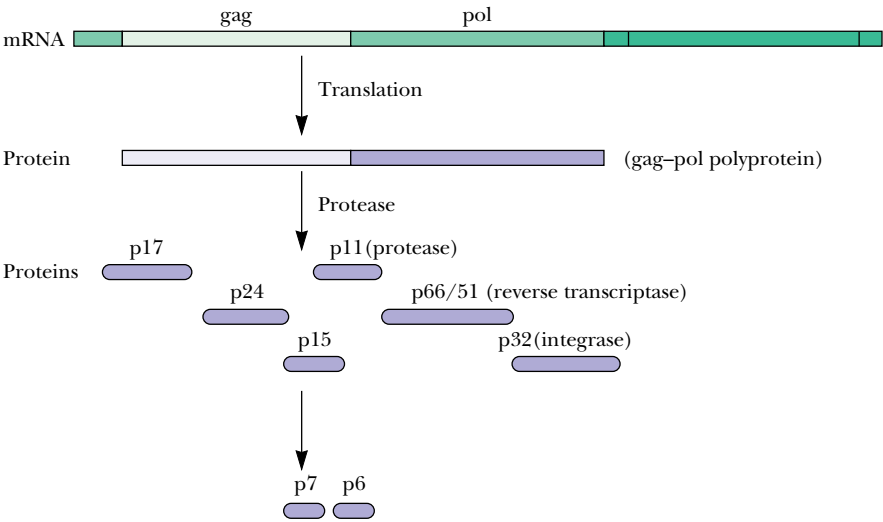
Recent research on acquired immune deficiency syndrome (AIDS) and its causative viral agent, the human immunodeficiency virus (HIV-1), has brought a new aspartic protease to light. **HIV-1 protease** cleaves the polypeptide products of the HIV-1 genome, producing several proteins necessary for viral growth and cellular infection. HIV-1 protease cleaves several different peptide linkages in the HIV-1 polyproteins, including those shown in Figure 16.28. For example, the protease cleaves between the Tyr and Pro residues of the sequence Ser-Gln-Asn-Tyr-Pro-Ile-Val, which joins the p17 and p24 HIV-1 proteins.

The HIV-1 protease is a remarkable viral imitation of mammalian aspartic proteases: It is a **dimer of identical subunits** that mimics the two-lobed monomeric structure of pepsin and other aspartic proteases. The HIV-1 protease subunits are 99-residue polypeptides that are homologous with the individual domains of the monomeric proteases. Structures determined by X-ray diffraction studies reveal that the active site of HIV-1 protease is formed at the interface of the homodimer and consists of two aspartate residues, designated Asp²⁵ and Asp^{25'}, one contributed by each subunit (Figure 16.29). In the homodimer, the active site is covered by two identical “flaps,” one from each subunit, in contrast to the monomeric aspartic proteases, which possess only a single active-site flap.

Enzyme kinetic measurements by Thomas Meek and his collaborators at SmithKline Beecham Pharmaceuticals have shown that the mechanism of HIV-1 protease is very similar to those of other aspartic proteases (Figure 16.30). Two concerted proton transfers by the aspartate carboxyls facilitate nucleophilic attack by water on the carbonyl carbon of the peptide substrate. If the protease-substrate complex is incubated with H₂¹⁸O, incorporation of ¹⁸O into the peptide carbonyl group can be measured. *Thus, not only is the formation of the amide dihydrate reversible, and the two hydroxyl groups of the dihydrate equivalent, but the protonation states of the active-site carboxyl groups must interchange* (Figure 16.30). The simplest model would involve direct proton exchange across the Asp-Asp dyad as shown. The symmetrical nature of the active-site aspartyl groups is consistent with the idea of facile exchange of the proton between the two Asp residues.

The observation that ¹⁸O can accumulate in the substrate from H₂¹⁸O also implies that formation and reversal of the amide dihydrate must be faster than

FIGURE 16.28 • HIV mRNA provides the genetic information for synthesis of a polyprotein. Proteolytic cleavage of this polyprotein by HIV protease produces the individual proteins required for viral growth and cellular infection.



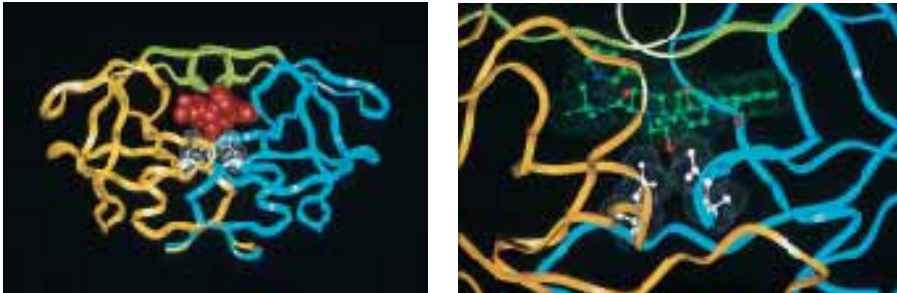


FIGURE 16.29 • (left) HIV-1 protease complexed with the inhibitor Crixivan® (red) made by Merck. The flaps (residues 46–55 from each subunit) covering the active site are shown in green and the active site aspartate residues involved in catalysis are shown in white. (right) The close-up of the active site shows the interaction of Crixivan® with the carboxyl groups of the essential aspartate residues.



its breakdown to form product. Kinetic studies by Meek and his coworkers are consistent with a model in which breakdown of the dihydrate intermediate is the rate-determining step for the protease reaction. These studies also show that the transition state of this step involves two proton transfers—with one aspartate acting as a general acid to donate a proton to the departing proline and the other aspartate acting as a general base to abstract a proton from one of the hydroxyl groups, facilitating collapse of the dihydrate to a carboxyl group. Note that the events of the breakdown to form product are simply the reversal of the steps that formed the intermediate.

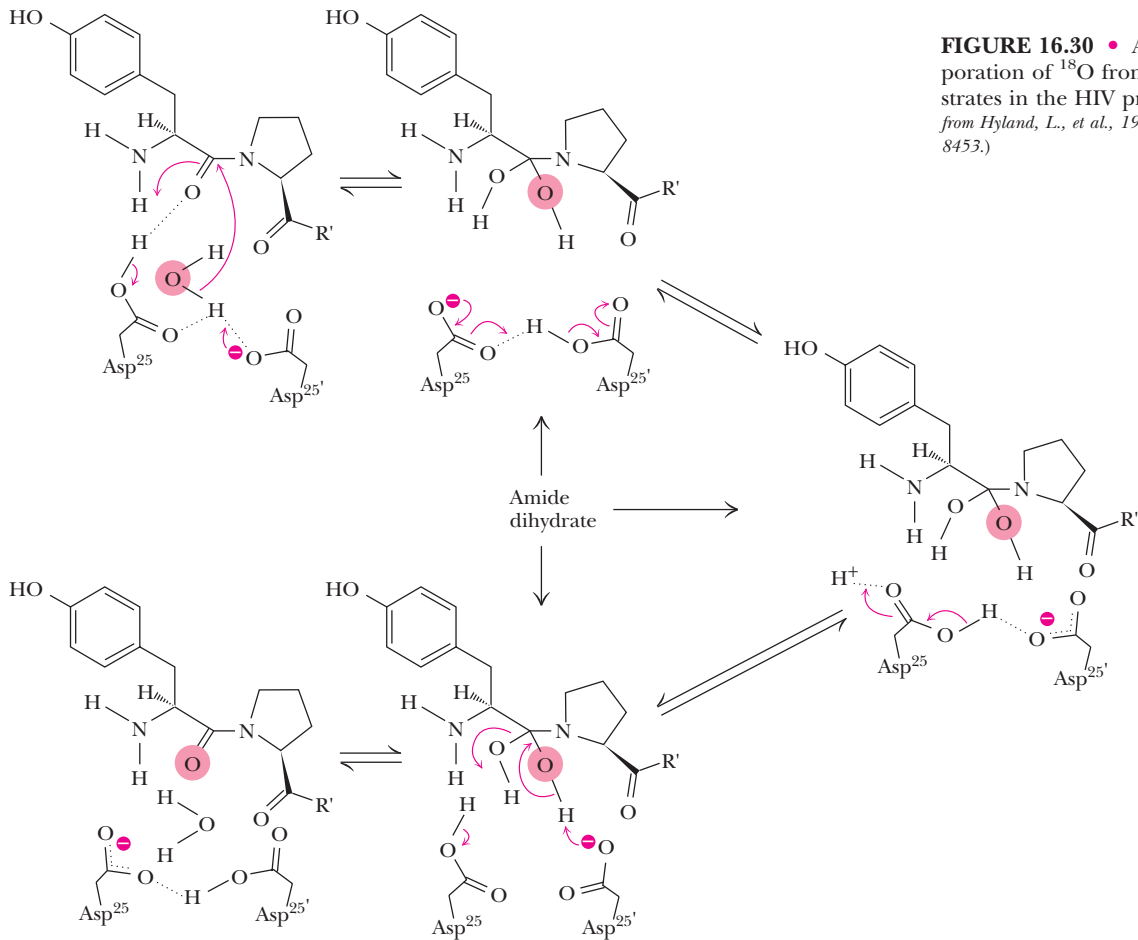


FIGURE 16.30 • A mechanism for the incorporation of ^{18}O from H_2^{18}O into peptide substrates in the HIV protease reaction. (Adapted from Hyland, L., et al., 1991. *Biochemistry* 30:8441–8453.)

HUMAN BIOCHEMISTRY

Protease Inhibitors Give Life to AIDS Patients

Infection by HIV was once considered a death sentence, but the emergence of a new family of drugs called protease inhibitors has made it possible for some AIDS patients to improve their overall health and extend their lives. These drugs are all specific inhibitors of the HIV protease. By inhibiting the protease, they prevent the development of new virus particles in the cells of infected patients. Clinical testing has shown that a combination of drugs—including a protease inhibitor together with a reverse transcriptase inhibitor like AZT—can reduce the human immunodeficiency virus (HIV) to undetectable levels in about 40 to 50% of infected individuals. Patients who respond successfully to this combination therapy have experienced dramatic improvement in their overall health and a substantially lengthened lifespan.

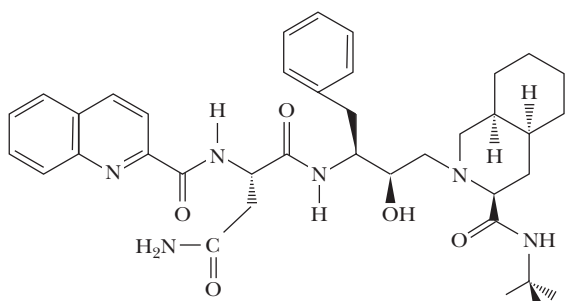
Four of the protease inhibitors approved for use in humans by the U.S. Food and Drug Administration are shown below: Crixivan[®] by Merck, Invirase[®] by Hoffman-LaRoche, Norvir[®] by Abbott, and Viracept[®] by Agouron. These drugs were all developed from a “structure-based” design strategy; that is, the drug molecules were designed to bind tightly to the active site of the HIV-1 protease. The backbone OH-group in all these substances inserts between the two active-site carboxyl groups of the protease.

In the development of an effective drug, it is not sufficient merely to show that a candidate compound can cause the desired

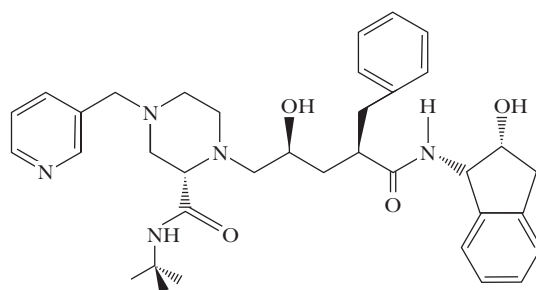
biochemical effect. It must also be demonstrated that the drug can be effectively delivered in sufficient quantities to the desired site(s) of action in the organism, and that the drug does not cause undesirable side effects. The HIV-1 protease inhibitors shown here fulfill all of these criteria. Other drug candidates have been found that are even better inhibitors of HIV-1 protease in cell cultures, but many of these fail the test of bioavailability—the ability of a drug to be delivered to the desired site of action in the organism.

Candidate protease inhibitor drugs must be relatively specific for the HIV-1 protease. Many other aspartic proteases exist in the human body and are essential to a variety of body functions, including digestion of food and processing of hormones. An ideal drug thus must strongly inhibit the HIV-1 protease, must be delivered effectively to the lymphocytes where the protease must be blocked, and should not adversely affect the activities of the essential human aspartic proteases.

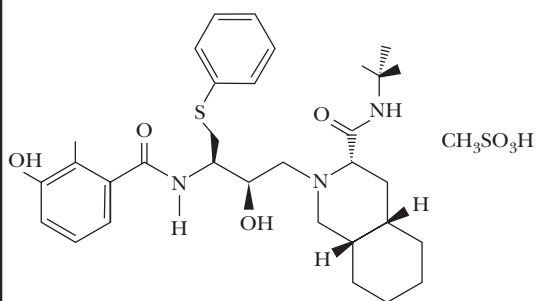
A final but important consideration is viral mutation. Certain mutant HIV strains are resistant to one or more of the protease inhibitors, and even for patients who respond initially to protease inhibitors it is possible that mutant viral forms may eventually thrive in the infected individual. The search for new and more effective protease inhibitors is ongoing.



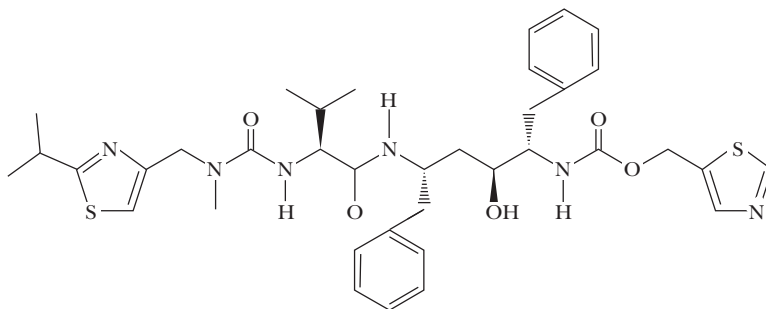
Invirase (Saquinavir)



Crixivan (Indinavir)



Viracept (Nelfinavir mesylate)



Norvir (Ritonavir)

The pH Dependence of Aspartic Proteases and HIV-1 Protease

The first hint that two active-site carboxyl groups—one protonated and one ionized—might be involved in the catalytic activity of the aspartic proteases came from studies of the pH dependence of enzymatic activity. If an ionizable group in an enzyme active site is essential for activity, a plot of enzyme activity versus pH may look like one of the plots at right.

If activity increases dramatically as pH is increased, catalysis may depend on a deprotonated group that may normally act as a general base, accepting a proton from the substrate or a water molecule, for example (a). Protonation of this group at lower pH prevents it from accepting another proton (from the substrate or water, for example).

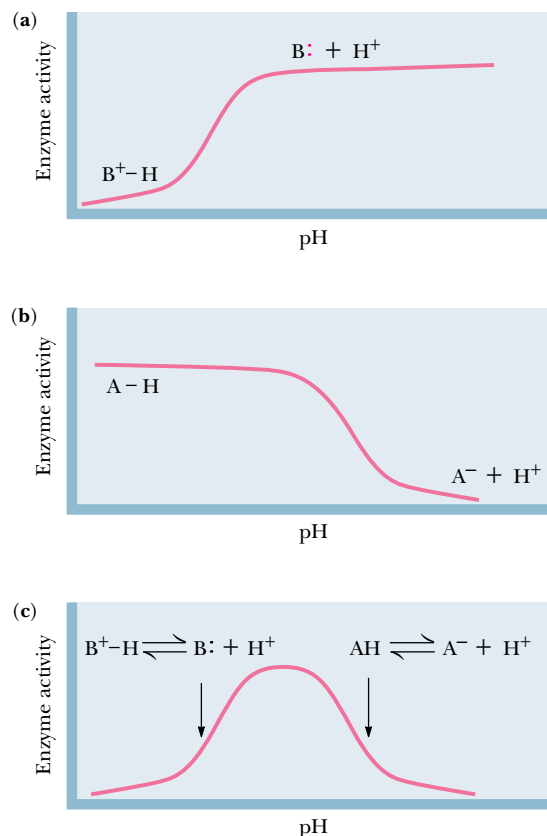
On the other hand, if activity decreases sharply as pH is raised, activity may depend on a protonated group, which may act as a general acid, donating a proton to the substrate or a catalytic water molecule (b). At high pH, the proton dissociates and is not available in the catalytic events.

The bell-shaped curve in part (c) of the figure combines both of these behaviors, so that activity first increases, then decreases, as pH is increased. This is consistent with the involvement of two ionizable groups—one with a low pK_a that acts as a base above its pK_a , and a second group with a higher pK_a that acts as an acid below its pK_a .

Kinetic studies with pepsin have produced bell-shaped curves for a variety of substrate peptides; see below, (a).

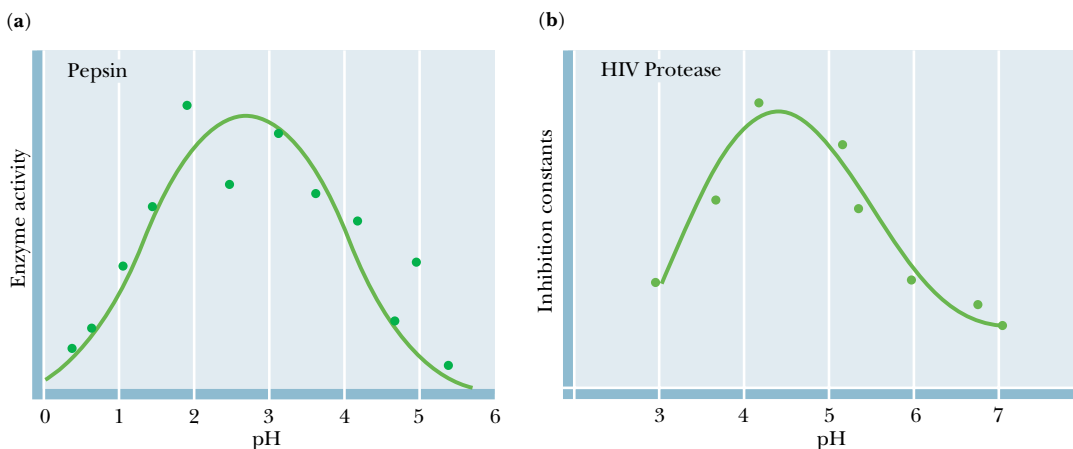
As such data are fitted to calculated curves, dissociation constants can be estimated for the ionizable groups at the active site. In pepsin, the general base exhibits a pK_a of approximately 1.4, whereas the general acid displays a pK_a of about 4.3. Compare these values with the expected pK_a for an aspartate carboxyl group in a protein of 4.2 to 4.6. The value for the aspartyl group acting as a general acid in pepsin is typical, but that for the aspartyl group acting as a general base is unusually low.

The pH dependence of HIV-1 protease has been assessed by measuring the apparent inhibition constant for a synthetic substrate analog (b). The data are consistent with the catalytic involvement of ionizable groups with pK_a values of 3.3 and 5.3. Maximal enzymatic activity occurs in the pH range between these two values. On the basis of the accumulated kinetic and structural data on HIV-1 protease, these pK_a values have been ascribed to the



Bell-shaped activity versus pH profiles arise from two separate active-site ionizations. (a) Enzyme activity increases upon deprotonation of $B^+ - H$. (b) Enzyme activity decreases upon deprotonation of $A - H$. (c) Enzyme activity is maximal in the pH range where one ionizable group is deprotonated (as $B:$) and the other group is protonated (as $A - H$).

two active-site aspartate carboxyl groups. Note that the value of 3.3 is somewhat low for an aspartate side chain in a protein, whereas the value of 5.3 is somewhat high.



pH-rate profiles for (a) pepsin and (b) HIV protease. (Adapted from Denburg, J., et al., 1968. *Journal of the American Chemical Society* 90:479–486, and Hyland, J., et al., 1991. *Biochemistry* 30:8454–8463.)

16.13 • Lysozyme

Lysozyme is an enzyme that hydrolyzes polysaccharide chains. It ruptures certain bacterial cells by cleaving the polysaccharide chains that make up their cell wall. Lysozyme is found in many body fluids, but the most thoroughly studied form is from hen egg whites. The Russian scientist P. Laschtchenko first described the bacteriolytic properties of hen egg white lysozyme in 1909. In 1922, Alexander Fleming, the London bacteriologist who later discovered penicillin, gave the name *lysozyme* to the agent in mucus and tears that destroyed certain bacteria, because it was an *enzyme* that caused bacterial *lysis*.

As seen in Chapter 9, bacterial cells are surrounded by a rigid, strong wall of peptidoglycan, a copolymer of two sugar units, *N*-acetylmuramic acid (NAM) and *N*-acetylglucosamine (NAG). Both of these sugars are *N*-acetylated analogs of glucosamine, and in bacterial cell wall polysaccharides, they are joined in $\beta(1 \rightarrow 4)$ glycosidic linkages (Figure 16.31). Lysozyme hydrolyzes the glycosidic bond between C-1 of NAM and C-4 of NAG, as shown in Figure 16.31, but does not act on the $\beta(1 \rightarrow 4)$ linkages between NAG and NAM.

Lysozyme is a small globular protein composed of 129 amino acids (14 kD) in a single polypeptide chain. It has eight cysteine residues linked in four disulfide bonds. The structure of this very stable protein was determined by X-ray crystallographic methods in 1965 by David Phillips (Figure 16.32). Although X-ray structures had previously been reported for proteins (hemoglobin and myoglobin), lysozyme was the first enzyme structure to be solved by crystallographic (or any other) methods. Although the location of the active site was not obvious from the X-ray structure of the protein alone, X-ray studies of lysozyme-inhibitor complexes soon revealed the location and nature of the active site. Since it is an enzyme, lysozyme cannot form stable ES complexes for structural studies, because the substrate is rapidly transformed into products. On the other hand, several substrate analogs have proven to be good com-

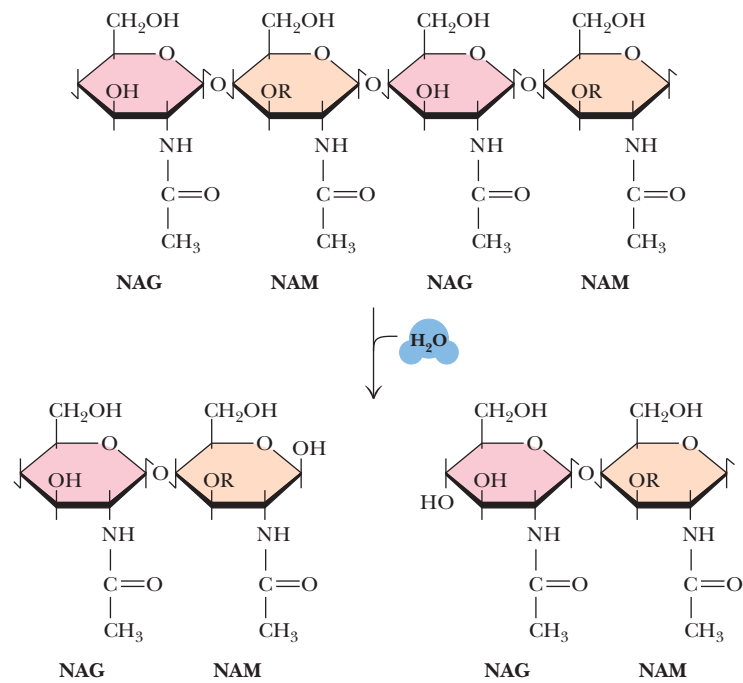


FIGURE 16.31 • The lysozyme reaction.

petitive inhibitors of lysozyme that can form complexes with the enzyme stable enough to be characterized by X-ray crystallography and other physical techniques. One of the best is a trimer of *N*-acetylglucosamine, (NAG)₃ (Figure 16.33), which is hydrolyzed by lysozyme at a rate only 1/60,000 that of the native substrate (Table 16.4). (NAG)₃ binds at the enzyme active site by forming five hydrogen bonds with residues located in one-half of a depression or crevice that spans the surface of the enzyme (Figure 16.34). The few hydrophobic residues that exist on the surface of lysozyme are located in this depression, and they may participate in hydrophobic and van der Waals interactions with (NAG)₃, as well as the normal substrate. The absence of charged groups on (NAG)₃ precludes the involvement of electrostatic interactions with the enzyme. Comparisons of the X-ray structures of the native lysozyme and the lysozyme-(NAG)₃ complex reveal that several amino acid residues at the active site move slightly upon inhibitor binding, including Trp⁶², which moves about 0.75 Å to form a hydrogen bond with a hydroxymethyl group (Figure 16.35).

Model Studies Reveal a Strain-Induced Destabilization of a Bound Substrate on Lysozyme

One of the premises of lysozyme models is that the native substrate would occupy the rest of the crevice or depression running across the surface of the enzyme, because there is room to fit three more sugar residues into the crevice and because the hexamer (NAG)₆ is in fact a good substrate for lysozyme (Table 16.4). The model building studies refer to the six sugar residue-binding subsites in the crevice with the letters A through F, with A, B, and C representing the part of the crevice occupied by the (NAG)₃ inhibitor (Figure 16.35). Modeling studies clearly show that NAG residues fit nicely into subsites A, B, C, E, and F of the crevice, but that fitting a residue of the (NAG)₆ hexamer into site D requires a substantial distortion of the sugar (out of its preferred chair conformation) to prevent steric crowding and overlap between atoms C-6 and O-6 of the sugar at the D site and Ile⁹⁸ of the enzyme. This distorted sugar residue is adjacent to the glycosidic bond to be cleaved (between sites D and E), and the inference is made that this distortion or strain brings the substrate closer to the transition state for hydrolysis. This is a good example of strain-induced destabilization of an otherwise favorably binding substrate (Section 16.4). Thus, the overall binding interaction of the rest of the sugar substrate would be favorable ($\Delta G < 0$), but distortion of the ring at the D site uses some of this binding energy to raise the substrate closer to the transition state for hydrolysis, *an example of stabilization of a transition state (relative to the simple enzyme-substrate complex)*. As noted in Section 16.4, distortion is one of the molecular mechanisms that can lead to such transition-state stabilization.

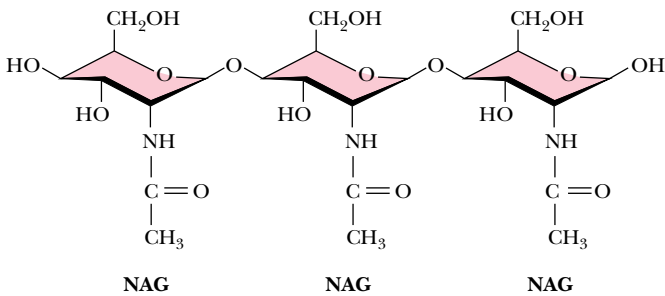


FIGURE 16.33 • (NAG)₃, a substrate analog, forms stable complexes with lysozyme.

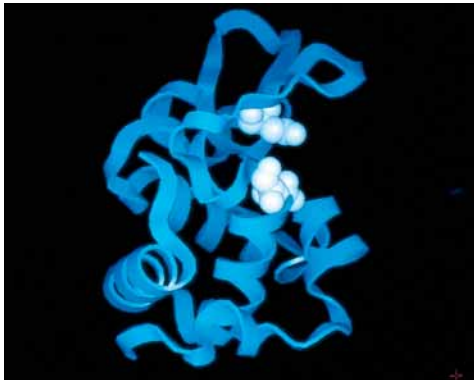


FIGURE 16.32 • The structure of lysozyme. Glu³⁵ and Asp⁵² are shown in white.

Table 16.4
Hydrolysis Rate Constants for Model Oligosaccharides with Lysozyme

Oligosaccharide	Rate Constant, $k_{cat}(s^{-1})$
(NAG-NAM) ₃	0.5
(NAG) ₆	0.25
(NAG) ₅	0.033
(NAG) ₄	7×10^{-5}
(NAG) ₃	8×10^{-6}
(NAG) ₂	2.5×10^{-8}

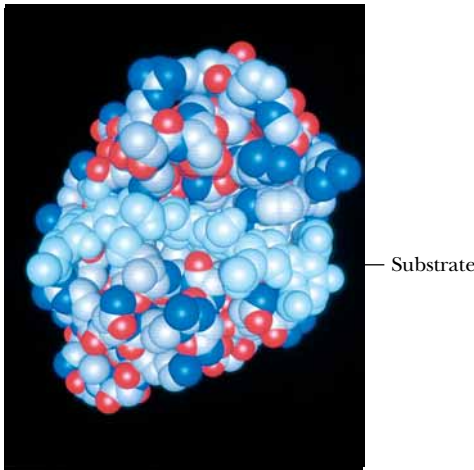
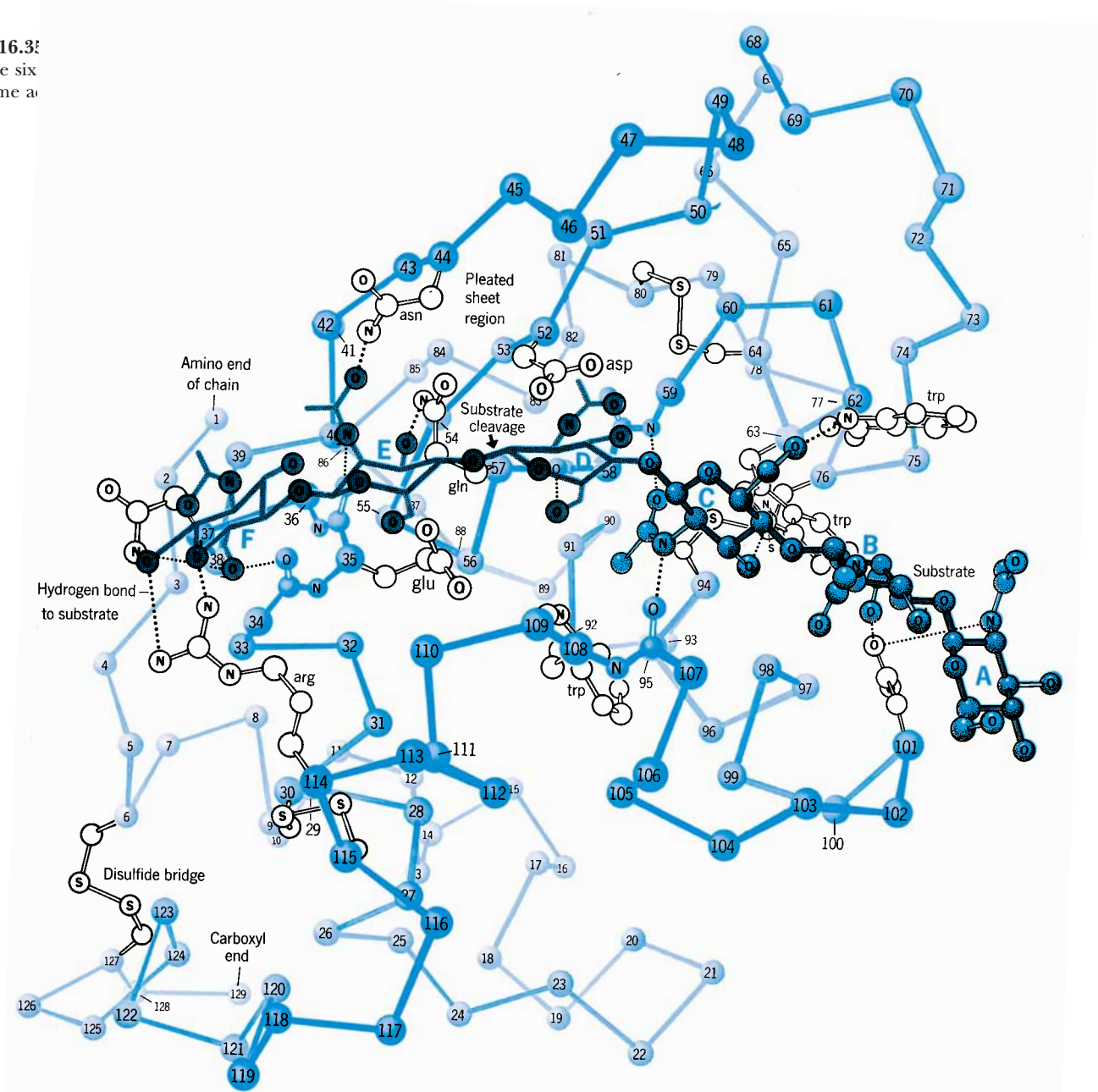
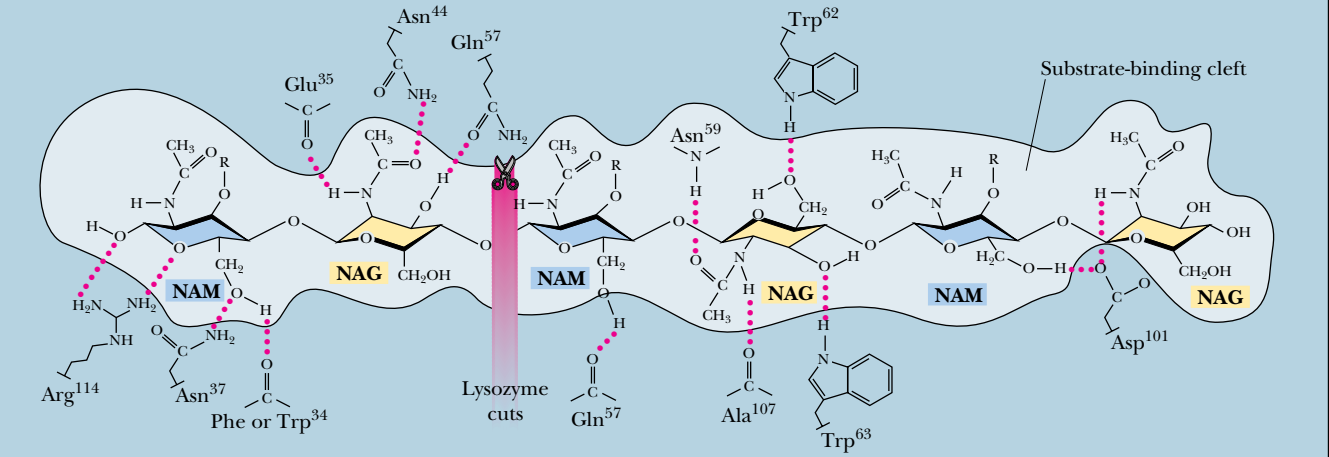


FIGURE 16.34 • The lysozyme-enzyme-substrate complex. (Photo courtesy of John Rupley, University of Arizona)

FIGURE 16.3
tions at the six
the lysozyme a



Lysozyme



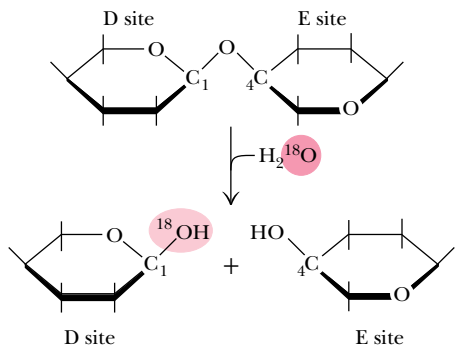


FIGURE 16.36 • The C₁–O bond, not the O–C₄ bond, is cleaved in the lysozyme reaction. ¹⁸O from H₂¹⁸O is thus incorporated at the C₁ position.

The Lysozyme Mechanism Involves General Acid–Base Catalysis

The mechanism of the lysozyme reaction is shown in Figures 16.36 and 16.37. Studies using ¹⁸O-enriched water showed that the C₁–O bond is cleaved on the substrate between the D and E sites. Hydrolysis under these conditions incorporates ¹⁸O into the C₁ position of the sugar at the D site, not into the oxygen at C₄ at the E site (Figure 16.36). Model building studies place the cleaved bond approximately between protein residues Glu³⁵ and Asp⁵². Glu³⁵ is in a nonpolar or hydrophobic region of the protein, whereas Asp⁵² is located in a much more polar environment. Glu³⁵ is protonated, but Asp⁵² is ionized (Figure 16.37). Glu³⁵ may thus act as a general acid, donating a proton to the oxygen atom of the glycosidic bond and accelerating the reaction. Asp⁵², on the other hand, probably stabilizes the carbonium ion generated at the D site upon bond cleavage. Formation of the carbonium ion may also be enhanced by the strain on the ring at the D site. Following bond cleavage, the product formed at the E site diffuses away, and the carbonium ion intermediate can then react with H₂O from the solution. Glu³⁵ can now act as a general base, accepting a proton from the attacking water. The tetramer of NAG thus formed at sites A through D can now be dissociated from the enzyme.

On the basis of the above, the rate acceleration afforded by lysozyme appears to be due to (a) general acid catalysis by Glu³⁵; (b) distortion of the sugar ring at the D site, which may stabilize the carbonium ion (*and the transition state*); and (c) electrostatic stabilization of the carbonium ion by nearby Asp⁵². The overall *k*_{cat} for lysozyme is about 0.5/sec, which is quite slow (Table

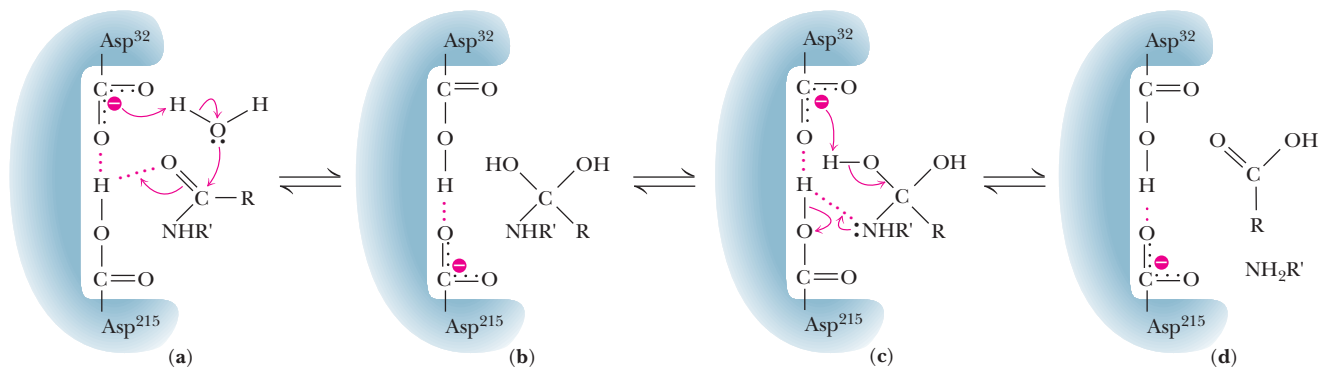
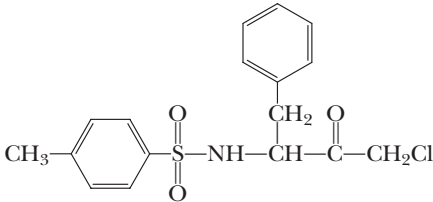


FIGURE 16.37 • A mechanism for the lysozyme reaction.

14.4) compared with that for other enzymes. On the other hand, the destruction of a bacterial cell wall may only require hydrolysis of a few polysaccharide chains. The high osmotic pressure of the cell ensures that cell rupture will follow rapidly. Thus, lysozyme can accomplish cell lysis without a particularly high k_{cat} .

PROBLEMS

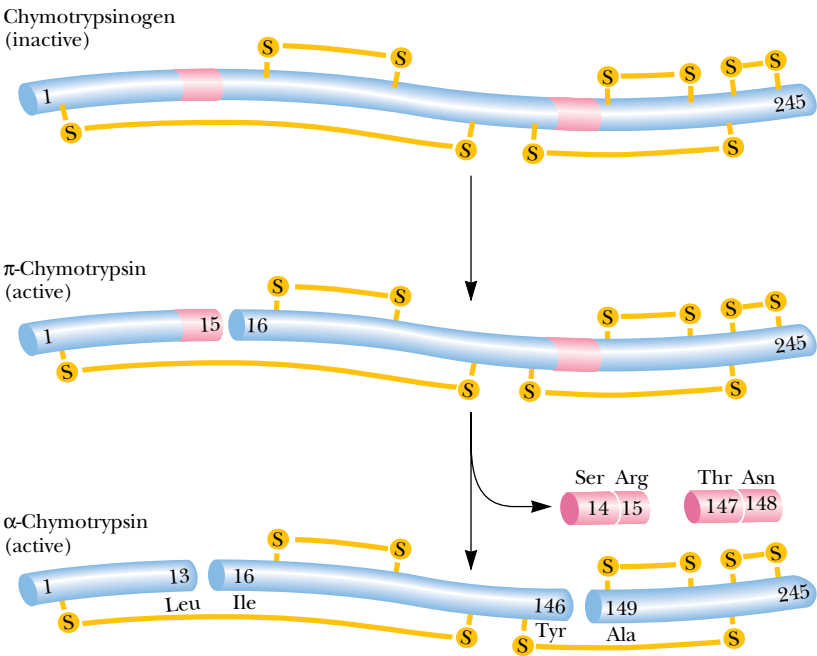
1. Tosyl-L-phenylalanine chloromethyl ketone (TPCK) specifically inhibits chymotrypsin by covalently labeling His⁵⁷.



Tosyl-L-phenylalanine chloromethyl ketone (TPCK)

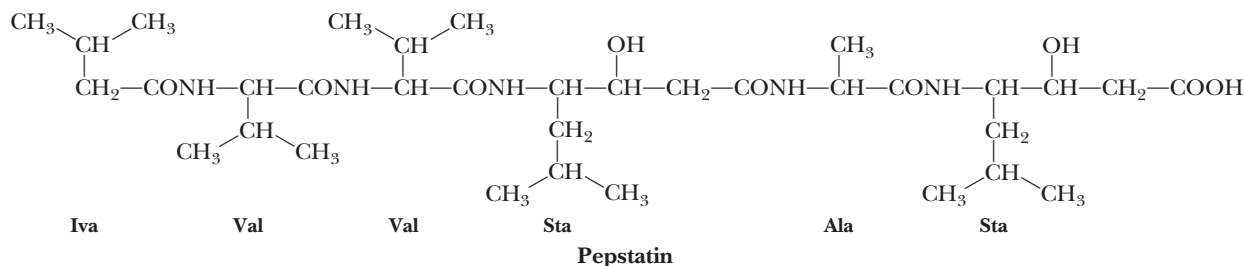
- a. Propose a mechanism for the inactivation reaction, indicating the structure of the product(s).
b. State why this inhibitor is specific for chymotrypsin.
c. Propose a reagent based on the structure of TPCK that might be an effective inhibitor of trypsin.
2. In this chapter, the experiment in which Craik and Rutter replaced Asp¹⁰² with Asn in trypsin (reducing activity 10,000-fold) was discussed.

- a. On the basis of your knowledge of the catalytic triad structure in trypsin, suggest a structure for the “uncatalytic triad” of Asn-His-Ser in this mutant enzyme.
b. Explain why the structure you have proposed explains the reduced activity of the mutant trypsin.
c. See the original journal articles (Sprang, et al., 1987. *Science* **237**:905–909 and Craik, et al., 1987. *Science* **237**:909–913) to see what Craik and Rutter’s answer to this question was.
3. Pepstatin (see page 531) is an extremely potent inhibitor of the monomeric aspartic proteases, with K_i values of less than 1 nM.
- a. On the basis of the structure of pepstatin, suggest an explanation for the strongly inhibitory properties of this peptide.
b. Would pepstatin be expected to also inhibit the HIV-1 protease? Explain your answer.
4. The k_{cat} for alkaline phosphatase-catalyzed hydrolysis of methylphosphate is approximately 14/sec at pH 8 and 25°C. The rate constant for the uncatalyzed hydrolysis of methylphosphate under the same conditions is approximately 1×10^{-15} /sec. What is the difference in the free energies of activation of these two reactions?



5. Active α -chymotrypsin is produced from chymotrypsinogen, an inactive precursor, as shown in the color figure on page 530.

The first intermediate— π -chymotrypsin—displays chymotrypsin activity. Suggest proteolytic enzymes that might carry out these cleavage reactions effectively.



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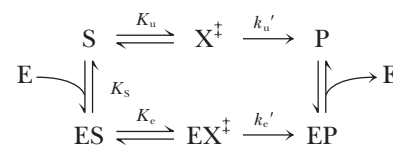
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6. Based on the reaction scheme shown below, derive an expression for k_e/k_u , the ratio of the rate constants for the catalyzed and uncatalyzed reactions, respectively, in terms of the free energies of activation for the catalyzed ($\Delta G_e^\ddagger$) and the uncatalyzed ($\Delta G_u^\ddagger$) reactions.



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Chapter 17

Molecular Motors



Michaelangelo's "David" epitomizes the musculature of the human form. (The Firenze Academia/photo by Stephanie Colasanti/Corbis)

Movement is an intrinsic property associated with all living things. Within cells, molecules undergo coordinated and organized movements, and cells themselves may move across a surface. At the tissue level, **muscle contraction** allows higher organisms to carry out and control crucial internal functions, such as peristalsis in the gut and the beating of the heart. Muscle contraction also enables the organism to carry out organized and sophisticated movements, such as walking, running, flying, and swimming.

17.1 • Molecular Motors

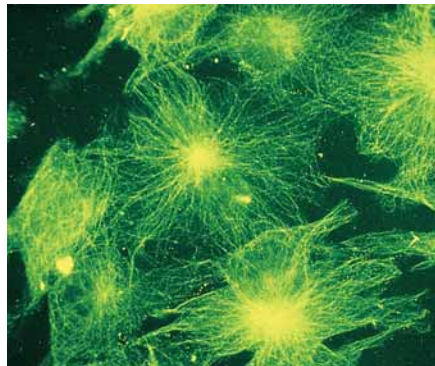
Motor proteins, also known as **molecular motors**, use chemical energy (ATP) to orchestrate all these movements, transforming ATP energy into the mechanical energy of motion. In all cases, ATP hydrolysis is presumed to drive and

Under the spreading chestnut tree
The village smithy stands;
The smith a mighty man is he
With large and sinewy hands.
And the muscles of his brawny arms
Are strong as iron bands.

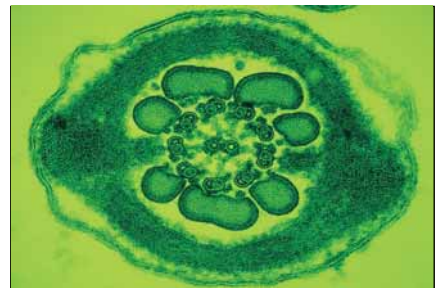
HENRY WADSWORTH LONGFELLOW, "The Village Blacksmith"

OUTLINE

- 17.1 • Molecular Motors
- 17.2 • Microtubules and Their Motors
- 17.3 • Skeletal Muscle Myosin and Muscle Contraction
- 17.4 • A Proton Gradient Drives the Rotation of Bacterial Flagella



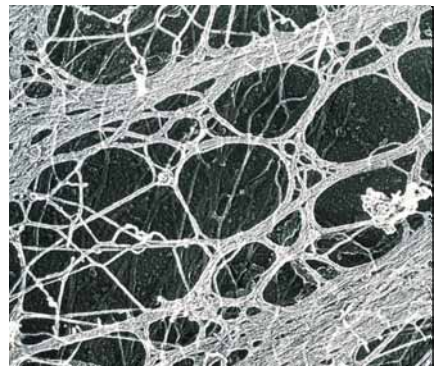
(a)



(b)



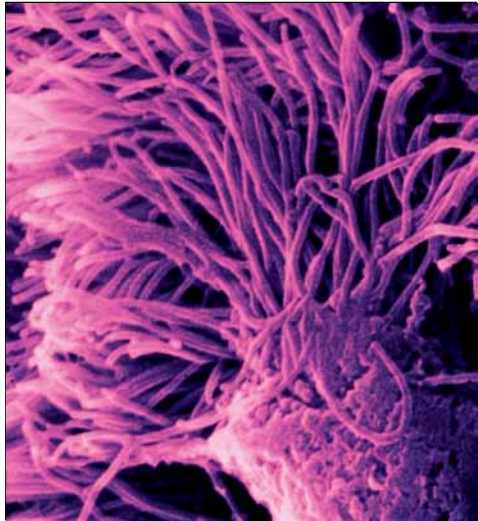
(c)



(d)



(e)



(f)

control protein conformational changes that result in sliding or walking movements of one molecule relative to another. To carry out directed movements, molecular motors must be able to associate and dissociate reversibly with a polymeric protein array, a surface or substructure in the cell. ATP hydrolysis drives the process by which the motor protein ratchets along the protein array or surface. As fundamental and straightforward as all this sounds, elucidation of these basically simple processes has been extremely challenging for biochemists, involving the application of many sophisticated chemical and physical methods in many different laboratories. This chapter describes the structures and chemical functions of molecular motor proteins and some of the experiments by which we have come to understand them.

17.2 • Microtubules and Their Motors

One of the simplest self-assembling structures found in biological systems is the *microtubule*, one of the fundamental components of the eukaryotic *cytoskeleton* and the primary structural element of cilia and flagella (Figure 17.1). **Microtubules** are hollow, cylindrical structures, approximately 30 nm in diameter, formed from **tubulin**, a dimeric protein composed of two similar 55-kD subunits known as α -tubulin and β -tubulin. Eva Nogales, Sharon Wolf, and Kenneth Downing have determined the structure of the bovine tubulin $\alpha\beta$ dimer to 3.7 Å resolution (Figure 17.2a). Tubulin dimers polymerize as shown in Figure 17.2b to form microtubules, which are essentially helical structures, with 13 tubulin monomer “residues” per turn. Microtubules grown *in vitro* are dynamic structures that are constantly being assembled and disassembled.

FIGURE 17.1 • Micrographs and electron micrographs of cytoskeletal elements, cilia, and flagella: (a) microtubules, (b) rat sperm tail microtubules (cross-section), (c) *Stylonychia*, a ciliated protozoan (undergoing division), (d) cytoskeleton of a eukaryotic cell, (e) *Pseudomonas fluorescens* (aerobic soil bacterium), showing flagella, (f) nasal cilia. (a, K. G. Murti/Visuals Unlimited; b, David Phillips/Visuals Unlimited; c, Eric Grave/Phototake; d, Fawcett and Heuser/Photo Researchers, Inc.; e, Dr. Tony Brain/Custom Medical Stock; f, Veronika Burmeister, Visuals Unlimited)

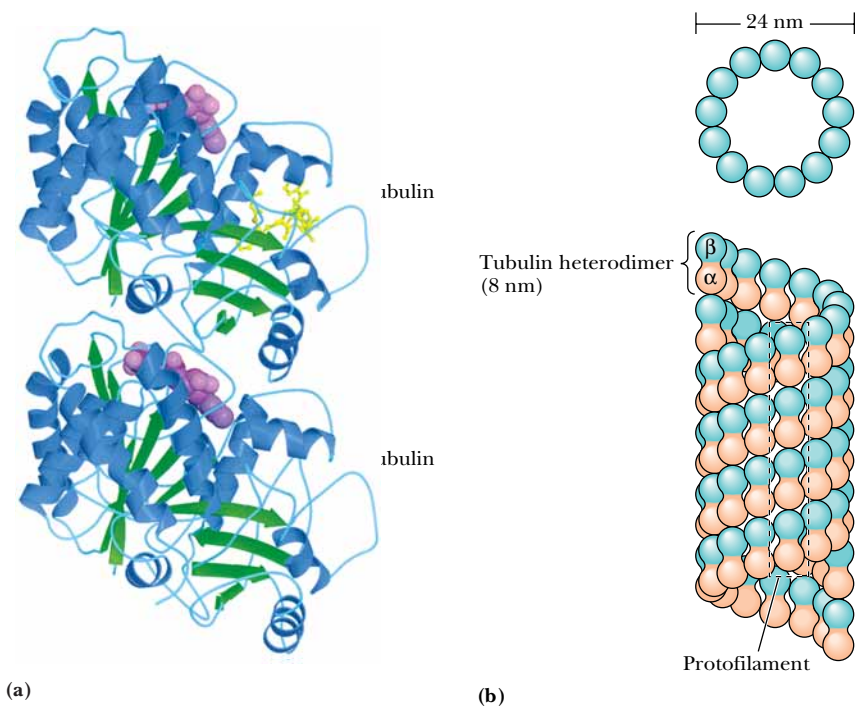


FIGURE 17.2 • (a) The structure of the tubulin $\alpha\beta$ heterodimer. (b) Microtubules may be viewed as consisting of 13 parallel, staggered protofilaments of alternating α -tubulin and β -tubulin subunits. The sequences of the α and β subunits of tubulin are homologous, and the $\alpha\beta$ tubulin dimers are quite stable if Ca^{2+} is present. The dimer is dissociated only by strong denaturing agents.

Because all tubulin dimers in a microtubule are oriented similarly, microtubules are polar structures. The end of the microtubule at which growth occurs is the **plus end**, and the other is the **minus end**. Microtubules *in vitro* carry out a GTP-dependent process called **treadmilling**, in which tubulin dimers are added to the plus end at about the same rate at which dimers are removed from the minus end (Figure 17.3).

Microtubules Are Constituents of the Cytoskeleton

Although composed only of 55-kD tubulin subunits, microtubules can grow sufficiently large to span a eukaryotic cell or to form large structures such as cilia and flagella. Inside cells, networks of microtubules play many functions, including formation of the mitotic spindle that segregates chromosomes during cell division, the movement of organelles and various vesicular structures through the cell, and the variation and maintenance of cell shape. Microtubules are, in fact, a significant part of the **cytoskeleton**, a sort of intracellular scaffold formed of microtubules, *intermediate filaments*, and *microfilaments* (Figure 17.4). In most cells, microtubules are oriented with their minus ends toward the centrosome and their plus ends toward the cell periphery. This consistent orientation is important for mechanisms of intracellular transport.

Microtubules Are the Fundamental Structural Units of Cilia and Flagella

As already noted, microtubules are also the fundamental building blocks of cilia and flagella. **Cilia** are short, cylindrical, hairlike projections on the surfaces of the cells of many animals and lower plants. The beating motion of cilia functions either to move cells from place to place or to facilitate the movement of extracellular fluid over the cell surface. Flagella are much longer structures found singly or a few at a time on certain cells (such as sperm cells). They pro-

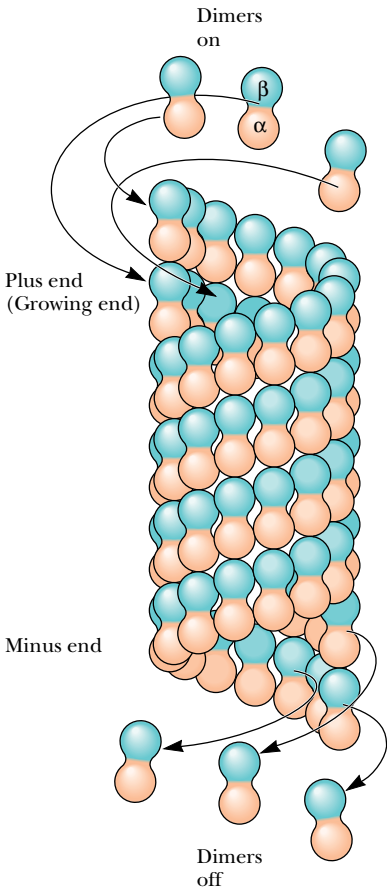


FIGURE 17.3 • A model of the GTP-dependent treadmilling process. Both α - and β -tubulin possess two different binding sites for GTP. The polymerization of tubulin to form microtubules is driven by GTP hydrolysis in a process that is only beginning to be understood in detail.

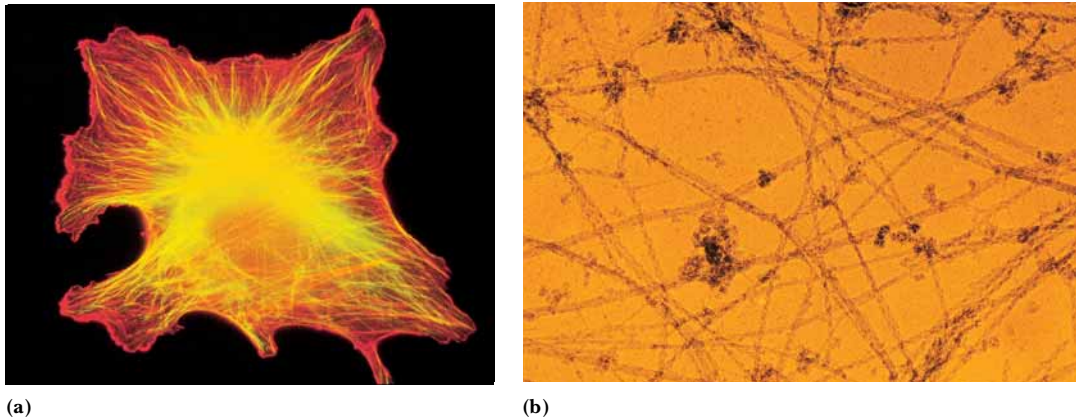


FIGURE 17.4 • Intermediate filaments have diameters of approximately 7 to 12 nm, whereas microfilaments, which are made from actin, have diameters of approximately 7 nm. The intermediate filaments appear to play only a structural role (maintaining cell shape), but the microfilaments and microtubules play more dynamic roles. Microfilaments are involved in cell motility, whereas microtubules act as long filamentous tracks, along which cellular components may be rapidly transported by specific mechanisms. (a) Cytoskeleton, double-labeled with actin in red and tubulin in green. (b) Cytoskeletal elements in a eukaryotic cell, including microtubules (thickest strands), intermediate filaments, and actin microfilaments (smallest strands). (a, b, M. Schliwa/Visuals Unlimited)

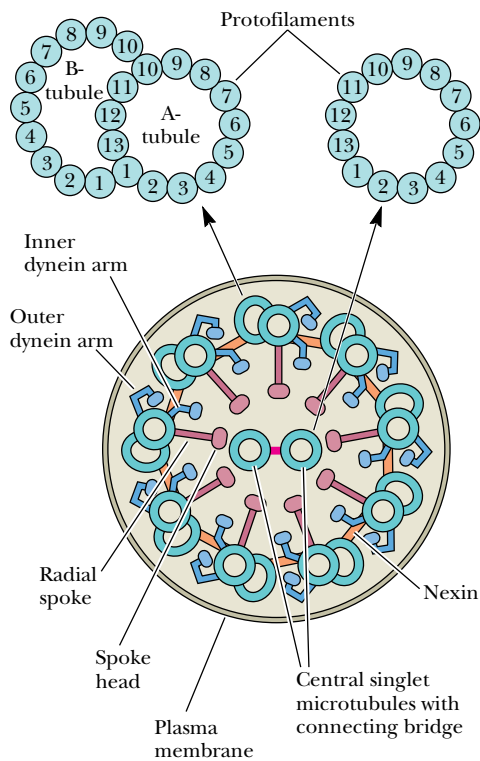


FIGURE 17.5 • The structure of an axoneme. Note the manner in which two microtubules are joined in the nine outer pairs. The smaller-diameter tubule of each pair, which is a true cylinder, is called the A-tubule and is joined to the center sheath of the axoneme by a spoke structure. Each outer pair of tubules is joined to adjacent pairs by a nexin bridge. The A-tubule of each outer pair possesses an outer dynein arm and an inner dynein arm. The larger-diameter tubule is known as the B-tubule.

pel cells through fluids. Cilia and flagella share a common design (Figure 17.5). The **axoneme** is a complex bundle of microtubule fibers that includes two central, separated microtubules surrounded by nine pairs of joined microtubules. The axoneme is surrounded by a plasma membrane that is continuous with the plasma membrane of the cell. Removal of the plasma membrane by detergent and subsequent treatment of the exposed axonemes with high concentrations of salt releases the **dynein** molecules (Figure 17.6), which form the **dynein arms**.

The Mechanism of Ciliary Motion

The motion of cilia results from the ATP-driven sliding or walking of dyneins along one microtubule while they remain firmly attached to an adjacent microtubule. The flexible stems of the dyneins remain permanently attached to A-tubules (Figure 17.6). However, the projections on the globular heads form transient attachments to adjacent B-tubules. Binding of ATP to the dynein heavy chain causes dissociation of the projections from the B-tubules. These projections then reattach to the B-tubules at a position closer to the minus end. Repetition of this process causes the sliding of A-tubules relative to B-tubules. The cross-linked structure of the axoneme dictates that this sliding motion will occur in an asymmetric fashion, resulting in a bending motion of the axoneme, as shown in Figure 17.7

Microtubules Also Mediate Intracellular Motion of Organelles and Vesicles

The ability of dyneins to effect **mechano-chemical coupling**—i.e., motion coupled with a chemical reaction—is also vitally important *inside* eukaryotic cells, which, as already noted, contain microtubule networks as part of the cytoskeleton. The mechanisms of intracellular, microtubule-based transport of organelles and vesicles were first elucidated in studies of **axons**, the long pro-

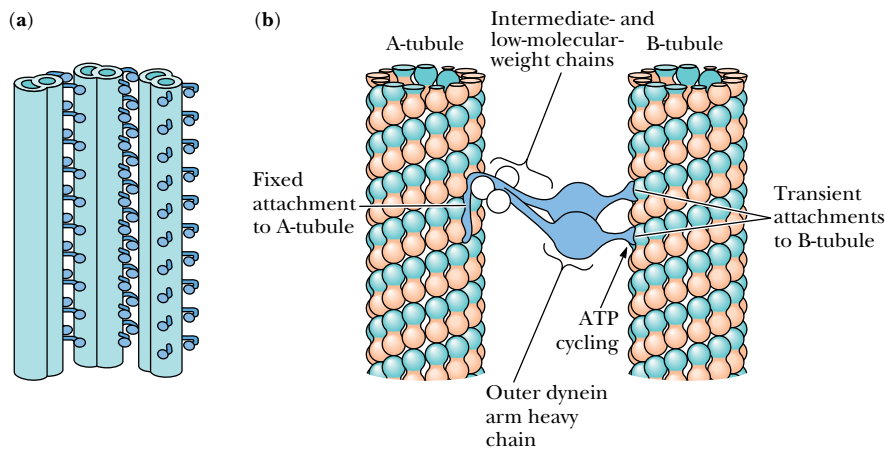


FIGURE 17.6 • (a) Diagram showing dynein interactions between adjacent microtubule pairs. (b) Detailed views of dynein crosslinks between the A-tubule of one microtubule pair and the B-tubule of a neighboring pair. (The B-tubule of the first pair and the A-tubule of the neighboring pair are omitted for clarity.) Isolated axonemal dyneins, which possess ATPase activity, consist of two or three “heavy chains” with molecular masses of 400 to 500 kD, referred to as α and β (and γ when present), as well as several chains with intermediate (40 to 120 kD) and low (15 to 25 kD) molecular masses. Each outer-arm heavy chain consists of a globular domain with a flexible stem on one end and a shorter projection extending at an angle with respect to the flexible stem. In a dynein arm, the flexible stems of several heavy chains are joined in a common base, where the intermediate- and low-molecular-weight proteins are located.

jections of neurons that extend great distances away from the body of the cell. In these cells, it was found that subcellular organelles and vesicles could travel at surprisingly fast rates—as great as 2 to 5 $\mu\text{m}/\text{sec}$ —in either direction. Unraveling the molecular mechanism for this rapid transport turned out to be a challenging biochemical problem. The early evidence that these movements occur by association with specialized proteins on the microtubules was met with some resistance, for two reasons. First, the notion that a network of microtubules could mediate transport was novel and, like all novel ideas, difficult to accept. Second, many early attempts to isolate dyneins from neural tissue were unsuccessful, and the dynein-like proteins that were first isolated from cytosolic fractions were thought to represent contaminations from axoneme structures. However, things changed dramatically in 1985 with a report by Michael Sheetz and his coworkers of a new ATP-driven, force-generating protein, different from myosin and dynein, which they called *kinesin*. Then, in 1987, Richard McIntosh and Mary Porter described the isolation of *cytosolic dynein* proteins from *Caenorhabditis elegans*, a nematode worm that never makes motile axonemes at any stage of its life cycle. Kinesins have now been found in many eukaryotic cell types, and similar cytosolic dyneins have been found in fruit flies, amoebae, and slime molds; in vertebrate brain and testes; and in HeLa cells (a unique human tumor cell line).

**Dyneins Move Organelles in a Plus-to-Minus Direction;
Kinesins, in a Minus-to-Plus Direction**

The cytosolic dyneins bear many similarities to axonemal dynein. The protein isolated from *C. elegans* includes a “heavy chain” with a molecular mass of approximately 400 kD, as well as smaller peptides with molecular mass ranging from 53 kD to 74 kD. The protein possesses a microtubule-activated ATPase

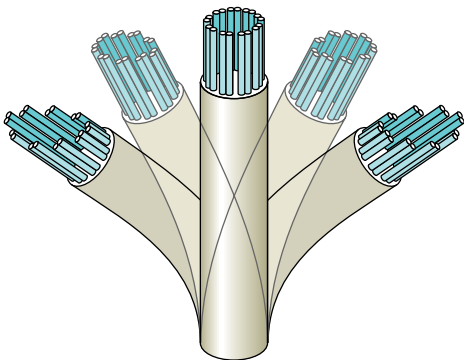


FIGURE 17.7 • A mechanism for ciliary motion. The sliding motion of dyneins along one microtubule while attached to an adjacent microtubule results in a bending motion of the axoneme.

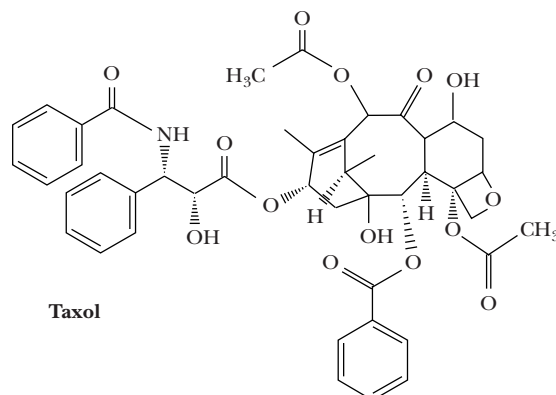
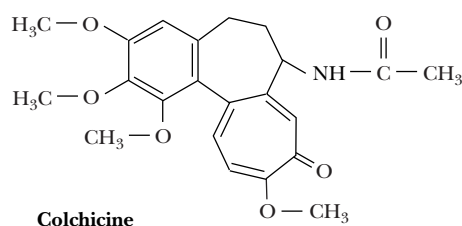
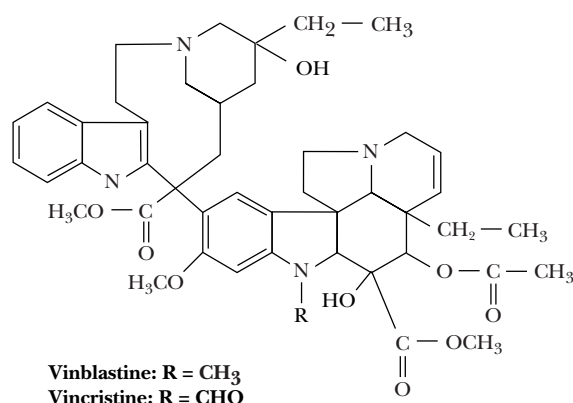
CRITICAL DEVELOPMENTS IN BIOCHEMISTRY

Effectors of Microtubule Polymerization as Therapeutic Agents

Microtubules in eukaryotic cells are important for the maintenance and modulation of cell shape and the disposition of intracellular elements during the growth cycle and mitosis. It may thus come as no surprise that the *inhibition of microtubule polymerization* can block many normal cellular processes. The alkaloid **colchicine** (see figure), a constituent of the swollen, underground stems of the autumn crocus (*Colchicum autumnale*) and meadow saffron, inhibits the polymerization of tubulin into microtubules. This effect blocks the mitotic cycle of plants and animals. Colchicine also inhibits cell motility and intracellular transport of vesicles and organelles (which in turn blocks secretory processes of cells). Colchicine has been used for hundreds of years to alleviate some of the acute pain of gout and rheumatism. In gout, white cell lysosomes surround and engulf small crystals of uric acid. The subsequent rupture of the lysosomes and the attendant lysis of the white cells initiate an inflammatory response that causes intense pain. The mechanism of pain alleviation by colchicine is not known for certain, but appears to involve inhibition of white cell movement in tissues. Interestingly, colchicine's ability to inhibit mitosis has given it an important role in the commercial development of new varieties of agricultural and ornamental plants. When mitosis is blocked by colchicine, the treated cells may be left with an extra set of chromosomes. Plants with extra sets of chromosomes are typically larger and more vigorous than normal plants. Flowers developed in this way may grow with double the normal number of petals, and fruits may produce much larger amounts of sugar.

Another class of alkaloids, the **vinca alkaloids** from *Vinca rosea*, the Madagascar periwinkle, can also bind to tubulin and inhibit microtubule polymerization. **Vinblastine** and **vincristine** are used as potent agents for cancer chemotherapy, owing to their ability to inhibit the growth of fast-growing tumor cells. For reasons that are not well understood, colchicine is not an effective chemotherapeutic agent, though it appears to act similarly to the vinca alkaloids in inhibiting tubulin polymerization.

A new antitumor drug, **taxol**, has been isolated from the bark of *Taxus brevifolia*, the Pacific yew tree. Like vinblastine and colchicine, taxol inhibits cell replication by acting on microtubules. Unlike these other antimitotic drugs, however, taxol stimulates microtubule polymerization and stabilizes microtubules. The remarkable success of taxol in treatment of breast and ovarian cancers stimulated research efforts to synthesize taxol directly and to identify new antimitotic agents that, like taxol, stimulate microtubule polymerization.



The structures of vinblastine, vincristine, colchicine, and taxol.

activity, and, when anchored to a glass surface *in vitro*, these proteins, in the presence of ATP, can bind microtubules and move them through the solution. In the cell, cytosolic dyneins specifically move organelles and vesicles from the plus end of a microtubule to the minus end. Thus, as shown in Figure 17.8, dyneins move vesicles and organelles from the cell periphery toward the centrosome (or, in an axon, from the synaptic termini toward the cell body). The

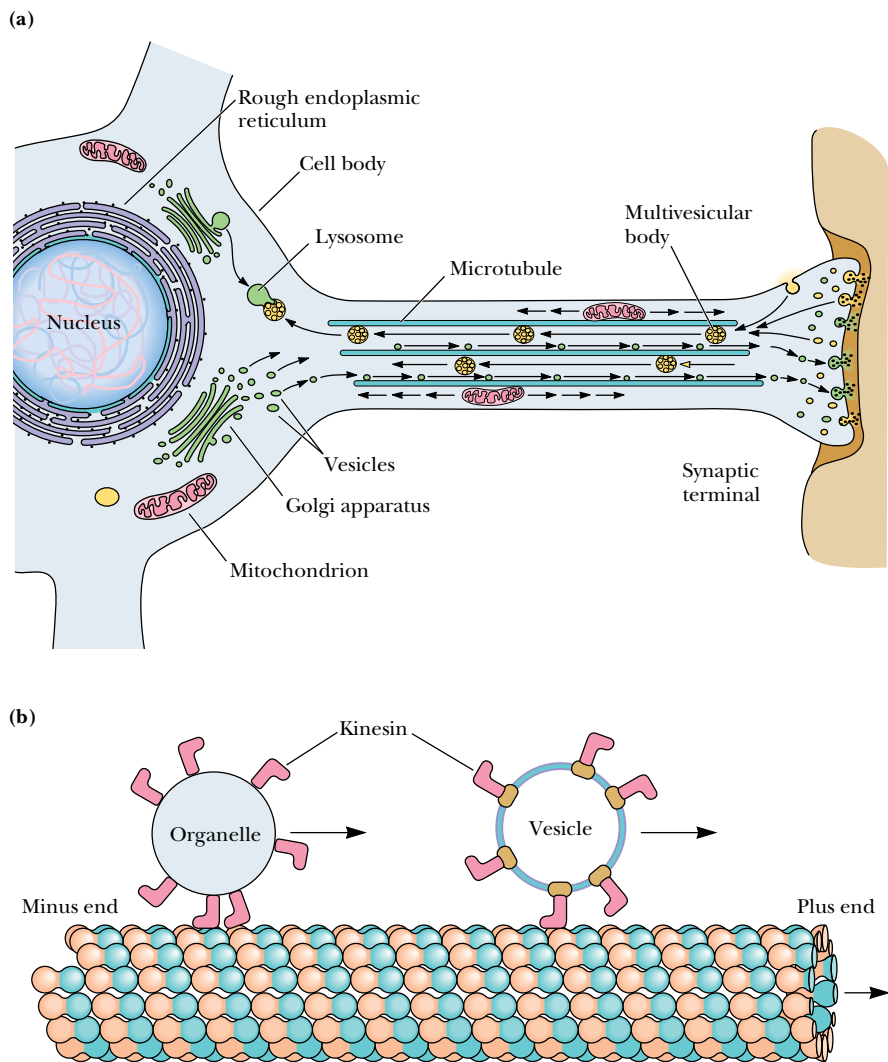
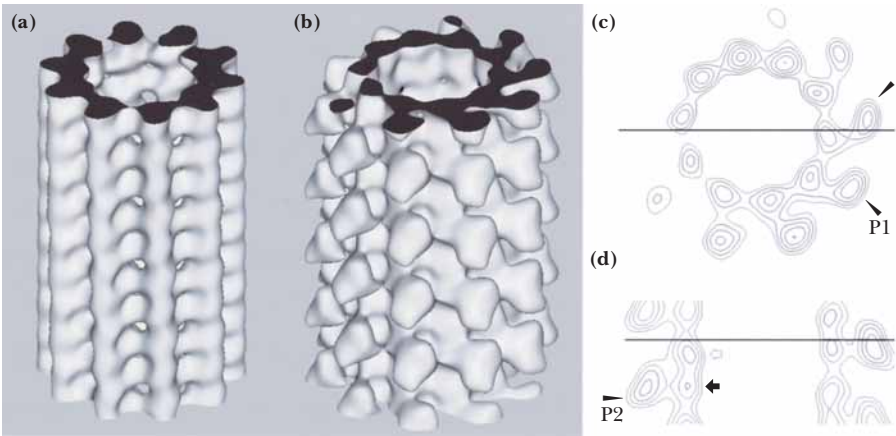


FIGURE 17.8 • (a) Rapid axonal transport along microtubules permits the exchange of material between the synaptic terminal and the body of the nerve cell. (b) Vesicles, multivesicular bodies, and mitochondria are carried through the axon by this mechanism. (Adapted from a drawing by Ronald Vale)

kinesins, on the other hand, assist the movement of organelles and vesicles from the minus end to the plus end of microtubules, resulting in outward movement of organelles and vesicles. Kinesin is similar to cytosolic dyneins but smaller in size (360 kD), and contains subunits of 110 kD and 65 to 70 kD. Its length is 100 nm. Like dyneins, kinesins possess ATPase activity in their globular heads, and it is the free energy of ATP hydrolysis that drives the movement of vesicles along the microtubules.

The N-terminal domain of the kinesin heavy chain (38 kD, approximately 340 residues) contains the ATP- and microtubule-binding sites and is the domain responsible for movement. Electron microscopy and image analysis of tubulin–kinesin complexes reveals (Figure 17.9) that the kinesin head domain is compact and primarily contacts a single tubulin subunit on a microtubule surface, inducing a conformational change in the tubulin subunit. Optical trapping experiments (see page 554) demonstrate that kinesin heads move in 8-nm (80-Å) steps along the long axis of a microtubule. Kenneth Johnson and his coworkers have shown that the ability of a single kinesin tetramer to move unidirectionally for long distances on a microtubule depends upon cooperative interactions between the two mechanochemical head domains of the protein.

FIGURE 17.9 • The structure of the tubulin–kinesin complex, as revealed by image analysis of cryoelectron microscopy data. (a) The computed, three-dimensional map of a microtubule, (b) the kinesin globular head domain–microtubule complex, (c) a contour plot of a horizontal section of the kinesin–microtubule complex, and (d) a contour plot of a vertical section of the same complex. (Taken from Kikkawa *et al.*, 1995. *Nature* 376:274–277. Photo courtesy of Nobutaka Hirokawa.)



17.3 • Skeletal Muscle Myosin and Muscle Contraction

The Morphology of Muscle

Four different kinds of muscle are found in animals (Figure 17.10). They are **skeletal** muscle, **cardiac (heart)** muscle, **smooth** muscle, and **myoepithelial cells**. The cells of the latter three types contain only a single nucleus and are called **myocytes**. The cells of skeletal muscle are long and multinucleate and are referred to as **muscle fibers**. At the microscopic level, skeletal muscle and cardiac muscle display alternating light and dark bands, and for this reason are often referred to as **striated** muscles. The different types of muscle cells vary widely in structure, size, and function. In addition, the times required for contractions and relaxations by various muscle types vary considerably. The fastest responses (on the order of milliseconds) are observed for **fast-twitch** skeletal

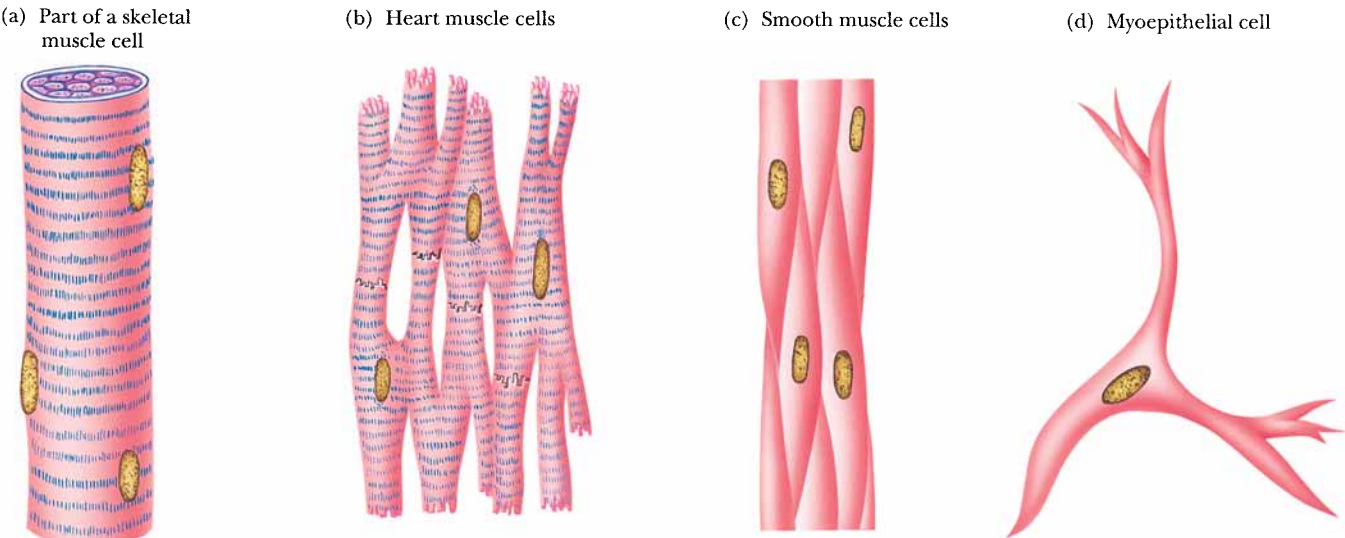


FIGURE 17.10 • The four classes of muscle cells in mammals. Skeletal muscle and cardiac muscle are striated. Cardiac muscle, smooth muscle, and myoepithelial cells are mononucleate, whereas skeletal muscle is multinucleate.

muscle, and the slowest responses (on the order of seconds) are found in smooth muscle. **Slow-twitch** skeletal muscle tissue displays an intermediate response time.

Structural Features of Skeletal Muscle

Skeletal muscles in higher animals consist of 100- μm -diameter **fiber bundles**, some as long as the muscle itself. Each of these muscle fibers contains hundreds of **myofibrils** (Figure 17.11), each of which spans the length of the fiber and is about 1 to 2 μm in diameter. Myofibrils are linear arrays of cylindrical **sarcomeres**, the basic structural units of muscle contraction. The sarcomeres are surrounded on each end by a membrane system that is actually an elaborate extension of the muscle fiber plasma membrane or **sarcolemma**. These extensions of the sarcolemma, which are called **transverse tubules** or **t-tubules**, enable the sarcolemmal membrane to contact the ends of each myofibril in the muscle fiber (Figure 17.11). This topological feature is crucial to the initiation of contractions. In between the t-tubules, the sarcomere is covered with a specialized endoplasmic reticulum called the **sarcoplasmic reticulum**, or **SR**. The SR contains high concentrations of Ca^{2+} , and the release of Ca^{2+} from the SR and its interactions within the sarcomeres trigger muscle contraction, as we will see. Each SR structure consists of two domains. **Longitudinal tubules** run the length of the sarcomere and are capped on either end by the **terminal cisternae** (Figure 17.11). The structure at the end of each sarcomere, which consists of a t-tubule and two apposed terminal cisternae, is called a **triad**, and the intervening gaps of approximately 15 nm are called **triad junctions**. The junctional face of each terminal cisterna is joined to its respective t-tubule by a **foot structure**. Skeletal muscle contractions are initiated by nerve stimuli that act directly on the muscle. Nerve impulses produce an electrochemical signal

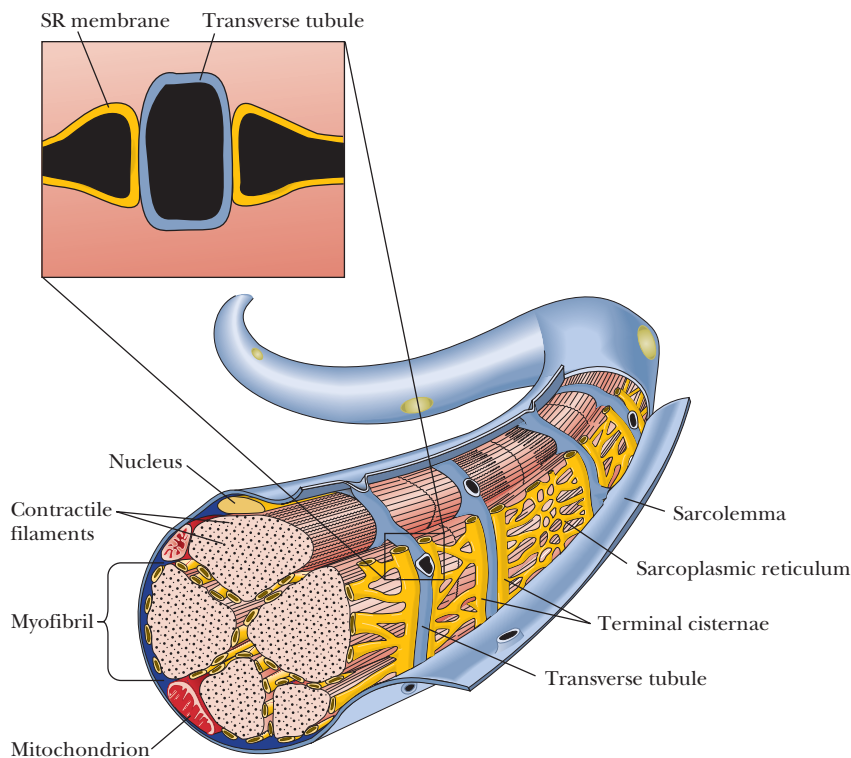


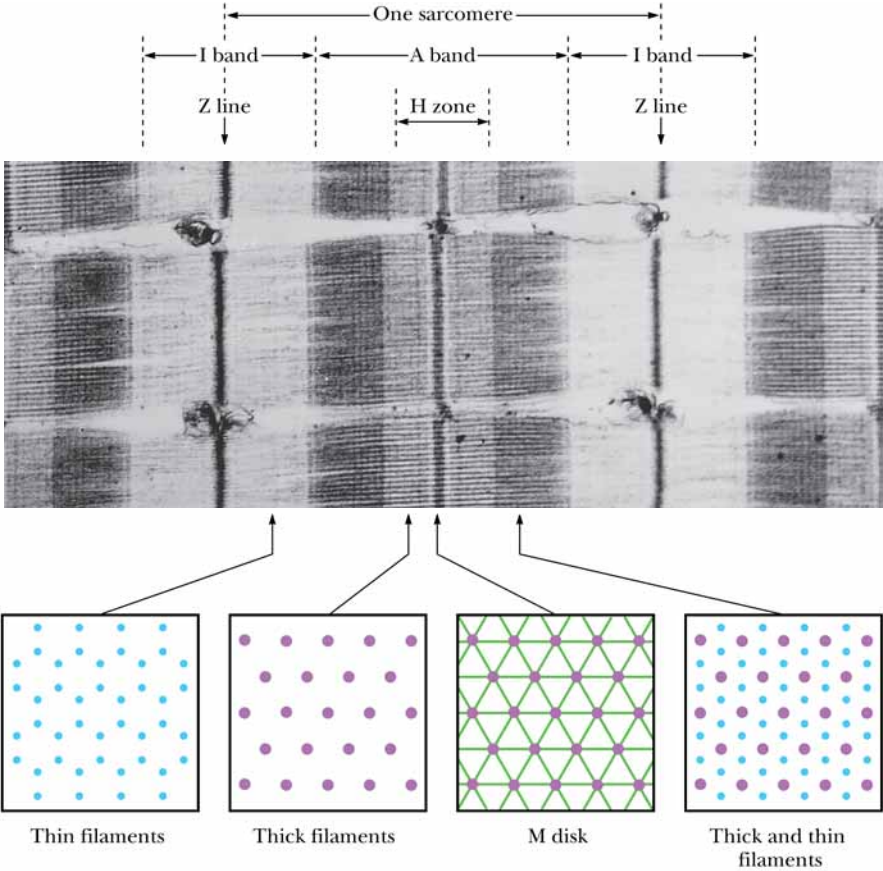
FIGURE 17.11 • The structure of a skeletal muscle cell, showing the manner in which t-tubules enable the sarcolemmal membrane to contact the ends of each myofibril in the muscle fiber. The foot structure is shown in the box.

(see Chapter 34) called an action potential that spreads over the sarcolemmal membrane and into the fiber along the t-tubule network. This signal is passed across the triad junction and induces the release of Ca^{2+} ions from the SR. These Ca^{2+} ions bind to the muscle fibers and induce contraction.

The Molecular Structure of Skeletal Muscle

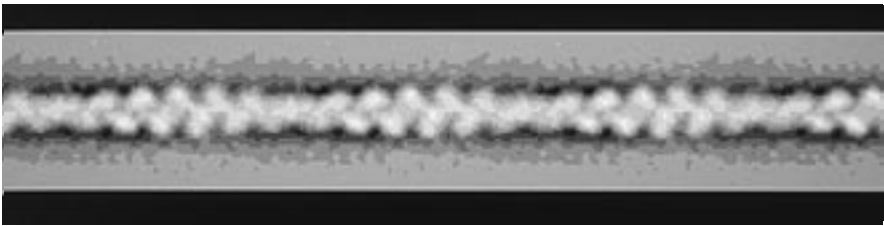
Examination of myofibrils in the electron microscope reveals a banded or striated structure. The bands are traditionally identified by letters (Figure 17.12). Regions of high electron density, denoted **A bands**, alternate with regions of low electron density, the **I bands**. Small, dark **Z lines** lie in the middle of the I bands, marking the ends of the sarcomere. Each A band has a central region of slightly lower electron density called the **H zone**, which contains a central **M disk** (also called an **M line**). Electron micrographs of cross-sections of each of these regions reveal molecular details. The H zone shows a regular, hexagonally arranged array of **thick filaments** (15 nm diameter), whereas the I band shows a regular, hexagonal array of **thin filaments** (7 nm diameter). In the dark regions at the ends of each A band, the thin and thick filaments interdigitate, as shown in Figure 17.12. The thin filaments are composed primarily of three proteins called **actin**, **troponin**, and **tropomyosin**. The thick filaments consist mainly of a protein called **myosin**. The thin and thick filaments are joined by **cross-bridges**. These cross-bridges are actually extensions of the myosin molecules, and muscle contraction is accomplished by the sliding of the cross-bridges along the thin filaments, a mechanical movement driven by the free energy of ATP hydrolysis.

FIGURE 17.12 • Electron micrograph of a skeletal muscle myofibril (in longitudinal section). The length of one sarcomere is indicated, as are the A and I bands, the H zone, the M disk, and the Z lines. Cross-sections from the H zone show a hexagonal array of thick filaments, whereas the I band cross-section shows a hexagonal array of thin filaments. (Photo courtesy of Hugh Huxley, Brandeis University)

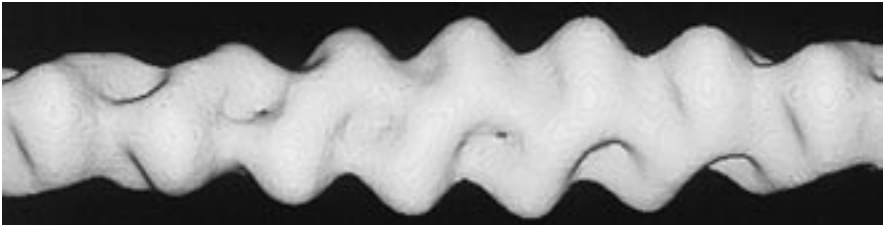


The Composition and Structure of Thin Filaments

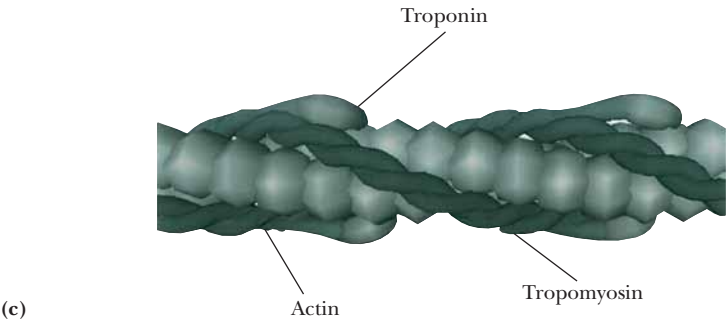
Actin, the principal component of thin filaments, can be isolated in two forms. Under conditions of low ionic strength, actin exists as a 42-kD globular protein, denoted **G-actin**. G-actin consists of two principal lobes or domains (Figure 17.13). Under physiological conditions (higher ionic strength), G-actin polymerizes to form a *fibrous* form of actin, called **F-actin**. As shown in Figure 17.14, F-actin is a right-handed helical structure, with a helix pitch of about 72 nm per turn. The F-actin helix is the core of the thin filament, to which *tropomyosin* and the **troponin complex** also add. Tropomyosin is a dimer of homologous but nonidentical 33-kD subunits. These two subunits form long α -helices that intertwine, creating 38- to 40-nm-long coiled coils, which join in head-to-tail fashion to form long rods. These rods bind to the F-actin polymer and lie almost parallel to the long axis of the F-actin helix (Figure 17.15a–c). Each tropomyosin heterodimer contacts approximately seven actin subunits. The troponin complex consists of three different proteins: **troponin T**, or **TnT** (37 kD); **troponin I**, or **TnI** (24 kD); and **troponin C**, or **TnC** (18 kD). TnT binds to tropomyosin, specifically at the head-to-tail junction. Troponin I binds both to tropomyosin and to actin. Troponin C is a Ca^{2+} -binding protein that binds to TnI. TnC shows 70% homology with the important Ca^{2+} signaling protein, calmodulin (Chapter 34). The release of Ca^{2+} from the SR, which signals a contraction, raises the cytosolic Ca^{2+} concentration high enough to saturate the Ca^{2+} sites on TnC. Ca^{2+} binding induces a conformational change in the amino-terminal domain of TnC, which in turn causes a rearrangement of the troponin complex and tropomyosin with respect to the actin fiber.



(a)



(b)



(c)

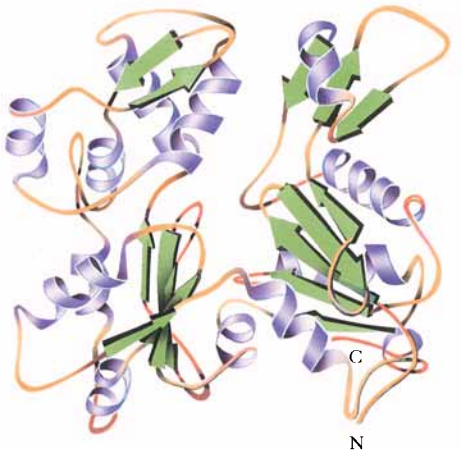


FIGURE 17.13 • The three-dimensional structure of an actin monomer from skeletal muscle. This view shows the two domains (*left and right*) of actin.

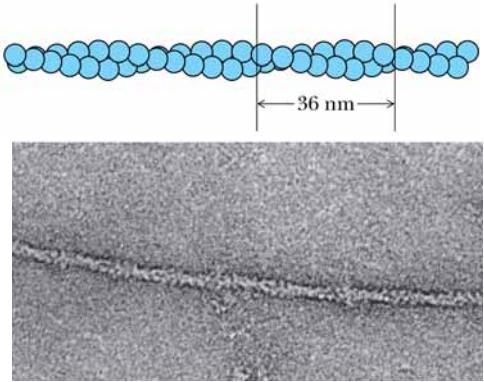


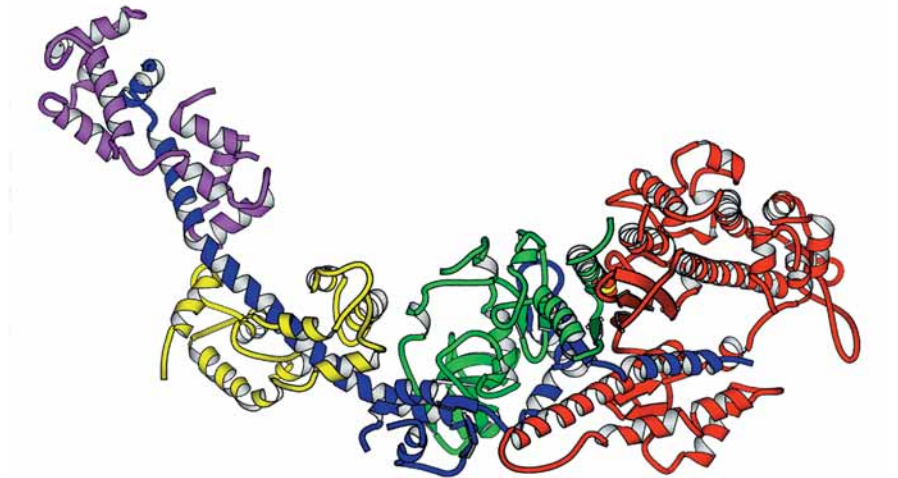
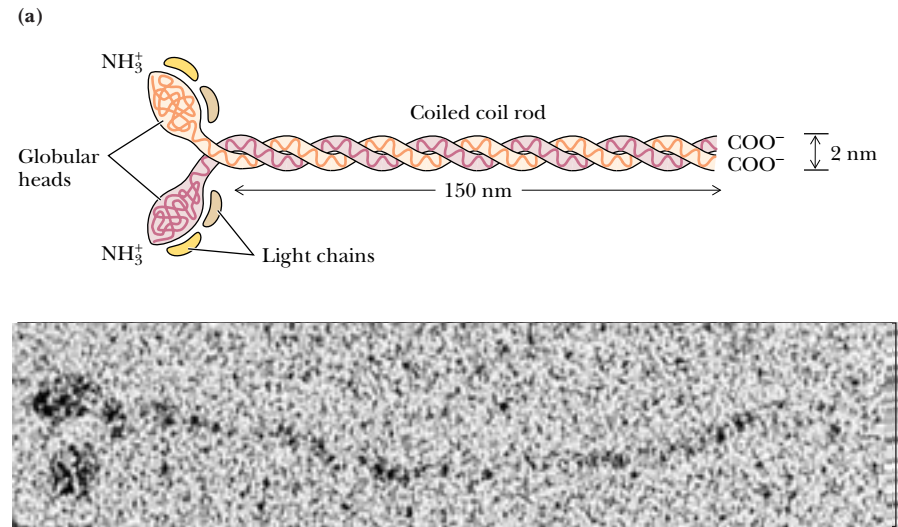
FIGURE 17.14 • The helical arrangement of actin monomers in F-actin. The F-actin helix has a pitch of 72 nm and a repeat distance of 36 nm. (*Electron micrograph courtesy of Hugh Huxley, Brandeis University*)

FIGURE 17.15 • (a) An electron micrograph of a thin filament, (b) a corresponding image reconstruction, and (c) a schematic drawing based on the images in (a) and (b). The tropomyosin coiled coil winds around the actin helix, each tropomyosin dimer interacting with seven consecutive actin monomers. Troponin T binds to tropomyosin at the head-to-tail junction. (*a and b, courtesy of Linda Rost and David DeRosier, Brandeis University; c, courtesy of George Phillips, Rice University*)

The Composition and Structure of Thick Filaments

Myosin, the principal component of muscle thick filaments, is a large protein consisting of six polypeptides, with an aggregate molecular weight of approximately 540 kD. As shown in Figure 17.16, the six peptides include two 230-kD **heavy chains**, as well as two pairs of different 20-kD **light chains**, denoted **LC1** and **LC2**. The heavy chains consist of globular amino-terminal **myosin heads**, joined to long α -helical carboxy-terminal segments, the **tails**. These tails are intertwined to form a left-handed coiled coil approximately 2 nm in diameter and 130 to 150 nm long. Each of the heads in this dimeric structure is associated with an LC1 and an LC2. The myosin heads exhibit **ATPase activity**, and hydrolysis of ATP by the myosin heads drives muscle contraction. LC1 is also known as the **essential light chain**, and LC2 is designated the **regulatory light chain**. Both light chains are homologous to calmodulin and TnC. Dissociation of LC1 from the myosin heads by alkali cations results in loss of the myosin ATPase activity.

FIGURE 17.16 • (a) An electron micrograph of a myosin molecule and a corresponding schematic drawing. The tail is a coiled coil of intertwined α -helices extending from the two globular heads. One of each of the myosin light chain proteins, LC1 and LC2, is bound to each of the globular heads. (b) A ribbon diagram shows the structure of the S1 myosin head (green, red, and purple segments) and its associated essential (yellow) and regulatory (magenta) light chains. (a, Electron micrograph courtesy of Henry Slayter, Harvard Medical School; b, courtesy of Ivan Rayment and Hazel M. Holden, University of Wisconsin, Madison)



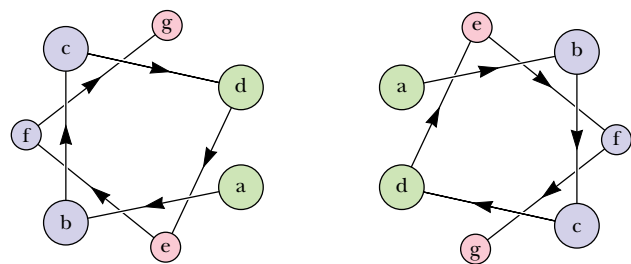


FIGURE 17.17 • An axial view of the two-stranded, α -helical coiled coil of a myosin tail. Hydrophobic residues a and d of the seven-residue repeat sequence align to form a hydrophobic core. Residues b, c, and f face the outer surface of the coiled coil and are typically ionic.



Approximately 500 of the 820 amino acid residues of the myosin head are highly conserved between various species. One conserved region, located approximately at residues 170 to 214, constitutes part of the ATP-binding site. Whereas many ATP-binding proteins and enzymes employ a β -sheet- α -helix- β -sheet motif, this region of myosin forms a related α - β - α structure, beginning with an Arg at (approximately) residue 192. The β -sheet in this region of all myosins includes the amino acid sequence

Gly-Glu-Ser-Gly-Ala-Gly-Lys-Thr

The Gly-X-X-Gly-X-Gly found in this segment is found in many ATP- and nucleotide-binding enzymes. The Lys of this segment is thought to interact with the α -phosphate of bound ATP.

Repeating Structural Elements Are the Secret of Myosin’s Coiled Coils

Myosin tails show less homology than the head regions, but several key features of the tail sequence are responsible for the α -helical coiled coils formed by myosin tails. *Several orders of repeating structure* are found in all myosin tails, including 7-residue, 28-residue, and 196-residue repeating units. Large stretches of the tail domain are composed of 7-residue repeating segments. The first and fourth residues of these 7-residue units are generally small, hydrophobic amino acids, whereas the second, third, and sixth are likely to be charged residues. The consequence of this arrangement is shown in Figure 17.17. Seven residues form two turns of an α -helix, and, in the coiled coil structure of the myosin tails, the first and fourth residues face the interior contact region of the coiled coil. Residues b, c, and f (2, 3, and 6) of the 7-residue repeat face the periphery, where charged residues can interact with the water solvent. Groups of four 7-residue units with distinct patterns of alternating side-chain charge form 28-residue repeats that establish alternating regions of positive and negative charge on the surface of the myosin coiled coil. These alternating charged regions interact with similar regions in the tails of adjacent myosin molecules to assist in stabilizing the thick filament.

At a still higher level of organization, groups of seven of these 28-residue units—a total of 196 residues—also form a repeating pattern, and this large-scale repeating motif contributes to the packing of the myosin molecules in the thick filament. The myosin molecules in thick filaments are offset (Figure 17.18) by approximately 14 nm, a distance that corresponds to 98 residues of a coiled coil, or exactly half the length of the 196-residue repeat. Thus, several layers of repeating structure play specific roles in the formation and stabilization of the myosin coiled coil and the thick filament formed from them.

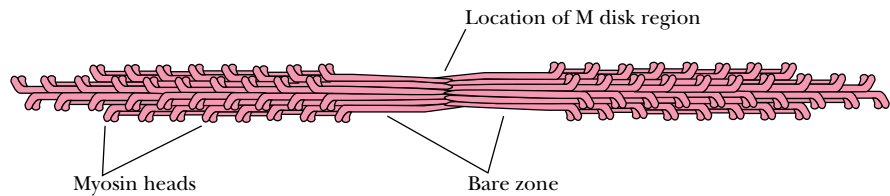


FIGURE 17.18 • The packing of myosin molecules in a thick filament. Adjoining molecules are offset by approximately 14 nm, a distance corresponding to 98 residues of the coiled coil.

The Associated Proteins of Striated Muscle

In addition to the major proteins of striated muscle (myosin, actin, tropomyosin, and the troponins), numerous other proteins play important roles in the maintenance of muscle structure and the regulation of muscle contraction. Myosin and actin together account for 65% of the total muscle protein, and tropomyosin and the troponins each contribute an additional 5% (Table 17.1). The other regulatory and structural proteins thus comprise approximately 25% of the myofibrillar protein. The regulatory proteins can be classified as either **myosin-associated proteins** or **actin-associated proteins**.

The myosin-associated proteins include three proteins found in the M disks. The M disks consist primarily of **M protein** (165 kD), **myomesin** (185 kD), and **creatine kinase** (a dimer of 42-kD subunits). Creatine kinase facilitates rapid regeneration of the ATP consumed during muscle contraction. The association of M protein, myomesin, and creatine kinase in the M disk maintains the structural integrity of the myosin filaments. Several other myosin-associated proteins have also been identified, including **C protein** (135 kD), **F protein** (121 kD), **H protein** (74 kD), and **I protein** (50 kD). The C protein is localized to several regularly spaced stripes in the A band. C protein inhibits myosin ATPase activity at low ionic strength but activates it at physiological ionic strength. The roles of F, H, and I proteins are not yet understood.

Actin-associated proteins (other than tropomyosin and the troponins) include **α -actinin** (a homodimer of 95-kD subunits), **β -actinin** (a heterodimer of 37-kD and 34-kD subunits), **γ -actinin** (a 35-kD monomer), and **paratropomyosin** (a homodimer of 34-kD subunits). α -Actinin is found in the Z lines and activates contraction of actomyosin. It is thought to play a role in attachment of actin to the Z lines. α -Actinin consists of three domains: an N-terminal, actin-binding domain; a central domain consisting of four repeats of a 122-residue sequence; and a C-terminal domain that contains two EF-hand, calcium-binding domains (Figure 17.19). The four central repeats in α -actinin are highly homologous with the 106-residue repeat sequences of **spectrin**, the major structural protein of the red blood cell cytoskeleton. The repeating segments of both α -actinin and spectrin are thought to consist of bundles of four α -helices (Figure 17.20). β -Actinin acts as an actin-capping protein, specifically binding to the end of an actin filament. γ -Actinin also inhibits actin polymerization, but its location in thin filaments is not known with certainty. Paratropomyosin is similar to tropomyosin, but appears to be located only at the A band–I band junction.

Two cytoskeletal proteins, **titin** (also known as **connectin**) and **nebulin**, account for 15% of the total protein in the myofibril. Together these proteins form a flexible filamentous network that surrounds the myofibrils. Titin is an **elastic protein** and can stretch under tension. Its discovery and characteriza-

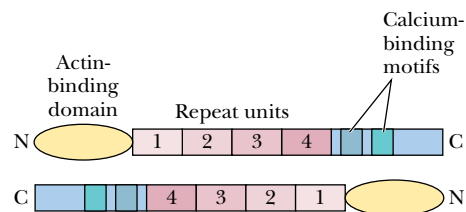


FIGURE 17.19 • α -Actinin exists as a homodimer of antiparallel subunits, illustrated here in terms of their primary structure. The N-terminal, actin-binding domain and the C-terminal, EF-hand domains are separated by a central domain consisting of four repeats of a 122-residue sequence.

Table 17.1
Myofibrillar Structural Proteins of Rabbit Skeletal Muscle

Protein	Molecular Mass (kD)	Content (wt %)	Localization	Function
Contractile proteins				
Myosin	520	43	A band	Contracts with actin
Actin	42	22	I band	Contracts with myosin
Regulatory proteins				
Major				
Tropomyosin	33 × 2	5	I band	Binds to actin and locates troponin
Troponin	70	5	I band	Ca regulation
Troponin C	18			Ca binding
Troponin I	21			Inhibits actin–myosin interaction
Troponin T	31			Binds to tropomyosin
Minor				
M protein	165	2	M line	Binds to myosin
Myomesin	185	<1	M line	Binds to myosin
Creatine kinase	42	<1	M line	Binds to myosin
C protein	135	2	A band	Binds to myosin
F protein	121	<1	A band	Binds to myosin
H protein	74	<1	Near M line	Binds to myosin
I protein	50	<1	A band	Inhibits actin–myosin interaction
α-Actinin	95 × 2	2	Z line	Gelates actin filaments
β-Actinin	37 + 34	<1	Free end of actin filament	Caps actin filaments
γ-Actinin	35	<1	?	Inhibits actin polymerization
eu-Actinin	42	<1	Z line	Binds to actin
ABP (filamin)	240 × 2	<1	Z line	Gelates actin filaments
Paratropomyosin	34 × 2	<1	A–I junction	Inhibits actin–myosin interaction
Cytoskeletal proteins				
Titin 1	2800	10	A–I	Links myosin filament to Z line
Titin 2	2100			
Nebulin	800	5	N ₂ line*	
Vinculin	130	<1	Under sarcolemma	
Desmin (skeleton)	53	<1	Periphery of Z line	Intermediate filament
Vimentin	55	<1	Periphery of Z line	Intermediate filament
Synemin	220	<1	Z line	
Z protein	50	<1	Z line	Forms lattice structure
Z-nin	400	<1	Z line	

*A structure within the I band.
Adapted from Ohtsuki, I., Maruyama, K., and Ebashi, S., 1986. Regulatory and cytoskeletal proteins of vertebrate skeletal muscle. *Advances in Protein Chemistry* 38:1–67.

HUMAN BIOCHEMISTRY

The Molecular Defect in Duchenne Muscular Dystrophy Involves an Actin-Anchoring Protein

Discovery of a new actinin/spectrin-like protein has provided insights into the molecular basis for at least one form of muscular dystrophy. Duchenne muscular dystrophy is a degenerative and fatal disorder of muscle affecting approximately 1 in 3500 boys. Victims of Duchenne dystrophy show early abnormalities in walking and running. By the age of five, the victim cannot run and has difficulty standing, and by early adolescence, walking is difficult or impossible. The loss of muscle function progresses upward in the body, affecting next the arms and the diaphragm. Respiratory problems or infections usually result in death by the age of 30. Louis Kunkel and his coworkers identified the Duchenne muscular dystrophy gene in 1986. This gene produces a protein called **dystrophin**, which is highly homologous to α -actinin and spectrin. A defect in dystrophin is responsible for the muscle degeneration of Duchenne dystrophy.

Dystrophin is located on the cytoplasmic face of the muscle

plasma membrane, linked to the plasma membrane via an integral membrane glycoprotein. Dystrophin has a high molecular mass (427 kD), but constitutes less than 0.01% of the total muscle protein. It folds into four principal domains (figure, part a), including an N-terminal domain similar to the actin-binding domains of actinin and spectrin, a long repeat domain, a cysteine-rich domain, and a C-terminal domain that is unique to dystrophin. The repeat domain consists of 24 repeat units of approximately 109 residues each. “Spacer sequences” high in proline content, which do not align with the repeat consensus sequence, occur at the beginning and end of the repeat domain. Spacer segments are found between repeat elements 3 and 4 and 19 and 20. The high proline content of the spacers suggests that they may represent hinge domains. The spacer/hinge segments are sensitive to proteolytic enzymes, indicating that they may represent more exposed regions of the polypeptide.

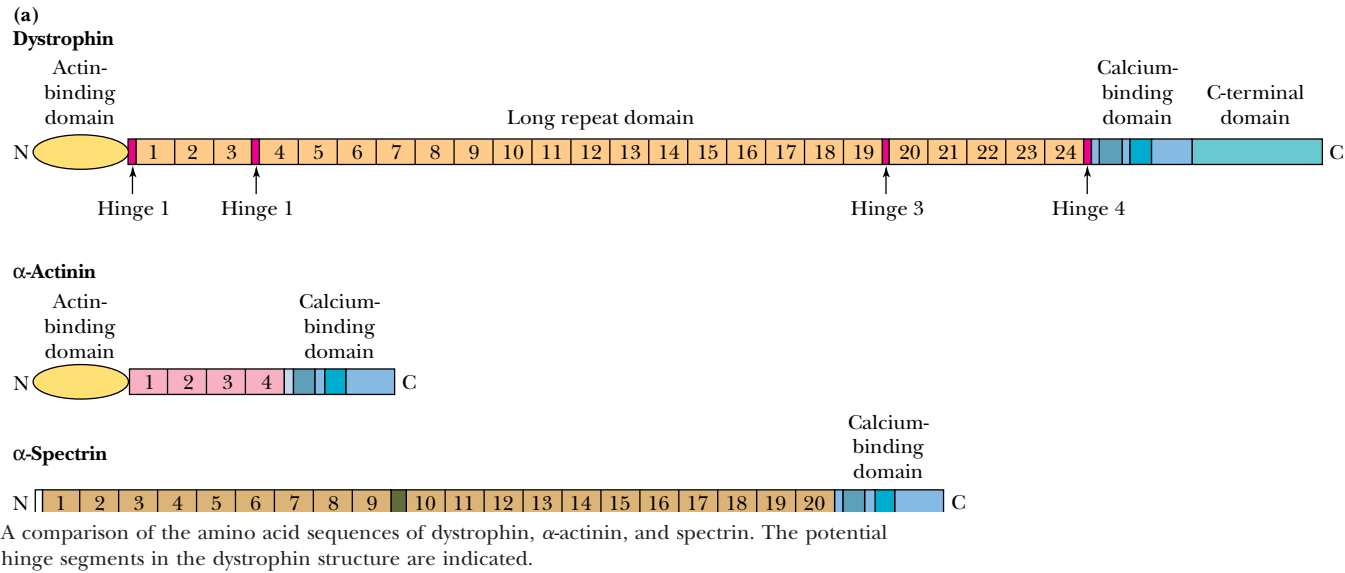
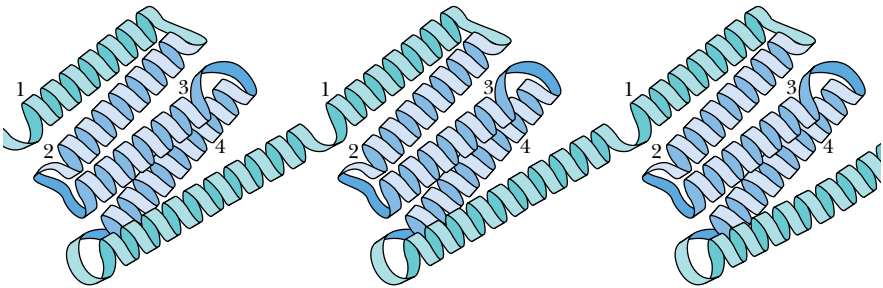
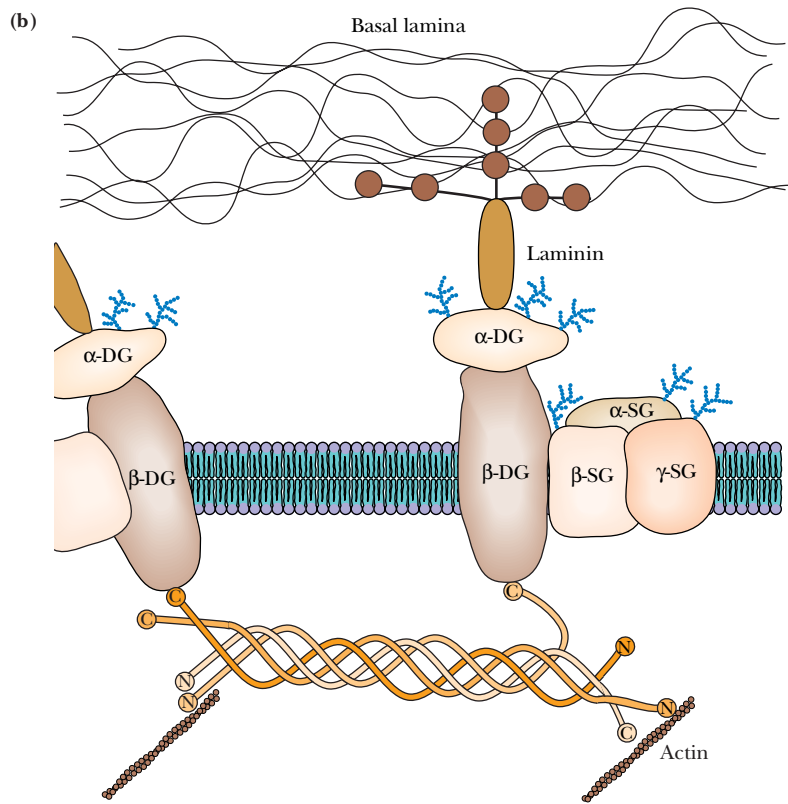


FIGURE 17.20 • A schematic drawing of the four-helix cluster model for α -actinin and spectrin. Helix 1 is long and is postulated to lie at an angle with respect to the long axis of the repeated domain.



Dystrophin itself appears to be part of an elaborate protein/glycoprotein complex that bridges the inner cytoskeleton (actin filaments) and the extracellular matrix (via a matrix protein called laminin (see figure)). It is now clear that defects in one or more of the proteins in this complex are responsible for many of the other forms of muscular dystrophy. The glycoprotein complex is composed of two subcomplexes, the dystroglycan complex and the sarcoglycan complex. The dystroglycan complex consists of α -dystroglycan, an extracellular protein that binds to merosin, a laminin subunit and component of the extracellular

matrix, and β -dystroglycan, a transmembrane protein that binds the C-terminal domain of dystrophin inside the cell (see figure). The sarcoglycan complex is composed of α -, β -, and γ -sarcoglycans, all of which are transmembrane glycoproteins. Alterations of the sarcoglycan proteins are linked to limb-girdle muscular dystrophy and autosomal recessive muscular dystrophy. Mutations in the gene for merosin, which binds to α -dystroglycan, are linked to severe congenital muscular dystrophy, yet another form of the disease.



A model for the actin–dystrophin–glycoprotein complex in skeletal muscle. Dystrophin is postulated to form tetramers of antiparallel monomers that bind actin at their N-termini and a family of dystrophin-associated glycoproteins at their C-termini. This dystrophin-anchored complex may function to stabilize the sarcolemmal membrane during contraction–relaxation cycles, link the contractile force generated in the cell (fiber) with the extracellular environment, or maintain local organization of key proteins in the membrane. The dystrophin-associated membrane proteins (dystroglycans and sarcoglycans) range from 25 to 154 kD. (Adapted from Ahn, A. H., and Kunkel, L. M., 1993. *Nature Genetics* 3:283–291, and Worton, R., 1995. *Science* 270:755–756.)

tion ended a century-long debate over the possible existence of an elastic component in muscle.

Titin is the largest known protein, consisting of nearly 27,000 amino acids and having a molecular mass of 2,993 kD. The sequencing of titin was accomplished by Siegfried Labeit and Bernhard Kolmerer in 1995. (To accomplish this prodigious feat, Labeit and Kolmerer pieced together the full sequence from 50 overlapping cDNA fragments!) Titin’s sequence is composed mainly (90%) of 244 repeats of the immunoglobulin (Ig) and fibronectin 3 (FN3) domains. In the center of the titin molecule is a novel protein motif (not known in other proteins), consisting of repeats of the sequence **PEVK** (proline-gluta-

mate-valine-lysine). PEVK motifs may act as spring devices to pull muscles back into shape after they have been stretched and may also play a role in regulating the stiffness and elasticity of muscle fibers. In characteristically stiff muscles, such as cardiac muscle, the PEVK region is only 163 residues in length, whereas in the more elastic skeletal muscle the PEVK domain is over 2000 residues in length.

Titin forms long, flexible, thin filaments in muscle fibers. A single titin filament spans 1000 nm in its relaxed state, and a titin filament under tension can stretch to a length of over 3000 nm! Titin filaments in muscle originate at the periphery of the M band and extend along the myosin filaments all the way to the Z line (Figure 17.21). They appear to function by linking the myosin filaments to the Z lines and by acting as a template to regulate the assembly of myosin filaments and the spacing of myosin monomers in the filaments. When myofibrils are stretched beyond the overlap of the thick and thin filaments, titin filaments passively generate tension. This effect is provided by a relatively small number of titin molecules. The ratio of myosin to titin filaments is approximately 24 to 1. With 300 myosin molecules per thick filament, only 6 or so titin filaments are present in each half of a myosin thick filament.

The Mechanism of Muscle Contraction

When muscle fibers contract, the thick myosin filaments slide or walk along the thin actin filaments. The basic elements of the **sliding filament model** were first described in 1954 by two different research groups, Hugh Huxley and his colleague Jean Hanson, and the physiologist Andrew Huxley and his colleague Ralph Niedergerke. Several key discoveries paved the way for this model. Electron microscopic studies of muscle revealed that sarcomeres decreased in length during contraction, and that this decrease was due to decreases in the width of both the I band and the H zone (Figure 17.22). At the same time, the width of the A band (which is the length of the thick filaments) and the distance from the Z disks to the nearby H zone (that is, the length of the thin filaments) did not change. These observations made it clear that the lengths of both the thin and thick filaments were constant during contraction. This conclusion was consistent with a sliding filament model.

The Sliding Filament Model

The shortening of a sarcomere (Figure 17.22) involves sliding motions in opposing directions at the two ends of a myosin thick filament. Net sliding motions in a specific direction occur because the thin and thick filaments both have

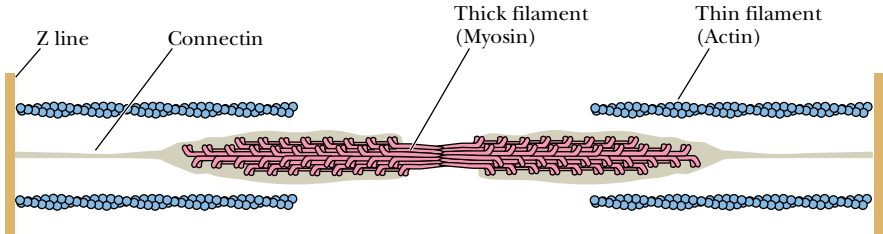


FIGURE 17.21 • A drawing of the arrangement of the elastic protein titin in the skeletal muscle sarcomere. Titin filaments originate at the periphery of the M band and extend along the myosin filaments to the Z lines. These titin filaments produce the passive tension existing in myofibrils that have been stretched so that the thick and thin filaments no longer overlap and cannot interact. (Adapted from Ohtsuki, I., Maruyama, K., and Ebashi, S., 1986. *Advances in Protein Chemistry* 38:1–67.)

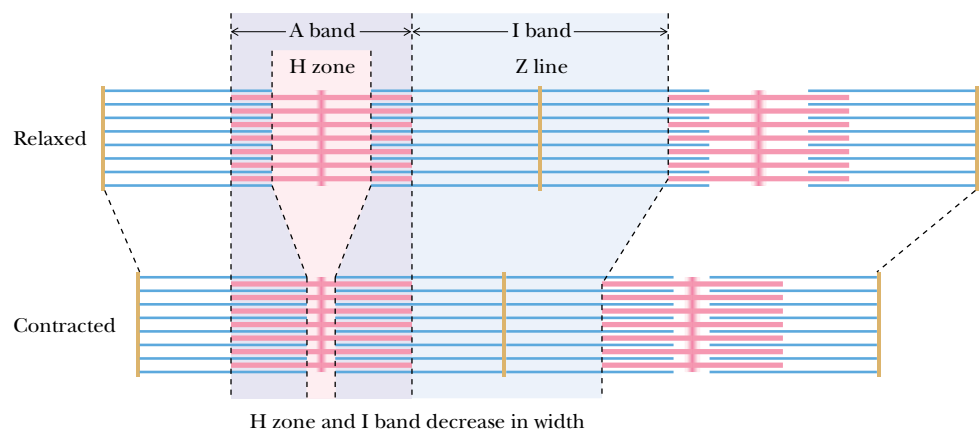


FIGURE 17.22 • The sliding filament model of skeletal muscle contraction. The decrease in sarcomere length is due to decreases in the width of the I band and H zone, with no change in the width of the A band. These observations mean that the lengths of both the thick and thin filaments do not change during contraction. Rather, the thick and thin filaments slide along one another.

directional character. The organization of the thin and thick filaments in the sarcomere takes particular advantage of this directional character. Actin filaments always extend outward from the Z lines in a uniform manner. Thus, between any two Z lines, the two sets of actin filaments point in opposing directions. The myosin thick filaments, on the other hand, also assemble in a directional manner. The polarity of myosin thick filaments reverses at the M disk. The nature of this reversal is not well understood, but presumably involves structural constraints provided by proteins in the M disk, such as the M protein and myomesin described above. The reversal of polarity at the M disk means that actin filaments on either side of the M disk are pulled toward the M disk during contraction by the sliding of the myosin heads, causing net shortening of the sarcomere.

Albert Szent-Györgyi’s Discovery of the Effects of Actin on Myosin

The molecular events of contraction are powered by the ATPase activity of myosin. Much of our present understanding of this reaction and its dependence on actin can be traced to several key discoveries by Albert Szent-Györgyi at the University of Szeged in Hungary in the early 1940s. Szent-Györgyi showed that solution viscosity is dramatically increased when solutions of myosin and actin are mixed. Increased viscosity is a manifestation of the formation of an **actomyosin complex**.

A

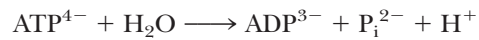
DEEPER LOOK

Viscous Solutions Reflect Long-Range Molecular Interactions

High viscosity in an aqueous solution is a sign of long-range molecular interactions, defined as interactions that extend through and connect many molecules. Concentrated sugar solutions (molasses, for example) are viscous because of the extensive hydrogen-bonding networks established by the multiple hydroxyl groups of sugar molecules. Solutions of DNA are highly viscous because isolated DNA fibers are extremely long (often in the millimeter size range) and highly hydrated. As Szent-Györgyi discovered, the extensive aggregates formed by myosin and actin also produce highly viscous solutions.

Szent-Györgyi further showed that the viscosity of an actomyosin solution was lowered by the addition of ATP, indicating that ATP decreases myosin's affinity for actin. Kinetic studies demonstrated that myosin ATPase activity was increased substantially by actin. (For this reason, Szent-Györgyi gave the name **actin** to the thin filament protein.) The ATPase turnover number of pure myosin is 0.05/sec. In the presence of actin, however, the turnover number increases to about 10/sec, a number more like that of intact muscle fibers.

The specific effect of actin on myosin ATPase becomes apparent if the *product release* steps of the reaction are carefully compared. In the absence of actin, the addition of ATP to myosin produces a rapid release of H^+ , one of the products of the ATPase reaction:



However, release of ADP and P_i from myosin is much slower. Actin activates myosin ATPase activity by stimulating the release of P_i and then ADP. Product release is followed by the binding of a new ATP to the actomyosin complex, which causes actomyosin to dissociate into free actin and myosin. The cycle of ATP hydrolysis then repeats, as shown in Figure 17.23a. The crucial point of this model is that *ATP hydrolysis and the association and dissociation of actin and myosin are coupled*. It is this coupling that enables ATP hydrolysis to power muscle contraction.

The Coupling Mechanism: ATP Hydrolysis Drives Conformation Changes in the Myosin Heads

The only remaining piece of the puzzle is this: How does the close coupling of actin-myosin binding and ATP hydrolysis result in the shortening of myofibrils? Put another way, how are the model for ATP hydrolysis and the sliding filament model related? The answer to this puzzle is shown in Figure 17.23b. The free energy of ATP hydrolysis is translated into a conformation change in the myosin head, so that dissociation of myosin and actin, hydrolysis of ATP, and rebinding of myosin and actin occur with stepwise movement of the myosin S1 head along the actin filament. The conformation change in the myosin head is driven by the hydrolysis of ATP.

As shown in the cycle in Figure 17.23a, the myosin heads—with the hydrolysis products ADP and P_i bound—are mainly dissociated from the actin filaments in resting muscle. When the signal to contract is presented (see following discussion), the myosin heads move out from the thick filaments to bind to actin on the thin filaments (Step 1). Binding to actin stimulates the release of phosphate, and this is followed by the crucial conformational change by the S1 myosin heads—the so-called **power stroke**—and ADP dissociation. In this step (Step 2), the thick filaments *move* along the thin filaments as the myosin heads relax to a lower energy conformation. In the power stroke, the myosin heads tilt by approximately 45 degrees and the conformational energy of the myosin heads is lowered by about 29 kJ/mol. This moves the thick filament approximately 10 nm along the thin filament (Step 3). Subsequent binding (Step 4) and hydrolysis (Step 5) of ATP cause dissociation of the heads from the thin filaments and also cause the myosin heads to shift back to their high-energy conformation with the heads' long axis nearly perpendicular to the long axis of the thick filaments. The heads may then begin another cycle by binding to actin filaments. This cycle is repeated at rates up to 5/sec in a typical skeletal muscle contraction. The conformational changes occurring in this cycle are the secret of the energy coupling that allows ATP binding and hydrolysis to drive muscle contraction.

The conformation change in the power stroke has been studied in two ways: (1) cryoelectron microscopy together with computerized image analysis

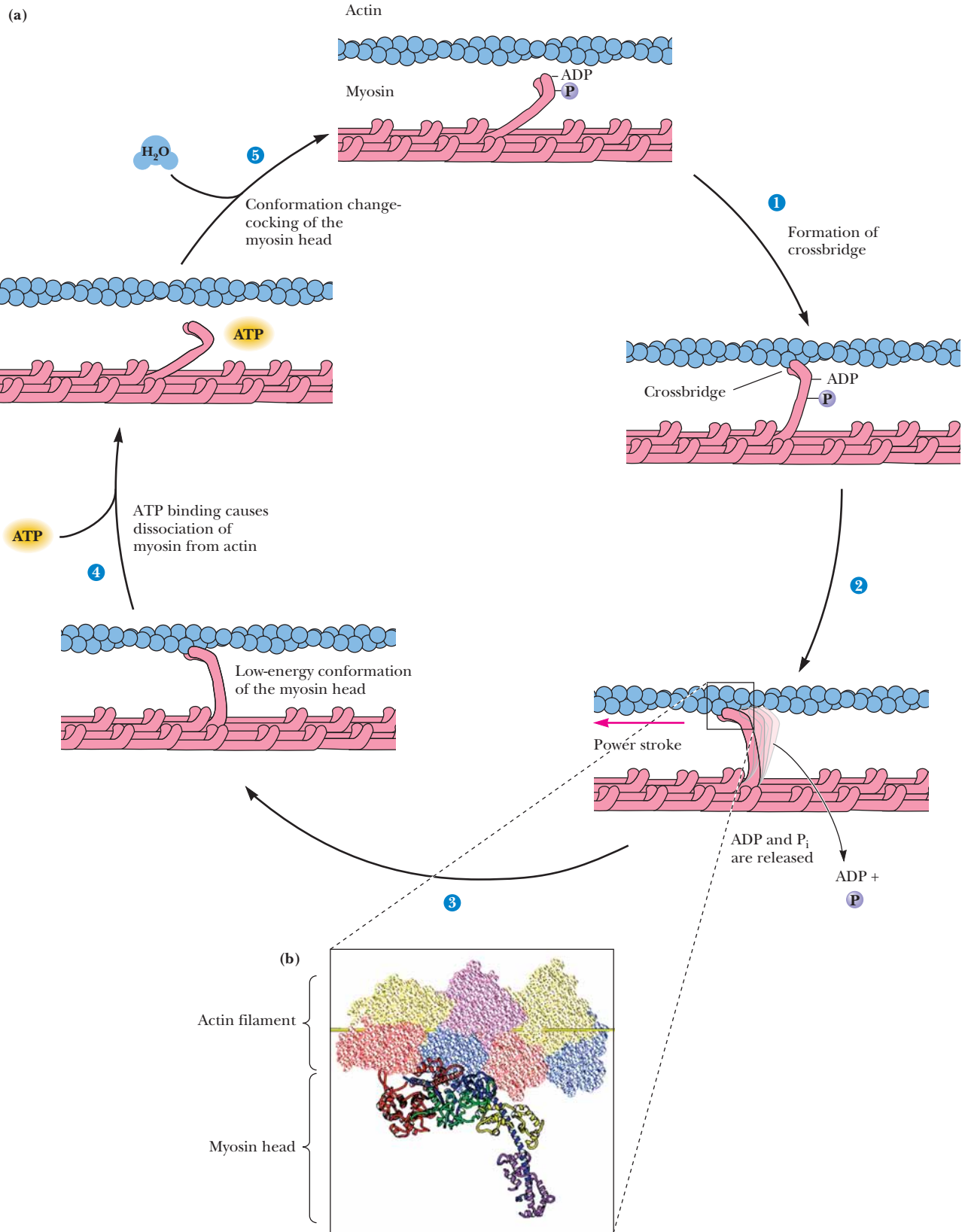


FIGURE 17.23 • The mechanism of skeletal muscle contraction. The free energy of ATP hydrolysis drives a conformational change in the myosin head, resulting in net movement of the myosin heads along the actin filament. (*Inset*) A ribbon and space-filling representation of the actin–myosin interaction. (*S1 myosin image courtesy of Ivan Rayment and Hazel M. Holden, University of Wisconsin, Madison.*)

has yielded low-resolution images of S1-decorated actin in the presence and absence of MgADP (corresponding approximately to the states before and after the power stroke), and (2) feedback-enhanced laser optical trapping experiments have measured the movements and forces exerted during single turnovers of single myosin molecules along an actin filament. The images of myosin, when compared with the X-ray crystal structure of myosin S1, show that the long α -helix of S1 that binds the light chains (ELC and RLC) may behave as a lever arm, and that this arm swings through an arc of 23 degrees upon release of ADP. (A glycine residue at position 770 in the S1 myosin head lies at the N-terminal end of this helix/lever arm and may act as a hinge.) *This results in a 3.5-nm (35-Å) movement of the last myosin heavy chain residue of the X-ray structure in a direction nearly parallel to the actin filament.* These two imaging “snapshots” of the myosin S1 conformation may represent only part of the working power stroke of the contraction cycle, and the total movement of a myosin head with respect to the apposed actin filament may thus be more than 3.5 nm.

CRITICAL DEVELOPMENTS IN BIOCHEMISTRY

Molecular “Tweezers” of Light Take the Measure of a Muscle Fiber’s Force

The optical trapping experiment involves the attachment of myosin molecules to silica beads that are immobilized on a microscope coverslip (see figure). Actin filaments are then prepared such that a polystyrene bead is attached to each end of the filament. These beads can be “caught” and held in place in solution by a pair of “optical traps”—two high-intensity infrared laser beams, one focused on the polystyrene bead at one end of the actin filament and the other focused on the bead at the other end of the actin filament. The force acting on each bead in such a trap is proportional to the position of the bead in the “trap,” so that displacement and forces acting on the bead (and thus on the actin filament) can both be measured. When the “trapped” actin filament is brought close to the silica bead, one or a few myosin molecules may interact with sites on the actin, and ATP-induced interactions of individual myosin molecules with the trapped actin filament can be measured and quantitated. Such optical trapping experiments have shown that *a single cycle or turnover of a single myosin molecule along an actin filament involves an average movement of 4 to 11 nm (40–110 Å) and generates an average force of 1.7 to 4 × 10^{−12} newton (1.7–4 piconewtons (pN)).*

The magnitudes of the movements observed in the optical trapping experiments are consistent with the movements predicted by the cryoelectron microscopy imaging data. Can the movements and forces detected in a single contraction cycle by optical trapping also be related to the energy available from hydrolysis of a single ATP molecule? The energy required for a contraction cycle is defined by the “work” accomplished by contraction, and work (w) is defined as force (F) times distance (d):

$$w = F \cdot d$$

For a movement of 4 nm against a force of 1.7 pN, we have

$$w = (1.7 \text{ pN}) \cdot (4 \text{ nm}) = 0.68 \times 10^{-20} \text{ J}$$

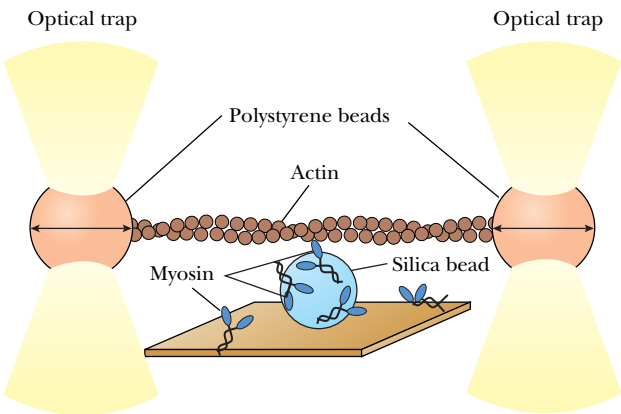
For a movement of 11 nm against a force of 4 pN, the energy requirement is larger:

$$w = (4 \text{ pN}) \cdot (11 \text{ nm}) = 4.4 \times 10^{-20} \text{ J}$$

If the cellular free energy of hydrolysis of ATP is taken as −50 kJ/mol, the free energy available from the hydrolysis of a single ATP molecule is

$$\Delta G = (-50 \text{ kJ/mol}) / 6.02 \times 10^{23} \text{ molecules/mol} = 8.3 \times 10^{-20} \text{ J}$$

Thus, *the free energy of hydrolysis of a single ATP molecule is sufficient to drive the observed movements against the forces that have been measured.*



Movements of single myosin molecules along an actin filament can be measured by means of an optical trap consisting of laser beams focused on polystyrene beads attached to the ends of actin molecules. (Adapted from *Finer et al.*, 1994. *Nature* **368**:113–119. See also *Block*, 1995. *Nature* **378**:132–133.)

Control of the Contraction–Relaxation Cycle
by Calcium Channels and Pumps

The trigger for all muscle contraction is an increase in Ca^{2+} concentration in the vicinity of the muscle fibers of skeletal muscle or the myocytes of cardiac and smooth muscle. In all these cases, this increase in Ca^{2+} is due to the flow of Ca^{2+} through **calcium channels** (Figure 17.24). A muscle contraction ends when the Ca^{2+} concentration is reduced by specific calcium pumps (such as the SR Ca^{2+} -ATPase, Chapter 10). The sarcoplasmic reticulum, t-tubule, and sarcolemmal membranes all contain Ca^{2+} channels. As we shall see, the Ca^{2+} channels of the SR function together with the t-tubules in a remarkable coupled process.

Ca^{2+} release in skeletal and heart muscle has been characterized through the use of specific antagonist molecules that block Ca^{2+} channel activity. The **dihydropyridine (DHP) receptors** of t-tubules, for example, are blocked by **dihydropyridine** derivatives, such as **nifedipine** (Figure 17.25). The purified DHP receptor of heart muscle can be incorporated into liposomes, whereupon it shows calcium channel activity. The channel displays voltage-dependent gating and is selective for divalent cations over monovalent cations. Thus, the heart muscle DHP receptor is a voltage-dependent Ca^{2+} channel. Other evidence suggests that the skeletal muscle DHP receptor is a voltage-sensing protein; it presumably undergoes voltage-dependent conformation changes.

The DHP receptor from t-tubules consists of five different polypeptides, designated α_1 (150 to 173 kD), α_2 (120 to 150 kD), β (50 to 65 kD), γ (30 to 35 kD), and δ (22 to 27 kD). The α_2 - and δ -subunits are linked by a disulfide bond. The α_1 , α_2 - δ , β , and γ stoichiometry is 1:1:1:1. The α_2 -subunit is glycosylated, but α_1 is not. α_1 is homologous with the α -subunit of the voltage-sensitive sodium channel (Chapter 34). The sequence of α_1 contains four internal sequence repeats, each containing six transmembrane helices, one of which is positively charged and is believed to be a voltage sensor (Figure 17.26). The loop between helices 5 and 6 contributes to the pore. These six segments share many similarities with the corresponding segments of the sodium channel. The α_1 -subunit of the DHP receptor in heart muscle is implicated in channel formation and voltage-dependent gating.

The Ca^{2+} -release channel from the terminal cisternae of sarcoplasmic reticulum has been identified by virtue of its high affinity for **ryanodine**, a toxic alkaloid (Figure 17.25). The purified receptor consists of oligomers, containing four or more subunits of a single large polypeptide (565 kD). Electron microscopy reveals that the purified ryanodine receptor (Figure 17.27) is in fact the **foot structure** observed in native muscle tissue. Image reconstructions reveal that the receptor is a square structure with fourfold symmetry, containing a central pore with four radially extending canals (Figure 17.28). These

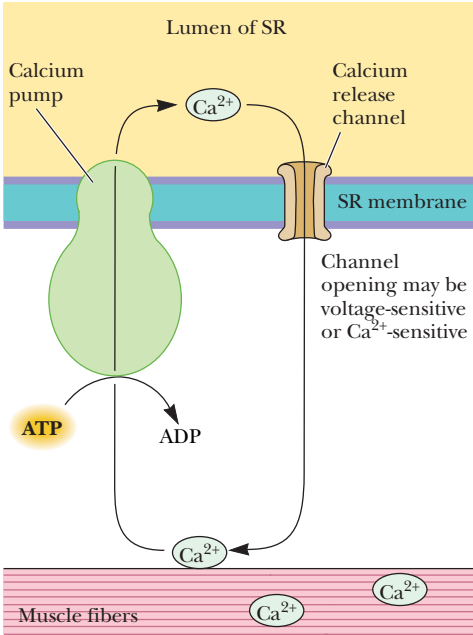


FIGURE 17.24 • Ca^{2+} is the trigger signal for muscle contraction. Release of Ca^{2+} through voltage- or Ca^{2+} -sensitive channels activates contraction. Ca^{2+} pumps induce relaxation by reducing the concentration of Ca^{2+} available to the muscle fibers.

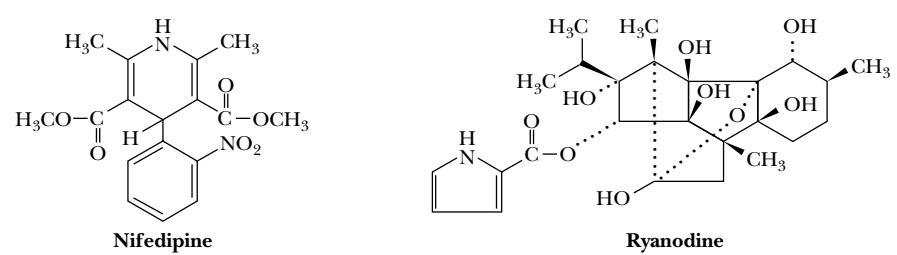


FIGURE 17.25 • The structures of nifedipine and ryanodine. Nifedipine binds with high affinity to the Ca^{2+} -release channels of t-tubules. Ryanodine binds with high affinity to the Ca^{2+} channels of SR terminal cisternae.

DHP-sensitive calcium channel

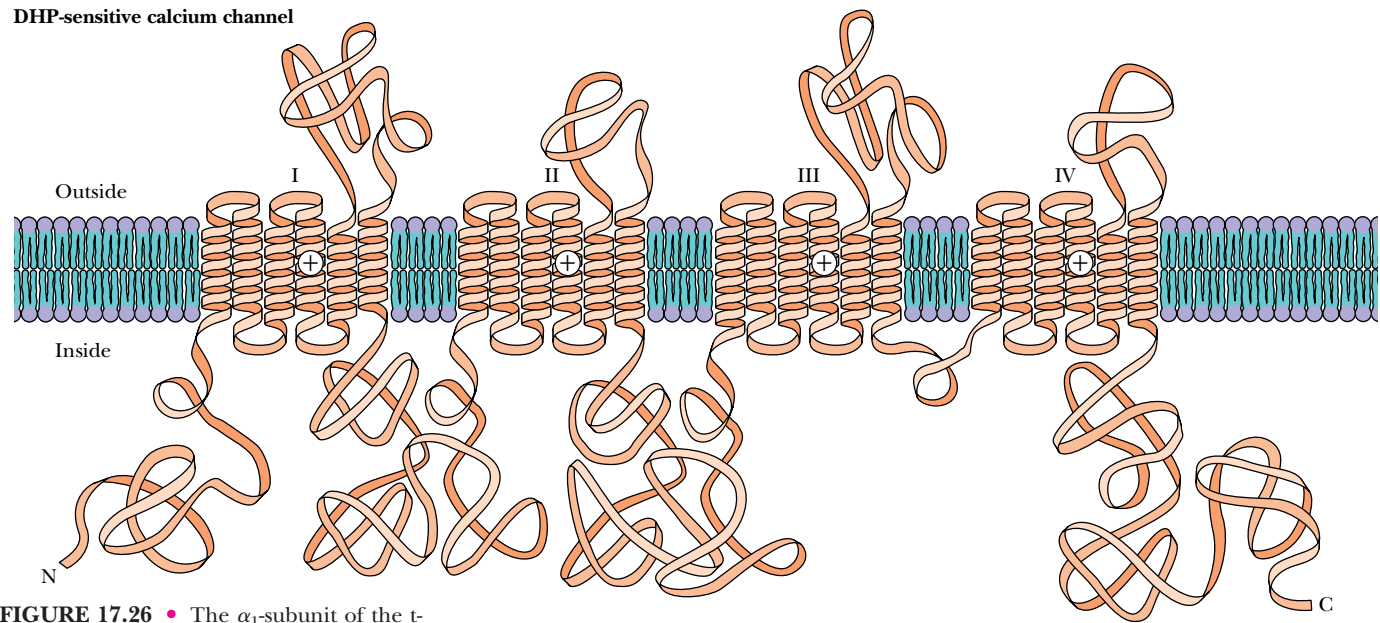


FIGURE 17.26 • The α_1 -subunit of the t-tubule Ca^{2+} channel/DHP receptor contains six peptide segments that may associate to form the Ca^{2+} channel. This Ca^{2+} channel polypeptide is homologous with the voltage-sensitive Na^+ channel of neuronal tissue.

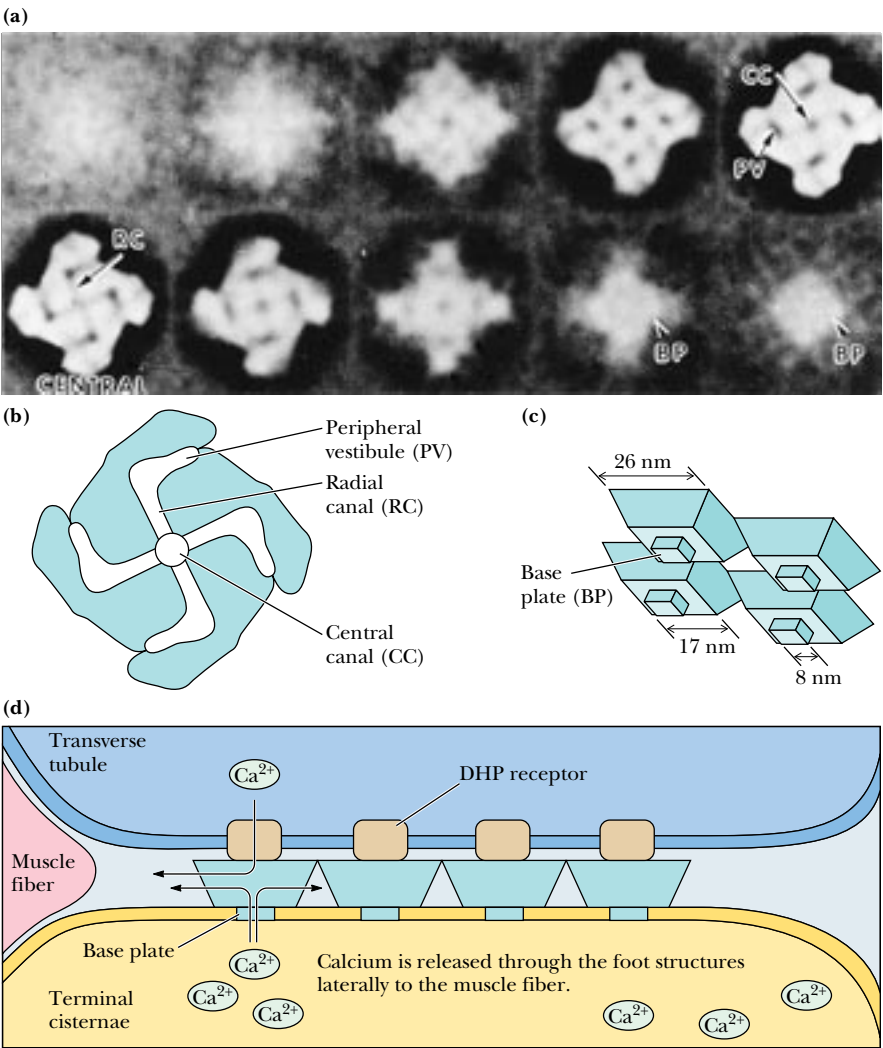


FIGURE 17.27 • (a) Electron micrograph images of foot structures of terminal cisternae. (b, c) Foot structures appear as trapezoids and diamonds on the surface of the membrane. The central canal (CC), radial canals (RC), and peripheral vestibules (PV) are indicated. (d) The relationship between the foot structures, t-tubule, terminal cisternae, and muscle fiber. (Photo courtesy of Sidney Fleischer, Vanderbilt University)



FIGURE 17.28 • Image reconstructions of the junctional channel complex of a foot structure. (Photo courtesy of Sidney Fleischer, Vanderbilt University)

radial canals each extend to openings in the periphery of the structure and are therefore contiguous with the myoplasm.

So how do the foot structures effect the release of Ca^{2+} from the terminal cisternae of the SR? The feet that join the t-tubules and the terminal cisternae of the SR are approximately 16 nm thick. The feet apparently function by first sensing either a voltage-dependent conformation change (skeletal muscle) or the transport of Ca^{2+} across the voltage-sensitive Ca^{2+} channel (heart muscle) of the t-tubule and then facilitating the release of large amounts of Ca^{2+} through the foot structure from the SR. The reconstructed image (Figure 17.28) for the foot structure suggests a possible pathway for Ca^{2+} transport from the lumen of the SR to the myoplasm via the ryanodine receptor. A Ca^{2+} - or voltage-dependent conformation change may serve to gate open the central canal of the foot structure. On entering the central canal, calcium ions move outward through the radial canals to the outer vestibule regions and into the myoplasm adjacent to the triad junctions, where binding to the muscle fibers induces contraction.

Regulation of Contraction by Ca^{2+}

Early in this chapter, the importance of Ca^{2+} ion as the triggering signal for muscle contraction was described. Ca^{2+} is the intermediary signal that allows striated muscle to respond to motor nerve impulses (Figure 17.24). The importance of Ca^{2+} as a contraction signal was understood in the 1940s, but it remained for Setsuro Ebashi, a pioneer of muscle research, to show in the early 1960s that the Ca^{2+} signal is correctly interpreted by muscle only when tropomyosin and the troponins are present. Specifically, actomyosin prepared from pure preparations of actin and myosin (thus containing no tropomyosin and troponins) was observed to contract when ATP was added, even in the absence of Ca^{2+} . However, actomyosin prepared directly from whole muscle would contract in the presence of ATP only when Ca^{2+} was added. Clearly the muscle extracts contained a factor that conferred normal Ca^{2+} sensitivity to actomyosin. The factor turned out to be the tropomyosin–troponin complex.

Actin thin filaments consist of actin, tropomyosin, and the troponins in a 7:1:1 ratio (Figure 17.15). Each tropomyosin molecule spans seven actin molecules, lying along the thin filament groove, between pairs of actin monomers. As shown in a cross-section view in Figure 17.29, in the absence of Ca^{2+} , troponin I is thought to interact directly with actin to prevent the interaction of actin with myosin S1 heads. Troponin I and troponin T interact with tropomyosin to keep tropomyosin away from the groove between adjacent actin

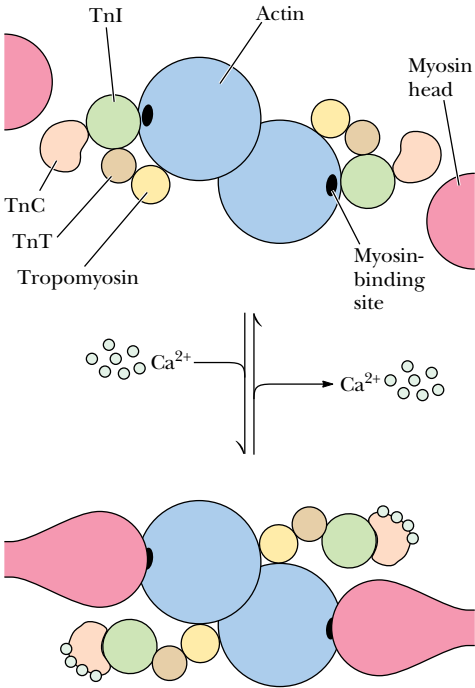


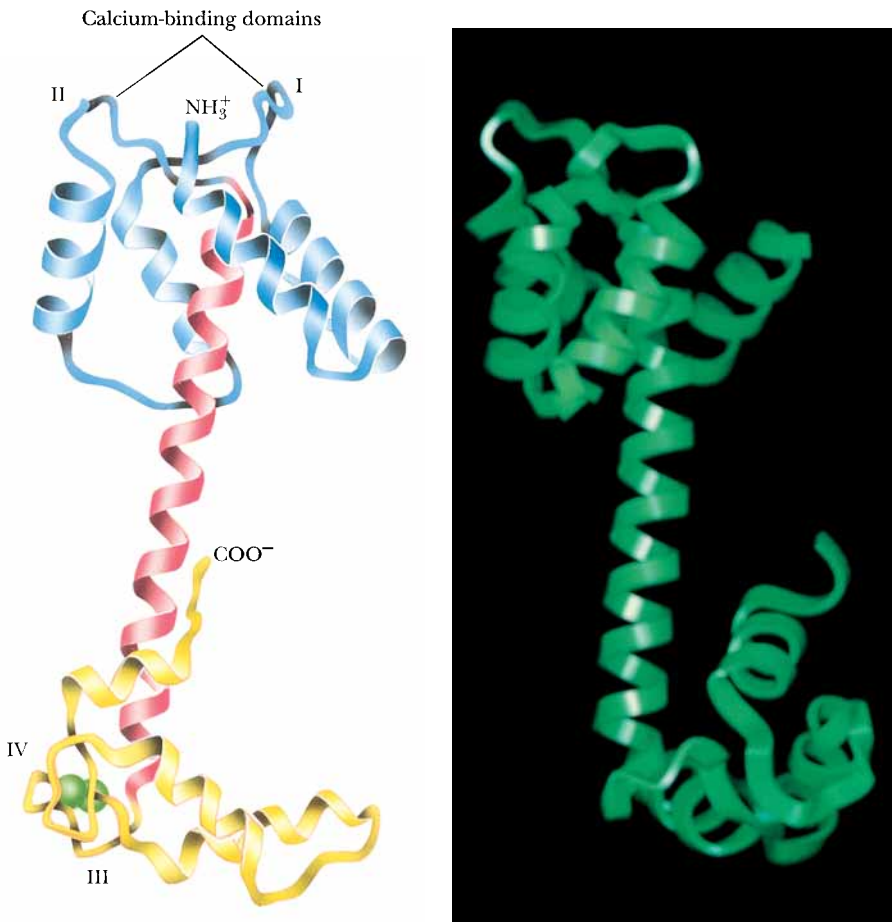
FIGURE 17.29 • A drawing of the thick and thin filaments of skeletal muscle in cross-section showing the changes that are postulated to occur when Ca^{2+} binds to troponin C.

monomers. However, the binding of Ca^{2+} ions to troponin C appears to increase the binding of troponin C to troponin I, simultaneously decreasing the interaction of troponin I with actin. As a result, tropomyosin slides deeper into the actin-thin filament groove, exposing myosin-binding sites on actin, and initiating the muscle contraction cycle (Figure 17.23). Because the troponin complexes can interact only with every seventh actin in the thin filament, the conformational changes that expose myosin-binding sites on actin may well be cooperative. Binding of an S1 head to an actin may displace tropomyosin and the troponin complex from myosin-binding sites on adjacent actin subunits.

The Interaction of Ca^{2+} with Troponin C

There are four Ca^{2+} -binding sites on troponin C—two high-affinity sites on the carboxy-terminal end of the molecule, labeled III and IV in Figure 17.30, and two low-affinity sites on the amino-terminal end, labeled I and II. Ca^{2+} binding to sites III and IV is sufficiently strong ($K_D = 0.1 \mu\text{M}$) that these sites are presumed to be filled under resting conditions. Sites I and II, however, where the K_D is approximately $10 \mu\text{M}$, are empty in resting muscle. The rise of Ca^{2+} levels when contraction is signaled leads to the filling of sites I and II, causing a conformational change in the amino-terminal domain of TnC. This conformational change apparently facilitates a more intimate binding of TnI to TnC that involves the C helix, and also possibly the E helix of TnC. The increased interaction between TnI and TnC results in a decreased interaction between TnI and actin.

FIGURE 17.30 • (a) A ribbon diagram and (b) a molecular graphic showing two slightly different views of the structure of troponin C. Note the long α -helical domain connecting the N-terminal and C-terminal lobes of the molecule.



The Structure of Cardiac and Smooth Muscle

The structure of heart myocytes is different from that of skeletal muscle fibers. Heart myocytes are approximately 50 to 100 μm long and 10 to 20 μm in diameter. The t-tubules found in heart tissue have a fivefold larger diameter than those of skeletal muscle. The number of t-tubules found in cardiac muscle differs from species to species. Terminal cisternae of mammalian cardiac muscle can associate with other cellular elements to form **dyads** as well as triads. The association of terminal cisternae with the sarcolemma membrane in a dyad structure is called a **peripheral coupling**. The terminal cisternae may also form dyad structures with t-tubules that are called **internal couplings** (Figure 17.31). As with skeletal muscle, foot structures form the connection between the terminal cisternae and t-tubule membranes.

In higher animals, large percentages of the terminal cisternae of cardiac muscle are not associated with t-tubules at all. For SR of this type, Ca^{2+} release must occur by a different mechanism from that found in skeletal muscle. In this case, it appears that Ca^{2+} leaking through sarcolemmal Ca^{2+} channels can trigger the release of even more Ca^{2+} from the SR. This latter process is called **Ca^{2+} -induced Ca^{2+} release** (abbreviated **CICR**).

The Structure of Smooth Muscle Myocytes

The myocytes of smooth muscle are approximately 100 to 500 μm in length and only 2 to 6 μm in diameter. Smooth muscle contains very few t-tubules and much less SR than skeletal muscle. The Ca^{2+} that stimulates contraction in smooth muscle cells is predominantly extracellular in origin. This Ca^{2+} enters the cell through Ca^{2+} channels in the sarcolemmal membrane that can be opened by electrical stimulation, or by the binding of hormones or drugs. The contraction response time of smooth muscle cells is very slow compared with that of skeletal and cardiac muscle.

The Mechanism of Smooth Muscle Contraction

Vertebrate organisms employ smooth muscle myocytes for long, slow, and involuntary contractions in various organs, including large blood vessels; intestinal walls; and, in the female, the uterus. Smooth muscle contains no troponin complex; thin filaments consist only of actin and tropomyosin. Despite the absence of troponins, smooth muscle contraction is dependent on Ca^{2+} , which activates **myosin light chain kinase (MLCK)**, an enzyme that phosphorylates LC2, the regulatory light chain of myosin. Contraction of smooth muscle is initiated by phosphorylation of LC2, and dephosphorylation causes relaxation of smooth muscle tissue.

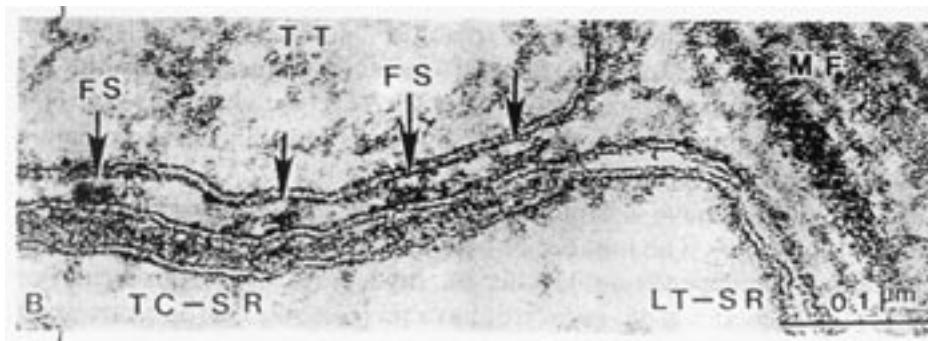


FIGURE 17.31 • Electron micrograph of a dog heart muscle. The terminal cisterna of the SR (TC-SR) is associated with the t-tubule (TT) by means of foot structures (FS), forming a dyad junction. MF indicates the location of myofilaments. LT-SR signifies the longitudinal tubule of the SR. (From Fleischer, S., and Inui, M., 1989. *Annual Review of Biophysics and Biophysical Chemistry* 18:333–364.)

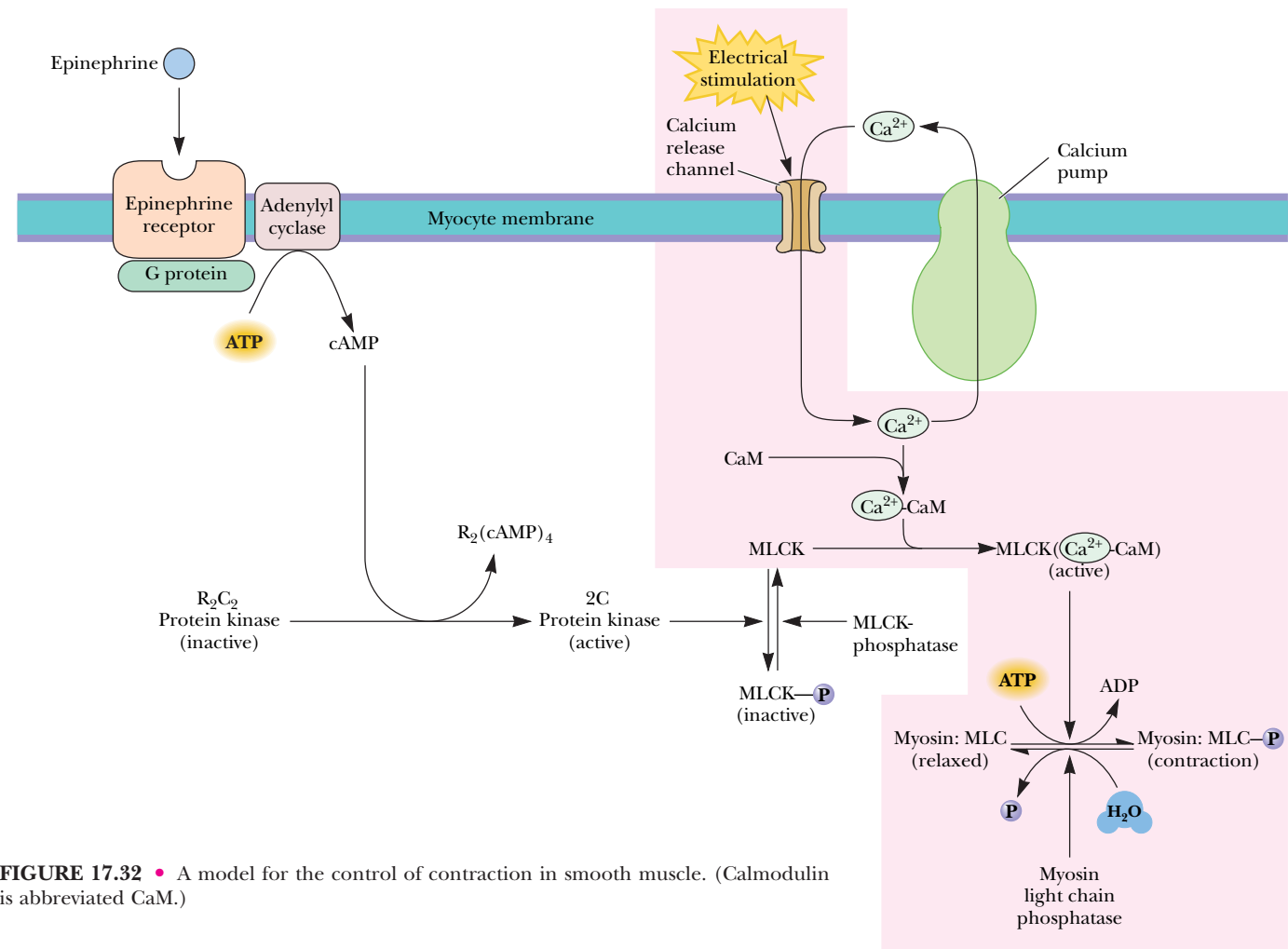


FIGURE 17.32 • A model for the control of contraction in smooth muscle. (Calmodulin is abbreviated CaM.)

The mechanism of this contraction process is shown in Figure 17.32. Smooth muscle myocytes have a resting $[Ca^{2+}]$ of approximately $0.1\ \mu M$. Electrical stimulation (by the autonomic or involuntary nervous system) opens Ca^{2+} channels in the sarcolemmal membrane, allowing $[Ca^{2+}]$ to rise to about $10\ \mu M$, a concentration at which Ca^{2+} binds readily to **calmodulin** (see Chapter 34). Binding of the Ca^{2+} -calmodulin complex to MLCK activates the kinase reaction, phosphorylating LC2 and stimulating smooth muscle contraction. Export of Ca^{2+} by the plasma membrane Ca^{2+} -ATPase returns Ca^{2+} to its resting level, deactivating MLCK. Smooth muscle relaxation then occurs through the action of **myosin light chain phosphatase**, which dephosphorylates LC2. This reaction is relatively slow, and smooth muscle contractions are typically more sustained and dissipate more slowly than those of striated muscle.

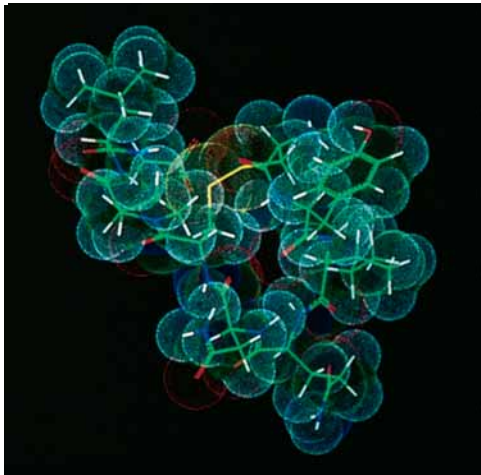
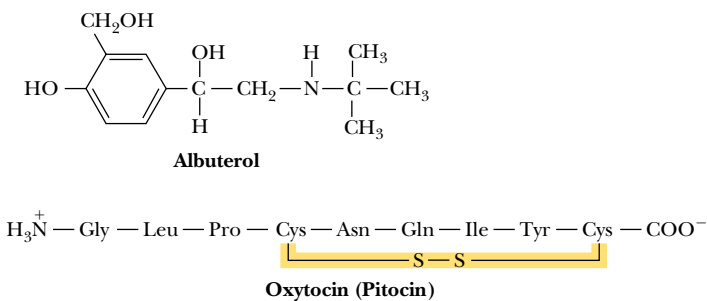
Smooth muscle contractions are subject to the actions of hormones and related agents. As shown in Figure 17.32, binding of the hormone **epinephrine** to smooth muscle receptors activates an intracellular **adenylyl cyclase** reaction that produces cyclic AMP (cAMP). The cAMP serves to activate a protein kinase that phosphorylates the myosin light chain kinase. The phosphorylated MLCK has a lower affinity for the Ca^{2+} -calmodulin complex and thus is physiologically inactive. Reversal of this inactivation occurs via **myosin light chain kinase phosphatase**.

HUMAN BIOCHEMISTRY

Smooth Muscle Effectors Are Useful Drugs

The action of epinephrine and related agents forms the basis of therapeutic control of smooth muscle contraction. Breathing disorders, including asthma and various allergies, can result from excessive contraction of bronchial smooth muscle tissue. Treatment with epinephrine, whether by tablets or aerosol inhalation, inhibits MLCK and relaxes bronchial muscle tissue. More specific **bronchodilators**, such as **albuterol** (see figure), act more selec-

tively on the lungs and avoid the undesirable side effects of epinephrine on the heart. Albuterol is also used to prevent premature labor in pregnant women, owing to its relaxing effect on uterine smooth muscle. Conversely, **oxytocin**, known also as **pitocin**, stimulates contraction of uterine smooth muscle. This natural secretion of the pituitary gland is often administered to induce labor.



 The structure of oxytocin.

17.4 • A Proton Gradient Drives the Rotation of Bacterial Flagella

Bacterial cells swim and move by rotating their flagella. The flagella of *Escherichia coli* are helical filaments about 10,000 nm (10 μm) in length and 15 nm in diameter. The direction of rotation of these filaments affects the movements of the cell. When the half-dozen filaments on the surface of the bacterial cell rotate in a counterclockwise direction, they twist and bundle together and rotate in a concerted fashion, propelling the cell through the medium. (On the other hand, clockwise-rotating flagella cannot bundle together and under such conditions the cell merely tumbles and moves erratically.)

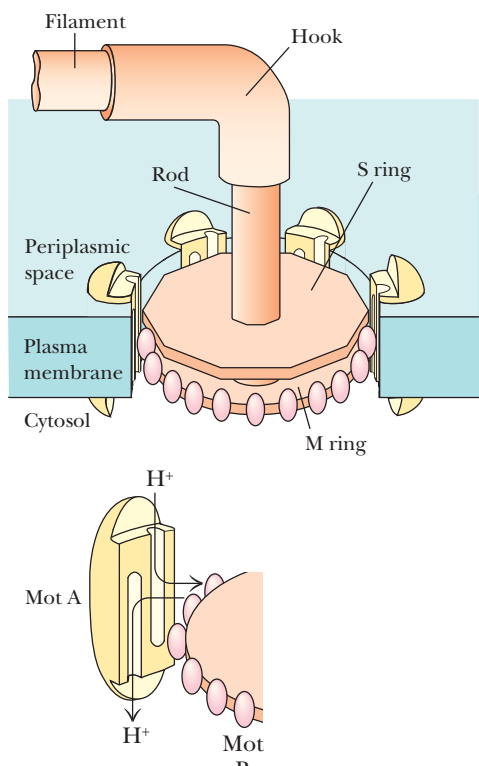


FIGURE 17.33 • A model of the flagellar motor assembly of *Escherichia coli*. The M ring carries an array of about 100 motB proteins at its periphery. These juxtapose with motA proteins in the protein complex that surrounds the ring assembly. Motion of protons through the motA/motB complexes drives the rotation of the rings and the associated rod and helical filament.

The rotations of bacterial flagellar filaments are the result of the rotation of motor protein complexes in the bacterial plasma membrane. The flagellar motor consists of at least two rings (including the M ring and the S ring) with diameters of about 25 nm assembled around and connected rigidly to a rod attached in turn to the helical filament (Figure 17.33). The rings are surrounded by a circular array of membrane proteins. In all, at least 40 genes appear to code for proteins involved in this magnificent assembly. One of these, the motB protein, lies on the edge of the M ring, where it interacts with the motA protein, located in the membrane protein array and facing the M ring.

In contrast to the many other motor proteins described in this chapter, a proton gradient, not ATP hydrolysis, drives the flagellar motor. The concentration of protons, $[H^+]$, outside the cell is typically higher than that inside the cell. Thus, there is a thermodynamic tendency for protons to move into the cell. The motA and motB proteins together form a proton shuttling device that is coupled to motion of the motor disks. Proton movement into the cell through this protein complex or “channel” drives the rotation of the flagellar motor. A model for this coupling has been proposed by Howard Berg and his coworkers (Figure 17.34). In this model, the motB proteins possess proton exchanging sites—for example, carboxyl groups on aspartate or glutamate residues or imidazole moieties on histidine residues. The motA proteins, on the other hand, possess a pair of “half-channels,” with one half-channel facing the inside of the cell and the other facing the outside. In Berg’s model, the outside edges of the motA channel protein cannot move past a proton-exchanging site on motB when that site has a proton bound, and the center of the channel protein cannot move past an exchange site when that site is empty. As shown in Figure 17.34, these constraints lead to coupling between proton translocation and rotation of the flagellar filament. For example, imagine that a proton has entered the outside channel of motA and is bound to an exchange site on motB (Figure 17.34a). An oscillation by motA, linked elastically to the cell wall, can then position the inside channel over the proton at the exchange site (Figure 17.34b), whereupon the proton can travel through the inside channel and into the cell, while another proton travels up the outside channel to bind to an adjacent exchange site. The restoring force acting on the channel protein then pulls the motA/motB complex to the left as shown (Figure 17.34c), leading to counterclockwise rotation of the disk, rod, and helical filament. The flagellar motor is driven entirely by the proton gradient. Thus, a reversal of the proton gradient (which would occur, for example, if the external medium became alkaline) would drive the flagellar filaments in a clockwise direction. Extending this picture of a single motA/motB complex to the

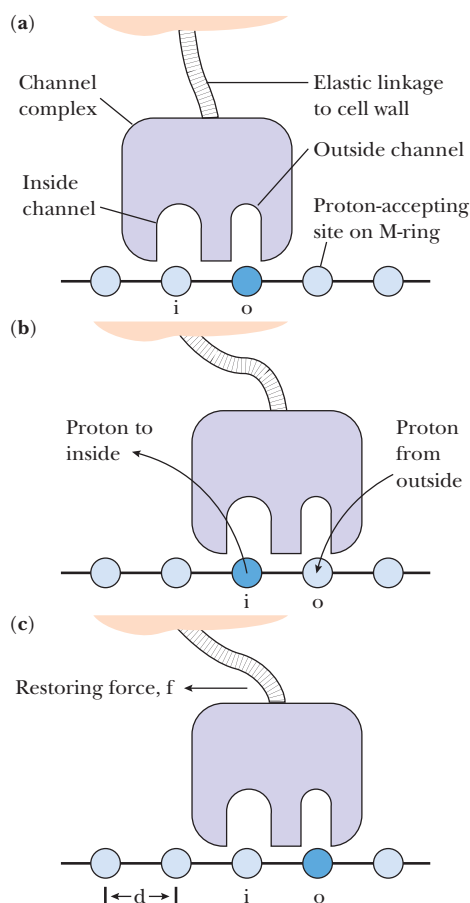


FIGURE 17.34 • Howard Berg’s model for coupling between transmembrane proton flow and rotation of the flagellar motor. A proton moves through an outside channel to bind to an exchange site on the M ring. When the channel protein slides one step around the ring, the proton is released and flows through an inside channel and into the cell, while another proton flows into the outside channel to bind to an adjacent exchange site. When the motA channel protein returns to its original position under an elastic restoring force, the associated motB protein moves with it, causing a counterclockwise rotation of the ring, rod, and helical filament. (Adapted from Meister, M., Caplan, S. R., and Berg, H. C., 1989. Dynamics of a tightly coupled mechanism for flagellar rotation. *Biophysical Journal* 55:905–914)

whole motor disk array, one can imagine the torrent of protons that pass through the motor assembly to drive flagellar rotation at a typical speed of 100 rotations per second. Berg estimates that the M ring carries 100 motB proton-exchange sites, and various models predict that 800 to 1200 protons must flow through the complex during a single rotation of the flagellar filament!

PROBLEMS

1. The cheetah is generally regarded as nature's fastest mammal, but another amazing athlete in the animal kingdom (and almost as fast as the cheetah) is the pronghorn antelope, which roams the plains of Wyoming. Whereas the cheetah can maintain its top speed of 70 mph for only a few seconds, the pronghorn antelope can run at 60 mph for about an hour! (It is thought to have evolved to do so in order to elude now-extinct ancestral cheetahs that lived in North America.) What differences would you expect in the muscle structure and anatomy of pronghorn antelopes that could account for their remarkable speed and endurance?
2. An ATP analog, β,γ -methylene-ATP, in which a $-\text{CH}_2-$ group replaces the oxygen atom between the β - and γ -phosphorus atoms, is a potent inhibitor of muscle contraction. At which step in the contraction cycle would you expect β,γ -methylene-ATP to block contraction?
3. ATP stores in muscle are augmented or supplemented by stores of phosphocreatine. During periods of contraction, phosphocreatine is hydrolyzed to drive the synthesis of needed ATP in the creatine kinase reaction:



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Muscle cells contain two different isozymes of creatine kinase, one in the mitochondria and one in the sarcoplasm. Explain.

4. *Rigor* is a muscle condition in which muscle fibers, depleted of ATP and phosphocreatine, develop a state of extreme rigidity and cannot be easily extended. (In death, this state is called *rigor mortis*, the rigor of death.) From what you have learned about muscle contraction, explain the state of rigor in molecular terms.

5. Skeletal muscle can generate approximately 3 to 4 kg of tension or force per square centimeter of cross-sectional area. This number is roughly the same for all mammals. Because many human muscles have large cross-sectional areas, the force that these muscles can (and must) generate is prodigious. The gluteus maximus (on which you are probably sitting as you read this) can generate a tension of 1200 kg! Estimate the cross-sectional area of all of the muscles in your body and the total force that your skeletal muscles could generate if they all contracted at once.

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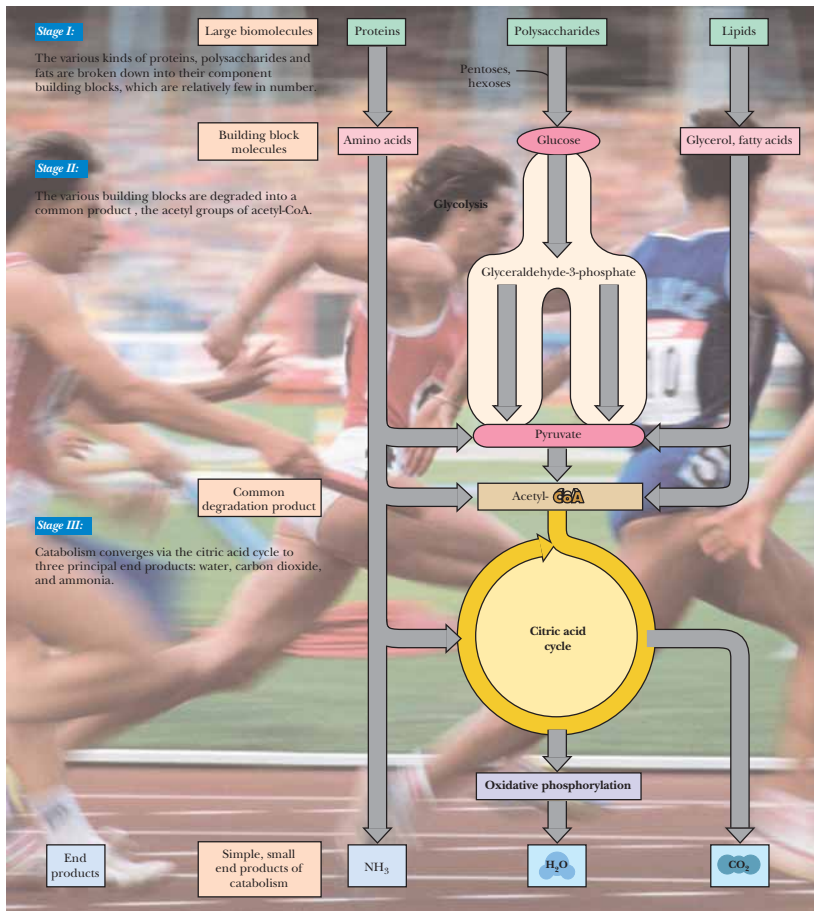
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Part III

Metabolism and Its Regulation



Metabolism accomplishes, among other things, the conversion of food energy into the energy of motion. Regulation of metabolism allows the abrupt transition from a state of rest to the breath-taking power and grace of athletic competition.
(European champs by Paul J. Sutton/Duomo; line art by J/B Woolsey Associates)

All is flux, nothing stays still.
Nothing endures but change.
HERACLITUS (c. 540–c. 480 B.C.)

OUTLINE

- 18.1 • Virtually All Organisms Have the Same Basic Set of Metabolic Pathways
 - 18.2 • Metabolism Consists of Catabolism (Degradative Pathways) and Anabolism (Biosynthetic Pathways)
 - 18.3 • Experimental Methods To Reveal Metabolic Pathways
 - 18.4 • Nutrition
- Special Focus: Vitamins

Chapter 18

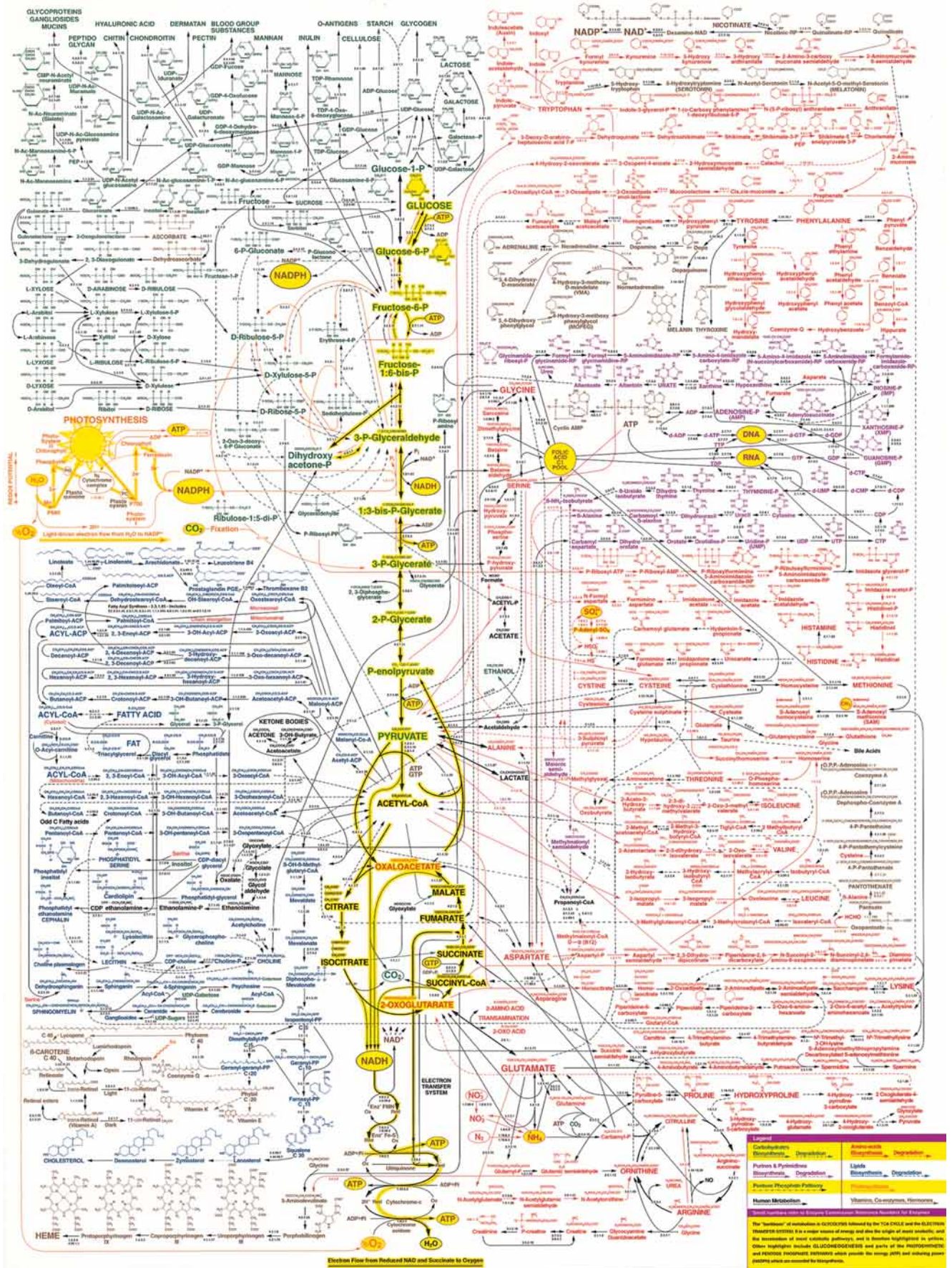
Metabolism—An Overview



Anise swallowtail butterfly (*Papilio zelicanus*) with its pupal case. Metamorphosis of butterflies is a dramatic example of metabolic change.
(© 1986 Peter Bryant/Biological Photo Service)

The word *metabolism* derives from the Greek word for “change.” **Metabolism** represents the sum of the chemical changes that convert **nutrients**, the “raw materials” necessary to nourish living organisms, into energy and the chemically complex finished products of cells. Metabolism consists of literally hundreds of enzymatic reactions organized into discrete pathways. These pathways proceed in a stepwise fashion, transforming substrates into end products through many specific chemical **intermediates**. Metabolism is sometimes referred to as **intermediary metabolism** to reflect this aspect of the process. Metabolic maps (Figure 18.1) portray virtually all of the principal reactions of the intermediary metabolism of carbohydrates, lipids, amino acids, nucleotides,

FIGURE 18.1 • A metabolic map, indicating the reactions of intermediary metabolism and the enzymes that catalyze them. Over 500 different chemical intermediates, or metabolites, and a greater number of enzymes are represented here. (© 1997 20th edition, designed by and courtesy of D. E. Nicholson, University of Leeds, U. K., and the Sigma Chemical Co.)



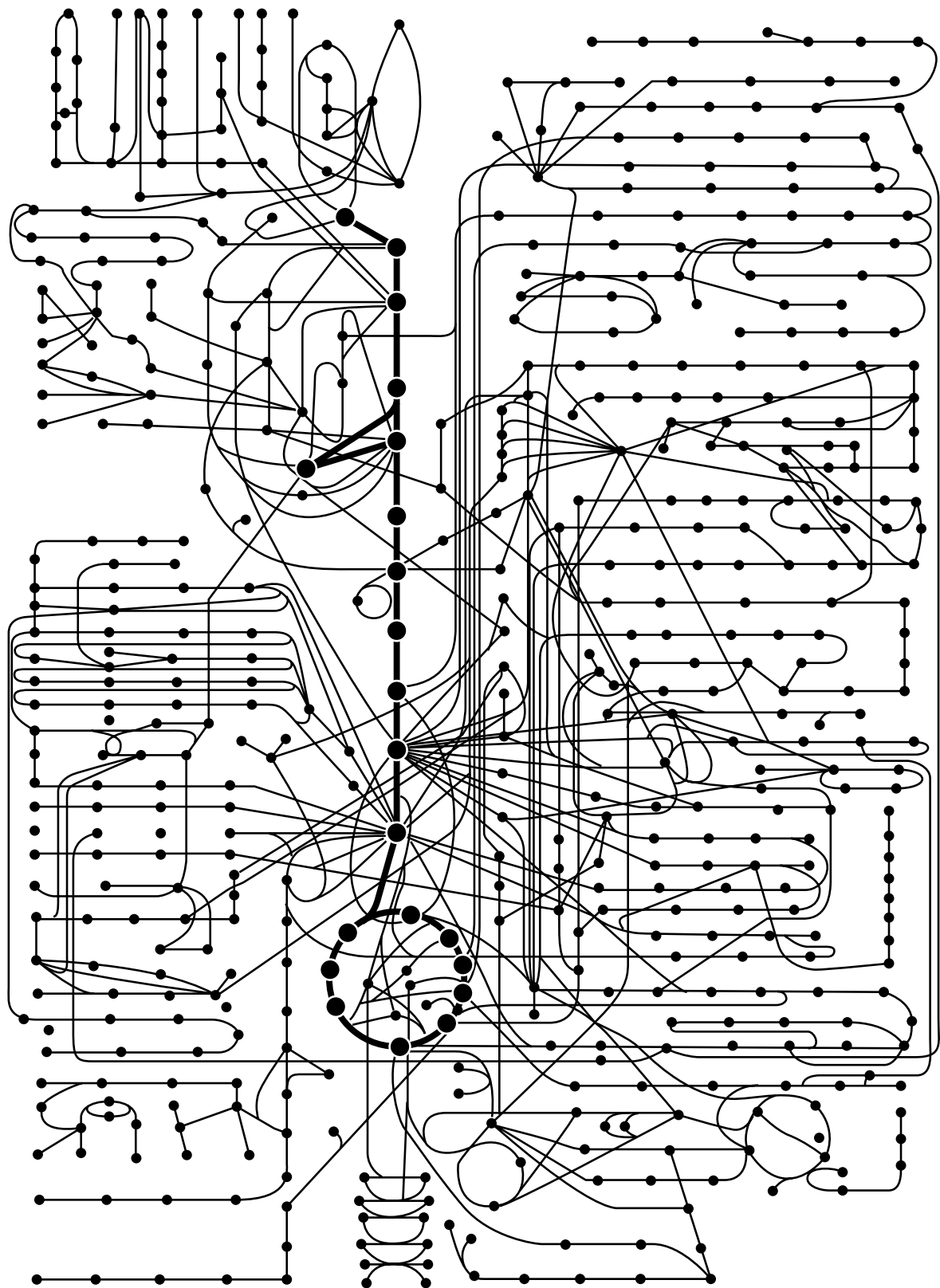


FIGURE 18.2 • The metabolic map as a set of dots and lines. The heavy dots and lines trace the central energy-releasing pathways known as glycolysis and the citric acid cycle. (Adapted from Alberts, B., et al., 1989. *Molecular Biology of the Cell*, 2nd ed. New York: Garland Publishing Co.)

and their derivatives. These maps are very complex at first glance and seem to be virtually impossible to learn easily. Despite their appearance, these maps become easy to follow once the major metabolic routes are known and their functions are understood. The underlying order of metabolism and the important interrelationships between the various pathways then appear as simple patterns against the seemingly complicated background.

The Metabolic Map as a Set of Dots and Lines

One interesting transformation of the intermediary metabolism map is to represent each intermediate as a black dot and each enzyme as a line (Figure 18.2). Then, the more than 1000 different enzymes and substrates are represented by just two symbols. This chart has about 520 dots (intermediates). Table 18.1 lists the numbers of dots that have one or two or more lines (enzymes) associated with them. Thus, this table classifies intermediates by the number of enzymes that act upon them. A dot connected to just a single line must be either a nutrient, a storage form, an end product, or an excretory product of metabolism. Also, because many pathways tend to proceed in only one direction (that is, they are essentially irreversible under physiological conditions), a dot connected to just two lines is probably an intermediate in only one pathway and has only one fate in metabolism. If three lines are connected to a dot, that intermediate has at least two possible metabolic fates; four lines, three fates; and so on. Note that about 80% of the intermediates connect to only one or two lines and thus have only a limited purpose in the cell. However, many intermediates are subject to a variety of fates. In such instances, the pathway followed is an important regulatory choice. Indeed, whether any substrate is routed down a particular metabolic pathway is the consequence of a regulatory decision made in response to the cell’s (or organism’s) momentary requirements for energy or nutrition. The regulation of metabolism is an interesting and important subject to which we will return often.

Table 18.1

Number of Dots (Intermediates) in the Metabolic Map of Figure 18.2, and the Number of Lines Associated with Them

Lines	Dots
1 or 2	410
3	71
4	20
5	11
6 or more	8

18.1 • Virtually All Organisms Have the Same Basic Set of Metabolic Pathways

One of the great unifying principles of modern biology is that organisms show marked similarity in their major pathways of metabolism. Given the almost unlimited possibilities within organic chemistry, this generality would appear most unlikely. Yet it’s true, and it provides strong evidence that all life has descended from a common ancestral form. All forms of nutrition and almost all metabolic pathways evolved in early prokaryotes prior to the appearance of eukaryotes 1 billion years ago. For example, **glycolysis**, the metabolic pathway by which energy is released from glucose and captured in the form of ATP under anaerobic conditions, is common to almost every cell. It is believed to be the most ancient of metabolic pathways, having arisen prior to the appearance of oxygen in abundance in the atmosphere. All organisms, even those that can synthesize their own glucose, are capable of glucose degradation and ATP synthesis via glycolysis. Other prominent pathways are also virtually ubiquitous among organisms.

Metabolic Diversity

Although most cells have the same basic set of central metabolic pathways, different cells (and, by extension, different organisms) are characterized by the alternative pathways they might express. These pathways offer a wide diversity

of metabolic possibilities. For instance, organisms are often classified according to the major metabolic pathways they exploit to obtain carbon or energy. Classification based on carbon requirements defines two major groups, autotrophs and heterotrophs. **Autotrophs** are organisms that can use just carbon dioxide as their sole source of carbon. **Heterotrophs** require an organic form of carbon, such as glucose, in order to synthesize other essential carbon compounds.

Classification based on energy sources also gives two groups: phototrophs and chemotrophs. **Phototrophs** are *photosynthetic organisms*, which use light as a source of energy. **Chemotrophs** use organic compounds such as glucose or, in some instances, oxidizable inorganic substances such as Fe^{2+} , NO_2^- , NH_4^+ , or elemental sulfur as sole sources of energy. Typically, the energy is extracted through oxidation–reduction reactions. Based on these characteristics, every organism falls into one of four categories (Table 18.2).

Metabolic Diversity Among the Five Kingdoms

Prokaryotes (the kingdom Monera—bacteria) show a greater metabolic diversity than all the four eukaryotic kingdoms (Protocista [previously called Protozoa], Fungi, Plants, and Animals) put together. Prokaryotes are variously chemoheterotrophic, photoautotrophic, photoheterotrophic, or chemoautotrophic. No protocista are chemoautotrophs; fungi and animals are exclusively chemoheterotrophs; plants are characteristically photoautotrophs, although some are heterotrophic in their mode of carbon acquisition.

The Role of O₂ in Metabolism

A further metabolic distinction among organisms is whether or not they can use oxygen as an electron acceptor in energy-producing pathways. Those that can are called **aerobes** or *aerobic organisms*; others, termed **anaerobes**, can subsist without O₂. Organisms for which O₂ is obligatory for life are called **obligate aerobes**; humans are an example. Some species, the so-called **facultative anaerobes**, can adapt to anaerobic conditions by substituting other electron acceptors for O₂ in their energy-producing pathways; *Escherichia coli* is an exam-

Table 18.2

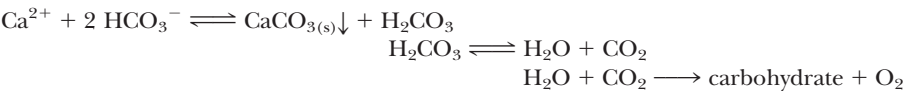
Metabolic Classification of Organisms According to Their Carbon and Energy Requirements				
Classification	Carbon Source	Energy Source	Electron Donors	Examples
<i>Photoautotrophs</i>	CO ₂	Light	H ₂ O, H ₂ S, S, other inorganic compounds	Green plants, algae, cyanobacteria, photosynthetic bacteria
<i>Photoheterotrophs</i>	Organic compounds	Light	Organic compounds	Nonsulfur purple bacteria
<i>Chemoautotrophs</i>	CO ₂	Oxidation–reduction reactions	Inorganic compounds: H ₂ , H ₂ S, NH ₄ ⁺ , NO ₂ [−] , Fe ²⁺ , Mn ²⁺	Nitrifying bacteria; hydrogen, sulfur, and iron bacteria
<i>Chemoheterotrophs</i>	Organic compounds	Oxidation–reduction reactions	Organic compounds, e.g., glucose	All animals, most microorganisms, nonphotosynthetic plant tissue such as roots, photosynthetic cells in the dark

A
D E E P E R L O O K

Calcium Carbonate—A Biological Sink for CO₂

A major biological sink for CO₂ that is often overlooked is the calcium carbonate shells of corals, molluscs, and crustacea. These invertebrate animals deposit CaCO₃ in the form of protective exoskeletons. In some invertebrates, such as the *scleractinians* (hard corals) of tropical seas, photosynthetic dinoflagellates (kingdom Protocista) known as *zooxanthellae* live within the ani-

mal cells as **endosymbionts**. These phototrophic cells use light to drive the resynthesis of organic molecules from CO₂ released (as bicarbonate ion) by the animal’s metabolic activity. In the presence of Ca²⁺, the photosynthetic CO₂ fixation “pulls” the deposition of CaCO₃, as summarized in the following coupled reactions:



ple. Yet others cannot use oxygen at all and are even poisoned by it; these are the **obligate anaerobes**. *Clostridium botulinum*, the bacterium that produces botulin toxin, is representative.

The Flow of Energy in the Biosphere and the Carbon and Oxygen Cycles Are Intimately Related

The primary source of energy for life is the sun. Photoautotrophs utilize light energy to drive the synthesis of organic molecules, such as carbohydrates, from atmospheric CO₂ and water (Figure 18.3). Heterotrophic cells then use these organic products of photosynthetic cells both as fuels and as building blocks, or precursors, for the biosynthesis of their own unique complement of biomolecules. Ultimately, CO₂ is the end product of heterotrophic carbon metabolism, and CO₂ is returned to the atmosphere for reuse by the photoautotrophs. In effect, solar energy is converted to the chemical energy of organic molecules by photoautotrophs, and heterotrophs recover this energy by metabolizing the organic substances. The flow of energy in the biosphere is thus conveyed within the carbon cycle, and the impetus driving the cycle is light energy.

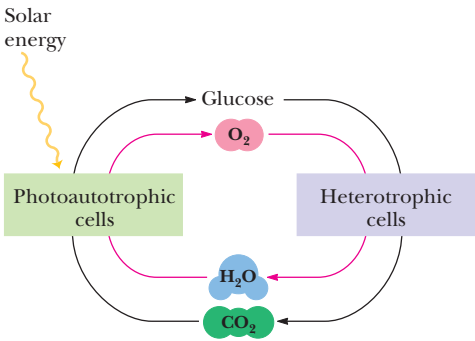


FIGURE 18.3 • The flow of energy in the biosphere is coupled primarily to the carbon and oxygen cycles.

18.2 • Metabolism Consists of Catabolism (Degradative Pathways) and Anabolism (Biosynthetic Pathways)

Metabolism serves two fundamentally different purposes: the generation of energy to drive vital functions and the synthesis of biological molecules. To achieve these ends, metabolism consists largely of two contrasting processes, catabolism and anabolism. *Catabolic pathways are characteristically energy-yielding, whereas anabolic pathways are energy-requiring.* **Catabolism** involves the oxidative degradation of complex nutrient molecules (carbohydrates, lipids, and proteins) obtained either from the environment or from cellular reserves. The breakdown of these molecules by catabolism leads to the formation of simpler molecules such as lactic acid, ethanol, carbon dioxide, urea, or ammonia. Catabolic reactions are usually exergonic, and often the chemical energy released is captured in the form of ATP (Chapter 3). Because catabolism is oxidative for the most part, part of the chemical energy may be conserved as

energy-rich electrons transferred to the coenzymes NAD^+ and NADP^+ . These two reduced coenzymes have very different metabolic roles: *NAD^+ reduction is part of catabolism; NADPH oxidation is an important aspect of anabolism.* The energy released upon oxidation of NADH is coupled to the phosphorylation of ADP in aerobic cells, and so NADH oxidation back to NAD^+ serves to generate more ATP ; in contrast, NADPH is the source of the reducing power needed to drive reductive biosynthetic reactions.

Thermodynamic considerations demand that the energy necessary for biosynthesis of any substance exceed the energy available from its catabolism. Otherwise, organisms could achieve the status of perpetual motion machines: A few molecules of substrate whose catabolism yielded more ATP than required for its resynthesis would allow the cell to cycle this substance and harvest an endless supply of energy.

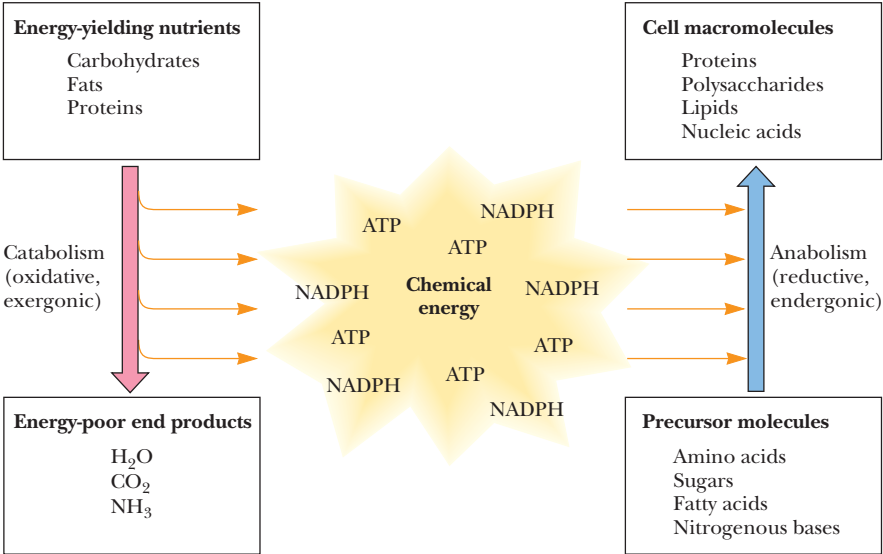
Anabolism Is Biosynthesis

Anabolism is a synthetic process in which the varied and complex biomolecules (proteins, nucleic acids, polysaccharides, and lipids) are assembled from simpler precursors. Such biosynthesis involves the formation of new covalent bonds, and an input of chemical energy is necessary to drive such endergonic processes. The ATP generated by catabolism provides this energy. Furthermore, NADPH is an excellent donor of high-energy electrons for the reductive reactions of anabolism. Despite their divergent roles, anabolism and catabolism are interrelated in that the products of one provide the substrates of the other (Figure 18.4). Many metabolic intermediates are shared between the two processes, and the precursors needed by anabolic pathways are found among the products of catabolism.

Anabolism and Catabolism Are Not Mutually Exclusive

Interestingly, anabolism and catabolism occur simultaneously in the cell. The conflicting demands of concomitant catabolism and anabolism are managed by cells in two ways. First, the cell maintains tight and separate regulation of both catabolism and anabolism, so that metabolic needs are served in an immediate and orderly fashion. Second, competing metabolic pathways are often

FIGURE 18.4 • Energy relationships between the pathways of catabolism and anabolism. Oxidative, exergonic pathways of catabolism release free energy and reducing power that are captured in the form of ATP and NADPH , respectively. Anabolic processes are endergonic, consuming chemical energy in the form of ATP and using NADPH as a source of high-energy electrons for reductive purposes.



localized within different cellular compartments. Isolating opposing activities within distinct compartments, such as separate organelles, avoids interference between them. For example, the enzymes responsible for catabolism of fatty acids, the *fatty acid oxidation pathway*, are localized within mitochondria. In contrast, *fatty acid biosynthesis* takes place in the cytosol. In subsequent chapters, we shall see that the particular molecular interactions responsible for the regulation of metabolism become important to an understanding and appreciation of metabolic biochemistry.

Modes of Enzyme Organization in Metabolic Pathways

The individual metabolic pathways of anabolism and catabolism consist of sequential enzymatic steps (Figure 18.5). Several types of organization are possible. The enzymes of some multienzyme systems may exist as physically separate, soluble entities, with diffusing intermediates (Figure 18.5a). In other instances, the enzymes of a pathway are collected to form a discrete *multienzyme complex*, and the substrate is sequentially modified as it is passed along from enzyme to enzyme (Figure 18.5b). This type of organization has the advantage that intermediates are not lost or diluted by diffusion. In a third pattern of organization, the enzymes common to a pathway reside together as a *membrane-bound system* (Figure 18.5c). In this case, the enzyme participants (and perhaps the substrates as well) must diffuse in just the two dimensions of the membrane to interact with their neighbors.

As research reveals the ultrastructural organization of the cell in ever greater detail, more and more of the so-called soluble enzyme systems are found to be physically united into functional complexes. Thus, in many (perhaps all) metabolic pathways, the consecutively acting enzymes are associated into stable multienzyme complexes that are sometimes referred to as **metabolons**, a word meaning “units of metabolism.”

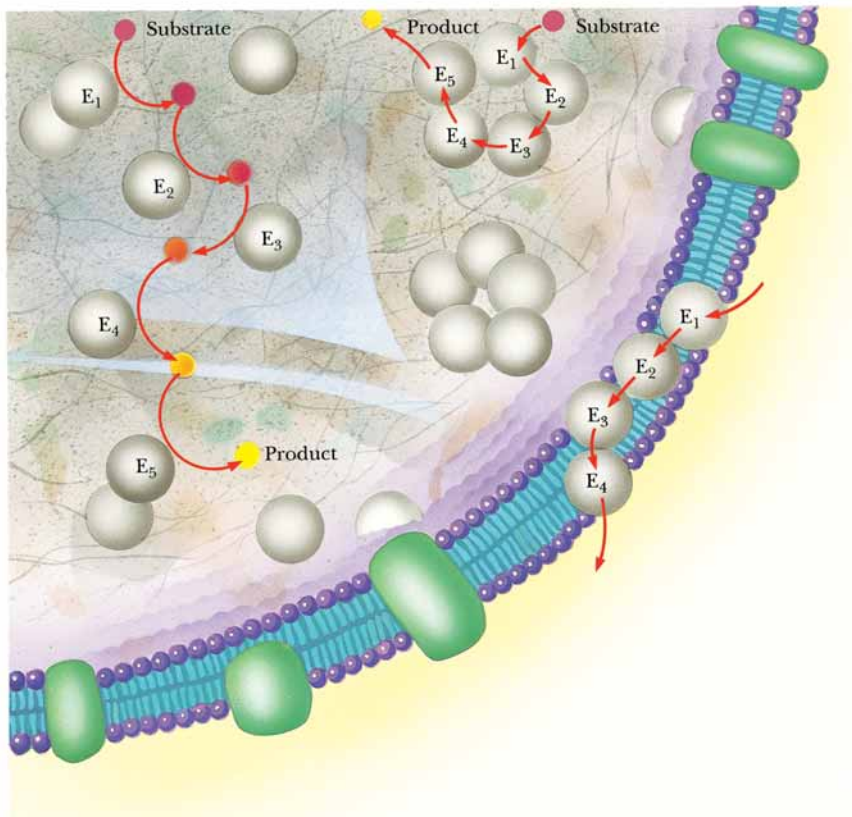


FIGURE 18.5 • Schematic representation of types of multienzyme systems carrying out a metabolic pathway: (a) Physically separate, soluble enzymes with diffusing intermediates. (b) A multienzyme complex. Substrate enters the complex, becomes covalently bound and then sequentially modified by enzymes E₁ to E₅ before product is released. No intermediates are free to diffuse away. (c) A membrane-bound multienzyme system.

The Pathways of Catabolism Converge to a Few End Products

If we survey the catabolism of the principal energy-yielding nutrients (carbohydrates, lipids, and proteins) in a typical heterotrophic cell, we see that the degradation of these substances involves a succession of enzymatic reactions. In the presence of oxygen (*aerobic catabolism*), these molecules are degraded ultimately to carbon dioxide, water, and ammonia. Aerobic catabolism consists of three distinct stages. In **stage 1**, the nutrient macromolecules are broken down into their respective building blocks. Given the great diversity of macromolecules, these building blocks represent a rather limited number of products. Proteins yield up their 20 component amino acids, polysaccharides give rise to carbohydrate units that are convertible to glucose, and lipids are broken down into glycerol and fatty acids (Figure 18.6).

In **stage 2**, the collection of product building blocks generated in stage 1 is further degraded to yield an even more limited set of simpler metabolic intermediates. The deamination of amino acids leaves α -keto acid carbon skeletons. Several of these α -keto acids are citric acid cycle intermediates and are fed directly into stage 3 catabolism via this cycle. Others are converted either to the three-carbon α -keto acid *pyruvate* or to the acetyl groups of *acetyl-coenzyme A* (acetyl-CoA). Glucose and the glycerol from lipids also generate pyruvate, whereas the fatty acids are broken into two-carbon units that appear as *acetyl-CoA*. Because pyruvate also gives rise to acetyl-CoA, we see that the degradation of macromolecular nutrients converges to a common end product, acetyl-CoA (Figure 18.6).

The combustion of the acetyl groups of acetyl-CoA by the *citric acid cycle* and *oxidative phosphorylation* to produce CO_2 and H_2O represents **stage 3** of catabolism. The end products of the citric acid cycle, CO_2 and H_2O , are the ultimate waste products of aerobic catabolism. As we shall see in Chapter 20, the oxidation of acetyl-CoA during stage 3 metabolism generates most of the energy produced by the cell.

Anabolic Pathways Diverge, Synthesizing an Astounding Variety of Biomolecules from a Limited Set of Building Blocks

A rather limited collection of simple precursor molecules is sufficient to provide for the biosynthesis of virtually any cellular constituent, be it protein, nucleic acid, lipid, or polysaccharide. All of these substances are constructed from appropriate building blocks via the pathways of anabolism. In turn, the building blocks (amino acids, nucleotides, sugars, and fatty acids) can be generated from metabolites in the cell. For example, amino acids can be formed by amination of the corresponding α -keto acid carbon skeletons, and pyruvate can be converted to hexoses for polysaccharide biosynthesis.

Amphibolic Intermediates

Certain of the central pathways of intermediary metabolism, such as the citric acid cycle, and many metabolites of other pathways have dual purposes—they serve in both catabolism and anabolism. This dual nature is reflected in the designation of such pathways as **amphibolic** rather than solely catabolic or anabolic. In any event, in contrast to catabolism—which converges to the common intermediate, acetyl-CoA—the pathways of anabolism diverge from a small group of simple metabolic intermediates to yield a spectacular variety of cellular constituents.

amphi • from the Greek for “on both sides”

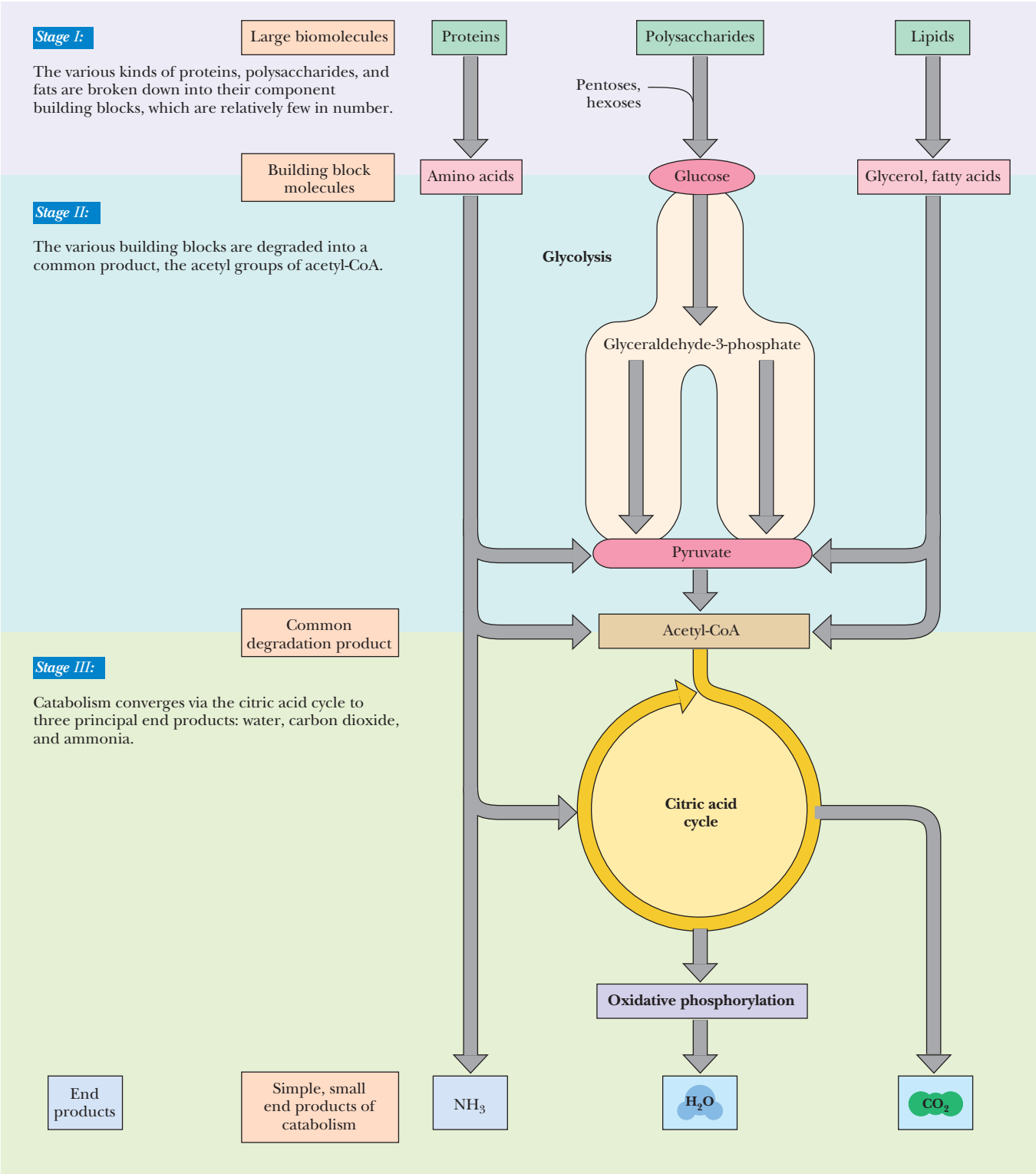


FIGURE 18.6 • The three stages of catabolism. **Stage I:** Proteins, polysaccharides, and lipids are broken down into their component building blocks, which are relatively few in number. **Stage II:** The various building blocks are degraded into the common product, the acetyl groups of acetyl-CoA. **Stage III:** Catabolism converges to three principal end products: water, carbon dioxide, and ammonia.

Corresponding Pathways of Catabolism and Anabolism Differ in Important Ways

The anabolic pathway for synthesis of a given end product usually does not precisely match the pathway used for catabolism of the same substance. Some of the intermediates may be common to steps in both pathways, but different enzymatic reactions and unique metabolites characterize other steps. A good example of these differences is found in a comparison of the catabolism of glucose to pyruvic acid by the pathway of glycolysis and the biosynthesis of glucose from pyruvate by the pathway called *gluconeogenesis*. The glycolytic pathway from glucose to pyruvate consists of 10 enzymes. Although it may seem efficient for glucose synthesis from pyruvate to proceed by a reversal of all 10 steps, gluconeogenesis uses only seven of the glycolytic enzymes in reverse, replacing the remaining three with four enzymes specific to glucose biosynthesis. In similar fashion, the pathway responsible for degrading proteins to amino acids differs from the protein synthesis system, and the oxidative degradation of fatty acids to two-carbon acetyl-CoA groups does not follow the same reaction path as the biosynthesis of fatty acids from acetyl-CoA.

Metabolic Regulation Favors Different Pathways for Oppositely Directed Metabolic Sequences

A second reason for different pathways serving in opposite metabolic directions is that such pathways must be independently regulated. If catabolism and anabolism passed along the same set of metabolic tracks, equilibrium considerations would dictate that slowing the traffic in one direction by inhibiting a particular enzymatic reaction would necessarily slow traffic in the opposite direction. Independent regulation of anabolism and catabolism can be accomplished only if these two contrasting processes move along different routes or, in the case of shared pathways, the rate-limiting steps serving as the points of regulation are catalyzed by enzymes that are unique to each opposing sequence (Figure 18.7).

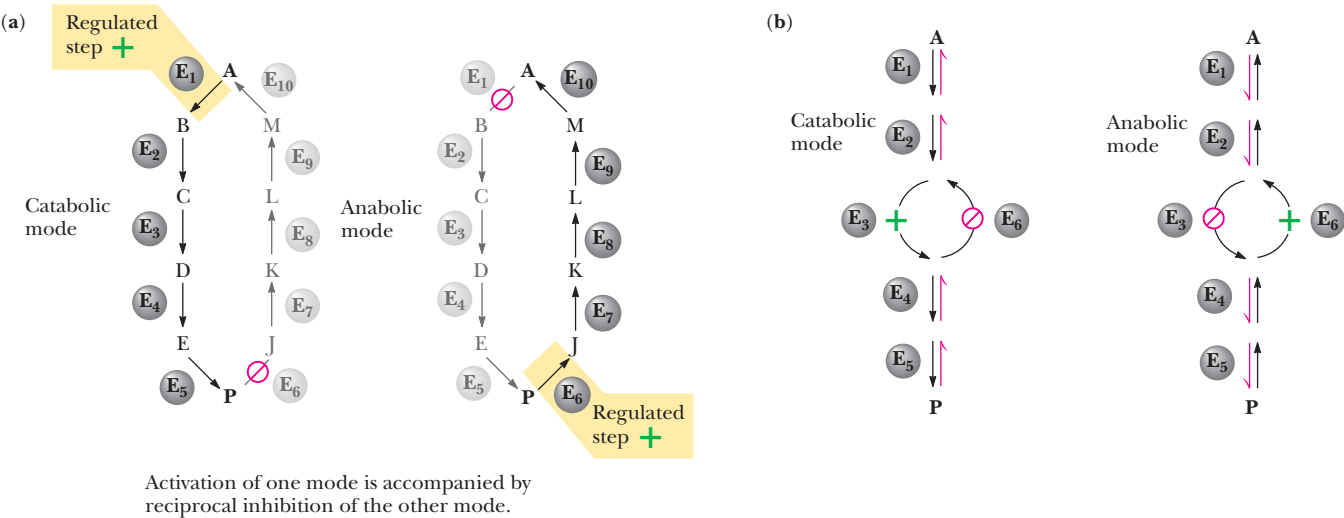


FIGURE 18.7 • Parallel pathways of catabolism and anabolism must differ in at least one metabolic step in order that they can be regulated independently. Shown here are two possible arrangements of opposing catabolic and anabolic sequences between A and P. (a) The parallel sequences proceed via independent routes. (b) Only one reaction has two different enzymes, a catabolic one (E₃) and its anabolic counterpart (E₆). These provide sites for regulation.

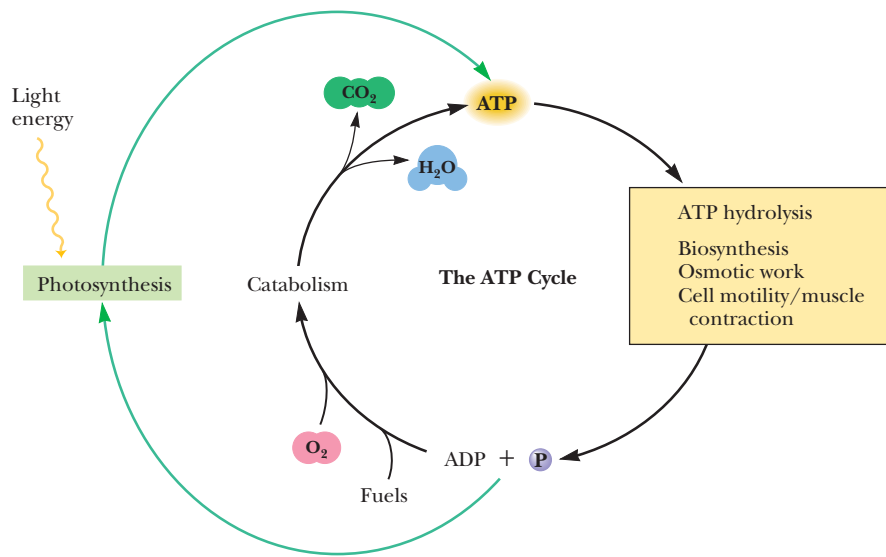


FIGURE 18.8 • The ATP cycle in cells. ATP is formed via photosynthesis in phototrophic cells or catabolism in heterotrophic cells. Energy-requiring cellular activities are powered by ATP hydrolysis, liberating ADP and P_i.

The ATP Cycle

We saw in Chapter 3 that ATP is the energy currency of cells. In phototrophs, ATP is one of the two energy-rich primary products resulting from the transformation of light energy into chemical energy. (The other is NADPH; see the following discussion.) In heterotrophs, the pathways of catabolism have as their major purpose the release of free energy that can be captured in the form of energy-rich phosphoric anhydride bonds in ATP. In turn, ATP provides the energy that drives the manifold activities of all living cells—the synthesis of complex biomolecules, the osmotic work involved in transporting substances into cells, the work of cell motility, the work of muscle contraction. These diverse activities are all powered by energy released in the hydrolysis of ATP to ADP and P_i. Thus, there is an energy cycle in cells where ATP serves as the vessel carrying energy from photosynthesis or catabolism to the energy-requiring processes unique to living cells (Figure 18.8).

NAD⁺ Collects Electrons Released in Catabolism

The substrates of catabolism—proteins, carbohydrates, and lipids—are good sources of chemical energy because the carbon atoms in these molecules are in a relatively reduced state (Figure 18.9). In the oxidative reactions of catabolism, reducing equivalents are released from these substrates, often in the form of **hydride ions** (a proton coupled with two electrons, H[−]). These hydride ions are transferred in enzymatic **dehydrogenase** reactions from the substrates

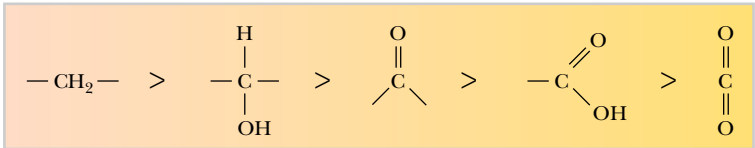


FIGURE 18.9 • Comparison of the state of reduction of carbon atoms in biomolecules: —CH₂— (fats) > —CHOH— (carbohydrates) > —C=O (carbonyls) > —COOH (carboxyls) > CO₂ (carbon dioxide, the final product of catabolism).

to NAD^+ molecules, reducing them to NADH. A second proton accompanies these reactions, appearing in the overall equation as H^+ (Figure 18.10). In turn, NADH is oxidized back to NAD^+ when it transfers its reducing equivalents to electron acceptor systems that are part of the metabolic apparatus of the mitochondria. The ultimate oxidizing agent (e^- acceptor) is O_2 , becoming reduced to H_2O .

Oxidation reactions are exergonic, and the energy released is coupled with the formation of ATP in a process called **oxidative phosphorylation**. The NAD^+ –NADH system can be viewed as a *shuttle* that carries the electrons released from catabolic substrates to the mitochondria, where they are transferred to O_2 , the ultimate electron acceptor in catabolism. In the process, the free energy released is trapped in ATP. The NADH cycle is an important player in the transformation of the chemical energy of carbon compounds into the chemical energy of phosphoric anhydride bonds. Such transformations of energy from one form to another are referred to as **energy transduction**. Oxidative phosphorylation is one cellular mechanism for energy transduction. Chapter 21 is devoted to electron transport reactions and oxidative phosphorylation.

NADPH Provides the Reducing Power for Anabolic Processes

Whereas catabolism is fundamentally an oxidative process, anabolism is, by its contrasting nature, reductive. The biosynthesis of the complex constituents of the cell begins at the level of intermediates derived from the degradative pathways of catabolism; or, less commonly, biosynthesis begins with oxidized substances available in the inanimate environment, such as carbon dioxide. When the hydrocarbon chains of fatty acids are assembled from acetyl-CoA units, activated hydrogens are needed to reduce the carbonyl ($\text{C}=\text{O}$) carbon of acetyl-CoA into a $-\text{CH}_2-$ at every other position along the chain. When glucose is

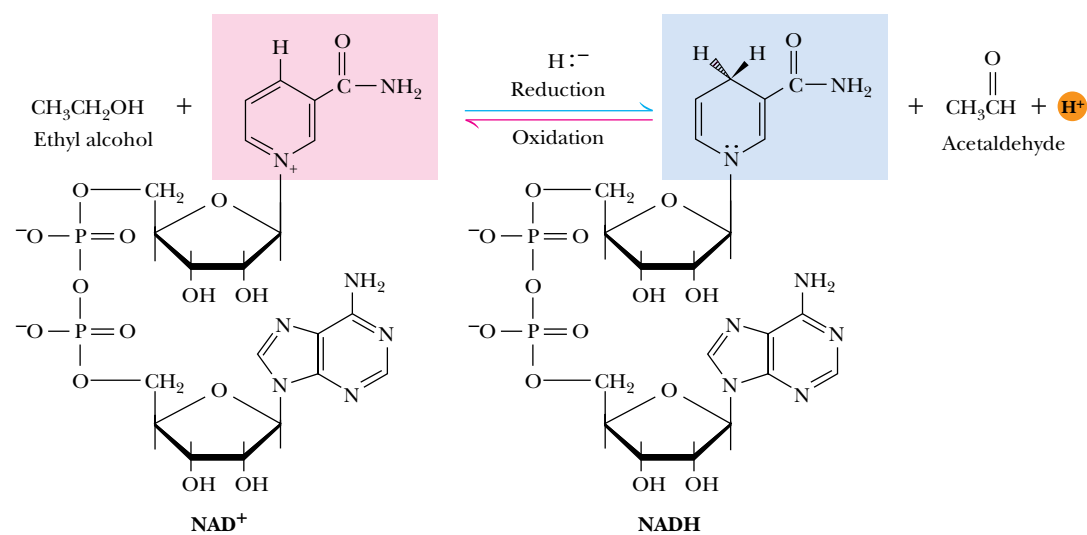
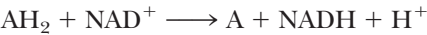


FIGURE 18.10 • Hydrogen and electrons released in the course of oxidative catabolism are transferred as hydride ions to the pyridine nucleotide, NAD^+ , to form $\text{NADH} + \text{H}^+$ in dehydrogenase reactions of the type



The reaction shown is catalyzed by alcohol dehydrogenase.

synthesized from CO₂ during photosynthesis in plants, reducing power is required. These reducing equivalents are provided by NADPH, the usual source of high-energy hydrogens for reductive biosynthesis. NADPH is generated when NADP⁺ is reduced with electrons in the form of hydride ions. In heterotrophic organisms, these electrons are removed from fuel molecules by NADP⁺-specific dehydrogenases. In these organisms, NADPH can be viewed as the carrier of electrons from catabolic reactions to anabolic reactions (Figure 18.11). In photosynthetic organisms, the energy of light is used to pull electrons from water and transfer them to NADP⁺; O₂ is a by-product of this process.

18.3 • Experimental Methods to Reveal Metabolic Pathways

Armed with the knowledge that metabolism is organized into pathways of successive reactions, we can appreciate by hindsight the techniques employed by early biochemists to reveal their sequence. A major intellectual advance took place at the end of the 19th century when Eduard Buchner showed that the fermentation of glucose to yield ethanol and carbon dioxide can occur in extracts of broken yeast cells. Until this discovery, many thought that metabolism was a vital property, unique to intact cells; even the eminent microbiologist Louis Pasteur, who contributed so much to our understanding of fermentation, was a *vitalist*, one of those who believed that the processes of living substance transcend the laws of chemistry and physics. After Buchner’s revelation, biochemists searched for intermediates in the transformation of glucose and soon learned that inorganic phosphate was essential to glucose breakdown. This observation gradually led to the discovery of a variety of phosphorylated organic compounds that serve as intermediates along the fermentative pathway.

An important tool for elucidating the steps in the pathway was the use of *metabolic inhibitors*. Adding an enzyme inhibitor to a cell-free extract caused an accumulation of intermediates in the pathway prior to the point of inhibition (Figure 18.12). Each inhibitor was specific for a particular site in the sequence of metabolic events. As the arsenal of inhibitors was expanded, the individual steps in metabolism were revealed.

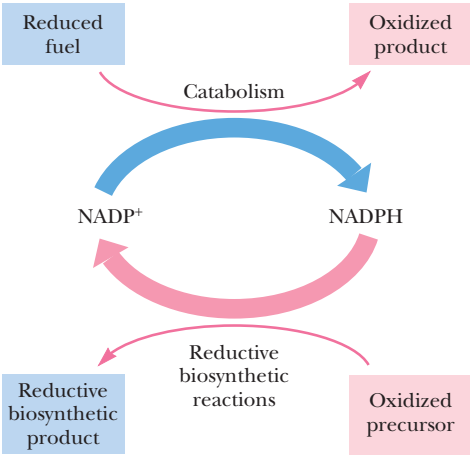


FIGURE 18.11 • Transfer of reducing equivalents from catabolism to anabolism via the NADPH cycle.

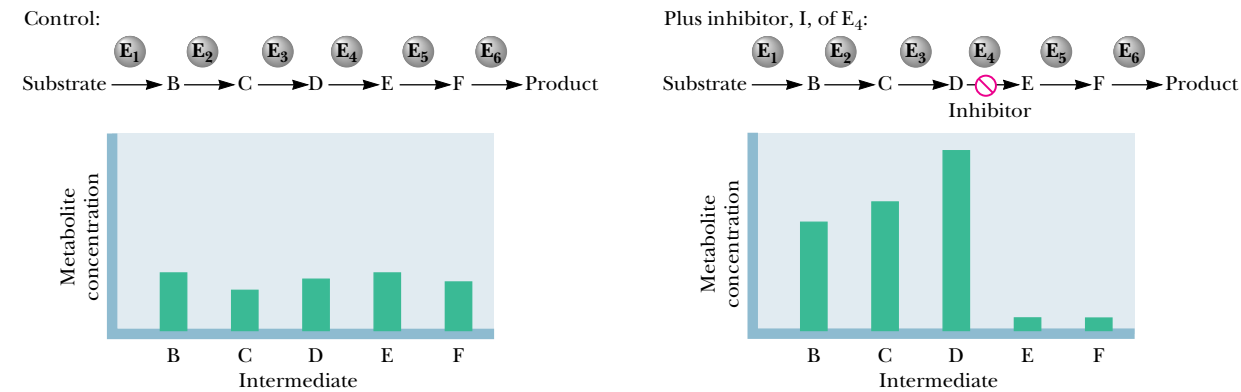


FIGURE 18.12 • The use of inhibitors to reveal the sequence of reactions in a metabolic pathway. (a) **Control:** Under normal conditions, the steady-state concentrations of a series of intermediates will be determined by the relative activities of the enzymes in the pathway. (b) **Plus inhibitor:** In the presence of an inhibitor (in this case, an inhibitor of enzyme 4), intermediates upstream of the metabolic block (B, C, and D) accumulate, revealing themselves as intermediates in the pathway. The concentration of intermediates lying downstream (E and F) will fall.

Mutations Create Specific Metabolic Blocks

Genetics provides an approach to the identification of intermediate steps in metabolism that is somewhat analogous to inhibition. Mutation in a gene encoding an enzyme often results in an inability to synthesize the enzyme in an active form. Such a defect leads to a block in the metabolic pathway at the point where the enzyme acts, and the enzyme’s substrate accumulates. Such genetic disorders are lethal if the end product of the pathway is essential or if the accumulated intermediates have toxic effects. In microorganisms, however, it is often possible to manipulate the growth medium so that essential end products are provided. Then the biochemical consequences of the mutation can be investigated. Studies on mutations in genes of the filamentous fungus *Neurospora crassa* led G. W. Beadle and E. L. Tatum to hypothesize in 1941 that genes are units of heredity that encode enzymes (a principle referred to as the “one gene–one enzyme” hypothesis).

Isotopic Tracers as Metabolic Probes

Another widely used approach to the elucidation of metabolic sequences is to “feed” cells a substrate or metabolic intermediate labeled with a particular isotopic form of an element that can be traced. Two sorts of isotopes are useful in this regard: radioactive isotopes, such as ¹⁴C, and stable “heavy” isotopes, such as ¹⁸O or ¹⁵N (Table 18.3). Because the chemical behavior of isotopically labeled compounds is rarely distinguishable from that of their unlabeled counterparts, isotopes provide reliable “tags” for observing metabolic changes. The metabolic fate of a radioactively labeled substance can be traced by determining the presence and position of the radioactive atoms in intermediates derived from the labeled compound (Figure 18.13).

Table 18.3

Properties of Radioactive and Stable “Heavy” Isotopes Used as Tracers in Metabolic Studies				
Isotope	Type	Radiation Type	Half-Life	Relative Abundance*
² H	Stable			0.0154%
³ H	Radioactive	β [−]	12.1 years	
¹³ C	Stable			1.1%
¹⁴ C	Radioactive	β [−]	5700 years	
¹⁵ N	Stable			0.365%
¹⁸ O	Stable			0.204%
²⁴ Na	Radioactive	β [−] , γ	15 hours	
³² P	Radioactive	β [−]	14.3 days	
³⁵ S	Radioactive	β [−]	87.1 days	
³⁶ Cl	Radioactive	β [−]	310,000 years	
⁴² K	Radioactive	β [−]	12.5 hours	
⁴⁵ Ca	Radioactive	β [−]	152 days	
⁵⁹ Fe	Radioactive	β [−] , γ	45 days	
¹³¹ I	Radioactive	β [−] , γ	8 days	

*The relative natural abundance of a stable isotope is important because, in tracer studies, the amount of stable isotope is typically expressed in terms of atoms percent excess over the natural abundance of the isotope.

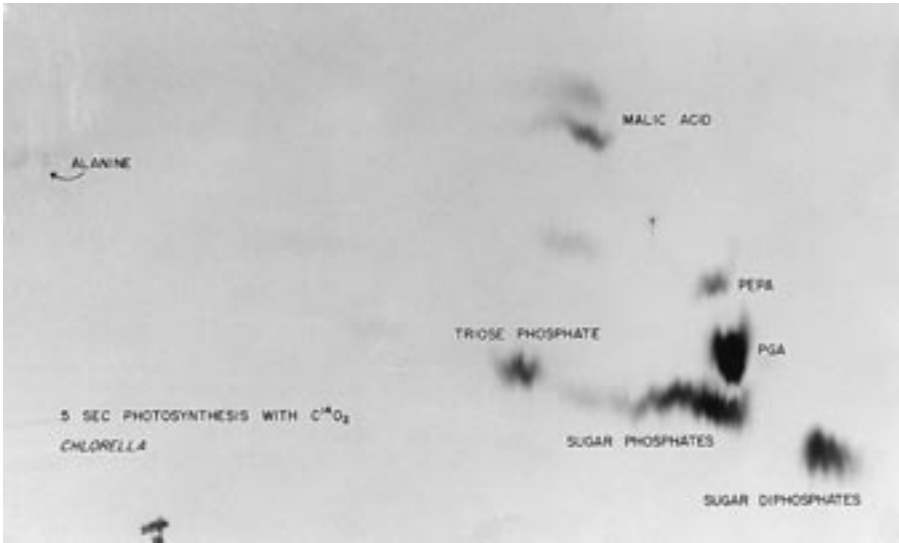
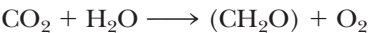


FIGURE 18.13 • One of the earliest experiments using a radioactive isotope as a metabolic tracer. Cells of *Chlorella* (a green alga) synthesizing carbohydrate from carbon dioxide were exposed briefly (5 sec) to ¹⁴C-labeled CO₂. The products of CO₂ incorporation were then quickly isolated from the cells, separated by two-dimensional paper chromatography, and observed via autoradiographic exposure of the chromatogram. Such experiments identified radioactive 3-phosphoglycerate (PGA) as the primary product of CO₂ fixation. The 3-phosphoglycerate was labeled in the 1-position (in its carboxyl group). Radioactive compounds arising from the conversion of 3-phosphoglycerate to other metabolic intermediates included phosphoenolpyruvate (PEP), malic acid, triose phosphate, alanine, and sugar phosphates and diphosphates. (Photograph courtesy of Professor Melvin Calvin, Lawmann Berkeley Laboratory, University of California, Berkeley.)

Heavy Isotopes

Heavy isotopes endow the compounds in which they appear with slightly greater masses than their unlabeled counterparts. These compounds can be separated and quantitated by mass spectrometry (or density gradient centrifugation, if they are macromolecules). For example, ¹⁸O was used in separate experiments as a tracer of the fate of the oxygen atoms in water and carbon dioxide to determine whether the atmospheric oxygen produced in photosynthesis arose from H₂O, CO₂, or both:



If ¹⁸O-labeled CO₂ was presented to a green plant carrying out photosynthesis, none of the ¹⁸O was found in O₂. Curiously, it was recovered as H₂¹⁸O. In contrast, when plants fixing CO₂ were equilibrated with H₂¹⁸O, ¹⁸O₂ was evolved. These latter labeling experiments established that photosynthesis is best described by the equation



That is, in the process of photosynthesis, the two oxygen atoms in O₂ come from two H₂O molecules. One O is lost from CO₂ and appears in H₂O, and the other O of CO₂ is retained in the carbohydrate product. Two of the four H atoms are accounted for in (CH₂O), and two reduce the O lost from CO₂ to H₂O.

NMR as a Metabolic Probe

A technology analogous to isotopic tracers is provided by **nuclear magnetic resonance (NMR) spectroscopy**. The atomic nuclei of certain isotopes, such as the naturally occurring isotope of phosphorus, ³¹P, have *magnetic moments*. The resonance frequency of a magnetic moment is influenced by the local chemical environment. That is, the NMR signal of the nucleus is influenced in an identifiable way by the chemical nature of its neighboring atoms in the compound. In many ways, these nuclei are ideal tracers because their signals contain a great deal of structural information about the environment around the atom, and thus the nature of the compound containing the atom. Transformations of sub-

strates and metabolic intermediates labeled with magnetic nuclei can be traced by following changes in NMR spectra. Furthermore, NMR spectroscopy is a noninvasive procedure. Whole-body NMR spectrometers are being used today in hospitals to directly observe the metabolism (and clinical condition) of living subjects (Figure 18.14). NMR promises to be a revolutionary tool for clinical diagnosis and for the investigation of metabolism *in situ* (literally “in site,” meaning, in this case, “where and as it happens”).

Metabolic Pathways Are Compartmentalized Within Cells

Although the interior of a prokaryotic cell is not subdivided into compartments by internal membranes, the cell still shows some segregation of metabolism. For example, certain metabolic pathways, such as phospholipid synthesis and oxidative phosphorylation, are localized in the plasma membrane. Also, protein biosynthesis is carried out on ribosomes.

In contrast, eukaryotic cells are extensively compartmentalized by an endomembrane system. Each of these cells has a true nucleus bounded by a double membrane called the *nuclear envelope*. The nuclear envelope is continuous with the endomembrane system, which is composed of differentiated regions: the endoplasmic reticulum; the Golgi complex; various membrane-bounded vesicles such as lysosomes, vacuoles, and microbodies; and, ultimately, the plasma membrane itself. Eukaryotic cells also possess mitochondria and, if they are photosynthetic, chloroplasts. Disruption of the cell membrane and fractionation of the cell contents into the component organelles have allowed an analysis of their respective functions (Figure 18.15). Each compartment is dedicated to specialized metabolic functions, and the enzymes appropriate to these specialized functions are confined together within the organelle. In many instances, the enzymes of a metabolic sequence occur together within the organellar membrane. Thus, *the flow of metabolic intermediates in the cell is spa-*

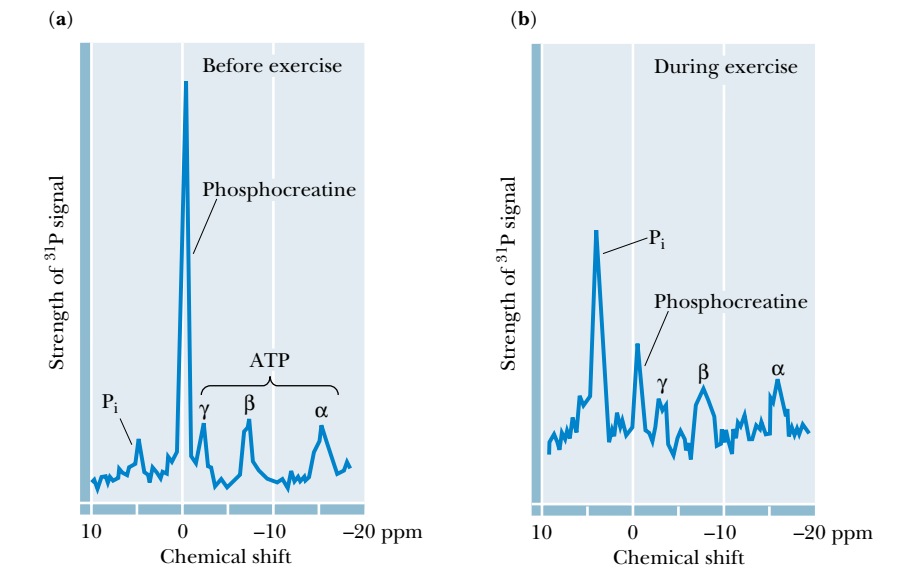


FIGURE 18.14 • With NMR spectroscopy one can observe the metabolism of a living subject in real time. These NMR spectra show the changes in ATP, creatine-P (phosphocreatine), and P_i levels in the forearm muscle of a human subjected to 19 minutes of exercise. Note that the three P atoms of ATP (α, β, and γ) have different chemical shifts, reflecting their different chemical environments.

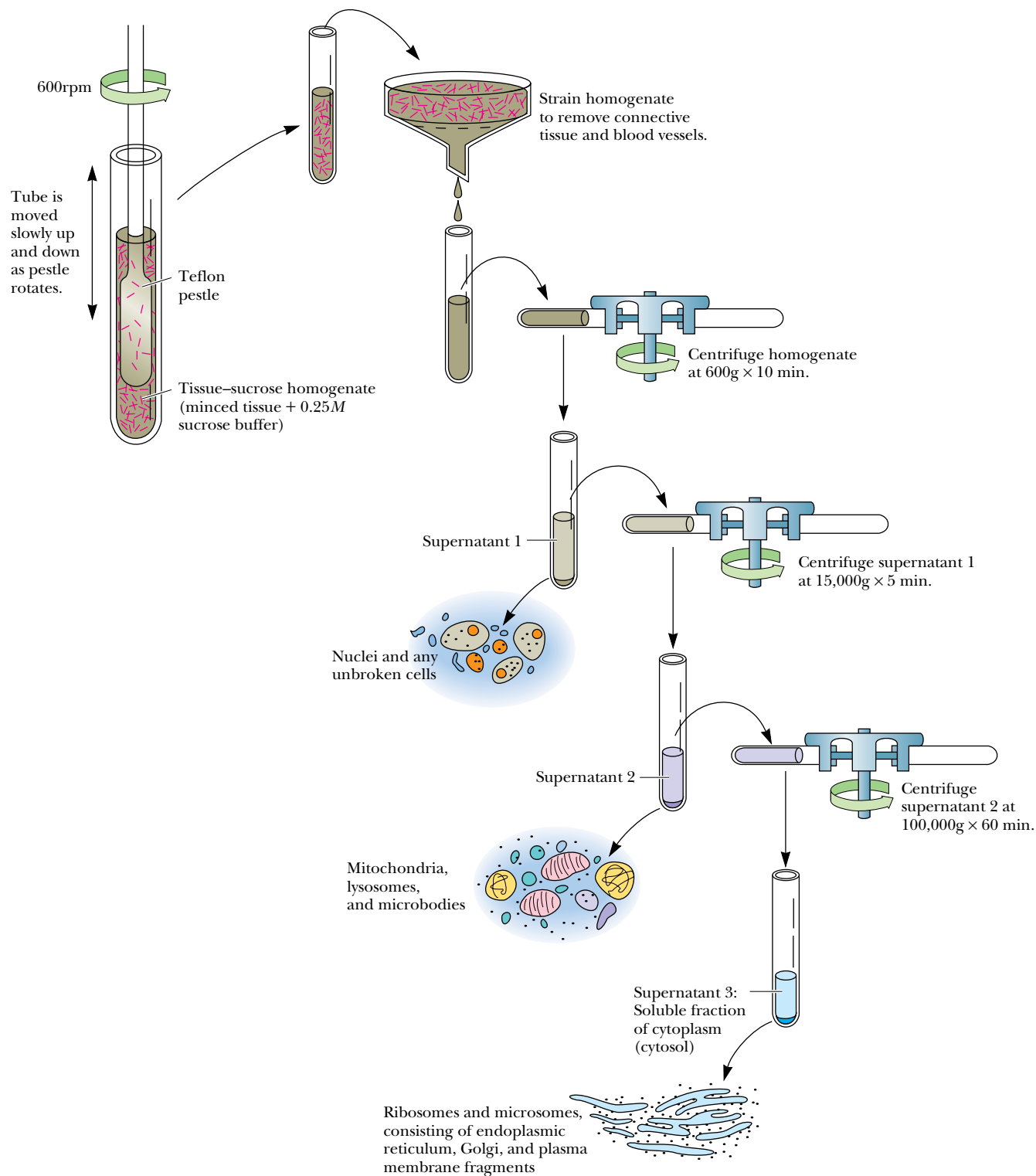


FIGURE 18.15 • Fractionation of a cell extract by differential centrifugation. It is possible to separate organelles and subcellular particles in a centrifuge because their inherent size and density differences give them different rates of sedimentation in an applied centrifugal field. Nuclei are pelleted in relatively weak centrifugal fields, mitochondria in somewhat stronger fields, whereas very strong centrifugal fields are necessary to pellet ribosomes and fragments of the endomembrane system.

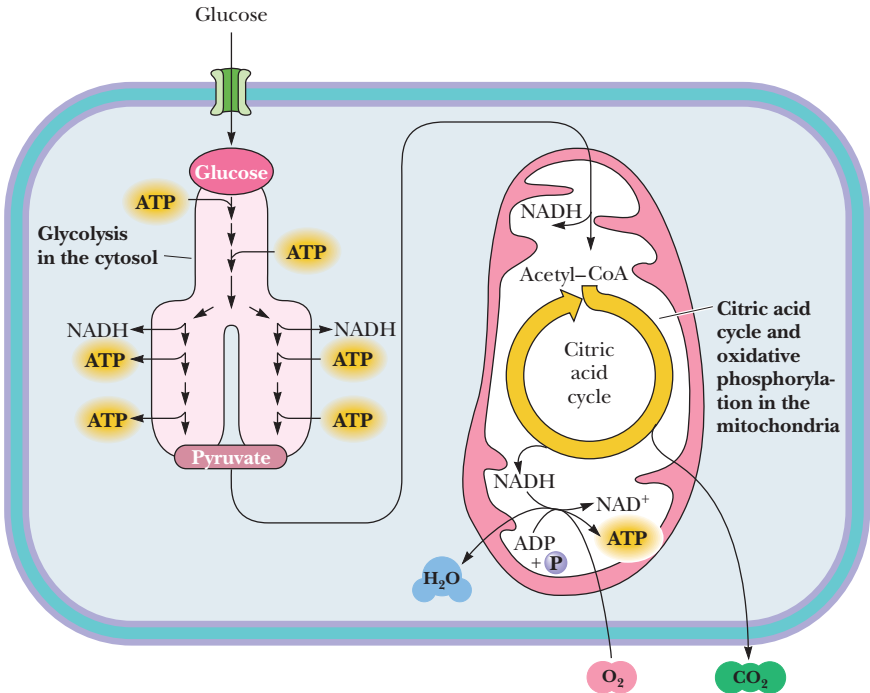


FIGURE 18.16 • Compartmentalization of glycolysis, the citric acid cycle, and oxidative phosphorylation.

tially as well as chemically segregated. For example, the 10 enzymes of glycolysis are found in the cytosol, but pyruvate, the product of glycolysis, is fed into the mitochondria. These organelles contain the citric acid cycle enzymes, which oxidize pyruvate to CO₂. The great amount of energy released in the process is captured by the oxidative phosphorylation system of mitochondrial membranes and used to drive the formation of ATP (Figure 18.16).

18.4 • Nutrition

The use of foods by organisms is termed **nutrition**. The ability of an organism to use a particular food material depends upon its chemical composition and upon the metabolic pathways available to the organism. In addition to essential fiber, food includes the macronutrients—protein, carbohydrate, and lipid—and the micronutrients—including vitamins and minerals.

Protein

Higher organisms must consume protein in order to make new proteins. Dietary protein is a rich source of nitrogen, and certain amino acids—the so-called **essential amino acids**—cannot be synthesized by higher organisms and can be obtained only in the diet. The average adult in the United States consumes far more protein than required for synthesis of essential proteins. Excess dietary protein is then merely a source of metabolic energy. Some of the amino acids (termed **glucogenic**) can be converted into glucose, whereas others, the **ketogenic** amino acids, can be converted to fatty acids and/or keto acids. If fat and carbohydrate are already adequate for the energy needs of the organism, then both kinds of amino acids will be converted to triacylglycerol and stored in adipose tissue.

A certain percentage of an organism’s own protein undergoes a constant process of degradation and resynthesis. Together with dietary protein, this recycled protein material participates in a nitrogen equilibrium or **nitrogen balance**. A positive nitrogen balance occurs whenever there is a net increase in the organism’s protein content, such as during periods of growth. A negative nitrogen balance exists when dietary intake of nitrogen is insufficient.

Carbohydrate

The principal purpose of carbohydrate in the diet is production of metabolic energy. Simple sugars are metabolized in the glycolytic pathway (see Chapter 19). Complex carbohydrates are degraded into simple sugars, which then can enter the glycolytic pathway. Carbohydrates are also essential components of nucleotides, nucleic acids, glycoproteins, and glycolipids. Human metabolism can adapt to a wide range of dietary carbohydrate levels, but the brain requires glucose for fuel. When dietary carbohydrate consumption exceeds the energy needs of the organism, excess carbohydrate is converted to triacylglycerols and glycogen for long-term energy storage. On the other hand, when dietary carbohydrate intake is low, **ketone bodies** are formed from acetate units to provide metabolic fuel for the brain.

Lipid

Fatty acids and triacylglycerols can be used as fuel by many tissues in the human body, and phospholipids are essential components of all biological membranes. Even though the human body can tolerate a wide range of fat intake levels, there are disadvantages in either extreme. Excess dietary fat is stored as triacylglycerols in adipose tissue, but high levels of dietary fat can also increase the risk of atherosclerosis and heart disease. Moreover, high dietary fat levels are also correlated with increased risk for colon, breast, and prostate cancers. When dietary fat consumption is low, there is a risk of **essential fatty acid** deficiencies. As seen in Chapter 25, the human body cannot synthesize linoleic and linolenic acids, so these must be acquired in the diet. Additionally, arachidonic

DEEPER LOOK

A Fad Diet—Low Carbohydrates, High Protein, High Fat

Possibly the most serious nutrition problem in the United States is excessive food consumption, and many people have experimented with fad diets in the hope of losing excess weight. One of the most popular of the fad diets has been the high-protein, high-fat (low-carbohydrate) diet. The premise for such diets is tantalizing: because the tricarboxylic acid (TCA) cycle (see Chapter 20) is the primary site of fat metabolism, and because glucose is usually needed to replenish intermediates in the TCA cycle, if carbohydrates are restricted in the diet, dietary fat should merely be converted to ketone bodies and excreted. This so-called diet appears to work at first because a low-carbohydrate diet results in an initial water (and weight) loss. This occurs because

glycogen reserves are depleted by the diet and because about 3 grams of water of hydration are lost for every gram of glycogen. However, the long-term results of this diet are usually disappointing for several reasons. First, ketone body excretion by the human body usually does not exceed 20 grams (400 kJ) per day. Second, amino acids can function effectively to replenish TCA cycle intermediates, making the reduced carbohydrate regimen irrelevant. Third, the typical fare in a high-protein, high-fat, low-carbohydrate diet is expensive but not very tasty, and it is thus difficult to maintain. Finally, a high-fat diet is a high risk factor for atherosclerosis and coronary artery disease.

acid can be synthesized in humans only from linoleic acid, so it too is classified as essential. The essential fatty acids are key components of biological membranes, and arachidonic acid is the precursor to prostaglandins, which mediate a variety of processes in the body.

Fiber

The components of food materials that cannot be broken down by human digestive enzymes are referred to as **dietary fiber**. There are several kinds of dietary fiber, each with its own chemical and biological properties. **Cellulose** and **hemicellulose** are insoluble fiber materials that stimulate regular function of the colon. They may play a role in reducing the risk of colon cancer. **Lignins** make up another class of insoluble fibers which absorb organic molecules in the digestive system. Lignins bind cholesterol and clear it from the digestive system, reducing the risk of heart disease. Pectins and gums are water-soluble fiber materials that form viscous gel-like suspensions in the digestive system, slowing the rate of absorption of many nutrients, including carbohydrates, and lowering serum cholesterol in many cases. The insoluble fibers are prevalent in vegetable grains. Water-soluble fiber is a component of fruits, legumes, and oats.

SPECIAL FOCUS:

VITAMINS

Vitamins are essential nutrients that are required in the diet, usually in trace amounts, because they cannot be synthesized by the organism itself. The requirement for any given vitamin depends on the organism. Not all “vitamins” are required by all organisms. Vitamins required in the human diet are listed in Table 18.4. These important substances are traditionally distinguished as being either water-soluble or fat-soluble. Except for vitamin C (ascorbic acid), the water-soluble vitamins are all components or precursors of important biological substances known as **coenzymes**. These are low-molecular-weight molecules that bring unique chemical functionality to certain enzyme reactions. Coenzymes may also act as *carriers* of specific functional groups, such as methyl groups and acyl groups. The side chains of the common amino acids provide only a limited range of chemical reactivities and carrier properties. Coenzymes, acting in concert with appropriate enzymes, provide a broader range of catalytic properties for the reactions of metabolism. Coenzymes are typically modified by these reactions and are then converted back to their original forms by other enzymes, so that small amounts of these substances can be used repeatedly. The coenzymes derived from the water-soluble vitamins are listed in Table 18.4. Each of these will be discussed in this chapter. The fat-soluble vitamins are not directly related to coenzymes, but they play essential roles in a variety of critical biological processes, including vision, maintenance of bone structure, and blood coagulation. The mechanisms of action of fat-soluble vitamins are not as well understood as their water-soluble counterparts, but modern research efforts are gradually closing this gap.

Vitamin B₁: Thiamine and Thiamine Pyrophosphate

As shown in Figure 18.17, thiamine is composed of a substituted *thiazole* ring joined to a substituted pyrimidine by a methylene bridge. It is the precursor of **thiamine pyrophosphate (TPP)**, a coenzyme involved in reactions of carbo-

Table 18.4

Vitamins and Coenzymes

Vitamin	Coenzyme Form
Water-Soluble	
Thiamine (vitamin B ₁)	Thiamine pyrophosphate
Niacin (nicotinic acid)	Nicotinamide adenine dinucleotide (NAD ⁺) Nicotinamide adenine dinucleotide phosphate (NADP ⁺)
Riboflavin (vitamin B ₂)	Flavin adenine dinucleotide (FAD) Flavin mononucleotide (FMN)
Pantothenic acid	Coenzyme A
Pyridoxal, pyridoxine, pyridoxamine (vitamin B ₆)	Pyridoxal phosphate
Cobalamin (vitamin B ₁₂)	5'-Deoxyadenosylcobalamin Methylcobalamin
Biotin	Biotin-lysine complexes (biocytin)
Lipoic acid	Lipoyl-lysine complexes (lipoamide)
Folic acid	Tetrahydrofolate
Fat-Soluble	
Retinol (vitamin A)	
Ergocalciferol (vitamin D ₂)	
Cholecalciferol (vitamin D ₃)	
α-Tocopherol (vitamin E)	
Vitamin K	

hydrate metabolism in which bonds to carbonyl carbons (aldehydes or ketones) are synthesized or cleaved. In particular, the *decarboxylations of α-keto acids and the formation and cleavage of α-hydroxyketones* depend on thiamine pyrophosphate. The first of these is illustrated in Figure 18.18 by (a) the decarboxylation of pyruvate by **yeast pyruvate decarboxylase** to yield carbon dioxide and acetaldehyde. An example of the formation and cleavage of α-hydroxyketones is presented in Figure 18.18 (b) the condensation of two molecules of pyruvate in the **acetolactate synthase** reaction. Another example is provided by a reaction from the pentose phosphate pathway (Chapters 22 and 23) called the **trans-ketolase** reaction. This latter reaction is referred to as an **α-ketol transfer** for obvious reasons.

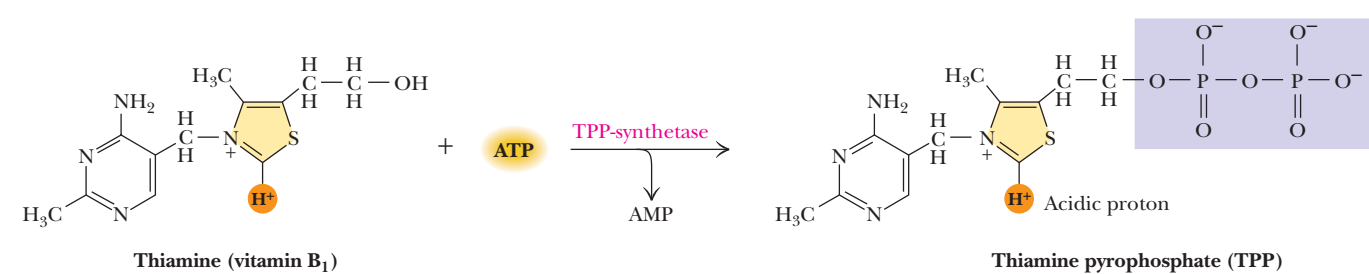


FIGURE 18.17 • Thiamine pyrophosphate (TPP), the active form of vitamin B₁, is formed by the action of TPP-synthetase.

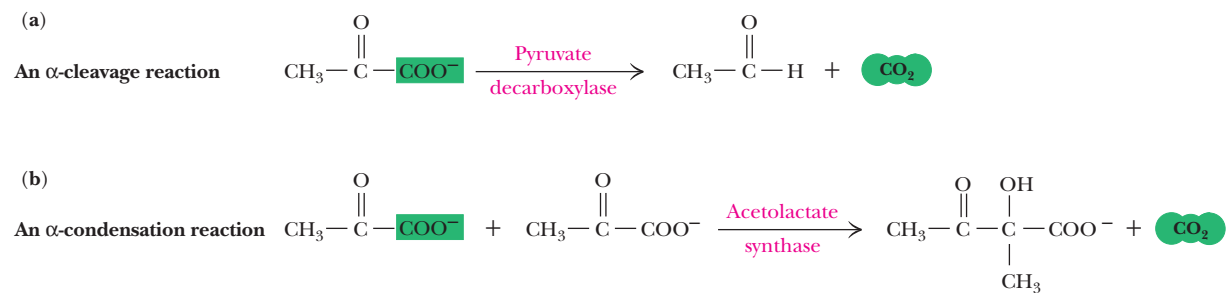


FIGURE 18.18 • Thiamine pyrophosphate participates in (a) the decarboxylation of α -keto acids and (b) the formation and cleavage of α -hydroxyketones.

Vitamins Containing Adenine Nucleotides

Several classes of vitamins are related to, or are precursors of, coenzymes that contain adenine nucleotides as part of their structure. These coenzymes include the flavin dinucleotides, the pyridine dinucleotides, and coenzyme A. The adenine nucleotide portion of these coenzymes does not participate actively in the reactions of these coenzymes; rather, it enables the proper enzymes to recognize the coenzyme. Specifically, the adenine nucleotide greatly increases both the *affinity* and the *specificity* of the coenzyme for its site on the enzyme, owing to its numerous sites for hydrogen bonding, and also the hydrophobic and ionic bonding possibilities it brings to the coenzyme structure.

Nicotinic Acid and the Nicotinamide Coenzymes

Nicotinamide is an essential part of two important coenzymes: **nicotinamide adenine dinucleotide (NAD⁺)** and **nicotinamide adenine dinucleotide phosphate (NADP⁺)** (Figure 18.19). The reduced forms of these coenzymes are NADH and NADPH. *The nicotinamide coenzymes (also known as pyridine nucleotides) are electron carriers.* They play vital roles in a variety of enzyme-catalyzed oxidation–reduction reactions. (NAD⁺ is an electron acceptor in oxidative (catabolic) pathways and NADPH is an electron donor in reductive (biosynthetic) pathways.) These reactions involve direct transfer of hydride anion either to NAD(P)⁺ or from NAD(P)H. The enzymes that facilitate such

HUMAN BIOCHEMISTRY

Thiamine and Beriberi

Thiamine, whose structure is shown in Figure 18.17, is known as vitamin B₁ and is essential for the prevention of **beriberi**, a nervous system disease that has occurred in the Far East for centuries and has resulted in considerable sickness and death in these countries. (As recently as 1958, it was the fourth leading cause of death in the Philippine Islands.) It was shown in 1882 by the director-general of the medical department of the Japanese navy that beriberi could be prevented by dietary modifications. Ten years later, Christiaan Eijkman, a Dutch medical scientist working in Java, began research that eventually showed that thiamine was the

“anti-beriberi” substance. He found that chickens fed polished rice exhibited paralysis and head retractions and that these symptoms could be reversed if the rice polishings (the outer layers and embryo of the rice kernel) were fed to the birds. In 1911, Casimir Funk prepared a crystalline material from rice bran that cured beriberi in birds. He named it **beriberi vitamin**, because he viewed it as a “vital amine,” and thus he is credited with coining the word *vitamin*. The American biochemist R. R. Williams and his research group were the first to establish the structure of thiamine (in 1935) and a route for its synthesis.

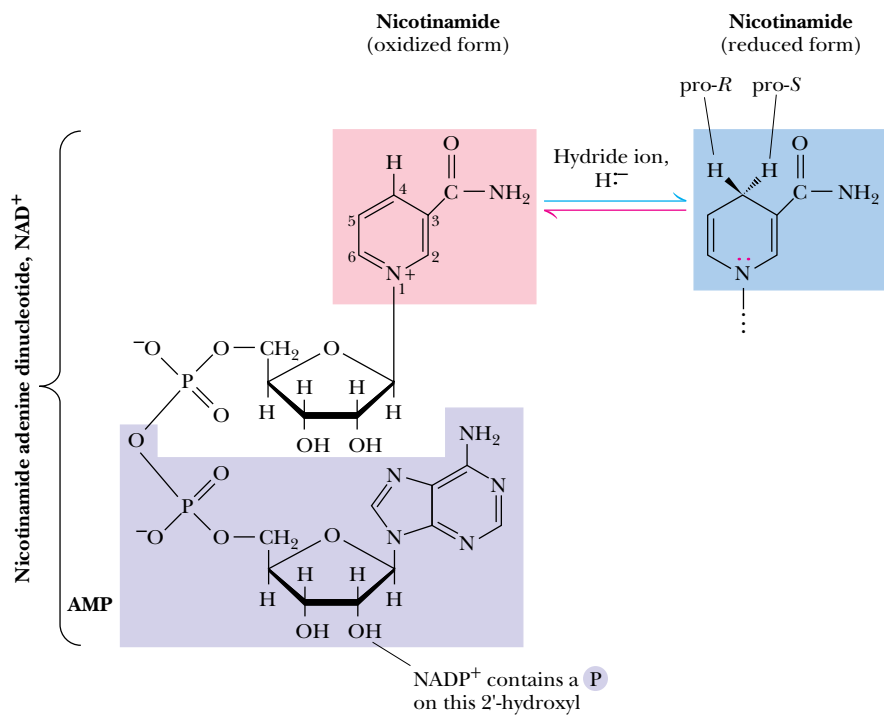


FIGURE 18.19 • The structures and redox states of the nicotinamide coenzymes. Hydride ion (H^- , a proton with two electrons) transfers to NAD^+ to produce NADH .

transfers are thus known as **dehydrogenases**. The hydride anion contains two electrons, and thus NAD^+ and NADP^+ act exclusively as **two-electron carriers**. The C-4 position of the pyridine ring, which can either accept or donate hydride ion, is the reactive center of both NAD and NADP . The quaternary nitrogen of the nicotinamide ring functions as an electron sink to facilitate hydride transfer to NAD^+ , as shown in Figure 18.20. The adenine portion of the molecule is not directly involved in redox processes.

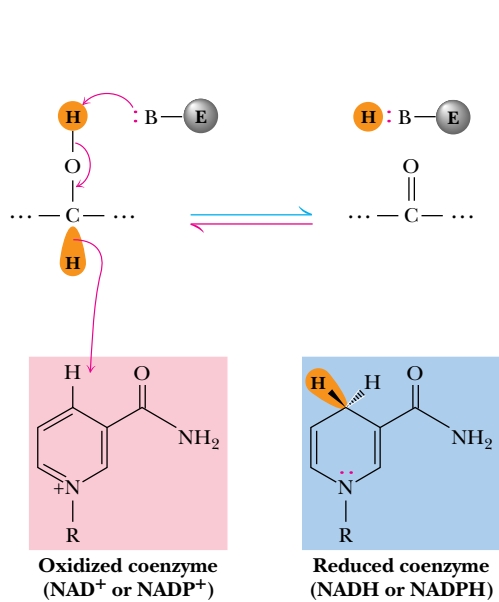


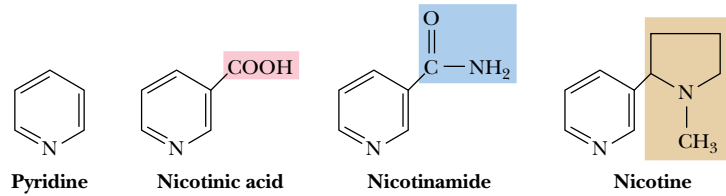
FIGURE 18.20 • NAD^+ and NADP^+ participate exclusively in two-electron transfer reactions. For example, alcohols can be oxidized to ketones or aldehydes via hydride transfer to NAD(P)^+ .

HUMAN BIOCHEMISTRY

Niacin and Pellegra

Pellegra, a disease characterized by dermatitis, diarrhea, and dementia, has been known for centuries. It was once prevalent in the southern part of the United States and is still a common problem in some parts of Spain, Italy, and Romania. Pellegra was once thought to be an infectious disease, but Joseph Goldberger showed early in this century that it could be cured by dietary actions. Soon thereafter, it was found that brewer’s yeast would prevent pellegra in humans. Studies of a similar disease in dogs, called **blacktongue**, eventually led to the identification of **nicotinic acid** as the relevant dietary factor. Elvehjem and his colleagues at the University of Wisconsin in 1937 isolated **nicotinamide** from liver, and showed that it and nicotinic acid could prevent and cure blacktongue in dogs. That same year, nicotin-

amide and nicotinic acid were both shown to be able to cure pellegra in humans. Interestingly, plants and many animals can synthesize nicotinic acid from tryptophan and other precursors, and nicotinic acid is thus not a true vitamin for these species. However, if dietary intake of tryptophan is low, nicotinic acid is required for optimal health. Nicotinic acid, which is beneficial to humans and animals, is structurally related to **nicotine**, a highly toxic tobacco alkaloid. In order to avoid confusion of nicotinic acid and nicotinamide with nicotine itself, **niacin** was adopted as a common name for nicotinic acid. Cowgill, at Yale University, suggested the name from the letters of three words—*nicotinic*, *acid*, and *vitamin*.



The structures of pyridine, nicotinic acid, nicotinamide, and nicotine.

Examination of the structures of NADH and NADPH reveals that the 4-position of the nicotinamide ring is **pro-chiral**, meaning that while this carbon is not chiral, it *would* be if either of its hydrogens were replaced by something else. As shown in Figure 18.20, the hydrogen “projecting” out of the page toward you is the “pro-*R*” hydrogen because, if a deuterium is substituted at this position, the molecule would have the *R*-configuration. Substitution of the other hydrogen would yield an *S*-configuration. An interesting aspect of the enzymes that require nicotinamide coenzymes is that they are **stereospecific** and withdraw hydrogen from either the pro-*R* or the pro-*S* position selectively. This stereospecificity arises from the fact that enzymes (and the active sites of enzymes) are inherently asymmetric structures. These same enzymes are stereospecific with respect to the substrates as well.

The NAD- and NADP-dependent dehydrogenases catalyze at least six different types of reactions: simple hydride transfer, deamination of an amino acid to form an α -keto acid, oxidation of β -hydroxy acids followed by decarboxylation of the β -keto acid intermediate, oxidation of aldehydes, reduction of isolated double bonds, and the oxidation of carbon–nitrogen bonds (as with dihydrofolate reductase).

Riboflavin and the Flavin Coenzymes

Riboflavin, or **vitamin B₂**, is a constituent and precursor of both **riboflavin 5’-phosphate**, also known as **flavin mononucleotide (FMN)**, and **flavin adenine dinucleotide (FAD)**. The name *riboflavin* is a synthesis of the names for the molecule’s component parts, **ribitol** and **flavin**. The structures of riboflavin,

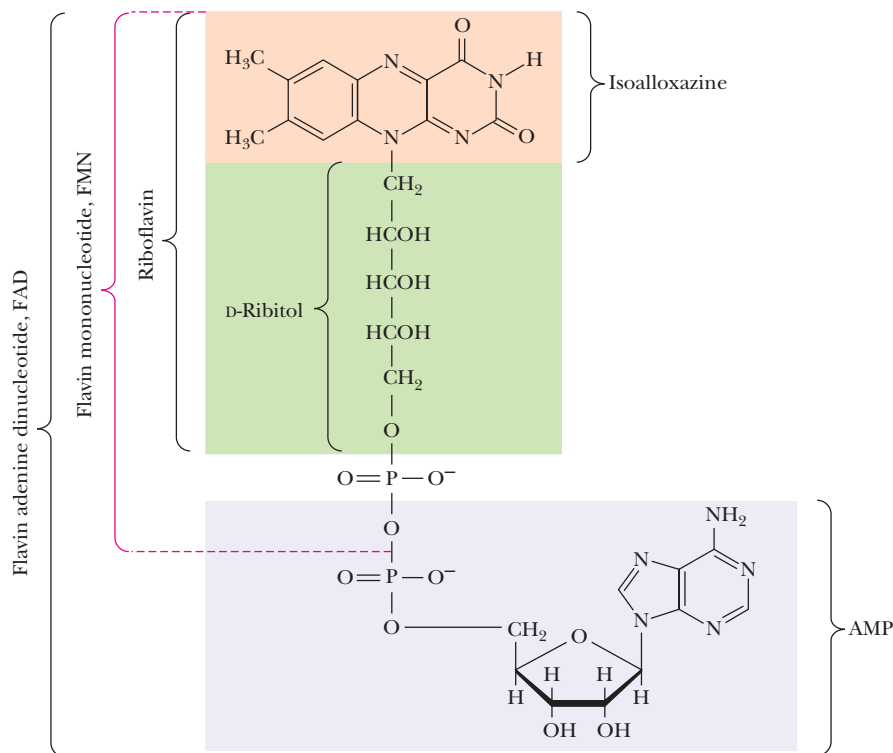


FIGURE 18.21 • The structures of riboflavin, flavin mononucleotide (FMN), and flavin adenine dinucleotide (FAD). Flavin coenzymes bind tightly to the enzymes that use them, with typical dissociation constants in the range of 10^{-8} to 10^{-11} M, so that only very low levels of free flavin coenzymes occur in most cells. Even in organisms that rely on the nicotinamide coenzymes (NADH and NADPH) for many of their oxidation–reduction cycles, the flavin coenzymes fill essential roles. Flavins are stronger oxidizing agents than NAD^+ and NADP^+ . They can be reduced by both one-electron and two-electron pathways and can be reoxidized easily by molecular oxygen. Enzymes that use flavins to carry out their reactions—*flavoenzymes*—are involved in many kinds of oxidation–reduction reactions.

FMN, and FAD are shown in Figure 18.21. The *isoalloxazine ring* is the core structure of the various flavins. Because the ribityl group is not a true pentose sugar (it is a sugar alcohol) and is not joined to riboflavin in a glycosidic bond, the molecule is not truly a “nucleotide,” and the terms *flavin mononucleotide* and *dinucleotide* are incorrect. Nonetheless, these designations are so deeply ingrained in common biochemical usage that the erroneous nomenclature persists. The flavins have a characteristic bright yellow color and take their name from the Latin *flavus* for “yellow.” As shown in Figure 18.22, the oxidized form of the isoalloxazine structure absorbs light around 450 nm (in the visible region) and also at 350 to 380 nm. The color is lost, however, when the ring is reduced or “bleached.” Similarly, the enzymes that bind flavins, known as **flavoenzymes**, can be yellow, red, or green in their oxidized states. Nevertheless, these enzymes also lose their color on reduction of the bound flavin group.

Flavin coenzymes can exist in any of three different redox states. Fully oxidized flavin is converted to a **semiquinone** by a one-electron transfer, as shown in Figure 18.22. At physiological pH, the semiquinone is a neutral radical, blue in color, with a λ_{max} of 570 nm. The semiquinone possesses a $\text{p}K_{\text{a}}$ of about 8.4. When it loses a proton at higher pH values, it becomes a radical anion, displaying a red color with a λ_{max} of 490 nm. The semiquinone radical is particularly stable, owing to extensive delocalization of the unpaired electron across the π -electron system of the isoalloxazine. A second one-electron transfer converts the semiquinone to the completely reduced dihydroflavin as shown in Figure 18.22.

Access to three different redox states allows flavin coenzymes to participate in *one-electron transfer* and *two-electron transfer reactions*. Partly because of this, flavoproteins catalyze many different reactions in biological systems and work together with many different electron acceptors and donors. These include two-electron acceptor/donors, such as NAD^+ and NADP^+ , one- or two-electron

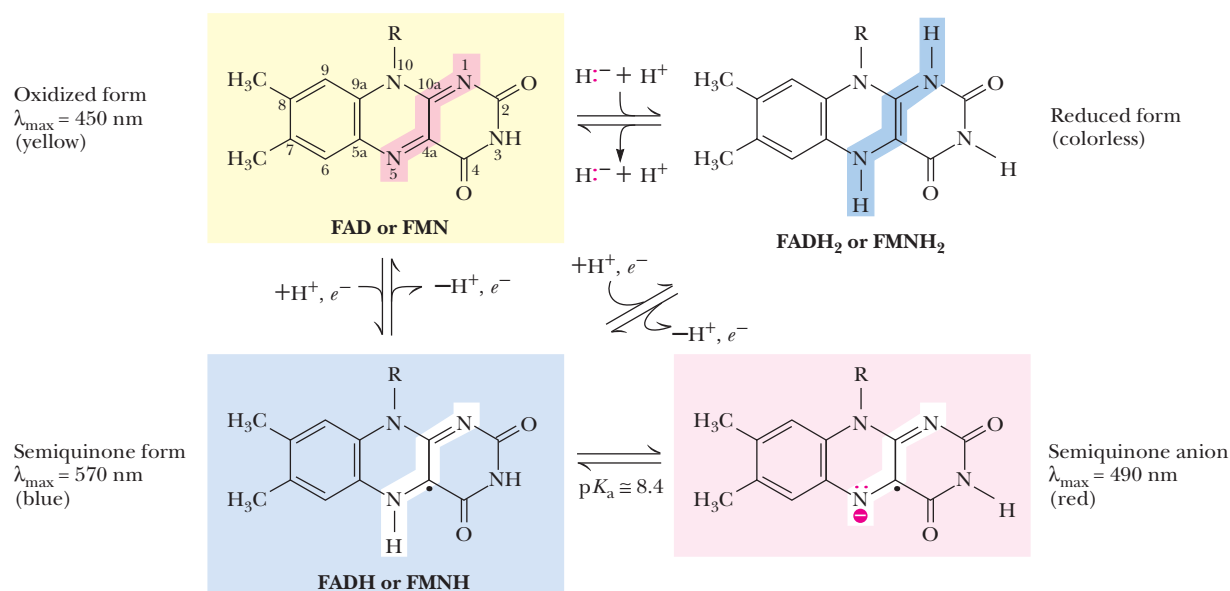


FIGURE 18.22 • The redox states of FAD and FMN. The boxes correspond to the colors of each of these forms. The atoms primarily involved in electron transfer are indicated by red shading in the oxidized form, white in the semiquinone form, and blue in the reduced form.



tron carriers, such as quinones, and a variety of one-electron acceptor/donors, such as cytochrome proteins. Many of the components of the respiratory electron transport chain are one-electron acceptor/donors. The stability of the flavin semiquinone state allows flavoproteins to function as effective electron carriers in respiration processes (Chapter 21).

A DEEPER LOOK

Riboflavin and Old Yellow Enzyme

Riboflavin was first isolated from whey in 1879 by Blyth, and the structure was determined by Kuhn and coworkers in 1933. For the structure determination, this group isolated 30 mg of pure riboflavin from the whites of about 10,000 eggs. The discovery of the actions of riboflavin in biological systems arose from the work of Otto Warburg in Germany and Hugo Theorell in Sweden, both of whom identified yellow substances bound to a yeast enzyme involved in the oxidation of pyridine nucleotides. Theorell showed that riboflavin 5'-phosphate was the source of the yellow color in this *old yellow enzyme*. By 1938, Warburg had identified FAD, the second common form of riboflavin, as the coenzyme in D-amino acid oxidase, another yellow protein. Riboflavin deficiencies are not at all common. Humans require only about 2 mg per day, and the vitamin is prevalent in many foods. This vitamin

is extremely light sensitive, and it is degraded in foods (milk, for example) left in the sun.

The milling and refining of wheat, rice, and other grains causes a loss of riboflavin and other vitamins. In order to correct and prevent dietary deficiencies, the Committee on Food and Nutrition of the National Research Council began in the 1940s to recommend enrichment of cereal grains sold in the United States. Thiamine, riboflavin, niacin, and iron were the first nutrients originally recommended for enrichment by this group. As a result of these actions, generations of American children have become accustomed to reading (on their cereal boxes and bread wrappers) that their foods contain certain percentages of the "U.S. Recommended Daily Allowance" of various vitamins and nutrients.

FIGURE 18.23 • The structure of coenzyme A. Acyl groups form thioester linkages with the —SH group of the β-mercaptoethylamine moiety.

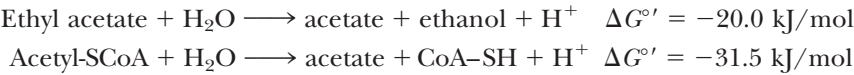
Pantothenic Acid and Coenzyme A

Pantothenic acid, sometimes called vitamin B₃, is a vitamin that makes up one part of a complex coenzyme called **coenzyme A (CoA)** (Figure 18.23). Pantothenic acid is also a constituent of **acyl carrier proteins**. Coenzyme A consists of 3',5'-adenosine bisphosphate joined to 4-phosphopantetheine in a phosphoric anhydride linkage. Phosphopantetheine in turn consists of three parts: β-mercaptoethylamine linked to β-alanine, which makes an amide bond with a branched-chain dihydroxy acid. As was the case for the nicotinamide and flavin coenzymes, the adenine nucleotide moiety of CoA acts as a recognition site, increasing the affinity and specificity of CoA binding to its enzymes.

- The two main functions of coenzyme A are
- (a) *activation of acyl groups for transfer by nucleophilic attack* and
 - (b) *activation of the α-hydrogen of the acyl group for abstraction as a proton.*

Both of these functions are mediated by the reactive sulfhydryl group on CoA, which forms **thioester** linkages with acyl groups.

The activation of acyl groups for transfer by CoA can be appreciated by comparing the hydrolysis of the thioester bond of acetyl-CoA with hydrolysis of a simple oxygen ester:



Hydrolysis of the thioester is more favorable than that of oxygen esters, presumably because the carbon–sulfur bond has less double bond character than the corresponding carbon–oxygen bond. This means that transfer of the acetyl group from acetyl-CoA to a given nucleophile (Figure 18.24) will be more spontaneous than transfer of an acetyl group from an oxygen ester. For this reason, acetyl-CoA is said to have a high group-transfer potential.

The 4-phosphopantetheine group of CoA is also utilized (for essentially the same purposes) in **acyl carrier proteins (ACPs)** involved in fatty acid biosynthesis (see Chapter 25). In acyl carrier proteins, the 4-phosphopantetheine is covalently linked to a serine hydroxyl group. Pantothenic acid is an essential factor for the metabolism of fat, protein, and carbohydrates in the tricarboxylic acid cycle and other pathways. In view of its universal importance in metabolism, it is surprising that pantothenic acid deficiencies are not a more serious problem in humans, but this vitamin is abundant in almost all foods, so that deficiencies are rarely observed.

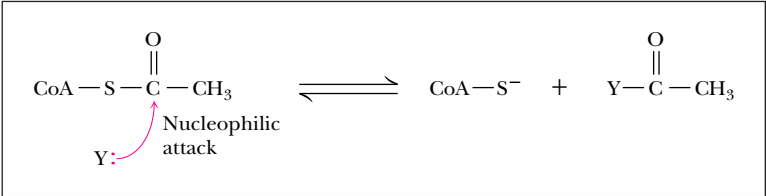
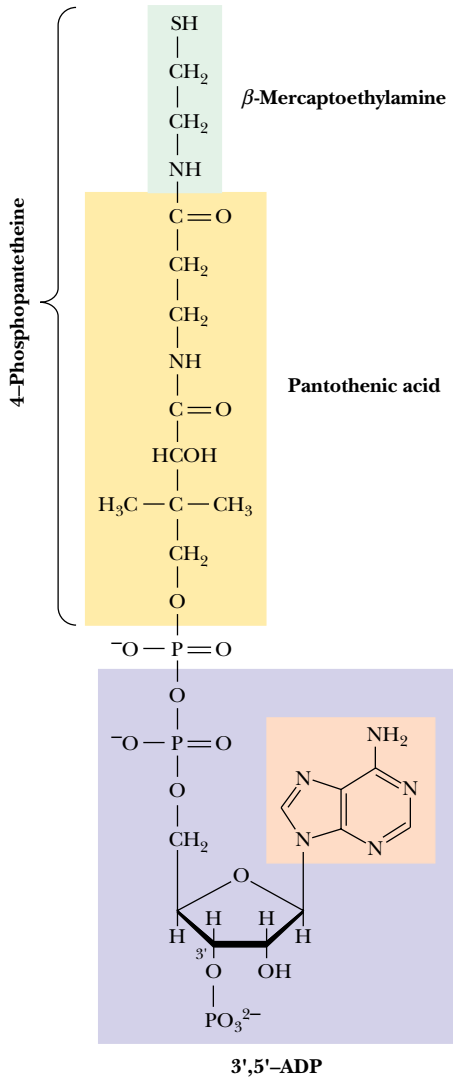


FIGURE 18.24 • Acyl transfer from acyl-CoA to a nucleophile is more favorable than transfer of an acyl group from an oxygen ester.

A DEEPER LOOK

Fritz Lipmann and Coenzyme A

Pantothenic acid is found in extracts from nearly all plants, bacteria, and animals, and the name derives from the Greek *pantos*, meaning “everywhere.” It is required in the diet of all vertebrates, but some microorganisms produce it in the rumens of animals such as cattle and sheep. This vitamin is widely distributed in foods common to the human diet, and deficiencies are only observed in cases of severe malnutrition. The eminent German-born biochemist Fritz Lipmann was the first to show that a coenzyme was required to facilitate biological acetylation reactions. (The “A” in

coenzyme A in fact stands for *acetylation*.) In studies of acetylation of sulfanilic acid (chosen because of a favorable colorimetric assay) by liver extracts, Lipmann found that a heat-stable cofactor was required. Eventually Lipmann isolated and purified the required cofactor—coenzyme A—from both liver and yeast. For his pioneering work in elucidating the role of this important coenzyme, Fritz Lipmann received the Nobel Prize in physiology or medicine in 1953.

Vitamin B₆: Pyridoxine and Pyridoxal Phosphate

The biologically active form of vitamin B₆ is **pyridoxal-5-phosphate (PLP)**, a coenzyme that exists under physiological conditions in two tautomeric forms (Figure 18.25). PLP participates in the catalysis of a wide variety of reactions involving amino acids, including transaminations, α - and β -decarboxylations, β - and γ -eliminations, racemizations, and aldol reactions (Figure 18.26). Note that these reactions include cleavage of any of the bonds to the amino acid alpha carbon, as well as several bonds in the side chain. The remarkably versatile chemistry of PLP is due to its ability to

- (a) *form stable Schiff base (aldimine) adducts with α -amino groups of amino acids,* and
- (b) *act as an effective electron sink to stabilize reaction intermediates.*

The Schiff base formed by PLP and its role as an electron sink are illustrated in Figure 18.27. In nearly all PLP-dependent enzymes, PLP in the absence of substrate is bound in a Schiff base linkage with the ϵ -NH₂ group of an active site lysine. Rearrangement to a Schiff base with the arriving substrate is a **transaldiminization** reaction. One key to PLP chemistry is the protonation of the Schiff base, which is stabilized by H bonding to the ring oxygen, increasing the acidity of the C _{α} proton [as shown in (3) of Figure 18.27]. The carbanion formed by loss of the C _{α} proton is stabilized by electron delocalization into the pyridinium ring, with the positively charged ring nitrogen acting as an electron sink. Another important intermediate is formed by protonation of the aldehyde carbon of PLP. As shown, this produces a new substrate–PLP Schiff base, which plays a role in transamination reactions and increases the acidity of the proton at C _{β} , a feature important in γ -elimination reactions.

The versatile chemistry of pyridoxal phosphate offers a rich learning experience for the student of mechanistic chemistry. William Jencks, in his classic text, *Catalysis in Chemistry and Enzymology*, writes:

It has been said that God created an organism especially adapted to help the biologist find an answer to every question about the physiology of living systems; if this is so it must be concluded that pyridoxal phosphate was created to provide satisfaction and enlightenment to those enzymologists and chemists who enjoy pushing electrons, for no other coenzyme is involved in such a wide variety of reactions, in both enzyme and model systems, which can be reasonably interpreted in terms of the chemical properties of the coenzyme. Most of

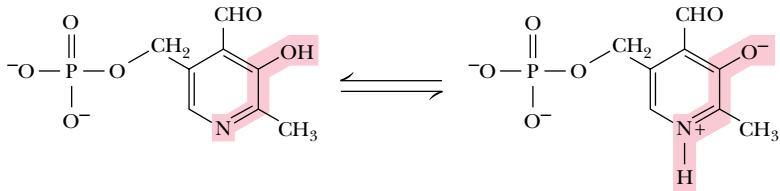


FIGURE 18.25 • The tautomeric forms of pyridoxal-5-phosphate (PLP).

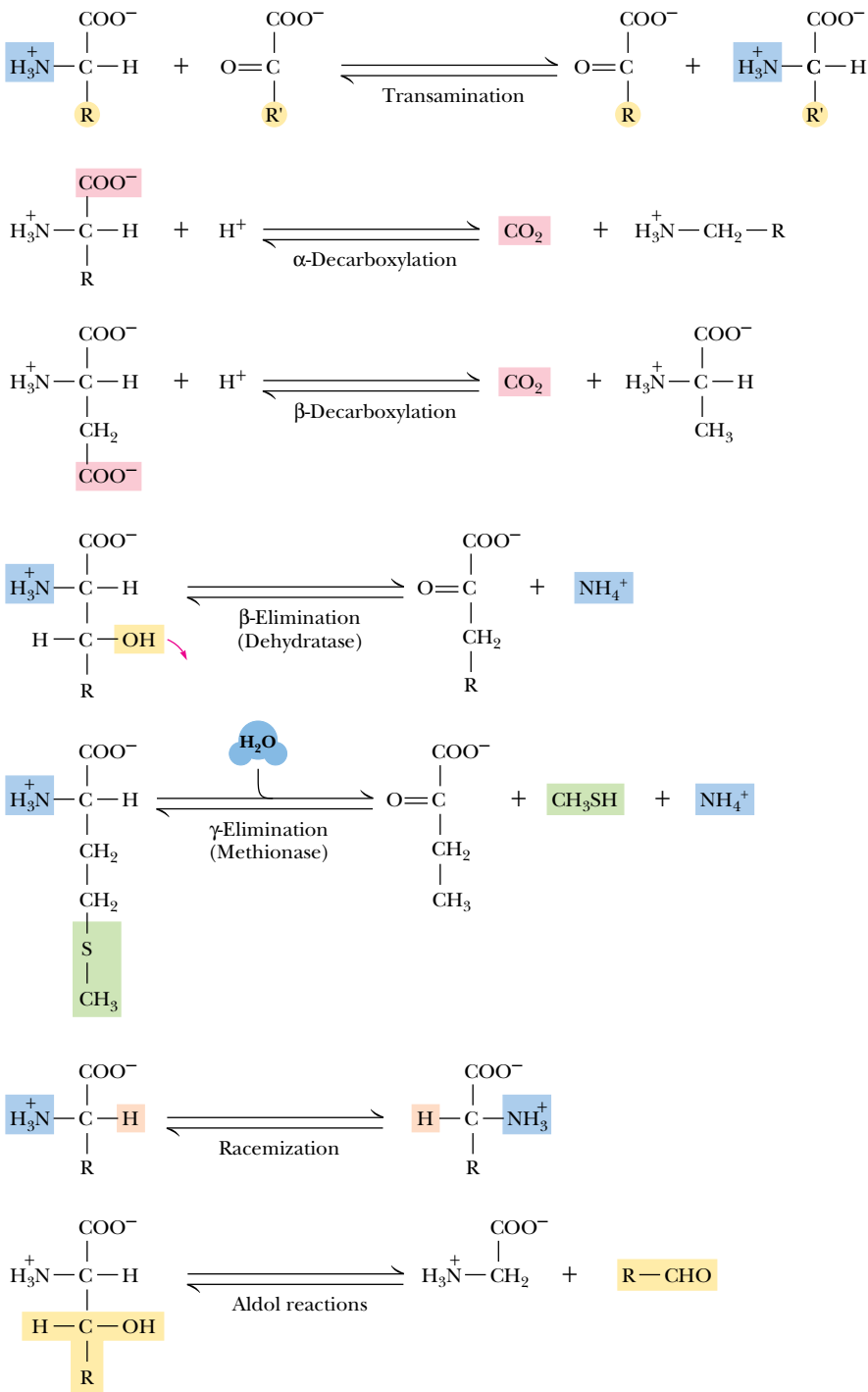


FIGURE 18.26 • The seven classes of reactions catalyzed by pyridoxal-5-phosphate.

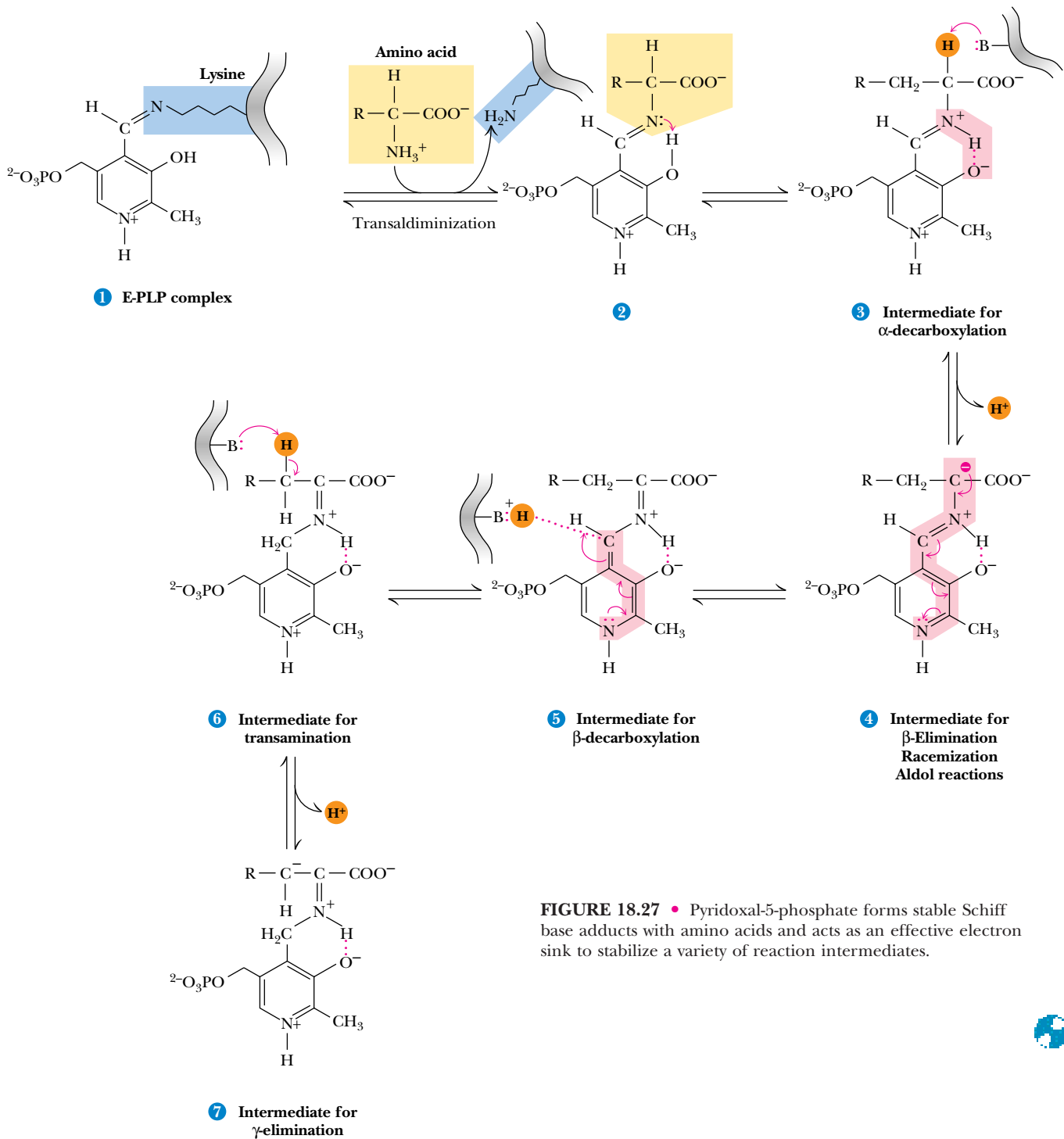


FIGURE 18.27 • Pyridoxal-5-phosphate forms stable Schiff base adducts with amino acids and acts as an effective electron sink to stabilize a variety of reaction intermediates.

these reactions are made possible by a common structural feature. That is, electron withdrawal toward the cationic nitrogen atom of the imine and into the electron sink of the pyridoxal ring from the α carbon atom of the attached amino acid activates all three of the substituents on this carbon atom for reactions which require electron withdrawal from this atom.¹

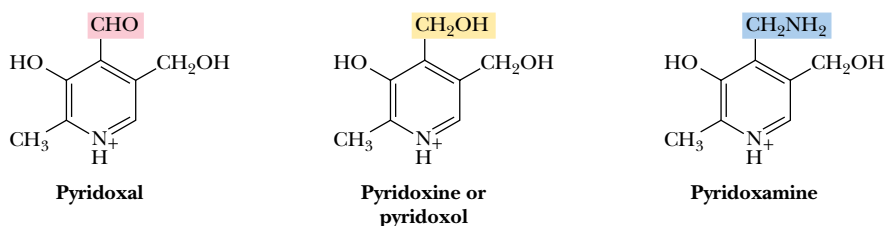
¹Jencks, William P., 1969. *Catalysis in Chemistry and Enzymology*. New York: McGraw-Hill.

DEEPER LOOK

Vitamin B₆

Goldberger and Lillie in 1926 found that rats fed certain nutritionally deficient diets developed **dermatitis acrodynia**, a skin disorder characterized by edema and lesions of the ears, paws, nose, and tail. Szent-Györgyi later found that a factor he had isolated prevented these skin lesions in the rat. He proposed the name **vitamin B₆** for his factor. **Pyridoxine**, a form of this vitamin found in plants (and the form of B₆ sold commercially), was isolated in 1938 by three research groups working independently. **Pyridoxal** and **pyridoxamine**, the forms that predominate in animals, were

identified in 1945. A metabolic role for pyridoxal was postulated by Esmond Snell, who had shown that when pyridoxal was heated with glutamate (in the absence of any enzymes), the amino group of glutamate was transferred to pyridoxal, forming pyridoxamine. Snell postulated (correctly) that pyridoxal might be a component of a coenzyme needed for transamination reactions in which the α -amino group of an amino acid is transferred to the α -carbon of an α -keto acid.



The structures of pyridoxal, pyridoxine, and pyridoxamine.

Vitamin B₁₂: Cyanocobalamin

Vitamin B₁₂, or **cyanocobalamin**, is converted in the body into two coenzymes. The predominant coenzyme form is **5'-deoxyadenosylcobalamin** (Figure 18.28), but smaller amounts of **methylcobalamin** also exist in liver, for example. The crystal structure of 5'-deoxyadenosylcobalamin was determined by X-ray diffraction in 1961 by Dorothy Hodgkin and coworkers in England. The structure consists of a *corrin ring* with a *cobalt ion* in the center. The corrin ring, with four *pyrrole groups*, is similar to the heme prophyrin ring, except that two of the pyrrole rings are linked directly. Methylene bridges form the other pyrrole-pyrrole linkages, as for porphyrin. The cobalt is coordinated to the four (planar) pyrrole nitrogens. One of the axial cobalt ligands is a nitrogen of the dimethylbenzimidazole group. The other axial cobalt ligand may be —CN, —CH₃, —OH, or the 5'-carbon of a 5'-deoxyadenosyl group, depending on the form of the coenzyme. The most striking feature of Hodgkin's structure of 5'-deoxyadenosylcobalamin is the cobalt-carbon bond distance of 0.205 nm. *This bond is predominantly covalent* and the structure is actually an **alkyl cobalt**. Such alkyl cobalts were thought to be highly unstable until Hodgkin's pioneering X-ray study. The Co-carbon-carbon bond angle of 130 degrees indicates partial ionic character.

The B₁₂ coenzymes participate in three types of reactions (Figure 18.29):

1. *Intramolecular rearrangements*
2. *Reductions of ribonucleotides to deoxyribonucleotides* (in certain bacteria)
3. *Methyl group transfers*

FIGURE 18.28 • The structure of cyanocobalamin (*top*) and simplified structures showing several coenzyme forms of vitamin B₁₂. The Co—C bond of 5'-deoxyadenosylcobalamin is predominantly covalent (note the short bond length of 0.205 nm) but with some ionic character. Note that the convention of writing the cobalt atom as Co³⁺ attributes the electrons of the Co—C and Co—N bonds to carbon and nitrogen, respectively.

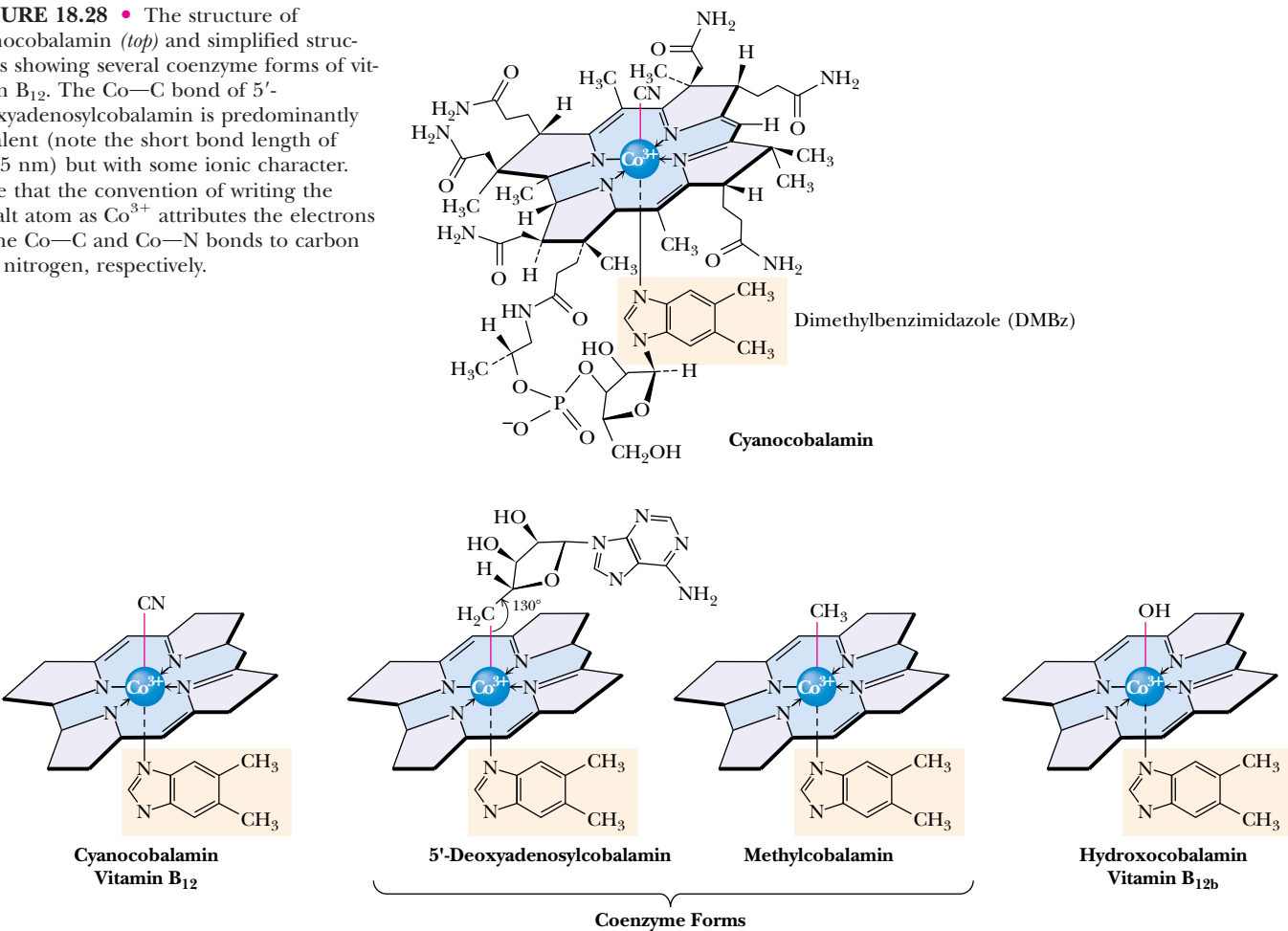
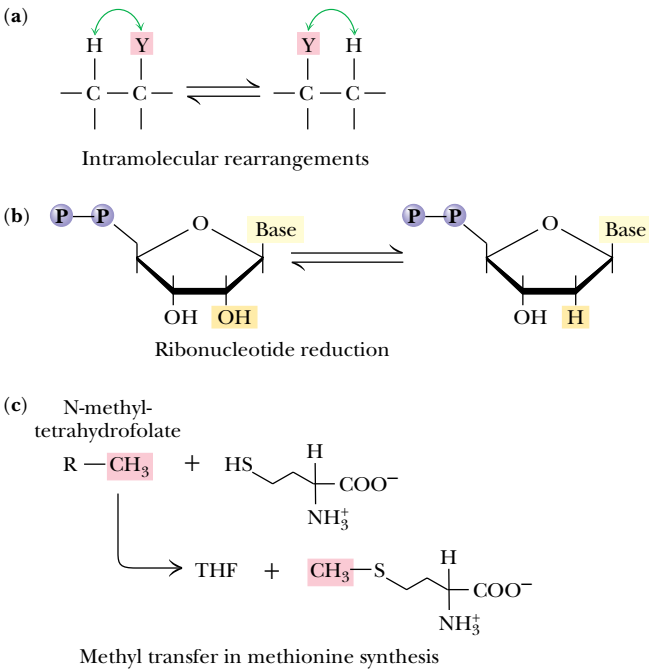


FIGURE 18.29 • Vitamin B₁₂ functions as a coenzyme in intramolecular rearrangements, reduction of ribonucleotides, and methyl group transfers.



HUMAN BIOCHEMISTRY

Vitamin B₁₂ and Pernicious Anemia

The most potent known vitamin (that is, the one needed in the smallest amounts) was the last to be discovered. Vitamin B₁₂ is best known as the vitamin that prevents **pernicious anemia**. Minot and Murphy in 1926 demonstrated that such anemia could be prevented by eating large quantities of liver, but the active agent was not identified for many years. In 1948, Rickes and coworkers (in the United States) and Smith (in England) both reported the first successful isolation of vitamin B₁₂. West showed that injections of the vitamin induced dramatic beneficial responses in pernicious anemia patients. Eventually, two different crystalline preparations of the vitamin were distinguished. The first appeared to be true cyanocobalamin. The second showed the same biological activity as a cyanocobalamin, but had a different spectrum and was named **vitamin B_{12b}** and also **hydroxocobalamin**. It was eventually found that the cyanide group in cyanocobalamin originated from the charcoal used in the purification process!

Vitamin B₁₂ is not synthesized by animals or by plants. Only a few species of bacteria synthesize this complex substance. Carnivorous animals easily acquire sufficient amounts of B₁₂ from meat in their diet, but herbivorous creatures typically depend on intestinal bacteria to synthesize B₁₂ for them. This is sometimes not sufficient, and certain animals, including rabbits, occasionally eat their feces in order to accumulate the necessary quantities of B₁₂.

The nutritional requirement for vitamin B₁₂ is low. Adult humans require only about 3 micrograms per day, an amount easily acquired with normal eating habits. However, because plants do not synthesize vitamin B₁₂, pernicious anemia symptoms are sometimes observed in strict vegetarians.

The first two of these are mediated by 5'-deoxyadenosylcobalamin, whereas methyl transfers are effected by methylcobalamin. The mechanism of ribonucleotide reductase is discussed in Chapter 27. Methyl group transfers that employ *tetrahydrofolate* as a coenzyme are described later in this chapter.

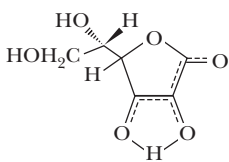
Vitamin C: Ascorbic Acid

L-Ascorbic acid, better known as **vitamin C**, has the simplest chemical structure of all the vitamins (Figure 18.30). It is widely distributed in the animal and plant kingdoms, and only a few vertebrates—humans and other primates, guinea pigs, fruit-eating bats, certain birds, and some fish (rainbow trout, carp, and Coho salmon, for example)—are unable to synthesize it. In all these organisms, the inability to synthesize ascorbic acid stems from a lack of a liver enzyme, L-gulonolactone oxidase.

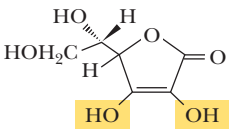
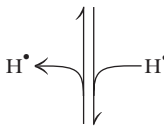
Ascorbic acid is a reasonably strong reducing agent. The biochemical and physiological functions of ascorbic acid most likely derive from its reducing properties—it functions as an electron carrier. Loss of one electron due to interactions with oxygen or metal ions leads to **semidehydro-L-ascorbate**, a reactive free radical (Figure 18.30) that can be reduced back to L-ascorbic acid by various enzymes in animals and plants. A characteristic reaction of ascorbic acid is its oxidation to *dehydro-L-ascorbic acid*. Ascorbic acid and dehydroascorbic acid form an effective redox system.

In addition to its role in preventing scurvy (see *Human Biochemistry box: Ascorbic Acid and Scurvy* and also Chapter 6), ascorbic acid also plays important roles in the brain and nervous system. It also mobilizes iron in the body, prevents anemia, ameliorates allergic responses, and stimulates the immune system.

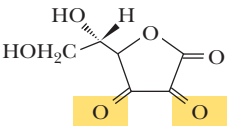
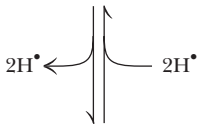
FIGURE 18.30 • The physiological effects of ascorbic acid (vitamin C) are the result of its action as a reducing agent. A two-electron oxidation of ascorbic acid yields dehydroascorbic acid.



L-Ascorbate free radical



Ascorbic acid (Vitamin C)



Dehydro-L-ascorbic acid

HUMAN BIOCHEMISTRY

Ascorbic Acid and Scurvy

Ascorbic acid is effective in the treatment and prevention of **scurvy**, a potentially fatal disorder characterized by anemia, alteration of protein metabolism, and weakening of collagenous structures in bone, cartilage, teeth, and connective tissues (see Chapter 6). Western world diets are now routinely so rich in vitamin C that it is easy to forget that scurvy affected many people in ancient Egypt, Greece, and Rome, and that, in the Middle Ages, it was endemic in northern Europe in winter when fresh fruits and vegetables were scarce. Ascorbic acid is a vitamin that has routinely altered the course of history, ending ocean voyages and military

campaigns when food supplies became depleted of vitamin C and fatal outbreaks of scurvy occurred. The isolation of ascorbic acid was first reported by Albert Szent-Györgyi (who called it *hexuronic acid*) in 1928. The structure was determined by Hirst and Haworth in 1933, and, simultaneously, Reichstein reported its synthesis. Haworth and Szent-Györgyi, who together suggested that the name be changed to L-ascorbic acid to describe its **antiscorbutic** (antiscurvy) activity, were awarded the Nobel Prize in 1937 for their studies of vitamin C.

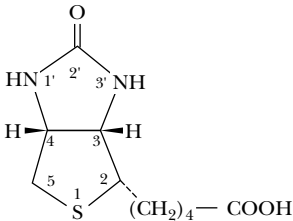


FIGURE 18.31 • The structure of biotin.

Biotin

Biotin (Figure 18.31) acts as a **mobile carboxyl group carrier** in a variety of enzymatic carboxylation reactions. In each of these, biotin is bound covalently to the enzyme as a prosthetic group via the ϵ -amino group of a lysine residue on the protein (Figure 18.32). The biotin-lysine function is referred to as a **biocytin** residue. The result is that *the biotin ring system is tethered to the protein by a long, flexible chain*. The 10 atoms in this chain separate the biotin ring and the lysine α -carbon by approximately 1.5 nm. This chain allows biotin to acquire carboxyl groups at one subsite of the enzyme active site and deliver them to a substrate acceptor at another subsite.

Most biotin-dependent carboxylations (Table 18.5) use *bicarbonate* as the carboxylating agent and transfer the carboxyl group to a *substrate carbanion*. Bicarbonate is plentiful in biological fluids, but it is a poor electrophile at carbon and must be “activated” for attack by the substrate carbanion.

Table 18.5	
Principal Biotin-Dependent Carboxylations	
ATP-dependent	$\text{ATP} + \text{HCO}_3^- + \text{H}_3\text{C}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{COO}^- \longrightarrow \text{}^-\text{OOC}-\underset{\underset{\text{H}}{\mid}}{\text{C}}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{COO}^- + \text{ADP} + \text{P}$ Pyruvate Oxaloacetate
	$\text{ATP} + \text{HCO}_3^- + \text{H}_3\text{C}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{SCoA} \longrightarrow \text{}^-\text{OOC}-\underset{\underset{\text{H}}{\mid}}{\text{C}}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{SCoA} + \text{ADP} + \text{P}$ Acetyl-CoA Malonyl-CoA
	$\text{ATP} + \text{HCO}_3^- + \text{H}_3\text{C}-\underset{\underset{\text{H}}{\mid}}{\overset{\overset{\text{H}}{\mid}}{\text{C}}}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{SCoA} \longrightarrow \text{H}_3\text{C}-\underset{\underset{\text{H}}{\mid}}{\overset{\overset{\text{}^-\text{OOC}}{\mid}}{\text{C}}}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{SCoA} + \text{ADP} + \text{P}$ Propionyl-CoA Methylmalonyl-CoA

DEEPER LOOK

Biotin

Early in the 1900s, it was observed that certain strains of yeast required a material called **bios** for growth. Bios was eventually found to contain four different substances: myoinositol, β -alanine, pantothenic acid, and a compound later shown to be *biotin*. Kögl and Tönnis first isolated biotin from egg yolk in 1936. Boas, in 1927, and Szent-György, in 1931, found substances in liver that were capable of curing and preventing the dermatitis, loss of hair, and paralysis that occurred in rats fed large amounts of raw egg whites (a condition known as *egg white injury*). Boas called the factor “protective factor X” and Szent-György named the substance *vitamin H* (from the German *haut*, meaning “skin”), but both were

soon shown to be identical to biotin. It is now known that egg white contains a basic protein called **avidin**, which has an extremely high affinity for biotin ($K_d = 10^{-15} M$). The sequestering of biotin by avidin is the cause of the egg white injury condition.

The structure of biotin was determined in the early 1940s by Kögl in Europe and by du Vigneaud and coworkers in the United States. Interestingly, the biotin molecule contains three asymmetric carbon atoms, and biotin could thus exist as eight different stereoisomers. Only one of these shows biological activity.

Lipoic Acid

Lipoic acid exists as a mixture of two structures: a closed-ring disulfide form and an open-chain reduced form (Figure 18.33). Oxidation–reduction cycles interconvert these two species. As is the case for biotin, lipoic acid does not often occur free in nature, but rather is covalently attached in amide linkage with lysine residues on enzymes. The enzyme that catalyzes the formation of the *lipoamide* linkage requires ATP and produces lipoamide-enzyme conjugates, AMP, and pyrophosphate as products of the reaction.

Lipoic acid is an **acyl group carrier**. It is found in *pyruvate dehydrogenase* and *α -ketoglutarate dehydrogenase*, two multienzyme complexes involved in carbohydrate metabolism (Figure 18.34). *Lipoic acid functions to couple acyl-group transfer and electron transfer during oxidation and decarboxylation of α -keto acids*.

The special properties of lipoic acid arise from the ring strain experienced by oxidized lipoic acid. The closed ring form is approximately 20 kJ higher in energy than the open-chain form, and this results in a strong negative reduction potential of about $-0.30 V$. The oxidized form readily oxidizes cyanides to isothiocyanates and sulfhydryl groups to mixed disulfides.

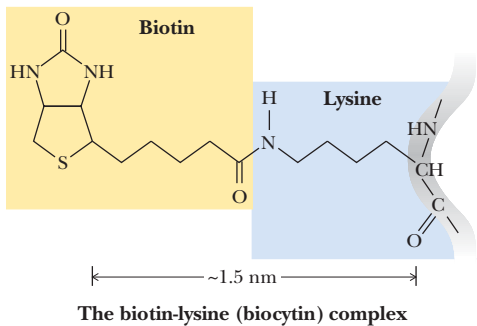


FIGURE 18.32 • Biotin is covalently linked to a protein via the ϵ -amino group of a lysine residue. The biotin ring is thus tethered to the protein by a 10-atom chain. It functions by carrying carboxyl groups between distant sites on biotin-dependent enzymes.

DEEPER LOOK

Lipoic Acid

Lipoic acid (6,8-dithiooctanoic acid) was isolated and characterized in 1951 in studies that showed that it was required for the growth of certain bacteria and protozoa. This accomplishment was one of the most impressive feats of isolation in the early history of biochemistry. Eli Lilly and Co., in cooperation with Lester J. Reed at the University of Texas and I. C. Gunsalus at the

University of Illinois, isolated just 30 mg of lipoic acid from approximately 10 tons of liver! No evidence exists of a dietary lipoic acid requirement by humans; strictly speaking, it is not considered a vitamin. Nevertheless, it is an essential component of several enzymes of intermediary metabolism and is present in body tissues in small amounts.

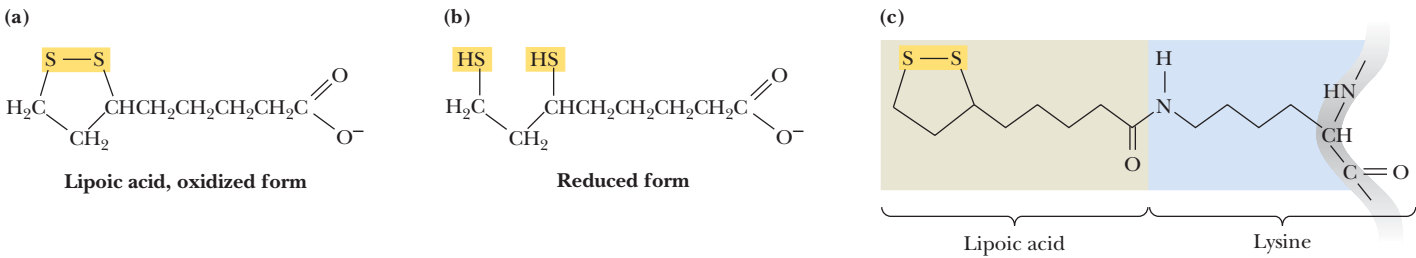


FIGURE 18.33 • The oxidized and reduced forms of lipoic acid and the structure of the lipoic acid–lysine conjugate.

Folic Acid

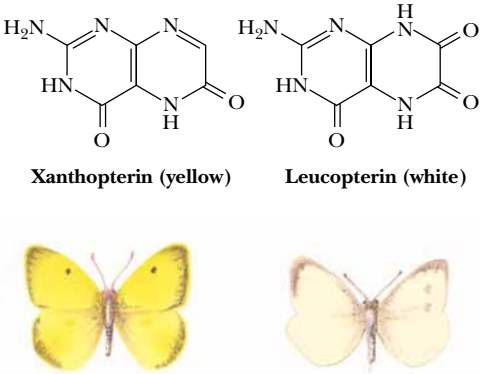
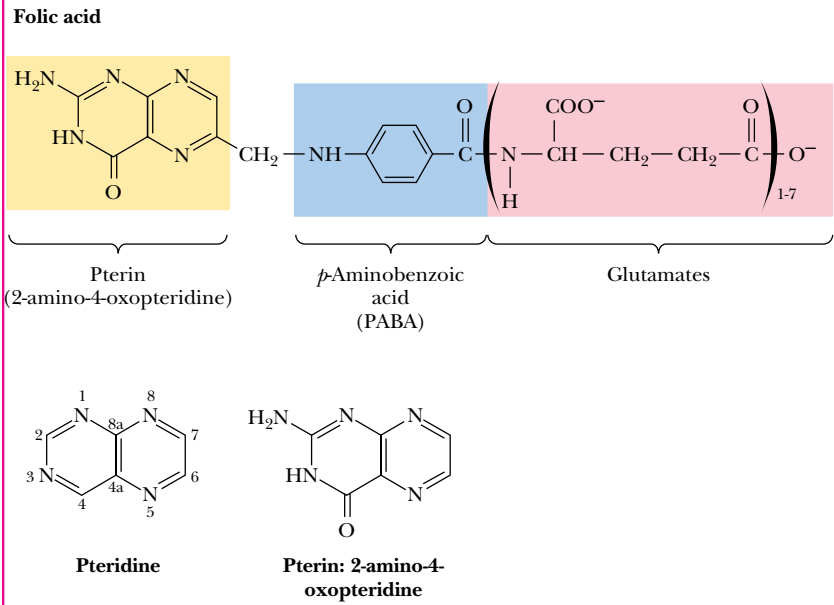
Folic acid derivatives (folates) are acceptors and donors of one-carbon units for all oxidation levels of carbon except that of CO₂ (where biotin is the relevant carrier). The active coenzyme form of folic acid is **tetrahydrofolate (THF)**. THF is formed via two successive reductions of folate by *dihydrofolate reductase* (Figure 18.35). One-carbon units in three different oxidation states may be bound to tetrahydrofolate at the N⁵ and/or N¹⁰ nitrogens (Table 18.6). These one-carbon units

DEEPER LOOK

Folic Acid, Pterins, and Insect Wings

Folic acid is a member of the vitamin B complex found in green plants, fresh fruit, yeast, and liver. Folic acid takes its name from *folium*, Latin for “leaf.” Pterin compounds are named from the Greek word for “wing” because these substances were first identified in insect wings. Two pterins are familiar to any child who has seen (and chased) the common yellow sulfur butterfly and its white counterpart, the cabbage butterfly. *Xanthopterin* and *leuco-*

copterin are the respective pigments in these butterflies’ wings. Mammalian organisms cannot synthesize pterins; they derive folates from their diet or from microorganisms active in the intestines. Folic acid derives its name from *folium*, Latin for “leaf.” Pterin compounds are named from the Greek *πτέρυξ*, for “wing,” since these substances were first identified in insect wings.



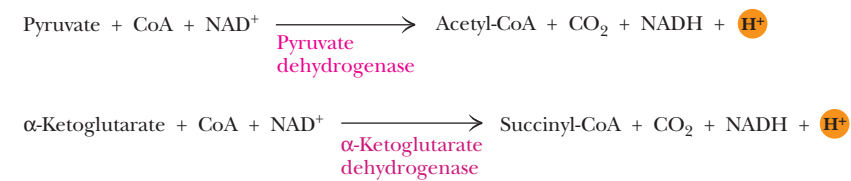


FIGURE 18.34 • The enzyme reactions catalyzed by lipoic acid.

may exist at the oxidation levels of methanol, formaldehyde, or formate (carbon atom oxidation states of −2, 0, and 2, respectively). The biosynthetic pathways for methionine and homocysteine (Chapter 26), purines (Chapter 27), and the pyrimidine thymine (Chapter 27) rely on the incorporation of one-carbon units from THF derivatives.

The Vitamin A Group

Vitamin A or **retinol** (Figure 18.36) often occurs in the form of esters, called **retinyl esters**. The aldehyde form is called **retinal** or **retinaldehyde**. Like all the fat-soluble vitamins, retinol is an *isoprenoid* molecule and is biosynthesized from isoprene building blocks (Chapter 8). Retinol can be absorbed in the diet from animal sources or synthesized from β-carotene from plant sources. The absorption by the body of fat-soluble vitamins proceeds by mechanisms different from those of the water-soluble vitamins. Once ingested, preformed vitamin A or β-carotene and its analogs are released from proteins by the action of proteolytic enzymes in the stomach and small intestine. The free carotenoids and retinyl esters aggregate in fatty globules that enter the duodenum. The detergent actions of bile salts break these globules down into small aggregates that can be digested by pancreatic lipase, cholesteryl ester hydrolase, retinyl ester hydrolase, and similar enzymes. The product compounds form *mixed micelles* (see Chapter 8) containing the retinol, carotenoids, and other lipids, which are absorbed into mucosal cells in the upper half of the intestinal tract. Retinol is esterified (usually with palmitic acid) and transported to the liver in a lipoprotein complex.

The retinol that is delivered to the retinas of the eyes in this manner is accumulated by **rod** and **cone cells**. In the rods (which are the better characterized of the two cell types), retinol is oxidized by a specific **retinol dehydrogenase** to become all-*trans* retinal and then converted to 11-*cis* retinal by **reti-**

Table 18.6

Oxidation States of Carbon in 1-Carbon Units Carried by Tetrahydrofolate

Oxidation Number*	Oxidation Level	One-Carbon Form†	Tetrahydrofolate Form
−2	Methanol (most reduced)	—CH ₃	N ⁵ -Methyl-THF
0	Formaldehyde	—CH ₂ —	N ⁵ ,N ¹⁰ -Methylene-THF
2	Formate (most oxidized)	—CH=O	N ⁵ -Formyl-THF
		—CH=O	N ¹⁰ -Formyl-THF
		—CH=NH	N ⁵ -Formimino-THF
		—CH=	N ⁵ ,N ¹⁰ -Methenyl-THF

*Calculated by assigning valence bond electrons to the more electronegative atom and then counting the charge on the quasi ion. A carbon assigned four valence electrons would have an oxidation number of 0. The carbon in N⁵-methyl-THF is assigned six electrons from the three C—H bonds and thus has an oxidation number of −2.

† Note: All vacant bonds in the structures shown are to atoms more electronegative than C.

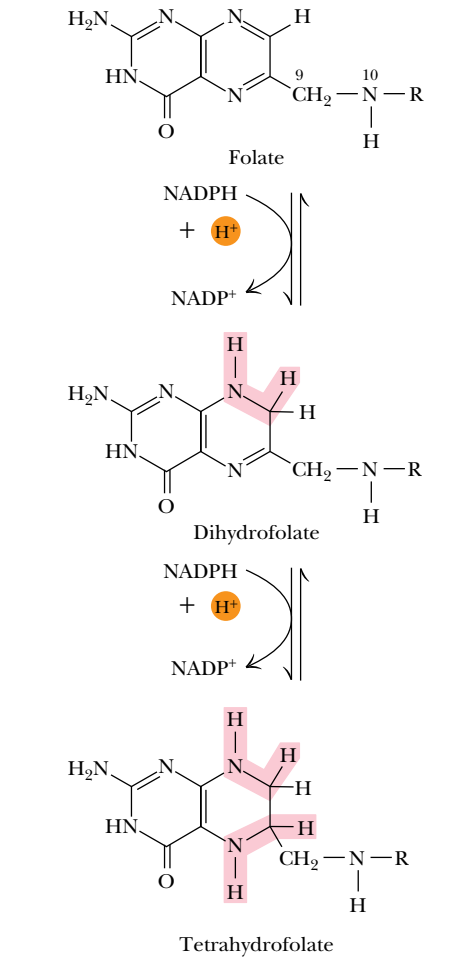


FIGURE 18.35 • Formation of THF from folic acid by the dihydrofolate reductase reaction. The R group on these folate molecules symbolizes the one to seven (or more) glutamate units that folates characteristically contain. All of these glutamates are bound in γ-carboxyl amide linkages (as in the folic acid structure shown in the box *A Deeper Look: Folic Acid, Pterins, and Insect Wings*). The one-carbon units carried by THF are bound at N⁵, or at N¹⁰, or as a single carbon attached to both N⁵ and N¹⁰.

A
D E E P E R L O O K

β-Carotene and Vision

Night blindness was probably the first disorder to be ascribed to a nutritional deficiency. The ancient Egyptians left records as early as 1500 B.C. of recommendations that the juice squeezed from cooked liver could cure night blindness if applied topically, and the method may have been known much earlier. Frederick Gowland Hopkins, working in England in the early 1900s, found that alcoholic extracts of milk contained a growth-stimulating fac-

tor. Marguerite Davis and Elmer McCollum at Wisconsin showed that egg yolk and butter contain a similar growth-stimulating lipid, which, in 1915, they called “fat soluble A.” Moore in England showed that **β-carotene**, the plant pigment, could be converted to the colorless form of the liver-derived vitamin. In 1935, George Wald of Harvard showed that *retinene* found in visual pigments of the eye was identical with *retinaldehyde*, a derivative of vitamin A.

nal isomerase (Figure 18.36). The aldehyde group of retinal forms a Schiff base with a lysine on **opsin**, to form light-sensitive **rhodopsin**.

The Vitamin D Group

The two most prominent members of the **vitamin D** family are **ergocalciferol** (known as vitamin D₂) and **cholecalciferol** (vitamin D₃). Cholecalciferol is produced in the skin of animals by the action of ultraviolet light (sunlight, for example) on its precursor molecule, 7-dehydrocholesterol (Figure 18.37). The absorption of light energy induces a photoisomerization via an excited singlet state, which results in breakage of the 9,10 carbon bond and formation of **pre-vitamin D₃**. The next step is a spontaneous isomerization to yield vitamin D₃, cholecalciferol. Ergocalciferol, which differs from cholecalciferol only in the side-chain structure, is similarly produced by the action of sunlight on the plant sterol **ergosterol**. (Ergosterol is so named because it was first isolated from

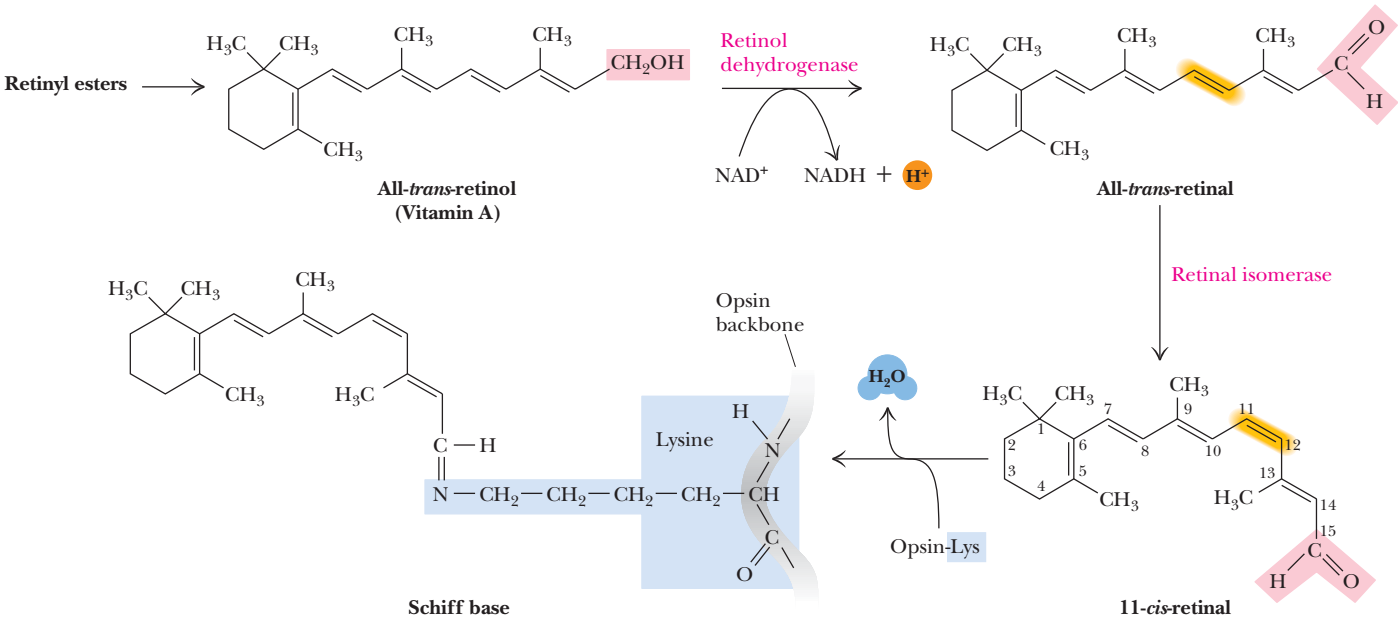


FIGURE 18.36 • The incorporation of retinal into the light-sensitive protein rhodopsin involves several steps. All-*trans*-retinol is oxidized by retinol dehydrogenase and then isomerized to 11-*cis*-retinal, which forms a Schiff base linkage with opsin to form light-sensitive rhodopsin.

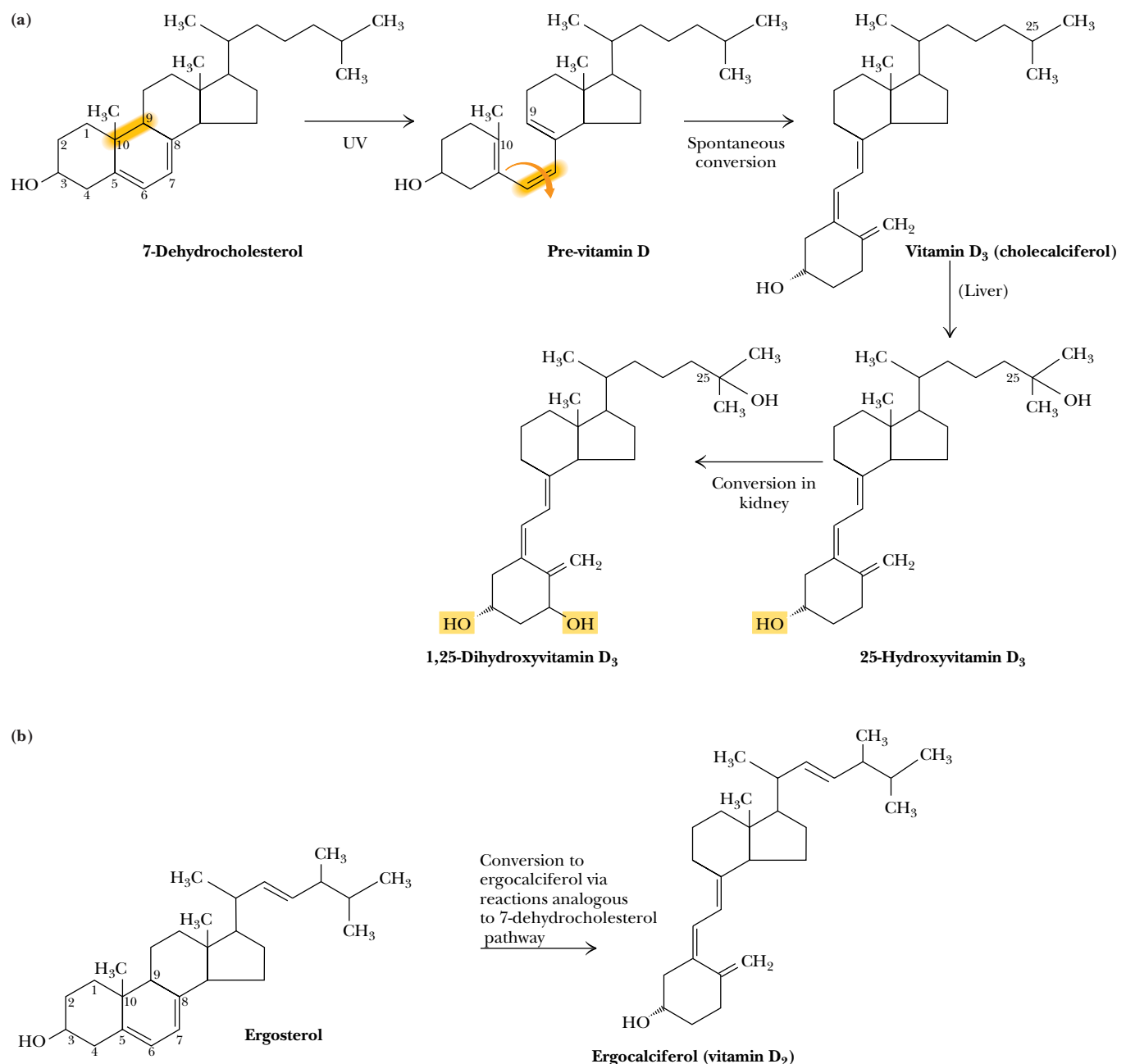


FIGURE 18.37 • (a) Vitamin D₃ (cholecalciferol) is produced in the skin by the action of sunlight on 7-dehydrocholesterol. The successive action of mixed-function oxidases in the liver and kidney produces 1,25-dihydroxyvitamin D₃, the active form of vitamin D. (b) Ergocalciferol is produced in analogous fashion from ergosterol.

HUMAN BIOCHEMISTRY

Vitamin D and Rickets

Vitamin D is a family of closely related molecules that prevent **rickets**, a childhood disease characterized by inadequate intestinal absorption and kidney reabsorption of calcium and phosphate. These inadequacies eventually lead to the demineralization of bones. The symptoms of rickets include bowlegs,

knock-knees, curvature of the spine, and pelvic and thoracic deformities, the results of normal mechanical stresses on demineralized bones. Vitamin D deficiency in adults leads to a weakening of bones and cartilage known as **osteomalacia**.

A D E E P E R L O O K

Vitamin E

In a study of the effect of nutrition on reproduction in the rat in the 1920s, Herbert Evans and Katherine Bishop found that rats failed to reproduce on a diet of rancid lard, unless lettuce or whole wheat was added to the diet. The essential factor was traced to a vitamin in the wheat germ oil. Named *vitamin E* by Evans (using the next available letter following on the discovery of vita-

min D), the factor was purified by Emerson, who named it *tocopherol*, from the Greek *tokos* for “childbirth,” and *pherein*, for “to bring forth.” Vitamin E is now recognized as a generic term for a family of substances, all of them similar in structure to the most active form, *α-tocopherol*.

ergot, a rye fungus.) Because humans can produce vitamin D₃ from 7-dehydrocholesterol by the action of sunlight on the skin, “vitamin D” is not strictly speaking a vitamin at all.

On the basis of its mechanism of action in the body, cholecalciferol should be called a **prohormone**, a hormone precursor. Dietary forms of vitamin D are absorbed through the aid of bile salts in the small intestine. Whether absorbed in the intestine or photosynthesized in the skin, cholecalciferol is then transported to the liver by a specific **vitamin D-binding protein (DBP)**, also known as **transcalciferin**. In the liver, cholecalciferol is hydroxylated at the C-25 position by a mixed-function oxidase to form *25-hydroxyvitamin D* (that is, *25-hydroxycholecalciferol*). Although this is the major circulating form of vitamin D in the body, 25-hydroxyvitamin D possesses far less biological activity than the final active form. To form this latter species, 25-hydroxyvitamin D is returned to the circulatory system and transported to the kidneys. There it is hydroxylated at the C-1 position by a mitochondrial mixed-function oxidase to form *1,25-dihydroxyvitamin D₃* (that is, *1,25-dihydroxycholecalciferol*), the active form of vitamin D. 1,25-Dihydroxycholecalciferol is then transported to target tissues, where it acts like a hormone to regulate calcium and phosphate metabolism.

1,25-Dihydroxyvitamin D₃, together with two peptide hormones, *calcitonin* and *parathyroid hormone (PTH)*, functions to regulate calcium homeostasis and plays a role in phosphorus homeostasis. As described elsewhere in this text, calcium is important for many processes, including muscle contraction, nerve impulse transmission, blood clotting, and membrane structure. Phosphorus, of course, is of critical importance to DNA, RNA, lipids, and many metabolic processes. Phosphorylation of proteins is an important regulatory signal for many biological processes. Phosphorus and calcium are also critically important for the formation of bones. Any disturbance of normal serum phosphorus and calcium levels will result in alterations of bone structure, as in rickets. The mechanism of calcium homeostasis involves precise coordination of calcium (a) absorption in the intestine, (b) deposition in the bones, and (c) excretion by the kidneys. If a decrease in serum calcium occurs, vitamin D is converted to its active form, which acts in the intestine to increase calcium absorption. PTH and vitamin D act on bones to enhance absorption of calcium, and PTH acts on the kidney to cause increased calcium reabsorption. If serum calcium levels get too high, calcitonin induces calcium excretion from the kidneys and inhibits calcium mobilization from bone, while inhibiting vitamin D metabolism and PTH secretion.

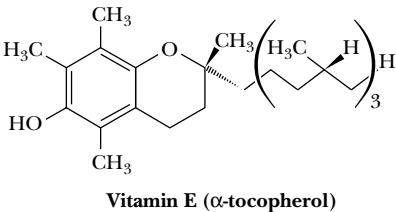


FIGURE 18.38 • The structure of vitamin E (α-tocopherol).

Vitamin E: Tocopherol

The structure of **vitamin E** in its most active form, **α-tocopherol**, is shown in Figure 18.38. α-Tocopherol is a potent antioxidant, and its function in animals and humans is often ascribed to this property. On the other hand, the molecular details of its function are almost entirely unknown. One possible role for

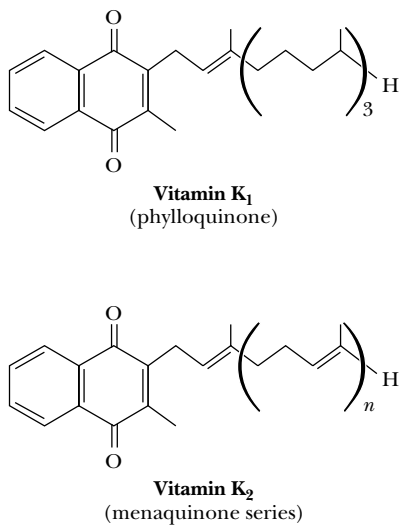


FIGURE 18.39 • The structures of the K vitamins.

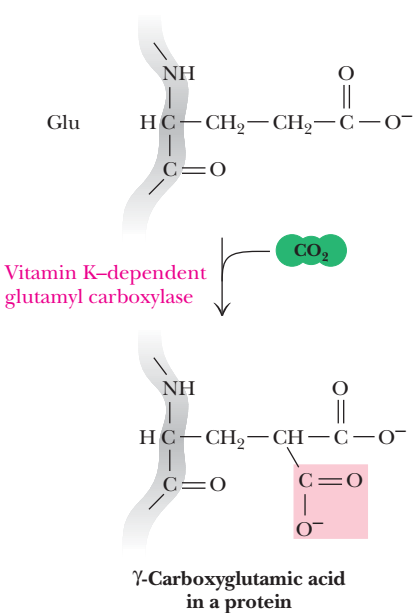


FIGURE 18.40 • The glutamyl carboxylase reaction is vitamin K-dependent. This enzyme activity is essential for the formation of γ-carboxyglutamyl residues in several proteins of the blood-clotting cascade (Figure 15.5), accounting for the vitamin K dependence of coagulation.

vitamin E may relate to the protection of unsaturated fatty acids in membranes because these fatty acids are particularly susceptible to oxidation. When human plasma levels of α-tocopherol are low, red blood cells are increasingly subject to oxidative hemolysis. Infants, especially premature infants, are deficient in vitamin E. When low-birth-weight infants are exposed to high oxygen levels for the purpose of alleviating respiratory distress, the risk of oxygen-induced retina damage can be reduced with vitamin E administration. The mechanism(s) of action of vitamin E remain obscure.

Vitamin K: Naphthoquinone

The function of **vitamin K** (Figure 18.39) in the activation of blood clotting was not elucidated until the early 1970s, when it was found that animals and humans treated with coumarin-type anticoagulants contained an inactive form of **prothrombin** (an essential protein in the coagulation cascade). It was soon shown that a post-translational modification of prothrombin is essential to its function. In this modification, 10 glutamic acid residues on the amino terminal end of prothrombin are carboxylated to form γ-carboxyglutamyl residues. These residues are effective in the coordination of calcium, which is required for the coagulation process. The enzyme responsible for this modification, a liver microsomal *glutamyl carboxylase*, requires vitamin K for its activity (Figure 18.40). Not only prothrombin (called “factor II” in the clotting pathway) but also clotting factors VII, IX, and X and several plasma proteins—proteins C, M, S, and Z—contain γ-carboxyglutamyl residues in a manner similar to prothrombin. Other examples of γ-carboxyglutamyl residues in proteins are known.

HUMAN BIOCHEMISTRY

Vitamin K and Blood Clotting

In studies in Denmark in the 1920s, Henrik Dam noticed that chicks fed a diet extracted with nonpolar solvents developed hemorrhages. Moreover, blood taken from such animals clotted slowly. Further studies by Dam led him to conclude in 1935 that the anti-hemorrhage factor was a new fat-soluble vitamin, which he called *vitamin K* (from *koagulerig*, the Danish word for “coagulation”).

Dam, along with Karrar of Zurich, isolated the pure vitamin from alfalfa as a yellow oil. Another form, which was crystalline at room temperature, was soon isolated from fish meal. These two compounds were named *vitamins K₁* and *K₂*. Vitamin *K₂* can actually occur as a family of structures with different chain lengths at the C-3 position.

PROBLEMS

1. If 3×10^{14} kg of CO₂ are cycled through the biosphere annually, how many human equivalents (70-kg persons composed of 18% carbon by weight) could be produced each year from this amount of CO₂?
2. Define the differences in carbon and energy metabolism between *photoautotrophs* and *photoheterotrophs*, and between *chemoautotrophs* and *chemoheterotrophs*.
3. Name three principal inorganic sources of oxygen atoms that are commonly available in the inanimate environment and readily accessible to the biosphere.
4. What are the features that generally distinguish pathways of catabolism from pathways of anabolism?
5. Name the three principal modes of enzyme organization in metabolic pathways.
6. Why do metabolic pathways have so many different steps?
7. Why is the pathway for the biosynthesis of a biomolecule at least partially different from the pathway for its catabolism? Why is the pathway for the biosynthesis of a biomolecule inherently more complex than the pathway for its degradation?
8. What are the metabolic roles of ATP, NAD⁺, and NADPH?
9. Metabolic regulation is achieved via regulating enzyme activity in three prominent ways: allosteric regulation, covalent modification, and enzyme synthesis and degradation. Which of these three modes of regulation is likely to be the quickest; which the slowest? For each of these general enzyme regulatory mechanisms, cite conditions in which cells might employ that mode in preference to either of the other two.
10. What are the advantages of compartmentalizing particular metabolic pathways within specific organelles?
11. Maple-syrup urine disease (MSUD) is an autosomal recessive genetic disease characterized by progressive neurological dysfunction and a sweet, burnt-sugar or maple-syrup smell in the urine. Affected individuals carry high levels of branched-chain amino acids (leucine, isoleucine, and valine) and their respective branched-chain α -keto acids in cells and body fluids. The genetic defect has been traced to the mitochondrial branched-chain α -keto acid dehydrogenase (BCKD). Affected individuals exhibit mutations in their BCKD, but these mutant enzymes exhibit normal levels of activity. Nonetheless, treatment of MSUD patients with substantial doses of thiamine can alleviate the symptoms of the disease. Suggest an explanation for the symptoms described and for the role of thiamine in ameliorating the symptoms of MSUD.

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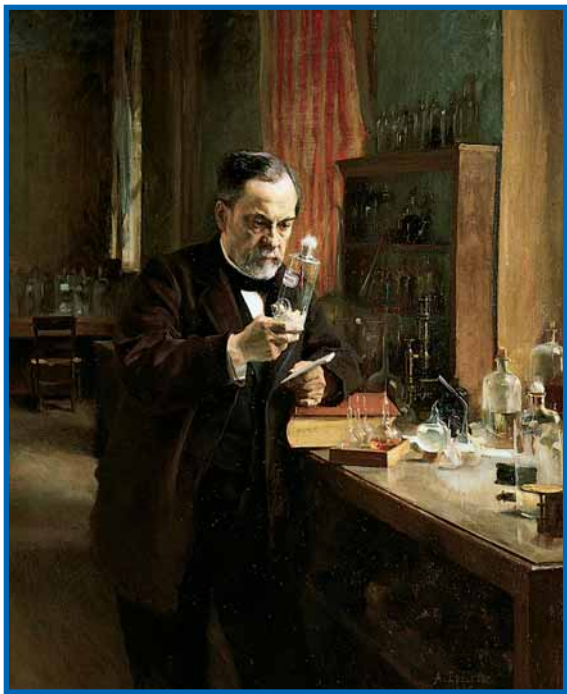
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Chapter 19

Glycolysis



Louis Pasteur in his laboratory. Pasteur's scientific investigations into fermentation of sugar were sponsored by the French wine industry. (Albert Edelfelt, Musee d'Orsay, Paris; Giraudon/Art Resource, New York)

Nearly every living cell carries out a catabolic process known as **glycolysis**—the stepwise degradation of glucose (and other simple sugars). Glycolysis is a paradigm of metabolic pathways. Carried out in the cytosol of cells, it is basically an anaerobic process; its principal steps occur with no requirement for oxygen. Living things first appeared in an environment lacking O_2 , and glycolysis was an early and important pathway for extracting energy from nutrient molecules. It played a central role in anaerobic metabolic processes during the first 2 billion years of biological evolution on earth. Modern organisms still employ glycolysis to provide precursor molecules for aerobic catabolic pathways (such as the tricarboxylic acid cycle) and as a short-term energy source when oxygen is limiting.

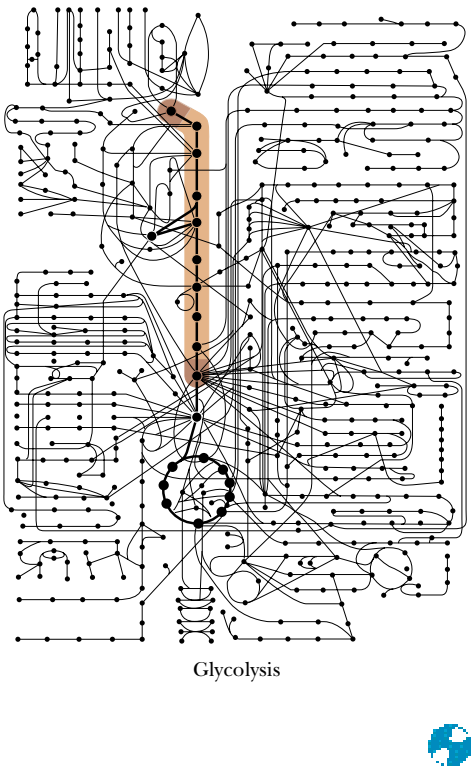
Living organisms, like machines, conform to the law of conservation of energy, and must pay for all their activities in the currency of catabolism.

ERNEST BALDWIN, *Dynamic Aspects of Biochemistry* (1952)

OUTLINE

- 19.1 • Overview of Glycolysis
- 19.2 • The Importance of Coupled Reactions in Glycolysis
- 19.3 • The First Phase of Glycolysis
- 19.4 • The Second Phase of Glycolysis
- 19.5 • The Metabolic Fates of NADH and Pyruvate—The Products of Glycolysis
- 19.6 • Anaerobic Pathways for Pyruvate
- 19.7 • The Energetic Elegance of Glycolysis
- 19.8 • Utilization of Other Substrates in Glycolysis

glycolysis • from the Greek *glyk-*, sweet, and *lysis*, splitting



19.1 • Overview of Glycolysis

An overview of the glycolytic pathway is presented in Figure 19.1. Most of the details of this pathway (the first metabolic pathway to be elucidated) were worked out in the first half of the 20th century by the German biochemists Otto Warburg, G. Embden, and O. Meyerhof. In fact, the sequence of reactions in Figure 19.1 is often referred to as the **Embden–Meyerhof pathway**.

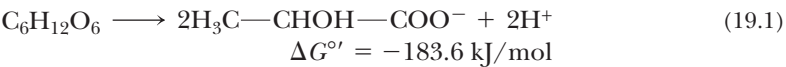
Glycolysis consists of two phases. In the first, a series of five reactions, glucose is broken down to two molecules of glyceraldehyde-3-phosphate. In the second phase, five subsequent reactions convert these two molecules of glyceraldehyde-3-phosphate into two molecules of pyruvate. Phase 1 consumes two molecules of ATP (Figure 19.2). The later stages of glycolysis result in the production of four molecules of ATP. The net is $4 - 2 = 2$ molecules of ATP produced per molecule of glucose.

Rates and Regulation of Glycolytic Reactions Vary Among Species

Microorganisms, plants, and animals (including humans) carry out the 10 reactions of glycolysis in more or less similar fashion, although the rates of the individual reactions and the means by which they are regulated differ from species to species. The most significant difference among species, however, is the way in which the product pyruvate is utilized. The three possible paths for pyruvate are shown in Figure 19.1. In aerobic organisms, including humans, pyruvate is oxidized (with loss of the carboxyl group as CO_2), and the remaining two-carbon unit becomes the acetyl group of acetyl-coenzyme A. This acetyl group is metabolized by the tricarboxylic acid cycle (and fully oxidized) to yield CO_2 . The electrons removed in this oxidation process are subsequently passed through the mitochondrial electron transport system and used to generate molecules of ATP by oxidative phosphorylation, thus capturing most of the metabolic energy available in the original glucose molecule.

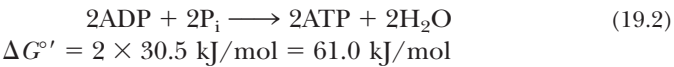
19.2 • The Importance of Coupled Reactions in Glycolysis

The process of glycolysis converts some, but not all, of the metabolic energy of the glucose molecule into ATP. The free energy change for the conversion of glucose to two molecules of lactate (the anaerobic route shown in Figure 19.1) is -183.6 kJ/mol :



This process occurs with no net oxidation or reduction. Although several individual steps in the pathway involve oxidation or reduction, these steps compensate each other exactly. Thus, the conversion of a molecule of glucose to two molecules of lactate involves simply a rearrangement of bonds, with no net loss or gain of electrons. The energy made available through this rearrangement into a more stable (lower energy) form is a relatively small part of the total energy obtainable from glucose.

The production of two molecules of ATP in glycolysis is an energy-requiring process:



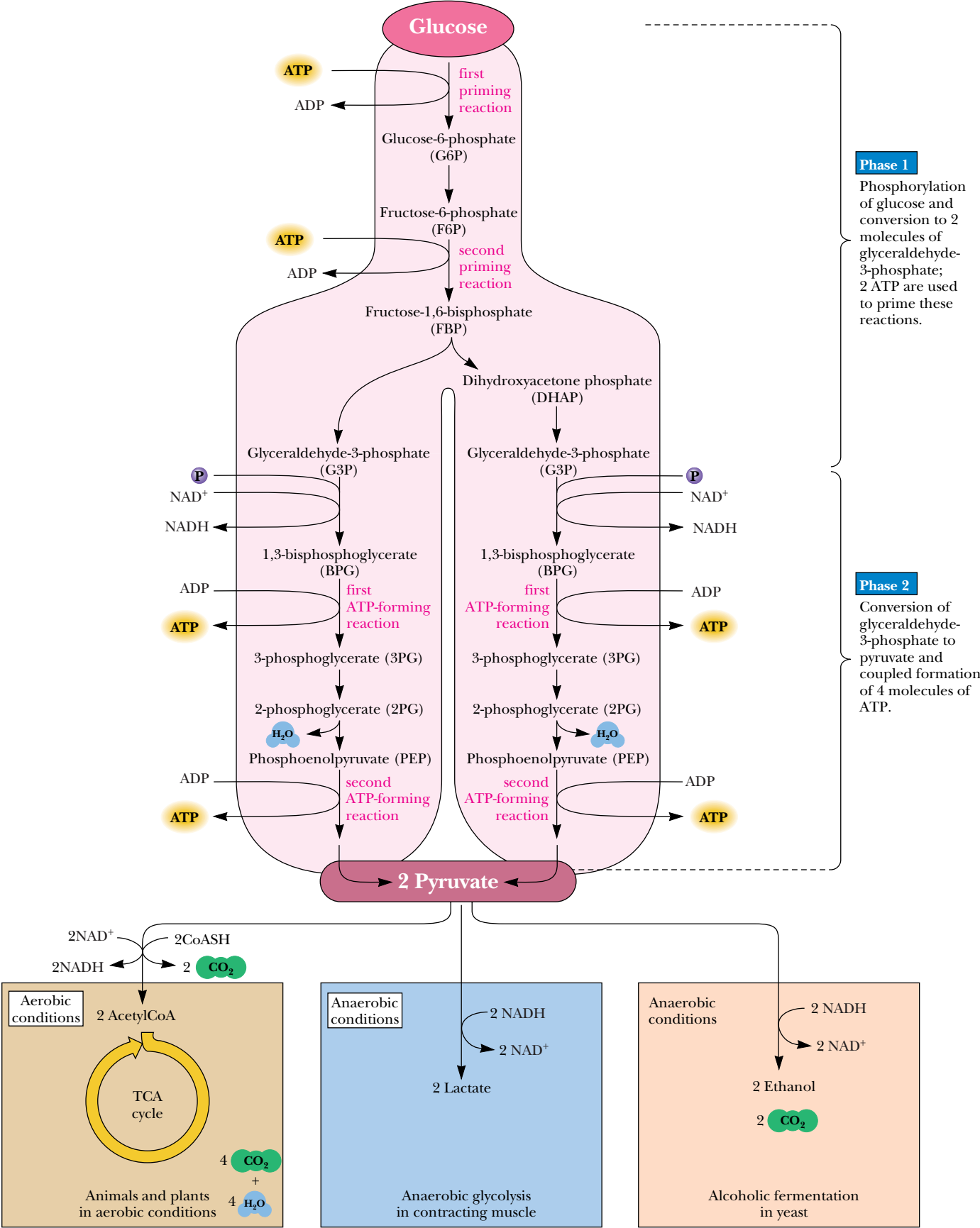
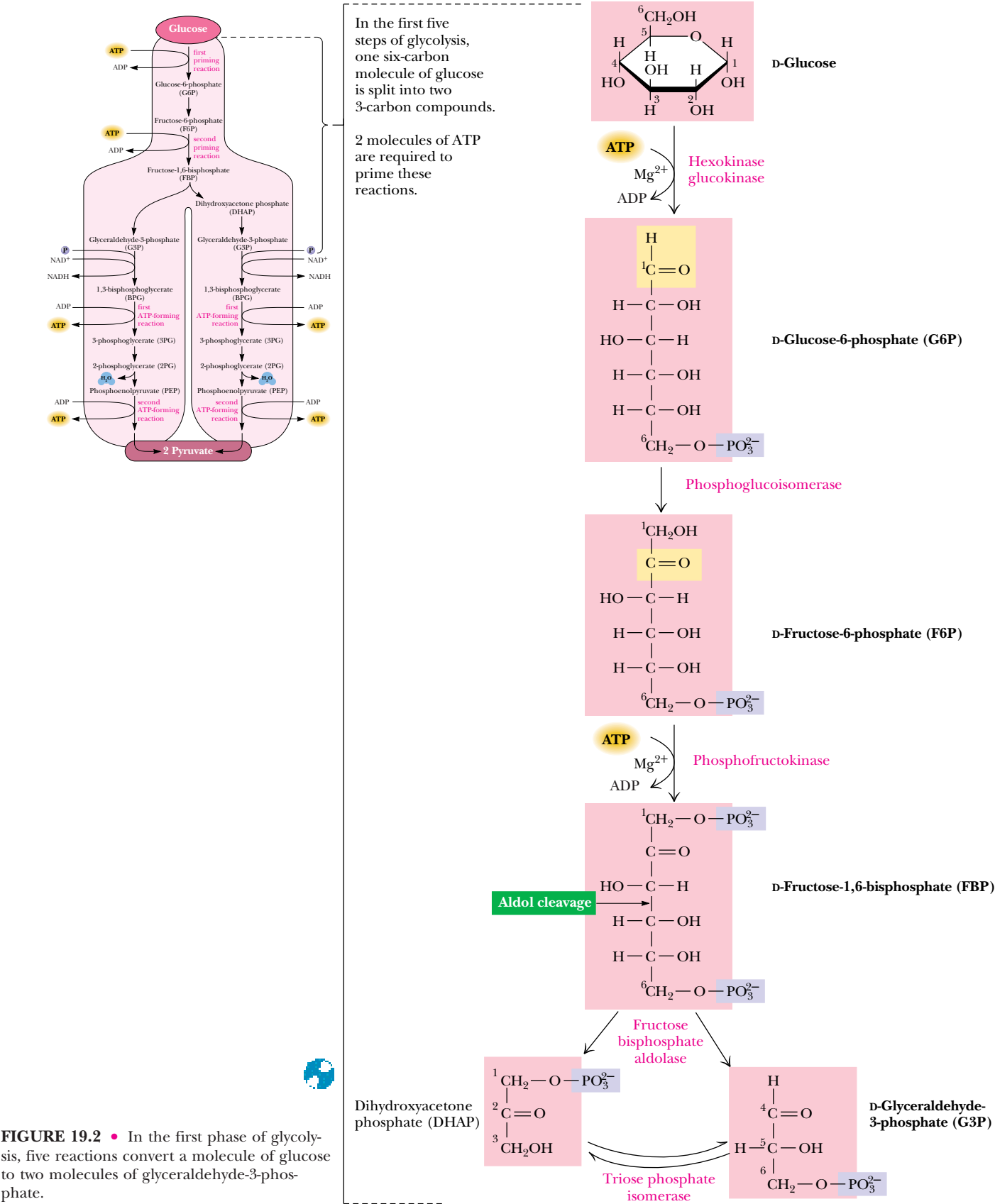
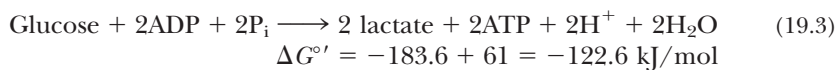


FIGURE 19.1 • The glycolytic pathway.



Glycolysis couples these two reactions:



Thus, under standard-state conditions, $(61/183.6) \times 100\%$, or 33%, of the free energy released is preserved in the form of ATP in these reactions. However, as we discussed in Chapter 3, the various solution conditions, such as pH, concentration, ionic strength, and presence of metal ions, can substantially alter the free energy change for such reactions. Under actual cellular conditions, the free energy change for the synthesis of ATP (Equation 19.2) is much larger, and approximately 50% of the available free energy is converted into ATP. Clearly, then, more than enough free energy is available in the conversion of glucose into lactate to drive the synthesis of two molecules of ATP.

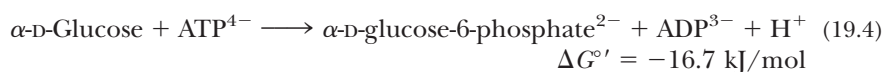
19.3 • The First Phase of Glycolysis

One way to synthesize ATP using the metabolic free energy contained in the glucose molecule would be to convert glucose into one (or more) of the high-energy phosphates in Table 3.3 that have standard-state free energies of hydrolysis more negative than that of ATP. Those molecules in Table 3.3 that can be synthesized easily from glucose are phosphoenolpyruvate, 1,3-bisphosphoglycerate, and acetyl phosphate. In fact, in the first stage of glycolysis, glucose is converted into two molecules of glyceraldehyde-3-phosphate. Energy released from this high-energy molecule in the second phase of glycolysis is then used to synthesize ATP.

Reaction 1: Phosphorylation of Glucose by Hexokinase or Glucokinase—The First Priming Reaction

The initial reaction of the glycolysis pathway involves phosphorylation of glucose at carbon atom 6 by either hexokinase or glucokinase. The formation of such a phosphoester is thermodynamically unfavorable and requires energy input to operate in the forward direction (Chapter 3). The energy comes from ATP, a requirement that at first seems counterproductive. Glycolysis is designed to *make* ATP, not consume it. However, the hexokinase, glucokinase reaction (Figure 19.2) is one of two **priming reactions** in the cycle. Just as old-fashioned, hand-operated water pumps (Figure 19.3) have to be primed with a small amount of water to deliver more water to the thirsty pumper, the glycolysis pathway requires two priming ATP molecules to start the sequence of reactions and delivers four molecules of ATP in the end.

The complete reaction for the first step in glycolysis is



The hydrolysis of ATP makes 30.5 kJ/mol available in this reaction, and the phosphorylation of glucose “costs” 13.8 kJ/mol (see Table 19.1). Thus, the reaction liberates 16.7 kJ/mol under standard-state conditions (1M concentrations), and the equilibrium of the reaction lies far to the right ($K_{\text{eq}} = 850$ at 25°C; see Table 19.1).

Under cellular conditions, this first reaction of glycolysis is even more favorable than at standard state. As pointed out in Chapter 3, the free energy change for any reaction depends on the concentrations of reactants and products.



FIGURE 19.3 • Just as a water pump must be “primed” with water to get more water out, the glycolytic pathway is primed with ATP in steps 1 and 3 in order to achieve net production of ATP in the second phase of the pathway.

(Michelle Sassi/The Stock Market)

Table 19.1

Reactions and Thermodynamics of Glycolysis	
Reaction	Enzyme
$\alpha\text{-D-Glucose} + \text{ATP}^{4-} \rightleftharpoons \text{glucose-6-phosphate}^{2-} + \text{ADP}^{3-} + \text{H}^+$	Hexokinase Hexokinase Glucokinase
$\text{Glucose-6-phosphate}^{2-} \rightleftharpoons \text{fructose-6-phosphate}^{2-}$	Phosphoglucoisomerase
$\text{Fructose-6-phosphate}^{2-} + \text{ATP}^{4-} \rightleftharpoons \text{fructose-1,6-bisphosphate}^{4-} + \text{ADP}^{3-} + \text{H}^+$	Phosphofructokinase
$\text{Fructose-1,6-bisphosphate}^{4-} \rightleftharpoons \text{dihydroxyacetone-P}^{2-} + \text{glyceraldehyde-3-P}^{2-}$	Fructose bisphosphate aldolase
$\text{Dihydroxyacetone-P}^{2-} \rightleftharpoons \text{glyceraldehyde-3-P}^{2-}$	Triose phosphate isomerase
$\text{Glyceraldehyde-3-P}^{2-} + \text{P}_i^{2-} + \text{NAD}^+ \rightleftharpoons \text{1,3-bisphosphoglycerate}^{4-} + \text{NADH} + \text{H}^+$	Glyceraldehyde-3-P dehydrogenase
$\text{1,3-Bisphosphoglycerate}^{4-} + \text{ADP}^{3-} \rightleftharpoons \text{3-P-glycerate}^{3-} + \text{ATP}^{4-}$	Phosphoglycerate kinase
$\text{3-Phosphoglycerate}^{3-} \rightleftharpoons \text{2-phosphoglycerate}^{3-}$	Phosphoglycerate mutase
$\text{2-Phosphoglycerate}^{3-} \rightleftharpoons \text{phosphoenolpyruvate}^{3-} + \text{H}_2\text{O}$	Enolase
$\text{Phosphoenolpyruvate}^{3-} + \text{ADP}^{3-} + \text{H}^+ \rightleftharpoons \text{pyruvate}^- + \text{ATP}^{4-}$	Pyruvate kinase
$\text{Pyruvate}^- + \text{NADH} + \text{H}^+ \rightleftharpoons \text{lactate}^- + \text{NAD}^+$	Lactate dehydrogenase

continued

Equation 3.12 in Chapter 3 and the data in Table 19.2 can be used to calculate a value for ΔG for the hexokinase, glucokinase reaction in erythrocytes:

$$\Delta G = \Delta G^\circ + RT \ln \left(\frac{[\text{G-6-P}][\text{ADP}]}{[\text{Glu}][\text{ATP}]} \right)$$

(19.5)

$$\Delta G = -16.7 \text{ kJ/mol} + (8.314 \text{ J/mol} \cdot \text{K})(310 \text{ K}) \ln \left(\frac{[0.083][0.14]}{[5.0][1.85]} \right)$$

$$\Delta G = -33.9 \text{ kJ/mol}$$

Thus, ΔG is even more favorable under cellular conditions than at standard state. As we will see later in this chapter, the hexokinase, glucokinase reaction is one of several that drive glycolysis forward.

The Cellular Advantages of Phosphorylating Glucose

The incorporation of a phosphate into glucose in this energetically favorable reaction is important for several reasons. First, phosphorylation keeps the substrate in the cell. Glucose is a neutral molecule and could diffuse across the cell membrane, but phosphorylation confers a negative charge on glucose, and the plasma membrane is essentially impermeable to glucose-6-phosphate (Figure 19.4). Moreover, rapid conversion of glucose to glucose-6-phosphate keeps the *intracellular* concentration of glucose low, favoring diffusion of glucose *into* the cell. In addition, because regulatory control can be imposed only on reactions not at equilibrium, the favorable thermodynamics of this first reaction makes it an important site for regulation.

Hexokinase

In most animal, plant, and microbial cells, the enzyme that phosphorylates glucose is **hexokinase**. Magnesium ion (Mg^{2+}) is required for this reaction, as for the other kinase enzymes in the glycolytic pathway. The true substrate for the hexokinase reaction is MgATP^{2-} . The apparent K_m for glucose of the animal

Table 19.2

Steady-State Concentrations of Glycolytic Metabolites in Erythrocytes	
Metabolite	mM
Glucose	5.0
Glucose-6-phosphate	0.083
Fructose-6-phosphate	0.014
Fructose-1,6-bisphosphate	0.031
Dihydroxyacetone phosphate	0.14
Glyceraldehyde-3-phosphate	0.019
1,3-Bisphosphoglycerate	0.001
2,3-Bisphosphoglycerate	4.0
3-Phosphoglycerate	0.12
2-Phosphoglycerate	0.030
Phosphoenolpyruvate	0.023
Pyruvate	0.051
Lactate	2.9
ATP	1.85
ADP	0.14
P _i	1.0

Adapted from Minakami, S., and Yoshikawa, H., 1965.
Biochemical and Biophysical Research Communications 18:345.

Table 19.1

Continued

Source	Subunit Molecular Weight (M _r)	Oligomeric Composition	ΔG°' (kJ/mol)	K _{eq} at 25°C	ΔG (kJ/mol)
Mammals	100,000	Monomer	−16.7	850	−33.9*
Yeast	55,000	Dimer			
Mammalian liver	50,000	Monomer			
Human	65,000	Dimer	+1.67	0.51	−2.92
Rabbit muscle	78,000	Tetramer	−14.2	310	−18.8
Rabbit muscle	40,000	Tetramer	+23.9	6.43 × 10 ^{−5}	−0.23
Chicken muscle	27,000	Dimer	+7.56	0.0472	+2.41
Rabbit muscle	37,000	Tetramer	+6.30	0.0786	−1.29
Rabbit muscle	64,000	Monomer	−18.9	2060	+0.1
Rabbit muscle	27,000	Dimer	+4.4	0.169	+0.83
Rabbit muscle	41,000	Dimer	+1.8	0.483	+1.1
Rabbit muscle	57,000	Tetramer	−31.7	3.63 × 10 ⁵	−23.0
Rabbit muscle	35,000	Tetramer	−25.2	2.63 × 10 ⁴	−14.8

*ΔG values calculated for 310 K (37°C) using the data in Table 19.2 for metabolite concentrations in erythrocytes. ΔG°' values are assumed to be the same at 25°C and 37°C.

skeletal muscle enzyme is approximately 0.1 mM, and the enzyme thus operates efficiently at normal blood glucose levels of 4 mM or so. Different body tissues possess different isozymes of hexokinase, each exhibiting somewhat different kinetic properties. The animal enzyme is allosterically inhibited by the product, glucose-6-phosphate. High levels of glucose-6-phosphate inhibit hexokinase activity until consumption by glycolysis lowers its concentration. The hexokinase reaction is one of three points in the glycolysis pathway that are *regulated*. As the generic name implies, hexokinase can phosphorylate a variety of hexose sugars, including glucose, mannose, and fructose.

Glucokinase

Liver contains an enzyme called **glucokinase**, which also carries out the reaction in Figure 19.4 but is highly specific for D-glucose, has a much higher *K_m* for glucose (approximately 10.0 mM), and is not product-inhibited. With such

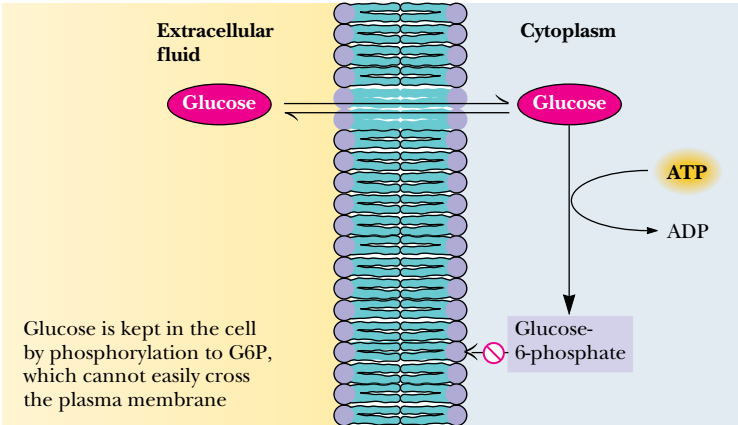


FIGURE 19.4 • Phosphorylation of glucose to glucose-6-phosphate by ATP creates a charged molecule that cannot easily cross the plasma membrane.

a high K_m for glucose, glucokinase becomes important metabolically only when liver glucose levels are high (for example, when the individual has consumed large amounts of sugar). When glucose levels are low, hexokinase is primarily responsible for phosphorylating glucose. However, when glucose levels are high, glucose is converted by glucokinase to glucose-6-phosphate and is eventually stored in the liver as glycogen. Glucokinase is an *inducible* enzyme—the amount present in the liver is controlled by *insulin* (secreted by the pancreas). (Patients with **diabetes mellitus** produce insufficient insulin. They have low levels of glucokinase, cannot tolerate high levels of blood glucose, and produce little liver glycogen.) Because glucose-6-phosphate is common to several metabolic pathways (Figure 19.5), it occupies a branch point in glucose metabolism.

Reaction 2: Phosphoglucose isomerase catalyzes the isomerization of glucose-6-phosphate

The second step in glycolysis is a common type of metabolic reaction: the isomerization of a sugar. In this particular case, the carbonyl oxygen of glucose-6-phosphate is shifted from C-1 to C-2. This amounts to isomerization of an aldose (glucose-6-phosphate) to a ketose—fructose-6-phosphate (Figure 19.6). The reaction is necessary for two reasons. First, the next step in glycolysis is phosphorylation at C-1, and the hemiacetal —OH of glucose would be more difficult to phosphorylate than a simple primary hydroxyl. Second, the isomerization to fructose (with a carbonyl group at position 2 in the linear form) activates carbon C-3 for cleavage in the fourth step of glycolysis. The enzyme responsible for this isomerization is **phosphoglucose isomerase**, also known as **glucose phosphate isomerase**. In humans, the enzyme requires Mg^{2+} for activity and is highly specific for glucose-6-phosphate. The ΔG° is 1.67 kJ/mol, and the value of ΔG under cellular conditions (Table 19.1) is -2.92 kJ/mol. This small value means that the reaction operates near equilibrium in the cell and is readily reversible. Phosphoglucose isomerase proceeds through an *enediol* intermediate, as shown in Figure 19.6. Although the predominant forms of glucose-6-phosphate and fructose-6-phosphate in solution are the ring forms (Figure 19.6), the isomerase interconverts the open-chain form of G-6-P with the open-chain form of F-6-P. The first reaction catalyzed by the isomerase is the opening of the pyranose ring (Figure 19.6, Step A). In the next step, the C-2 proton is removed from the substrate by a basic residue on the enzyme, facilitating formation of the enediol intermediate (Figure 19.6, Step B). This process then

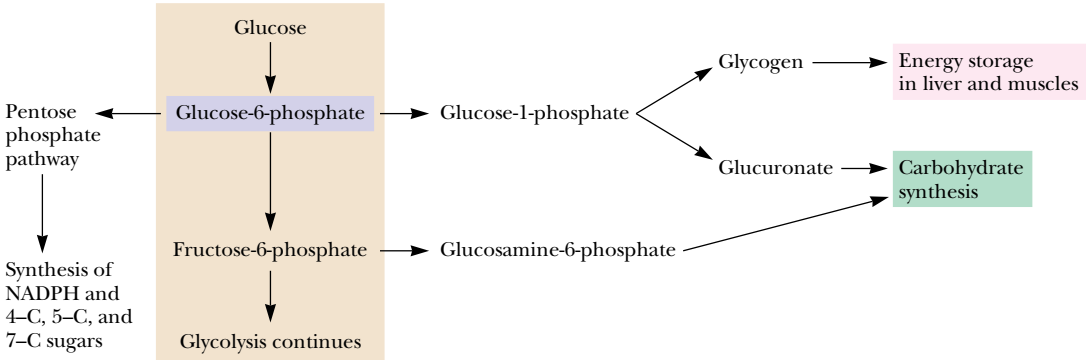


FIGURE 19.5 • Glucose-6-phosphate is the branch point for several metabolic pathways.

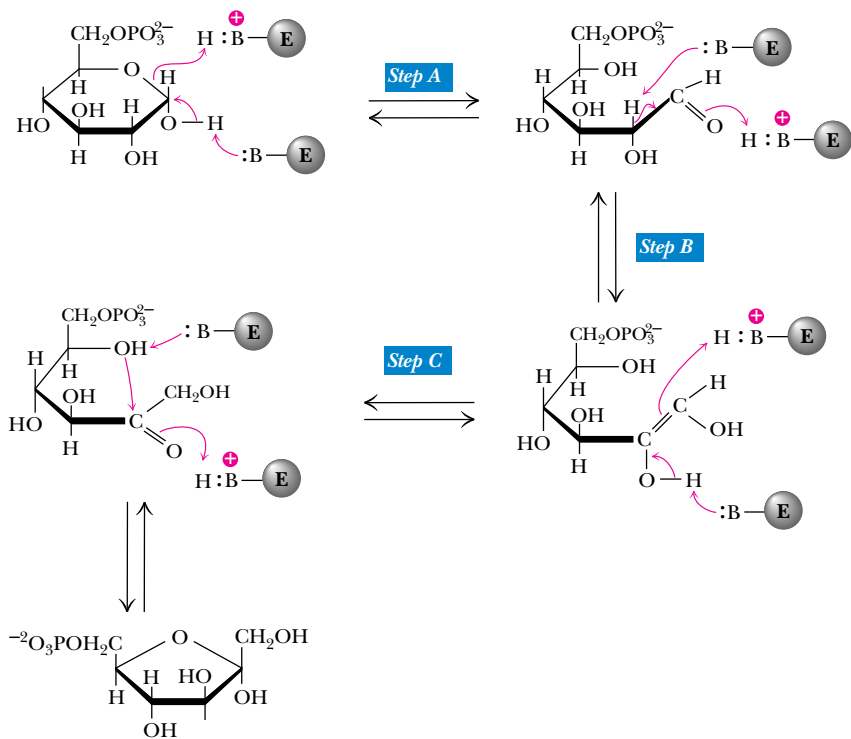
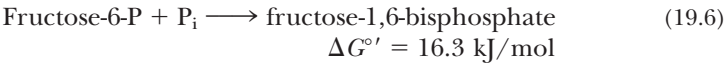


FIGURE 19.6 • The phosphoglucose isomerase mechanism involves opening of the pyranose ring (Step A), proton abstraction leading to enediol formation (Step B), and proton addition to the double bond, followed by ring closure (Step C).

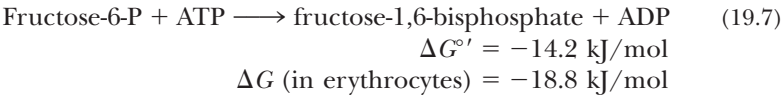
operates somewhat in reverse (Figure 19.6, Step C), creating a carbonyl group at C-2 to complete the formation of fructose-6-phosphate. The furanose form of the product is formed in the usual manner by attack of the C-5 hydroxyl on the carbonyl group, as shown.

Reaction 3: Phosphofructokinase—The Second Priming Reaction

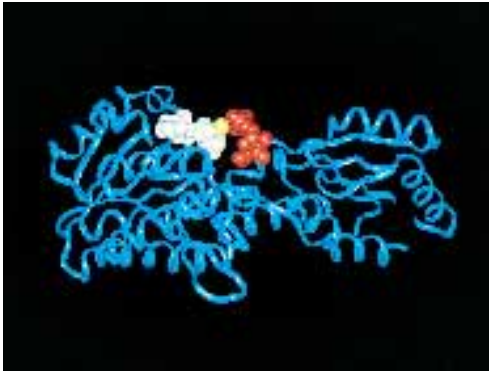
The action of phosphoglucose isomerase, “moving” the carbonyl group from C-1 to C-2, creates a new primary alcohol function at C-1 (see Figure 19.5). The next step in the glycolytic pathway is the phosphorylation of this group by **phosphofructokinase**. Once again, the substrate that provides the phosphoryl group is ATP. Like the hexokinase, glucokinase reaction, the phosphorylation of fructose-6-phosphate is a priming reaction and is endergonic:



When coupled (by phosphofructokinase) with the hydrolysis of ATP, the overall reaction (Figure 19.7) is strongly exergonic:



At pH 7 and 37°C, the phosphofructokinase reaction equilibrium lies far to the right. Just as the hexokinase reaction commits the cell to taking up glucose, *the phosphofructokinase reaction commits the cell to metabolizing glucose* rather than converting it to another sugar or storing it. Similarly, just as the large free energy change of the hexokinase reaction makes it a likely candidate for regulation, so the phosphofructokinase reaction is an important site of regulation—indeed, the most important site in the glycolytic pathway.



Phosphofructokinase with ADP shown in white and fructose-6-P in red.

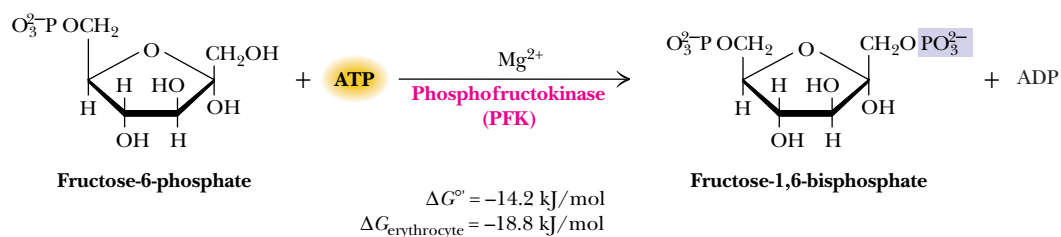


FIGURE 19.7 • The phosphofructokinase reaction.

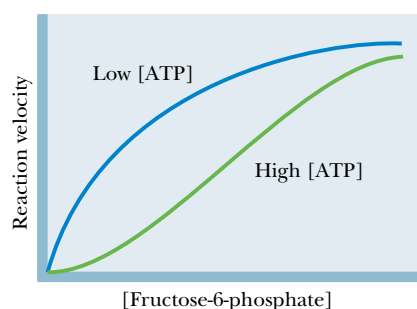


FIGURE 19.8 • At high [ATP], phosphofructokinase (PFK) behaves cooperatively, and the plot of enzyme activity versus [fructose-6-phosphate] is sigmoid. High [ATP] thus inhibits PFK, decreasing the enzyme's affinity for fructose-6-phosphate.

Regulation of Phosphofructokinase

Phosphofructokinase is the “valve” controlling the rate of glycolysis. ATP is an allosteric inhibitor of this enzyme. In the presence of high ATP concentrations, phosphofructokinase behaves cooperatively, plots of enzyme activity versus fructose-6-phosphate are sigmoid, and the K_m for fructose-6-phosphate is increased (Figure 19.8). Thus, when ATP levels are sufficiently high in the cytosol, glycolysis “turns off.” Under most cellular conditions, however, the ATP concentration does not vary over a large range. The ATP concentration in muscle during vigorous exercise, for example, is only about 10% lower than that during the resting state. The rate of glycolysis, however, varies much more. A large range of glycolytic rates cannot be directly accounted for by only a 10% change in ATP levels.

AMP reverses the inhibition due to ATP, and AMP levels in cells *can* rise dramatically when ATP levels decrease, due to the action of the enzyme *adenylate kinase*, which catalyzes the reaction



with the equilibrium constant:

$$K_{\text{eq}} = \frac{[\text{ATP}][\text{AMP}]}{[\text{ADP}]^2} = 0.44 \quad (19.8)$$

Adenylate kinase rapidly interconverts ADP, ATP, and AMP to maintain this equilibrium. ADP levels in cells are typically 10% of ATP levels, and AMP levels are often less than 1% of the ATP concentration. Under such conditions, a small net change in ATP concentration due to ATP hydrolysis results in a much larger relative increase in the AMP levels because of adenylate kinase activity.

EXAMPLE

Calculate the change in concentration in AMP that would occur if 8% of the ATP in an erythrocyte (red blood cell) were suddenly hydrolyzed to ADP. In erythrocytes (Table 19.2), the concentration of ATP is typically $1850\ \mu\text{M}$, the concentration of ADP is $145\ \mu\text{M}$, and the concentration of AMP is $5\ \mu\text{M}$. The total adenine nucleotide concentration is $2000\ \mu\text{M}$.

SOLUTION

The problem can be solved using the equilibrium expression for the adenylate kinase reaction:

$$K_{\text{eq}} = 0.44 = \frac{[\text{ATP}][\text{AMP}]}{[\text{ADP}]^2}$$

If 8% of the ATP is hydrolyzed to ADP, then [ATP] becomes $1850(0.92) = 1702\ \mu\text{M}$, and $[\text{AMP}] + [\text{ADP}]$ becomes $2000 - 1702 = 298\ \mu\text{M}$, and [AMP] may be calculated from the adenylate kinase equilibrium:

$$0.44 = \frac{[1702\ \mu\text{M}][\text{AMP}]}{[\text{ADP}]^2}$$

Since $[\text{AMP}] = 298\ \mu\text{M} - [\text{ADP}]$,

$$0.44 = \frac{1702(298 - [\text{ADP}])}{[\text{ADP}]^2}$$

$$[\text{ADP}] = 278\ \mu\text{M}$$

$$[\text{AMP}] = 20\ \mu\text{M}$$

Thus, an 8% decrease in [ATP] results in a 20/5 or *fourfold increase in the concentration of AMP*.

Clearly, the activity of phosphofructokinase depends both on ATP and AMP levels and is a function of the cellular energy status. Phosphofructokinase activity is increased when the energy status falls and is decreased when the energy status is high. The rate of glycolysis activity thus decreases when ATP is plentiful and increases when more ATP is needed.

Glycolysis and the citric acid cycle (to be discussed in Chapter 20) are coupled via phosphofructokinase, because *citrate*, an intermediate in the citric acid cycle, is an allosteric inhibitor of phosphofructokinase. When the citric acid cycle reaches saturation, glycolysis (which “feeds” the citric acid cycle under aerobic conditions) slows down. The citric acid cycle directs electrons into the electron transport chain (for the purpose of ATP synthesis in oxidative phosphorylation) and also provides precursor molecules for biosynthetic pathways. Inhibition of glycolysis by citrate ensures that glucose will not be committed to these activities if the citric acid cycle is already saturated.

Phosphofructokinase is also regulated by **β -D-fructose-2,6-bisphosphate**, a potent allosteric activator that increases the affinity of phosphofructokinase for the substrate fructose-6-phosphate (Figure 19.9). Stimulation of phosphofructokinase is also achieved by decreasing the inhibitory effects of ATP (Figure 19.10). Fructose-2,6-bisphosphate increases the net flow of glucose through glycolysis by stimulating phosphofructokinase and, as we shall see in Chapter 23, by inhibiting fructose-1,6-bisphosphatase, the enzyme that catalyzes this reaction in the opposite direction.

Reaction 4: Cleavage of Fructose-1,6-bisP by Fructose Bisphosphate Aldolase

Fructose bisphosphate aldolase cleaves fructose-1,6-bisphosphate between the C-3 and C-4 carbons to yield two triose phosphates. The products are dihydroxyacetone phosphate (DHAP) and glyceraldehyde-3-phosphate. The reaction (Figure 19.11) has an equilibrium constant of approximately $10^{-4}\ M$, and a corresponding ΔG° of $+23.9\ \text{kJ/mol}$. These values might imply that the reaction does not proceed effectively from left to right as written. However, the reaction makes two molecules (glyceraldehyde-3-P and dihydroxyacetone-P) from one molecule (fructose-1,6-bisphosphate), and the equilibrium is thus greatly influenced by concentration. The value of ΔG in erythrocytes is actually $-0.23\ \text{kJ/mol}$ (see Table 19.1). At physiological concentrations, the reaction is essentially at equilibrium.

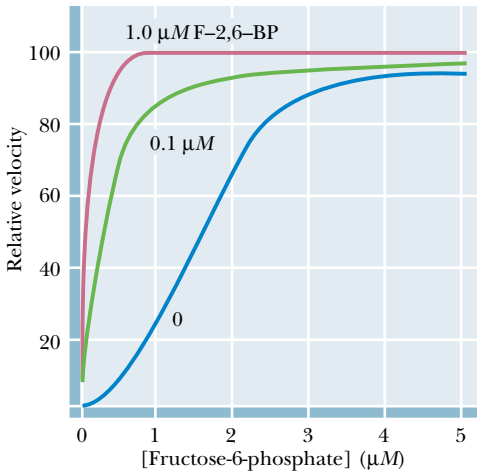


FIGURE 19.9 • Fructose-2,6-bisphosphate activates phosphofructokinase, increasing the affinity of the enzyme for fructose-6-phosphate and restoring the hyperbolic dependence of enzyme activity on substrate.

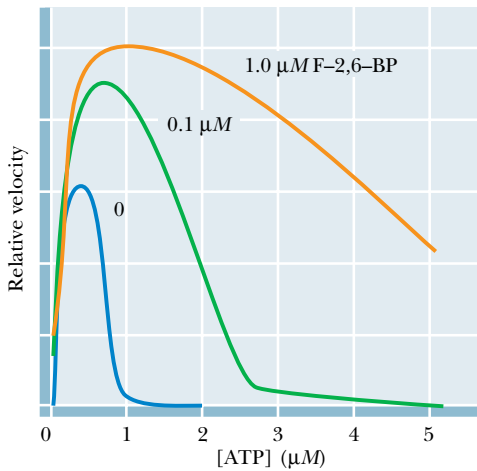
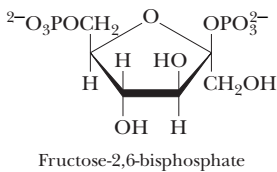


FIGURE 19.10 • Fructose-2,6-bisphosphate decreases the inhibition of phosphofructokinase due to ATP.



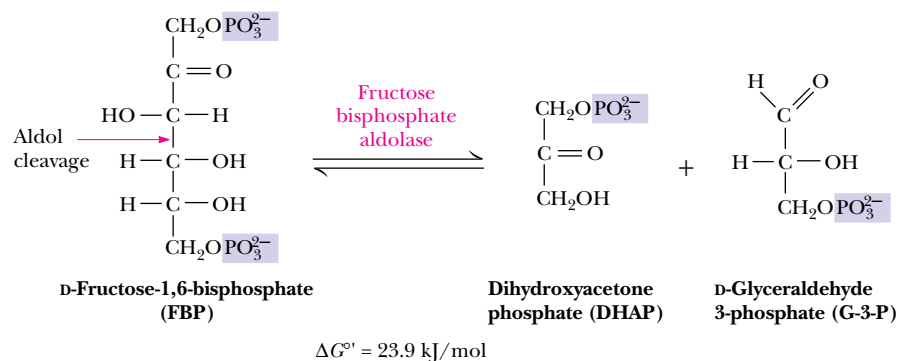


FIGURE 19.11 • The fructose-1,6-bisphosphate aldolase reaction.

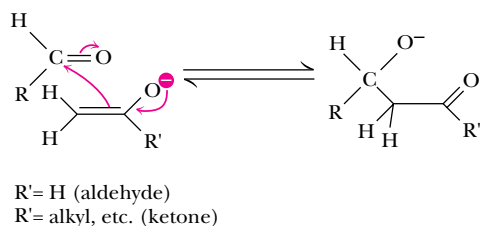


FIGURE 19.12 • An aldol condensation reaction.

Two classes of aldolase enzymes are found in nature. Animal tissues produce a Class I aldolase, characterized by the formation of a covalent Schiff base intermediate between an active-site lysine and the carbonyl group of the substrate. Class I aldolases do not require a divalent metal ion (and thus are not inhibited by EDTA) but are inhibited by sodium borohydride, NaBH_4 , in the presence of substrate (see A Deeper Look, page 622). Class II aldolases are produced mainly in bacteria and fungi and are not inhibited by borohydride, but do contain an active-site metal (normally zinc, Zn^{2+}) and are inhibited by EDTA. Cyanobacteria and some other simple organisms possess both classes of aldolase.

The aldolase reaction is merely the reverse of the **aldol condensation** well known to organic chemists. The latter reaction involves an attack by a nucleophilic enolate anion of an aldehyde or ketone on the carbonyl carbon of an aldehyde (Figure 19.12). The opposite reaction, aldol cleavage, begins with removal of a proton from the β -hydroxyl group, which is followed by the elimination of the enolate anion. A mechanism for the aldol cleavage reaction of fructose-1,6-bisphosphate in the Class I-type aldolases is shown in Figure 19.13a. In Class II aldolases, an active-site metal such as Zn^{2+} behaves as an electrophile, polarizing the carbonyl group of the substrate and stabilizing the enolate intermediate (Figure 19.13b).



Triose phosphate isomerase with substrate analog 2-phosphoglycerate shown in red.

Reaction 5: Triose Phosphate Isomerase

Of the two products of the aldolase reaction, only glyceraldehyde-3-phosphate goes directly into the second phase of glycolysis. The other triose phosphate, dihydroxyacetone phosphate, must be converted to glyceraldehyde-3-phosphate by the enzyme **triose phosphate isomerase** (Figure 19.14). This reaction thus permits both products of the aldolase reaction to continue in the glycolytic pathway, and in essence makes the C-1, C-2, and C-3 carbons of the starting glucose molecule equivalent to the C-6, C-5, and C-4 carbons, respectively. The reaction mechanism involves an enediol intermediate that can donate either of its hydroxyl protons to a basic residue on the enzyme and thereby become either dihydroxyacetone phosphate or glyceraldehyde-3-phosphate (Figure 19.15). Triose phosphate isomerase is one of the enzymes that have evolved to a state of “catalytic perfection,” with a turnover number near the diffusion limit (Chapter 14, Table 14.5).

The triose phosphate isomerase reaction completes the first phase of glycolysis, each glucose that passes through being converted to two molecules of glyceraldehyde-3-phosphate. Although the last two steps of the pathway are

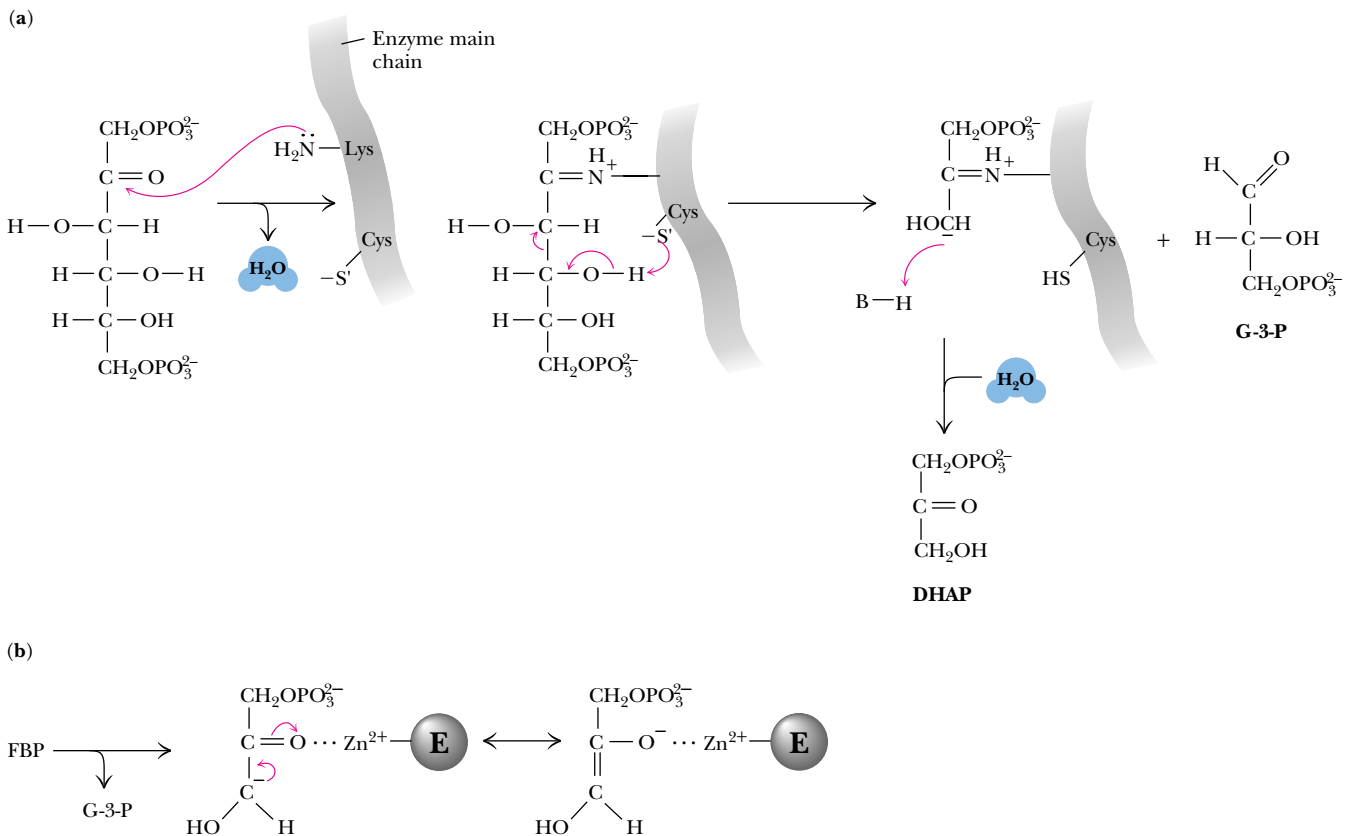


FIGURE 19.13 • (a) A mechanism for the fructose-1,6-bisphosphate aldolase reaction. The Schiff base formed between the substrate carbonyl and an active-site lysine acts as an electron sink, increasing the acidity of the β -hydroxyl group and facilitating cleavage as shown. (b) In class II aldolases, an active-site Zn^{2+} stabilizes the enolate intermediate, leading to polarization of the substrate carbonyl group.

energetically unfavorable, the overall five-step reaction sequence has a net ΔG° of $+2.2 \text{ kJ/mol}$ ($K_{\text{eq}} \approx 0.43$). It is the free energy of hydrolysis from the two priming molecules of ATP that brings the overall equilibrium constant close to 1 under standard-state conditions. The net ΔG under cellular conditions is quite negative (-53.4 kJ/mol in erythrocytes).

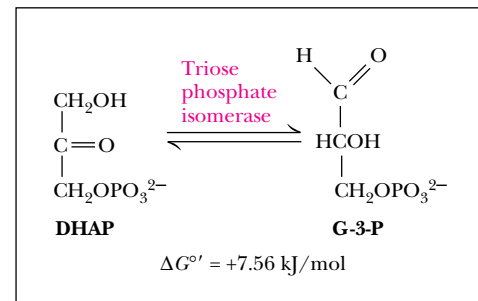


FIGURE 19.14 • The triose phosphate isomerase reaction.

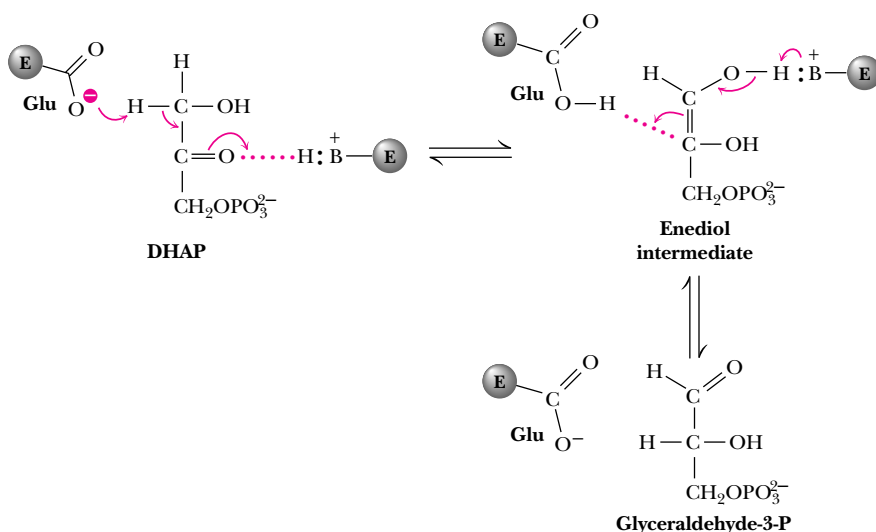


FIGURE 19.15 • A reaction mechanism for triose phosphate isomerase.

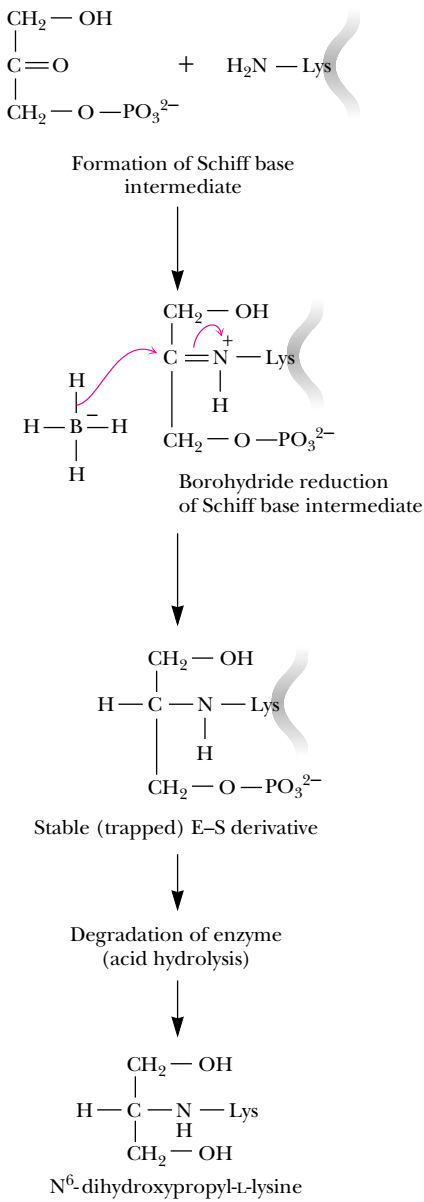
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The Chemical Evidence for the Schiff Base Intermediate in Class I Aldolases

Fructose biphosphate aldolase of animal muscle is a Class I aldolase, which forms a Schiff base or *imine* intermediate between the substrate (fructose-1,6-bisP or dihydroxyacetone-P) and a lysine amino group at the enzyme active site. The chemical evidence for this intermediate comes from studies with the aldolase and the reducing agent sodium borohydride, NaBH₄. Incubation of fructose biphosphate aldolase with dihydroxyacetone-P and NaBH₄ inactivates the enzyme. Interestingly, no inactivation is observed if NaBH₄ is added to the enzyme in the absence of substrate.

These observations are explained by the mechanism shown in the figure. NaBH₄ inactivates Class I aldolases by transfer of a hydride ion (H[−]) to the imine carbon atom of the enzyme–substrate adduct. The resulting secondary amine is stable to hydrolysis, and the active-site lysine is thus permanently modified and inactivated. NaBH₄ inactivates Class I aldolases in the presence of either dihydroxyacetone-P or fructose-1,6-bisP, but inhibition doesn’t occur in the presence of glyceraldehyde-3-P.

Definitive identification of lysine as the modified active-site residue has come from radioisotope-labeling studies. NaBH₄ reduction of the aldolase Schiff base intermediate formed from ¹⁴C-labeled dihydroxyacetone-P yields an enzyme covalently labeled with ¹⁴C. Acid hydrolysis of the inactivated enzyme liberates a novel ¹⁴C-labeled amino acid, *N*⁶-dihydroxypropyl-L-lysine. This is the product anticipated from reduction of the Schiff base formed between a lysine residue and the ¹⁴C-labeled dihydroxyacetone-P. (The phosphate group is lost during acid hydrolysis of the inactivated enzyme.) The use of ¹⁴C labeling in a case such as this facilitates the separation and identification of the telltale amino acid.



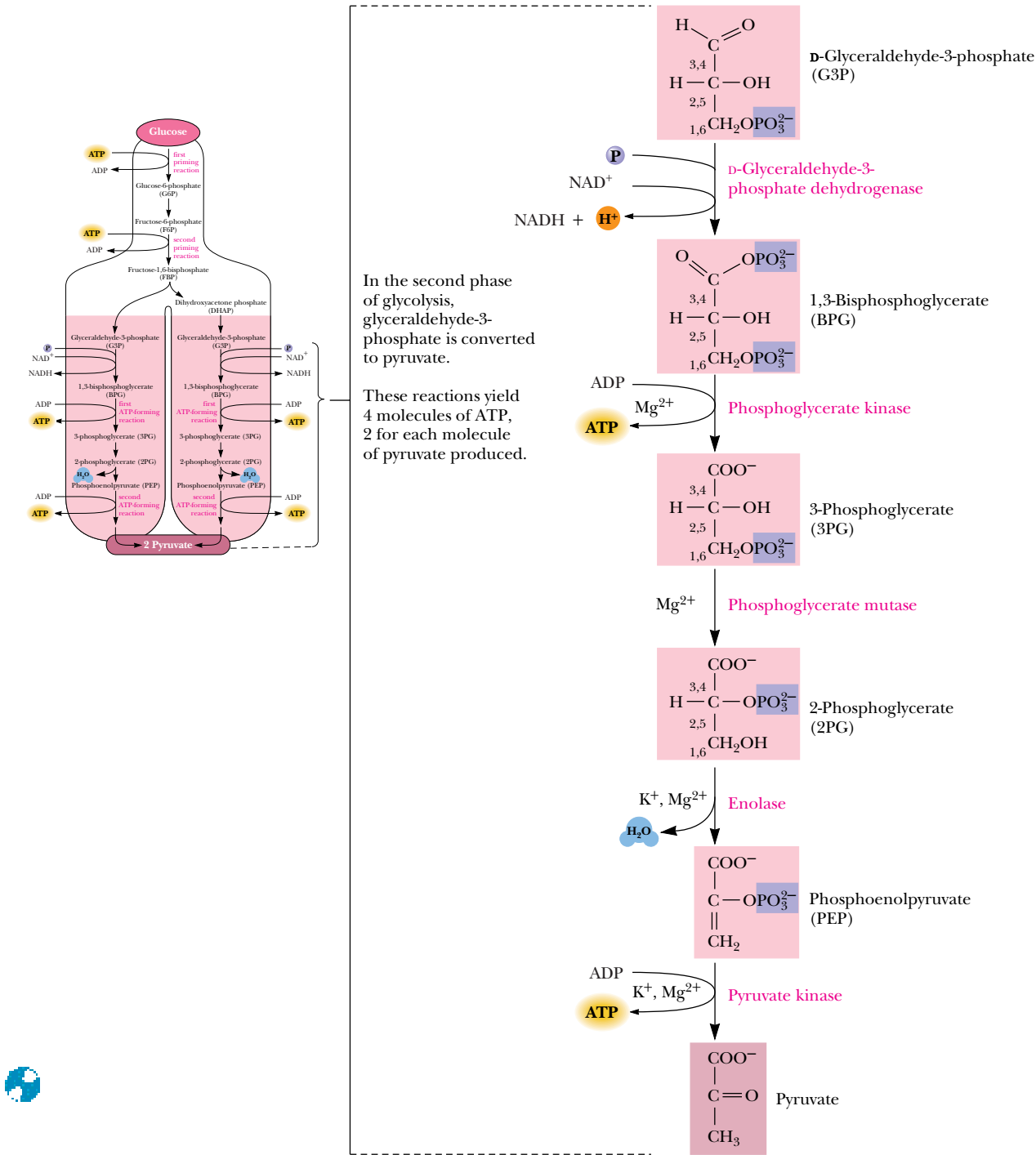


FIGURE 19.16 • The second phase of glycolysis. Carbon atoms are numbered to show their original positions in glucose.

enough energy “kick” to produce a high-energy phosphate, namely, 1,3-bisphosphoglycerate (Figure 19.16). Phosphoryl transfer from 1,3-BPG to ADP to make ATP is highly favorable. The product, 3-phosphoglycerate, is converted via several steps to phosphoenolpyruvate (PEP), another high-energy phosphate. PEP readily transfers its phosphoryl group to ADP in the pyruvate kinase reaction to make another ATP.

Reaction 6: Glyceraldehyde-3-Phosphate Dehydrogenase

In the first glycolytic reaction to involve oxidation–reduction, glyceraldehyde-3-phosphate is oxidized to 1,3-bisphosphoglycerate by **glyceraldehyde-3-phosphate dehydrogenase**. Although the oxidation of an aldehyde to a carboxylic acid is a highly exergonic reaction, the overall reaction (Figure 19.17) involves both formation of a carboxylic-phosphoric anhydride and the reduction of NAD^+ to NADH and is therefore slightly endergonic at standard state, with a ΔG° of +6.30 kJ/mol. The free energy that might otherwise be released as heat in this reaction is directed into the formation of a high-energy phosphate compound, 1,3-bisphosphoglycerate, and the reduction of NAD^+ . The reaction mechanism involves nucleophilic attack by a cysteine —SH group on the carbonyl carbon of glyceraldehyde-3-phosphate to form a hemithioacetal (Figure 19.18). The hemithioacetal intermediate decomposes by hydride (H^-) transfer to NAD^+ to form a high-energy thioester. Nucleophilic attack by phosphate displaces the product, 1,3-bisphosphoglycerate, from the enzyme. The enzyme can be inactivated by reaction with iodoacetate, which reacts with and blocks the essential cysteine sulfhydryl.

The glyceraldehyde-3-phosphate dehydrogenase reaction is the site of action of *arsenate* (AsO_4^{3-}), an anion analogous to phosphate. Arsenate is an effective substrate in this reaction, forming *1-arseno-3-phosphoglycerate* (Figure 19.19), but acyl arsenates are quite unstable and are rapidly hydrolyzed. 1-Arseno-3-phosphoglycerate breaks down to yield *3-phosphoglycerate*, the product of the seventh reaction of glycolysis. The result is that glycolysis continues in the presence of arsenate, but the molecule of ATP formed in reaction 7 (phosphoglycerate kinase) is not made because this step has been bypassed. The lability of 1-arseno-3-phosphoglycerate effectively uncouples the oxidation and phosphorylation events, which are normally tightly coupled in the glyceraldehyde-3-phosphate dehydrogenase reaction.

Reaction 7: Phosphoglycerate Kinase

The glycolytic pathway breaks even in terms of ATPs consumed and produced with this reaction. The enzyme **phosphoglycerate kinase** transfers a phosphoryl group from 1,3-bisphosphoglycerate to ADP to form an ATP (Figure 19.20). Because each glucose molecule sends two molecules of glyceraldehyde-3-phosphate into the second phase of glycolysis and because two ATPs were consumed per glucose in the first-phase reactions, the phosphoglycerate kinase reaction “pays off” the ATP debt created by the priming reactions. As might be expected for a phosphoryl transfer enzyme, Mg^{2+} ion is required for activity, and the true nucleotide substrate for the reaction is MgADP^- . It is appropriate to view the sixth and seventh reactions of glycolysis as a coupled pair, with 1,3-bis-

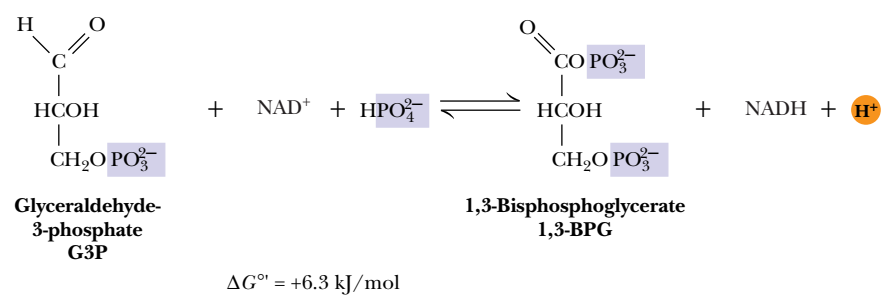


FIGURE 19.17 • The glyceraldehyde-3-phosphate dehydrogenase reaction.

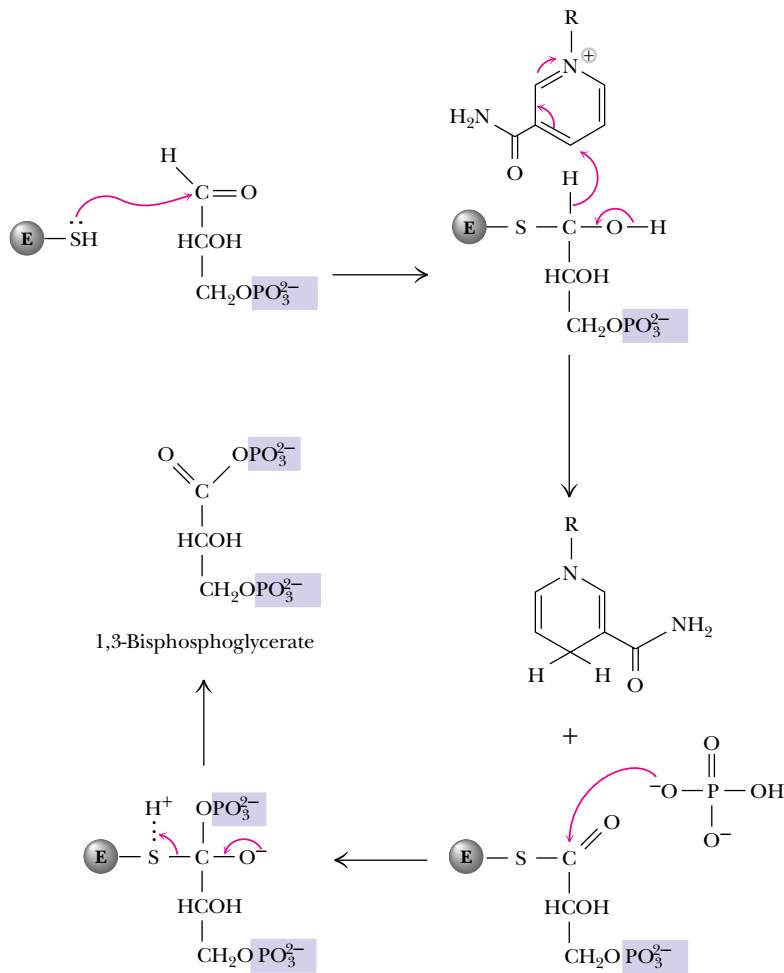
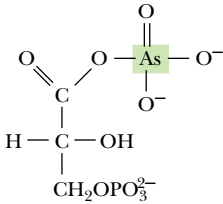
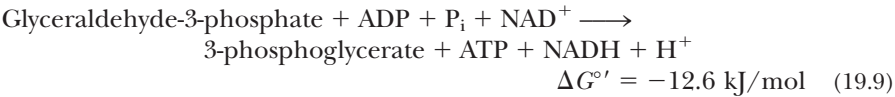


FIGURE 19.18 • A mechanism for the glyceraldehyde-3-phosphate dehydrogenase reaction. Reaction of an enzyme sulfhydryl with the carbonyl carbon of glyceraldehyde-3-P forms a thiohemiacetal, which loses a hydride to NAD⁺ to become a thioester. Phosphorolysis of this thioester releases 1,3-bisphosphoglycerate.

phosphoglycerate as an intermediate. The phosphoglycerate kinase reaction is sufficiently exergonic at standard state to pull the G-3-P dehydrogenase reaction along. (In fact, the aldolase and triose phosphate isomerase are also pulled forward by phosphoglycerate kinase.) The net result of these coupled reactions is



1-Arseno-3-phosphoglycerate

FIGURE 19.19

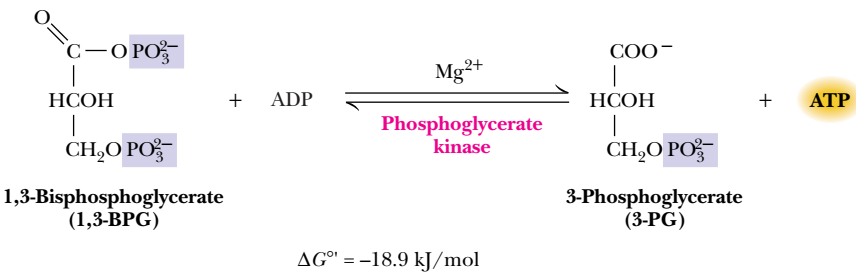


FIGURE 19.20 • The phosphoglycerate kinase reaction.

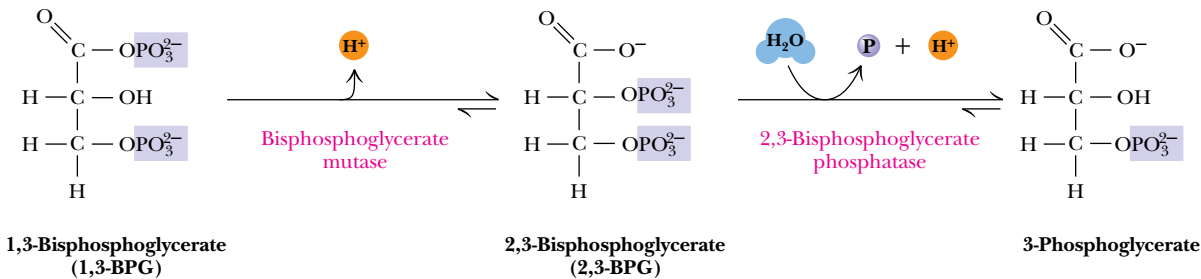


FIGURE 19.21 • Formation and decomposition of 2,3-bisphosphoglycerate.

Another reflection of the coupling between these reactions lies in their values of ΔG under cellular conditions (Table 19.1). In spite of its strongly negative ΔG° , the phosphoglycerate kinase reaction operates at equilibrium in the erythrocyte ($\Delta G = 0.1$ kJ/mol). In essence, the free energy available in the phosphoglycerate kinase reaction is used to bring the three previous reactions closer to equilibrium. Viewed in this context, it is clear that ADP has been phosphorylated to form ATP at the expense of a substrate, namely, glyceraldehyde-3-phosphate. This is an example of **substrate-level phosphorylation**, a concept that will be encountered again. (The other kind of phosphorylation, *oxidative phosphorylation*, is driven energetically by the transport of electrons from appropriate coenzymes and substrates to oxygen. Oxidative phosphorylation will be covered in detail in Chapter 21). Even though the coupled reactions exhibit a very favorable ΔG° , there are conditions (i.e., high ATP and 3-phosphoglycerate levels) under which Equation 19.9 can be reversed, so that 3-phosphoglycerate is phosphorylated from ATP.

An important regulatory molecule, *2,3-bisphosphoglycerate*, is synthesized and metabolized by a pair of reactions that make a detour around the phosphoglycerate kinase reaction. 2,3-BPG, which stabilizes the deoxy form of hemoglobin and is primarily responsible for the cooperative nature of oxygen binding by hemoglobin (see Chapter 15), is formed from 1,3-bisphosphoglycerate by **bisphosphoglycerate mutase** (Figure 19.21). Interestingly, 3-phosphoglycerate is required for this reaction, which involves phosphoryl transfer from the C-1 position of 1,3-bisphosphoglycerate to the C-2 position of 3-phosphoglycerate (Figure 19.22). Hydrolysis of 2,3-BPG is carried out by *2,3-bisphosphoglycerate phosphatase*. Although other cells contain only a trace of 2,3-BPG, erythrocytes typically contain 4 to 5 mM 2,3-BPG.

Reaction 8: Phosphoglycerate Mutase

The remaining steps in the glycolytic pathway prepare for synthesis of the second ATP equivalent. This begins with the **phosphoglycerate mutase** reaction (Figure 19.23), in which the phosphoryl group of 3-phosphoglycerate is moved



FIGURE 19.22 • The mutase that forms 2,3-BPG from 1,3-BPG requires 3-phosphoglycerate. The reaction is actually an intermolecular phosphoryl transfer from C-1 of 1,3-BPG to C-2 of 3-PG.

from C-3 to C-2. (The term *mutase* is applied to enzymes that catalyze migration of a functional group within a substrate molecule.) The free energy change for this reaction is very small under cellular conditions ($\Delta G = 0.83 \text{ kJ/mol}$ in erythrocytes). Phosphoglycerate mutase enzymes isolated from different sources exhibit different reaction mechanisms. As shown in Figure 19.24, the

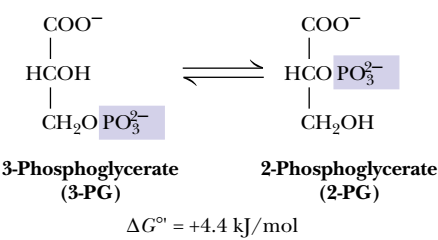


FIGURE 19.23 • The phosphoglycerate mutase reaction.

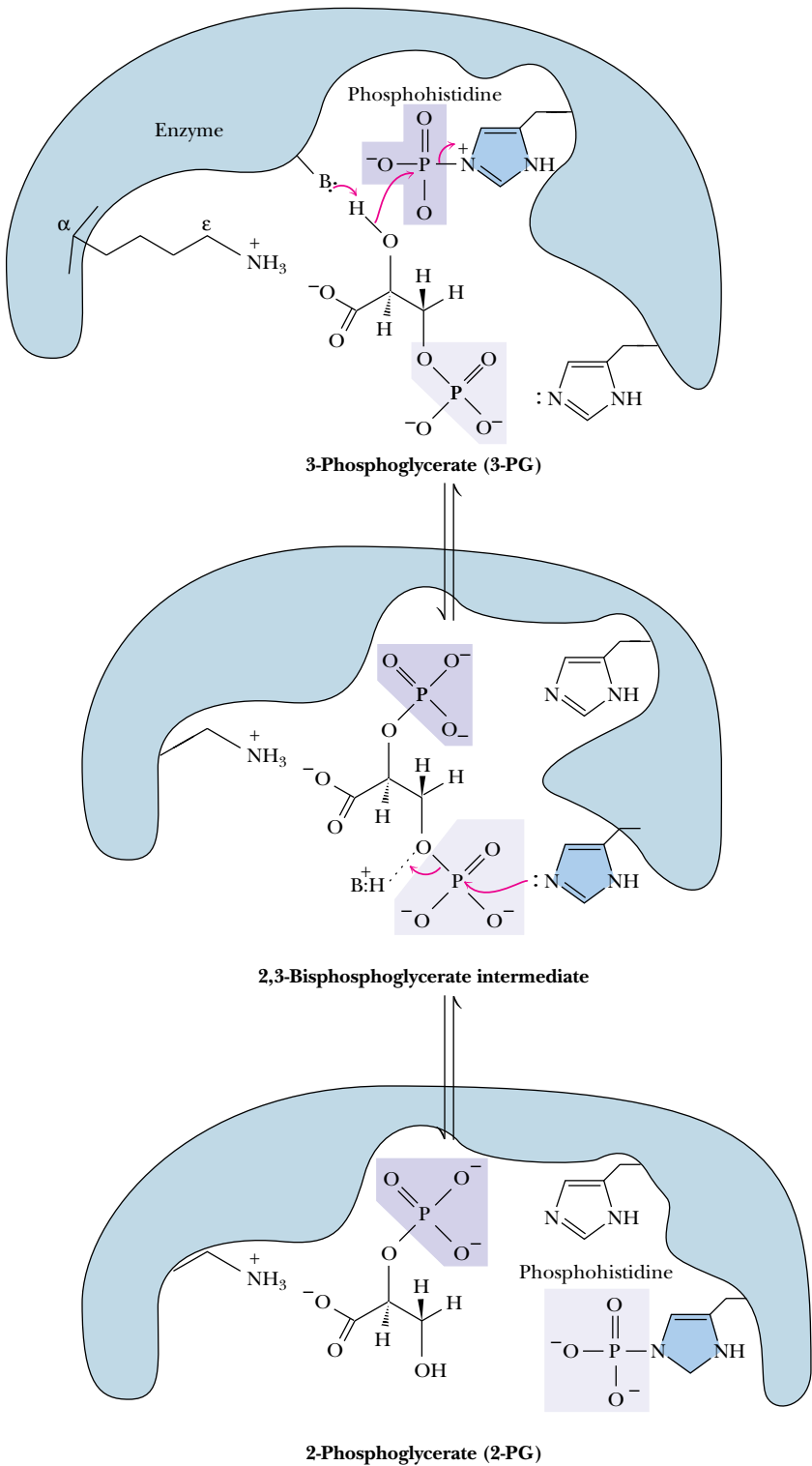


FIGURE 19.24 • A mechanism for the phosphoglycerate mutase reaction in rabbit muscle and in yeast. Zelda Rose of the Institute for Cancer Research in Philadelphia showed that the enzyme requires a small amount of 2,3-BPG to phosphorylate the histidine residue before the mechanism can proceed. Prior to her work, the role of the phosphohistidine in this mechanism was not understood.

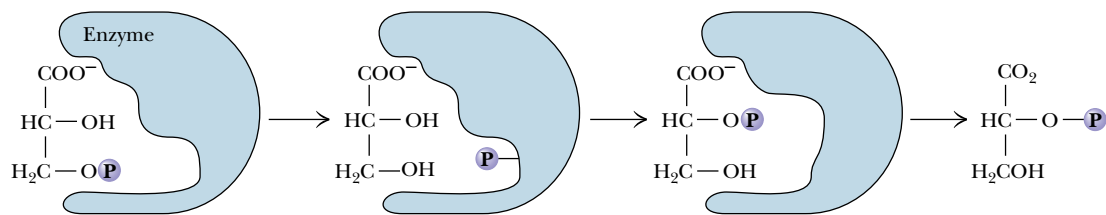


FIGURE 19.25 • The phosphoglycerate mutase of wheat germ catalyzes an intramolecular phosphoryl transfer.

enzymes isolated from yeast and from rabbit muscle form *phosphoenzyme* intermediates, use *2,3-bisphosphoglycerate* as a cofactor, and undergo *intermolecular* phosphoryl group transfers (in which the phosphate of the product 2-phosphoglycerate is not that from the 3-phosphoglycerate substrate). The prevalent form of phosphoglycerate mutase is a *phosphoenzyme*, with a phosphoryl group covalently bound to a histidine residue at the active site. This phosphoryl group is transferred to the C-2 position of the substrate to form a transient, enzyme-bound 2,3-bisphosphoglycerate, which then decomposes by a second phosphoryl transfer from the C-3 position of the intermediate to the histidine residue on the enzyme. About once in every 100 enzyme turnovers, the intermediate, 2,3-bisphosphoglycerate, dissociates from the active site, leaving an inactive, unphosphorylated enzyme. The unphosphorylated enzyme can be reactivated by binding 2,3-BPG. For this reason, maximal activity of phosphoglycerate mutase requires the presence of small amounts of 2,3-BPG.

A different mechanism operates in the wheat germ enzyme. 2,3-Bisphosphoglycerate is not a cofactor. Instead, the enzyme carries out *intramolecular* phosphoryl group transfer (Figure 19.25). The C-3 phosphate is transferred to an active-site residue and then to the C-2 position of the original substrate molecule to form the product, 2-phosphoglycerate.

Reaction 9: Enolase

Recall that, prior to synthesizing ATP in the phosphoglycerate kinase reaction, it was necessary to first make a substrate having a high-energy phosphate. Reaction 9 of glycolysis similarly makes a high-energy phosphate in preparation for ATP synthesis. **Enolase** catalyzes the formation of *phosphoenolpyruvate* from 2-phosphoglycerate (Figure 19.26). The reaction in essence involves a dehydration—the removal of a water molecule—to form the enol structure of PEP. The $\Delta G^{\circ'}$ for this reaction is relatively small at 1.8 kJ/mol ($K_{eq} = 0.5$); and, under cellular conditions, ΔG is very close to zero. In light of this condition, it may be difficult at first to understand how the enolase reaction transforms a substrate with a relatively low free energy of hydrolysis into a product (PEP) with a very high free energy of hydrolysis. This puzzle is clarified by real-

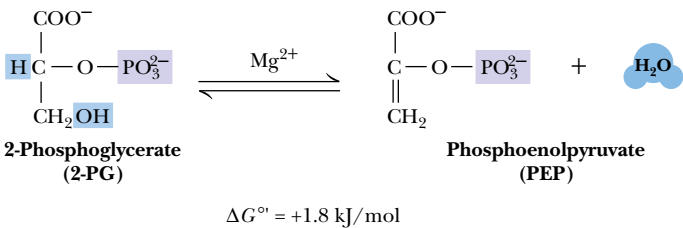


FIGURE 19.26 • The enolase reaction.

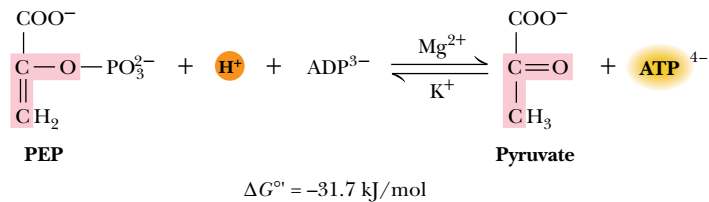


FIGURE 19.27 • The pyruvate kinase reaction.

izing that 2-phosphoglycerate and PEP contain about the same amount of *potential* metabolic energy, with respect to decomposition to P_i, CO₂, and H₂O. What the enolase reaction does is rearrange the substrate into a form from which more of this potential energy can be released upon hydrolysis. The enzyme is strongly inhibited by fluoride ion in the presence of phosphate. Inhibition arises from the formation of *fluorophosphate* (FPO_3^{2-}), which forms a complex with Mg²⁺ at the active site of the enzyme.

Reaction 10: Pyruvate Kinase

The second ATP-synthesizing reaction of glycolysis is catalyzed by **pyruvate kinase**, which brings the pathway at last to its pyruvate branch point. Pyruvate kinase mediates the transfer of a phosphoryl group from phosphoenolpyruvate to ADP to make ATP and pyruvate (Figure 19.27). The reaction requires Mg^{2+} ion and is stimulated by K^{+} and certain other monovalent cations.

	$\Delta G^{\circ'}$ (kJ/mol)
Phosphoenolpyruvate ³⁻ + H ₂ O $\longrightarrow$ pyruvate ⁻ + HPO ₄ ²⁻	-62.2
ADP ³⁻ + HPO ₄ ²⁻ + H ⁺ $\longrightarrow$ ATP ⁴⁻ + H ₂ O	+30.5
Phosphoenolpyruvate ³⁻ + ADP ³⁻ + H ⁺ $\longrightarrow$ pyruvate ⁻ + ATP ⁴⁻	-31.7

The corresponding K_{eq} at 25°C is 3.63×10^5 , and it is clear that the pyruvate kinase reaction equilibrium lies very far to the right. Concentration effects reduce the magnitude of the free energy change somewhat in the cellular environment, but the ΔG in erythrocytes is still quite favorable at -23.0 kJ/mol. The high free energy change for the conversion of PEP to pyruvate is due largely to the highly favorable and spontaneous conversion of the enol tautomer of pyruvate to the more stable keto form (Figure 19.28) following the phosphoryl group transfer step.

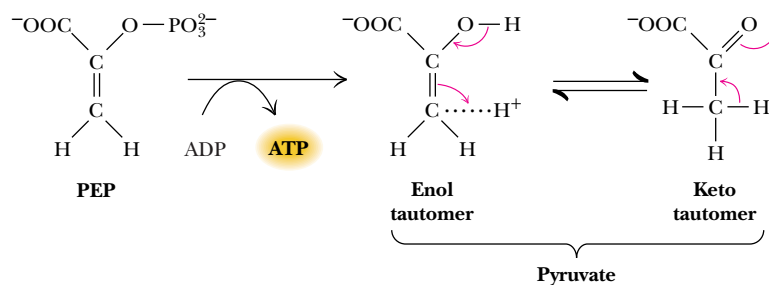
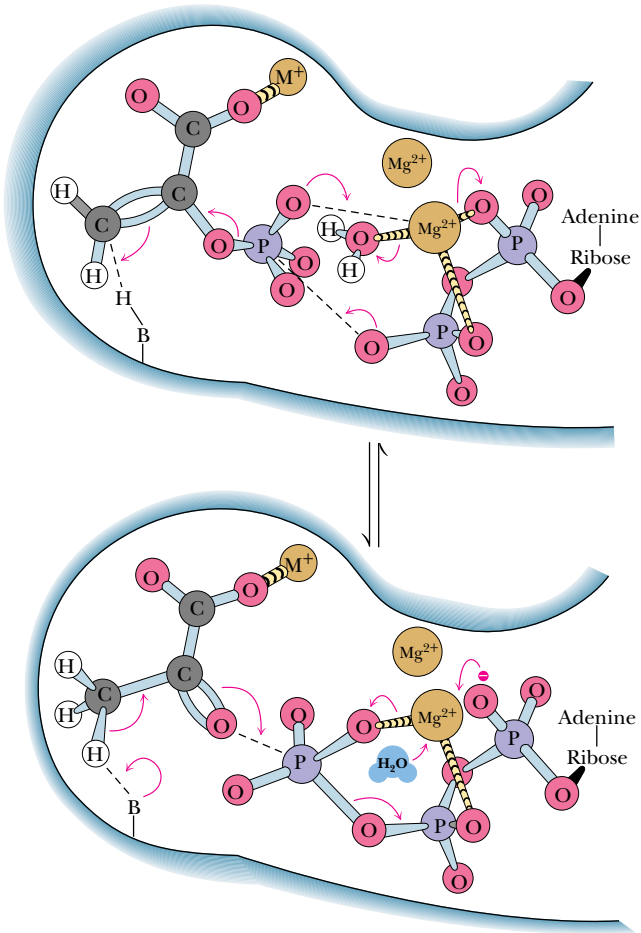


FIGURE 19.28 • The conversion of phosphoenolpyruvate (PEP) to pyruvate may be viewed as involving two steps: phosphoryl transfer followed by an enol-keto tautomerization. The tautomerization is spontaneous ($\Delta G' \approx -35$ – 40 kJ/mol) and accounts for much of the free energy change for PEP hydrolysis.

The large negative ΔG of this reaction makes pyruvate kinase a suitable target site for regulation of glycolysis. For each glucose molecule in the glycolysis pathway, two ATPs are made at the pyruvate kinase stage (because two triose molecules were produced per glucose in the aldolase reaction). Because the pathway broke even in terms of ATP at the phosphoglycerate kinase reaction (two ATPs consumed and two ATPs produced), the two ATPs produced by pyruvate kinase represent the “payoff” of glycolysis—a net yield of two ATP molecules.

Pyruvate kinase possesses allosteric sites for numerous effectors. It is activated by AMP and fructose-1,6-bisphosphate and inhibited by ATP, acetyl-CoA, and alanine. (Note that alanine is the α -amino acid counterpart of the α -keto acid, pyruvate.) Furthermore, liver pyruvate kinase is regulated by covalent modification. Hormones such as *glucagon* activate a cAMP-dependent protein kinase, which transfers a phosphoryl group from ATP to the enzyme. The phosphorylated form of pyruvate kinase is more strongly inhibited by ATP and alanine and has a higher K_m for PEP, so that, in the presence of physiological levels of PEP, the enzyme is inactive. Then PEP is used as a substrate for glucose synthesis in the *gluconeogenesis* pathway (to be described in Chapter 23), instead of going on through glycolysis and the citric acid cycle (or fermentation routes). A suggested active-site geometry for pyruvate kinase, based on NMR and EPR studies by Albert Mildvan and colleagues, is presented in Figure 19.29. The carbonyl oxygen of pyruvate and the γ -phosphorus of ATP lie within 0.3 nm of each other at the active site, consistent with direct transfer of the phosphoryl group without formation of a phosphoenzyme intermediate.

FIGURE 19.29 • A mechanism for the pyruvate kinase reaction, based on NMR and EPR studies by Albert Mildvan and colleagues. Phosphoryl transfer from phosphoenolpyruvate (PEP) to ADP occurs in four steps: (a) a water on the Mg^{2+} ion coordinated to ADP is replaced by the phosphoryl group of PEP; (b) Mg^{2+} dissociates from the α -P of ADP; (c) the phosphoryl group is transferred; and (d) the enolate of pyruvate is protonated. (Adapted from Mildvan, A., 1979. *Advances in Enzymology* 49:103–126.)



19.5 • The Metabolic Fates of NADH and Pyruvate—
The Products of Glycolysis

In addition to ATP, the products of glycolysis are NADH and pyruvate. Their processing depends upon other cellular pathways. NADH must be recycled to NAD^+ , lest NAD^+ become limiting in glycolysis. NADH can be recycled by both aerobic and anaerobic paths, either of which results in further metabolism of pyruvate. What a given cell does with the pyruvate produced in glycolysis depends in part on the availability of oxygen. Under aerobic conditions, pyruvate can be sent into the citric acid cycle (also known as the tricarboxylic acid cycle; see Chapter 20), where it is oxidized to CO_2 with the production of additional NADH (and FADH_2). Under aerobic conditions, the NADH produced in glycolysis and the citric acid cycle is reoxidized to NAD^+ in the mitochondrial electron transport chain (Chapter 21).

19.6 • Anaerobic Pathways for Pyruvate

Under anaerobic conditions, the pyruvate produced in glycolysis is processed differently. In yeast, it is reduced to ethanol; in other microorganisms and in animals, it is reduced to lactate. These processes are examples of **fermentation**—the production of ATP energy by reaction pathways in which organic molecules function as donors and acceptors of electrons. In either case, reduction of pyruvate provides a means of reoxidizing the NADH produced in the glyceraldehyde-3-phosphate dehydrogenase reaction of glycolysis (Figure 19.30). In yeast, **alcoholic fermentation** is a two-step process. Pyruvate is decarboxylated to acetaldehyde by **pyruvate decarboxylase** in an essentially irreversible reaction. Thiamine pyrophosphate is a required cofactor for this enzyme. The second step, the reduction of acetaldehyde to ethanol by NADH, is catalyzed

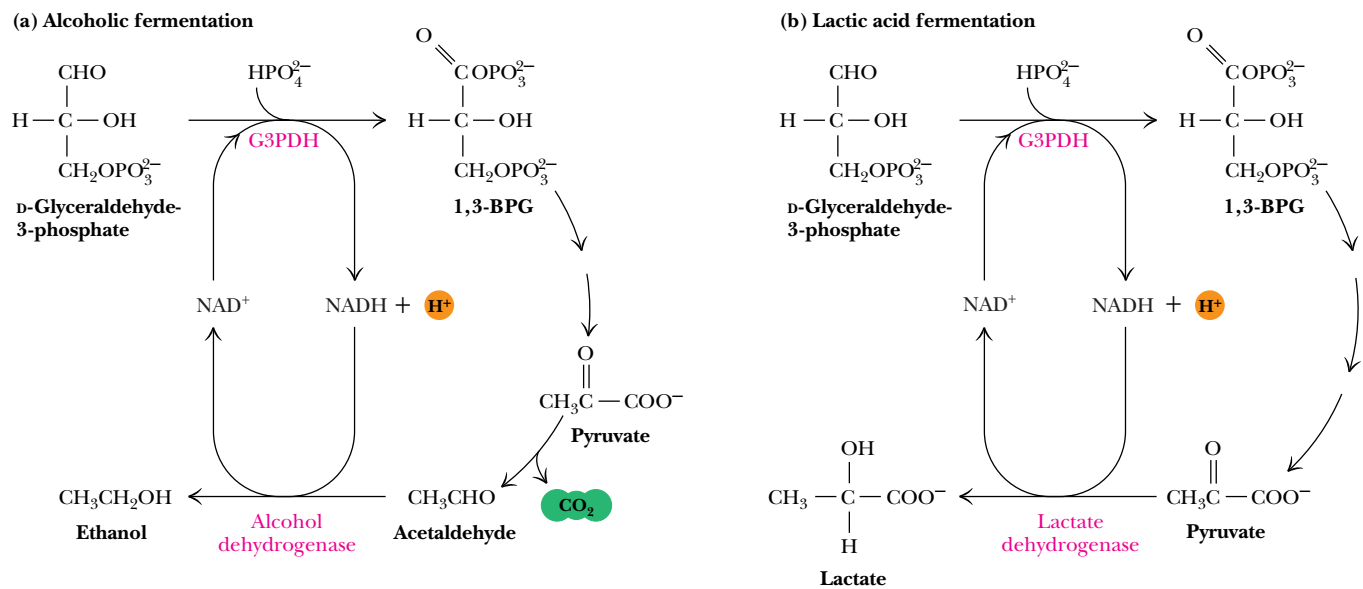


FIGURE 19.30 • (a) Pyruvate reduction to ethanol in yeast provides a means for regenerating NAD^+ consumed in the glyceraldehyde-3-P dehydrogenase reaction. (b) In oxygen-depleted muscle, NAD^+ is regenerated in the lactate dehydrogenase reaction.

by **alcohol dehydrogenase** (Figure 19.30). At pH 7, the reaction equilibrium strongly favors ethanol. The end products of alcoholic fermentation are thus ethanol and carbon dioxide. Alcoholic fermentations are the basis for the brewing of beers and the fermentation of grape sugar in wine making. Lactate produced by anaerobic microorganisms during **lactic acid fermentation** is responsible for the taste of sour milk and for the characteristic taste and fragrance of sauerkraut, which in reality is fermented cabbage.

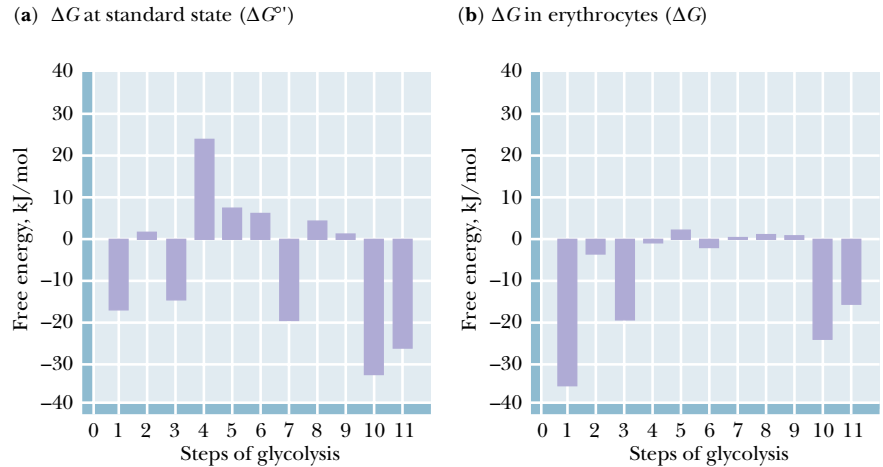
Lactate Accumulates Under Anaerobic Conditions in Animal Tissues

In animal tissues experiencing anaerobic conditions, pyruvate is reduced to lactate. Pyruvate reduction occurs in tissues that normally experience minimal access to blood flow (e.g., the cornea of the eye) and also in rapidly contracting skeletal muscle. When skeletal muscles are exercised strenuously, the available tissue oxygen is consumed, and the pyruvate generated by glycolysis can no longer be oxidized in the TCA cycle. Instead, excess pyruvate is reduced to lactate by **lactate dehydrogenase** (Figure 19.30). In anaerobic muscle tissue, lactate represents the end of glycolysis. Anyone who exercises to the point of consuming all available muscle oxygen stores knows the cramps and muscle fatigue associated with the buildup of lactic acid in the muscle. Most of this lactate must be carried out of the muscle by the blood and transported to the liver, where it can be resynthesized into glucose in gluconeogenesis. Moreover, because glycolysis generates only a fraction of the total energy available from the breakdown of glucose (the rest is generated by the TCA cycle and oxidative phosphorylation), the onset of anaerobic conditions in skeletal muscle also means a reduction in the energy available from the breakdown of glucose.

19.7 • The Energetic Elegance of Glycolysis

The elegance of nature’s design for the glycolytic pathway may be appreciated through an examination of Figure 19.31. The standard-state free energy changes for the 10 reactions of glycolysis and the lactate dehydrogenase reaction (Figure 19.31a) are variously positive and negative and, taken together, offer little insight into the coupling that occurs in the cellular milieu. On the other hand, the values of ΔG under cellular conditions (Figure 19.31b) fall into two distinct classes. For reactions 2 and 4 through 9, ΔG is very close to zero, so that these reactions operate essentially at equilibrium. Small changes

FIGURE 19.31 • A comparison of free energy changes for the reactions of glycolysis (step 1 = hexokinase) under (a) standard-state conditions and (b) actual intracellular conditions in erythrocytes. The values of ΔG° provide little insight into the actual free energy changes that occur in glycolysis. On the other hand, under intracellular conditions, seven of the glycolytic reactions operate near equilibrium (with ΔG near zero). The driving force for glycolysis lies in the hexokinase (1), phosphofructokinase (3), and pyruvate kinase (10) reactions. The lactate dehydrogenase (step 11) reaction also exhibits a large negative ΔG under cellular conditions.



in the concentrations of reactants and products could “push” any of these reactions either forward or backward. By contrast, the hexokinase, phosphofructokinase, and pyruvate kinase reactions all exhibit large negative ΔG values under cellular conditions. These reactions are thus the sites of glycolytic regulation. When these three enzymes are active, glycolysis proceeds and glucose is readily metabolized to pyruvate or lactate. Inhibition of the three key enzymes by allosteric effectors brings glycolysis to a halt. When we consider **gluconeogenesis**—the biosynthesis of glucose—in Chapter 23, we will see that different enzymes are used to carry out reactions 1, 3, and 10 in reverse, effecting the net synthesis of glucose. The maintenance of reactions 2 and 4 through 9 at or near equilibrium permits these reactions (and their respective enzymes!) to operate effectively in *either* the forward or reverse direction.

19.8 • Utilization of Other Substrates in Glycolysis

The glycolytic pathway described in this chapter begins with the breakdown of glucose, but other sugars, both simple and complex, can enter the cycle if they can be converted by appropriate enzymes to one of the intermediates of glycolysis. Figure 19.32 shows the mechanisms by which several simple metabolites can enter the glycolytic pathway. **Fructose**, for example, which is pro-

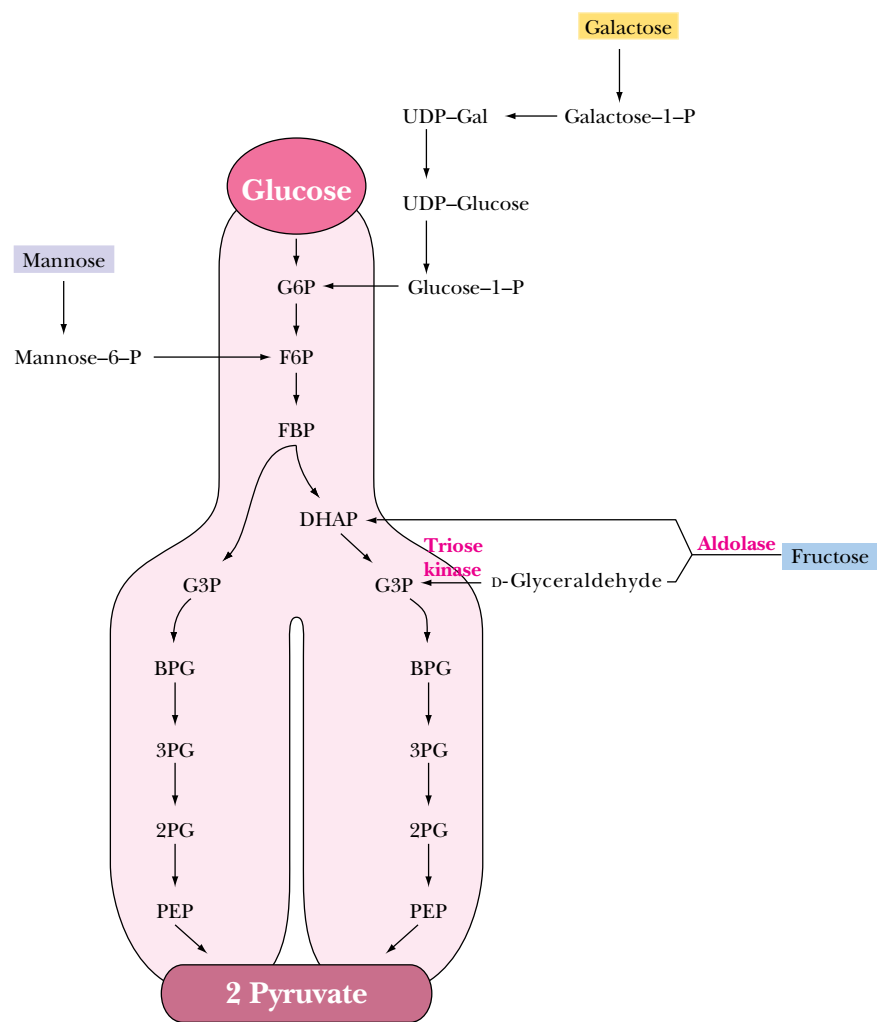
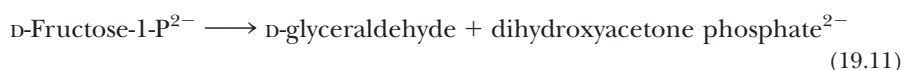


FIGURE 19.32 • Mannose, galactose, fructose, and other simple metabolites can enter the glycolytic pathway.

duced by breakdown of sucrose, may participate in glycolysis by at least two different routes. In the liver, fructose is phosphorylated at C-1 by the enzyme **fructokinase**:

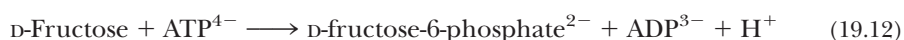


Subsequent action by **fructose-1-phosphate aldolase** cleaves fructose-1-P in a manner like the fructose biphosphate aldolase reaction to produce dihydroxyacetone phosphate and D-glyceraldehyde:



Dihydroxyacetone phosphate is of course an intermediate in glycolysis. D-Glyceraldehyde can be phosphorylated by **triose kinase** in the presence of ATP to form D-glyceraldehyde-3-phosphate, another glycolytic intermediate.

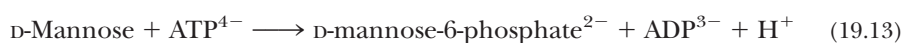
In the kidney and in muscle tissues, fructose is readily phosphorylated by hexokinase, which, as pointed out above, can utilize several different hexose substrates. The free energy of hydrolysis of ATP drives the reaction forward:



Fructose-6-phosphate generated in this way enters the glycolytic pathway directly in step 3, the second priming reaction. This is the principal means for channeling fructose into glycolysis in adipose tissue, which contains high levels of fructose.

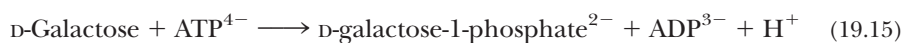
The Entry of Mannose into Glycolysis

Another simple sugar that enters glycolysis at the same point as fructose is **mannose**, which occurs in many glycoproteins, glycolipids, and polysaccharides (Chapter 7). Mannose is also phosphorylated from ATP by hexokinase, and the mannose-6-phosphate thus produced is converted to fructose-6-phosphate by **phosphomannoisomerase**.



The Special Case of Galactose

A somewhat more complicated route into glycolysis is followed by **galactose**, another simple hexose sugar. The process, called the **Leloir pathway** after Luis Leloir, its discoverer, begins with phosphorylation from ATP at the C-1 position by **galactokinase**:



Galactose-1-phosphate is then converted into *UDP-galactose* (a sugar nucleotide) by **galactose-1-phosphate uridylyltransferase** (Figure 19.33), with concurrent production of glucose-1-phosphate and consumption of a molecule of UDP-glucose. The uridylyltransferase reaction proceeds via a “ping-pong” mechanism (Figure 19.34) with a covalent enzyme-UMP intermediate. The glucose-1-phosphate produced by the transferase reaction is a substrate for the **phosphoglucomutase** reaction (Figure 19.33), which produces glucose-6-phosphate, a glycolytic substrate. The other transferase product, UDP-galactose, is converted to UDP-glucose by **UDP-glucose-4-epimerase**. The combined action of the uridylyltransferase and epimerase thus produces glucose-1-P from galactose-1-P, with regeneration of UDP-glucose.

A rare hereditary condition known as **galactosemia** involves defects in galactose-1-P uridylyltransferase that render the enzyme inactive. Toxic levels of

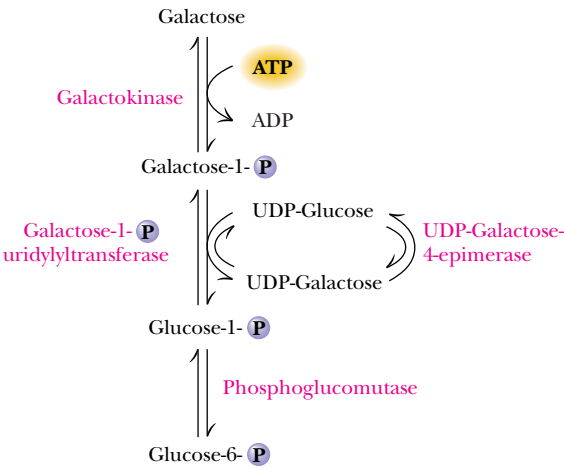


FIGURE 19.33 • Galactose metabolism via the Leloir pathway.

galactose accumulate in afflicted individuals, causing cataracts and permanent neurological disorders. These problems can be prevented by removing galactose and lactose from the diet. In adults, the toxicity of galactose appears to be less severe, due in part to the metabolism of galactose-1-P by **UDP-glucose pyrophosphorylase**, which apparently can accept galactose-1-P in place of glucose-1-P (Figure 19.35). The levels of this enzyme may increase in galactosemic individuals, in order to accommodate the metabolism of galactose.

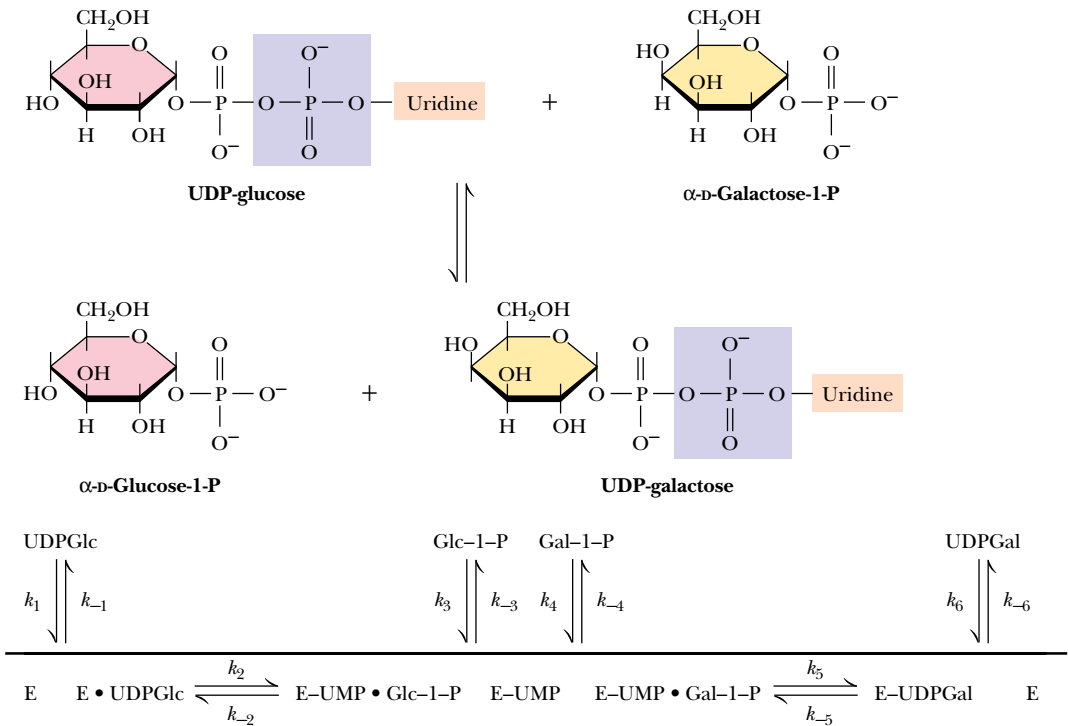
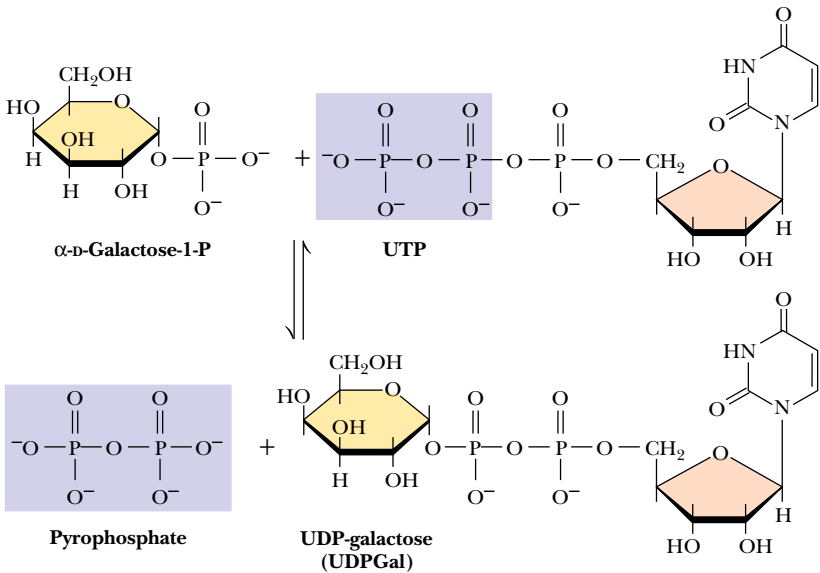


FIGURE 19.34 • The galactose-1-phosphate uridylyltransferase reaction involves a “ping-pong” kinetic mechanism.

FIGURE 19.35 • The UDP–glucose pyrophosphorylase reaction.



Lactose Intolerance

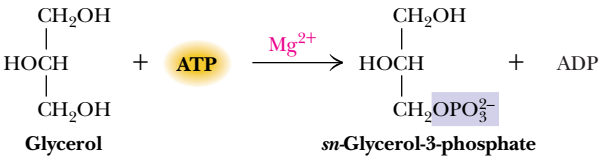
A much more common metabolic disorder, **lactose intolerance**, occurs commonly in most parts of the world (notable exceptions being some parts of Africa and northern Europe). Lactose intolerance is an inability to digest lactose because of the absence of the enzyme **lactase** in the intestines of adults. The symptoms of this disorder, which include diarrhea and general discomfort, can be relieved by eliminating milk from the diet.

Glycerol Can Also Enter Glycolysis

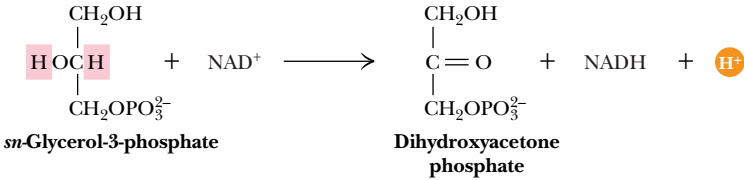
Glycerol is the last important simple substance whose ability to enter the glycolytic pathway must be considered. This metabolite, which is produced in substantial amounts by the decomposition of *triacylglycerols* (see Chapter 24) can be converted to glycerol-3-phosphate by the action of **glycerol kinase** and then oxidized to dihydroxyacetone phosphate by the action of **glycerol phosphate dehydrogenase**, with NAD^+ as the required coenzyme (Figure 19.36). The dihydroxyacetone phosphate thereby produced enters the glycolytic pathway as a substrate for triose phosphate isomerase.

FIGURE 19.36

The glycerol kinase reaction

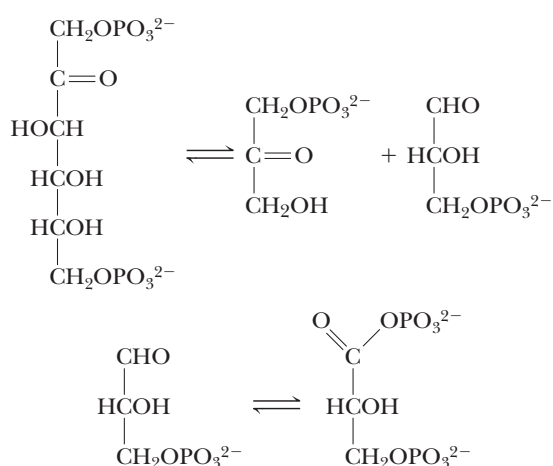


The glycerol phosphate dehydrogenase reaction

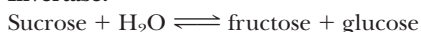


PROBLEMS

- List the reactions of glycolysis that
 - are energy-consuming (under standard-state conditions).
 - are energy-yielding (under standard-state conditions).
 - consume ATP.
 - yield ATP.
 - are strongly influenced by changes in concentration of substrate and product because of their molecularity.
 - are at or near equilibrium in the erythrocyte (see Table 19.2).
- Determine the anticipated location in pyruvate of labeled carbons if glucose molecules labeled (in separate experiments) with ^{14}C at each position of the carbon skeleton proceed through the glycolytic pathway.
- In an erythrocyte undergoing glycolysis, what would be the effect of a sudden increase in the concentration of
 - ATP?
 - AMP?
 - fructose-1,6-bisphosphate?
 - fructose-2,6-bisphosphate?
 - citrate?
 - glucose-6-phosphate?
- Discuss the cycling of NADH and NAD^+ in glycolysis and the related fermentation reactions.
- For each of the following reactions, name the enzyme that carries out this reaction in glycolysis and write a suitable mechanism for the reaction.



- Write the reactions that permit galactose to be utilized in glycolysis. Write a suitable mechanism for one of these reactions.
- How might iodoacetic acid affect the glyceraldehyde-3-phosphate dehydrogenase reaction in glycolysis? Justify your answer.
- If ^{32}P -labeled inorganic phosphate were introduced to erythrocytes undergoing glycolysis, would you expect to detect ^{32}P in glycolytic intermediates? If so, describe the relevant reactions and the ^{32}P incorporation you would observe.
- Sucrose can enter glycolysis by either of two routes:

Sucrose phosphorylase:**Invertase:**

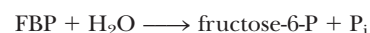
Would either of these reactions offer an advantage over the other in the preparation of hexoses for entry into glycolysis?

- What would be the consequences of a Mg^{2+} ion deficiency for the reactions of glycolysis?
- Triose phosphate isomerase catalyzes the conversion of dihydroxyacetone-P to glyceraldehyde-3-P. The standard free energy change, ΔG° , for this reaction is +7.6 kJ/mol. However, the observed free energy change (ΔG) for this reaction in erythrocytes is +2.4 kJ/mol.
 - Calculate the ratio of [dihydroxyacetone-P]/[glyceraldehyde-3-P] in erythrocytes from ΔG .
 - If [dihydroxyacetone-P] = 0.2 mM, what is [glyceraldehyde-3-P]?
- Enolase catalyzes the conversion of 2-phosphoglycerate to phosphoenolpyruvate + H_2O . The standard free energy change, ΔG° , for this reaction is +1.8 kJ/mol. If the concentration of 2-phosphoglycerate is 0.045 mM and the concentration of phosphoenolpyruvate is 0.034 mM, what is ΔG , the free energy change for the enolase reaction, under these conditions?
- The standard free energy change (ΔG°) for hydrolysis of phosphoenolpyruvate (PEP) is -61.9 kJ/mol. The standard free energy change (ΔG°) for ATP hydrolysis is -30.5 kJ/mol.

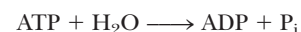
a. What is the standard free energy change for the pyruvate kinase reaction:



- What is the equilibrium constant for this reaction?
 - Assuming the intracellular concentrations of [ATP] and [ADP] remain fixed at 8 mM and 1 mM, respectively, what will be the ratio of [pyruvate]/[phosphoenolpyruvate] when the pyruvate kinase reaction reaches equilibrium?
- The standard free energy change (ΔG°) for hydrolysis of fructose-1,6-bisphosphate (FBP) to fructose-6-phosphate (F-6-P) and P_i is -16.7 kJ/mol:



The standard free energy change (ΔG°) for ATP hydrolysis is -30.5 kJ/mol:



a. What is the standard free energy change for the phosphofructokinase reaction:



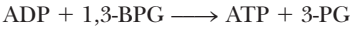
- What is the equilibrium constant for this reaction?
 - Assuming the intracellular concentrations of [ATP] and [ADP] are maintained constant at 4 mM and 1.6 mM, respectively, in a rat liver cell, what will be the ratio of [FBP]/[fructose-6-P] when the phosphofructokinase reaction reaches equilibrium?
- The standard free energy change (ΔG°) for hydrolysis of 1,3-bisphosphoglycerate (1,3-BPG) to 3-phosphoglycerate (3-PG) and P_i is -49.6 kJ/mol:



The standard free energy change ($\Delta G^\circ'$) for ATP hydrolysis is -30.5 kJ/mol :



a. What is the standard free energy change for the phosphoglycerate kinase reaction:



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- b. What is the equilibrium constant for this reaction?
 c. If the steady-state concentrations of [1,3-BPG] and [3-PG] in an erythrocyte are $1\text{ }\mu\text{M}$ and $120\text{ }\mu\text{M}$, respectively, what will be the ratio of [ATP]/[ADP], assuming the phosphoglycerate kinase reaction is at equilibrium?

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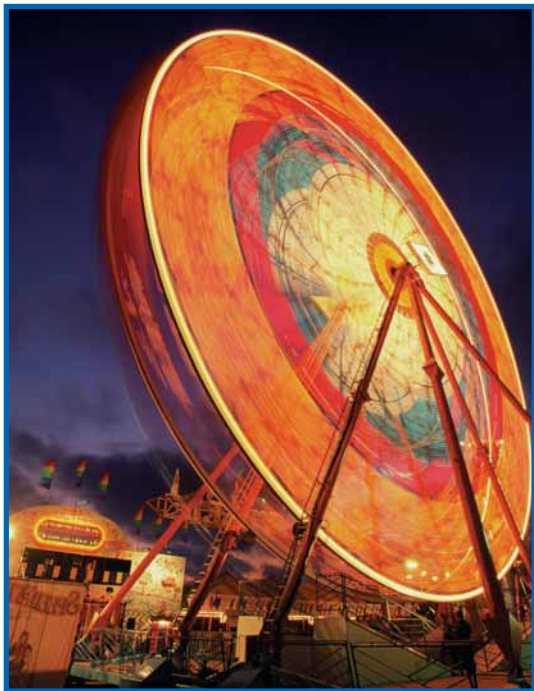
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Chapter 20

The Tricarboxylic Acid Cycle



A time-lapse photograph of a ferris wheel at night. Aerobic cells use a metabolic wheel—the tricarboxylic acid cycle—to generate energy by acetyl-CoA oxidation. (Ferns Wheel, DelMar Fair © Corbis/Richard Cummins)

The glycolytic pathway converts glucose to pyruvate and produces two molecules of ATP per glucose—only a small fraction of the potential energy available from glucose. Under anaerobic conditions, pyruvate is reduced to lactate in animals and to ethanol in yeast, and much of the potential energy of the glucose molecule remains untapped. *In the presence of oxygen*, however, a much more interesting and thermodynamically complete story unfolds. Under aerobic conditions, NADH is oxidized in the electron transport chain, rather than becoming oxidized through reduction of pyruvate to lactate or acetaldehyde to ethanol, for example. Further, pyruvate is converted to *acetyl-coenzyme A* and oxidized to CO₂ in the **tricarboxylic acid (TCA) cycle** (also called the **citric acid**

Thus times do shift, each thing his turn
does hold;
New things succeed, as former things grow
old.

ROBERT HERRICK (*Hesperides* [1648], “Ceremonies for Christmas Eve”)

OUTLINE

- 20.1 • Hans Krebs and the Discovery of the TCA Cycle
- 20.2 • The TCA Cycle—A Brief Summary
- 20.3 • The Bridging Step: Oxidative Decarboxylation of Pyruvate
- 20.4 • Entry into the Cycle: The Citrate Synthase Reaction
- 20.5 • The Isomerization of Citrate by Aconitase
- 20.6 • Isocitrate Dehydrogenase—The First Oxidation in the Cycle
- 20.7 • α -Ketoglutarate Dehydrogenase—A Second Decarboxylation
- 20.8 • Succinyl-CoA Synthetase—A Substrate-Level Phosphorylation
- 20.9 • Succinate Dehydrogenase—An Oxidation Involving FAD
- 20.10 • Fumarase Catalyzes *Trans*-Hydration of Fumarate
- 20.11 • Malate Dehydrogenase—Completing the Cycle
- 20.12 • A Summary of the Cycle
- 20.13 • The TCA Cycle Provides Intermediates for Biosynthetic Pathways
- 20.14 • The Anaplerotic, or “Filling Up,” Reactions
- 20.15 • Regulation of the TCA Cycle
- 20.16 • The Glyoxylate Cycle of Plants and Bacteria

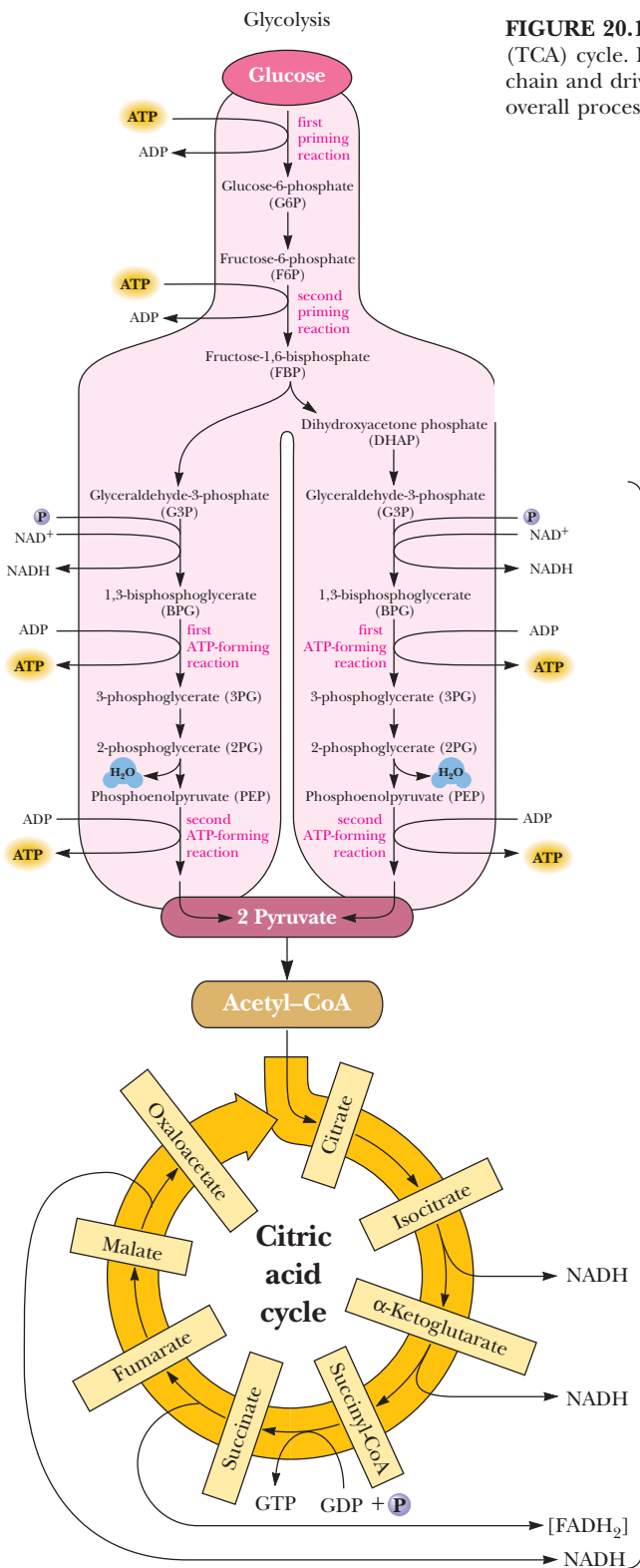
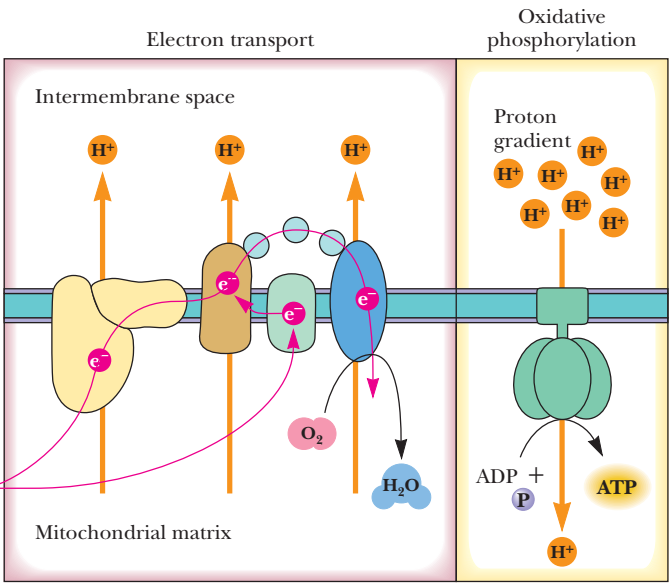


FIGURE 20.1 • Pyruvate produced in glycolysis is oxidized in the tricarboxylic acid (TCA) cycle. Electrons liberated in this oxidation flow through the electron transport chain and drive the synthesis of ATP in oxidative phosphorylation. In eukaryotic cells, this overall process occurs in mitochondria.



cycle). The electrons liberated by this oxidative process are then passed through an elaborate, membrane-associated **electron transport pathway** to O₂, the final electron acceptor. Electron transfer is coupled to creation of a proton gradient across the membrane. Such a gradient represents an energized state, and the energy stored in this gradient is used to drive the synthesis of many equivalents of ATP.

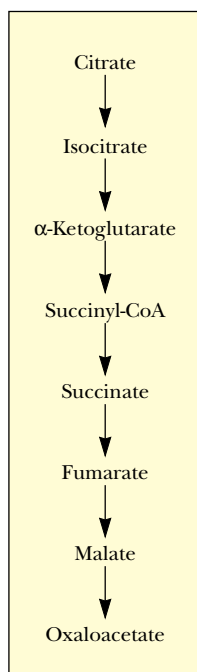


FIGURE 20.3 • Martius and Knoop's observation that citrate could be converted to isocitrate and then α -ketoglutarate provided a complete pathway from citrate to oxaloacetate.

Another important piece of the puzzle came from the work of Carl Martius and Franz Knoop, who showed that citric acid could be converted to isocitrate and then to α -ketoglutarate. This finding was significant because it was already known that α -ketoglutarate could be enzymatically oxidized to succinate. At this juncture, the pathway from citrate to oxaloacetate seemed to be as shown in Figure 20.3. Whereas the pathway made sense, the *catalytic* effect of succinate and the other dicarboxylic acids from Szent-Györgyi's studies remained a puzzle.

In 1937 Krebs found that citrate could be formed in muscle suspensions if oxaloacetate and either pyruvate or acetate were added. He saw that he now had a cycle, not a simple pathway, and that addition of any of the intermediates could generate all of the others. The existence of a cycle, together with the entry of pyruvate into the cycle in the synthesis of citrate, provided a clear explanation for the accelerating properties of succinate, fumarate, and malate. If all these intermediates led to oxaloacetate, which combined with pyruvate from glycolysis, they could stimulate the oxidation of many substances besides themselves. (Kreb's conceptual leap to a cycle was not his first. Together with medical student Kurt Henseleit, he had already elucidated the details of the *urea cycle* in 1932.) The complete tricarboxylic acid (Krebs) cycle, as it is now understood, is shown in Figure 20.4.

20.2 • The TCA Cycle—A Brief Summary

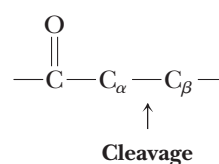
The entry of new carbon units into the cycle is through acetyl-CoA. This entry metabolite can be formed either from pyruvate (from glycolysis) or from oxidation of fatty acids (discussed in Chapter 25). Transfer of the two-carbon acetyl group from acetyl-CoA to the four-carbon oxaloacetate to yield six-carbon citrate is catalyzed by *citrate synthase*. A dehydration–rehydration rearrangement of citrate yields isocitrate. Two successive decarboxylations produce α -ketoglutarate and then succinyl-CoA, a CoA conjugate of a four-carbon unit. Several steps later, oxaloacetate is regenerated and can combine with another two-carbon unit of acetyl-CoA. Thus, carbon enters the cycle as acetyl-CoA and exits as CO_2 . In the process, metabolic energy is captured in the form of ATP, NADH, and enzyme-bound FADH_2 (symbolized as $[\text{FADH}_2]$).

The Chemical Logic of the TCA Cycle

The cycle shown in Figure 20.4 at first appears to be a complicated way to oxidize acetate units to CO_2 , but there is a chemical basis for the apparent complexity. Oxidation of an acetyl group to a pair of CO_2 molecules requires C—C cleavage:



In many instances, C—C cleavage reactions in biological systems occur between carbon atoms α - and β - to a carbonyl group:



A good example of such a cleavage is the fructose biphosphate aldolase reaction (see Chapter 19, Figure 19.14a).

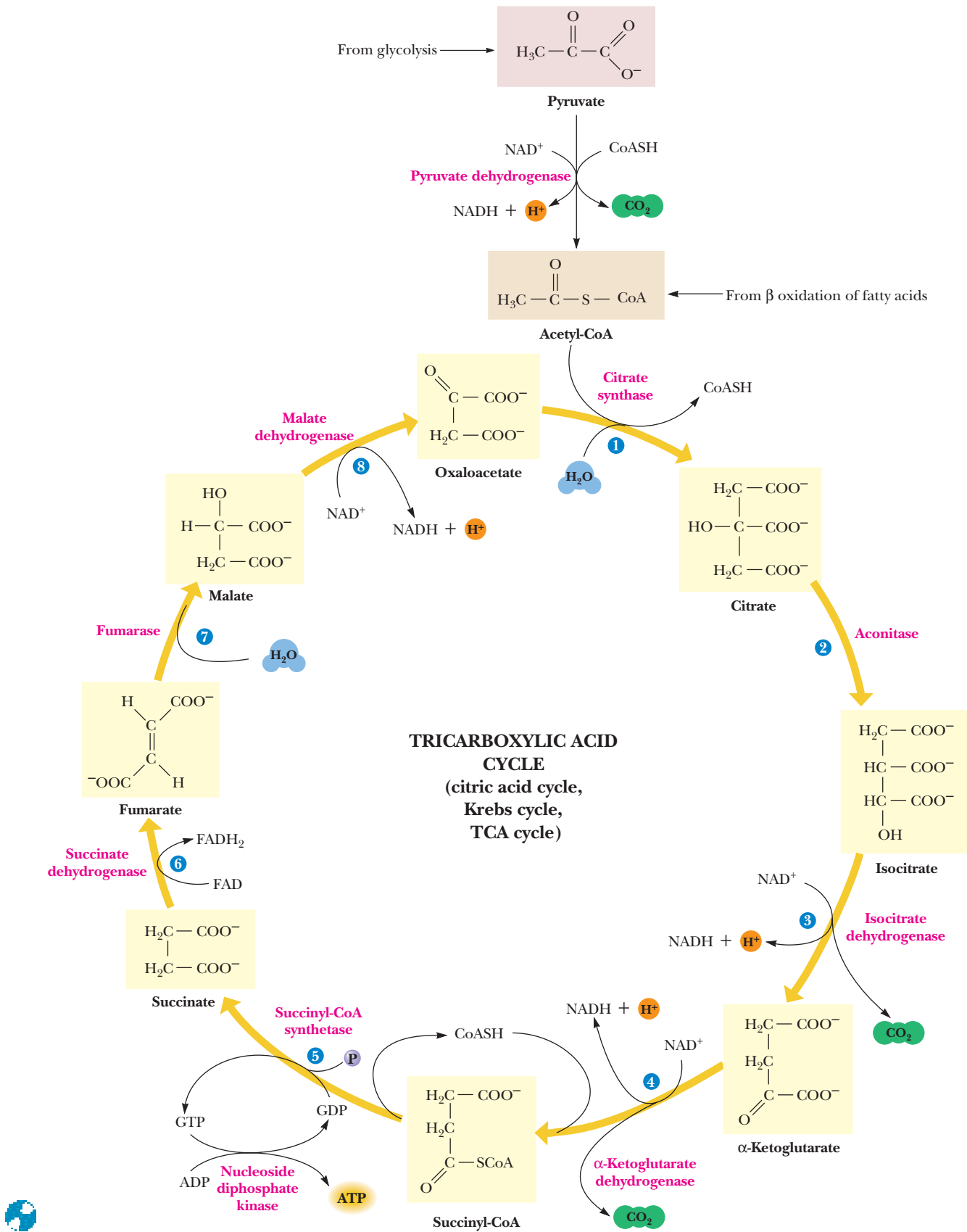
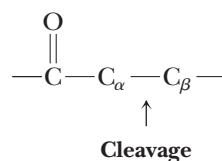


FIGURE 20.4 • The tricarboxylic acid cycle.

Another common type of C—C cleavage is α -cleavage of an α -hydroxyketone:



(We see this type of cleavage in the *transketolase* reaction described in Chapter 23.)

Neither of these cleavage strategies is suitable for acetate. It has no β -carbon, and the second method would require hydroxylation—not a favorable reaction for acetate. Instead, living things have evolved the clever chemistry of condensing acetate with oxaloacetate and *then* carrying out a β -cleavage. *The TCA cycle combines this β -cleavage reaction with oxidation to form CO_2 , regenerate oxaloacetate, and capture the liberated metabolic energy in NADH and ATP.*

20.3 • The Bridging Step: Oxidative Decarboxylation of Pyruvate

Pyruvate produced by glycolysis is a significant source of acetyl-CoA for the TCA cycle. Because, in eukaryotic cells, glycolysis occurs in the cytoplasm, whereas the TCA cycle reactions and all subsequent steps of aerobic metabolism take place in the mitochondria, pyruvate must first enter the mitochondria to enter the TCA cycle. The oxidative decarboxylation of pyruvate to acetyl-CoA,



is the connecting link between glycolysis and the TCA cycle. The reaction is catalyzed by pyruvate dehydrogenase, a multienzyme complex.

The **pyruvate dehydrogenase complex (PDC)** is a noncovalent assembly of three different enzymes operating in concert to catalyze successive steps in the conversion of pyruvate to acetyl-CoA. The active sites of all three enzymes are not far removed from one another, and the product of the first enzyme is passed directly to the second enzyme and so on, without diffusion of substrates and products through the solution. The overall reaction (see *A Deeper Look: “Reaction Mechanism of the Pyruvate Dehydrogenase Complex”*) involves a total of five coenzymes: thiamine pyrophosphate, coenzyme A, lipoic acid, NAD^+ , and FAD.

20.4 • Entry into the Cycle: The Citrate Synthase Reaction

The first reaction within the TCA cycle, the one by which carbon atoms are introduced, is the **citrate synthase reaction** (Figure 20.5). Here acetyl-CoA reacts with oxaloacetate in a **Perkin condensation** (a carbon–carbon condensation between a ketone or aldehyde and an ester). The acyl group is activated in two ways in an acyl-CoA molecule: the carbonyl carbon is activated for attack by nucleophiles, and the C_α carbon is more acidic and can be deprotonated to form a carbanion. The citrate synthase reaction depends upon the latter mode of activation. As shown in Figure 20.5, a general base on the enzyme accepts a proton from the methyl group of acetyl-CoA, producing a stabilized α -carbanion of acetyl-CoA. This strong nucleophile attacks the α -carbonyl of oxaloacetate, yielding citryl-CoA. This part of the reaction has an equilibrium constant

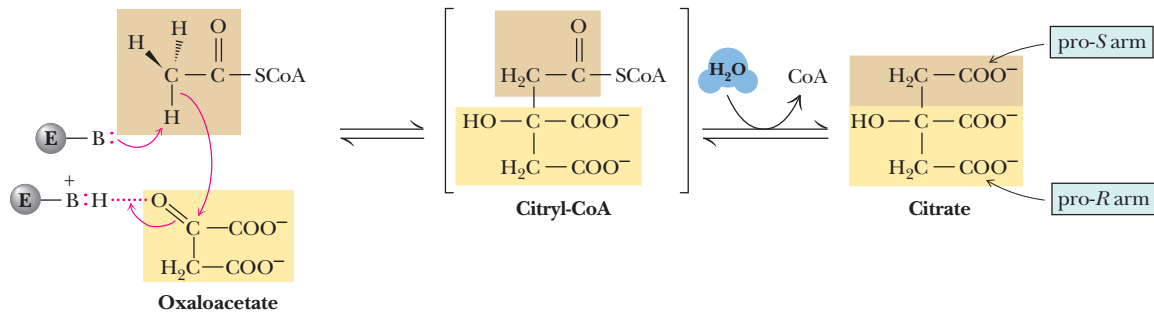


FIGURE 20.5 • Citrate is formed in the citrate synthase reaction from oxaloacetate and acetyl-CoA. The mechanism involves nucleophilic attack by the carbanion of acetyl-CoA on the carbonyl carbon of oxaloacetate, followed by thioester hydrolysis.

near 1, but the overall reaction is driven to completion by the subsequent hydrolysis of the high-energy thioester to citrate and free CoA. The overall $\Delta G^{\circ'}$ is -31.4 kJ/mol, and under standard conditions the reaction is essentially irreversible. Although the mitochondrial concentration of oxaloacetate is very low (much less than $1\text{ }\mu\text{M}$ —see example in Section 20.11), the strong, negative $\Delta G^{\circ'}$ drives the reaction forward.

The Structure of Citrate Synthase

Citrate synthase in mammals is a dimer of 49-kD subunits (Table 20.1). On each subunit, oxaloacetate and acetyl-CoA bind to the active site, which lies in a cleft between two domains and is surrounded mainly by α -helical segments (Figure 20.6). Binding of oxaloacetate induces a conformational change that facilitates the binding of acetyl-CoA and closes the active site, so that the reactive carbanion of acetyl-CoA is protected from protonation by water.

Regulation of Citrate Synthase

Citrate synthase is the first step in this metabolic pathway, and as stated the reaction has a large negative $\Delta G^{\circ'}$. As might be expected, it is a highly regulated enzyme. NADH, a product of the TCA cycle, is an allosteric inhibitor of citrate synthase, as is succinyl-CoA, the product of the fifth step in the cycle (and an acetyl-CoA analog).

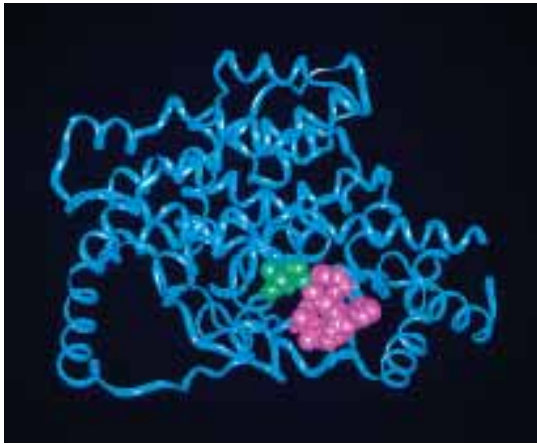


FIGURE 20.6 • Citrate synthase. In the monomer shown here, citrate is shown in green, and CoA is pink.



Text continued on page 648.

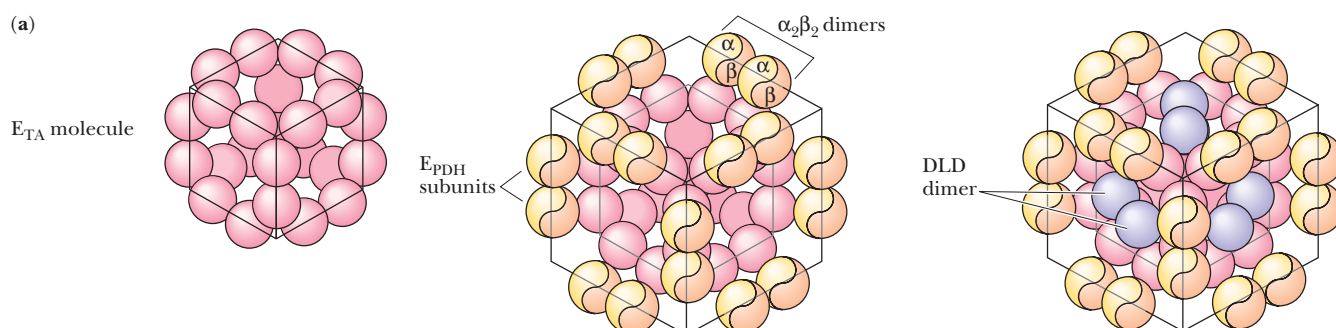
Reaction Mechanism of the Pyruvate Dehydrogenase Complex

The mechanism of the pyruvate dehydrogenase reaction is a *tour de force* of mechanistic chemistry, involving as it does a total of three enzymes (a) and five different coenzymes—thiamine pyrophosphate, lipoic acid, coenzyme A, FAD, and NAD^+ (b).

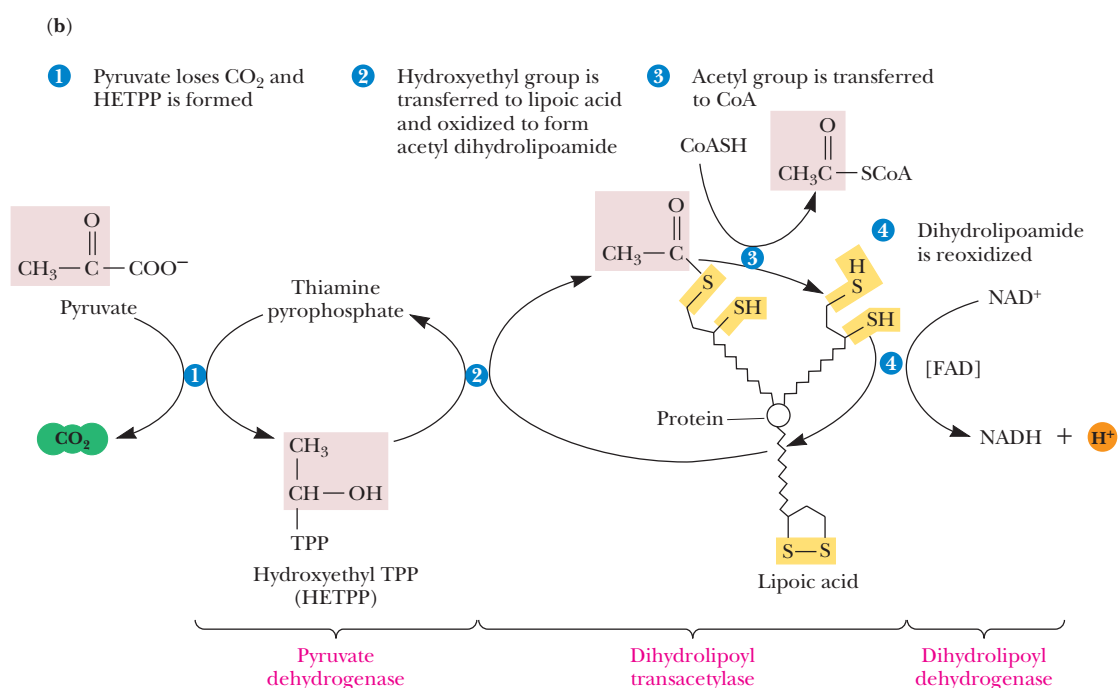
The first step of this reaction, decarboxylation of pyruvate and transfer of the acetyl group to lipoic acid, depends on accumulation of negative charge on the carbonyl carbon of pyruvate. This is facilitated by the quaternary nitrogen on the *thiazolium* group of thiamine pyrophosphate. As shown in (c), this cationic

imine nitrogen plays two distinct and important roles in TPP-catalyzed reactions:

1. It provides electrostatic stabilization of the carbanion formed upon removal of the C-2 proton. (The sp^2 hybridization and the availability of vacant d orbitals on the adjacent sulfur probably also facilitate proton removal at C-2.)
2. TPP attack on pyruvate leads to decarboxylation. The TPP cationic imine nitrogen can act as an effective electron sink to stabilize the negative charge that must develop on the carbon



(a) The structure of the pyruvate dehydrogenase complex. This complex consists of three enzymes: pyruvate dehydrogenase (PDH), dihydrolipoyl transacetylase (TA), and dihydrolipoyl dehydrogenase (DLD). (i) 24 dihydrolipoyl transacetylase subunits form a cubic core structure. (ii) 24 $\alpha\beta$ dimers of pyruvate dehydrogenase are added to the cube (two per edge). (iii) Addition of 12 dihydrolipoyl dehydrogenase subunits (two per face) completes the complex.

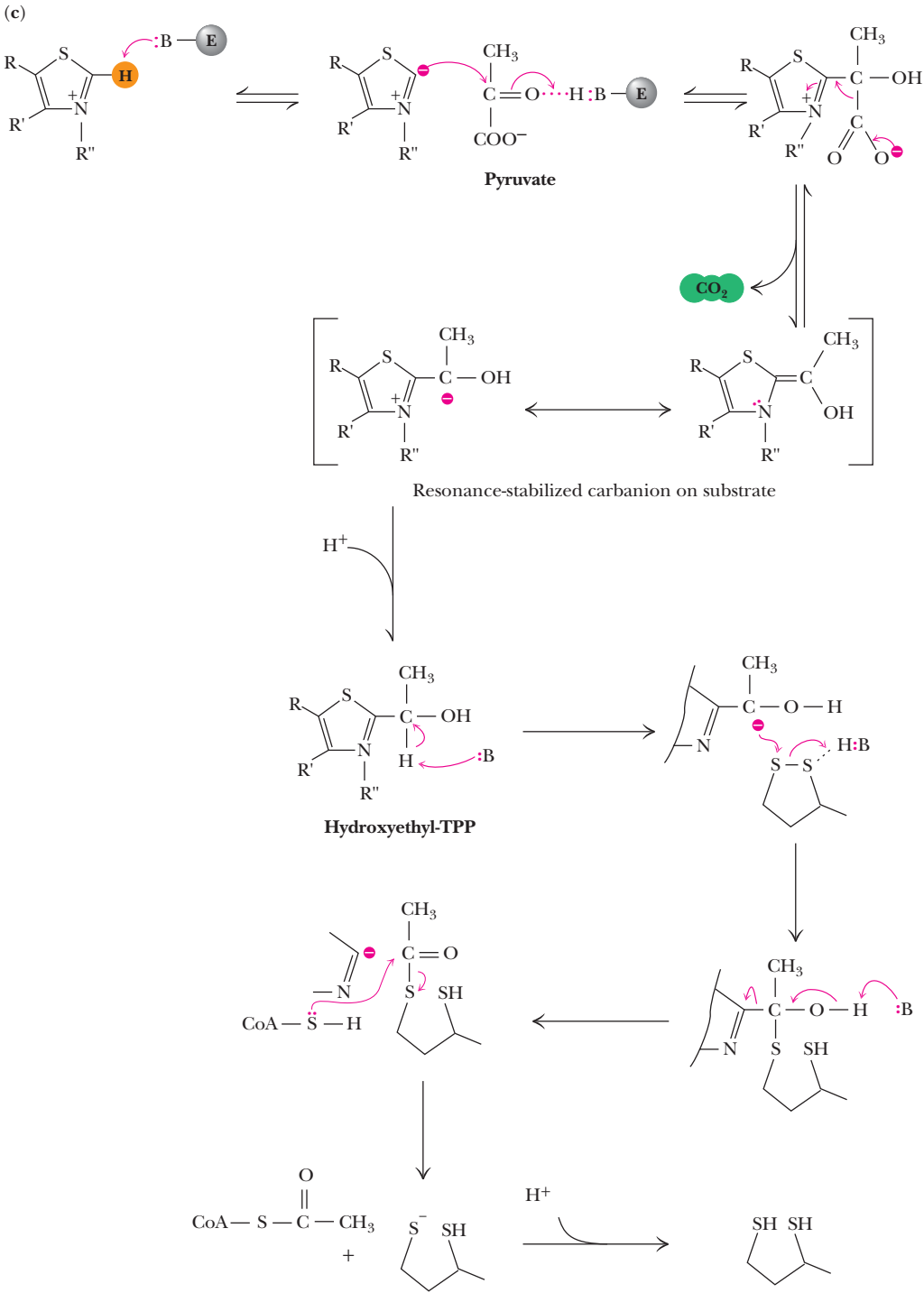


(b) The reaction mechanism of the pyruvate dehydrogenase complex. Decarboxylation of pyruvate occurs with formation of hydroxyethyl-TPP (Step 1). Transfer of the two-carbon unit to lipoic acid in Step 2 is followed by formation of acetyl-CoA in Step 3. Lipoic acid is reoxidized in Step 4 of the reaction.

that has been attacked. This stabilization takes place by resonance interaction through the double bond to the nitrogen atom.

This resonance-stabilized intermediate can be protonated to give **hydroxyethyl-TPP**. This well-characterized intermediate was once thought to be so unstable that it could not be synthesized or isolated. However, its synthesis and isolation are actually routine. (In fact, a substantial amount of the thiamine pyrophosphate in living things exists as the hydroxyethyl form.)

The reaction of hydroxyethyl-TPP with the oxidized form of lipoic acid yields the energy-rich thiol ester of reduced lipoic acid and results in oxidation of the hydroxyl-carbon of the two-carbon substrate unit (c). This is followed by nucleophilic attack by coenzyme A on the carbonyl-carbon (a characteristic feature of CoA chemistry). The result is transfer of the acetyl group from lipoic acid to CoA. The subsequent oxidation of lipoic acid is catalyzed by the FAD-dependent dihydrolipoyl dehydrogenase and NAD^+ is reduced.



(c) The mechanistic details of the first three steps of the pyruvate dehydrogenase complex reaction.

Table 20.1

continued					
Subunit M _r	Oligomeric Composition	Δ <i>G</i> ^{o'} (kJ/mol)	<i>K</i> _{eq} at 25°C	Δ <i>G</i> (kJ/mol)	
49,000*	Dimer	−31.4	3.2 × 10 ⁵	−53.9	
44,500	Dimer	+6.7	0.067	+0.8	
	α ₂ βγ	−8.4	29.7	−17.5	
<i>E</i> ₁ 96,000	Dimer				
<i>E</i> ₂ 70,000	24-mer	−30	1.8 × 10 ⁵	−43.9	
<i>E</i> ₃ 56,000	Dimer				
α 34,500	αβ	−3.3	3.8	≈0	
β 42,500					
α 70,000	αβ	+0.4	0.85	≠0	
β 27,000					
48,500	Tetramer	−3.8	4.6	≈0	
35,000	Dimer	+29.7	6.2 × 10 ^{−6}	≈0	
		−40		≈(−115)	
		−849			

*CS in mammals, A in pig heart, αKDC in *E. coli*, S-CoA S in pig heart, SD in bovine heart, F in pig heart, MD in pig heart. Δ*G* values from Newsholme, E. A., and Leech, A. R., 1983. *Biochemistry for the Medical Sciences*. New York: Wiley.

of water are first abstracted from citrate to yield aconitate, which is then rehydrated with H— and HO— adding back in opposite positions to produce isocitrate. The net effect is the conversion of a tertiary alcohol (citrate) to a secondary alcohol (isocitrate). Oxidation of the secondary alcohol of isocitrate involves breakage of a C—H bond, a simpler matter than the C—C cleavage required for the direct oxidation of citrate.

Inspection of the citrate structure shows a total of four chemically equivalent hydrogens, but only one of these—the pro-*R* H atom of the pro-*R* arm of citrate—is abstracted by aconitase, which is quite stereospecific. Formation of the double bond of aconitate following proton abstraction requires departure of hydroxide ion from the C-3 position. Hydroxide is a relatively poor leaving group, and its departure is facilitated in the aconitase reaction by coordination with an iron atom in an iron–sulfur cluster.

Aconitase Utilizes an Iron–Sulfur Cluster

Aconitase contains an **iron–sulfur cluster** consisting of three iron atoms and four sulfur atoms in a near-cubic arrangement (Figure 20.8). This cluster is bound to the enzyme via three cysteine groups from the protein. One corner of the cube is vacant and binds Fe²⁺, which activates aconitase. The iron atom in this position can coordinate the C-3 carboxyl and hydroxyl groups of citrate. This iron atom thus acts as a Lewis acid, accepting an unshared pair of electrons from the hydroxyl, making it a better leaving group. The equilibrium for the aconitase reaction favors citrate, and an equilibrium mixture typically contains about 90% citrate, 4% *cis*-aconitate, and 6% isocitrate. The Δ*G*^{o'} is +6.7 kJ/mol.

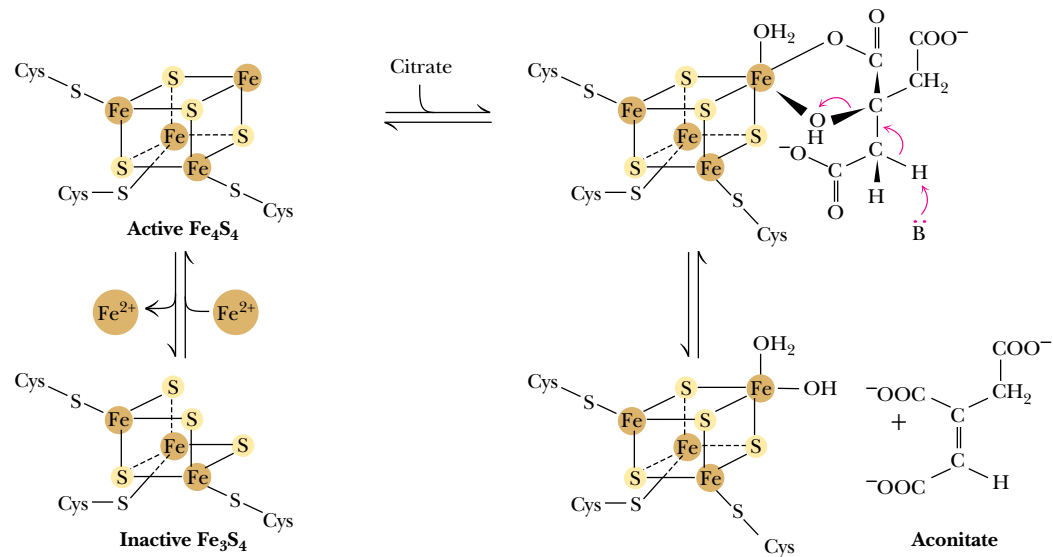


FIGURE 20.8 • The iron–sulfur cluster of aconitase. Binding of Fe^{2+} to the vacant position of the cluster activates aconitase. The added iron atom coordinates the C-3 carboxyl and hydroxyl groups of citrate and acts as a Lewis acid, accepting an electron pair from the hydroxyl group and making it a better leaving group.

Fluoroacetate Blocks the TCA Cycle

Fluoroacetate is an extremely poisonous agent that blocks the TCA cycle *in vivo*, although it has no apparent effect on any of the isolated enzymes. Its LD₅₀, the lethal dose for 50% of animals consuming it, is 0.2 mg per kilogram of body weight; it has been used as a rodent poison. The action of fluoroacetate has been traced to aconitase, which is inhibited *in vivo* by fluorocitrate, which is formed from fluoroacetate in two steps (Figure 20.9). Fluoroacetate readily crosses both the cellular and mitochondrial membranes, and in mitochondria it is converted to fluoroacetyl-CoA by *acetyl-CoA synthetase*. Fluoroacetyl-CoA is a substrate for citrate synthase, which condenses it with oxaloacetate to form fluorocitrate. Fluoroacetate may thus be viewed as a **trojan horse inhibitor**. Analogous to the giant Trojan Horse of legend—which the soldiers of Troy took into their city, not knowing that Greek soldiers were hidden inside it and waiting to attack—fluoroacetate enters the TCA cycle innocently enough, in the citrate synthase reaction. Citrate synthase converts fluoroacetate to inhibitory fluorocitrate for its TCA cycle partner, aconitase, blocking the cycle.

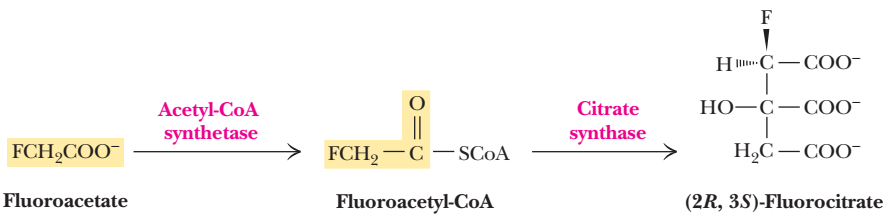


FIGURE 20.9 • The conversion of fluoroacetate to fluorocitrate.

20.6 • Isocitrate Dehydrogenase—The First Oxidation in the Cycle

In the next step of the TCA cycle, isocitrate is oxidatively decarboxylated to yield α -ketoglutarate, with concomitant reduction of NAD^+ to NADH in the isocitrate dehydrogenase reaction (Figure 20.10). The reaction has a net $\Delta G'$ of -8.4 kJ/mol, and it is sufficiently exergonic to pull the aconitase reaction forward. This two-step reaction involves (1) oxidation of the C-2 alcohol of isocitrate to form oxalosuccinate, followed by (2) a β -decarboxylation reaction that expels the central carboxyl group as CO_2 , leaving the product α -ketoglutarate. Oxalosuccinate, the β -keto acid produced by the initial dehydrogenation reaction, is unstable and thus is readily decarboxylated.

Isocitrate Dehydrogenase Links the TCA Cycle and Electron Transport

Isocitrate dehydrogenase provides the first connection between the TCA cycle and the electron transport pathway and oxidative phosphorylation, via its production of NADH. As a connecting point between two metabolic pathways, isocitrate dehydrogenase is a regulated reaction. NADH and ATP are allosteric inhibitors, whereas ADP acts as an allosteric activator, lowering the K_m for isocitrate by a factor of 10. The enzyme is virtually inactive in the absence of ADP. Also, the product, α -ketoglutarate, is a crucial α -keto acid for aminotransferase reactions (see Chapters 14 and 27), connecting the TCA cycle (that is, carbon metabolism) with nitrogen metabolism.

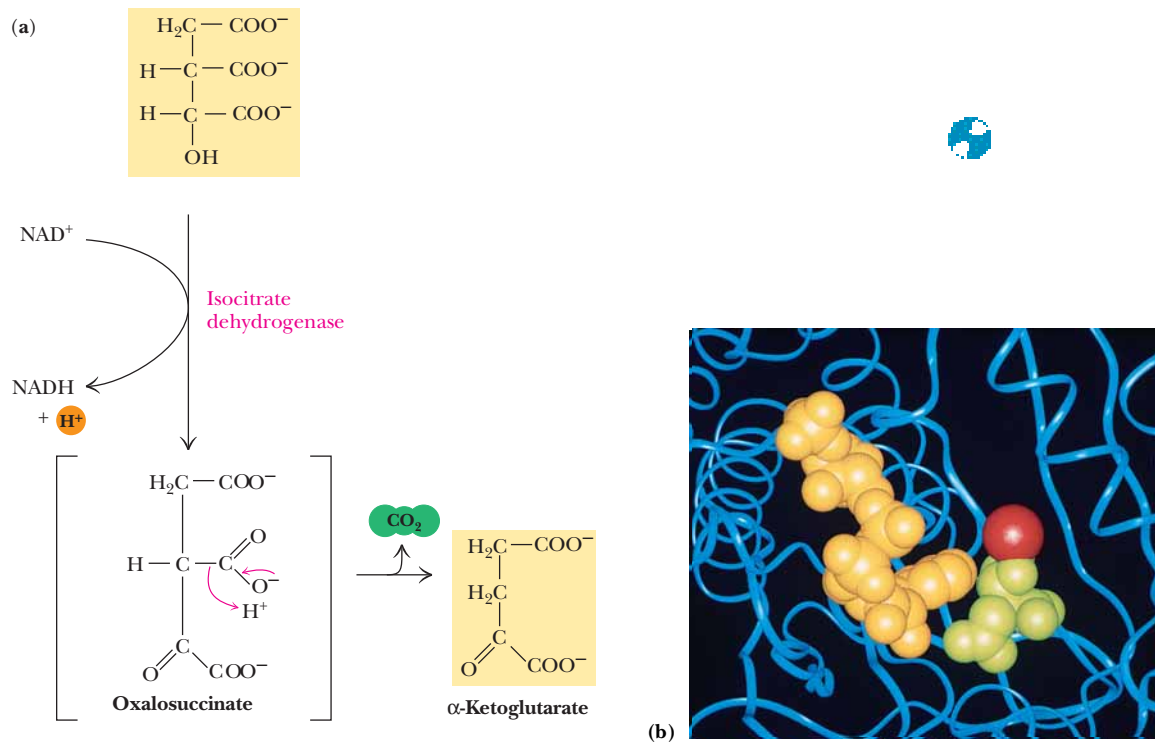


FIGURE 20.10 • (a) The isocitrate dehydrogenase reaction. (b) The active site of isocitrate dehydrogenase. Isocitrate is shown in green, NADP⁺ is shown in gold, with Ca²⁺ in red.

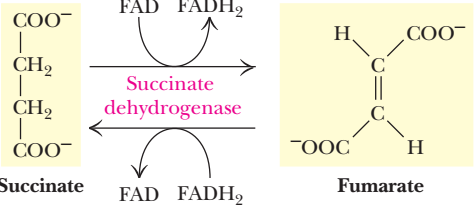


FIGURE 20.14 • The succinate dehydrogenase reaction. Oxidation of succinate occurs with reduction of [FAD]. Reoxidation of [FADH₂] transfers electrons to coenzyme Q.

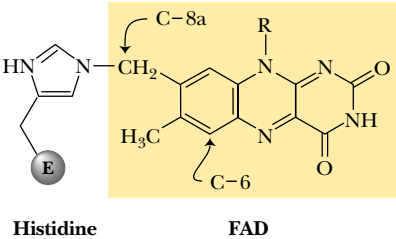


FIGURE 20.15 • The covalent bond between FAD and succinate dehydrogenase involves the C-8a methylene group of FAD and the N-3 of a histidine residue on the enzyme.

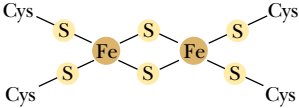


FIGURE 20.16 • The Fe₂S₂ cluster of succinate dehydrogenase.

20.9 • Succinate Dehydrogenase—An Oxidation Involving FAD

The oxidation of succinate to fumarate (Figure 20.14) is carried out by **succinate dehydrogenase**, a membrane-bound enzyme that is actually part of the electron transport chain. As will be seen in Chapter 21, succinate dehydrogenase is part of the succinate–coenzyme Q reductase of the electron transport chain. In contrast with all of the other enzymes of the TCA cycle, which are soluble proteins found in the mitochondrial matrix, succinate dehydrogenase is an integral membrane protein tightly associated with the inner mitochondrial membrane. Succinate oxidation involves removal of H atoms across a C—C bond, rather than a C—O or C—N bond, and produces the *trans*-unsaturated fumarate. This reaction (the oxidation of an alkane to an alkene) is not sufficiently exergonic to reduce NAD⁺, but it does yield enough energy to reduce [FAD]. (By contrast, oxidations of alcohols to ketones or aldehydes are more energetically favorable and provide sufficient energy to reduce NAD⁺.) This important point is illustrated and clarified in an example in Chapter 21.

Succinate dehydrogenase is a dimeric protein, with subunits of molecular masses 70 kD and 27 kD (see Table 20.1). FAD is covalently bound to the larger subunit; the bond involves a methylene group of C-8a of FAD and N-3 of a histidine on the protein (Figure 20.15). Succinate dehydrogenase also contains three different iron–sulfur clusters (Figure 20.16). Viewed from either end of the succinate molecule, the reaction involves dehydrogenation α,β to a carbonyl (actually, a carboxyl) group. The dehydrogenation is stereospecific (Figure 20.14), with the pro-*S* hydrogen removed from one carbon atom and the pro-*R* hydrogen removed from the other. The electrons captured by [FAD] in this reaction are passed directly into the iron–sulfur clusters of the enzyme and on to *coenzyme Q* (*UQ*). The covalently bound FAD is first reduced to [FADH₂] and then reoxidized to form [FAD] and the reduced form of coenzyme Q, *UQH*₂. Electrons captured by *UQH*₂ then flow through the rest of the electron transport chain in a series of events that is discussed in detail in Chapter 21.

Note that flavin coenzymes can carry out either one-electron or two-electron transfers. The succinate dehydrogenase reaction represents a net two-electron reduction of FAD.

20.10 • Fumarase Catalyzes *Trans*-Hydration of Fumarate

Fumarate is hydrated in a stereospecific reaction by fumarase to give L-malate (Figure 20.17). The reaction involves *trans*-addition of the elements of water across the double bond. Recall that aconitase carries out a similar reaction,

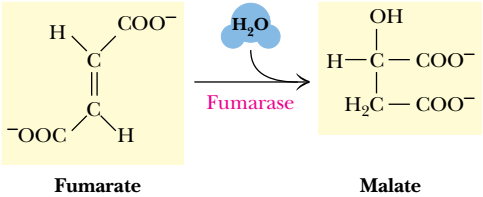


FIGURE 20.17 • The fumarase reaction.

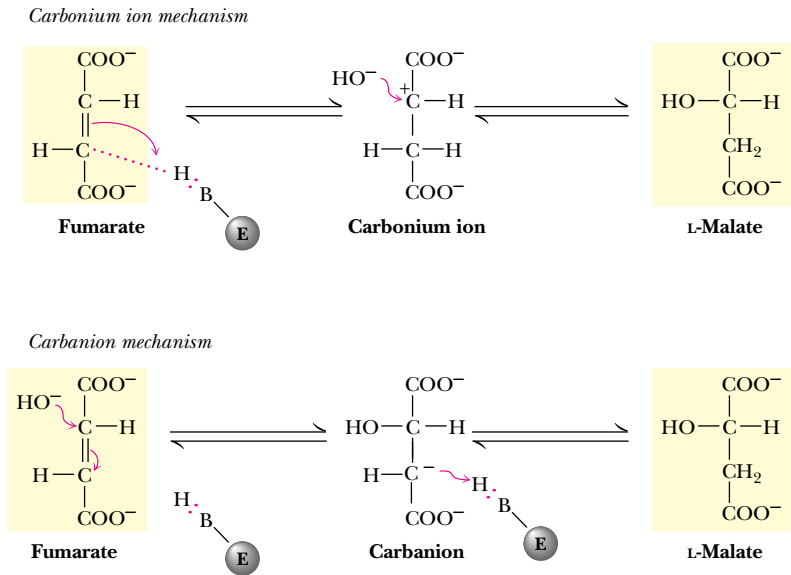


FIGURE 20.18 • Two possible mechanisms for the fumarase reaction.

and that *trans*-addition of —H and —OH occurs across the double bond of *cis*-aconitate. Though the exact mechanism is uncertain, it may involve protonation of the double bond to form an intermediate carbonium ion (Figure 20.18) or possibly attack by water or OH[−] anion to produce a carbanion, followed by protonation.

20.11 • Malate Dehydrogenase—Completing the Cycle

In the last step of the TCA cycle, L-malate is oxidized to oxaloacetate by **malate dehydrogenase** (Figure 20.19). This reaction is very endergonic, with a $\Delta G^{\circ'}$ of +30 kJ/mol. Consequently, the concentration of oxaloacetate in the mitochondrial matrix is usually quite low (see the following example). The reaction, however, is pulled forward by the favorable citrate synthase reaction. Oxidation of malate is coupled to reduction of yet another molecule of NAD⁺, the third one of the cycle. Counting the [FAD] reduced by succinate dehydrogenase, this makes the fourth coenzyme reduced through oxidation of a single acetate unit.

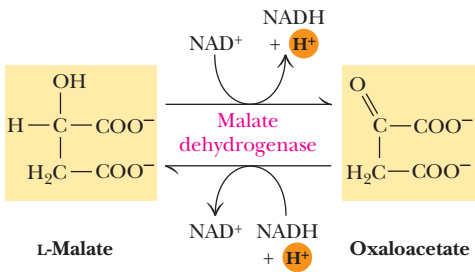


FIGURE 20.19 • The malate dehydrogenase reaction.

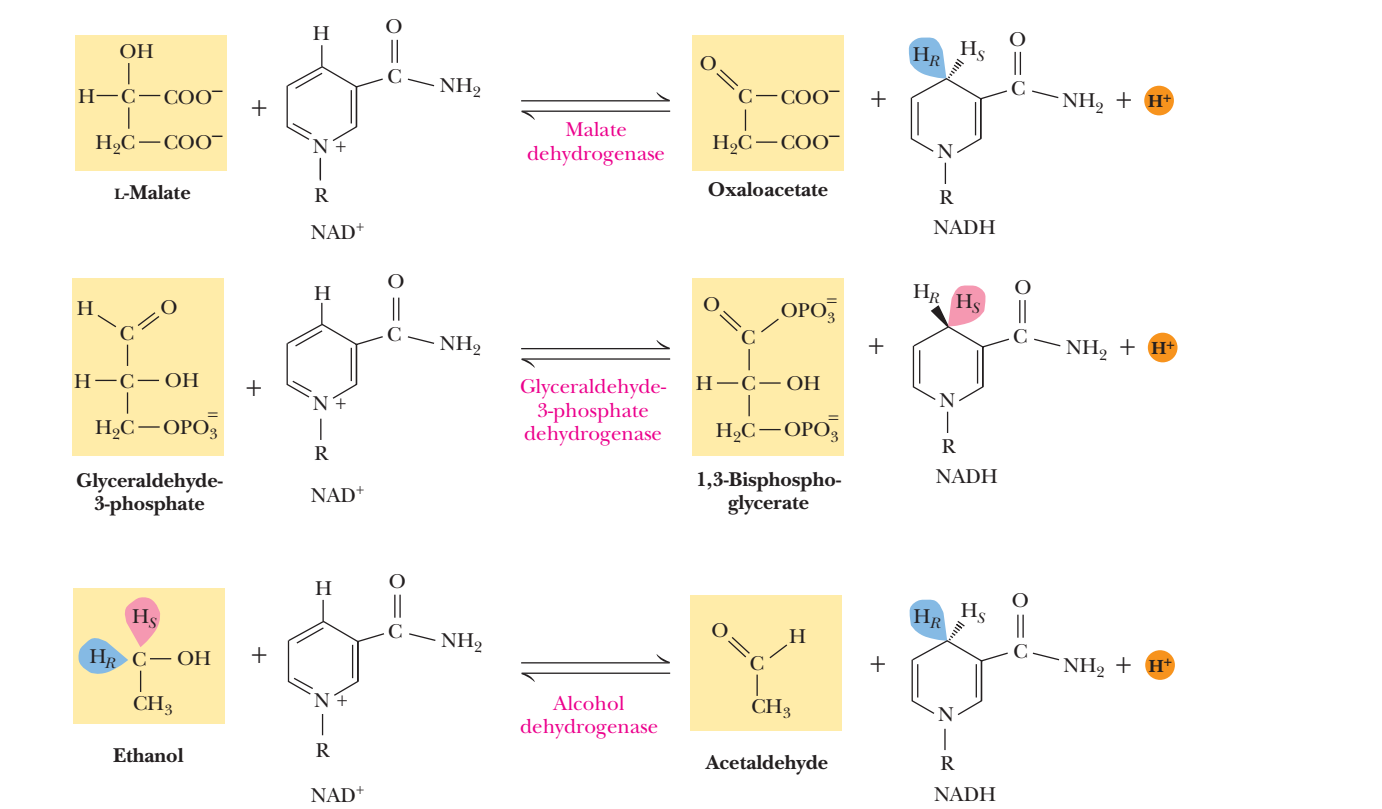
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Steric Preferences in NAD⁺-Dependent Dehydrogenases

As noted in Chapter 18, the enzymes that require nicotinamide coenzymes are stereospecific and transfer hydride to either the pro-*R* or the pro-*S* positions selectively. The table (facing page) lists the preferences of several dehydrogenases.

What accounts for this stereospecificity? It arises from the fact that the enzymes (and especially the active sites of enzymes) are inherently asymmetric structures. The nicotinamide coenzyme (and the substrate) fit the active site in only one way. Malate

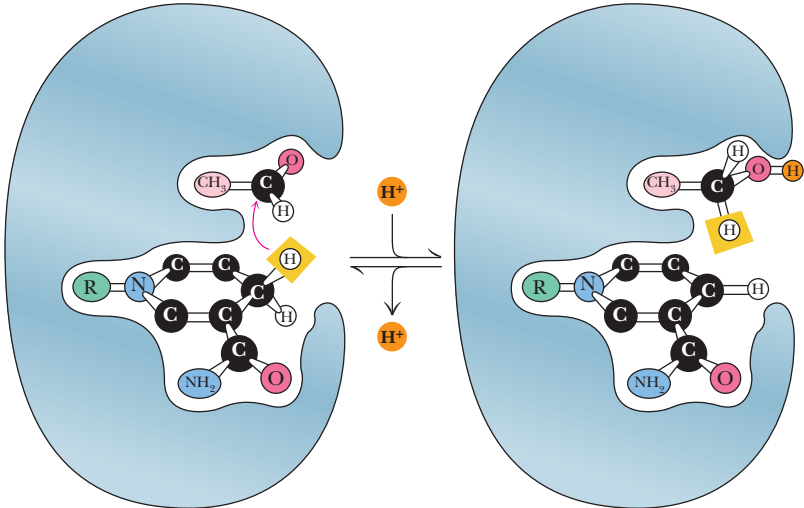
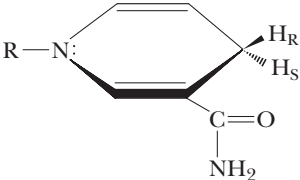
dehydrogenase, the citric acid cycle enzyme, transfers hydride to the H_{*R*} position of NADH, but glyceraldehyde-3-P dehydrogenase in the glycolytic pathway transfers hydride to the H_{*S*} position, as shown in the accompanying table. Dehydrogenases are stereospecific with respect to the substrates as well. Note that alcohol dehydrogenase removes hydrogen from the pro-*R* position of ethanol and transfers it to the pro-*R* position of NADH.



NAD(P)⁺-dependent enzymes are stereospecific. Malate dehydrogenase, for example, transfers a hydride to the pro-*R* position of NADH, whereas glyceraldehyde-3-phosphate dehydrogenase transfers a hydride to the pro-*S* position of the nicotinamide. Alcohol dehydrogenase removes a hydride from the pro-*R* position of ethanol and transfers it to the pro-*R* position of NADH.

Steric Specificity for NAD of Various Pyridine Nucleotide-Linked Enzymes

Dehydrogenase	Source	Steric Specificity
Alcohol (with ethanol)	Yeast, <i>Pseudomonas</i> , liver, wheat germ	H _R
Alcohol (with isopropyl alcohol)	Yeast	
Acetaldehyde	Liver	
L-Lactate	Heart muscle, <i>Lactobacillus</i>	
L-Malate	Pig heart, wheat germ	
D-Glycerate	Spinach	
Dihydroorotate	<i>Zymobacterium oroticum</i>	
α-Glycerophosphate	Muscle	H _S
Glyceraldehyde-3-P	Yeast, muscle	
L-Glutamate	Liver	
D-Glucose	Liver	
β-Hydroxysteroid	<i>Pseudomonas</i>	
NADH cytochrome c reductase	Rat liver mitochondria, pig heart	
NADPH transhydrogenase	<i>Pseudomonas</i>	
NADH diaphorase	Pig heart	
L-β-Hydroxybutyryl-CoA	Heart muscle	



The stereospecificity of hydride transfer in dehydrogenases is a consequence of the asymmetric nature of the active site.

Adapted from Kaplan, N. O., 1960. In *The Enzymes*, vol. 3, p. 115, edited by Boyer, Lardy, and Myrbäck. New York: Academic Press.

EXAMPLE

A typical intramitochondrial concentration of malate is 0.22 mM. If the $[NAD^+]/[NADH]$ ratio in mitochondria is 20 and if the malate dehydrogenase reaction is at equilibrium, calculate the intramitochondrial concentration of oxaloacetate at 25°C.

SOLUTION

For the malate dehydrogenase reaction,



with the value of $\Delta G^{\circ'}$ being +30 kJ/mol. Then

$$\begin{aligned} \Delta G^{\circ'} &= -RT \ln K_{eq} \\ &= -(8.314 \text{ J/mol}\cdot\text{K}) (298) \ln \left(\frac{[1]x}{[20][2.2 \times 10^{-4}]} \right) \\ \frac{-30,000 \text{ J/mol}}{2478 \text{ J/mol}} &= \ln (x/4.4 \times 10^{-3}) \\ -12.1 &= \ln (x/4.4 \times 10^{-3}) \\ x &= (5.6 \times 10^{-6})(4.4 \times 10^{-3}) \\ x &= [\text{oxaloacetate}] = 0.024 \mu\text{M} \end{aligned}$$

Malate dehydrogenase is structurally and functionally similar to other dehydrogenases, notably lactate dehydrogenase (Figure 20.20). Both consist of alternating β -sheet and α -helical segments. Binding of NAD^+ causes a conformational change in the 20-residue segment that connects the D and E strands of the β -sheet. The change is triggered by an interaction between the adenosine phosphate moiety of NAD^+ and an arginine residue in this loop region. Such a conformational change is consistent with an ordered single-displacement mechanism for NAD^+ -dependent dehydrogenases (Chapter 14).

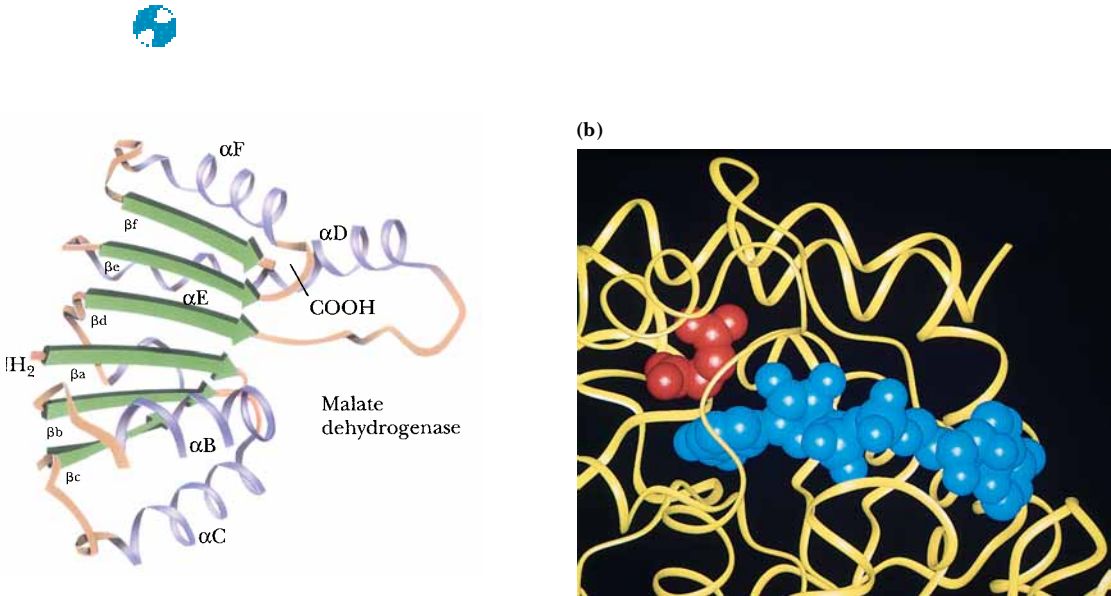
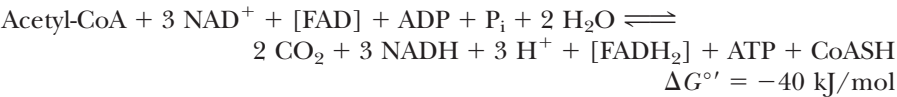


FIGURE 20.20 • (a) The structure of malate dehydrogenase. (b) The active site of malate dehydrogenase. Malate is shown in red; NAD^+ is blue.

20.12 • A Summary of the Cycle

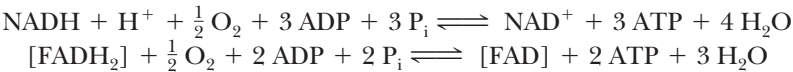
The net reaction accomplished by the TCA cycle, as follows, shows two molecules of CO₂, one ATP, and four reduced coenzymes produced per acetate group oxidized. The cycle is exergonic, with a net Δ*G*^{o'} for one pass around the cycle of approximately −40 kJ/mol. Table 20.1 compares the Δ*G*^{o'} values for the individual reactions with the overall Δ*G*^{o'} for the net reaction.



Glucose metabolized via glycolysis produces two molecules of pyruvate and thus two molecules of acetyl-CoA, which can enter the TCA cycle. Combining glycolysis and the TCA cycle gives the net reaction shown:



All six carbons of glucose are liberated as CO₂, and a total of four molecules of ATP are formed thus far in substrate-level phosphorylations. The 12 reduced coenzymes produced up to this point can eventually produce a maximum of 34 molecules of ATP in the electron transport and oxidative phosphorylation pathways. A stoichiometric relationship for these subsequent processes is



Thus, a total of 3 ATP per NADH and 2 ATP per FADH₂ may be produced through the processes of electron transport and oxidative phosphorylation.

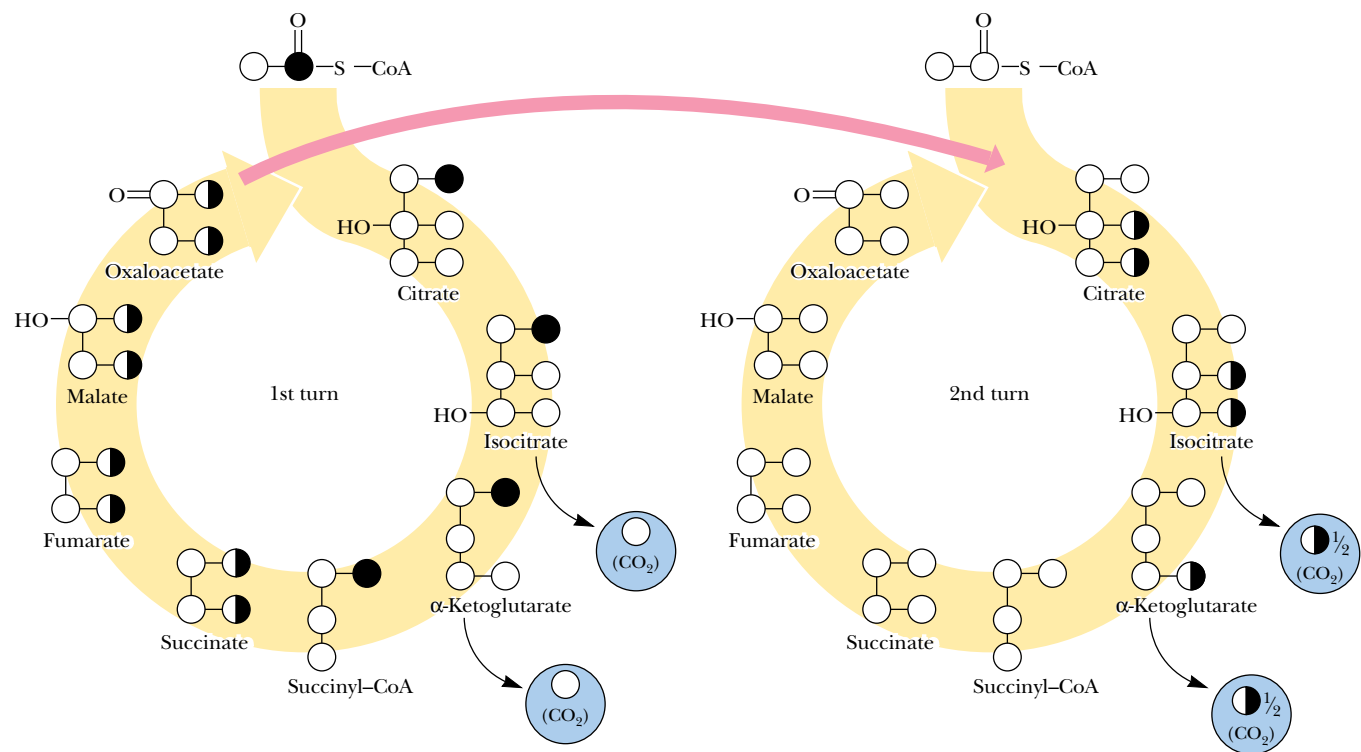
The Fate of the Carbon Atoms of Acetyl-CoA in the TCA Cycle

It is instructive to consider how the carbon atoms of a given acetate group are routed through several turns of the TCA cycle. As shown in Figure 20.21, neither of the carbon atoms of a labeled acetate unit is lost as CO₂ in the first turn of the cycle. The CO₂ evolved in any turn of the cycle derives from the carboxyl groups of the oxaloacetate acceptor (from the previous turn), not from incoming acetyl-CoA. On the other hand, succinate labeled on one end from the original labeled acetate forms two different labeled oxaloacetates. The carbonyl carbon of acetyl-CoA is evenly distributed between the two carboxyl carbons of oxaloacetate, and the labeled methyl carbon of incoming acetyl-CoA ends up evenly distributed between the methylene and carbonyl carbons of oxaloacetate.

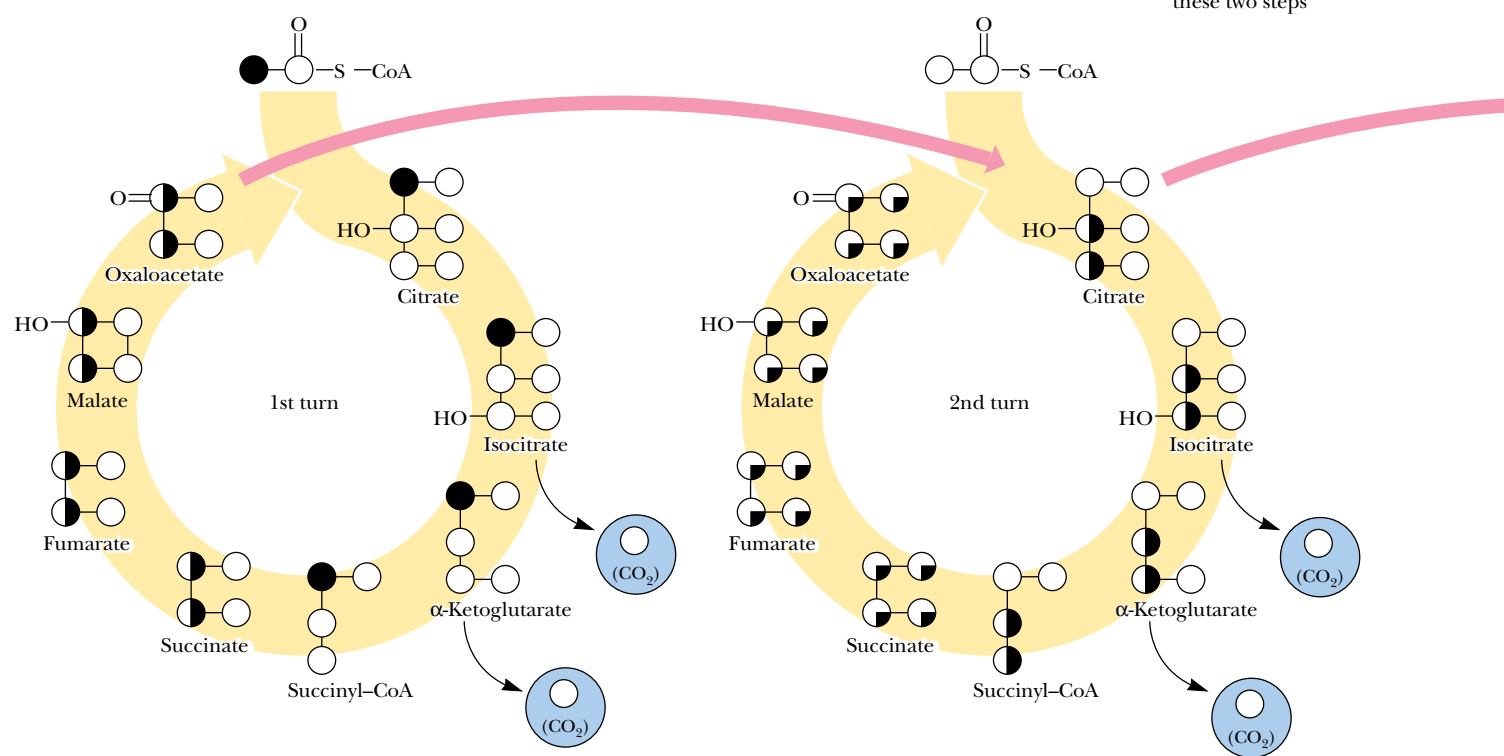
When these labeled oxaloacetates enter a second turn of the cycle, both of the carboxyl carbons are lost as CO₂, but the methylene and carbonyl carbons survive through the second turn. Thus, the methyl carbon of a labeled acetyl-CoA survives two full turns of the cycle. In the third turn of the cycle, one-half of the carbon from the original methyl group of acetyl-CoA has become one of the carboxyl carbons of oxaloacetate and is thus lost as CO₂. In the fourth turn of the cycle, further “scrambling” results in loss of half of the remaining labeled carbon (one-fourth of the original methyl carbon label of acetyl-CoA), and so on.

It can be seen that the carbonyl and methyl carbons of labeled acetyl-CoA have very different fates in the TCA cycle. The carbonyl carbon survives the first turn intact but is completely lost in the second turn. The methyl carbon

(a) Fate of the carboxyl carbon of acetate unit



(b) Fate of methyl carbon of acetate unit



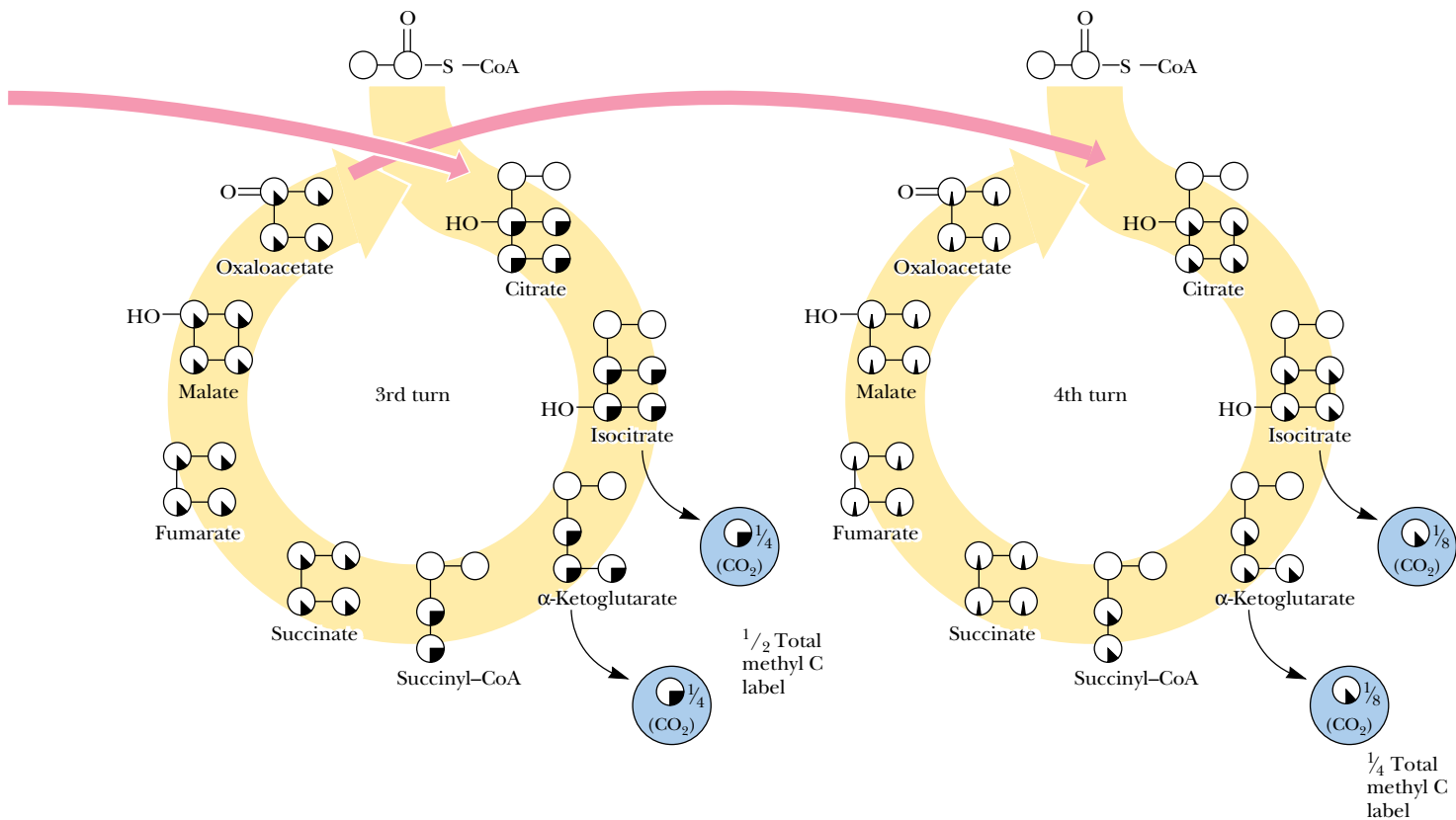
survives two full turns, then undergoes a 50% loss through each succeeding turn of the cycle.

It is worth noting that the carbon–carbon bond cleaved in the TCA pathway entered as an acetate unit in the previous turn of the cycle. Thus, the oxidative decarboxylations that cleave this bond are just a cleverly disguised acetate C—C cleavage and oxidation.

20.13 • The TCA Cycle Provides Intermediates for Biosynthetic Pathways

Until now we have viewed the TCA cycle as a catabolic process because it oxidizes acetate units to CO₂ and converts the liberated energy to ATP and reduced coenzymes. The TCA cycle is, after all, the end point for breakdown of food materials, at least in terms of carbon turnover. However, as shown in Figure 20.22, four-, five-, and six-carbon species produced in the TCA cycle also fuel a variety of **biosynthetic processes**. α-Ketoglutarate, succinyl-CoA, fumarate, and oxaloacetate are all precursors of important cellular species. (In order to par-

FIGURE 20.21 • The fate of the carbon atoms of acetate in successive TCA cycles. (a) The carbonyl carbon of acetyl-CoA is fully retained through one turn of the cycle but is lost completely in a second turn of the cycle. (b) The methyl carbon of a labeled acetyl-CoA survives two full turns of the cycle but becomes equally distributed among the four carbons of oxaloacetate by the end of the second turn. In each subsequent turn of the cycle, one-half of this carbon (the original labeled methyl group) is lost.



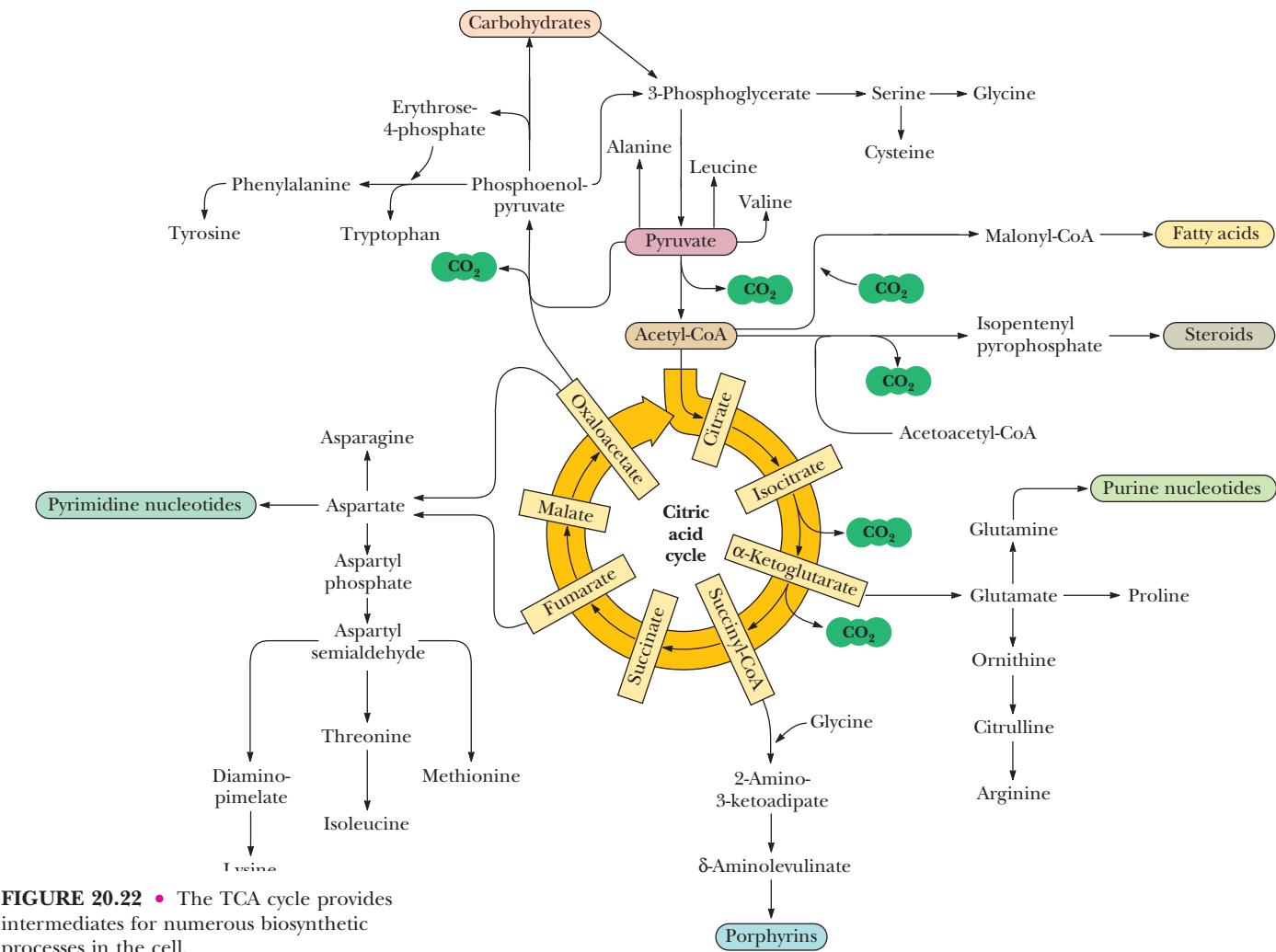


FIGURE 20.22 • The TCA cycle provides intermediates for numerous biosynthetic processes in the cell.

ticipate in eukaryotic biosynthetic processes, however, they must first be transported out of the mitochondria.) A transamination reaction converts α-ketoglutarate directly to glutamate, which can then serve as a versatile precursor for proline, arginine, and glutamine (as described in Chapter 26). Succinyl-CoA provides most of the carbon atoms of the porphyrins. Oxaloacetate can be transaminated to produce aspartate. Aspartic acid itself is a precursor of the pyrimidine nucleotides and, in addition, is a key precursor for the synthesis of asparagine, methionine, lysine, threonine, and isoleucine. Oxaloacetate can also be decarboxylated to yield PEP, which is a key element of several pathways, namely (1) synthesis (in plants and microorganisms) of the aromatic amino acids phenylalanine, tyrosine, and tryptophan; (2) formation of 3-phosphoglycerate and conversion to the amino acids serine, glycine, and cysteine; and (3) *gluconeogenesis*, which, as we will see in Chapter 23, is the pathway that synthesizes new glucose and many other carbohydrates.

Finally, citrate can be exported from the mitochondria and then broken down by **ATP-citrate lyase** to yield oxaloacetate and acetyl-CoA, a precursor of fatty acids (Figure 20.23). Oxaloacetate produced in this reaction is rapidly reduced to malate, which can then be processed in either of two ways: it may be transported into mitochondria, where it is reoxidized to oxaloacetate, or it may be oxidatively decarboxylated to pyruvate by **malic enzyme**, with subse-

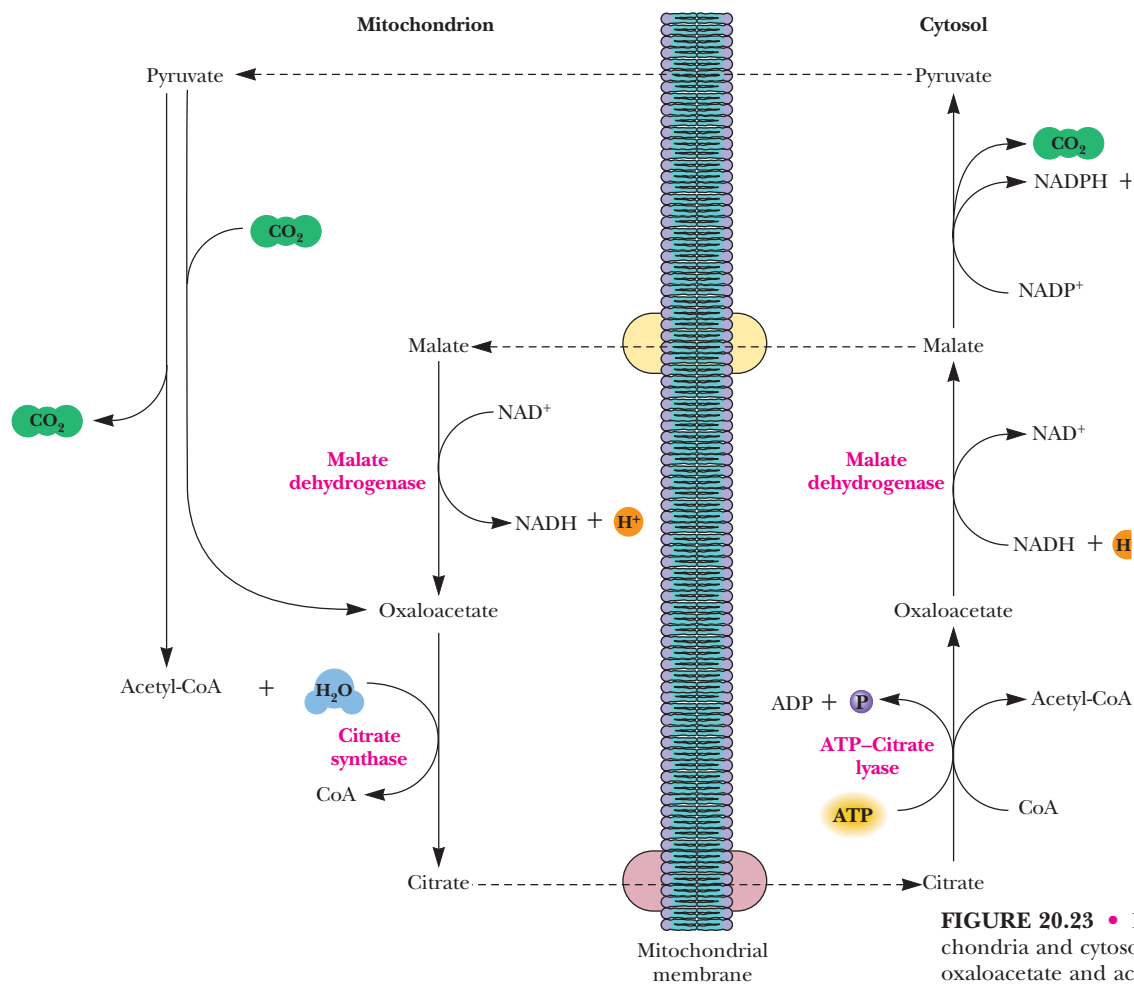


FIGURE 20.23 • Export of citrate from mitochondria and cytosolic breakdown produces oxaloacetate and acetyl-CoA. Oxaloacetate is recycled to malate or pyruvate, which re-enters the mitochondria. This cycle provides acetyl-CoA for fatty acid synthesis in the cytosol.

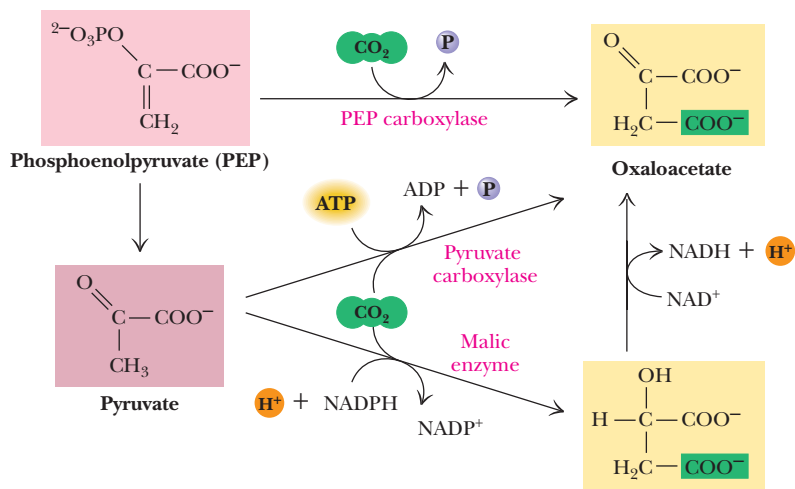
quent mitochondrial uptake of pyruvate. This cycle permits citrate to provide acetyl-CoA for biosynthetic processes, with return of the malate and pyruvate by-products to the mitochondria.

20.14 • The Anaplerotic, or “Filling Up,” Reactions

In a sort of reciprocal arrangement, the cell also feeds many intermediates back into the TCA cycle from other reactions. Since such reactions replenish the TCA cycle intermediates, Hans Kornberg proposed that they be called **anaplerotic reactions** (literally, the “filling up” reactions). Thus, **PEP carboxylase** and **pyruvate carboxylase** synthesize oxaloacetate from pyruvate (Figure 20.24).

Pyruvate carboxylase is the most important of the anaplerotic reactions. It exists in the mitochondria of animal cells but not in plants, and it provides a direct link between glycolysis and the TCA cycle. The enzyme is tetrameric and contains covalently bound biotin and an Mg²⁺ site on each subunit. (It is examined in greater detail in our discussion of gluconeogenesis in Chapter 23.) Pyruvate carboxylase has an absolute allosteric requirement for acetyl-CoA. Thus, when acetyl-CoA levels exceed the oxaloacetate supply, allosteric activation of pyruvate carboxylase by acetyl-CoA raises oxaloacetate levels, so that the excess acetyl-CoA can enter the TCA cycle.

FIGURE 20.24 • Phosphoenolpyruvate (PEP) carboxylase, pyruvate carboxylase, and malic enzyme catalyze anaplerotic reactions, replenishing TCA cycle intermediates.



DEEPER LOOK

Fool’s Gold and the Reductive Citric Acid Cycle—The First Metabolic Pathway?

How did life arise on the planet Earth? It was once supposed that a reducing atmosphere, together with random synthesis of organic compounds, gave rise to a prebiotic “soup,” in which the first living things appeared. However, certain key compounds, such as arginine, lysine, and histidine, the straight-chain fatty acids, porphyrins, and essential coenzymes, have not been convincingly synthesized under simulated prebiotic conditions. This and other problems have led researchers to consider other models for the evolution of life.

One of these alternate models, postulated by Günter Wächtershäuser, involves an archaic version of the TCA cycle running in the reverse (reductive) direction. Reversal of the TCA cycle results in assimilation of CO_2 and fixation of carbon as shown. For each turn of the reversed cycle, two carbons are fixed in the formation of isocitrate and two more are fixed in the reductive transformation of acetyl-CoA to oxaloacetate. Thus, for every succinate that enters the reversed cycle, two succinates are returned, making the cycle highly autocatalytic. Because TCA cycle intermediates are involved in many biosynthetic pathways (see Section 20.13), a reversed TCA cycle would be a bountiful and broad source of metabolic substrates.

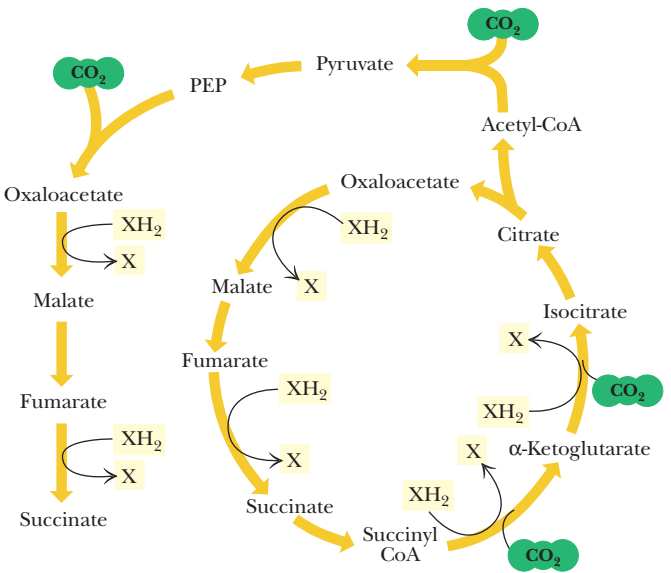
A reversed, reductive TCA cycle would require energy input to drive it. What might have been the thermodynamic driving force for such a cycle? Wächtershäuser hypothesizes that the anaerobic reaction of FeS and H_2S to form insoluble FeS_2 (pyrite, also known as fool’s gold) in the prebiotic milieu could have been the driving reaction:



This reaction is highly exergonic, with a standard-state free energy change (ΔG°) of -38 kJ/mol . Under the conditions that might have existed in a prebiotic world, this reaction would have been sufficiently exergonic to drive the reductive steps of a reversed TCA cycle. In addition, in an H_2S -rich prebiotic environment, organic compounds would have been in equilibrium with their thio-organic counterparts. High-energy thioesters formed in this

way may have played key roles in the energetics of early metabolic pathways.

Wächtershäuser has also suggested that early metabolic processes first occurred on the surface of pyrite and other related mineral materials. The iron–sulfur chemistry that prevailed on these mineral surfaces may have influenced the evolution of the iron–sulfur proteins that control and catalyze many reactions in modern pathways (including the succinate dehydrogenase and aconitase reactions of the TCA cycle).



A Reductive, Reversed TCA Cycle

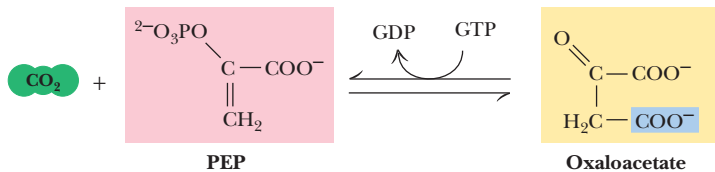


FIGURE 20.25 • The phosphoenolpyruvate carboxykinase reaction.

PEP carboxylase occurs in yeast, bacteria, and higher plants, but not in animals. The enzyme is specifically inhibited by aspartate, which is produced by transamination of oxaloacetate. Thus, organisms utilizing this enzyme control aspartate production by regulation of PEP carboxylase. Malic enzyme is found in the cytosol or mitochondria of many animal and plant cells and is an NADPH-dependent enzyme.

It is worth noting that the reaction catalyzed by **PEP carboxykinase** (Figure 20.25) could also function as an anaplerotic reaction, were it not for the particular properties of the enzyme. CO_2 binds weakly to PEP carboxykinase, whereas oxaloacetate binds very tightly ($K_D = 2 \times 10^{-6} \text{ M}$), and, as a result, the enzyme favors formation of PEP from oxaloacetate.

The catabolism of amino acids provides pyruvate, acetyl-CoA, oxaloacetate, fumarate, α -ketoglutarate, and succinate, all of which may be oxidized by the TCA cycle. In this way, proteins may serve as excellent sources of nutrient energy, as seen in Chapter 26.

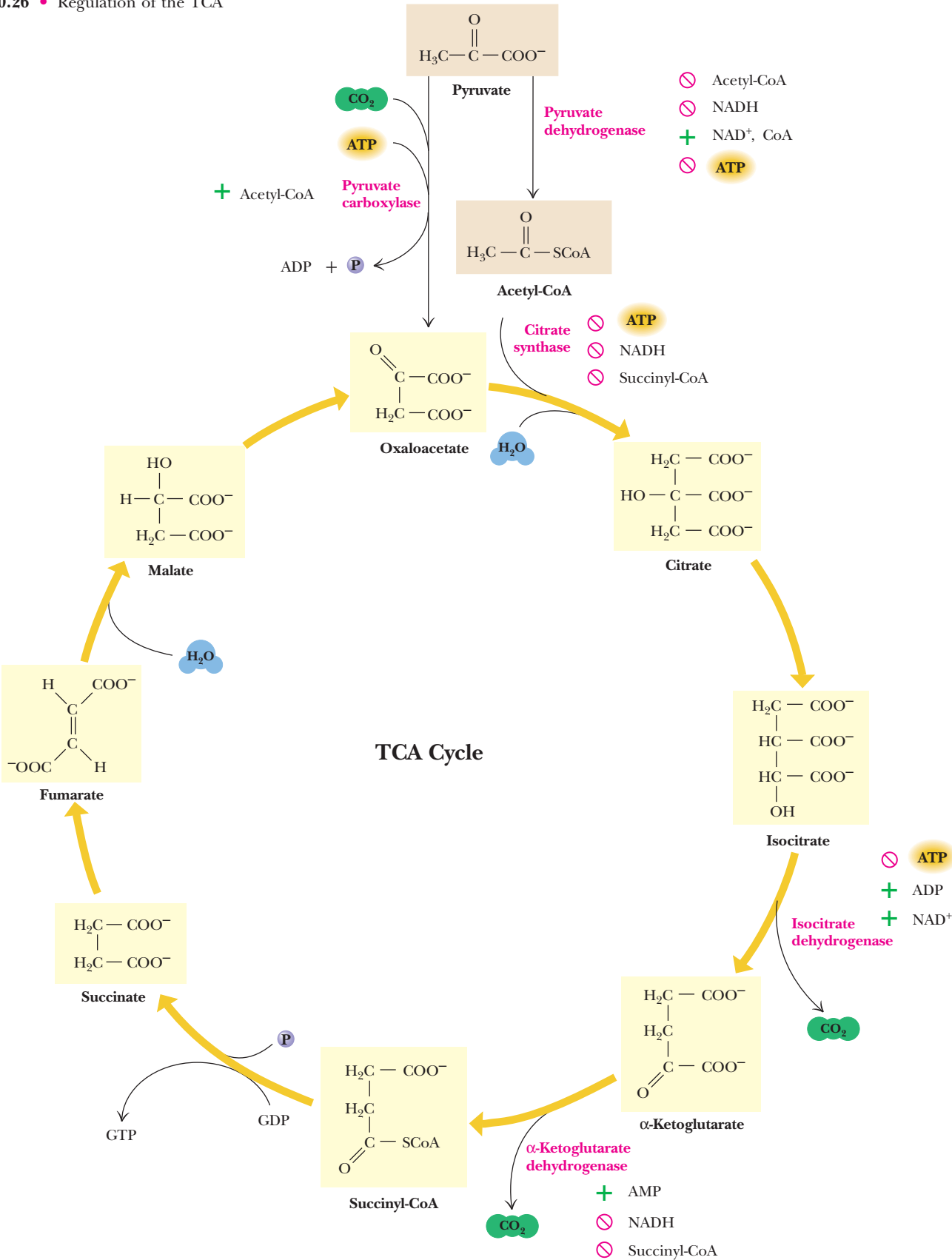
20.15 • Regulation of the TCA Cycle

Situated as it is between glycolysis and the electron transport chain, the TCA cycle must be carefully controlled by the cell. If the cycle were permitted to run unchecked, large amounts of metabolic energy could be wasted in overproduction of reduced coenzymes and ATP; conversely, if it ran too slowly, ATP would not be produced rapidly enough to satisfy the needs of the cell. Also, as just seen, the TCA cycle is an important source of precursors for biosynthetic processes and must be able to provide them as needed.

What are the sites of regulation in the TCA cycle? Based upon our experience with glycolysis (Figure 19.31), we might anticipate that some of the reactions of the TCA cycle would operate near equilibrium under cellular conditions (with $\Delta G \approx 0$), whereas others—the sites of regulation—would be characterized by large, negative ΔG values. Estimates for the values of ΔG in mitochondria, based on mitochondrial concentrations of metabolites, are summarized in Table 20.1. Three reactions of the cycle—citrate synthase, isocitrate dehydrogenase, and α -ketoglutarate dehydrogenase—operate with large, negative ΔG values under mitochondrial conditions and are thus the primary sites of regulation in the cycle.

The regulatory actions that control the TCA cycle are shown in Figure 20.26. As one might expect, the principal regulatory “signals” are the concentrations of acetyl-CoA, ATP, NAD^+ , and NADH, with additional effects provided by several other metabolites. The main sites of regulation are pyruvate dehydrogenase, citrate synthase, isocitrate dehydrogenase, and α -ketoglutarate dehydrogenase. All of these enzymes are inhibited by NADH, so that when the cell has produced all the NADH that can conveniently be turned into ATP, the cycle shuts down. For similar reasons, ATP is an inhibitor of pyruvate dehydrogenase and isocitrate dehydrogenase. The TCA cycle is turned on, however, when either the ADP/ATP or NAD^+/NADH ratio is high, an indication that the cell has run low on ATP or NADH. Regulation of the TCA cycle by NADH,

FIGURE 20.26 • Regulation of the TCA cycle.



NAD^+ , ATP, and ADP thus reflects the energy status of the cell. On the other hand, succinyl-CoA is an *intracycle regulator*, inhibiting citrate synthase and α -ketoglutarate dehydrogenase. Acetyl-CoA acts as a signal to the TCA cycle that glycolysis or fatty acid breakdown is producing two-carbon units. Acetyl-CoA activates pyruvate carboxylase, the anaplerotic reaction that provides oxaloacetate, the acceptor for increased flux of acetyl-CoA into the TCA cycle.

Regulation of Pyruvate Dehydrogenase

As we shall see in Chapter 23, most organisms can synthesize sugars such as glucose from pyruvate. However, animals cannot synthesize glucose from acetyl-CoA. For this reason, the pyruvate dehydrogenase complex, which converts pyruvate to acetyl-CoA, plays a pivotal role in metabolism. Conversion to acetyl-CoA commits nutrient carbon atoms either to oxidation in the TCA cycle or to fatty acid synthesis (see Chapter 25). Because this choice is so crucial to the organism, pyruvate dehydrogenase is a carefully regulated enzyme. It is subject to product inhibition and is further regulated by nucleotides. Finally, activity of pyruvate dehydrogenase is regulated by phosphorylation and dephosphorylation of the enzyme complex itself.

High levels of either product, acetyl-CoA or NADH, allosterically inhibit the pyruvate dehydrogenase complex. Acetyl-CoA specifically blocks dihydrolipoyl transacetylase, and NADH acts on dihydrolipoyl dehydrogenase. The mammalian pyruvate dehydrogenase is also regulated by covalent modifications. As shown in Figure 20.27, a Mg^{2+} -dependent **pyruvate dehydrogenase kinase** is associated with the enzyme in mammals. This kinase is allosterically activated by NADH and acetyl-CoA, and when levels of these metabolites rise in the mitochondrion, they stimulate phosphorylation of a serine residue on the pyruvate dehydrogenase subunit, blocking the first step of the pyruvate dehydrogenase reaction, the decarboxylation of pyruvate. Inhibition of the dehydrogenase in this manner eventually lowers the levels of NADH and acetyl-CoA in the matrix of the mitochondrion. Reactivation of the enzyme is carried out by **pyruvate dehydrogenase phosphatase**, a Ca^{2+} -activated enzyme that binds to the dehydrogenase complex and hydrolyzes the phosphoserine moiety on the dehydrogenase subunit. At low ratios of NADH to NAD^+ and low acetyl-CoA levels, the phosphatase maintains the dehydrogenase in an activated state, but a high level of acetyl-CoA or NADH once again activates the kinase and leads to the inhibition of the dehydrogenase. Insulin and Ca^{2+} ions activate dephosphorylation, and pyruvate inhibits the phosphorylation reaction.

Pyruvate dehydrogenase is also sensitive to the energy status of the cell. AMP activates pyruvate dehydrogenase, whereas GTP inhibits it. High levels of AMP are a sign that the cell may become energy-poor. Activation of pyruvate dehydrogenase under such conditions commits pyruvate to energy production.

Regulation of Isocitrate Dehydrogenase

The mechanism of regulation of isocitrate dehydrogenase is in some respects the reverse of pyruvate dehydrogenase. The mammalian isocitrate dehydrogenase is subject only to allosteric activation by ADP and NAD^+ and to inhibition by ATP and NADH. Thus, high NAD^+/NADH and ADP/ATP ratios stimulate isocitrate dehydrogenase and TCA cycle activity. The *Escherichia coli* enzyme, on the other hand, is regulated by covalent modification. Serine residues on each subunit of the dimeric enzyme are phosphorylated by a protein kinase, causing inhibition of the isocitrate dehydrogenase activity. Activity is restored by the action of a specific phosphatase. When TCA cycle and glycolytic intermediates—such as isocitrate, 3-phosphoglycerate, pyruvate, PEP, and oxaloacetate—are high, the kinase is inhibited, the phosphatase is acti-

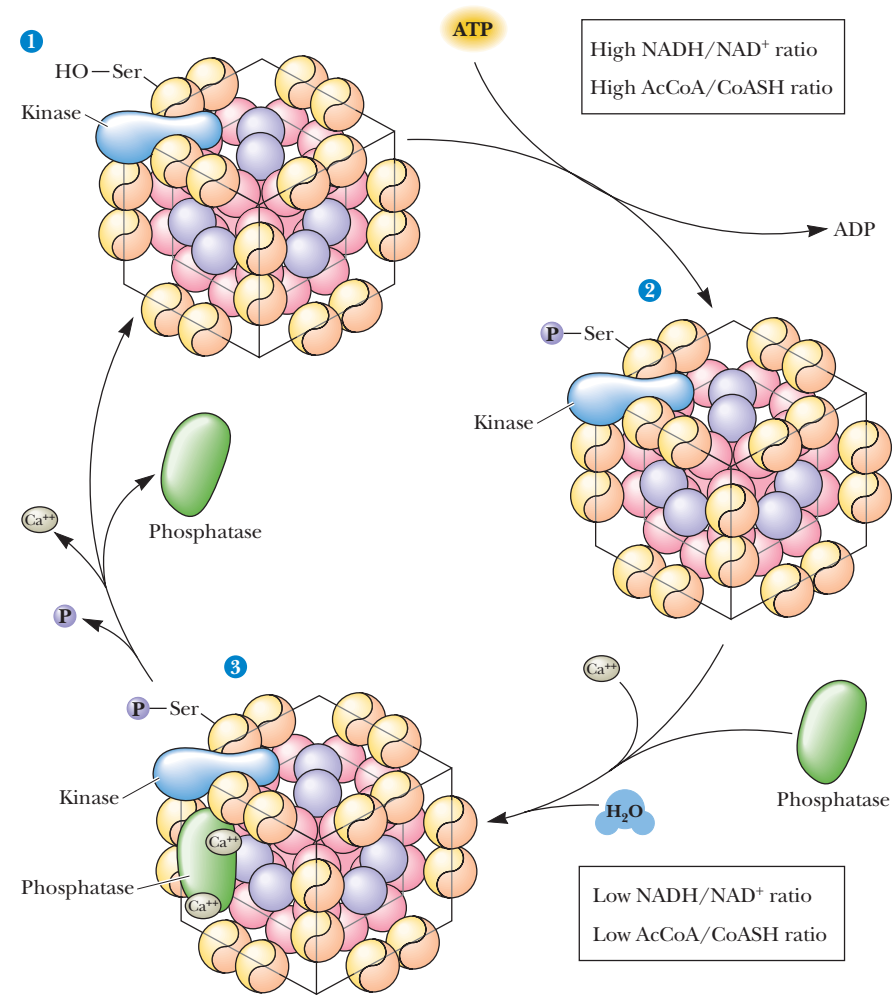


FIGURE 20.27 • Regulation of the pyruvate dehydrogenase reaction.

vated, and the TCA cycle operates normally. When levels of these intermediates fall, the kinase is activated, isocitrate dehydrogenase is inhibited, and isocitrate is diverted to the *glyoxylate pathway*, as explained in the next section.

It may seem surprising that isocitrate dehydrogenase is strongly regulated, because it is not an apparent branch point within the TCA cycle. However, the citrate/isocitrate ratio controls the rate of production of cytosolic acetyl-CoA, because acetyl-CoA in the cytosol is derived from citrate exported from the mitochondrion. (Breakdown of cytosolic citrate produces oxaloacetate and acetyl-CoA, which can be used in a variety of biosynthetic processes.) Thus, isocitrate dehydrogenase activity in the mitochondrion favors catabolic TCA cycle activity over anabolic utilization of acetyl-CoA in the cytosol.

20.16 • The Glyoxylate Cycle of Plants and Bacteria

Plants (particularly seedlings, which cannot yet accomplish efficient photosynthesis), as well as some bacteria and algae, can use acetate as the *only* source of carbon for all the carbon compounds they produce. Although we saw that the TCA cycle can supply intermediates for some biosynthetic processes, the

cycle gives off 2 CO₂ for every two-carbon acetate group that enters and cannot effect the *net synthesis* of TCA cycle intermediates. Thus, it would not be possible for the cycle to produce the massive amounts of biosynthetic intermediates needed for acetate-based growth unless alternative reactions were available. In essence, the TCA cycle is geared primarily to energy production, and it “wastes” carbon units by giving off CO₂. Modification of the cycle to support acetate-based growth would require eliminating the CO₂-producing reactions and enhancing the net production of four-carbon units (i.e., oxaloacetate). Plants and bacteria employ a modification of the TCA cycle called the **glyoxylate cycle** to produce four-carbon dicarboxylic acids (and eventually even sugars) from two-carbon acetate units. The glyoxylate cycle bypasses the two oxidative decarboxylations of the TCA cycle, and instead routes isocitrate through the **isocitrate lyase** and **malate synthase** reactions (Figure 20.28). *Glyoxylate* produced by isocitrate lyase reacts with a second molecule of acetyl-

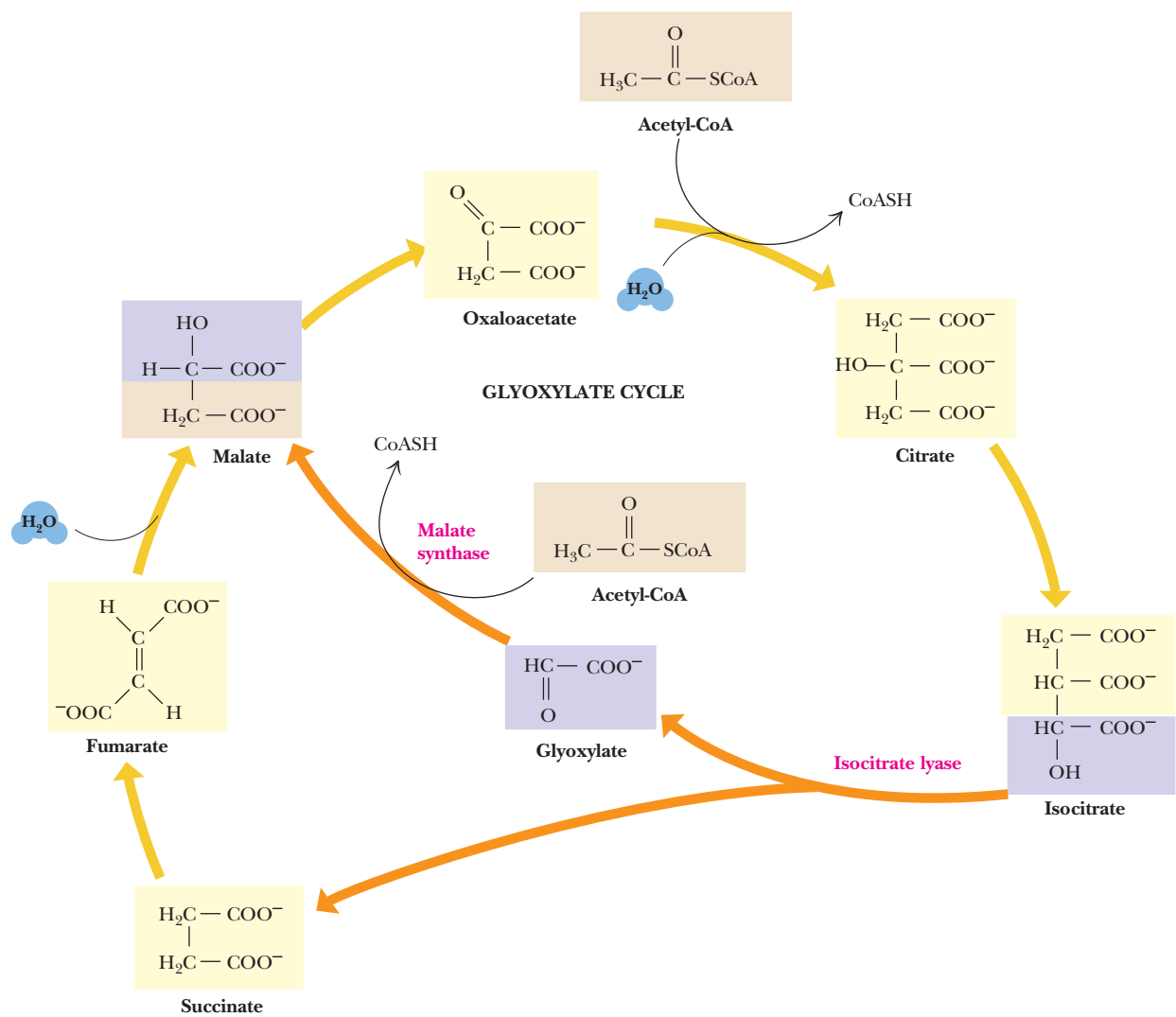


FIGURE 20.28 • The glyoxylate cycle. The first two steps are identical to TCA cycle reactions. The third step bypasses the CO₂-evolving steps of the TCA cycle to produce succinate and glyoxylate. The malate synthase reaction forms malate from glyoxylate and another acetyl-CoA. The result is that one turn of the cycle consumes one oxaloacetate and two acetyl-CoA molecules but produces two molecules of oxaloacetate. The net for this cycle is one oxaloacetate from two acetyl-CoA molecules.

CoA to form L-malate. The net effect is to conserve carbon units, using two acetyl-CoA molecules per cycle to generate oxaloacetate. Some of this is converted to PEP and then to glucose by pathways discussed in Chapter 23.

The Glyoxylate Cycle Operates in Specialized Organelles

The enzymes of the glyoxylate cycle in plants are contained in **glyoxysomes**, organelles devoted to this cycle. Yeast and algae carry out the glyoxylate cycle in the cytoplasm. The enzymes common to both the TCA and glyoxylate pathways exist as isozymes, with spatially and functionally distinct enzymes operating independently in the two cycles.

Isocitrate Lyase Short-Circuits the TCA Cycle by Producing Glyoxylate and Succinate

The **isocitrate lyase** reaction (Figure 20.29) produces succinate, a four-carbon product of the cycle, as well as glyoxylate, which can then combine with a second molecule of acetyl-CoA. Isocitrate lyase catalyzes an aldol cleavage and is similar to the reaction mediated by aldolase in glycolysis. The **malate synthase** reaction (Figure 20.30), a Claisen condensation of acetyl-CoA with the aldehyde of glyoxylate to yield malate, is quite similar to the citrate synthase reaction. Compared with the TCA cycle, the glyoxylate cycle (a) contains only five steps (as opposed to eight), (b) lacks the CO₂-liberating reactions, (c) consumes two molecules of acetyl-CoA per cycle, and (d) produces four-carbon units (oxaloacetate) as opposed to one-carbon units.

The Glyoxylate Cycle Helps Plants Grow in the Dark

The existence of the glyoxylate cycle explains how certain seeds grow underground (or in the dark), where photosynthesis is impossible. Many seeds (peanuts, soybeans, and castor beans, for example) are rich in lipids; and, as we see in Chapter 24, most organisms degrade the fatty acids of lipids to acetyl-CoA. Glyoxysomes form in seeds as germination begins, and the glyoxylate cycle uses the acetyl-CoA produced in fatty acid oxidation to provide large amounts of oxaloacetate and other intermediates for carbohydrate synthesis. Once the growing plant begins photosynthesis and can fix CO₂ to produce carbohydrates (see Chapter 22), the glyoxysomes disappear.

Glyoxysomes Must Borrow Three Reactions from Mitochondria

Glyoxysomes do not contain all the enzymes needed to run the glyoxylate cycle; succinate dehydrogenase, fumarase, and malate dehydrogenase are absent. Consequently, glyoxysomes must cooperate with mitochondria to run their cycle (Figure 20.31). Succinate travels from the glyoxysomes to the mitochondria, where it is converted to oxaloacetate. Transamination to aspartate follows

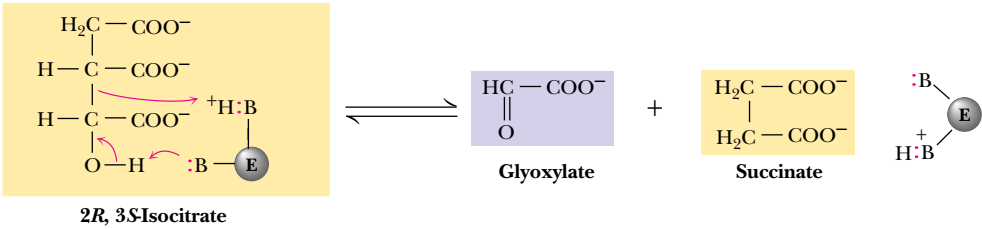


FIGURE 20.29 • The isocitrate lyase reaction.

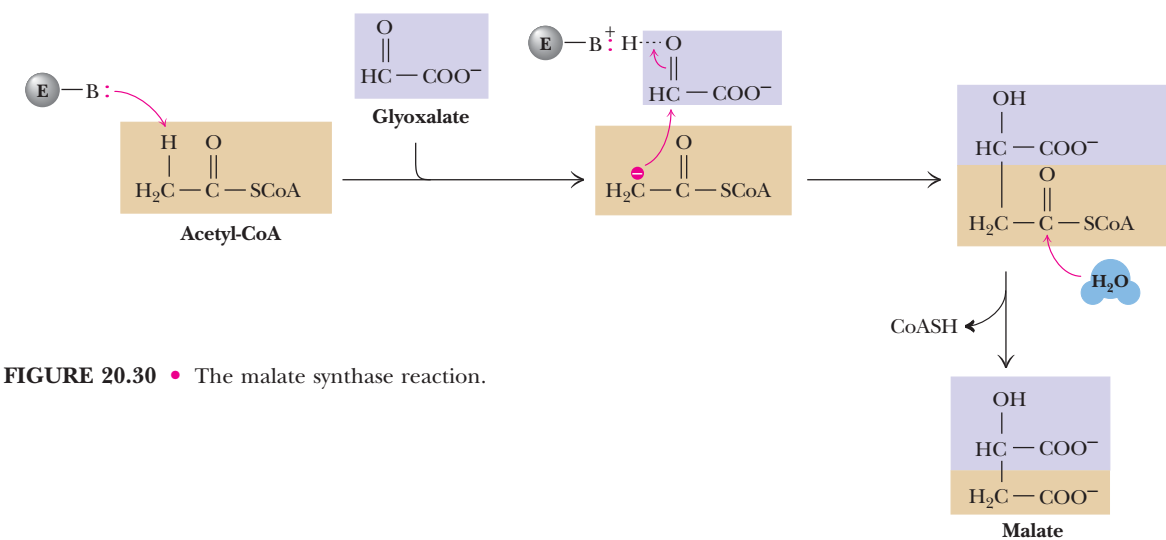


FIGURE 20.30 • The malate synthase reaction.

because oxaloacetate cannot be transported out of the mitochondria. Aspartate formed in this way then moves from the mitochondria back to the glyoxysomes, where a reverse transamination with α -ketoglutarate forms oxaloacetate, completing the shuttle. Finally, to balance the transaminations, glutamate shuttles from glyoxysomes to mitochondria.

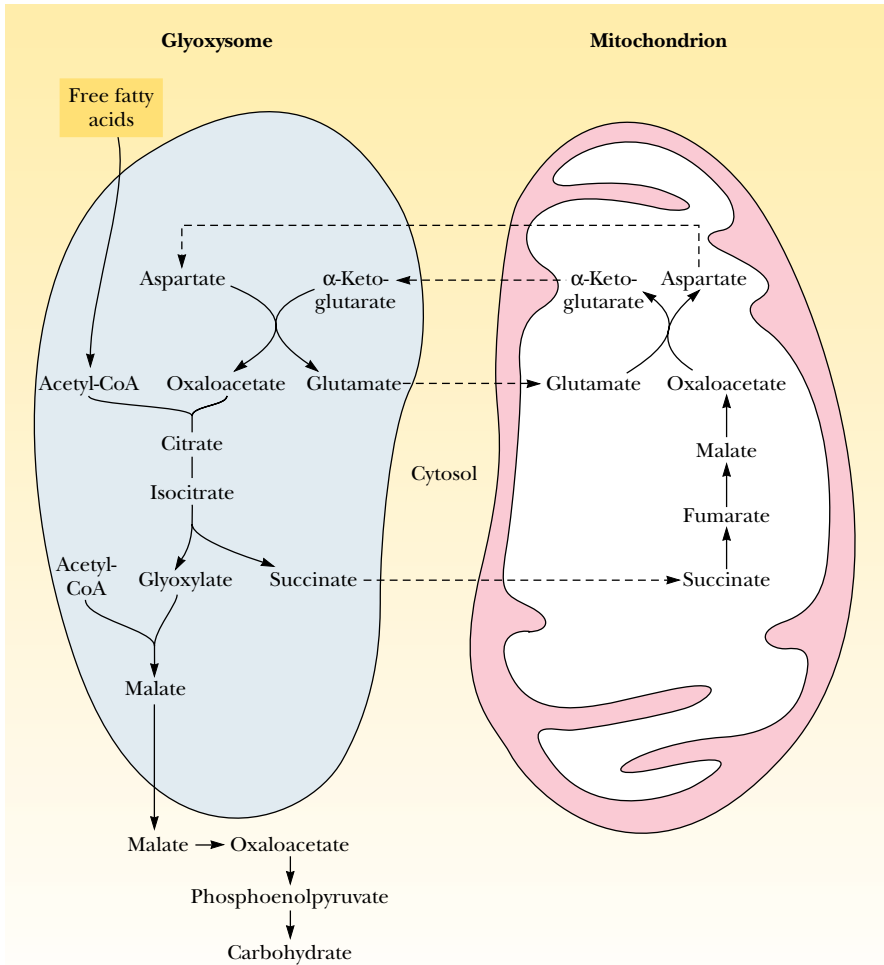


FIGURE 20.31 • Glyoxysomes lack three of the enzymes needed to run the glyoxylate cycle. Succinate dehydrogenase, fumarase, and malate dehydrogenase are all “borrowed” from the mitochondria in a shuttle in which succinate and glutamate are passed to the mitochondria, and α -ketoglutarate and aspartate are passed to the glyoxysome.

PROBLEMS

1. Describe the labeling pattern that would result from the introduction into the TCA cycle of glutamate labeled at C_γ with ¹⁴C.
2. Describe the effect on the TCA cycle of (a) increasing the concentration of NAD⁺, (b) reducing the concentration of ATP, and (c) increasing the concentration of isocitrate.
3. The serine residue of isocitrate dehydrogenase that is phosphorylated by protein kinase lies within the active site of the enzyme. This situation contrasts with most other examples of covalent modification by protein phosphorylation, where the phosphorylation occurs at a site remote from the active site. What direct effect do you think such active-site phosphorylation might have on the catalytic activity of isocitrate dehydrogenase? (See Barford, D., 1991. Molecular mechanisms for the control of enzymic activity by protein phosphorylation. *Biochimica et Biophysica Acta* **1133**:55–62.)
4. The first step of the α-ketoglutarate dehydrogenase reaction involves decarboxylation of the substrate and leaves a covalent TPP intermediate. Write a reasonable mechanism for this reaction.
5. In a tissue where the TCA cycle has been inhibited by fluoroacetate, what difference in the concentration of each TCA cycle metabolite would you expect, compared with a normal, uninhibited tissue?
6. On the basis of the description in Chapter 18 of the physical

- properties of FAD and FADH₂, suggest a method for the measurement of the enzyme activity of succinate dehydrogenase.
7. Starting with citrate, isocitrate, α-ketoglutarate, and succinate, state which of the individual carbons of the molecule undergo oxidation in the next step of the TCA cycle. Which molecules undergo a net oxidation?
 8. In addition to fluoroacetate, consider whether other analogs of TCA cycle metabolites or intermediates might be introduced to inhibit other, specific reactions of the cycle. Explain your reasoning.
 9. Based on the action of thiamine pyrophosphate in catalysis of the pyruvate dehydrogenase reaction, suggest a suitable chemical mechanism for the pyruvate decarboxylase reaction in yeast:



10. Aconitase catalyzes the citric acid cycle reaction:



The standard free energy change, Δ*G*^{o'}, for this reaction is +6.7 kJ/mol. However, the observed free energy change (Δ*G*) for this reaction in pig heart mitochondria is +0.8 kJ/mol. What is the ratio of [isocitrate]/[citrate] in these mitochondria? If [isocitrate] = 0.03 *mM*, what is [citrate]?

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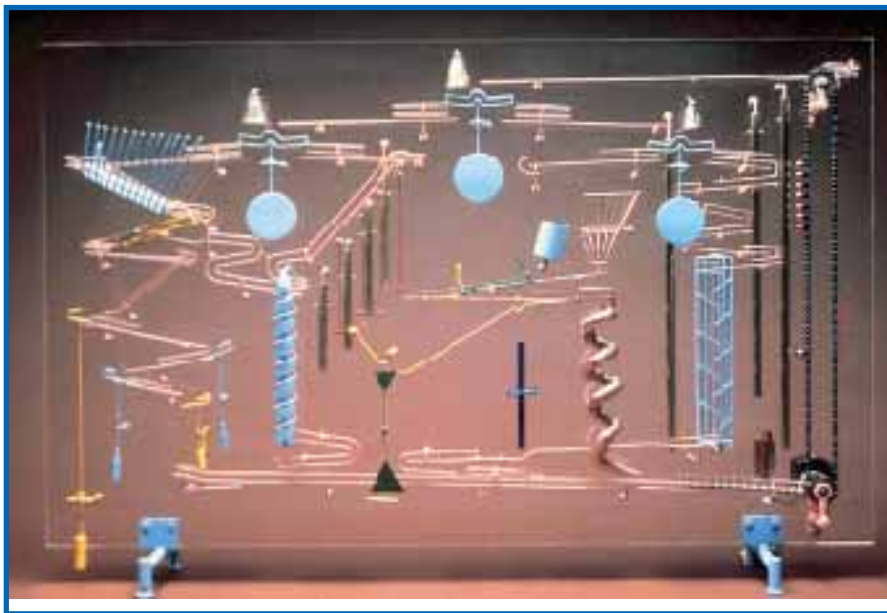
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Chapter 21

Electron Transport and Oxidative Phosphorylation



Wall Piece #IV (1985), a kinetic sculpture by George Rhoads. This complex mechanical art form can be viewed as a metaphor for the molecular apparatus underlying electron transport and ATP synthesis by oxidative phosphorylation. (1985 by George Rhoads)

Living cells save up metabolic energy predominantly in the form of fats and carbohydrates, and they “spend” this energy for biosynthesis, membrane transport, and movement. In both directions, energy is exchanged and transferred in the form of ATP. In Chapters 19 and 20 we saw that glycolysis and the TCA cycle convert some of the energy available from stored and dietary sugars directly to ATP. However, most of the metabolic energy that is obtainable from substrates entering glycolysis and the TCA cycle is funneled via oxidation–reduction reactions into NADH and reduced flavoproteins, the latter symbolized by [FADH₂]. We now embark on the discovery of how cells convert the stored metabolic energy of NADH and [FADH₂] into ATP.

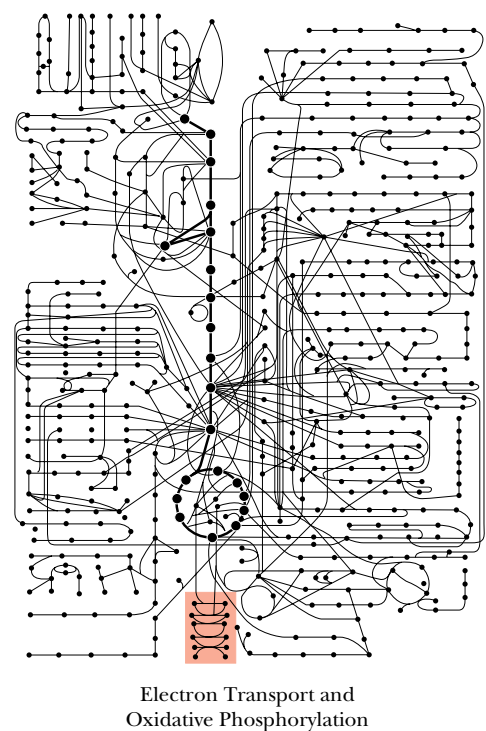
Whereas ATP made in glycolysis and the TCA cycle is the result of substrate-level phosphorylation, NADH-dependent ATP synthesis is the result of **oxidative phosphorylation**. Electrons stored in the form of the reduced coenzymes, NADH or [FADH₂], are passed through an elaborate and highly orga-

In all things of nature there is something of the marvelous.

ARISTOTLE (384–322 B.C.)

OUTLINE

- 21.1 • Electron Transport and Oxidative Phosphorylation Are Membrane-Associated Processes
- 21.2 • Reduction Potentials—An Accounting Device for Free Energy Changes in Redox Reactions
- 21.3 • The Electron Transport Chain—An Overview
- 21.4 • Complex I: NADH–Coenzyme Q Reductase
- 21.5 • Complex II: Succinate–Coenzyme Q Reductase
- 21.6 • Complex III: Coenzyme Q–Cytochrome *c* Reductase
- 21.7 • Complex IV: Cytochrome *c* Oxidase
- 21.8 • The Thermodynamic View of Chemiosmotic Coupling
- 21.9 • ATP Synthase
- 21.10 • Inhibitors of Oxidative Phosphorylation
- 21.11 • Uncouplers Disrupt the Coupling of Electron Transport and ATP Synthase
- 21.12 • ATP Exits the Mitochondria via an ATP–ADP Translocase
- 21.13 • What Is the P/O Ratio for Electron Transport and Oxidative Phosphorylation?
- 21.14 • Shuttle Systems Feed the Electrons of Cytosolic NADH into Electron Transport

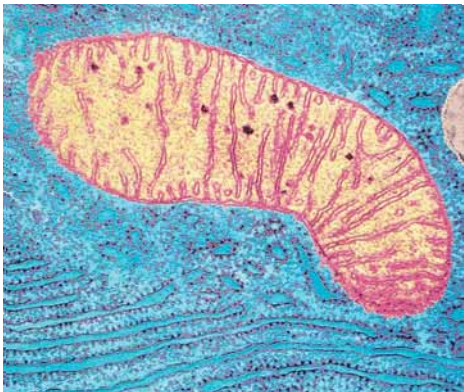


nized chain of proteins and coenzymes, the so-called **electron transport chain**, finally reaching O_2 (molecular oxygen), the terminal electron acceptor. Each component of the chain can exist in (at least) two oxidation states, and each component is successively reduced and reoxidized as electrons move through the chain from NADH (or $[\text{FADH}_2]$) to O_2 . In the course of electron transport, a proton gradient is established across the inner mitochondrial membrane. It is the energy of this proton gradient that drives ATP synthesis.

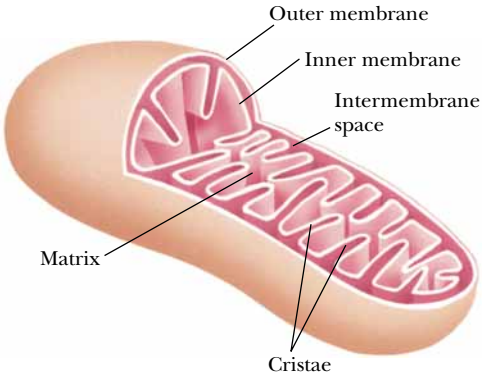
21.1 • Electron Transport and Oxidative Phosphorylation Are Membrane-Associated Processes

The processes of electron transport and oxidative phosphorylation are **membrane-associated**. Bacteria are the simplest life form, and bacterial cells typically consist of a single cellular compartment surrounded by a plasma membrane and a more rigid cell wall. In such a system, the conversion of energy from NADH and $[\text{FADH}_2]$ to the energy of ATP via electron transport and oxidative phosphorylation is carried out at (and across) the plasma membrane. In eukaryotic cells, electron transport and oxidative phosphorylation are localized in mitochondria, which are also the sites of TCA cycle activity and (as we shall see in Chapter 24) fatty acid oxidation. Mammalian cells contain from 800 to 2500 mitochondria; other types of cells may have as few as one or two or as many as half a million mitochondria. Human erythrocytes, whose purpose is simply to transport oxygen to tissues, contain *no* mitochondria at all. The typical mitochondrion is about 0.5 ± 0.3 microns in diameter and from 0.5 micron to several microns long; its overall shape is sensitive to metabolic conditions in the cell.

Mitochondria are surrounded by a simple **outer membrane** and a more complex **inner membrane** (Figure 21.1). The space between the inner and outer membranes is referred to as the **intermembrane space**. Several enzymes that utilize ATP (such as creatine kinase and adenylate kinase) are found in the intermembrane space. The smooth outer membrane is about 30 to 40% lipid and 60 to 70% protein, and has a relatively high concentration of phosphatidylinositol. The outer membrane contains significant amounts of **porin**—a transmembrane protein, rich in β -sheets, that forms large channels across the membrane, permitting free diffusion of molecules with molecular weights of about 10,000 or less. Apparently, the outer membrane functions mainly to



(a)



(b)

FIGURE 21.1 • (a) An electron micrograph of a mitochondrion. (b) A drawing of a mitochondrion with components labelled. (a, B. King/BPS)

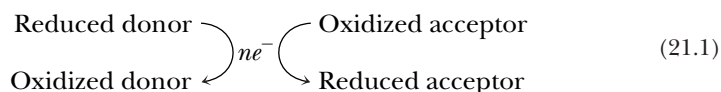
maintain the shape of the mitochondrion. The inner membrane is richly packed with proteins, which account for nearly 80% of its weight; thus, its density is higher than that of the outer membrane. The fatty acids of inner membrane lipids are highly unsaturated. Cardiolipin and diphosphatidylglycerol (Chapter 8) are abundant. The inner membrane lacks cholesterol and is quite impermeable to molecules and ions. Species that must cross the mitochondrial inner membrane—ions, substrates, fatty acids for oxidation, and so on—are carried by specific transport proteins in the membrane. Notably, the inner membrane is extensively folded (Figure 21.1). The folds, known as **cristae**, provide the inner membrane with a large surface area in a small volume. During periods of active respiration, the inner membrane appears to shrink significantly, leaving a comparatively large intermembrane space.

The Mitochondrial Matrix Contains the Enzymes of the TCA Cycle

The space inside the inner mitochondrial membrane is called the **matrix**, and it contains most of the enzymes of the TCA cycle and fatty acid oxidation. (An important exception, succinate dehydrogenase of the TCA cycle, is located in the inner membrane itself.) In addition, mitochondria contain circular DNA molecules, along with ribosomes and the enzymes required to synthesize proteins coded within the mitochondrial genome. Although some of the mitochondrial proteins are made this way, most are encoded by nuclear DNA and synthesized by cytosolic ribosomes.

21.2 • Reduction Potentials—An Accounting Device for Free Energy Changes in Redox Reactions

On numerous occasions in earlier chapters, we have stressed that NADH and reduced flavoproteins ($[FADH_2]$) are forms of metabolic energy. These reduced coenzymes have a strong tendency to be oxidized—that is, to transfer electrons to other species. The electron transport chain converts the energy of electron transfer into the energy of phosphoryl transfer stored in the phosphoric anhydride bonds of ATP. Just as the *group transfer potential* was used in Chapter 3 to quantitate the energy of phosphoryl transfer, the **standard reduction potential**, denoted by $\mathcal{E}_o'$, quantitates the tendency of chemical species to be reduced or oxidized. The standard reduction potential describing electron transfer between two species,



is related to the free energy change for the process by

$$\Delta G^{\circ'} = -n\mathcal{F}\Delta\mathcal{E}_o' \quad (21.2)$$

where n represents the number of electrons transferred; $\mathcal{F}$ is Faraday's constant, $96,485 \text{ J/V} \cdot \text{mol}$; and $\Delta\mathcal{E}_o'$ is the difference in reduction potentials between the donor and acceptor. This relationship is straightforward, but it depends on a *standard* of reference by which reduction potentials are defined.

Measurement of Standard Reduction Potentials

Standard reduction potentials are determined by measuring the voltages generated in **reaction half-cells** (Figure 21.2). A half-cell consists of a solution containing 1 M concentrations of both the oxidized and reduced forms of the substance whose reduction potential is being measured, and a simple electrode.

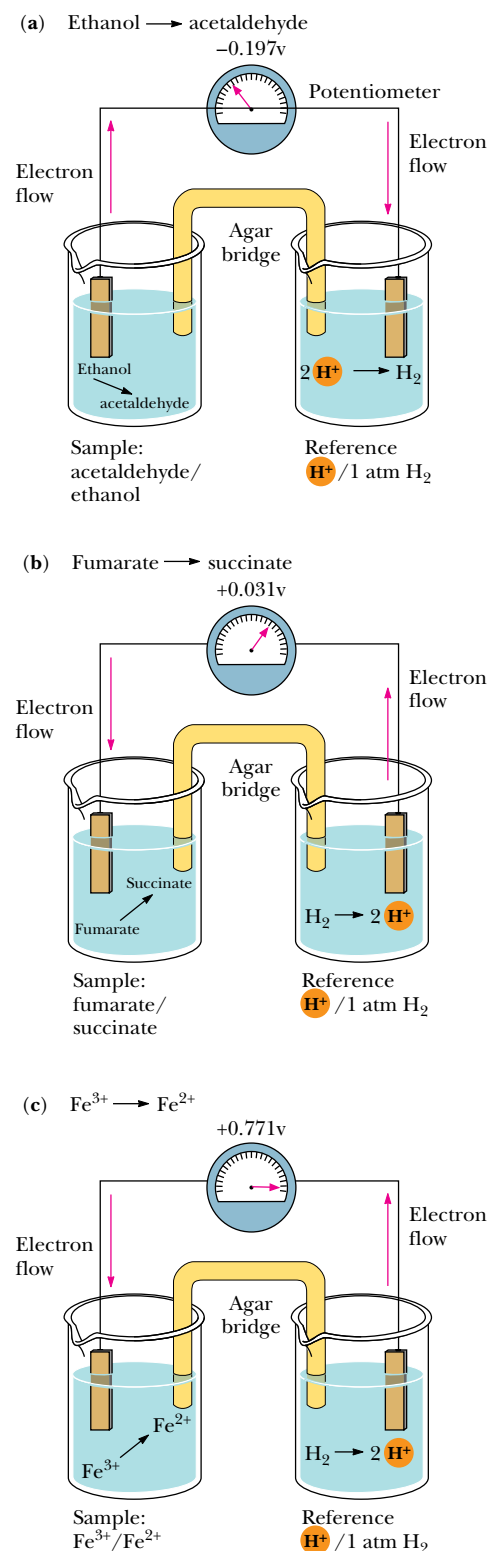


FIGURE 21.2 • Experimental apparatus used to measure the standard reduction potential of the indicated redox couples: (a) the acetalddehyde/ethanol couple, (b) the fumarate/succinate couple, (c) the $\text{Fe}^{3+}/\text{Fe}^{2+}$ couple.

(Together, the oxidized and reduced forms of the substance are referred to as a **redox couple**.) Such a **sample half-cell** is connected to a **reference half-cell** and electrode via a conductive bridge (usually a salt-containing agar gel). A sensitive potentiometer (voltmeter) connects the two electrodes so that the electrical potential (voltage) between them can be measured. The reference half-cell normally contains 1 M H^+ in equilibrium with H_2 gas at a pressure of 1 atm. The H^+/H_2 reference half-cell is arbitrarily assigned a standard reduction potential of 0.0 V. The standard reduction potentials of all other redox couples are defined relative to the H^+/H_2 reference half-cell on the basis of the sign and magnitude of the voltage (electromotive force, emf) registered on the potentiometer (Figure 21.2).

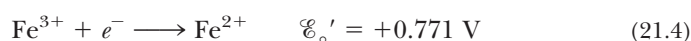
If electron flow between the electrodes is toward the sample half-cell, reduction occurs spontaneously in the sample half-cell, and the reduction potential is said to be positive. If electron flow between the electrodes is away from the sample half-cell and toward the reference cell, the reduction potential is said to be negative because electron loss (oxidation) is occurring in the sample half-cell. Strictly speaking, the standard reduction potential, $\mathcal{E}_o'$, is the electromotive force generated at 25°C and pH 7.0 by a sample half-cell (containing 1 M concentrations of the oxidized and reduced species) with respect to a reference half-cell. (Note that the reduction potential of the hydrogen half-cell is pH-dependent. The standard reduction potential, 0.0 V, assumes 1 M H^+ . The hydrogen half-cell measured at pH 7.0 has an $\mathcal{E}_o'$ of -0.421 V .)

Several Examples

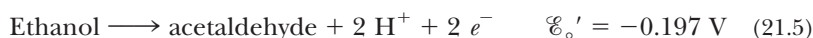
Figure 21.2a shows a sample/reference half-cell pair for measurement of the standard reduction potential of the acetaldehyde/ethanol couple. Because electrons flow toward the reference half-cell and away from the sample half-cell, the standard reduction potential is negative, specifically -0.197 V . In contrast, the fumarate/succinate couple and the $\text{Fe}^{3+}/\text{Fe}^{2+}$ couple both cause electrons to flow from the reference half-cell to the sample half-cell; that is, reduction occurs spontaneously in each system, and the reduction potentials of both are thus positive. The standard reduction potential for the $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cell is much larger than that for the fumarate/succinate half-cell, with values of $+0.771\text{ V}$ and $+0.031\text{ V}$, respectively. For each half-cell, a **half-cell reaction** describes the reaction taking place. For the fumarate/succinate half-cell coupled to a H^+/H_2 reference half-cell, the reaction occurring is indeed a reduction of fumarate.



Similarly, for the $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cell,



However, the reaction occurring in the acetaldehyde/ethanol half-cell is the oxidation of ethanol:



The Significance of $\mathcal{E}_o'$

Some typical half-cell reactions and their respective standard reduction potentials are listed in Table 21.1. Whenever reactions of this type are tabulated, they are uniformly written as *reduction* reactions, regardless of what occurs in the given half-cell. The sign of the standard reduction potential indicates which reaction really occurs when the given half-cell is combined with the reference hydrogen half-cell. Redox couples that have large positive reduction potentials

Table 21.1

Standard Reduction Potentials for Several Biological Reduction Half-Reactions

Reduction Half-Reaction	$\mathcal{E}_o'$ (V)
$\frac{1}{2} \text{O}_2 + 2 \text{H}^+ + 2 e^- \longrightarrow \text{H}_2\text{O}$	0.816
$\text{Fe}^{3+} + e^- \longrightarrow \text{Fe}^{2+}$	0.771
Photosystem P700	0.430
$\text{NO}_3^- + 2 \text{H}^+ + 2 e^- \longrightarrow \text{NO}_2^- + \text{H}_2\text{O}$	0.421
Cytochrome <i>f</i> (Fe ³⁺) + <i>e</i> [−] → cytochrome <i>f</i> (Fe ²⁺)	0.365
Cytochrome <i>a</i> ₃ (Fe ³⁺) + <i>e</i> [−] → cytochrome <i>a</i> ₃ (Fe ²⁺)	0.350
Cytochrome <i>a</i> (Fe ³⁺) + <i>e</i> [−] → cytochrome <i>a</i> (Fe ²⁺)	0.290
Rieske Fe-S(Fe ³⁺) + <i>e</i> [−] → Rieske Fe-S(Fe ²⁺)	0.280
Cytochrome <i>c</i> (Fe ³⁺) + <i>e</i> [−] → cytochrome <i>c</i> (Fe ²⁺)	0.254
Cytochrome <i>c</i> ₁ (Fe ³⁺) + <i>e</i> [−] → cytochrome <i>c</i> ₁ (Fe ²⁺)	0.220
$\text{UQH} \cdot + \text{H}^+ + e^- \longrightarrow \text{UQH}_2$ (UQ = coenzyme Q)	0.190
$\text{UQ} + 2 \text{H}^+ + 2 e^- \longrightarrow \text{UQH}_2$	0.060
Cytochrome <i>b</i> _H (Fe ³⁺) + <i>e</i> [−] → cytochrome <i>b</i> _H (Fe ²⁺)	0.050
Fumarate + 2 H ⁺ + 2 <i>e</i> [−] → succinate	0.031
$\text{UQ} + \text{H}^+ + e^- \longrightarrow \text{UQH} \cdot$	0.030
Cytochrome <i>b</i> ₅ (Fe ³⁺) + <i>e</i> [−] → cytochrome <i>b</i> ₅ (Fe ²⁺)	0.020
$[\text{FAD}] + 2 \text{H}^+ + 2 e^- \longrightarrow [\text{FADH}_2]$	0.003–0.091*
Cytochrome <i>b</i> _L (Fe ³⁺) + <i>e</i> [−] → cytochrome <i>b</i> _L (Fe ²⁺)	−0.100
Oxaloacetate + 2 H ⁺ + 2 <i>e</i> [−] → malate	−0.166
Pyruvate + 2 H ⁺ + 2 <i>e</i> [−] → lactate	−0.185
Acetaldehyde + 2 H ⁺ + 2 <i>e</i> [−] → ethanol	−0.197
$\text{FMN} + 2 \text{H}^+ + 2 e^- \longrightarrow \text{FMNH}_2$	−0.219
$\text{FAD} + 2 \text{H}^+ + 2 e^- \longrightarrow \text{FADH}_2$	−0.219
Glutathione (oxidized) + 2 H ⁺ + 2 <i>e</i> [−] → 2 glutathione (reduced)	−0.230
Lipoic acid + 2 H ⁺ + 2 <i>e</i> [−] → dihydrolipoic acid	−0.290
$1,3\text{-Bisphosphoglycerate} + 2 \text{H}^+ + 2 e^- \longrightarrow$ glyceraldehyde-3-phosphate + P _i	−0.290
$\text{NAD}^+ + 2 \text{H}^+ + 2 e^- \longrightarrow \text{NADH} + \text{H}^+$	−0.320
$\text{NADP}^+ + 2 \text{H}^+ + 2 e^- \longrightarrow \text{NADPH} + \text{H}^+$	−0.320
Lipoyl dehydrogenase [FAD] + 2 H ⁺ + 2 <i>e</i> [−] → lipoyl dehydrogenase [FADH ₂]	−0.340
$\alpha\text{-Ketoglutarate} + \text{CO}_2 + 2 \text{H}^+ + 2 e^- \longrightarrow \text{isocitrate}$	−0.380
$2 \text{H}^+ + 2 e^- \longrightarrow \text{H}_2$	−0.421
Ferredoxin (spinach) (Fe ³⁺) + <i>e</i> [−] → ferredoxin (spinach) (Fe ²⁺)	−0.430
$\text{Succinate} + \text{CO}_2 + 2 \text{H}^+ + 2 e^- \longrightarrow \alpha\text{-ketoglutarate} + \text{H}_2\text{O}$	−0.670

*Typical values for reduction of bound FAD in flavoproteins such as succinate dehydrogenase (see Bonomi, F., Pagani, S., Cerletti, P., and Giori, C., 1983. *European Journal of Biochemistry* 134:439–445).

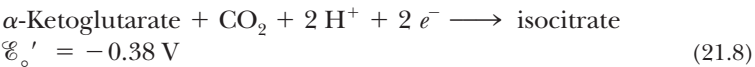
have a strong tendency to accept electrons, and the oxidized form of such a couple (O₂, for example) is a strong oxidizing agent. Redox couples with large negative reduction potentials have a strong tendency to undergo oxidation (that is, donate electrons), and the reduced form of such a couple (NADPH, for example) is a strong reducing agent.

Coupled Redox Reactions

The half-reactions and reduction potentials in Table 21.1 can be used to analyze energy changes in redox reactions. The oxidation of NADH to NAD⁺ can be coupled with the reduction of α-ketoglutarate to isocitrate:



This is the isocitrate dehydrogenase reaction of the TCA cycle. Writing the two half-cell reactions, we have



In a spontaneous reaction, electrons are donated by (flow away from) the half-reaction with the more negative reduction potential and are accepted by (flow toward) the half-reaction with the more positive reduction potential. Thus, in the present case, isocitrate donates electrons and NAD⁺ accepts electrons. The convention defines Δ $\mathcal{E}_\circ'$ as

$$\Delta\mathcal{E}_\circ' = \mathcal{E}_\circ'(\text{acceptor}) - \mathcal{E}_\circ'(\text{donor}) \quad (21.9)$$

In the present case, isocitrate is the donor and NAD⁺ the acceptor, so we write

$$\Delta\mathcal{E}_\circ' = -0.32 \text{ V} - (-0.38 \text{ V}) = +0.06 \text{ V} \quad (21.10)$$

From Equation 21.2, we can now calculate Δ*G*^{o'} as

$$\begin{aligned} \Delta G^{\circ'} &= -(2)(96.485 \text{ kJ/V} \cdot \text{mol})(0.06 \text{ V}) \\ \Delta G^{\circ'} &= -11.58 \text{ kJ/mol} \end{aligned} \quad (21.11)$$

Note that a reaction with a net positive Δ $\mathcal{E}_\circ'$ yields a negative Δ*G*^{o'}, indicating a spontaneous reaction.

The Dependence of the Reduction Potential on Concentration

We have already noted that the standard free energy change for a reaction, Δ*G*^{o'}, does not reflect the actual conditions in a cell, where reactants and products are not at standard-state concentrations (1 *M*). Equation 3.12 was introduced to permit calculations of actual free energy changes under non-standard-state conditions. Similarly, standard reduction potentials for redox couples must be modified to account for the actual concentrations of the oxidized and reduced species. For any redox couple,



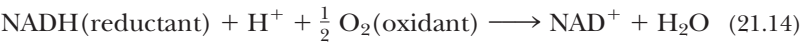
the actual reduction potential is given by

$$\mathcal{E} = \mathcal{E}_\circ' + (RT/n\mathcal{F}) \ln \frac{[\text{ox}]}{[\text{red}]} \quad (21.13)$$

Reduction potentials can also be quite sensitive to molecular environment. The influence of environment is especially important for flavins, such as FAD/FADH₂ and FMN/FMNH₂. These species are normally bound to their respective flavoproteins; the reduction potential of bound FAD, for example, can be very different from the value shown in Table 21.1 for the free FAD–FADH₂ couple of −0.219 V. A problem at the end of the chapter addresses this case.

21.3 • The Electron Transport Chain—An Overview

As we have seen, the metabolic energy from oxidation of food materials—sugars, fats, and amino acids—is funneled into formation of reduced coenzymes (NADH) and reduced flavoproteins ([FADH₂]). The electron transport chain reoxidizes the coenzymes, and channels the free energy obtained from these reactions into the synthesis of ATP. This reoxidation process involves the removal of both protons and electrons from the coenzymes. Electrons move from NADH and [FADH₂] to molecular oxygen, O₂, which is the terminal acceptor of electrons in the chain. The reoxidation of NADH,



involves the following half-reactions:



Here, half-reaction (21.16) is the electron acceptor and half-reaction (21.15) is the electron donor. Then

$$\Delta \mathcal{E}_\circ' = 0.816 - (-0.32) = 1.136 \text{ V}$$

and, according to Equation (21.2), the standard-state free energy change, Δ*G*°, is −219 kJ/mol. Molecules along the electron transport chain have reduction potentials between the values for the NAD⁺/NADH couple and the oxygen/H₂O couple, so that electrons move down the energy scale toward progressively more positive reduction potentials (Figure 21.3).

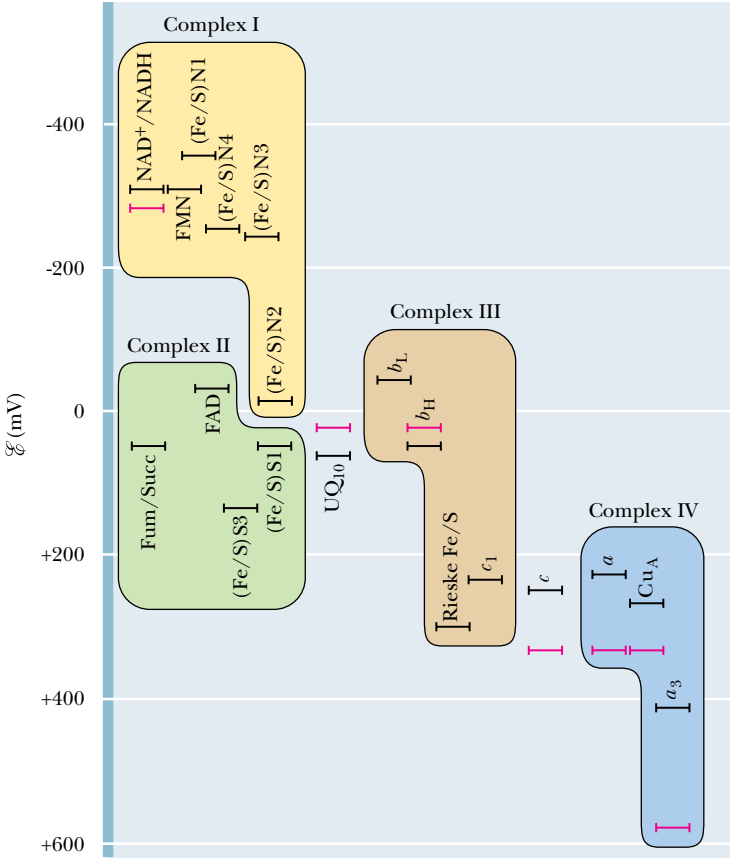


FIGURE 21.3 • $\mathcal{E}_\circ'$ and $\mathcal{E}$ values for the components of the mitochondrial electron transport chain. Values indicated are consensus values for animal mitochondria. Black bars represent $\mathcal{E}_\circ'$; red bars, $\mathcal{E}$.

Although electrons move from more negative to more positive reduction potentials in the electron transport chain, it should be emphasized that the electron carriers do not operate in a simple linear sequence. This will become evident when the individual components of the electron transport chain are discussed in the following paragraphs.

The Electron Transport Chain Can Be Isolated in Four Complexes

The electron transport chain involves several different molecular species, including:

- (a) **Flavoproteins**, which contain tightly bound FMN or FAD as prosthetic groups, and which (as noted in Chapter 20) may participate in one- or two-electron transfer events.
- (b) **Coenzyme Q**, also called **ubiquinone** (and abbreviated **CoQ** or **UQ**) (Figure 8.18), which can function in either one- or two-electron transfer reactions.
- (c) Several **cytochromes** (proteins containing heme prosthetic groups [see Chapter 5], which function by carrying or transferring electrons), including cytochromes *b*, *c*, *c*₁, *a*, and *a*₃. Cytochromes are one-electron transfer agents, in which the heme iron is converted from Fe²⁺ to Fe³⁺ and back.
- (d) A number of **iron–sulfur proteins**, which participate in one-electron transfers involving the Fe²⁺ and Fe³⁺ states.
- (e) Protein-bound **copper**, a one-electron transfer site, which converts between Cu⁺ and Cu²⁺.

All these intermediates except for cytochrome *c* are membrane-associated (either in the mitochondrial inner membrane of eukaryotes or in the plasma membrane of prokaryotes). All three types of proteins involved in this chain—flavoproteins, cytochromes, and iron–sulfur proteins—possess electron-transferring **prosthetic groups**.

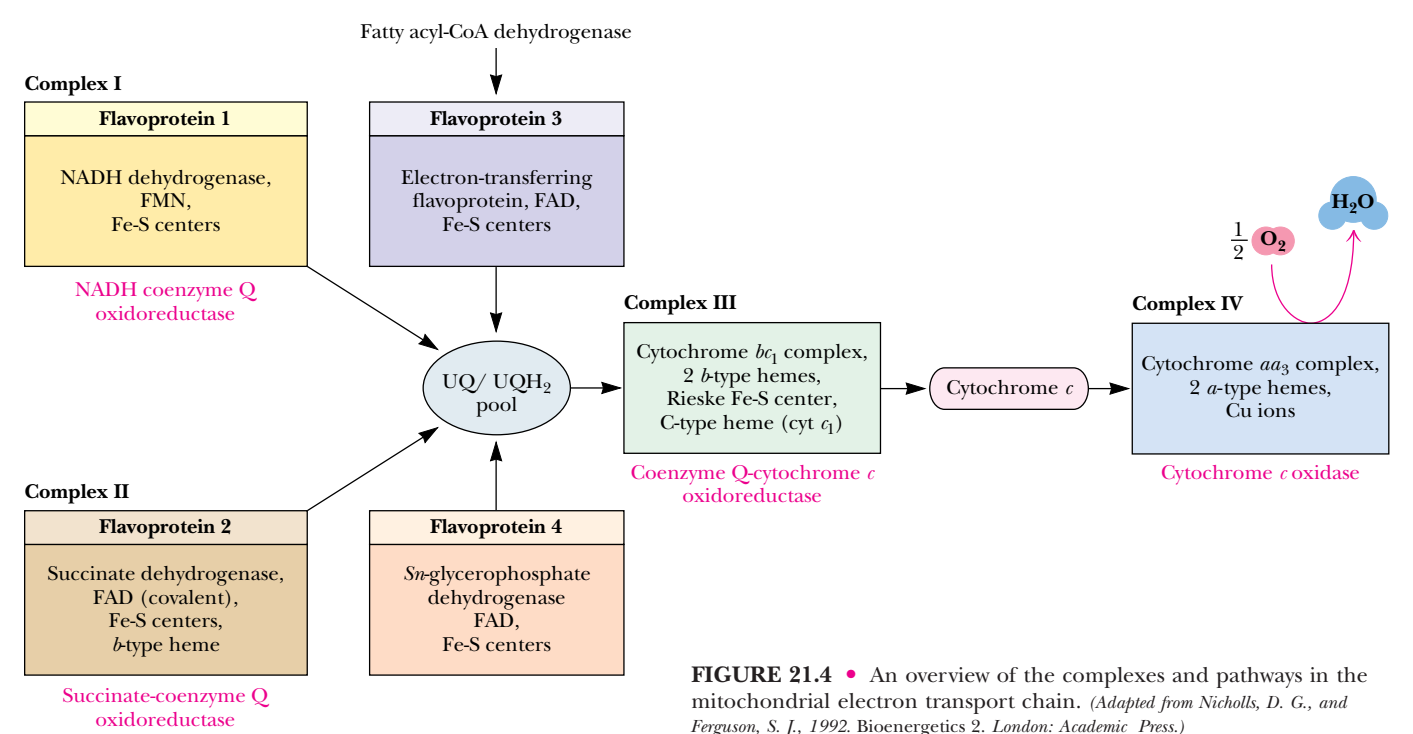


FIGURE 21.4 • An overview of the complexes and pathways in the mitochondrial electron transport chain. (Adapted from Nicholls, D. G., and Ferguson, S. J., 1992. Bioenergetics 2. London: Academic Press.)

The components of the electron transport chain can be purified from the mitochondrial inner membrane. Solubilization of the membranes containing the electron transport chain results in the isolation of four distinct protein complexes, and the complete chain can thus be considered to be composed of four parts: (I) **NADH–coenzyme Q reductase**, (II) **succinate–coenzyme Q reductase**, (III) **coenzyme Q–cytochrome c reductase**, and (IV) **cytochrome c oxidase** (Figure 21.4). Complex I accepts electrons from NADH, serving as a link between glycolysis, the TCA cycle, fatty acid oxidation, and the electron transport chain. Complex II includes succinate dehydrogenase and thus forms a direct link between the TCA cycle and electron transport. Complexes I and II produce a common product, reduced coenzyme Q (UQH₂), which is the substrate for coenzyme Q–cytochrome *c* reductase (Complex III). As shown in Figure 21.4, there are two other ways to feed electrons to UQ: the **electron-transferring flavoprotein**, which transfers electrons from the flavoprotein-linked step of fatty acyl-CoA dehydrogenase, and ***sn*-glycerophosphate dehydrogenase**. Complex III oxidizes UQH₂ while reducing cytochrome *c*, which in turn is the substrate for Complex IV, cytochrome *c* oxidase. Complex IV is responsible for reducing molecular oxygen. Each of the complexes shown in Figure 21.4 is a large multisubunit complex embedded within the inner mitochondrial membrane.

21.4 • Complex I: NADH–Coenzyme Q Reductase

As its name implies, this complex transfers a pair of electrons from NADH to coenzyme Q, a small, hydrophobic, yellow compound. Another common name for this enzyme complex is *NADH dehydrogenase*. The complex (with an estimated mass of 850 kD) involves more than 30 polypeptide chains, one molecule of flavin mononucleotide (FMN), and as many as seven Fe-S clusters, together containing a total of 20 to 26 iron atoms (Table 21.2). By virtue of its dependence on FMN, NADH–UQ reductase is a *flavoprotein*.

Protein Complexes of the Mitochondrial Electron-Transport Chain				
Complex	Mass (kD)	Subunits	Prosthetic Group	Binding Site for:
NADH–UQ reductase	850	>30	FMN Fe-S	NADH (matrix side) UQ (lipid core)
Succinate–UQ reductase	140	4	FAD Fe-S	Succinate (matrix side) UQ (lipid core)
UQ–Cyt <i>c</i> reductase	250	9–10	Heme <i>b_L</i> Heme <i>b_H</i> Heme <i>c₁</i> Fe-S	Cyt <i>c</i> (intermembrane space side)
Cytochrome <i>c</i>	13	1	Heme <i>c</i>	Cyt <i>c₁</i> Cyt <i>a</i>
Cytochrome <i>c</i> oxidase	162	>10	Heme <i>a</i> Heme <i>a₃</i> Cu _A Cu _B	Cyt <i>c</i> (intermembrane space side)

Adapted from: Hatefi, Y., 1985. The mitochondrial electron transport chain and oxidative phosphorylation system. *Annual Review of Biochemistry* **54**:1015–1069; and DePierre, J., and Ernster, L., 1977. Enzyme topology of intracellular membranes. *Annual Review of Biochemistry* **46**:201–262.

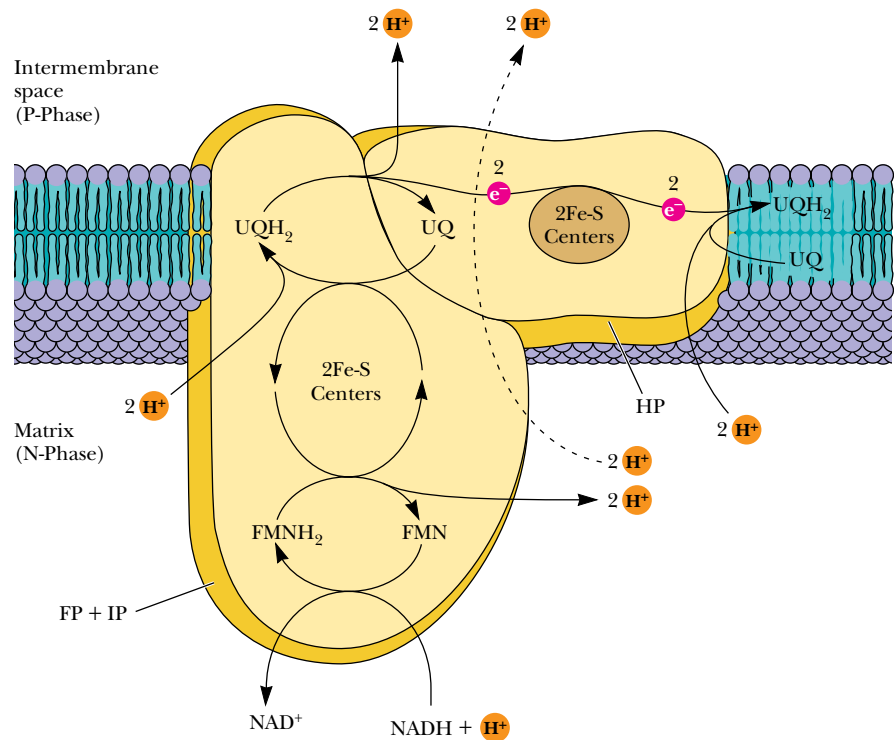
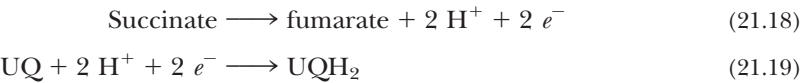


FIGURE 21.6 • Proposed structure and electron transport pathway for Complex I. Three protein complexes have been isolated, including the **flavoprotein (FP)**, **iron–sulfur protein (IP)**, and **hydrophobic protein (HP)**. FP contains three peptides (of mass 51, 24, and 10 kD) and bound FMN and has 2 Fe-S centers (a 2Fe-2S center and a 4Fe-4S center). IP contains six peptides and at least 3 Fe-S centers. HP contains at least seven peptides and one Fe-S center.

through this complex is used in a *coupled process* to drive the transport of protons across the membrane. (This is an example of *active transport*, a phenomenon examined in detail in Chapter 10.) Available experimental evidence suggests a stoichiometry of four H^+ transported per two electrons passed from NADH to UQ.

21.5 • Complex II: Succinate–Coenzyme Q Reductase

Complex II is perhaps better known by its other name—**succinate dehydrogenase**, the only TCA cycle enzyme that is an integral membrane protein in the inner mitochondrial membrane. This enzyme has a mass of approximately 100 to 140 kD and is composed of four subunits: two Fe-S proteins of masses 70 kD and 27 kD, and two other peptides of masses 15 kD and 13 kD. Also known as *flavoprotein 2 (FP₂)*, it contains an FAD covalently bound to a histidine residue (see Figure 20.15), and three Fe-S centers: a 4Fe-4S cluster, a 3Fe-4S cluster, and a 2Fe-2S cluster. When succinate is converted to fumarate in the TCA cycle, concomitant reduction of bound FAD to FADH₂ occurs in succinate dehydrogenase. This FADH₂ transfers its electrons immediately to Fe-S centers, which pass them on to UQ. Electron flow from succinate to UQ,



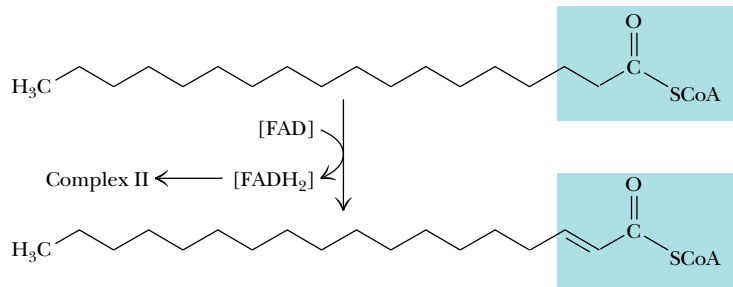


FIGURE 21.7 • The fatty acyl-CoA dehydrogenase reaction, emphasizing that the reaction involves reduction of enzyme-bound FAD (indicated by brackets).

Net rxn: Succinate + UQ $\longrightarrow$ fumarate + UQH₂ $\Delta\mathcal{E}_o' = 0.029\text{V}$ (21.20)
 $\Delta G^{\circ'} = -5.6\text{ kJ/mol}$

yields a net reduction potential of 0.029 V. (Note that the first half-reaction is written in the direction of the e^- flow. As always, $\Delta\mathcal{E}_o'$ is calculated according to Equation 21.9.) The small free energy change of this reaction is not sufficient to drive the transport of protons across the inner mitochondrial membrane.

This is a crucial point because (as we will see) proton transport is coupled with ATP synthesis. Oxidation of one FADH_2 in the electron transport chain results in synthesis of approximately two molecules of ATP, compared with the approximately three ATPs produced by the oxidation of one NADH. Other enzymes can also supply electrons to UQ, including mitochondrial *sn*-glycerophosphate dehydrogenase, an inner membrane-bound shuttle enzyme, and the fatty acyl-CoA dehydrogenases, three soluble matrix enzymes involved in fatty acid oxidation (Figure 21.7; also see Chapter 24). The path of electrons from succinate to UQ is shown in Figure 21.8.

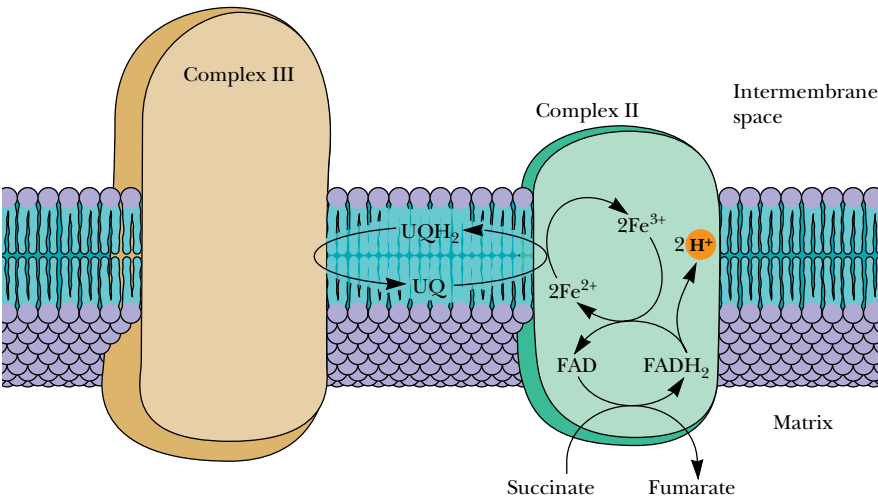


FIGURE 21.8 • A probable scheme for electron flow in Complex II. Oxidation of succinate occurs with reduction of [FAD]. Electrons are then passed to Fe-S centers and then to coenzyme Q (UQ). Proton transport does not occur in this complex.

21.6 • Complex III: Coenzyme Q–Cytochrome *c* Reductase

In the third complex of the electron transport chain, reduced coenzyme Q (UQH_2) passes its electrons to cytochrome *c* via a unique redox pathway known as the **Q cycle**. *UQ–cytochrome *c* reductase* (*UQ–cyt *c* reductase*), as this complex is known, involves three different cytochromes and an Fe-S protein. In the cytochromes of these and similar complexes, the iron atom at the center of the porphyrin ring cycles between the reduced Fe^{2+} (ferrous) and oxidized Fe^{3+} (ferric) states.

Cytochromes were first named and classified on the basis of their absorption spectra (Figure 21.9), which depend upon the structure and environment of their heme groups. The ***b* cytochromes** contain *iron–protoporphyrin IX* (Figure 21.10), the same heme found in hemoglobin and myoglobin. The ***c* cytochromes** contain *heme *c**, derived from iron–protoporphyrin IX by the covalent attachment of cysteine residues from the associated protein. *UQ–cyt *c**

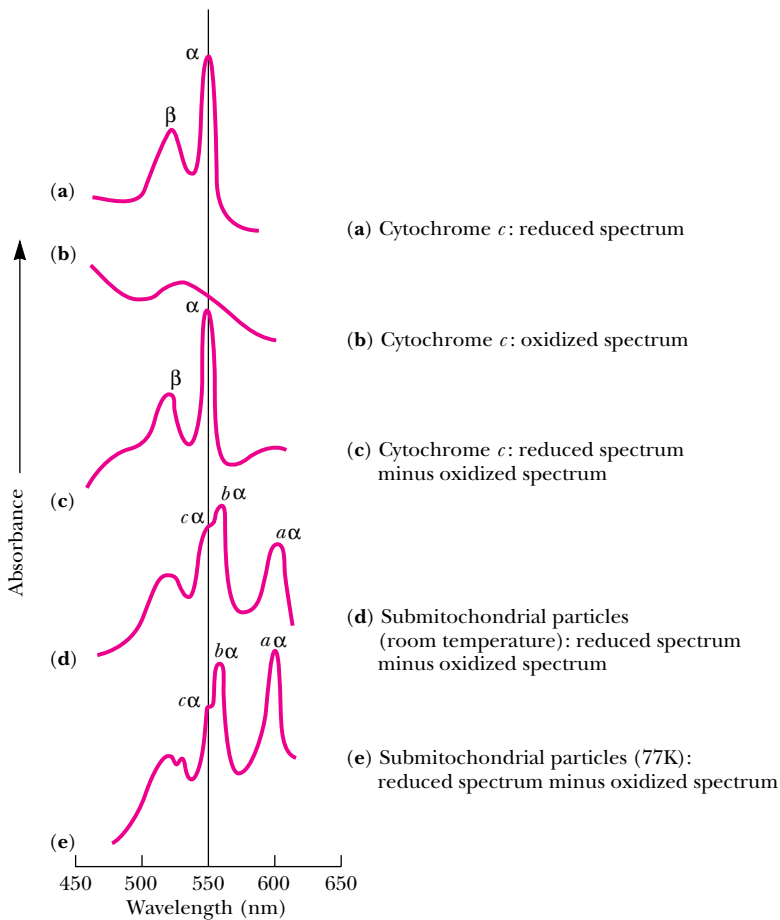


FIGURE 21.9 • Typical visible absorption spectra of cytochromes. (a) Cytochrome *c*, reduced spectrum; (b) cytochrome *c*, oxidized spectrum; (c) the difference spectrum: (a) minus (b); (d) beef heart mitochondrial particles: room temperature difference (reduced minus oxidized) spectrum; (e) beef heart submitochondrial particles: same as (d) but at 77 K. α - and β - bands are labeled, and in (d) and (e) the bands for cytochromes *a*, *b* and *c* are indicated.

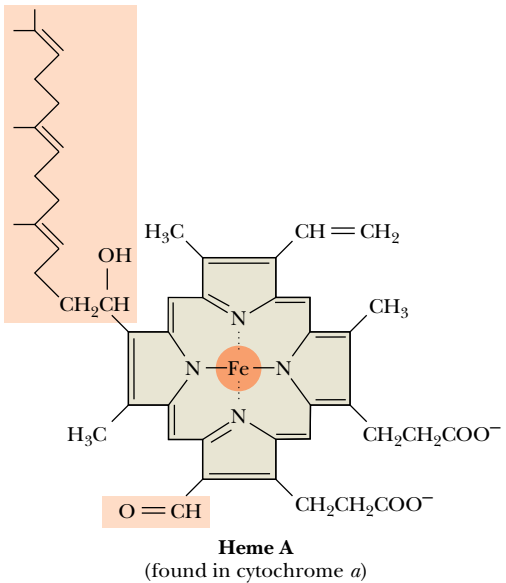
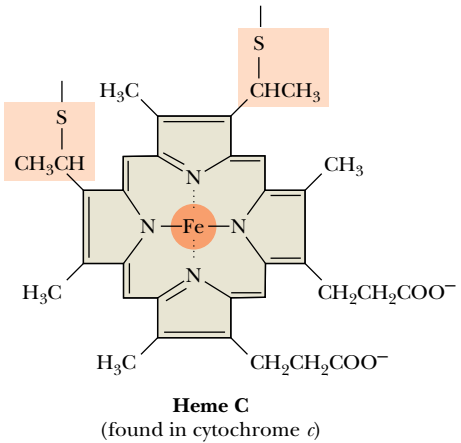
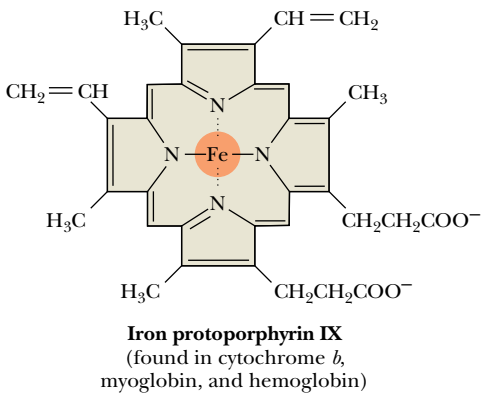
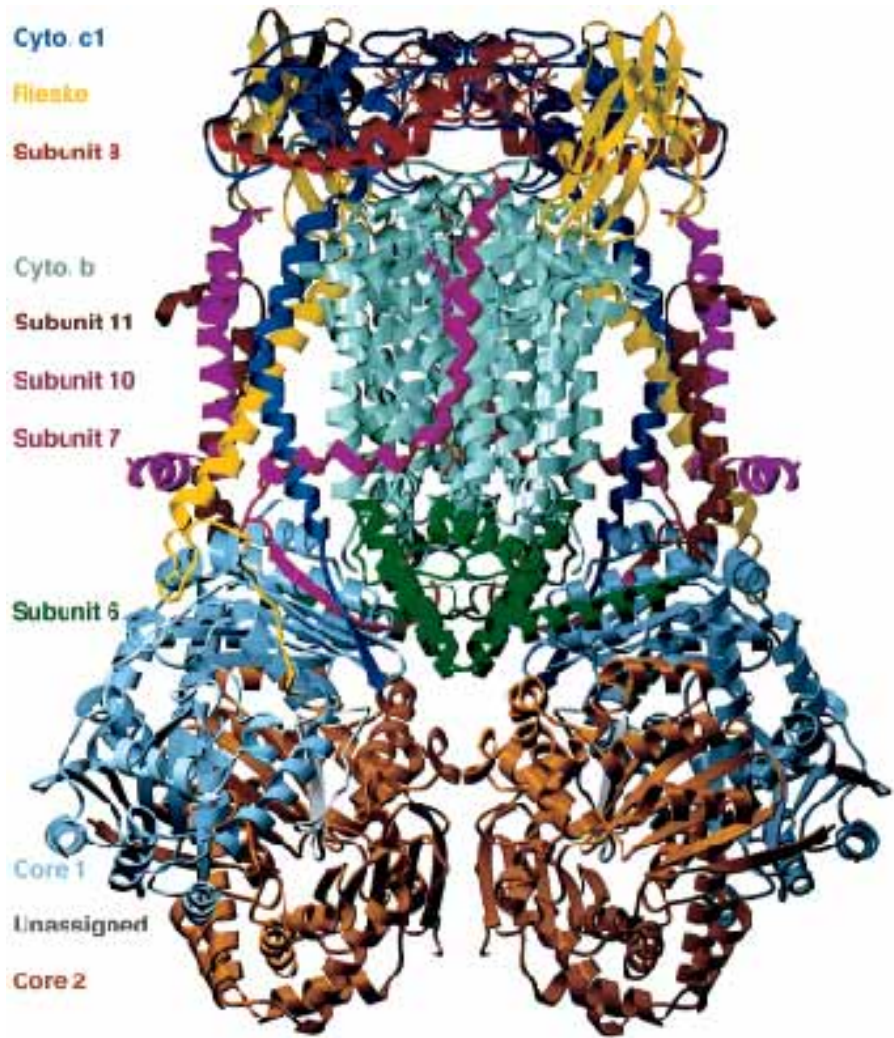


FIGURE 21.10 • The structures of iron protoporphyrin IX, heme *c*, and heme *a*.

FIGURE 21.11 • The structure of UQ-cyt *c* reductase, also known as the cytochrome *bc*₁ complex. The alpha helices of cytochrome *b* (pale green) define the transmembrane domain of the protein. The bottom of the structure as shown extends approximately 75 Å into the mitochondrial matrix, and the top of the structure as shown extends about 38 Å into the intermembrane space. (Photograph kindly provided by Di Xia and Johann Deisenhofer [From Xia, D., Yu, C.-A., Kim, H., Xia, J.-Z., Kachurin, A. M., Zhang, L., Yu, L., and Deisenhofer, J., 1997. The crystal structure of the cytochrome *bc*₁ complex from bovine heart mitochondria. *Science* 277:60–66.])



reductase contains a *b*-type cytochrome, of 30 to 40 kD, with two different heme sites (Figure 21.11) and one *c*-type cytochrome. (One other variation, heme *a*, contains a 15-carbon isoprenoid chain on a modified vinyl group, and a formyl group in place of one of the methyls [see Figure 21.10]. **Cytochrome *a*** is found in two forms in Complex IV of the electron transport chain, as we shall see.) The two hemes on the *b* cytochrome polypeptide in UQ-cyt *c* reductase are distinguished by their reduction potentials and the wavelength (λ_{max}) of the so-called **α band** (see Figure 21.9). One of these hemes, known as *b*_L or *b*₅₆₆, has a standard reduction potential, $\mathcal{E}_0'$, of -0.100 V and a wavelength of maximal absorbance (λ_{max}) of 566 nm. The other, known as *b*_H or *b*₅₆₂, has a standard reduction potential of $+0.050$ V and a λ_{max} of 562 nm. (*H* and *L* here refer to *high* and *low* reduction potential.)

The structure of the UQ-cyt *c* reductase, also known as the cytochrome *bc*₁ complex, has been determined by Johann Deisenhofer and his colleagues. (Deisenhofer was a co-recipient of the Nobel Prize in Chemistry for his work on the structure of a photosynthetic reaction center [see Chapter 22]). The complex is a dimer, with each monomer consisting of 11 protein subunits and 2165 amino acid residues (monomer mass, 248 kD). The dimeric structure is pear-shaped and consists of a large domain that extends 75 Å into the mito-

a hydrophobic segment. However, the hydrophobic segment is N-terminal in the Rieske protein and C-terminal in cytochrome *c*₁.

The electron on the *b*_L heme facing the cytosolic side of the membrane is now passed to the *b*_H heme on the matrix side of the membrane. This electron transfer occurs against a membrane potential of 0.15 V and is driven by the loss of redox potential as the electron moves from *b*_L ($\mathcal{E}_o' = -0.100\text{V}$) to *b*_H ($\mathcal{E}_o' = +0.050\text{V}$). The electron is then passed from *b*_H to a molecule of UQ at a second quinone-binding site, *Q*_n, converting this UQ to UQ · [−]. The resulting UQ · [−] remains firmly bound to the *Q*_n site. This completes the first half of the Q cycle (Figure 21.12a).

The second half of the cycle (Figure 21.12b) is similar to the first half, with a second molecule of UQH₂ oxidized at the *Q*_p site, one electron being passed to cytochrome *c*₁ and the other transferred to heme *b*_L and then to heme *b*_H. In this latter half of the Q cycle, however, the *b*_H electron is transferred to the semiquinone anion, UQ · [−], at the *Q*_n site. With the addition of two H⁺ from the mitochondrial matrix, this produces a molecule of UQH₂, which is released from the *Q*_n site and returns to the coenzyme Q pool, completing the Q cycle.

The Q Cycle Is an Unbalanced Proton Pump

Why has nature chosen this rather convoluted path for electrons in Complex III? First of all, Complex III takes up two protons on the matrix side of the inner membrane and releases four protons on the cytoplasmic side for each pair of electrons that passes through the Q cycle. The apparent imbalance of *two protons in* for *four protons out* is offset by proton translocations in Complex IV, the cytochrome oxidase complex. The other significant feature of this mechanism is that it offers a convenient way for a two-electron carrier, UQH₂, to interact with the *b*_L and *b*_H hemes, the Rieske protein Fe-S cluster, and cytochrome *c*₁, all of which are one-electron carriers.

Cytochrome *c* Is a Mobile Electron Carrier

Electrons traversing Complex III are passed through cytochrome *c*₁ to cytochrome *c*. Cytochrome *c* is the only one of the cytochromes that is water-soluble. Its structure, determined by X-ray crystallography (Figure 21.13), is globular; the planar heme group lies near the center of the protein, surrounded predominantly by hydrophobic protein residues. The iron in the porphyrin ring is coordinated both to a histidine nitrogen and to the sulfur atom of a methionine residue. Coordination with ligands in this manner on both sides of the porphyrin plane precludes the binding of oxygen and other ligands, a feature that distinguishes the cytochromes from hemoglobin (Chapter 15).

Cytochrome *c*, like UQ, is a mobile electron carrier. It associates loosely with the inner mitochondrial membrane (in the *intermembrane space* on the cytosolic side of the inner membrane) to acquire electrons from the Fe-S–cyt *c*₁ aggregate of Complex III, and then it migrates along the membrane surface in the reduced state, carrying electrons to *cytochrome c oxidase*, the fourth complex of the electron transport chain.

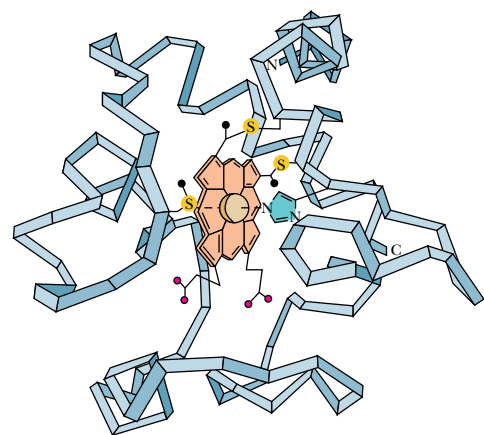


FIGURE 21.13 • The structure of mitochondrial cytochrome *c*. The heme is shown at the center of the structure, covalently linked to the protein via its two sulfur atoms (yellow). A third sulfur from a methionine residue coordinates the iron.

21.7 • Complex IV: Cytochrome *c* Oxidase

Complex IV is called cytochrome *c* oxidase because it accepts electrons from cytochrome *c* and directs them to the four-electron reduction of O₂ to form H₂O:

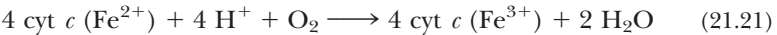


FIGURE 21.14 • An electrophoresis gel showing the complex subunit structure of bovine heart cytochrome *c* oxidase. The three largest subunits, I, II, and III, are coded for by mitochondrial DNA. The others are encoded by nuclear DNA. (Photo kindly provided by Professor Roderick Capaldi)

Thus, O₂ and cytochrome *c* oxidase are the final destination for the electrons derived from the oxidation of food materials. In concert with this process, cytochrome *c* oxidase also drives transport of protons across the inner mitochondrial membrane. These important functions are carried out by a transmembrane protein complex consisting of more than 10 subunits (Table 21.2).

An electrophoresis gel of the bovine heart complex is shown in Figure 21.14. The total mass of the protein in the complex, composed of 13 subunits, is 204 kD. Subunits I through III, the largest ones, are encoded by mitochondrial DNA, synthesized in the mitochondrion, and inserted into the inner membrane from the matrix side. The smaller subunits are coded by nuclear DNA and synthesized in the cytosol.

The structure of cytochrome *c* oxidase has been solved. The essential Fe and Cu sites are contained entirely within the structures of subunits I, II, and III. None of the 10 nuclear DNA–derived subunits directly impinges on the essential metal sites. The implication is that subunits I to III actively participate in the events of electron transfer, but that the other 10 subunits play regulatory roles in this process. Subunit I is cylindrical in shape and consists of 12 transmembrane helices, without any significant extramembrane parts (Figure 21.15). Hemes *a* and *a*₃, which lie perpendicular to the membrane plane, are cradled by the helices of subunit I. Subunits II and III lie on opposite sides of subunit I and do not contact each other. Subunit II has an extramembrane domain on the cytosolic face of the mitochondrial membrane. This domain consists of a 10-strand β -barrel that holds Cu_A 7 Å from the nearest surface atom of the subunit. Subunit III consists of 7 transmembrane helices with no significant extramembrane domains. Figure 21.16 presents a molecular graphic image of cytochrome *c* oxidase.

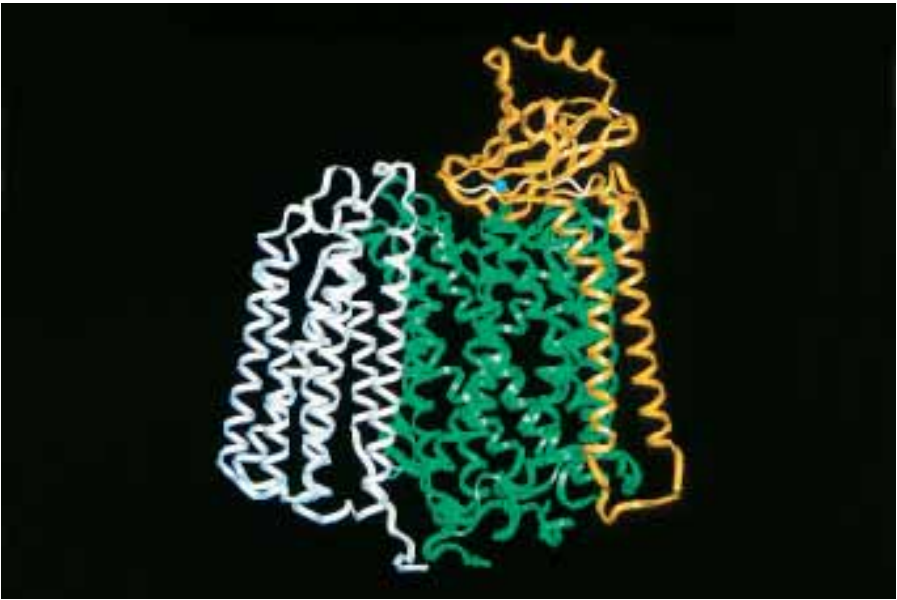


FIGURE 21.15 • Molecular graphic image of subunits I, II, and III of cytochrome *c* oxidase.



FIGURE 21.16 • Molecular graphic image of cytochrome *c* oxidase. Seven of the 10 nuclear DNA–derived subunits (IV, VIa, VIc, VIIa, VIIb, VIIc, and VIII) possess transmembrane segments. Three (Va, Vb, and Vlb) do not. Subunits IV and VIc are transmembrane and dumbbell-shaped. Subunit Va is globular and bound to the matrix side of the complex, whereas VIIb is a globular subunit on the cytosolic side of the membrane complex. Vb is globular and matrix-side associated as well, but it has an N-terminal extended domain. VIa has a transmembrane helix and a small globular domain. Subunit VIIa consists of a tilted transmembrane helix, with another short helical segment on the matrix side of the membrane. Subunits VIIa, VIIb, and VIII consist of transmembrane segments with short extended regions outside the membrane.

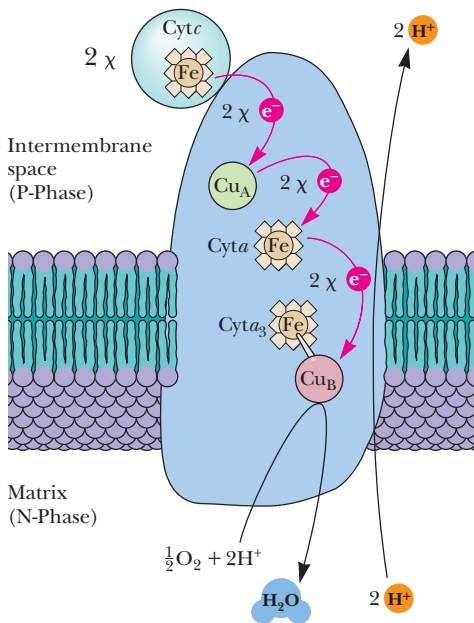


FIGURE 21.17 • The electron transfer pathway for cytochrome oxidase. Cytochrome *c* binds on the cytosolic side, transferring electrons through the copper and heme centers to reduce O₂ on the matrix side of the membrane.

Electron Transfer in Complex IV Involves Two Hemes and Two Copper Sites

Cytochrome *c* oxidase contains two heme centers (cytochromes *a* and *a*₃) as well as two copper atoms (Figure 21.17). The copper sites, Cu_A and Cu_B, are associated with cytochromes *a* and *a*₃, respectively. The copper sites participate in electron transfer by cycling between the reduced (*cuprous*) Cu⁺ state and the oxidized (*cupric*) Cu²⁺ state. (Remember, the cytochromes and copper sites are one-electron transfer agents.) Reduction of one oxygen molecule requires passage of four electrons through these carriers—one at a time (Figure 21.17).

Electrons from cytochrome *c* are transferred to Cu_A sites and then passed to the heme iron of cytochrome *a*. Cu_A is liganded by two cysteines and two histidines (Figure 21.18). The heme of cytochrome *a* is liganded by imidazole rings of histidine residues (Figure 21.18). The Cu_A and the Fe of cytochrome *a* are within 1.5 nm of each other.

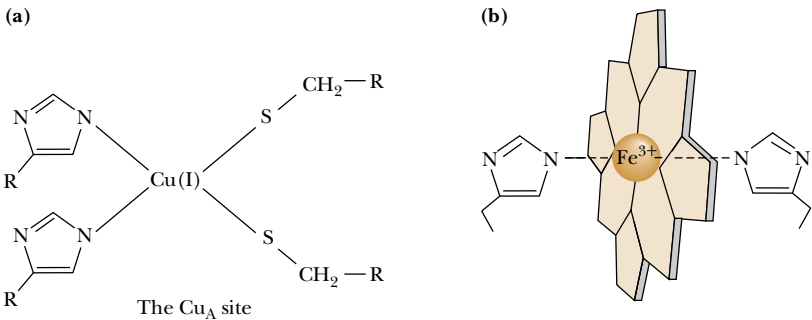
Cu_B and the iron atom of cytochrome *a*₃ are also situated close to each other and are thought to share a ligand, which may be a cysteine sulfur (Figure 21.19). This closely associated pair of metal ions is referred to as a **binuclear center**.

As shown in Figure 21.20, the electron pathway through Complex IV continues as Cu_B accepts a single electron from cytochrome *a* (state O → state H). A second electron then reduces the iron center to Fe²⁺ (H → R), leading to the binding of O₂ (R → A) and the formation of a peroxy bridge between heme *a*₃ and Cu_B (A → P). *This amounts to the transfer of two electrons from the binuclear center to the bound O₂.* The next step involves uptake of two H⁺ and a third electron (P → F), which leads to cleavage of the O—O bond and generation of Fe⁴⁺ at the heme. Uptake of a fourth *e*[−] facilitates formation of ferric hydroxide at the heme center (F → O′). In the final step of the cycle (O′ → O), protons from the mitochondrial matrix are accepted by the coordinated hydroxyl groups, and the resulting water molecules dissociate from the binuclear center.

Complex IV Also Transports Protons Across the Inner Mitochondrial Membrane

The reduction of oxygen in Complex IV is accompanied by transport of protons across the inner mitochondrial membrane. Transfer of four electrons through this complex drives the transport of approximately four protons. The mechanism of proton transport is unknown but is thought to involve the steps from state P to state O (Figure 21.20). Four protons are taken up on the matrix side for every two protons transported to the cytoplasm (see Figure 21.17).

FIGURE 21.18 • (a) The Cu_A site of cytochrome oxidase. Copper ligands include two histidine imidazole groups and two cysteine side chains from the protein. (b) The coordination of histidine imidazole ligands to the iron atom in the heme *a* center of cytochrome oxidase.



Independence of the Four Carrier Complexes

It should be emphasized here that the four major complexes of the electron transport chain operate quite independently in the inner mitochondrial membrane. Each is a multiprotein aggregate maintained by numerous strong associations between peptides of the complex, but there is no evidence that the complexes associate with one another in the membrane. Measurements of the lateral diffusion rates of the four complexes, of coenzyme Q, and of cytochrome *c* in the inner mitochondrial membrane show that the rates differ considerably, indicating that these complexes do not move together in the membrane. Kinetic studies with reconstituted systems show that electron transport does not operate by means of connected sets of the four complexes.

A Dynamic Model of Electron Transport

The model that emerges for electron transport is shown in Figure 21.21. The four complexes are independently mobile in the membrane. Coenzyme Q collects electrons from NADH–UQ reductase and succinate–UQ reductase and delivers them (by diffusion through the membrane core) to UQ–cyt *c* reductase. Cytochrome *c* is water-soluble and moves freely, carrying electrons from UQ–cyt *c* reductase to cytochrome *c* oxidase. In the process of these electron transfers, protons are driven across the inner membrane (from the matrix side to the intermembrane space). The proton gradient generated by electron transport represents an enormous source of potential energy. As seen in the next section, this potential energy is used to synthesize ATP as protons flow back into the matrix.

The $H^+/2e^-$ Ratio for Electron Transport Is Uncertain

In 1961, Peter Mitchell, a British biochemist, proposed that the energy stored in a proton gradient across the inner mitochondrial membrane by electron transport drives the synthesis of ATP in cells. The proposal became known as

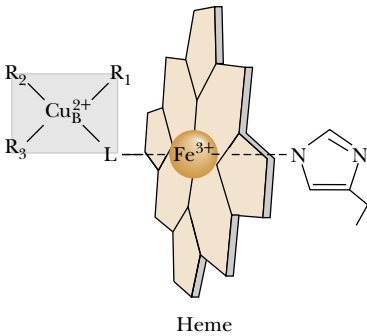


FIGURE 21.19 • The binuclear center of cytochrome oxidase. A ligand, L (probably a cysteine S), is shown bridging the Cu_B and Fe_{a₃} metal sites.

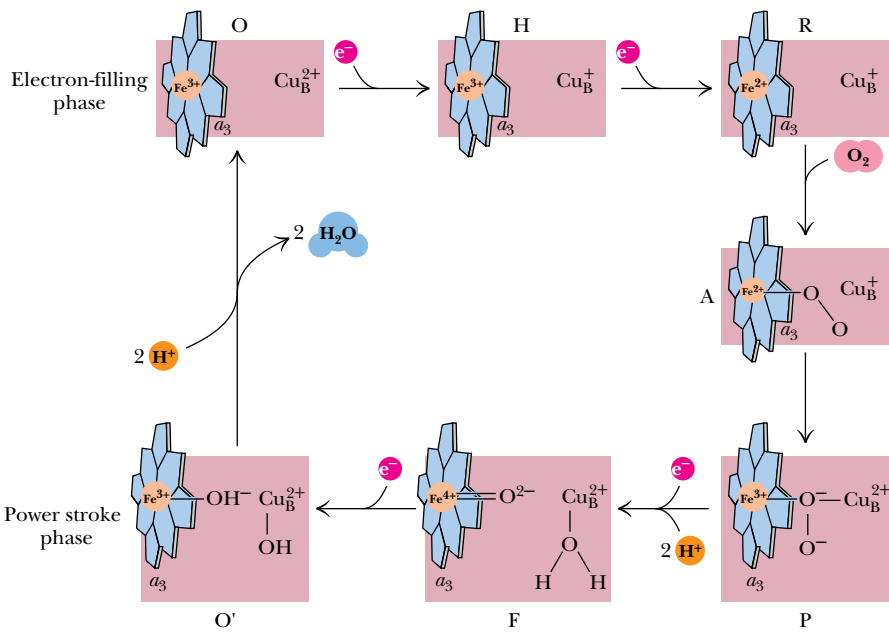


FIGURE 21.20 • A model for the mechanism of O₂ reduction by cytochrome oxidase. (Adapted from Nicholls, D. G., and Ferguson, S. J., 1992. Bioenergetics 2. London: Academic Press; and Babcock, G. T., and Wikström, M., 1992. Nature 356:301–309.)

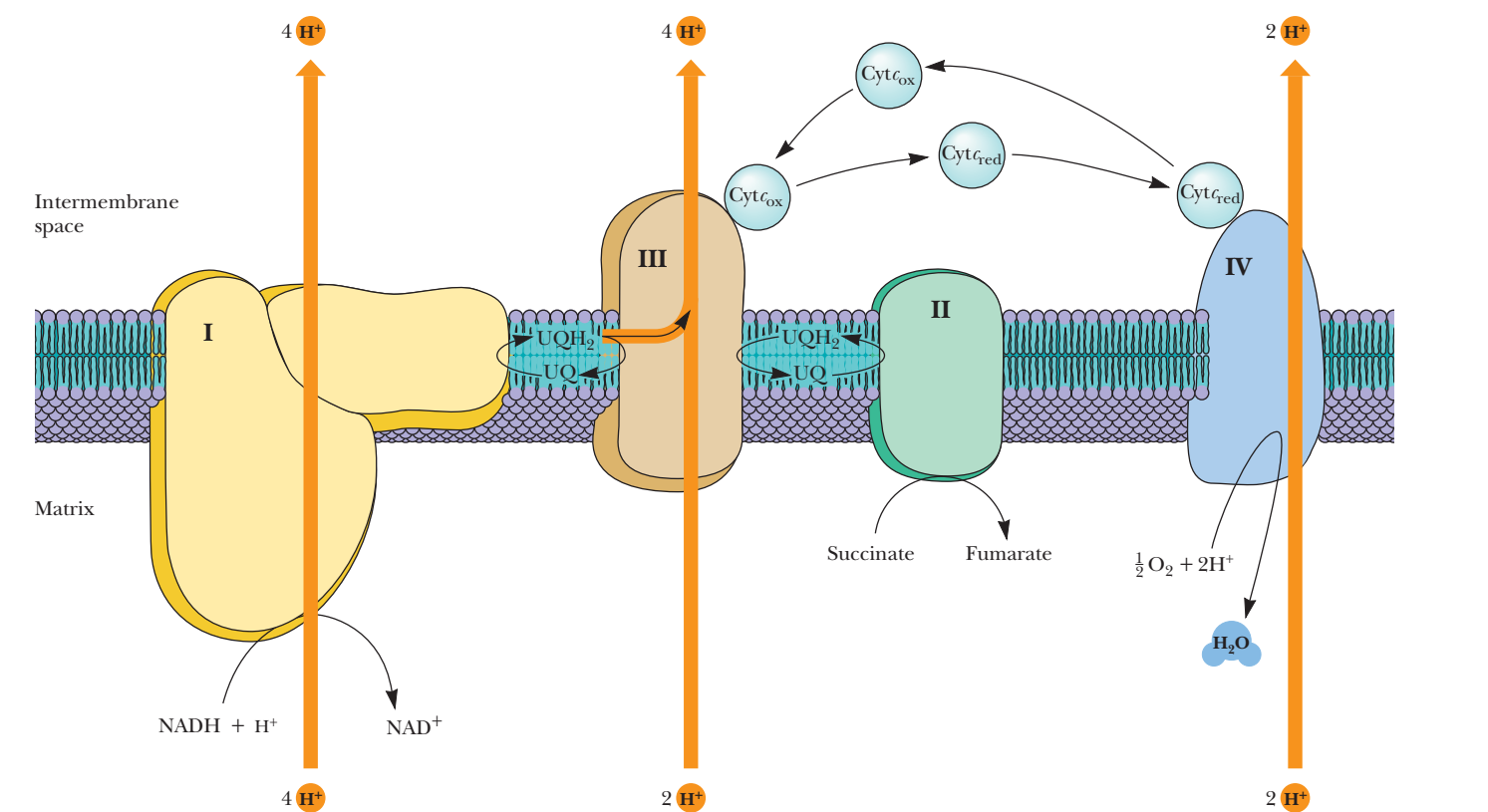


FIGURE 21.21 • A model for the electron transport pathway in the mitochondrial inner membrane. UQ/UQH₂ and cytochrome *c* are mobile electron carriers and function by transferring electrons between the complexes. The proton transport driven by Complexes I, III, and IV is indicated.

Mitchell’s chemiosmotic hypothesis. The ratio of protons transported per pair of electrons passed through the chain—the so-called **H⁺/2 e[−] ratio**—has been an object of great interest for many years. Nevertheless, the ratio has remained extremely difficult to determine. The consensus estimate for the electron transport pathway from succinate to O₂ is 6 H⁺/2 e[−]. The ratio for Complex I by itself remains uncertain, but recent best estimates place it as high as 4 H⁺/2 e[−]. On the basis of this value, the stoichiometry of transport for the pathway from NADH to O₂ is 10 H⁺/2 e[−]. Although this is the value assumed in Figure 21.21, it is important to realize that this represents a consensus drawn from many experiments.

21.8 • The Thermodynamic View of Chemiosmotic Coupling

Peter Mitchell’s chemiosmotic hypothesis revolutionized our thinking about the energy coupling that drives ATP synthesis by means of an electrochemical gradient. How much energy is stored in this electrochemical gradient? For the transmembrane flow of protons across the inner membrane (from inside [matrix] to outside), we could write



The free energy difference for protons across the inner mitochondrial membrane includes a term for the concentration difference and a term for the electrical potential. This is expressed as

$$\Delta G = RT \ln \frac{[c_2]}{[c_1]} + Z\mathcal{F}\Delta\Psi \tag{21.27}$$

Oxidative Phosphorylation—The Clash of Ideas and Energetic Personalities

For many years, the means by which electron transport and ATP synthesis are coupled was unknown. It is no exaggeration to say that the search for the coupling mechanism was one of the largest, longest, most bitter fights in the history of biochemical research. Since 1777, when the French chemist Lavoisier determined that foods undergo oxidative combustion in the body, chemists and biochemists have wondered how energy from food is captured by living things. A piece of the puzzle fell into place in 1929, when Fiske and Subbarow first discovered and studied adenosine 5'-triphosphate in muscle extracts. Soon it was understood that ATP hydrolysis provides the energy for muscle contraction and other processes.

Engelhardt's experiments in 1930 led to the notion that ATP is synthesized as the result of electron transport, and, by 1940, Severo Ochoa had carried out a measurement of the **P/O ratio**, the number of molecules of ATP generated per atom of oxygen consumed in the electron transport chain. Because two electrons are transferred down the chain per oxygen atom reduced, the P/O ratio also reflects the ratio of ATPs synthesized per pair of electrons consumed. After many tedious and careful measurements, scientists decided that the P/O ratio was 3 for NADH oxidation and 2 for succinate (that is, [FADH₂]) oxidation. Electron flow and ATP synthesis are very tightly coupled in the sense that, in normal mitochondria, neither occurs without the other.

A High-Energy Chemical Intermediate Coupling Oxidation and Phosphorylation Proved Elusive

Many models were proposed to account for the coupling of electron transport and ATP synthesis. A persuasive model, advanced by E. C. Slater in 1953, proposed that energy derived from electron transport was stored in a **high-energy intermediate** (symbolized as X~P). This chemical species—in essence an activated form of phosphate—functioned according to certain relations according to Equations (21.22)–(21.25) (see below) to drive ATP synthesis.

This hypothesis was based on the model of substrate-level phosphorylation in which a high-energy substrate intermediate is a precursor to ATP. A good example is the 3-phosphoglycerate kinase reaction of glycolysis, where 1,3-bisphosphoglycerate serves as a high-energy intermediate leading to ATP. Literally hundreds of attempts were made to isolate the high-energy intermediate, X~P. Among the scientists involved in the research, rumors that one group or another had isolated X~P circulated frequently, but none was substantiated. Eventually it became clear that the intermediate could not be isolated because it did not exist.

Peter Mitchell's Chemiosmotic Hypothesis

In 1961, Peter Mitchell proposed a novel coupling mechanism involving a proton gradient across the inner mitochondrial membrane. In Mitchell's **chemiosmotic hypothesis**, protons are driven across the membrane from the matrix to the intermembrane

space and cytosol by the events of electron transport. This mechanism stores the energy of electron transport in an **electrochemical potential**. As protons are driven out of the matrix, the pH rises and the matrix becomes negatively charged with respect to the cytosol (Figure 21.22). Proton pumping thus creates a pH gradient and an electrical gradient across the inner membrane, both of which tend to attract protons back into the matrix from the cytoplasm. Flow of protons down this electrochemical gradient, an energetically favorable process, then drives the synthesis of ATP.

Paul Boyer and the Conformational Coupling Model

Another popular model invoked what became known as **conformational coupling**. If the energy of electron transport was not stored in some high-energy intermediate, perhaps it was stored in a **high-energy protein conformation**. Proposed by Paul Boyer, this model suggested that reversible conformation changes transferred energy from proteins of the electron transport chain to the enzymes involved in ATP synthesis. This model was consistent with some of the observations made by others, and it eventually evolved into the **binding change mechanism** (the basis for the model in Figure 21.28). Boyer's model is supported by a variety of binding experiments and is essentially consistent with Mitchell's chemiosmotic hypothesis.

Electron transport drives H⁺ out and creates an electrochemical gradient

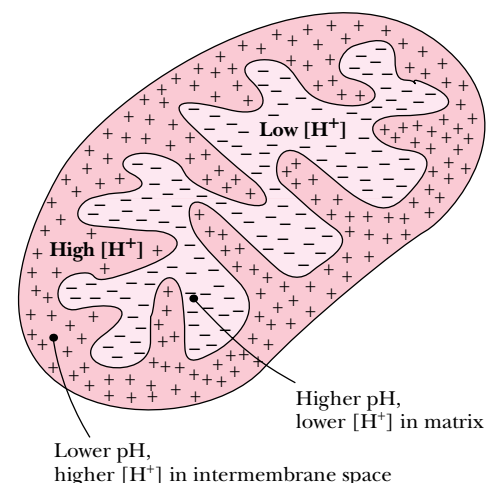
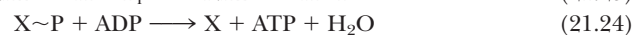
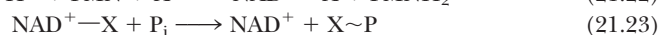


FIGURE 21.22 • The proton and electrochemical gradients existing across the inner mitochondrial membrane. The electrochemical gradient is generated by the transport of protons across the membrane.



Net reaction:



where c_1 and c_2 are the proton concentrations on the two sides of the membrane, Z is the charge on a proton, $\mathcal{F}$ is Faraday’s constant, and $\Delta\Psi$ is the potential difference across the membrane. For the case at hand, this equation becomes

$$\Delta G = RT \ln \frac{[H^+]_{out}}{[H^+]_{in}} + \mathcal{F}\Delta\Psi \tag{21.28}$$

In terms of the matrix and cytoplasm pH values, the free energy difference is

$$\Delta G = -2.303RT(\text{pH}_{out} - \text{pH}_{in}) + \mathcal{F}\Delta\Psi \tag{21.29}$$

Reported values for $\Delta\Psi$ and ΔpH vary, but the membrane potential is always found to be positive outside and negative inside, and the pH is always more acidic outside and more basic inside. Taking typical values of $\Delta\Psi = 0.18\text{ V}$ and $\Delta\text{pH} = 1$ unit, the free energy change associated with the movement of one mole of protons from inside to outside is

$$\Delta G = 2.3RT + \mathcal{F}(0.18\text{ V}) \tag{21.30}$$

With $\mathcal{F} = 96.485\text{ kJ/V}\cdot\text{mol}$, the value of ΔG at 37°C is

$$\Delta G = 5.9\text{ kJ} + 17.4\text{ kJ} = 23.3\text{ kJ} \tag{21.31}$$

which is the free energy change for movement of a mole of protons across a typical inner membrane. Note that the free energy terms for *both* the pH difference and the potential difference are unfavorable for the outward transport of protons, with the latter term making the greater contribution. On the other hand, the ΔG for *inward* flow of protons is -23.3 kJ/mol . It is this energy that drives the synthesis of ATP, in accord with Mitchell’s model. Peter Mitchell was awarded the Nobel Prize in chemistry in 1978.



FIGURE 21.23 • Electron micrograph of sub-mitochondrial particles showing the 8.5-nm projections or particles on the inner membrane, eventually shown to be F_1 –ATP synthase. (Parsons, D. F., 1963. *Science* **140**:985)

21.9 • ATP Synthase

The mitochondrial complex that carries out ATP synthesis is called **ATP synthase** or sometimes **F_1F_0 –ATPase** (for the reverse reaction it catalyzes). ATP synthase was observed in early electron micrographs of submitochondrial particles (prepared by sonication of inner membrane preparations) as round, 8.5-nm-diameter projections or particles on the inner membrane (Figure 21.23). In micrographs of native mitochondria, the projections appear on the matrix-facing surface of the inner membrane. Mild agitation removes the particles from isolated membrane preparations, and the isolated spherical particles catalyze ATP hydrolysis, the reverse reaction of the ATP synthase. Stripped of these particles, the membranes can still carry out electron transfer but cannot synthesize ATP. In one of the first *reconstitution* experiments with membrane proteins, Efraim Racker showed that adding the particles back to stripped membranes restored electron transfer–dependent ATP synthesis.

ATP Synthase Consists of Two Complexes— F_1 and F_0

ATP synthase actually consists of two principal complexes. The spheres observed in electron micrographs make up the **F_1 unit**, which catalyzes ATP synthesis. These F_1 spheres are attached to an integral membrane protein aggregate called the **F_0 unit**. F_1 consists of five polypeptide chains named α , β , γ , δ , and ϵ , with a subunit stoichiometry $\alpha_3\beta_3\gamma\delta\epsilon$ (Table 21.3). F_0 consists of three hydrophobic subunits denoted by a , b , and c , with an apparent stoichiometry of $a_1b_2c_{9-12}$. F_0 forms the transmembrane pore or channel through which protons move to drive ATP synthesis. The α , β , γ , δ , and ϵ subunits of F_1 contain 510, 482, 272, 146, and 50 amino acids, respectively, with a total molecular mass

Table 21.3
Escherichia coli F₁F₀ ATP Synthase Subunit Organization

Complex	Protein Subunit	Mass (kD)	Stoichiometry
F ₁	α	55	3
	β	52	3
	γ	30	1
	δ	15	1
	ε	5.6	1
F ₀	a	30	1
	b	17	2
	c	8	9–12

for F₁ of 371 kD. The α and β subunits are homologous, and each of these subunits bind a single ATP. The catalytic sites are in the β subunits; the function of the ATP sites in the α subunits is unknown (deletion of the sites does not affect activity).

John Walker and his colleagues have determined the structure of the F₁ complex (Figure 21.24). The F₁–ATPase is an inherently asymmetrical structure, with the three β subunits having three different conformations. In the structure solved by Walker, one of the β-subunit ATP sites contains AMP-PNP (a nonhydrolyzable analog of ATP), and another contains ADP, with the third site being empty. This state is consistent with the **binding change mechanism** for ATP synthesis proposed by Paul Boyer, in which three reaction sites cycle concertedly through the three intermediate states of ATP synthesis (take a look at Figure 21.28 on page 697).

How might such cycling occur? Important clues have emerged from several experiments that show that the γ subunit rotates with respect to the αβ complex. How such rotation might be linked to transmembrane proton flow and ATP synthesis is shown in Figure 21.25. In this model, the c subunits of F₀

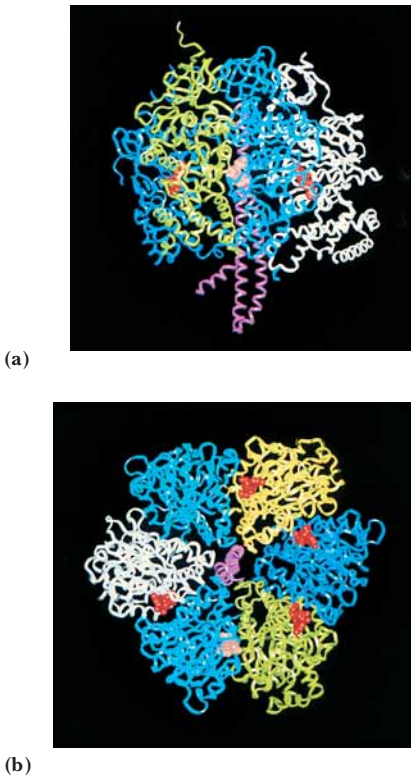


FIGURE 21.24 • Molecular graphic images (a) side view and (b) top view of the F₁–ATP synthase showing the individual component peptides. The γ-subunit is the pink structure visible in the center of view (b).

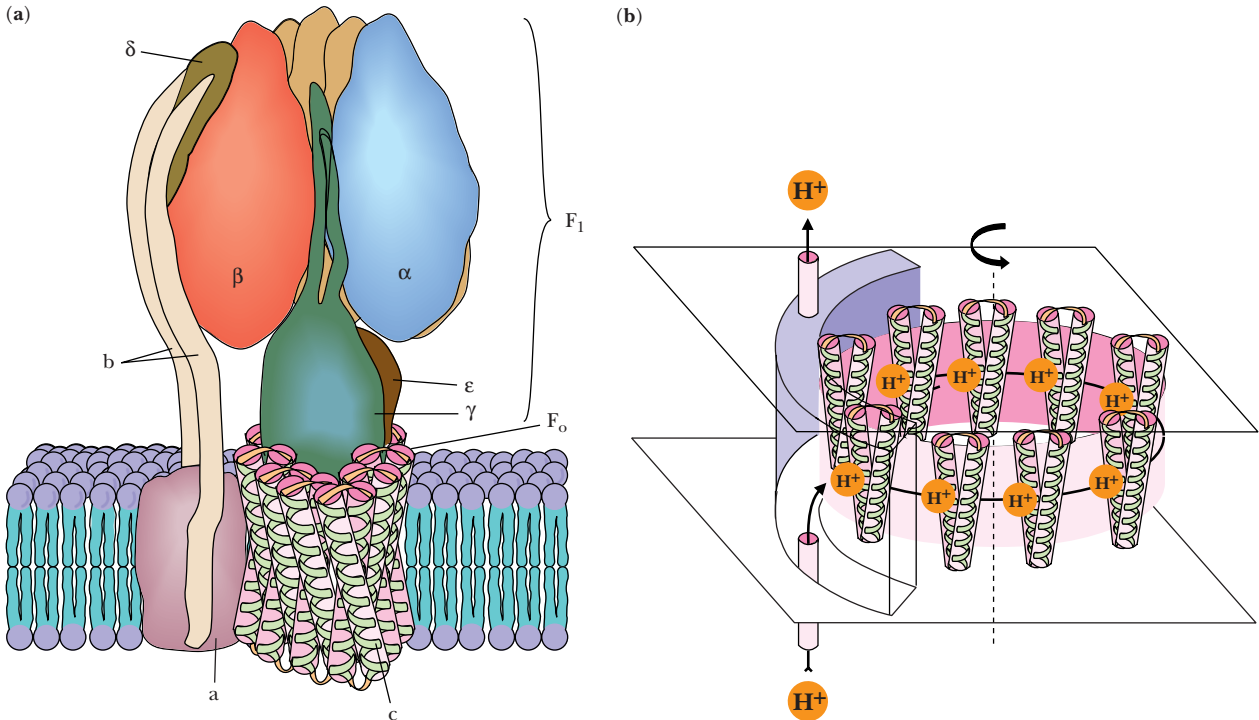


FIGURE 21.25 • A model of the F₁ and F₀ components of the ATP synthase, a rotating molecular motor. The a, b, α, β, and δ subunits constitute the stator of the motor, and the c, γ, and ε subunits form the rotor. Flow of protons through the structure turns the rotor and drives the cycle of conformational changes in α and β that synthesize ATP.

are arranged in a ring. Several lines of evidence suggest that each *c* subunit consists of a pair of antiparallel transmembrane helices with a short hairpin loop on the cytosolic side of the membrane. A ring of *c* subunits could form a **rotor** that turns with respect to the *a* subunit, a **stator** consisting of five transmembrane α -helices with proton access channels on either side of the membrane. The γ subunit is postulated to be the link between F_1 and F_0 . Several experiments have shown that γ rotates relative to the $(\alpha\beta)_3$ complex during ATP synthesis. If γ is anchored to the *c* subunit rotor, then the *c* rotor– γ complex can rotate together relative to the $(\alpha\beta)_3$ complex. Subunit *b* possesses a single transmembrane segment and a long hydrophilic head domain, and the complete stator may consist of the *b* subunits anchored at one end to the *a* subunit and linked at the other end to the $(\alpha\beta)_3$ complex via the δ subunit, as shown in Figure 21.25.

What, then, is the mechanism for ATP synthesis? The *c* rotor subunits each carry an essential residue, Asp⁶¹. (Changing this residue to Asn abolishes ATP synthase activity.) Rotation of the *c* rotor relative to the stator may depend upon neutralization of the negative charge on each *c* subunit Asp⁶¹ as the rotor turns. Protons taken up from the cytosol by one of the proton access channels in *a* could protonate an Asp⁶¹ and then ride the rotor until they reach the other proton access channel on *a*, from which they would be released into the matrix. Such rotation would cause the γ subunit to turn relative to the three β -subunit nucleotide sites of F_1 , changing the conformation of each in sequence, so that ADP is first bound, then phosphorylated, then released, according to Boyer’s binding change mechanism. Paul Boyer and John Walker shared in the 1997 Nobel Prize for chemistry for their work on the structure and mechanism of ATP synthase.

Boyer’s ¹⁸O Exchange Experiment Identified the Energy-Requiring Step

The elegant studies by Paul Boyer of ¹⁸O exchange in ATP synthase have provided other important insights into the mechanism of the enzyme. Boyer and his colleagues studied the ability of the synthase to incorporate labeled oxygen from H₂¹⁸O into P_i. This reaction (Figure 21.26) occurs via synthesis of ATP from ADP and P_i, followed by hydrolysis of ATP with incorporation of oxygen atoms from the solvent. Although *net production of ATP* requires coupling with a proton gradient, Boyer observed that this *exchange reaction* occurs readily, even in the absence of a proton gradient. His finding indicated that the formation of *enzyme-bound ATP* does not require energy. Indeed, movement of protons through the F_0 channel causes the *release of newly synthesized ATP* from the enzyme. Thus, the energy provided by electron transport creates a proton gradient that drives enzyme conformational changes resulting in the binding of

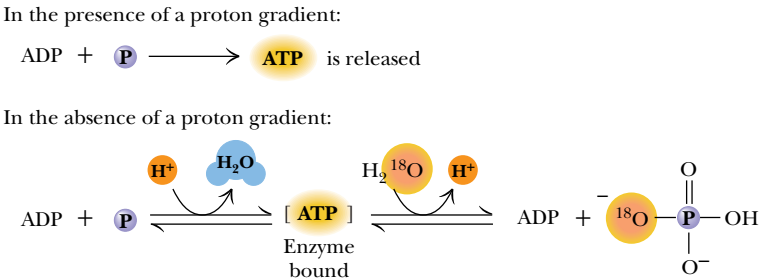


FIGURE 21.26 • ATP production in the presence of a proton gradient and ATP/ADP exchange in the absence of a proton gradient. Exchange leads to incorporation of ¹⁸O in phosphate as shown.

FIGURE 21.27 • The binding change mechanism for ATP synthesis by ATP synthase. This model assumes that F_1 has three interacting and conformationally distinct active sites. The open (O) conformation is inactive and has a low affinity for ligands; the L conformation (with “loose” affinity for ligands) is also inactive; the tight (T) conformation is active and has a high affinity for ligands. Synthesis of ATP is initiated (step 1) by binding of ADP and P_i to an L site. In the second step, an energy-driven conformational change converts the L site to a T conformation and also converts T to O and O to L. In the third step, ATP is synthesized at the T site and released from the O site. Two additional passes through this cycle produce two more ATPs and return the enzyme to its original state.

substrates on ATP synthase, ATP synthesis, and the release of products. The mechanism involves catalytic cooperativity between three interacting sites (Figure 21.27).

Racker and Stoeckenius Confirmed the Mitchell Model in a Reconstitution Experiment

When Mitchell first described his chemiosmotic hypothesis in 1961, little evidence existed to support it, and it was met with considerable skepticism by the scientific community. Eventually, however, considerable evidence accumulated to support this model. It is now clear that the electron transport chain generates a proton gradient, and careful measurements have shown that ATP is synthesized when a pH gradient is applied to mitochondria that cannot carry out electron transport. Even more relevant is a simple but crucial experiment reported in 1974 by Efraim Racker and Walther Stoeckenius, which provided specific confirmation of the Mitchell hypothesis. In this experiment, the bovine mitochondrial *ATP synthase* was reconstituted in simple lipid vesicles with **bacteriorhodopsin**, a light-driven proton pump from *Halobacterium halobium*. As shown in Figure 21.28, upon illumination, bacteriorhodopsin pumped protons

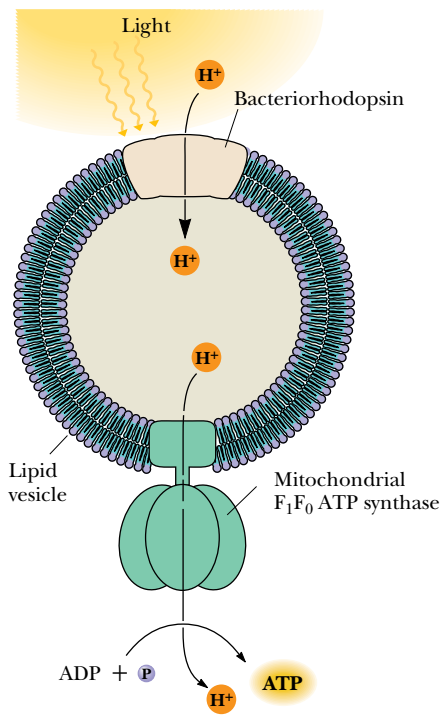
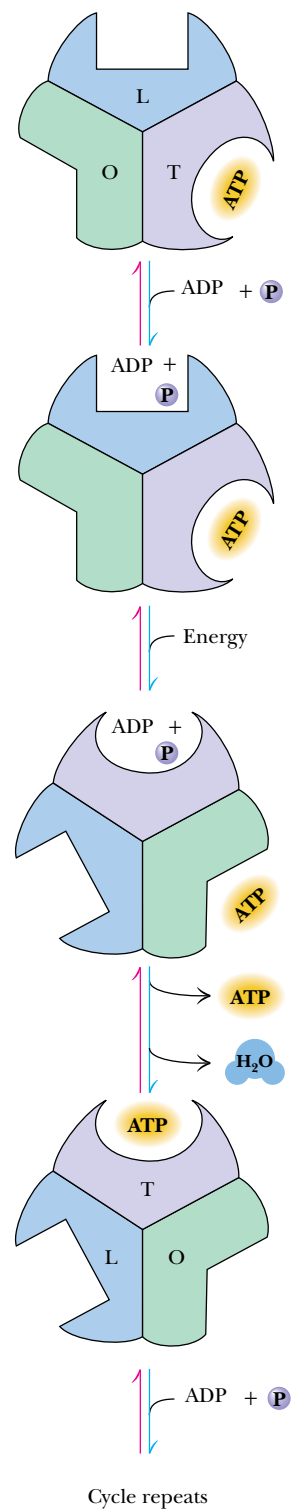


FIGURE 21.28 • The reconstituted vesicles containing ATP synthase and bacteriorhodopsin used by Stoeckenius and Racker to confirm the Mitchell chemiosmotic hypothesis.



into these vesicles, and the resulting proton gradient was sufficient to drive ATP synthesis by the ATP synthase. Because the only two kinds of proteins present were one that produced a proton gradient and one that used such a gradient to make ATP, this experiment essentially verified Mitchell’s chemiosmotic hypothesis.

21.10 • Inhibitors of Oxidative Phosphorylation

The unique properties and actions of an inhibitory substance can often help to identify aspects of an enzyme mechanism. Many details of electron transport and oxidative phosphorylation mechanisms have been gained from studying the effects of particular inhibitors. Figure 21.29 presents the structures of some electron transport and oxidative phosphorylation inhibitors. The sites of inhibition by these agents are indicated in Figure 21.30.

Inhibitors of Complexes I, II, and III Block Electron Transport

Rotenone is a common insecticide that strongly inhibits the NADH–UQ reductase. Rotenone is obtained from the roots of several species of plants. Tribes in certain parts of the world have made a practice of beating the roots of trees along riverbanks to release rotenone into the water, where it paralyzes fish and makes them easy prey. Ptericidin, Amytal, and other barbiturates, mercurial

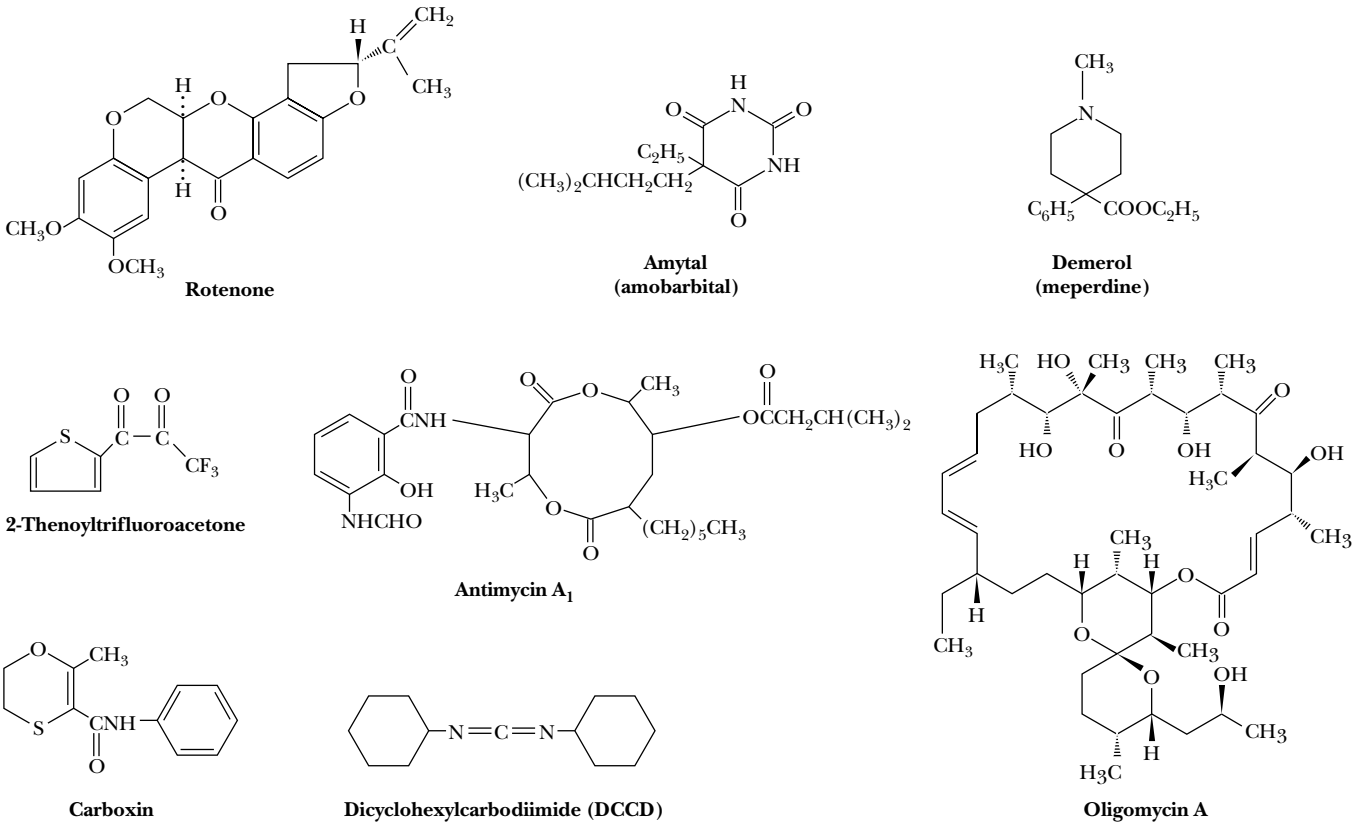


FIGURE 21.29 • The structures of several inhibitors of electron transport and oxidative phosphorylation.

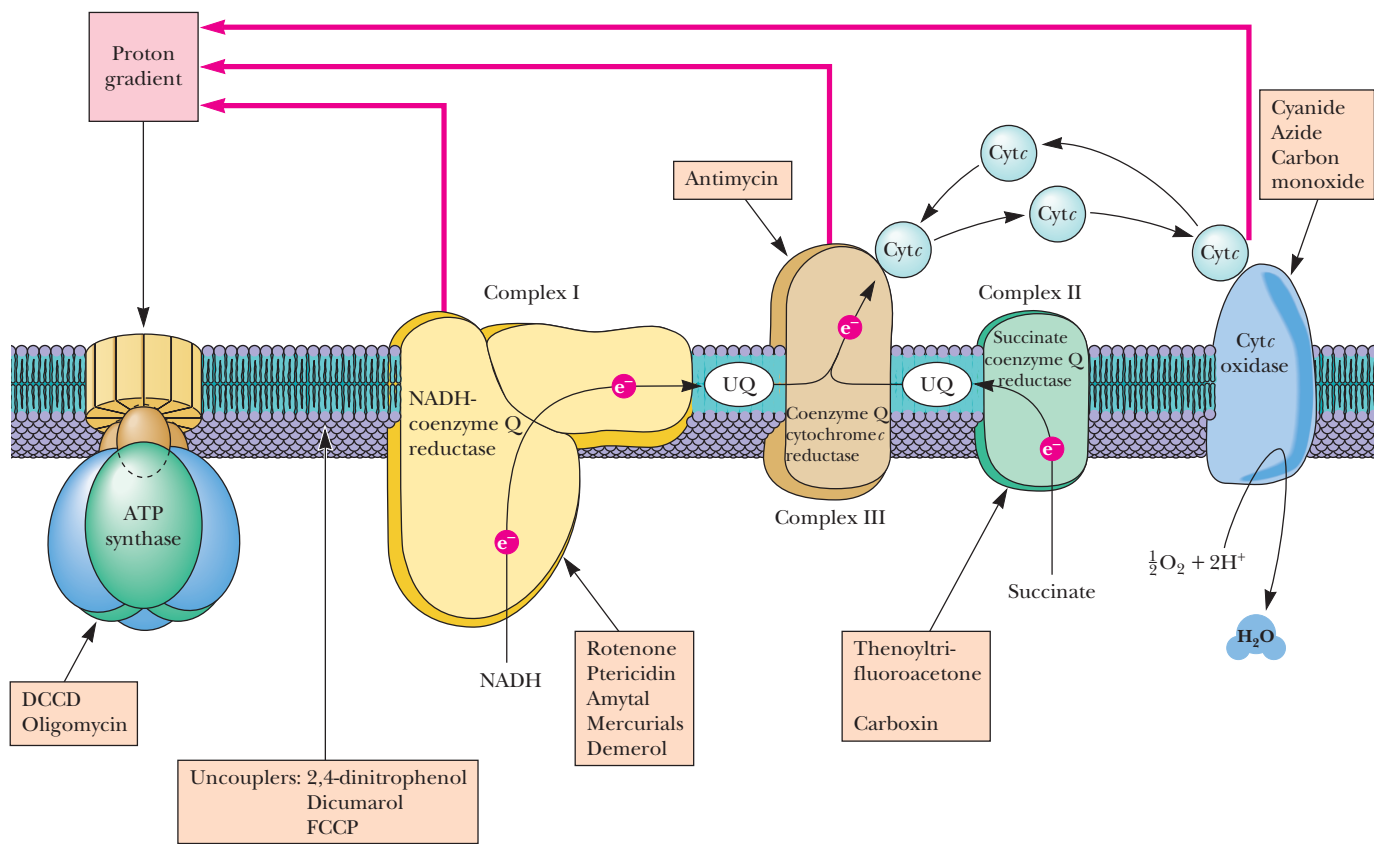


FIGURE 21.30 • The sites of action of several inhibitors of electron transport and/or oxidative phosphorylation.

agents, and the widely prescribed painkiller Demerol also exert inhibitory actions on this enzyme complex. All these substances appear to inhibit reduction of coenzyme Q and the oxidation of the Fe-S clusters of NADH–UQ reductase.

2-Thenoyltrifluoroacetone and carboxin and its derivatives specifically block Complex II, the succinate–UQ reductase. Antimycin, an antibiotic produced by *Streptomyces griseus*, inhibits the UQ–cytochrome *c* reductase by blocking electron transfer between b_H and coenzyme Q in the Q_n site. Myxothiazol inhibits the same complex by acting at the Q_p site.

Cyanide, Azide, and Carbon Monoxide Inhibit Complex IV

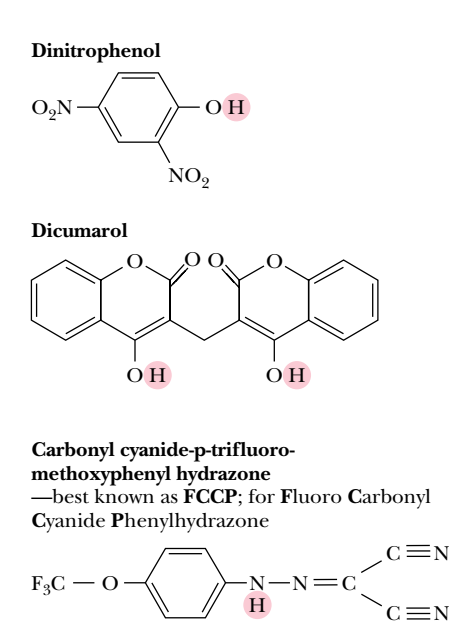
Complex IV, the cytochrome *c* oxidase, is specifically inhibited by cyanide (CN^-), azide (N_3^-), and carbon monoxide (CO). Cyanide and azide bind tightly to the ferric form of cytochrome a_3 , whereas carbon monoxide binds only to the ferrous form. The inhibitory actions of cyanide and azide at this site are very potent, whereas the principal toxicity of carbon monoxide arises from its affinity for the iron of hemoglobin. Herein lies an important distinction between the poisonous effects of cyanide and carbon monoxide. Because animals (including humans) carry many, many hemoglobin molecules, they must inhale a large quantity of carbon monoxide to die from it. These same organisms, however, possess comparatively few molecules of cytochrome a_3 . Consequently, a limited exposure to cyanide can be lethal. The sudden action of cyanide attests to the organism’s constant and immediate need for the energy supplied by electron transport.

Oligomycin and DCCD Are ATP Synthase Inhibitors

Inhibitors of ATP synthase include dicyclohexylcarbodiimide (DCCD) and oligomycin (Figure 21.29). DCCD bonds covalently to carboxyl groups in hydrophobic domains of proteins in general, and to a glutamic acid residue of the *c* subunit of F₀, the proteolipid forming the proton channel of the ATP synthase, in particular. If the *c* subunit is labeled with DCCD, proton flow through F₀ is blocked and ATP synthase activity is inhibited. Likewise, oligomycin acts directly on the ATP synthase. By binding to a subunit of F₀, oligomycin also blocks the movement of protons through F₀.

21.11 • Uncouplers Disrupt the Coupling of Electron Transport and ATP Synthase

Another important class of reagents affects ATP synthesis, but in a manner that does not involve direct binding to any of the proteins of the electron transport chain or the F₁F₀–ATPase. These agents are known as **uncouplers** because they disrupt the tight coupling between electron transport and the ATP synthase. Uncouplers act by dissipating the proton gradient across the inner mitochondrial membrane created by the electron transport system. Typical examples include 2,4-dinitrophenol, dicumarol, and carbonyl cyanide-*p*-trifluoromethoxyphenyl hydrazone (perhaps better known as fluorocarbonyl-cyanide phenylhydrazone or FCCP) (Figure 21.31). These compounds share two common features: hydrophobic character and a dissociable proton. As uncouplers, they function by carrying protons across the inner membrane. Their tendency is to acquire protons on the cytosolic surface of the membrane (where the proton concentration is high) and carry them to the matrix side, thereby destroying the proton gradient that couples electron transport and the ATP synthase. In mitochondria treated with uncouplers, electron transport continues, and protons are driven out through the inner membrane. However, they leak back in so rapidly via the uncouplers that ATP synthesis does not occur. Instead, the energy released in electron transport is dissipated as heat.



Endogenous Uncouplers Enable Organisms To Generate Heat

Ironically, certain cold-adapted animals, hibernating animals, and newborn animals generate large amounts of heat by uncoupling oxidative phosphorylation. Adipose tissue in these organisms contains so many mitochondria that it is called *brown adipose tissue* for the color imparted by the mitochondria. The inner membrane of brown adipose tissue mitochondria contains an endogenous protein called **thermogenin** (literally, “heat maker”), or *uncoupling protein*, that creates a passive proton channel through which protons flow from the cytosol to the matrix. Certain plants also use the heat of uncoupled proton transport for a special purpose. Skunk cabbage and related plants contain floral spikes that are maintained as much as 20 degrees above ambient temperature in this way. The warmth of the spikes serves to vaporize odiferous molecules, which attract insects that fertilize the flowers.

FIGURE 21.31 • Structures of several uncouplers, molecules that dissipate the proton gradient across the inner mitochondrial membrane and thereby destroy the tight coupling between electron transport and the ATP synthase reaction.

21.12 • ATP Exits the Mitochondria via an ATP–ADP Translocase

ATP, the cellular energy currency, must exit the mitochondria to carry energy throughout the cell, and ADP must be brought into the mitochondria for reprocessing. Neither of these processes occurs spontaneously because the highly charged ATP and ADP molecules do not readily cross biological membranes. Instead, these processes are mediated by a single transport system, the **ATP–ADP translocase**. This protein tightly couples the exit of ATP with the entry of ADP so that the mitochondrial nucleotide levels remain approximately constant. For each ATP transported out, one ADP is transported into the matrix. The translocase, which accounts for approximately 14% of the total mitochondrial membrane protein, is a homodimer of 30-kD subunits. Transport occurs via a single nucleotide-binding site, which alternately faces the matrix and the cytosol (Figure 21.32). It binds ATP on the matrix side, reorients to face the cytosol, and exchanges ATP for ADP, with subsequent movement back to the matrix face of the inner membrane.

Outward Movement of ATP Is Favored over Outward ADP Movement

The charge on ATP at pH 7.2 or so is about -4 , and the charge on ADP at the same pH is about -3 . Thus, net exchange of an ATP (out) for an ADP (in) results in the net movement of one negative charge from the matrix to the cytosol. (This process is equivalent to the movement of a proton from the cytosol to the matrix.) Recall that the inner membrane is positive outside (see Figure 21.22), and it becomes clear that outward movement of ATP is favored over outward ADP transport, ensuring that ATP will be transported out (Figure 21.32). Inward movement of ADP is favored over inward movement of ATP for the same reason. Thus, the membrane electrochemical potential itself controls the specificity of the ATP–ADP translocase. However, the electrochemical potential is diminished by the ATP–ADP translocase cycle and therefore operates with an energy cost to the cell. The cell must compensate by passing yet more electrons down the electron transport chain.

What is the cost of ATP–ADP exchange relative to the energy cost of ATP synthesis itself? We already noted that moving 1 ATP out and 1 ADP in is the equivalent of one proton moving from the cytosol to the matrix. *Synthesis of an*

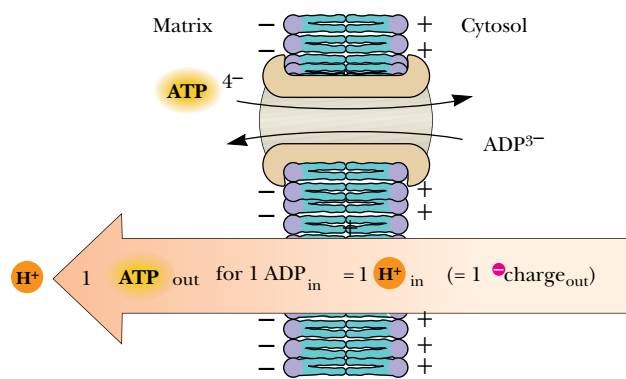


FIGURE 21.32 • Outward transport of ATP (via the ATP/ADP translocase) is favored by the membrane electrochemical potential.

ATP results from the movement of approximately three protons from the cytosol into the matrix through F_0 . Altogether this means that approximately four protons are transported into the matrix per ATP synthesized. Thus, approximately one-fourth of the energy derived from the respiratory chain (electron transport and oxidative phosphorylation) is expended as the electrochemical energy devoted to mitochondrial ATP–ADP transport.

21.13 • What Is the P/O Ratio for Mitochondrial Electron Transport and Oxidative Phosphorylation?

The **P/O ratio** is the number of molecules of ATP formed in oxidative phosphorylation per two electrons flowing through a defined segment of the electron transport chain. In spite of intense study of this ratio, its actual value remains a matter of contention. If we accept the value of 10 H^+ transported out of the matrix per 2 e^- passed from NADH to O_2 through the electron transport chain, and also agree (as above) that 4 H^+ are transported into the matrix per ATP synthesized (and translocated), then the mitochondrial P/O ratio is 10/4, or 2.5, for the case of electrons entering the electron transport chain as NADH. This is somewhat lower than earlier estimates, which placed the P/O ratio at 3 for mitochondrial oxidation of NADH. For the portion of the chain from succinate to O_2 , the $H^+/2 e^-$ ratio is 6 (as noted above), and the P/O ratio in this case would be 6/4, or 1.5; earlier estimates placed this number at 2. The consensus of experimental measurements of P/O ratios for these two cases has been closer to the more modern values of 2.5 and 1.5. Many chemists and biochemists, accustomed to the integral stoichiometries of chemical and metabolic reactions, have been reluctant to accept the notion of nonintegral P/O ratios. At some point, as we learn more about these complex coupled processes, it may be necessary to reassess the numbers.

21.14 • Shuttle Systems Feed the Electrons of Cytosolic NADH into Electron Transport

$$\left(\frac{1 \text{ ATP}}{4 H^+}\right)\left(\frac{10 H^+}{2 e^- [\text{NADH} \rightarrow \frac{1}{2}O_2]}\right) = \frac{10}{4} = \frac{P}{O}$$

Most of the NADH used in electron transport is produced in the mitochondrial matrix space, an appropriate site because NADH is oxidized by Complex I on the matrix side of the inner membrane. Furthermore, the inner mitochondrial membrane is impermeable to NADH. Recall, however, that NADH is produced in glycolysis by glyceraldehyde-3-P dehydrogenase in the cytosol. If this NADH were not oxidized to regenerate NAD^+ , the glycolytic pathway would cease to function due to NAD^+ limitation. Eukaryotic cells have a number of *shuttle systems* that harvest the electrons of cytosolic NADH for delivery to mitochondria without actually transporting NADH across the inner membrane (Figures 21.33 and 21.34).

The Glycerophosphate Shuttle Ensures Efficient Use of Cytosolic NADH

In the **glycerophosphate shuttle**, two different **glycerophosphate dehydrogenases**, one in the cytoplasm and one on the outer face of the mitochondrial inner membrane, work together to carry electrons into the mitochondrial matrix (Figure 21.32). NADH produced in the cytosol transfers its electrons to *dihydroxyacetone phosphate*, thus reducing it to *glycerol-3-phosphate*. This metabolite is reoxidized by the FAD^+ -dependent mitochondrial membrane enzyme to

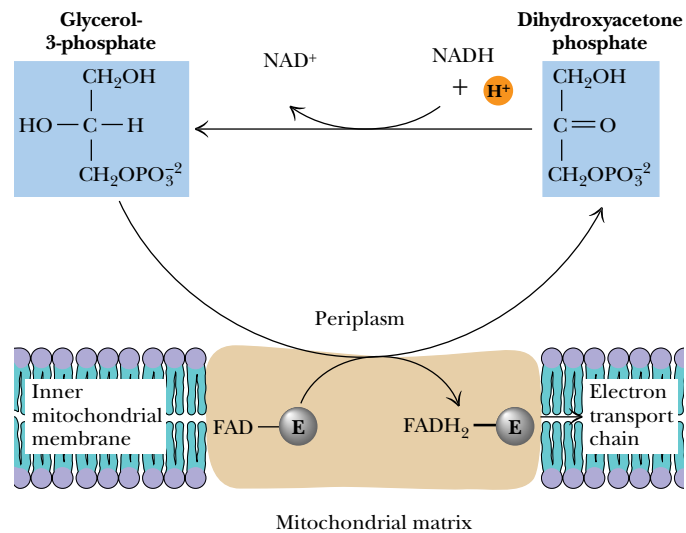


FIGURE 21.33 • The glycerophosphate shuttle (also known as the glycerol phosphate shuttle) couples the cytosolic oxidation of NADH with mitochondrial reduction of [FAD].

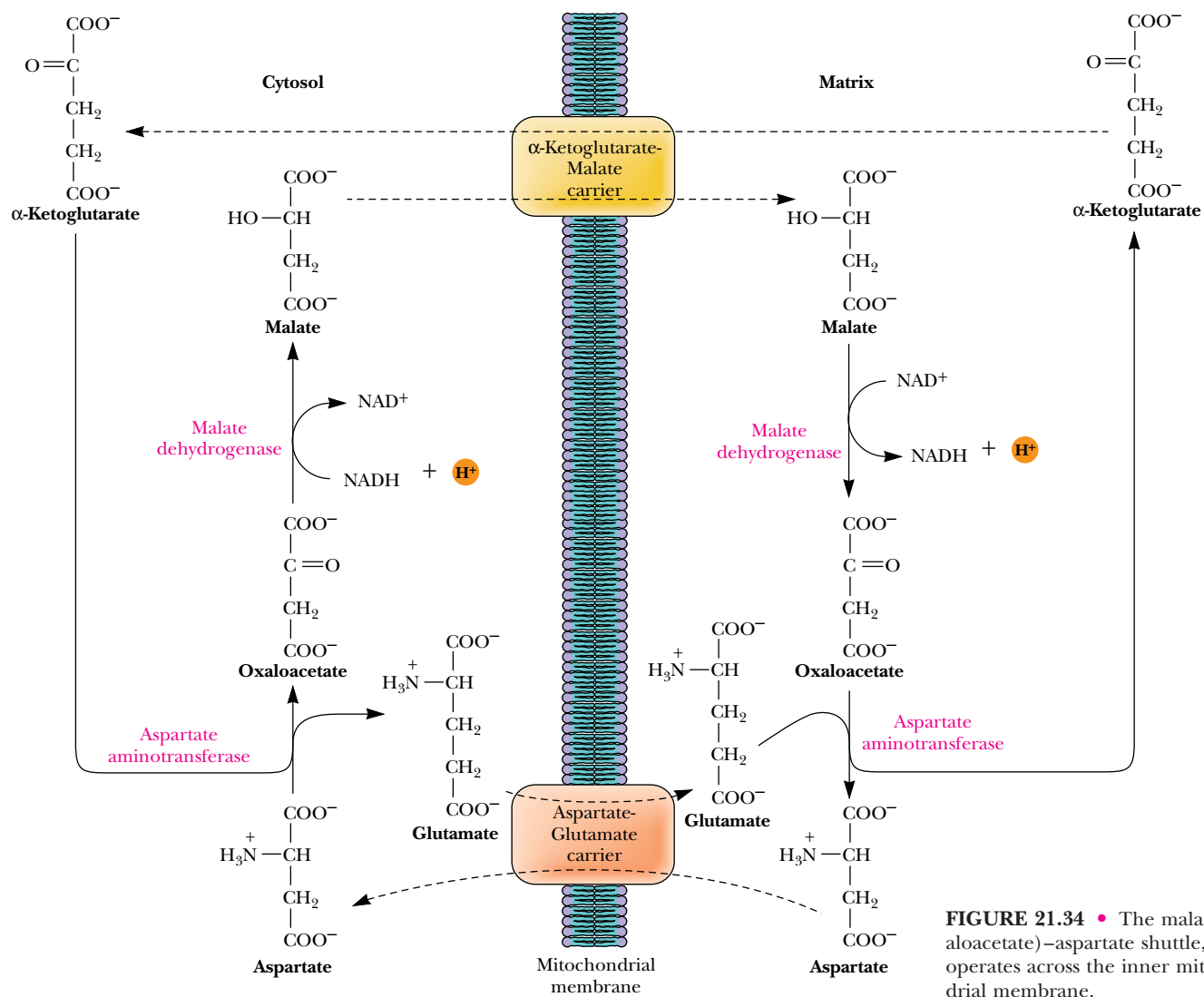


FIGURE 21.34 • The malate (oxaloacetate)-aspartate shuttle, which operates across the inner mitochondrial membrane.

reform dihydroxyacetone phosphate and enzyme-bound FADH_2 . The two electrons of $[\text{FADH}_2]$ are passed directly to UQ , forming UQH_2 . Thus, via this shuttle, cytosolic NADH can be used to produce mitochondrial $[\text{FADH}_2]$ and, subsequently, UQH_2 . As a result, cytosolic NADH oxidized via this shuttle route yields only 1.5 molecules of ATP . The cell “pays” with a potential ATP molecule for the convenience of getting cytosolic NADH into the mitochondria. Although this may seem wasteful, there is an important payoff. The glycerophosphate shuttle is essentially irreversible, and even when NADH levels are very low relative to NAD^+ , the cycle operates effectively.

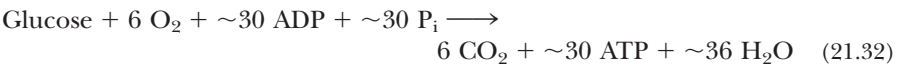
The Malate–Aspartate Shuttle Is Reversible

The second electron shuttle system, called the **malate–aspartate shuttle**, is shown in Figure 21.34. Oxaloacetate is reduced in the cytosol, acquiring the electrons of NADH (which is oxidized to NAD^+). Malate is transported across the inner membrane, where it is reoxidized by malate dehydrogenase, converting NAD^+ to NADH in the matrix. This mitochondrial NADH readily enters the electron transport chain. The oxaloacetate produced in this reaction cannot cross the inner membrane and must be transaminated to form *aspartate*, which can be transported across the membrane to the cytosolic side. Transamination in the cytosol recycles aspartate back to oxaloacetate. In contrast to the glycerol phosphate shuttle, the malate–aspartate cycle is reversible, and it operates as shown in Figure 21.34 only if the NADH/NAD^+ ratio in the cytosol is higher than the ratio in the matrix. Because this shuttle produces NADH in the matrix, the full 2.5 ATPs per NADH are recovered.

The Net Yield of ATP from Glucose Oxidation Depends on the Shuttle Used

The complete route for the conversion of the metabolic energy of glucose to ATP has now been described, in Chapters 19 through 21. Assuming appropriate P/O ratios, the number of ATP molecules produced by the complete oxidation of a molecule of glucose can be estimated. Keeping in mind that P/O ratios must be viewed as approximate, for all the reasons previously cited, we will assume the values of 2.5 and 1.5 for the mitochondrial oxidation of NADH and succinate, respectively. In eukaryotic cells, the combined pathways of glycolysis, the TCA cycle, electron transport, and oxidative phosphorylation then yield a net of approximately 30 to 32 molecules of ATP per molecule of glucose oxidized, depending on the shuttle route employed (Table 21.4).

The net stoichiometric equation for the oxidation of glucose, using the glycerol phosphate shuttle, is



Because the 2 NADH formed in glycolysis are “transported” by the glycerol phosphate shuttle in this case, they each yield only 1.5 ATP , as already described. On the other hand, if these 2 NADH take part in the malate–aspartate shuttle, each yields 2.5 ATP , giving a total (in this case) of 32 ATP formed per glucose oxidized. Most of the ATP —26 out of 30 or 28 out of 32—is produced by oxidative phosphorylation; only 4 ATP molecules result from direct synthesis during glycolysis and the TCA cycle.

The situation in bacteria is somewhat different. Prokaryotic cells need not carry out ATP/ADP exchange. Thus, bacteria have the potential to produce approximately 38 ATP per glucose.

Table 21.4

Yield of ATP from Glucose Oxidation

Pathway	ATP Yield per Glucose	
	Glycerol- Phosphate Shuttle	Malate- Aspartate Shuttle
Glycolysis: glucose to pyruvate (cytosol)		
Phosphorylation of glucose	−1	−1
Phosphorylation of fructose-6-phosphate	−1	−1
Dephosphorylation of 2 molecules of 1,3-BPG	+2	+2
Dephosphorylation of 2 molecules of PEP	+2	+2
Oxidation of 2 molecules of glyceraldehyde-3-phosphate yields 2 NADH		
Pyruvate conversion to acetyl-CoA (mitochondria)		
2 NADH		
Citric acid cycle (mitochondria)		
2 molecules of GTP from 2 molecules of succinyl-CoA	+2	+2
Oxidation of 2 molecules each of isocitrate, α-ketoglutarate, and malate yields 6 NADH		
Oxidation of 2 molecules of succinate yields 2 [FADH ₂]		
Oxidative phosphorylation (mitochondria)		
2 NADH from glycolysis yield 1.5 ATP each if NADH is oxidized by glycerol-phosphate shuttle; 2.5 ATP by malate-aspartate shuttle	+3	+5
Oxidative decarboxylation of 2 pyruvate to 2 acetyl-CoA: 2 NADH produce 2.5 ATP each	+5	+5
2 [FADH ₂] from each citric acid cycle produce 1.5 ATP each	+3	+3
6 NADH from citric acid cycle produce 2.5 ATP each	<u>+15</u>	<u>+15</u>
<i>Net Yield</i>	30	32

Note: These P/O ratios of 2.5 and 1.5 for mitochondrial oxidation of NADH and [FADH₂] are “consensus values.” Because they may not reflect actual values and because these ratios may change depending on metabolic conditions, these estimates of ATP yield from glucose oxidation are approximate.

3.5 Billion Years of Evolution Have Resulted in a System That Is 54% Efficient

Hypothetically speaking, how much energy does a eukaryotic cell extract from the glucose molecule? Taking a value of 50 kJ/mol for the hydrolysis of ATP under cellular conditions (Chapter 3), the production of 32 ATP per glucose oxidized yields 1600 kJ/mol of glucose. The cellular oxidation (combustion) of glucose yields Δ*G* = −2937 kJ/mol. We can calculate an efficiency for the pathways of glycolysis, the TCA cycle, electron transport, and oxidative phosphorylation of

$$\frac{1600}{2937} \times 100\% = 54\%$$

This is the result of approximately 3.5 billion years of evolution.

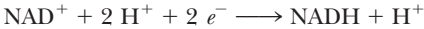
PROBLEMS

- For the following reaction,

$$[\text{FAD}] + 2 \text{ cyt } c (\text{Fe}^{2+}) + 2 \text{ H}^+ \rightleftharpoons [\text{FADH}_2] + 2 \text{ cyt } c (\text{Fe}^{3+})$$

determine which of the redox couples is the electron acceptor and which is the electron donor under standard-state conditions, calculate the value of $\Delta\mathcal{E}_\circ'$, and determine the free energy change for the reaction.

- Calculate the value of $\Delta\mathcal{E}_\circ'$ for the glyceraldehyde-3-phosphate dehydrogenase reaction, and calculate the free energy change for the reaction under standard-state conditions.
- For the following redox reaction,



suggest an equation (analogous to Equation 21.13) that predicts the pH dependence of this reaction, and calculate the reduction potential for this reaction at pH 8.

4. Sodium nitrite (NaNO_2) is used by emergency medical personnel as an antidote for cyanide poisoning (for this purpose, it must be administered immediately). Based on the discussion of cyanide poisoning in Section 21.10, suggest a mechanism for the life-saving effect of sodium nitrite.

5. A wealthy investor has come to you for advice. She has been approached by a biochemist who seeks financial backing for a company that would market dinitrophenol and dicumarol as weight-loss medications. The biochemist has explained to her that these agents are uncouplers and that they would dissipate metabolic energy as heat. The investor wants to know if you think she should invest in the biochemist's company. How do you respond?

6. Assuming that 3 H^+ are transported per ATP synthesized in the mitochondrial matrix, the membrane potential difference is 0.18 V (negative inside), and the pH difference is 1 unit (acid outside, basic inside), calculate the largest ratio of $[\text{ATP}]/[\text{ADP}][\text{P}_i]$ under which synthesis of ATP can occur.

7. Of the dehydrogenase reactions in glycolysis and the TCA cycle, all but one use NAD^+ as the electron acceptor. The lone exception is the succinate dehydrogenase reaction, which uses covalently bound FAD of a flavoprotein as the electron acceptor. The standard reduction potential for this bound FAD is in the range of 0.003 to 0.091 V (Table 21.1). Compared with the other dehydrogenase reactions of glycolysis and the TCA cycle, what is unique about succinate dehydrogenase? Why is bound FAD a more suitable electron acceptor in this case?

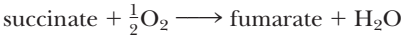
8. **a.** What is the standard free energy change ($\Delta G^\circ'$) for the reduction of coenzyme Q by NADH as carried out by complex I (NADH-coenzyme Q reductase) of the electron transport pathway if $\mathcal{E}_\circ'(\text{NAD}^+/\text{NADH} + \text{H}^+) = -0.320$ volts and $\mathcal{E}_\circ'(\text{CoQ}/\text{CoQH}_2) = +0.060$ volts.

- What is the equilibrium constant (K_{eq}) for this reaction?
- Assume that (1) the actual free energy release accompanying the NADH-coenzyme Q reductase reaction is equal to the amount released under standard conditions (as calculated above), (2) this energy can be converted into the synthesis of ATP with an efficiency = 0.75 (that is, 75% of the energy released upon NADH oxidation is captured in ATP synthesis), and (3) the oxidation of

1 equivalent of NADH by coenzyme Q leads to the phosphorylation of 1 equivalent of ATP.

Under these conditions, what is the maximum ratio of $[\text{ATP}]/[\text{ADP}]$ attainable by oxidative phosphorylation when $[\text{P}_i] = 1 \text{ mM}$? (Assume $\Delta G^\circ'$ for ATP synthesis = +30.5 kJ/mol.)

9. Consider the oxidation of succinate by molecular oxygen as carried out via the electron transport pathway

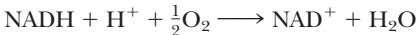


a. What is the standard free energy change ($\Delta G^\circ'$) for this reaction if $\mathcal{E}_\circ'(\text{fum/succ}) = +0.031$ volts and $\mathcal{E}_\circ'(\frac{1}{2}\text{O}_2/\text{H}_2\text{O}) = +0.816$ volts.

- What is the equilibrium constant (K_{eq}) for this reaction?
- Assume that (1) the actual free energy release accompanying succinate oxidation by the electron transport pathway is equal to the amount released under standard conditions (as calculated above), (2) this energy can be converted into the synthesis of ATP with an efficiency = 0.7 (that is, 70% of the energy released upon succinate oxidation is captured in ATP synthesis), (3) the oxidation of 1 succinate leads to the phosphorylation of 2 equivalents of ATP.

Under these conditions, what is the maximum ratio of $[\text{ATP}]/[\text{ADP}]$ attainable by oxidative phosphorylation when $[\text{P}_i] = 1 \text{ mM}$? (Assume $\Delta G^\circ'$ for ATP synthesis = +30.5 kJ/mol.)

10. Consider the oxidation of NADH by molecular oxygen as carried out via the electron transport pathway



a. What is the standard free energy change ($\Delta G^\circ'$) for this reaction if $\Delta\mathcal{E}_\circ'(\text{NAD}^+/\text{NADH}) = -0.320$ volts and $\Delta\mathcal{E}_\circ'(\frac{1}{2}\text{O}_2/\text{H}_2\text{O}) = +0.816$ volts.

- What is the equilibrium constant (K_{eq}) for this reaction?
- Assume that (1) the actual free energy release accompanying NADH oxidation by the electron transport pathway is equal to the amount released under standard conditions (as calculated above), (2) this energy can be converted into the synthesis of ATP with an efficiency = 0.75 (that is, 75% of the energy released upon NADH oxidation is captured in ATP synthesis), and (3) the oxidation of 1 NADH leads to the phosphorylation of 3 equivalents of ATP.

Under these conditions, what is the maximum ratio of $[\text{ATP}]/[\text{ADP}]$ attainable by oxidative phosphorylation when $[\text{P}_i] = 2 \text{ mM}$? (Assume $\Delta G^\circ'$ for ATP synthesis = +30.5 kJ/mol.)

11. Write a balanced equation for the reduction of molecular oxygen by reduced cytochrome *c* as carried out by complex IV (cytochrome oxidase) of the electron transport pathway.

a. What is the standard free energy change ($\Delta G^\circ'$) for this reaction if

$$\begin{aligned} \Delta\mathcal{E}_\circ' \text{ cyt } c(\text{Fe}^{3+})/\text{cyt } c(\text{Fe}^{2+}) &= +0.254 \text{ volts and} \\ \Delta\mathcal{E}_\circ'(\frac{1}{2}\text{O}_2/\text{H}_2\text{O}) &= 0.816 \text{ volts.} \end{aligned}$$

- What is the equilibrium constant (K_{eq}) for this reaction?

c. Assume that (1) the actual free energy release accompanying cytochrome *c* oxidation by the electron transport pathway is equal to the amount released under standard conditions (as calculated in section a.), (2) this energy can be converted into the synthesis of ATP with an efficiency = 0.6 (that is, 60% of the energy released upon cytochrome *c* oxidation is captured in ATP synthesis), and (3) the reduction of 1 molecule of O₂ by reduced cytochrome *c* leads to the phosphorylation of 2 equivalents of ATP.

Under these conditions, what is the maximum ratio of [ATP]/[ADP] attainable by oxidative phosphorylation when [P_i] = 3 mM? (Assume Δ*G*^{o'} for ATP synthesis = +30.5 kJ/mol.)

12. The standard reduction potential for (NAD⁺/NADH) is −0.320 volts, and the standard reduction potential for (pyruvate/lactate) is −0.185 volts.

a. What is the standard free energy change, Δ*G*^{o'}, for the lactate dehydrogenase reaction:



b. What is the equilibrium constant, *K*_{eq}, for this reaction?

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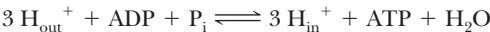
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c. If [pyruvate] = 0.05 mM and [lactate] = 2.9 mM and Δ*G* for the lactate dehydrogenase reaction = −15 kJ/mol in erythrocytes, what is the [NAD⁺]/[NADH] ratio under these conditions?

13. Assume that the free energy change, Δ*G*, associated with the movement of one mole of protons from the outside to the inside of a bacterial cell is −23 kJ/mol and 3 H⁺ must cross the bacterial plasma membrane per ATP formed by the bacterial F₁F₀–ATP synthase. ATP synthesis thus takes place by the coupled process:



a. If the overall free energy change (Δ*G*_{overall}) associated with ATP synthesis in these cells by the coupled process is −21 kJ/mol, what is the equilibrium constant, *K*_{eq}, for the process?

b. What is Δ*G*_{synthesis}, the free energy change for ATP synthesis, in these bacteria under these conditions?

c. The standard free energy change for ATP hydrolysis, Δ*G*^{o'}_{hydrolysis}, is −30.5 kJ/mol. If [P_i] = 2 mM in these bacterial cells, what is the [ATP]/[ADP] ratio in these cells?

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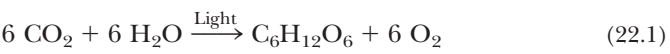
Chapter 22

Photosynthesis



“Sunflowers,” by Claude Monet (1840–1926), French, Metropolitan Museum of Art, New York City/Superstock, Inc.

The vast majority of energy consumed by living organisms stems from solar energy captured by the process of photosynthesis. Only chemolithotropic bacteria (Chapter 18) are independent of this energy source. Of the 1.5×10^{22} kJ of energy reaching the earth each day from the sun, 1% is absorbed by photosynthetic organisms and transduced into chemical energy.¹ This energy, in the form of biomolecules, becomes available to other members of the biosphere through food chains. The transduction of solar, or light, energy into chemical energy is often expressed in terms of **carbon dioxide fixation**, in which hexose is formed from carbon dioxide and oxygen is evolved:



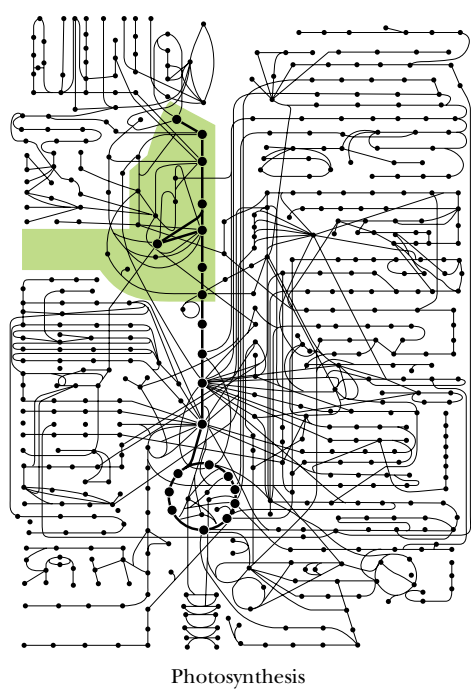
¹Of the remaining 99%, two-thirds is absorbed by the earth and oceans, thereby heating the planet; the remaining one-third is lost as light reflected back into space.

Is it so small a thing
To have enjoyed the sun,
To have lived light in the spring,
To have loved, to have thought, to have
done . . . ?

MATTHEW ARNOLD, *Empedocles on Etna* (1852)

OUTLINE

- 22.1 • General Aspects of Photosynthesis
- 22.2 • Photosynthesis Depends on the Photoreactivity of Chlorophyll
- 22.3 • Eukaryotic Phototrophs Possess Two Distinct Photosystems
- 22.4 • The Z Scheme of Photosynthetic Electron Transfer
- 22.5 • The Molecular Architecture of Photosynthetic Reaction Centers
- 22.6 • The Quantum Yield of Photosynthesis
- 22.7 • Light-Driven ATP Synthesis—Photophosphorylation
- 22.8 • Carbon Dioxide Fixation
- 22.9 • The Calvin–Benson Cycle
- 22.10 • Regulation of Carbon Dioxide Fixation
- 22.11 • The Ribulose Biphosphate Oxygenase Reaction: Photorespiration
- 22.12 • The C-4 Pathway of CO₂ Fixation
- 22.13 • Crassulacean Acid Metabolism



Estimates indicate that 10^{11} tons of carbon dioxide are fixed globally per year, of which one-third is fixed in the oceans, primarily by photosynthetic marine microorganisms.

Although photosynthesis is traditionally equated with CO_2 fixation, light energy (or rather the chemical energy derived from it) can be used to drive virtually any cellular process. The assimilation of inorganic forms of nitrogen and sulfur into organic molecules (Chapter 27) represents two other metabolic conversions driven by light energy in green plants. Our previous considerations of aerobic metabolism (Chapters 19 through 21) treated cellular respiration (precisely the reverse of Equation [22.1]) as the central energy-releasing process in life. It necessarily follows that the formation of hexose from carbon dioxide and water, the products of cellular respiration, must be endergonic. The necessary energy comes from light. Note that in the carbon dioxide fixation reaction described, light is used to drive a chemical reaction against its thermodynamic potential.

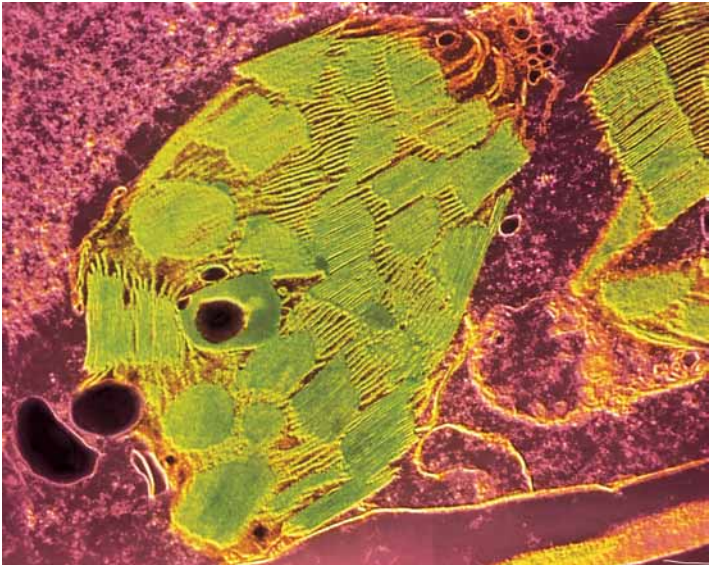
22.1 • General Aspects of Photosynthesis

Photosynthesis Occurs in Membranes

Organisms capable of photosynthesis are very diverse, ranging from simple prokaryotic forms to the largest organisms of all, *Sequoia gigantea*, the giant redwood trees of California. Despite this diversity, we find certain generalities regarding photosynthesis. An important one is that *photosynthesis occurs in membranes*. In photosynthetic prokaryotes, the photosynthetic membranes fill up the cell interior; in photosynthetic eukaryotes, the photosynthetic membranes are localized in large organelles known as **chloroplasts** (Figures 22.1 and 22.2). Chloroplasts are one member in a family of related plant-specific organelles known as **plastids**. Chloroplasts themselves show a range of diversity, from the single, spiral chloroplast that gives *Spirogyra* its name to the multitude of ellipsoidal plastids typical of higher plant cells (Figure 22.3).

Characteristic of all chloroplasts, however, is the organization of the inner membrane system, the so-called **thylakoid membrane**. The thylakoid membrane

FIGURE 22.1 • Electron micrograph of a representative chloroplast. (James Dennis/CNRI/Phototake NYC)



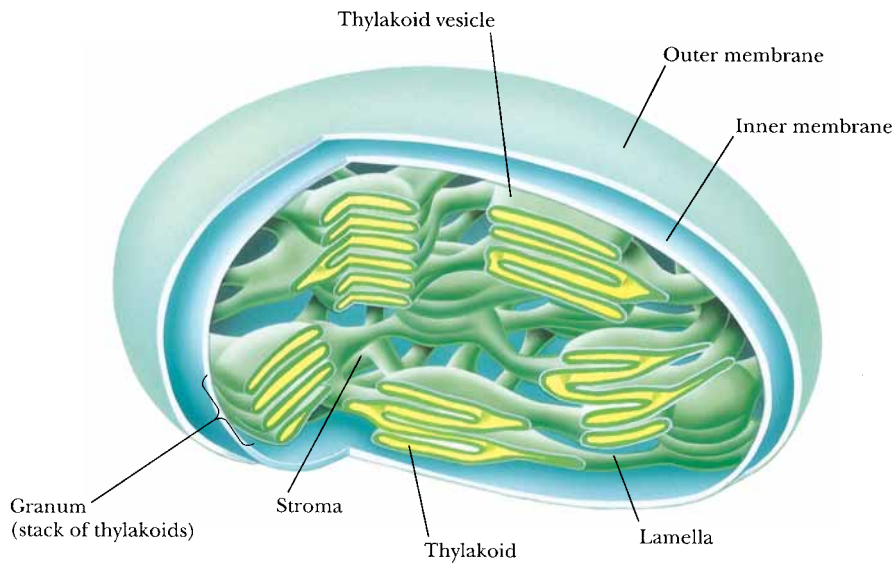


FIGURE 22.2 • Schematic diagram of an idealized chloroplast.

is organized into paired folds that extend throughout the organelle, as in Figure 22.1. These paired folds, or **lamellae**, give rise to flattened sacs or disks, **thylakoid vesicles** (from the Greek *thylakos*, meaning “sack”), which occur in stacks called **grana**. A single stack, or **granum**, may contain dozens of thylakoid vesicles, and different grana are joined by lamellae that run through the soluble portion, or **stroma**, of the organelle. Chloroplasts thus possess three membrane-bound aqueous compartments: the intermembrane space, the stroma, and the interior of the thylakoid vesicles, the so-called **thylakoid space** (also known as the **thylakoid lumen**). As we shall see, this third compartment serves an important function in the transduction of light energy into ATP formation. The thylakoid membrane has a highly characteristic lipid composition and, like the inner membrane of the mitochondrion, is impermeable to most ions and molecules. Chloroplasts, like their mitochondrial counterparts, possess DNA, RNA,

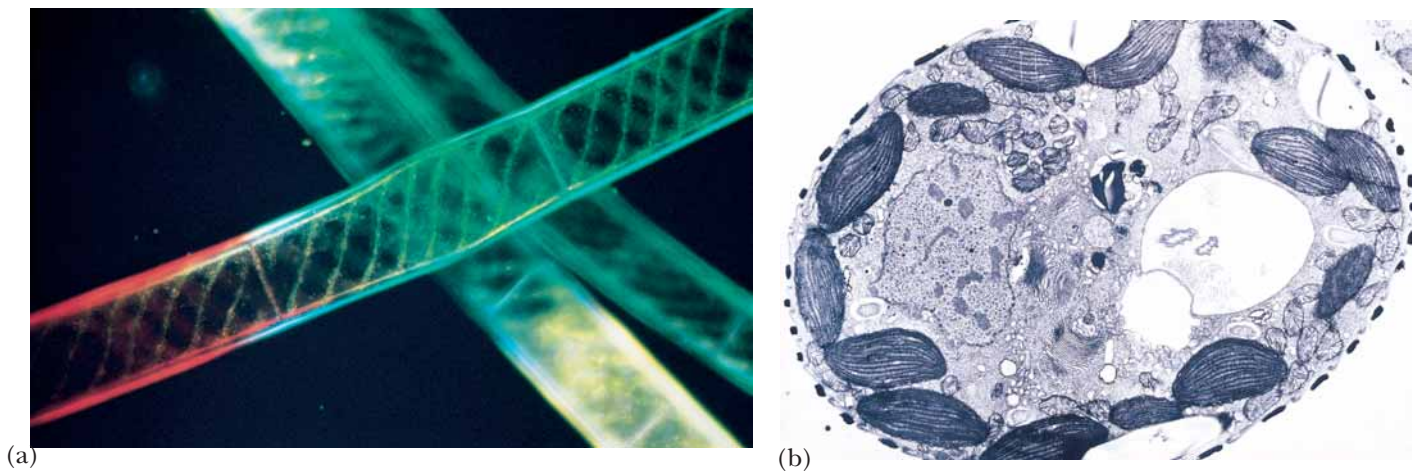


FIGURE 22.3 • (a) *Spirogyra*—a freshwater green alga. (b) A higher plant cell.
(a, Michael Siegel/Phototake NYC; b, Biophoto Associates/Science Source.)

and ribosomes and consequently display a considerable amount of autonomy. However, many critical chloroplast components are encoded by nuclear genes, so autonomy is far from absolute.

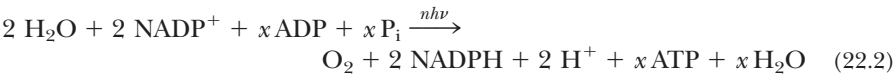
Photosynthesis Consists of Both Light Reactions and Dark Reactions

If a chloroplast suspension is illuminated in the absence of carbon dioxide, oxygen is evolved. Furthermore, if the illuminated chloroplasts are now placed in the dark and supplied with CO₂, net hexose synthesis can be observed (Figure 22.4). Thus, the evolution of oxygen can be temporally separated from CO₂ fixation and also has a light dependency that CO₂ fixation lacks. The **light reactions** of photosynthesis, of which O₂ evolution is only one part, are associated with the thylakoid membranes. In contrast, the light-independent reactions, or so-called **dark reactions**, notably CO₂ fixation, are located in the stroma. A concise summary of the photosynthetic process is that radiant electromagnetic energy (light) is transformed by a specific photochemical system located in the thylakoids to yield chemical energy in the form of reducing potential (NADPH) and high-energy phosphate (ATP). NADPH and ATP can then be used to drive the endergonic process of hexose formation from CO₂ by a series of enzymatic reactions found in the stroma (see Equation 22.3, which follows).

temporally • with regard to time

Water Is the Ultimate e[−] Donor for Photosynthetic NADP⁺ Reduction

In green plants, water serves as the ultimate electron donor for the photosynthetic generation of reducing equivalents. The reaction sequence



describes the process, where $nh\nu$ symbolizes *light energy* (n is some number of photons of energy $h\nu$, where h is Planck's constant and ν is the frequency of

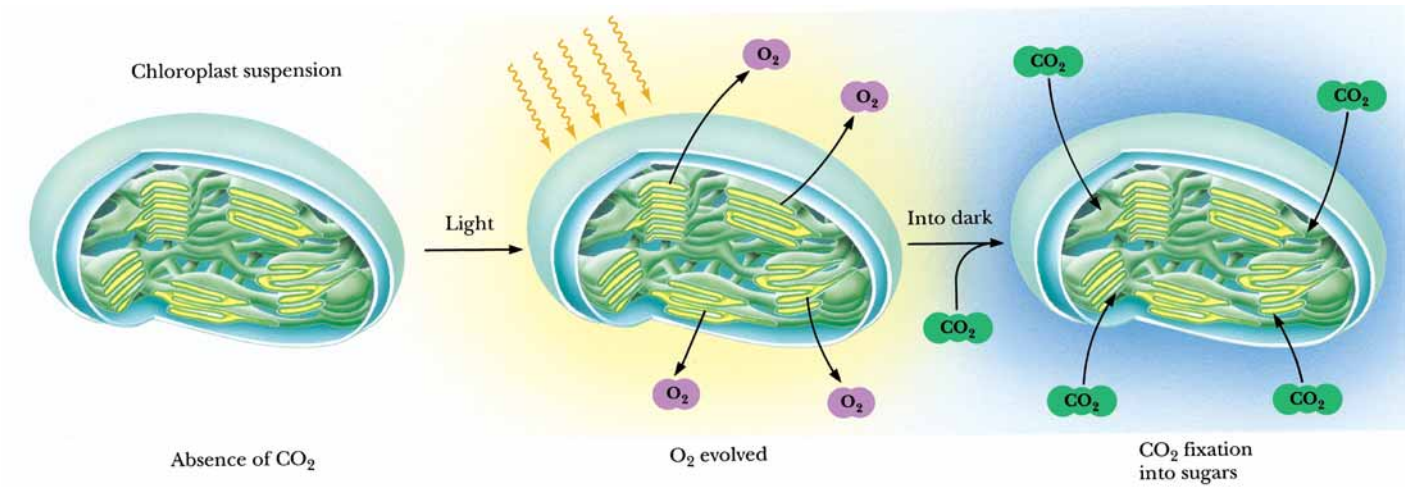
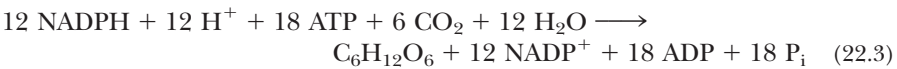


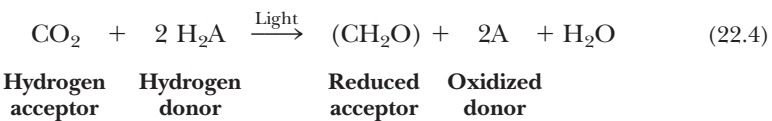
FIGURE 22.4 • The light-dependent and light-independent reactions of photosynthesis. Light reactions are associated with the thylakoid membranes, and light-independent reactions are associated with the stroma.

the light). Light energy is necessary to make the unfavorable reduction of NADP^+ by H_2O ($\Delta\mathcal{E}_\circ' = -1.136\text{ V}$; $\Delta G^\circ = +219\text{ kJ/mol NADP}^+$) thermodynamically favorable. Thus, the light energy input, $nh\nu$, must exceed 219 kJ/mol NADP^+ . The stoichiometry of ATP formation depends on the pattern of photophosphorylation operating in the cell at the time and on the ATP yield in terms of the chemiosmotic ratio, ATP/H^+ , as we will see later. Nevertheless, the stoichiometry of the metabolic pathway of CO_2 fixation is certain:

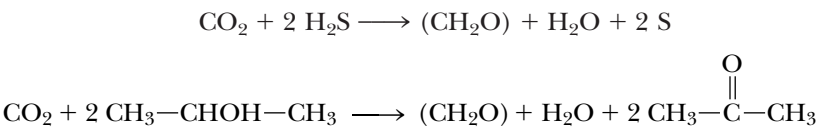


A More Generalized Equation for Photosynthesis

In 1931, comparative study of photosynthesis in bacteria led van Niel to a more general formulation of the overall reaction:



In photosynthetic bacteria, H_2A is variously H_2S (photosynthetic green and purple sulfur bacteria), isopropanol, or some similar oxidizable substrate. [(CH_2O) symbolizes a carbohydrate unit.]



In cyanobacteria and the eukaryotic photosynthetic cells of algae and higher plants, H_2A is H_2O , as implied earlier, and 2 A is O_2 . The accumulation of O_2 to constitute 20% of the earth’s atmosphere is the direct result of eons of global oxygenic photosynthesis.

22.2 • Photosynthesis Depends on the Photoreactivity of Chlorophyll

Chlorophylls are magnesium-containing substituted tetrapyrroles whose basic structure is reminiscent of heme, the iron-containing porphyrin (Chapters 5 and 21). Chlorophylls differ from heme in a number of properties: magnesium instead of iron is coordinated in the center of the planar conjugated ring structure; a long-chain alcohol, **phytol**, is esterified to a pyrrole ring substituent; and the methine bridge linking pyrroles III and IV is substituted and cross-linked to ring III, leading to the formation of a fifth five-membered ring. The structures of chlorophyll *a* and *b* are shown in Figure 22.5.

Chlorophylls are excellent light absorbers because of their aromaticity. That is, they possess delocalized π electrons above and below the planar ring structure. The energy differences between electronic states in these π orbitals correspond to the energies of visible light photons. When light energy is absorbed, an electron is promoted to a higher orbital, enhancing the potential for transfer of this electron to a suitable acceptor. Loss of such a photo-excited electron to an acceptor is an oxidation–reduction reaction. The net result is the transduction of light energy into the chemical energy of a redox reaction.

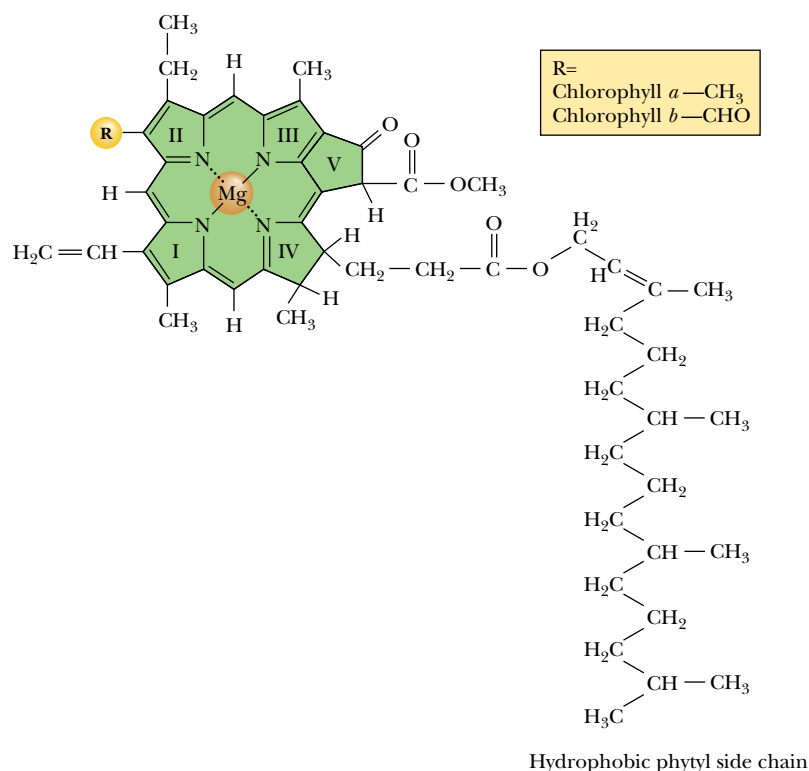


FIGURE 22.5 • Structures of chlorophyll *a* and *b*. Chlorophylls are structurally related to hemes, except Mg^{2+} replaces Fe^{2+} and ring II is more reduced than the corresponding ring of the porphyrins. The chlorophyll tetrapyrrole ring system is known as a chlorin. $\text{R} = \text{CH}_3$ in chlorophyll *a*; $\text{R} = \text{CHO}$ in chlorophyll *b*. Note that the aldehyde $\text{C}=\text{O}$ bond of chlorophyll *b* introduces an additional double bond into conjugation with the double bonds of the tetrapyrrole ring system. Ring V is the additional ring created by interaction of the substituent of the methine bridge between pyrroles III and IV with the side chain of ring III. The phytol side chain of ring IV provides a hydrophobic tail to anchor the chlorophyll in membrane protein complexes.

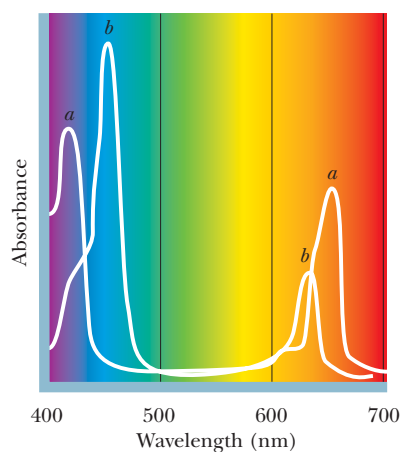


FIGURE 22.6 • Absorption spectra of chlorophylls *a* and *b*.

Chlorophylls and Accessory Light-Harvesting Pigments

The absorption spectra of chlorophylls *a* and *b* (Figure 22.6) differ somewhat. Plants that possess both chlorophylls can harvest a wider spectrum of incident energy. Other pigments in photosynthetic organisms, so-called **accessory light-harvesting pigments** (Figure 22.7), increase the possibility for absorption of incident light of wavelengths not absorbed by the chlorophylls. These accessory pigments, such as *carotenoids* and *phycobilins*, are also responsible for the magnificent colors of autumn. They persist longer after leaf death than the green chlorophylls, finally imparting their particular hues to the plant. These pigments, like chlorophyll, possess many conjugated double bonds and thus absorb visible light.

The Fate of Light Energy Absorbed by Photosynthetic Pigments

Each photon represents a quantum of light energy. A quantum of light energy absorbed by a photosynthetic pigment has four possible fates (Figure 22.8):

- A. Loss as heat.** The energy can be dissipated as heat through redistribution into atomic vibrations within the pigment molecule.
- B. Loss of light.** Energy of excitation reappears as **fluorescence** (light emission); a photon of fluorescence is emitted as the e^- returns to a lower orbital. This fate is common only in saturating light intensities. For thermodynamic reasons, the photon of fluorescence is of longer wavelength and hence lower energy than the quantum of excitation.
- C. Resonance energy transfer.** The excitation energy can be transferred by resonance energy transfer, a radiationless process, to a neighboring molecule if their energy level difference corresponds to the quantum of excitation energy. In this process, the quantum, or so-called **exciton**, is transferred,

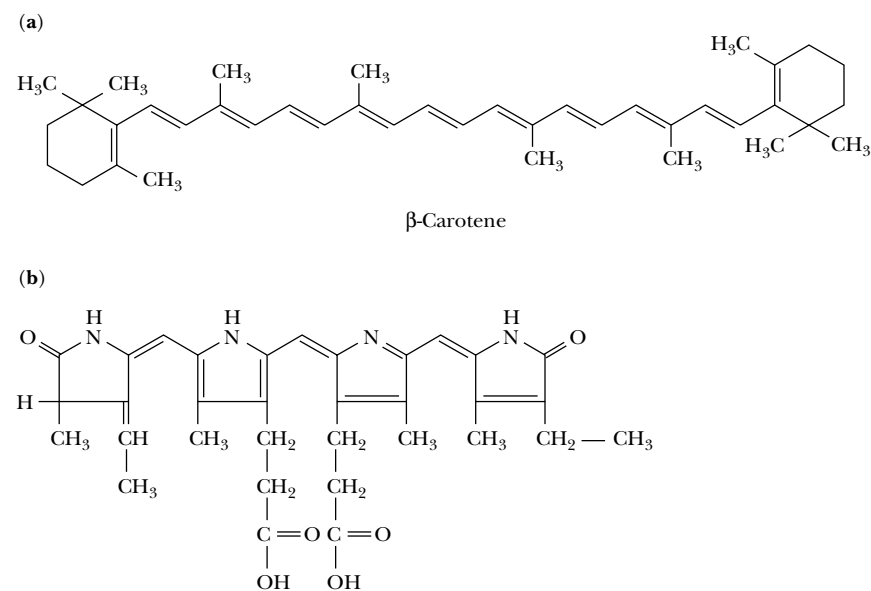


FIGURE 22.7 • Structures of representative accessory light-harvesting pigments in photosynthetic cells. (a) β -Carotene, an accessory light-harvesting pigment in leaves. Note the many conjugated double bonds. (b) Phycocyanobilin, a blue pigment found in cyanobacteria. It is a linear or open pyrrole.

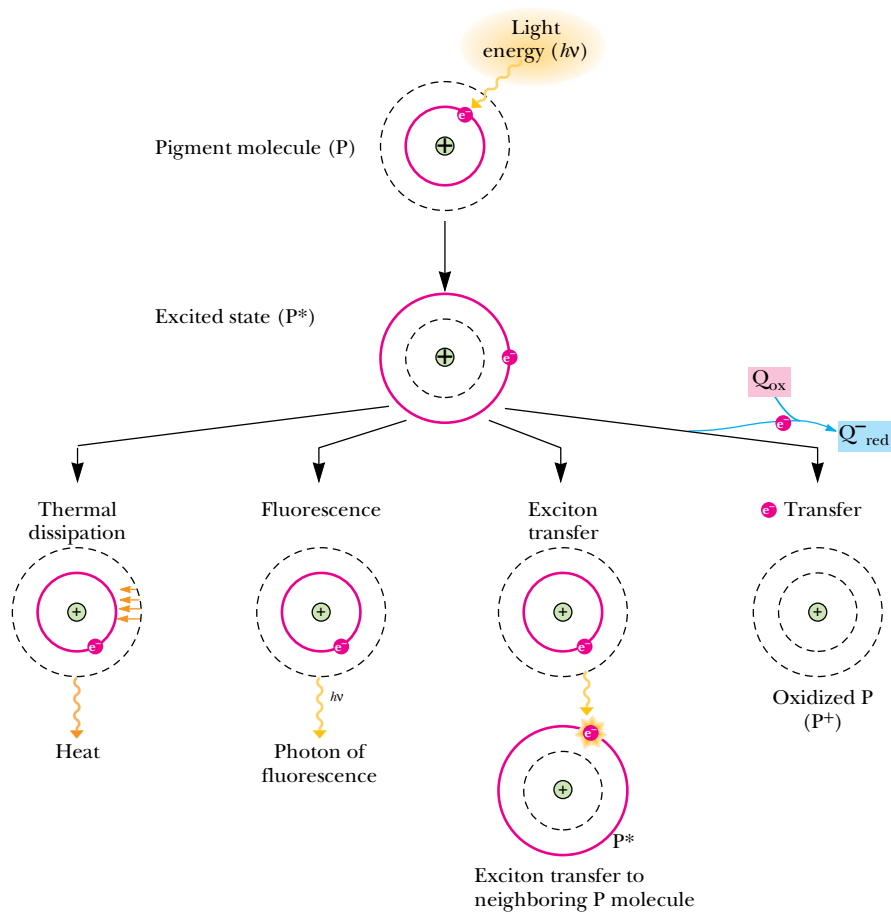


FIGURE 22.8 • Possible fates of the quantum of light energy absorbed by photosynthetic pigments.

raising an electron in the receptor molecule to a higher energy state as the photo-excited e^- in the original absorbing molecule returns to ground state. This so-called *Förster resonance energy transfer* is the mechanism whereby quanta of light falling anywhere within an array of pigment molecules can be transferred ultimately to specific photochemically reactive sites.

D. Energy transduction. The energy of excitation, in raising an electron to a higher energy orbital, dramatically changes the standard reduction potential, $\mathcal{E}_0'$, of the pigment such that it becomes a much more effective electron donor. That is, the excited-state species, by virtue of having an electron at a higher energy level through light absorption, has become a potent electron donor. Reaction of this excited-state electron donor with an electron acceptor situated in its vicinity leads to the transformation, or **transduction**, of light energy (photons) to chemical energy (reducing power, the potential for electron-transfer reactions). *Transduction of light energy into chemical energy, the photochemical event, is the essence of photosynthesis.*

Photosynthetic Units Consist of Many Chlorophyll Molecules but Only a Single Reaction Center

In the early 1930s, Emerson and Arnold investigated the relationship between the amount of incident light energy, the amount of chlorophyll present, and the amount of oxygen evolved by illuminated algal cells (this relationship is called the **quantum yield of photosynthesis**). Their studies gave an unexpected result: When algae were illuminated with very brief light flashes that could excite every chlorophyll molecule at least once, only one molecule of O_2 was evolved per 2400 chlorophyll molecules. This result implied that not all chlorophyll molecules are photochemically reactive, and it led to the concept that photosynthesis occurs in functionally discrete units. Chlorophyll serves two roles in photosynthesis. It is involved in light harvesting and the transfer of light energy to photoreactive sites by exciton transfer, and it participates directly in the photochemical events whereby light energy becomes chemical energy. A **photosynthetic unit** can be envisioned as an antenna of several hundred light-harvesting chlorophyll molecules plus a special pair of photochemically reactive chlorophyll *a* molecules called the **reaction center**. The purpose of the vast majority of chlorophyll in a photosynthetic unit is to harvest light incident within the unit and funnel it, via resonance energy transfer, to special reaction center chlorophyll molecules that are photochemically active. Most chlorophyll thus acts as a large light-collecting antenna, and it is at the reaction centers that the photochemical event occurs (Figure 22.9). Oxidation of chlorophyll leaves a **cationic free radical**, $Chl^{\cdot+}$, whose properties as an electron acceptor have important consequences for photosynthesis. Note that the Mg^{2+} ion does not change in valence during these redox reactions.

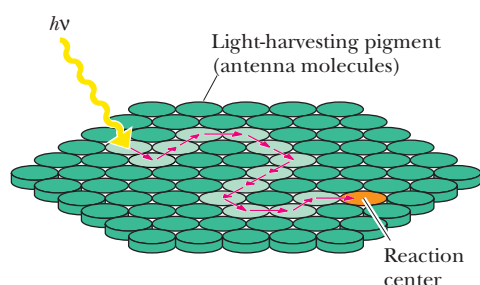


FIGURE 22.9 • Schematic diagram of a photosynthetic unit. The light-harvesting pigments, or antenna molecules (*green*), absorb and transfer light energy to the specialized chlorophyll dimer that constitutes the reaction center (*orange*).

22.3 • Eukaryotic Phototrophs Possess Two Distinct Photosystems

The existence of two separate but interacting photosystems in photosynthetic eukaryotes was demonstrated through analysis of the **photochemical action spectrum** of photosynthesis, in which the oxygen-evolving capacity as a function of light wavelength was determined (Figure 22.10).

Although chlorophyll *a* has some capacity to absorb 700-nm light, light of this wavelength is relatively inefficient in driving photosynthesis. However, if light of shorter wavelength (less than 680 nm) is used to supplement 700-nm

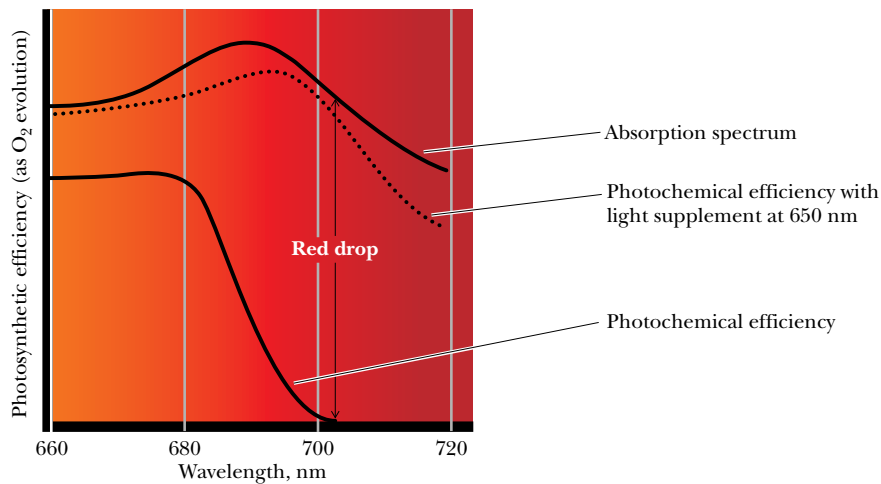


FIGURE 22.10 • The photochemical action spectrum of photosynthesis. The quantum yield of photosynthesis as a function of wavelength of incident light shows an abrupt decrease above 680 nm, the so-called *red drop*.

light, an enhancement of photosynthetic quantum yield, the so-called *Emerson enhancement effect*, is observed. In other words, these two wavelengths are synergistic: When given together, these wavelengths elicit more O_2 evolution than expected from the sum of the amounts when each wavelength of light is given alone. One interpretation is that two light reactions participate in oxygen-evolving photosynthetic cells, one using light of 700 nm and the other using light of wavelength 680 nm or less. The existence of two light reactions established the presence of two photosystems, I and II. **Photosystem I** (PSI) is defined as containing reaction center chlorophylls with maximal red light absorption at 700 nm; PSI is not involved in O_2 evolution. **Photosystem II** (PSII) functions in O_2 evolution, using reaction centers that exhibit maximal red light absorption at 680 nm.

All photosynthetic cells contain some form of photosystem. Photosynthetic bacteria, unlike cyanobacteria and eukaryotic phototrophs, have only one photosystem. Interestingly, bacterial photosystems resemble eukaryotic PSII more than PSI, even though photosynthetic bacteria lack O_2 -evolving capacity.

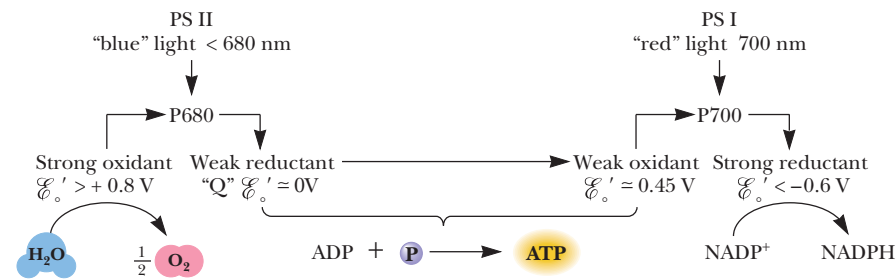
P700 and P680 Are the Reaction Centers of PSI and PSII, Respectively

Precise spectrophotometric measurements showed that a small amount of pigment absorbing 700-nm light (**P700**) is bleached when light of this wavelength is used to illuminate suspensions of eukaryotic photosynthetic cells. Because bleaching, or disappearance, of the 700-nm absorbance can be mimicked by adding an electron acceptor such as ferricyanide, bleaching is correlated with electron loss from P700. The concentration of P700 is small, only 0.25% of the total amount of chlorophyll in plants. However, this low concentration is consistent with the notion of **reaction centers** (specific photoreactive sites). P700 is the reaction center of photosystem I. Similar studies using shorter-wavelength light identified an analogous pigment, **P680**, which constitutes the reaction center of photosystem II. Both P700 and P680 are chlorophyll *a* dimers situated within specialized protein complexes.

Chlorophyll Exists in Plant Membranes in Association with Proteins

Detergent treatment of a suspension of thylakoids dissolves the membranes, releasing complexes containing both chlorophyll and protein. These chlorophyll–protein complexes represent integral components of the thylakoid membrane, and their organization reflects their roles as either **light-harvesting com-**

FIGURE 22.11 • Roles of the two photosystems, PSI and PSII.



plexes (LHC), **PSI complexes**, or **PSII complexes**. All chlorophyll is apparently localized within these three macromolecular assemblies.

The Roles of PSI and PSII

What are the roles of the two photosystems, and what is their relationship to each other? Photosystem I provides reducing power in the form of NADPH. Photosystem II splits water, producing O₂, and feeds the electrons released into an electron transport chain that couples PSII to PSI. Electron transfer between PSII and PSI pumps protons for chemiosmotic ATP synthesis. As summarized by Equation (22.2), photosynthesis involves the reduction of NADP⁺, using electrons derived from water and activated by light, *hν*. ATP is generated in the process. The standard reduction potential for the NADP⁺/NADPH couple is −0.32 V. Thus, a strong reductant with *E*_o' more negative than −0.32 V is required to reduce NADP⁺ under standard conditions. By similar reasoning, a very strong oxidant will be required to oxidize water to oxygen because *E*_o'(¹/₂ O₂/H₂O) is +0.82 V. Separation of the oxidizing and reducing aspects of Equation (22.2) is accomplished in nature by devoting PSI to NADP⁺ reduction and PSII to water oxidation. PSI and PSII are linked via an electron transport chain so that the weak reductant generated by PSII can provide an electron to reduce the weak oxidant side of P700 (Figure 22.11). Thus, *electrons flow from H₂O to NADP⁺*, driven by light energy absorbed at the reaction centers. Oxygen is a by-product of the **photolysis**, literally “light-splitting,” of water. Accompanying electron flow is production of a proton gradient and ATP synthesis (see Section 22.7). This light-driven phosphorylation is termed **photophosphorylation**.

22.4 • The Z Scheme of Photosynthetic Electron Transfer

Photosystems I and II contain unique complements of electron carriers, and these carriers mediate the stepwise transfer of electrons from water to NADP⁺. When the individual redox components of PSI and PSII are arranged as an *e*[−] transport chain according to their standard reduction potentials, the zigzag result resembles the letter Z laid sideways (Figure 22.12). The various electron carriers are indicated as follows: “Mn complex” symbolizes the manganese-containing oxygen-evolving complex; D is its *e*[−] acceptor and the immediate *e*[−] donor to P680⁺; Q_A and Q_B represent special plastoquinone molecules (see Figure 22.15) and PQ the plastoquinone pool; Fe-S stands for the Rieske iron–sulfur center, and *cyt f*, cytochrome *f*. PC is the abbreviation for plastocyanin, the immediate *e*[−] donor to P700[−]; and F_A, F_B, and F_X represent the membrane-

associated ferredoxins downstream from A_0 (a specialized Chl a) and A_1 (a specialized PSI quinone). Fd is the soluble ferredoxin pool that serves as the e^- donor to the flavoprotein (Fp), called **ferredoxin–NADP⁺ reductase**, which catalyzes reduction of NADP⁺ to NADPH. $Cyt(b_6)_m(b_6)_p$ symbolizes the cytochrome b_6 moieties functioning to transfer e^- from F_A/F_B back to P700⁺ during cyclic photophosphorylation (the pathway symbolized by the dashed arrow).

Overall photosynthetic electron transfer is accomplished by three **membrane-spanning supramolecular complexes**, composed of intrinsic and extrinsic polypeptides (shown as shaded boxes bounded by solid black lines in Figure 22.12). These complexes are the PSII complex, the cytochrome b_6 /cytochrome f complex, and the PSI complex. The PSII complex is aptly described as a light-driven **water:plastoquinone oxidoreductase**; it is the enzyme system responsible for photolysis of water, and as such, it is also referred to as the **oxygen-evolving complex**, or OEC. Within this complex, a manganese-containing protein is intimately involved in the evolution of oxygen, perhaps through formation of a tetrametallic center consisting of 4 Mn²⁺ coordinating two equivalents of water. Both protons and electrons are abstracted from these water molecules, and O₂ is released as P680 undergoes four cycles of oxidation (Figure 22.13).

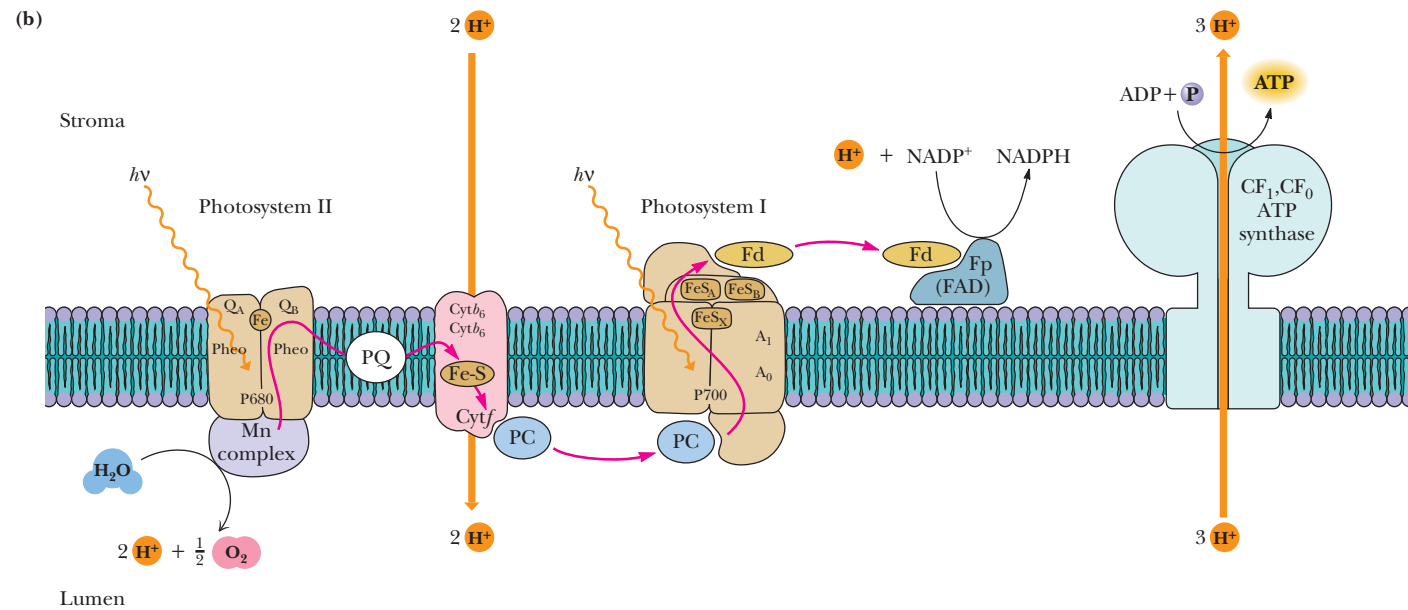
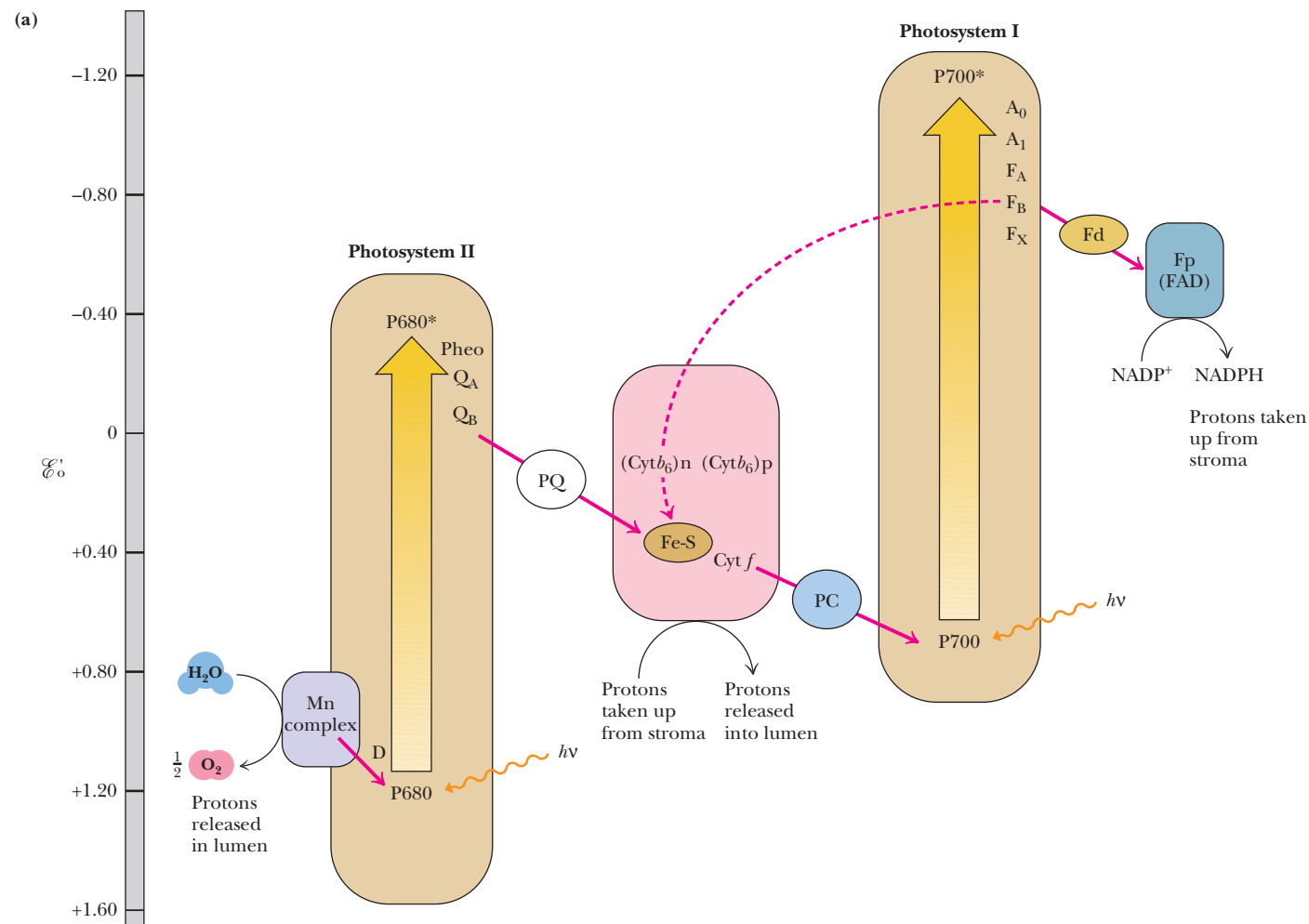
Oxygen Evolution Requires the Accumulation of Four Oxidizing Equivalents in PSII

When isolated chloroplasts that have been held in the dark are illuminated with very brief flashes of light, O₂ evolution reaches a peak on the third flash and every fourth flash thereafter (Figure 22.14a). The oscillation in O₂ evolved dampens over repeated flashes and converges to an average value. These data are interpreted to mean that the P680 reaction center complex cycles through five different oxidation states, numbered **S₀** to **S₄**. One electron and one proton are removed photochemically in each step. When **S₄** is attained, an O₂ molecule is released (Figure 22.14b) as PSII returns to oxidation state **S₀** and two new water molecules bind. The reason the first pulse of O₂ release occurred on the third flash (Figure 22.14a) is that the PSII reaction centers in the isolated chloroplasts were already poised at **S₁** reduction level.

Light-Driven Electron Flow from H₂O Through PSII

The events intervening between H₂O and P680 involve D , the name assigned to a specific protein **tyrosine residue** that mediates e^- transfer from H₂O via the Mn complex to P680⁺ (Figure 22.12). The oxidized form of D is a tyrosyl free radical species, $D^{\cdot+}$. To begin the cycle, an exciton of energy excites P680 to P680*, whereupon P680* donates an electron to a special molecule of **pheophytin**, symbolized by “Pheo” in Figure 22.12. Pheophytin is like chlorophyll a , except 2 H⁺ replace the centrally coordinated Mg²⁺ ion. This special pheophytin is the direct electron acceptor from P680*. Loss of an electron from P680* creates P680⁺, the electron acceptor for D . Electrons flow from Pheo via specialized molecules of **plastoquinone**, represented by “Q” in Figure 22.12, to a pool of plastoquinone within the membrane. Because of its lipid nature, plastoquinone is mobile within the membrane and hence serves to shuttle electrons from the PSII supramolecular complex to the cytochrome b_6 /cytochrome f complex. Alternate oxidation–reduction of plastoquinone to its hydroquinone form involves the uptake of protons (Figure 22.15). The asymmetry of the thylakoid membrane is designed to exploit this proton uptake and release so that protons (H⁺) accumulate within the thylakoid vesicle, establishing an electrochemical gradient. Note that plastoquinone is an analog of coenzyme Q, the mitochondrial electron carrier (Chapter 21).

(Text continues on page 722)



◀ **FIGURE 22.12** • The Z scheme of photosynthesis. (a) The Z scheme is a diagrammatic representation of photosynthetic electron flow from H_2O to NADP^+ . The energy relationships can be derived from the $\mathcal{E}_o'$ scale beside the Z diagram, with lower standard potentials and hence greater energy as you go from bottom to top. Energy input as light is indicated by two broad arrows, one photon appearing in P680 and the other in P700. P680* and P700* represent photoexcited states. Electron loss from P680* and P700* creates P680^+ and P700^+ . The representative components of the three supramolecular complexes (PSI, PSII, and the cytochrome b_6 /cytochrome f complex) are in shaded boxes enclosed by solid black lines. Proton translocations that establish the proton-motive force driving ATP synthesis are illustrated as well. (b) Figure showing the functional relationships among PSII, the cytochrome b /cytochrome f complex, PSI, and the photosynthetic CF_1CF_0 ATP synthase within the thylakoid membrane. Note that e^- acceptors Q_A (for PSII) and A_1 (for PSI) are at the stromal side of the thylakoid membrane, whereas the e^- donors to P680 $^+$ and P700 $^+$ are situated at the luminal side of the membrane. The consequence is charge separation ($-_{\text{stroma}}, +_{\text{lumen}}$) across the membrane. Also note that protons are translocated into the thylakoid lumen, giving rise to a chemiosmotic gradient that is the driving force for ATP synthesis by CF_1CF_0 ATP synthase.

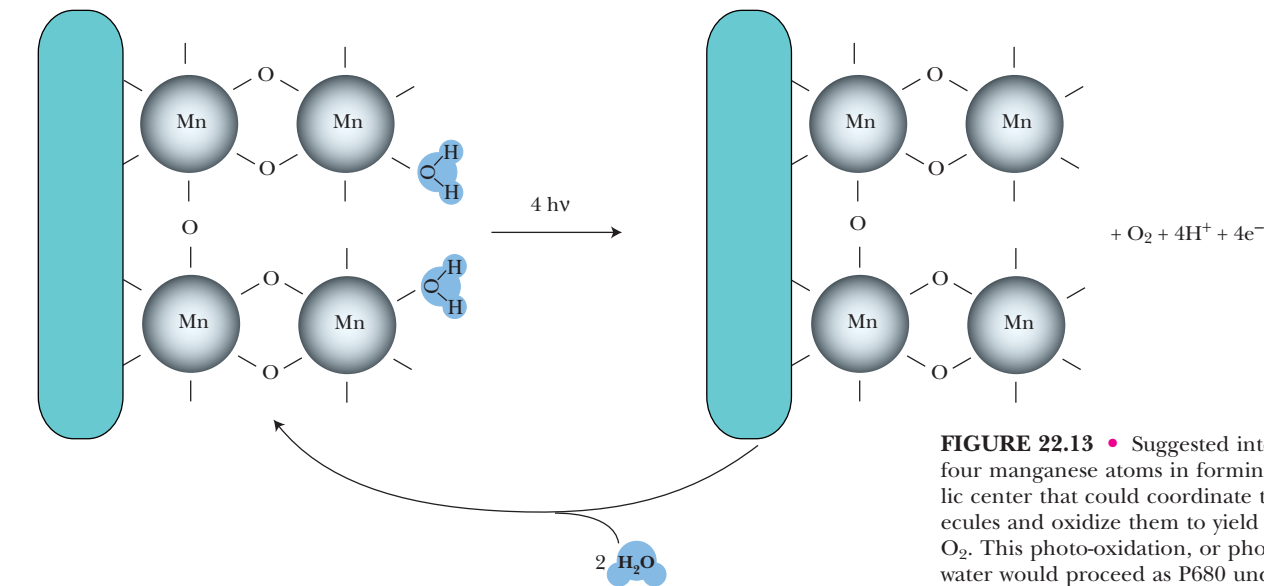


FIGURE 22.13 • Suggested interaction of four manganese atoms in forming a tetra-metallic center that could coordinate two water molecules and oxidize them to yield a molecule of O_2 . This photo-oxidation, or photolysis, of water would proceed as P680 undergoes four cycles of light-induced oxidation. The four oxidizing equivalents accumulate in the manganese-containing active site of the O_2 -evolving complex. (Adapted from Hoganson, C. W., and Babcock, G. T., 1997. A metalloradical mechanism for the generation of oxygen in photosynthesis. *Science* 277:1953–1956.)

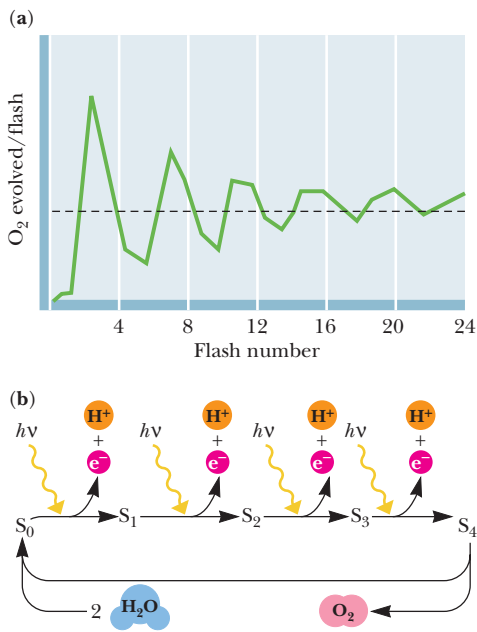


FIGURE 22.14 • Oxygen evolution requires the accumulation of four oxidizing equivalents in PSII. (a) Dark-adapted chloroplasts show little O_2 evolution after two brief light flashes. Oxygen evolution then shows a peak on the third flash and every fourth flash thereafter. The oscillation in O_2 evolution is dampened by repeated flashes and converges to an average value after 20 or so flashes. (b) The oscillation in O_2 evolution per light flash is due to the cycling of the PSII reaction center through five different oxidation states, S_0 to S_4 . When S_4 is reached, O_2 is released. One e^- is removed photochemically at each light flash, moving the reaction center successively through S_1 , S_2 , S_3 , and S_4 . S_4 decays spontaneously to S_0 by oxidizing $2 \text{H}_2\text{O}$ to O_2 . The peak of O_2 evolution at flash 3 in part (a) is due to the fact that the isolated chloroplast suspension is already at the S_1 stage.

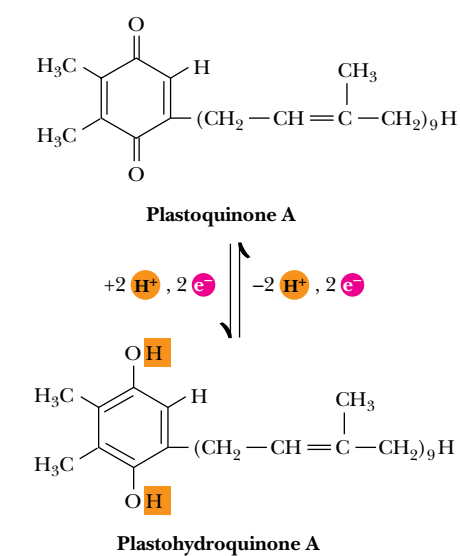


FIGURE 22.15 • The structures of plastoquinone and its reduced form, plastohydroquinone (or plastoquinol). The oxidation of the hydroquinone releases 2 H⁺ as well as 2 e⁻. The form shown (plastoquinone A) has nine isoprene units and is the most abundant plastoquinone in plants and algae. Other plastoquinones have different numbers of isoprene units and may vary in the substitutions on the quinone ring.

Electron Transfer Within the Cytochrome *b*₆/Cytochrome *f* Complex

The cytochrome *b*₆/cytochrome *f* or **plastoquinol:plastocyanin oxidoreductase** is a large (210 kD) multimeric protein possessing 22 to 24 transmembrane α-helices. It includes the two heme-containing electron transfer proteins for which it is named as well as *iron–sulfur clusters* (Chapter 21), which also participate in electron transport. The purpose of this complex is to mediate the transfer of electrons from PSII to PSI and to pump protons across the thylakoid membrane via a plastoquinone-mediated Q cycle, analogous to that found in mitochondrial *e*⁻ transport (Chapter 21). Cytochrome *f* (*f* from the Latin *folium*, meaning “foliage”) is a *c*-type cytochrome, with an α-absorbance band at 553 nm and a reduction potential of +0.365 V. Cytochrome *b*₆ apparently does not lie directly on the pathway of electron transfer from PSII to PSI. This cytochrome, whose α-absorbance band lies at 559 nm and whose standard reduction potential is −0.06 V, is thought to participate in an alternative ***cyclic e*⁻ transfer pathway**. Under certain conditions, electrons derived from P700* are not passed on to NADP⁺ but instead cycle down an alternative path via ferredoxins in the PSI complex to cytochrome *b*₆, plastoquinone, and ultimately back to P700⁺. This cyclic flow yields no O₂ evolution or NADP⁺ reduction but can lead to ATP synthesis via so-called *cyclic photophosphorylation*, discussed later.

Electron Transfer from the Cytochrome *b*₆/Cytochrome *f* Complex to PSI

Plastocyanin (“PC” in Figure 22.12) is an electron carrier capable of diffusion along the inside of the thylakoid and migration in and out of the membrane, aptly suited to its role in shuttling electrons between the cytochrome *b*₆/cytochrome *f* complex and PSI. Plastocyanin is a low-molecular-weight (10.4 kD) protein containing a single copper atom. PC functions as a single-electron carrier (*E*_o['] = +0.32 V) as its copper atom undergoes alternate oxidation–reduction between the cuprous (Cu⁺) and cupric (Cu²⁺) states. PSI is a light-driven **plastocyanin:ferredoxin oxidoreductase**. When P700, the specialized chlorophyll *a* dimer of PSI, is excited by light and oxidized by transferring its *e*⁻ to an adjacent chlorophyll *a* molecule that serves as its immediate *e*⁻ acceptor, P700⁺ is formed. (The standard reduction potential for the P700⁺/P700 couple lies near +0.45 V.) P700⁺ readily gains an electron from plastocyanin.

The immediate electron acceptor for P700* is a special molecule of chlorophyll. This unique Chl *a* (*A*₀) rapidly passes the electron to a specialized quinone (*A*₁), which in turn passes the *e*⁻ to the first in a series of *membrane-bound ferredoxins* (Fd, Chapter 21). This Fd series ends with a soluble form of ferredoxin, Fd_s, which serves as the immediate electron donor to the flavo-protein (Fp) that catalyzes NADP⁺ reduction, namely, **ferredoxin:NADP⁺ reductase**.

The Initial Events in Photosynthesis Are Very Rapid Electron-Transfer Reactions

Electron transfer from P680 to Q and from P700 to Fd occurs on a pico-second-to-microsecond time scale. The necessity for such rapid reaction becomes obvious when one realizes that light-induced Chl excitation followed by electron transfer leads to separation of opposite charges in close proximity, as in P700⁺:*A*₀⁻. Accordingly, subsequent electron transfer reactions occur rapidly in order to shuttle the electron away quickly, before the wasteful back reaction of charge recombination (and dissipation of excitation energy), as in return to P700:*A*₀, can happen.

22.5 • The Molecular Architecture of Photosynthetic Reaction Centers

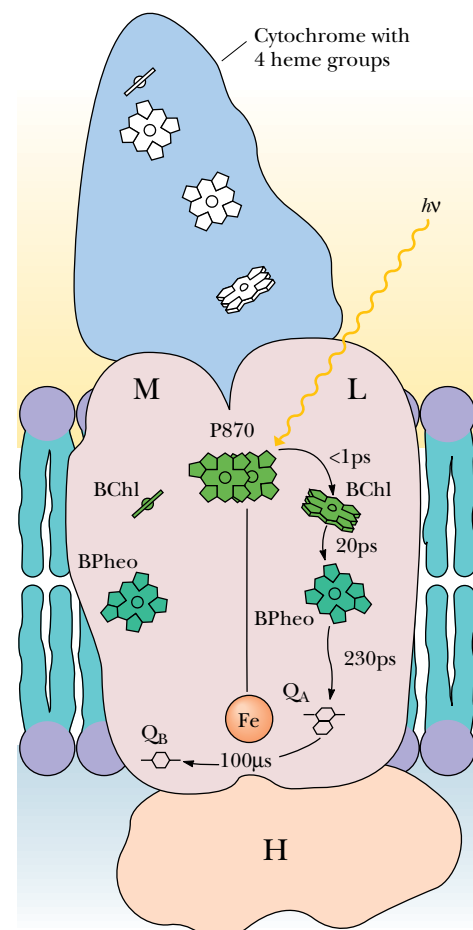
What molecular architecture couples the absorption of light energy to rapid electron-transfer events, in turn coupling these e^- transfers to proton translocations so that ATP synthesis is possible? Part of the answer to this question lies in the membrane-associated nature of the photosystems. Membrane proteins have been difficult to study due to their insolubility in the usual aqueous solvents employed in protein biochemistry. A major breakthrough occurred in 1984 when Johann Deisenhofer, Hartmut Michel, and Robert Huber reported the first X-ray crystallographic analysis of a membrane protein. To the great benefit of photosynthesis research, this protein was the reaction center from the photosynthetic purple bacterium *Rhodospseudomonas viridis*. This research earned these three scientists the 1984 Nobel Prize in chemistry.

Structure of the *R. viridis* Photosynthetic Reaction Center

Rhodospseudomonas viridis is a photosynthetic prokaryote with a single type of photosystem. The reaction center (145 kD) of the *R. viridis* photosystem is localized in the plasma membrane of these photosynthetic bacteria and is composed of four different polypeptides, designated *L* (273 amino acid residues), *M* (323 residues), *H* (258 residues), and *cytochrome* (333 amino acid residues). *L* and *M* each consist of five membrane-spanning α -helical segments; *H* has one such helix, the majority of the protein forming a globular domain in the cytoplasm (Figure 22.16). The cytochrome subunit contains four heme groups; the N-terminal amino acid of this protein is cysteine. This cytochrome is anchored to the periplasmic face of the membrane via the hydrophobic chains of two fatty acid groups that are esterified to a glycerol moiety joined via a thioether bond to the Cys (Figure 22.16). *L* and *M* each bear two *bacteriochlorophyll* molecules (the bacterial version of Chl) and one *bacteriopheophytin*. *L* also has a bound quinone molecule, Q_A . Together, *L* and *M* coordinate an Fe atom. The photochemically active species of the *R. viridis* reaction center, **P870**, is composed of two bacteriochlorophylls, one contributed by *L* and the other by *M*.

Photosynthetic Electron Transfer in the *R. viridis* Reaction Center

The prosthetic groups of the *R. viridis* reaction center (P870, BChl, BPheo, and the bound quinones) are fixed in a spatial relationship to one another that favors photosynthetic e^- transfer (Figure 22.16). Photoexcitation of P870 (creation of $P870^*$) leads to e^- loss ($P870^+$) via electron transfer to the nearby bacteriochlorophyll (BChl). The e^- is then transferred via the *L* bacteriopheophytin (BPheo) to Q_A , which is also an *L* prosthetic group. The corresponding site on *M* is occupied by a loosely bound quinone, Q_B , and electron transfer from Q_A to Q_B takes place. An interesting aspect of the system is that *no* elec-



Note: The cytochrome subunit is membrane-associated via a diacylglycerol moiety on its N-terminal Cys residue:

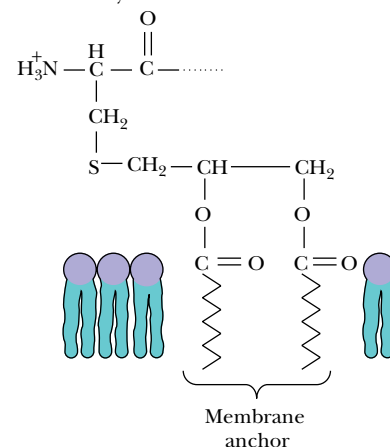


FIGURE 22.16 • Model of the structure and activity of the *R. viridis* reaction center. Four polypeptides (designated *cytochrome*, *M*, *L*, and *H*) make up the reaction center, an integral membrane complex. The cytochrome maintains its association with the membrane via a diacylglycerol group linked to its N-terminal Cys residue by a thioether bond. *M* and *L* both consist of five membrane-spanning α -helices; *H* has a single membrane-spanning α -helix. The prosthetic groups are spatially situated so that rapid e^- transfer from $P870^*$ to Q_B is facilitated. Photoexcitation of P870 leads in less than 1 picosecond (psec) to reduction of the *L*-branch BChl only. $P870^+$ is re-reduced via an e^- provided through the heme groups of the cytochrome.

tron transfer occurs through *M*, even though it has components apparently symmetrical to and identical with the *L* e^- transfer pathway.

The reduced quinone formed at the Q_B site is free to diffuse to a neighboring *cytochrome b/cytochrome c_1* membrane complex, where its oxidation is coupled to H^+ translocation (and, hence, ultimately to ATP synthesis) (Figure 22.17). *Cytochrome c_2* , a periplasmic protein, serves to cycle electrons back to $P870^+$ via the four hemes of the reaction center cytochrome subunit. A specific tyrosine residue of *L* (Try¹⁶²) is situated between $P870^+$ and the closest cytochrome heme. This Tyr is the immediate e^- donor to $P870^+$ and completes the light-driven electron transfer cycle. The structure of the *R. viridis* reaction center (derived from X-ray crystallographic data) is modeled in Figure 22.18.

Eukaryotic Reaction Centers: The Molecular Architecture of PSII

PSII of higher plants and green algae contains more than 20 subunits and is considerably more complex than the *R. viridis* reaction center. Nevertheless, the *R. viridis* reaction center is a fairly good model for the core structure of PSII. $P680$ and its two equivalents of pheophytin (Pheo) are located on a pair of integral membrane proteins designated **D1** (38 kD) and **D2** (39.4 kD) (Figure 22.19). The tyrosine species *D* is Tyr¹⁶¹ in the **D1** amino acid sequence. Complexed to **D2** is a tightly bound plastoquinone molecule, Q_A . Electrons flow from $P680^*$ to Pheo on **D1** and thence to Q_A on **D2** and then on to a second plastoquinone situated in the Q_B site on **D1**. Electron transfer from Q_A and Q_B is assisted by an iron atom located between them. Each plastoquinone (PQ) that enters the Q_B site accepts two electrons derived from water and two H^+ from the stroma before it is released into the membrane as the hydroquinone, PQH_2 . The stoichiometry of the overall reaction catalyzed by PSII is

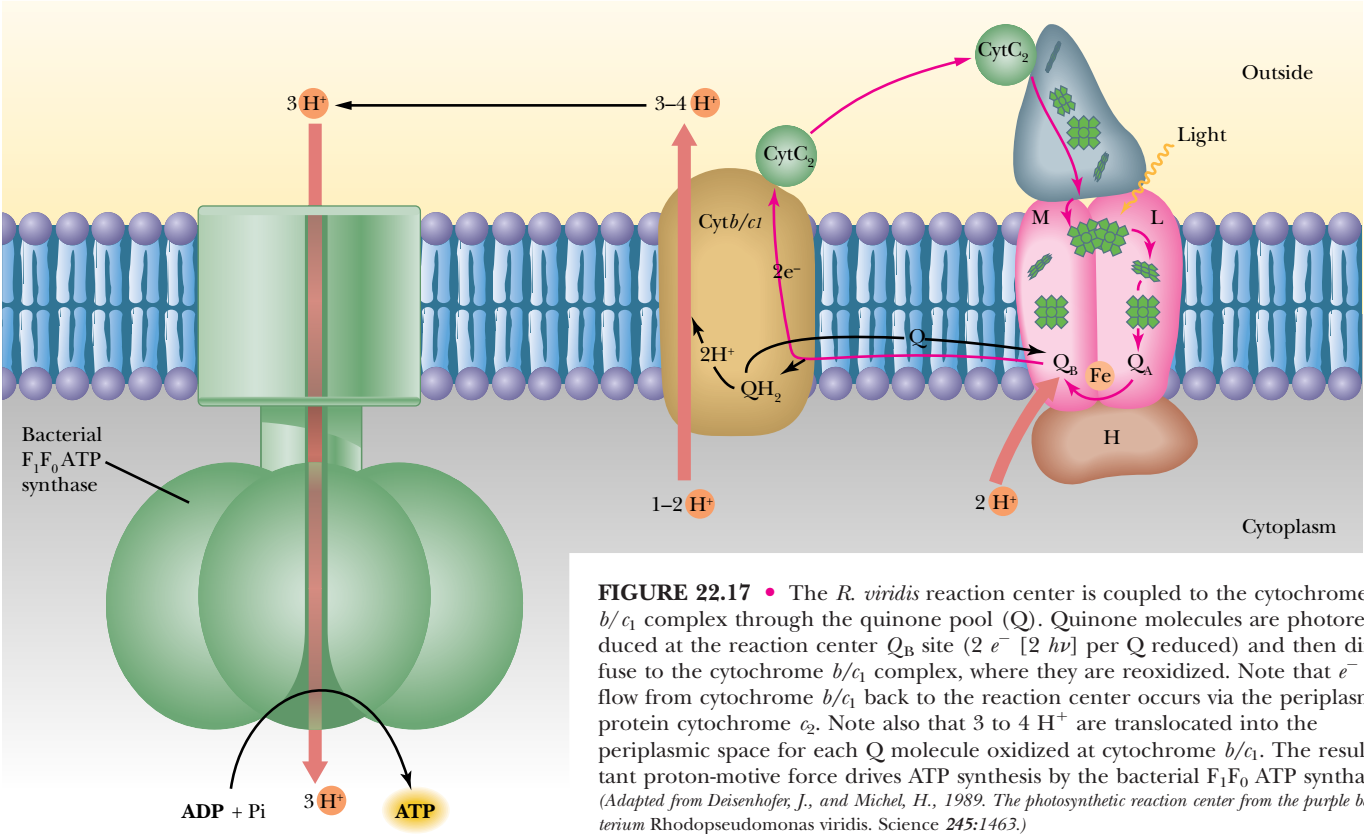


FIGURE 22.17 • The *R. viridis* reaction center is coupled to the cytochrome *b/c1* complex through the quinone pool (*Q*). Quinone molecules are photoreduced at the reaction center Q_B site ($2 e^- [2 h\nu]$ per *Q* reduced) and then diffuse to the cytochrome *b/c1* complex, where they are reoxidized. Note that e^- flow from cytochrome *b/c1* back to the reaction center occurs via the periplasmic protein cytochrome c_2 . Note also that 3 to 4 H^+ are translocated into the periplasmic space for each *Q* molecule oxidized at cytochrome *b/c1*. The resultant proton-motive force drives ATP synthesis by the bacterial F_1F_0 ATP synthase. (Adapted from Deisenhofer, J., and Michel, H., 1989. The photosynthetic reaction center from the purple bacterium *Rhodospseudomonas viridis*. *Science* 245:1463.)

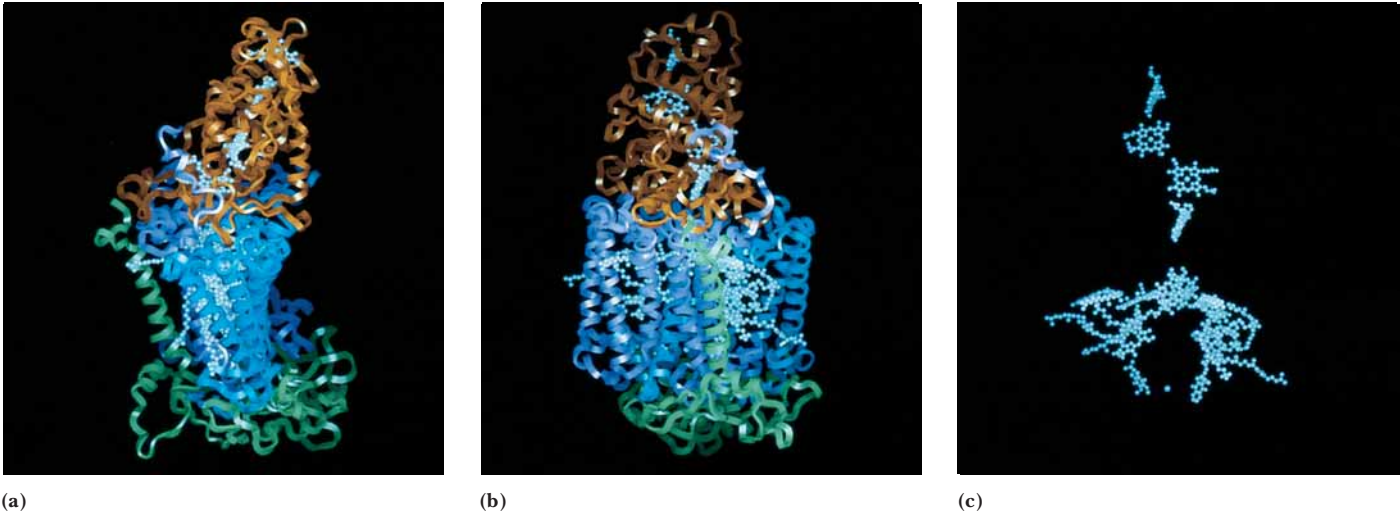


FIGURE 22.18 • Model of the *R. viridis* reaction center. (a, b) Two views of the ribbon diagram of the reaction center. *M* and *L* subunits appear in purple and blue, respectively. Cytochrome subunit is brown; H subunit is green. These proteins provide a scaffold upon which the prosthetic groups of the reaction center are situated for effective photosynthetic electron transfer. Panel (c) shows the spatial relationship between the various prosthetic groups (4 hemes, P870, 2 BChl, 2 BPheo, 2 quinones, and the Fe atom) in the same view as in (b), but with protein chains deleted.

$2\text{H}_2\text{O} + 2\text{PQ} + 4\text{h}\nu \rightarrow \text{O}_2 + 2\text{PQH}_2$. A cytochrome species, **cytochrome *b*₅₅₉**, composed of two polypeptides (4.4 kD and 9.3 kD), is associated with PSII; its function is as yet unclear. Two chlorophyll-binding proteins (47 and 43 kD) harvest light and deliver exciton energy to P680. A Mn-containing extrinsic membrane protein, the **OEC** or **oxygen-evolving complex** (whose principal subunits are 33-, 23-, and 16-kD polypeptides) is located on the luminal side of the thylakoid membrane.

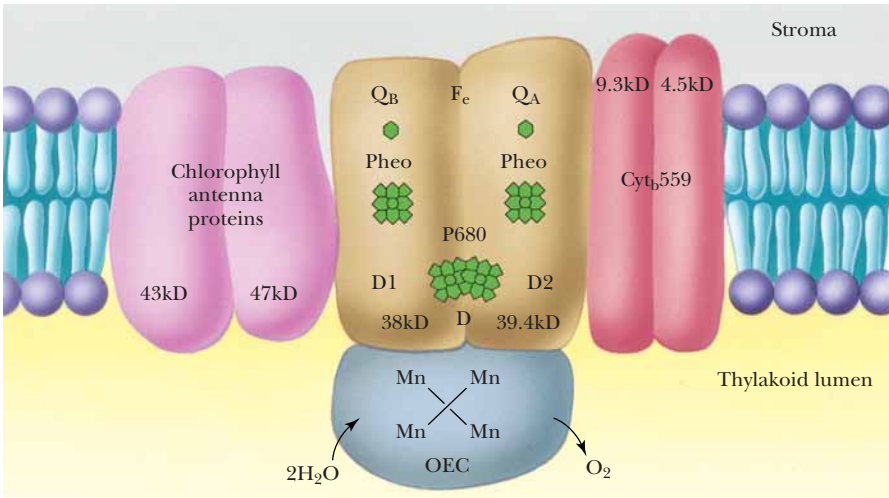


FIGURE 22.19 • The molecular architecture of PSII. The core of the PSII complex consists of the two polypeptides (D1 and D2) that bind P680, pheophytin (Pheo), and the quinones, *Q*_A and *Q*_B. Additional components of this complex include cytochrome *b*₅₅₉, two additional intrinsic proteins (47 and 43 kD) that serve an accessory light-harvesting function, and an extrinsic protein complex that is essential to O₂ evolution.

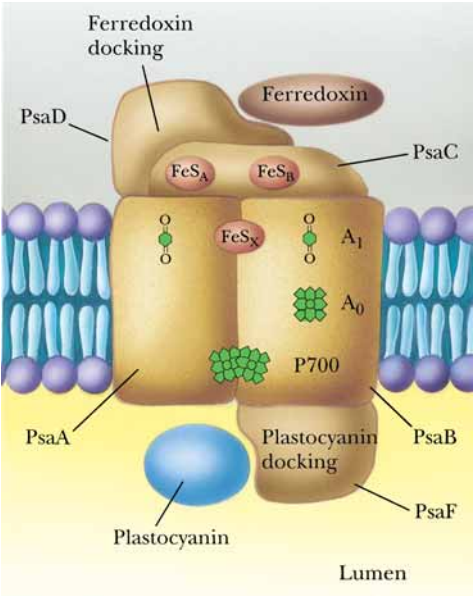


FIGURE 22.20 • The molecular architecture of PSI. PsaA and PsaB constitute the reaction center dimer, an integral membrane complex; P700 is located at the lumenal side of this dimer. PsaC, which bears Fe-S centers F_A and F_B , and PsaD, the interaction site for ferredoxin, are on the stromal side of the thylakoid membrane. PsaF, which provides the plastocyanin interaction site, is on the lumenal side. (Adapted from Golbeck, J. H., 1992. Annual Review of Plant Physiology and Plant Molecular Biology 43:293–324.)

The Molecular Architecture of PSI

The structure of PSI from the cyanobacterium *Synechococcus elongatus* has been solved by X-ray crystallography, and this structure shows strong similarities to the *R. viridis* reaction center and our emerging view of the eukaryotic PSII, both discussed earlier. Because of direct correlations with information about eukaryotic PSI, this cyanobacterial PSI provides a general model for all P700-dependent photosystems (Figure 22.20). Although PSI consists of 11 different protein subunits, all the electron-transferring prosthetic groups essential to PSI function are localized to just three polypeptides. Two of these, **PsaA** and **PsaB** (83 kD each), compose the reaction center heterodimer, a structural pattern that now seems universal in photosynthetic reaction centers. **PsaA** and **PsaB** each have 11 transmembrane α -helices. PSI has approximately 100 chlorophyll molecules, including the two composing P700 and two positioned some 16 Å from P700, one of which functions as A_0 , the immediate e^- acceptor for P700 (Figure 22.20). Quinones are found in association with PSI, including one that functions as A_1 , an intermediate e^- carrier. The Fe-S center designated F_x bridges **PsaA** and **PsaB**. The third protein, **PsaC** (9 kD), bears two additional Fe-S clusters designated F_A and F_B ; **PsaC** (along with two other proteins designated **PsaD** and **PsaE**) lies on the stromal face of the reaction center complex. **PsaD** is the site of ferredoxin binding in eukaryotic PSI systems. **PsaF**, with three transmembrane α -helices, provides the interaction site for plastocyanin (on the lumenal side of the membrane).

The overall structure of *S. elongatus* PSI thus consists of a core reaction center surrounded by and connected to a large Chl-based antenna system. Three equivalents of such PSI complexes occur together to form a trimeric structure. Photochemistry begins with exciton absorption at P700, almost instantaneous electron transfer and charge separation ($P700^+ : A_0^-$), followed by transfer of the electron from A_0 to A_1 and on to F_x and then F_A/F_B , where it is used to reduce a ferredoxin molecule at the “stromal” side of the membrane. The positive charge at $P700^+$ and the e^- at F_A/F_B represent a charge separation across the membrane, an energized condition created by light.

22.6 • The Quantum Yield of Photosynthesis

The **quantum yield** of photosynthesis, the amount of product formed per equivalent of light input, has traditionally been expressed as the ratio of CO_2 fixed or O_2 evolved per quantum absorbed. At each reaction center, one photon or quantum yields one electron. Interestingly, an overall stoichiometry of one H^+ translocated into the thylakoid vesicle for each photon has also been observed. Two photons per center would allow a pair of electrons to flow from H_2O to $NADP^+$ (Figure 22.12), resulting in the formation of 1 NADPH and $\frac{1}{2} O_2$. If one ATP were formed for every 3 H^+ translocated during photosynthetic electron transport, $1\frac{1}{3}$ ATP would be synthesized. More appropriately, $4 h\nu$ per center (8 quanta total) would drive the evolution of 1 O_2 , the reduction of 2 $NADP^+$, and the phosphorylation of $2\frac{2}{3}$ ATP.

The energy of a photon depends on its wavelength, according to the equation $E = h\nu = hc/\lambda$, where E is energy, c is the speed of light, and λ is its wavelength. Expressed in molar terms, an *Einstein* is the amount of energy in Avogadro’s number of photons: $E = Nhc/\lambda$. Light of 700-nm wavelength is the longest-wavelength and the lowest-energy light acting in the eukaryotic photosystems discussed here. An Einstein of 700-nm light is equivalent in energy to approximately 170 kJ. Eight Einsteins of this light, 1360 kJ, theoretically generate 2 moles of NADPH, $2\frac{2}{3}$ moles of ATP, and 1 mole of O_2 .

Photosynthetic Energy Requirements for Hexose Synthesis

The fixation of carbon dioxide to form hexose, *the dark reactions of photosynthesis*, requires considerable energy. The overall stoichiometry of this process (Eq. 22.3) involves 12 NADPH and 18 ATP. To generate 12 equivalents of NADPH necessitates the consumption of 48 Einsteins of light, minimally 170 kJ each. However, if the preceding ratio of $1\frac{1}{3}$ ATP per NADPH were correct, insufficient ATP for CO₂ fixation would be produced. Six additional Einsteins would provide the necessary two additional ATP. From 54 Einsteins, or 9180 kJ, one mole of hexose would be synthesized. The standard free energy change, ΔG° , for hexose formation from carbon dioxide and water (the exact reverse of cellular respiration) is +2870 kJ/mol.

22.7 • Light-Driven ATP Synthesis—Photophosphorylation

Light-driven ATP synthesis, termed **photophosphorylation**, is a fundamental part of the photosynthetic process. The conversion of light energy to chemical energy results in electron-transfer reactions leading to the generation of reducing power (NADPH). Coupled with these electron transfers, protons are driven across the thylakoid membranes from the stromal side to the lumenal side. These proton translocations occur in a manner analogous to the proton translocations accompanying mitochondrial electron transport that provide the driving force for oxidative phosphorylation (Chapter 21). Figure 22.12 indicates that proton translocations can occur at a number of sites. For example, protons may be translocated by reactions between H₂O and PSII as a consequence of the photolysis of water. The oxidation–reduction events as electrons pass through the plastoquinone pool and the Q cycle are another source of proton translocations. The proton transfer accompanying NADP⁺ reduction also can be envisioned as protons being taken from the stromal side of the thylakoid vesicle. The current view is that two protons are translocated for each electron that flows from H₂O to NADP⁺. Because this electron transfer requires two photons, one falling at PSII and one at PSI, the overall yield is one proton per quantum of light.

The Mechanism of Photophosphorylation Is Chemiosmotic

The thylakoid membrane is asymmetrically organized, or “sided,” like the mitochondrial membrane. It also shares the property of being a barrier to the passive diffusion of H⁺ ions. Photosynthetic electron transport thus establishes an electrochemical gradient, or proton-motive force, across the thylakoid membrane with the interior, or lumen, side accumulating H⁺ ions relative to the stroma of the chloroplast. Like oxidative phosphorylation, the mechanism of photophosphorylation is chemiosmotic.

A proton-motive force of approximately –250 mV is needed to achieve ATP synthesis. This proton-motive force, Δp , is composed of a membrane potential, $\Delta\Psi$, and a pH gradient, ΔpH (Chapter 21). The proton-motive force is defined as the free energy difference, ΔG , divided by $\mathcal{F}$, Faraday’s constant:

$$\Delta p = \Delta G / \mathcal{F} = \Delta\Psi - (2.3RT/\mathcal{F})\Delta\text{pH} \quad (22.5)$$

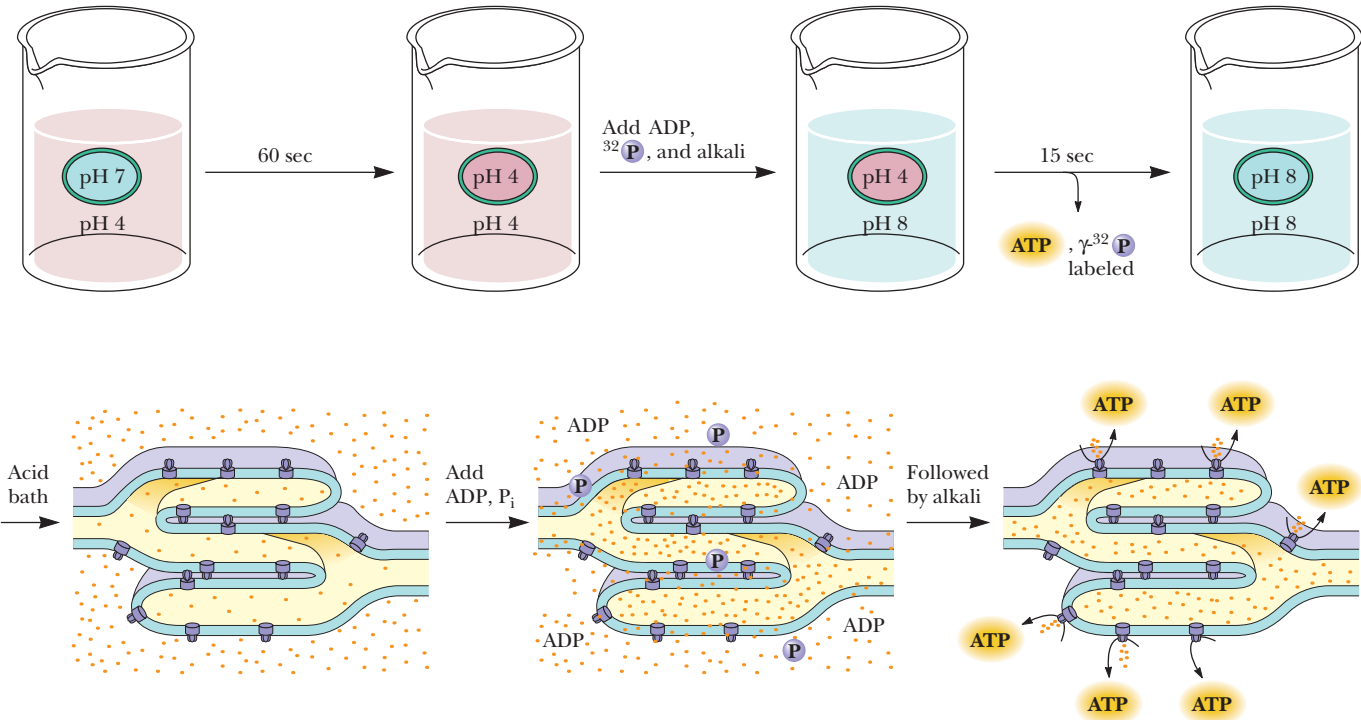
In chloroplasts, the value of $\Delta\Psi$ is typically –50 to –100 mV, and the pH gradient is equivalent to about 3 pH units, so that $-(2.3 RT/\mathcal{F})\Delta\text{pH} = -200$ mV. This situation contrasts with the mitochondrial proton-motive force, where the membrane potential contributes relatively more to Δp than does the pH gradient.

CRITICAL DEVELOPMENTS IN BIOCHEMISTRY

Experiments with Isolated Chloroplasts Provided the First Direct Evidence for the Chemiosmotic Hypothesis

Experimental proof that the mechanism of photophosphorylation is chemiosmotic was provided in an elegant experiment by Andre Jagendorf and Ernest Uribe in 1966 (see figure). Jagendorf and Uribe reasoned that if photophosphorylation were indeed driven by an electrochemical gradient established by photosynthetic electron transfer reactions, they might artificially generate such a gradient by first incubating chloroplasts in an acid bath in the dark and then quickly raising the pH of the external medium. The resulting inequality in hydrogen ion electrochemical activity across the membrane should mimic the conditions normally found upon illumination of chloroplasts and should provide the energized condition necessary to drive ATP formation. To test this interpretation, Jagendorf and Uribe bathed isolated chloroplasts in a weakly acidic (pH 4) medium for 60 seconds, allowing the pH inside the chloroplasts to equilibrate with the external medium. The pH of the solution was then quickly raised to slightly

alkaline pH (pH 8), artificially creating a pH gradient across the thylakoid membranes. When ADP and P_i were added, ATP synthesis was observed as the pH gradient collapsed. This classic experiment was the first real proof of Mitchell's chemiosmotic hypothesis and directed the scientific community to a greater acceptance of Mitchell's interpretations. Mitchell's chemiosmotic hypothesis for ATP synthesis now occupies the position of dogma as the weight of evidence has accumulated in its favor. Photophosphorylation then can be concisely summarized by noting that thylakoid vesicles accumulate H^+ upon illumination and that the electrochemical gradient thus created represents an energized state that can be tapped to drive ATP synthesis. Collapse of the gradient—that is, equilibration of the ion concentration difference across the membrane—is the energy-transducing mechanism: the chemical potential of a concentration difference is transduced into synthesis of ATP.



The mechanism of photophosphorylation is chemiosmotic. In 1966, Jagendorf and Uribe experimentally demonstrated for the first time that establishment of an electrochemical gradient across the membrane of an energy-transducing organelle could lead to ATP synthesis. They equilibrated isolated chloroplasts for 60 seconds in a pH 4 bath, adjusted the pH to 8 in the presence of ADP and P_i , and allowed phosphorylation to proceed for 15 seconds. The entire experiment was carried out in the dark.

CF₁CF₀ ATP Synthase Is the Chloroplast Equivalent of the Mitochondrial F₁F₀ ATP Synthase

The transduction of the electrochemical gradient into the chemical energy represented by ATP is carried out by the chloroplast ATP synthase, which is highly analogous to the mitochondrial F₁F₀ ATP synthase. The chloroplast enzyme complex is called **CF₁CF₀ ATP synthase**, “C” symbolizing chloroplast. Like the mitochondrial complex, CF₁CF₀ ATP synthase is a heteromultimer of α , β , γ , δ , ϵ , a, b, and c subunits (Chapter 21), consisting of a knoblike structure some 9 nm in diameter (CF₁) attached to a stalked base (CF₀) embedded in the thylakoid membrane. The mechanism of action of CF₁CF₀ ATP synthase in coupling ATP synthesis to the collapse of the pH gradient is similar to that of the mitochondrial ATP synthase described in Chapter 21. The mechanism of photophosphorylation is summarized schematically in Figure 22.21.

Cyclic and Noncyclic Photophosphorylation

Photosynthetic electron transport, which pumps H⁺ into the thylakoid lumen, can occur in two modes, both of which lead to the establishment of a transmembrane proton-motive force. Thus, both modes are coupled to ATP synthesis and are considered alternative mechanisms of photophosphorylation even though they are distinguished by differences in their electron transfer pathways. The two modes are cyclic and noncyclic photophosphorylation.

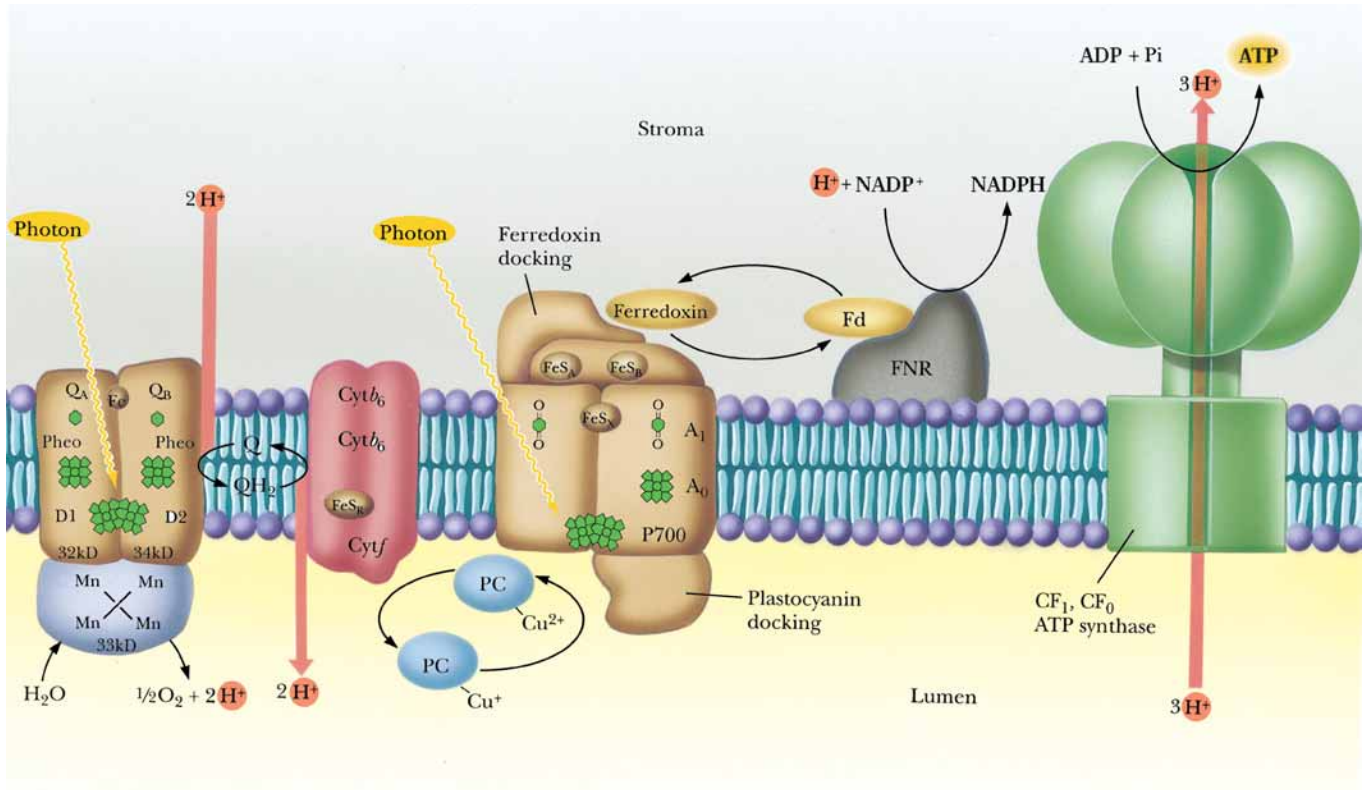


FIGURE 22.21 • The mechanism of photophosphorylation. Photosynthetic electron transport establishes a proton gradient that is tapped by the CF₁CF₀ ATP synthase to drive ATP synthesis. Critical to this mechanism is the fact that the membrane-bound components of light-induced electron transport and ATP synthesis are asymmetrical with respect to the thylakoid membrane so that vectorial discharge and uptake of H⁺ ensue, generating the proton-motive force.

Noncyclic photophosphorylation has been the focus of our discussion and is represented by the scheme in Figure 22.21, where electrons activated by quanta at PSII and PSI flow from H₂O to NADP⁺, with concomitant establishment of the proton-motive force driving ATP synthesis. Note that in noncyclic photophosphorylation, O₂ is evolved and NADP⁺ is reduced.

Cyclic Photophosphorylation

In **cyclic photophosphorylation**, the “electron hole” in P700⁺ created by electron loss from P700 is filled *not* by an electron derived from H₂O via PSII but by a cyclic pathway in which the photoexcited electron returns ultimately to P700⁺. This pathway is schematically represented in Figure 22.12 by the dashed line connecting F_B and cytochrome *b*₆. Thus, the function of cytochrome *b*₆ (*b*₅₆₃) is to couple the bound ferredoxin carriers of the PSI complex with the cytochrome *b*₆/cytochrome *f* complex via the plastoquinone pool. This pathway diverts the activated *e*[−] from NADP⁺ reduction back through plastocyanin to re-reduce P700⁺ (Figure 22.22).

Proton translocations accompany these cyclic electron transfer events, so ATP synthesis can be achieved. In cyclic photophosphorylation, ATP is the sole product of energy conversion. No NADPH is generated, and, because PSII is not involved, no oxygen is evolved. The maximal rate of cyclic photophosphorylation is less than 5% of the rate of noncyclic photophosphorylation. Cyclic photophosphorylation depends only on PSI.

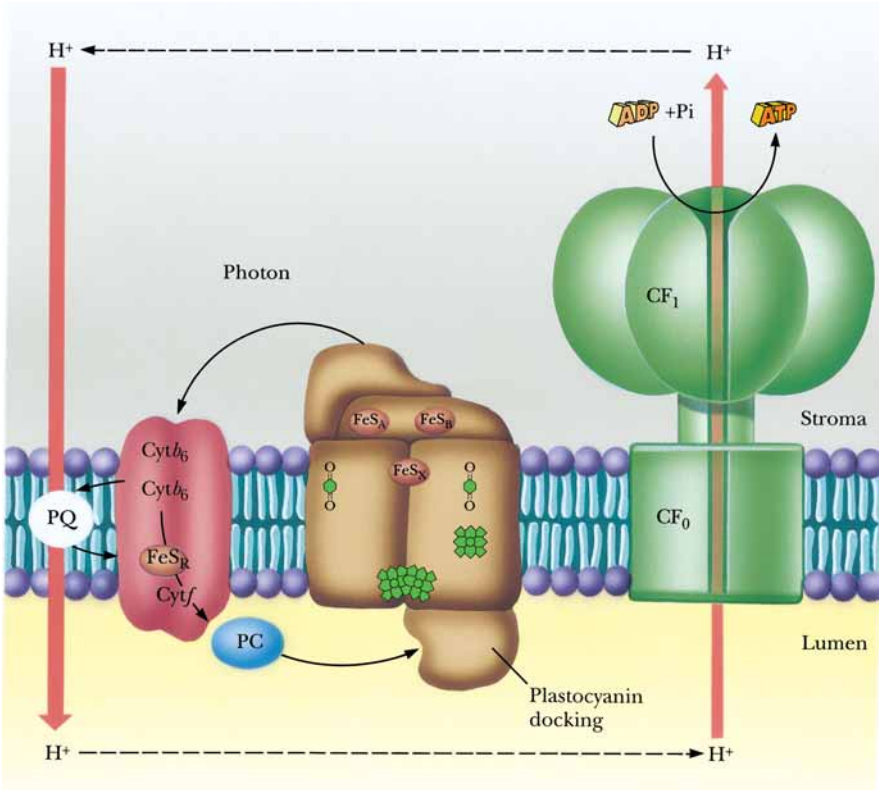
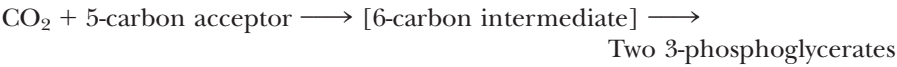


FIGURE 22.22 • The pathway of cyclic photophosphorylation by PSI. (Adapted from Arnon, D. I., 1984. Trends in Biochemical Sciences 9:258.)

22.8 • Carbon Dioxide Fixation

As we began this chapter, we saw that photosynthesis traditionally is equated with the process of CO₂ fixation, that is, the net synthesis of carbohydrate from CO₂. Indeed, the capacity to perform net accumulation of carbohydrate from CO₂ distinguishes the phototrophic (and autotrophic) organisms from heterotrophs. Although animals possess enzymes capable of linking CO₂ to organic acceptors, they cannot achieve a net accumulation of organic material by these reactions. For example, fatty acid biosynthesis is primed by covalent attachment of CO₂ to acetyl-CoA to form malonyl-CoA (Chapter 25). Nevertheless, this “fixed CO₂” is liberated in the very next reaction, so no net CO₂ incorporation occurs.

Elucidation of the pathway of CO₂ fixation represents one of the earliest applications of radioisotope tracers to the study of biology. In 1945, Melvin Calvin and his colleagues at the University of California, Berkeley, were investigating photosynthetic CO₂ fixation in *Chlorella*. Using ¹⁴CO₂, they traced the incorporation of radioactive ¹⁴C into organic products and found that the earliest labeled product was **3-phosphoglycerate** (see Figure 18.13). Although this result suggested that the CO₂ acceptor was a two-carbon compound, further investigation revealed that, in reality, two equivalents of 3-phosphoglycerate were formed following addition of CO₂ to a five-carbon (pentose) sugar:



Ribulose-1,5-Bisphosphate Is the CO₂ Acceptor in CO₂ Fixation

The five-carbon CO₂ acceptor was identified as **ribulose-1,5-bisphosphate** (RuBP), and the enzyme catalyzing this key reaction of CO₂ fixation is **ribulose biphosphate carboxylase/oxygenase**, or, in the jargon used by workers in this field, **rubisco**. The name *ribulose biphosphate carboxylase/oxygenase* reflects the fact that rubisco catalyzes the reaction of either CO₂ or, alternatively, O₂ with RuBP. Rubisco is found in the chloroplast stroma. It is a very abundant enzyme, constituting more than 15% of the total chloroplast protein. Given the preponderance of plant material in the biosphere, rubisco is probably the world’s most abundant protein. Rubisco is large: in higher plants, rubisco is a 550-kD heteromultimeric (α₈β₈) complex consisting of eight identical large subunits (55 kD) and eight small subunits (15 kD) (Figure 22.23). The large subunit is the catalytic unit of the enzyme. It binds both substrates (CO₂ and RuBP) and Mg²⁺ (a divalent cation essential for enzymatic activity). The small subunit modulates the activity of the enzyme, increasing *k*_{cat} more than 100-fold.²

The Ribulose-1,5-Bisphosphate Carboxylase Reaction

The addition of CO₂ to ribulose-1,5-bisphosphate results in the formation of an enzyme-bound intermediate, **2-carboxy,3-keto-arabinitol** (Figure 22.24). This intermediate arises when CO₂ adds to the enediol intermediate gener-

²The rubisco large subunit is encoded by a gene within the chloroplast DNA, whereas the small subunit is encoded by a multigene family in the nuclear DNA. Assembly of active rubisco heteromultimers occurs within chloroplasts following transit of the small subunit polypeptide across the chloroplast membrane.

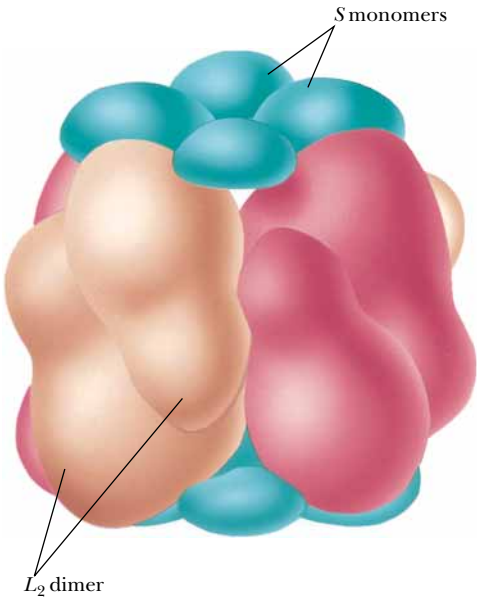


FIGURE 22.23 • Schematic diagram of the subunit organization of ribulose biphosphate carboxylase as revealed by X-ray crystallography. The enzyme consists of eight equivalents each of two types of subunits, large *L* (55 kD) and small *S* (15 kD). Clusters of four small subunits are located at each end of the symmetrical octamer formed by four *L*₂ dimers. (From Knight, S., Andersson, I., and Branden, C. I., 1990. *Journal of Molecular Biology* 215:113–160.)

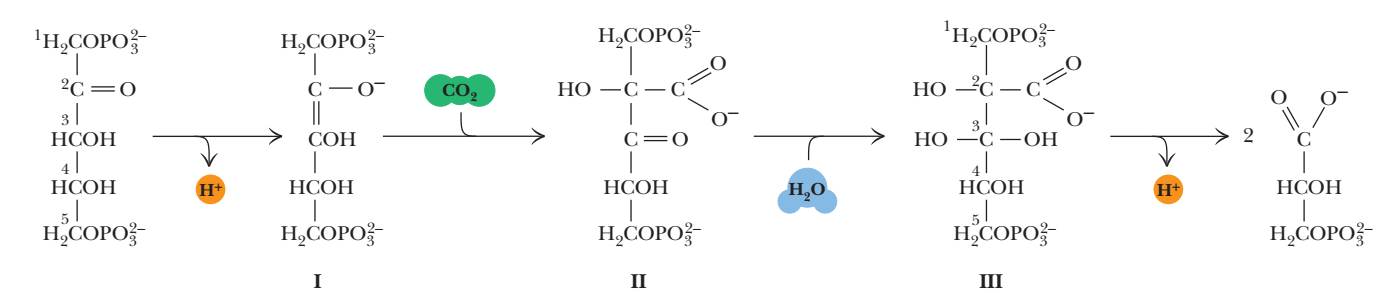


FIGURE 22.24 • The ribulose biphosphate carboxylase reaction. Enzymatic abstraction of the C-3 proton of RuBP yields a 2,3-enediol intermediate (I), which is stereospecifically carboxylated at C-2 to create the six-carbon β -keto acid intermediate (II) known as 2-carboxy,3-keto-arabinitol. Intermediate II is rapidly hydrated to give the gem-diol form (III). Deprotonation of the C-3 hydroxyl and cleavage yield two 3-phosphoglycerates. Mg^{2+} at the active site aids in stabilizing the 2,3-enediol transition state for CO_2 addition and in facilitating the carbon–carbon bond cleavage that leads to product formation. Note that CO_2 , not HCO_3^- (its hydrated form), is the true substrate.

ated from ribulose-1,5-bisphosphate. Hydrolysis of the C₂–C₃ bond of the intermediate generates two molecules of 3-phosphoglycerate. The CO₂ ends up as the carboxyl group of one of the two molecules.

Regulation of Ribulose-1,5-Bisphosphate Carboxylase Activity

Rubisco exists in three forms: an inactive form designated *E*; a carbamylated, but inactive, form designated *EC*; and an active form, *ECM*, which is carbamylated and has Mg^{2+} at its active sites as well. Carbamylation of rubisco takes place by addition of CO₂ to its Lys²⁰¹ $\epsilon\text{-NH}_2$ groups (to give $\epsilon\text{-NH-COO}^-$ derivatives). The CO₂ molecules used to carbamylate Lys residues do not become substrates. The carbamylation reaction is promoted by slightly alkaline pH (pH 8). Carbamylation of rubisco completes the formation of a binding site for the Mg^{2+} that participates in the catalytic reaction. Once Mg^{2+} binds to *EC*, rubisco achieves its active *ECM* form. Activated rubisco displays a K_m for CO₂ of 10 to 20 μM .³

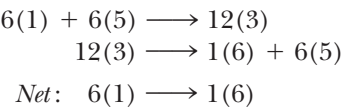
Substrate RuBP binds much more tightly to the inactive *E* form of rubisco ($K_D = 20\text{ nM}$) than to the active *ECM* form (K_m for RuBP = 20 μM). Thus, RuBP is also a potent inhibitor of rubisco activity. Release of RuBP from the active site of rubisco is mediated by **rubisco activase**. Rubisco activase is a *regulatory protein*; it binds to *E*-form rubisco and, in an ATP-dependent reaction, promotes the release of RuBP. Rubisco then becomes activated by carbamylation and Mg^{2+} binding. Rubisco activase itself is activated in an indirect manner by light. Thus, light is the ultimate activator of rubisco.

22.9 • The Calvin–Benson Cycle

The immediate product of CO₂ fixation, 3-phosphoglycerate, must undergo a series of transformations before the net synthesis of carbohydrate is realized. Among carbohydrates, hexoses (particularly glucose) occupy center stage. Glucose is the building block for both cellulose and starch synthesis. These plant polymers constitute the most abundant organic material in the living world, and thus, the central focus on glucose as the ultimate end product of CO₂ fixation is amply justified. Also, sucrose ($\alpha\text{-D-glucopyranosyl-(1} \rightarrow \text{2)-}\beta\text{-D-fructofuranoside}$) is the major carbon form translocated out of leaves to other plant tissues. In nonphotosynthetic tissues, sucrose is metabolized via glycolysis and the TCA cycle to produce ATP.

³The relative abundance of CO₂ in the atmosphere is low, about 0.03%. The concentration of CO₂ dissolved in aqueous solutions equilibrated with air is about 10 μM .

The set of reactions that transforms 3-phosphoglycerate into hexose is named the **Calvin–Benson cycle** (often referred to simply as the *Calvin cycle*) for its discoverers. The reaction series is indeed cyclic because not only must carbohydrate appear as an end product, but the 5-carbon acceptor, RuBP, must be regenerated to provide for continual CO₂ fixation. Balanced equations that schematically represent this situation are



Each number in parentheses represents the number of carbon atoms in a compound, and the number preceding the parentheses indicates the stoichiometry of the reaction. Thus, 6(1), or 6 CO₂, condense with 6(5) or 6 RuBP to give 12 3-phosphoglycerates. These 12(3)s are then rearranged in the Calvin cycle to form one hexose, 1(6), and regenerate the six 5-carbon (RuBP) acceptors.

The Enzymes of the Calvin Cycle

The Calvin cycle enzymes serve three important ends:

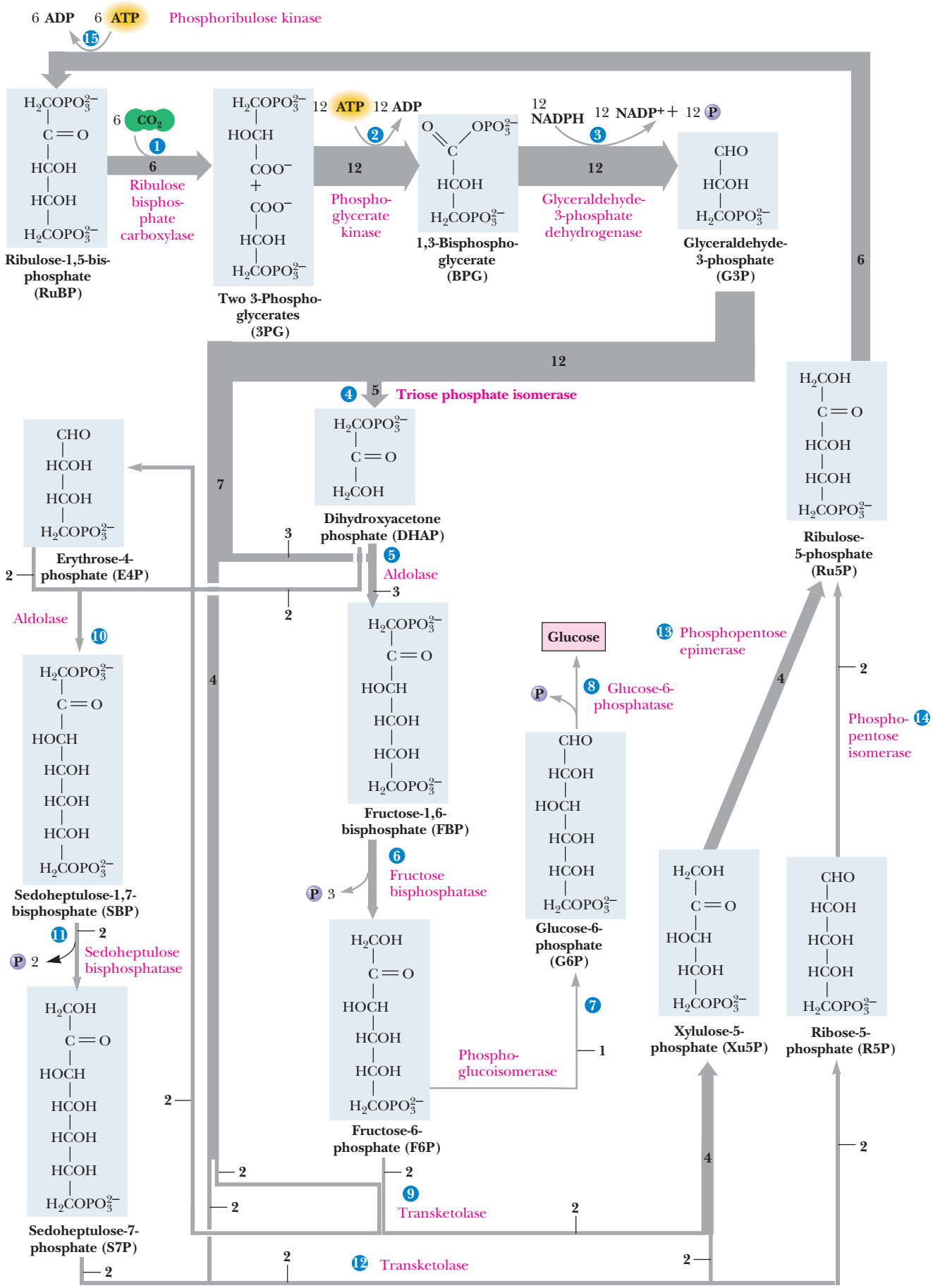
1. They constitute the predominant CO₂ fixation pathway in nature.
2. They accomplish the reduction of 3-phosphoglycerate, the primary product of CO₂ fixation, to glyceraldehyde-3-phosphate so that carbohydrate synthesis becomes feasible.
3. They catalyze reactions that transform 3-carbon compounds into 4-, 5-, 6-, and 7-carbon compounds.

Most of the enzymes mediating the reactions of the Calvin cycle also participate in either glycolysis (Chapter 19) or the pentose phosphate pathway (Chapter 23). The aim of the Calvin scheme is to account for hexose formation from 3-phosphoglycerate. In the course of this metabolic sequence, the NADPH and ATP produced in the light reactions are consumed, as indicated earlier in Equation (22.3).

The Calvin cycle of reactions starts with *ribulose biphosphate carboxylase* catalyzing formation of 3-phosphoglycerate from CO₂ and RuBP and concludes with **ribulose-5-phosphate kinase** (also called *phosphoribulose kinase*), which forms RuBP (Figure 22.25 and Table 22.1). The carbon balance is given at the right side of the table. Several features of the reactions in Table 22.1 merit discussion. Note that the 18 equivalents of ATP consumed in hexose formation are expended in reactions 2 and 15: 12 to form 12 equivalents of 1,3-bisphosphoglycerate from 3-phosphoglycerate by a reversal of the normal glycolytic reaction catalyzed by **3-phosphoglycerate kinase**, and six to phosphorylate Ru-5-P to regenerate 6 RuBP. All 12 NADPH equivalents are used in reaction 3. Plants possess an **NADPH-specific glyceraldehyde-3-phosphate dehydrogenase**, which contrasts with its glycolytic counterpart in its specificity for NADP over NAD and in the direction in which the reaction normally proceeds.

Balancing the Calvin Cycle Reactions To Account for Net Hexose Synthesis

When carbon rearrangements are balanced to account for net hexose synthesis, five of the glyceraldehyde-3-phosphate molecules are converted to dihydroxyacetone phosphate (DHAP). Three of these DHAPs then condense with three glyceraldehyde-3-P via the aldolase reaction to yield 3 hexoses in the form



◀ **FIGURE 22.25** • The Calvin–Benson cycle of reactions. The number associated with the arrow at each step indicates the number of molecules reacting in a turn of the cycle that produces one molecule of glucose. Reactions are numbered as in Table 22.1.

of fructose biphosphate (Figure 22.25). (Recall that the $\Delta G^\circ'$ for the aldolase reaction in the glycolytic direction is +23.9 kJ/mol. Thus, the aldolase reaction running “in reverse” in the Calvin cycle would be thermodynamically favored under standard-state conditions.) Taking one FBP to glucose, the desired product of this scheme, leaves 30 carbons, distributed as two fructose-6-phosphates, four glyceraldehyde-3-phosphates, and 2 DHAP. These 30 Cs are reorganized into 6 RuBP by reactions 9 through 15. Step 9 and steps 12 through 14 involve carbohydrate rearrangements like those in the pentose phosphate pathway (see Chapter 23). Reaction 11 is mediated by **sedoheptulose-1,7-bisphosphatase**. This phosphatase is unique to plants; it generates sedoheptulose-7-P, the seven-carbon sugar serving as the transketolase substrate. Likewise, **phosphoribulose kinase** carries out the unique plant function of providing RuBP from Ru-5-P (reaction 15). The net conversion accounts for the fixation of six equivalents of carbon dioxide into one hexose at the expense of 18 ATP and 12 NADPH.

Table 22.1

The Calvin Cycle Series of Reactions

Reactions 1 through 15 constitute the cycle that leads to the formation of one equivalent of glucose. The enzyme catalyzing each step, a concise reaction, and the overall carbon balance is given. Numbers in parentheses show the numbers of carbon atoms in the substrate and product molecules. Prefix numbers indicate in a stoichiometric fashion how many times each step is carried out in order to provide a balanced net reaction.

1. Ribulose biphosphate carboxylase: 6 CO ₂ + 6 H ₂ O + 6 RuBP → 12 3-PG	6(1) + 6(5) → 12(3)
2. 3-Phosphoglycerate kinase: 12 3-PG + 12 ATP → 12 1,3-BPG + 12 ADP	12(3) → 12(3)
3. NADP ⁺ -glyceraldehyde-3-P dehydrogenase: 12 1,3-BPG + 12 NADPH → 12 NADP ⁺ + 12 G3P + 12 P _i	12(3) → 12(3)
4. Triose-P isomerase: 5 G3P → 5 DHAP	5(3) → 5(3)
5. Aldolase: 3 G3P + 3 DHAP → 3 FBP	3(3) + 3(3) → 3(6)
6. Fructose biphosphatase: 3 FBP + 3 H ₂ O → 3 F6P + 3 P _i	3(6) → 3(6)
7. Phosphoglucoisomerase: 1 F6P → 1 G6P	1(6) → 1(6)
8. Glucose phosphatase: 1 G6P + 1 H ₂ O → 1 GLUCOSE + 1 P _i	1(6) → 1(6)
The remainder of the pathway involves regenerating six RuBP acceptors (= 30 C) from the leftover two F6P (12 C), four G3P (12 C), and two DHAP (6 C).	
9. Transketolase: 2 F6P + 2 G3P → 2 Xu5P + 2 E4P	2(6) + 2(3) → 2(5) + 2(4)
10. Aldolase: 2 E4P + 2 DHAP → 2 sedoheptulose-1,7-bisphosphate (SBP)	2(4) + 2(3) → 2(7)
11. Sedoheptulose biphosphatase: 2 SBP + 2 H ₂ O → 2 S7P + 2 P _i	2(7) → 2(7)
12. Transketolase: 2 S7P + 2 G3P → 2 Xu5P + 2 R5P	2(7) + 2(3) → 4(5)
13. Phosphopentose epimerase: 4 Xu5P → 4 Ru5P	4(5) → 4(5)
14. Phosphopentose isomerase: 2 R5P → 2 Ru5P	2(5) → 2(5)
15. Phosphoribulose kinase: 6 Ru5P + 6 ATP → 6 RuBP + 6 ADP	6(5) → 6(5)
Net: 6 CO ₂ + 18 ATP + 12 NADPH + 12 H ⁺ + 12 H ₂ O → glucose + 18 ADP + 18 P _i + 12 NADP ⁺	6(1) → 1(6)

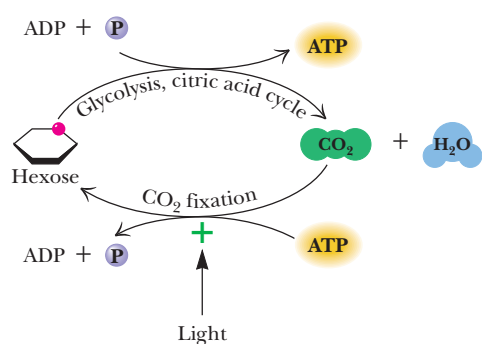


FIGURE 22.26 • Light regulation of CO₂ fixation prevents a substrate cycle between cellular respiration and hexose synthesis by CO₂ fixation. Because plants possess mitochondria and are capable of deriving energy from hexose catabolism (glycolysis and the citric acid cycle), regulation of photosynthetic CO₂ fixation by light activation controls the net flux of carbon between these opposing routes.

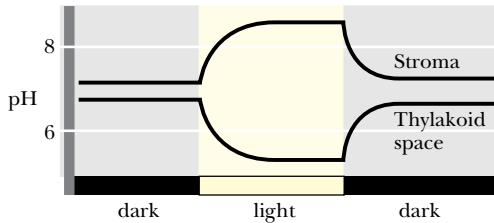


FIGURE 22.27 • Light-induced pH changes in chloroplast compartments. Illumination of chloroplasts leads to proton pumping and pH changes in the chloroplast, such that the pH within the thylakoid space falls and the pH of the stroma rises. These pH changes modulate the activity of key Calvin cycle enzymes.

FIGURE 22.28 • The pathway for light regulation of Calvin cycle enzymes. Light-generated reducing power (Fd_{red} = reduced ferredoxin) provides e[−] for reduction of thioredoxin (T) by FTR (ferredoxin–thioredoxin reductase). Several Calvin cycle enzymes have pairs of Cys residues that are involved in the disulfide–sulfhydryl transition between an inactive (—S—S—) form and an active (—SH HS—) form, as shown here. These enzymes include *fructose-1,6-bisphosphatase* (residues Cys¹⁷⁴ and Cys¹⁷⁹), *NADP⁺-malate dehydrogenase* (residues Cys¹⁰ and Cys¹⁵), and *ribulose-5-P kinase* (residues Cys¹⁶ and Cys⁵⁵).

22.10 • Regulation of Carbon Dioxide Fixation

Plant cells contain mitochondria and can carry out cellular respiration (glycolysis, the citric acid cycle, and oxidative phosphorylation) to provide energy in the dark. Futile cycling of carbohydrate to CO₂ by glycolysis and the citric acid cycle in one direction, and CO₂ to carbohydrate by the CO₂ fixation pathway in the opposite direction, is thwarted through regulation of the Calvin cycle (Figure 22.26). In this regulation, the activities of key Calvin cycle enzymes are coordinated with the output of photosynthesis. In effect, these enzymes respond indirectly to *light activation*. Thus, when light energy is available to generate ATP and NADPH for CO₂ fixation, the Calvin cycle proceeds. In the dark, when ATP and NADPH cannot be produced by photosynthesis, fixation of CO₂ ceases. The light-induced changes in the chloroplast which regulate key Calvin cycle enzymes include (1) *changes in stromal pH*, (2) *generation of reducing power*, and (3) *Mg²⁺ efflux from the thylakoid lumen*.

Light-Induced pH Changes in Chloroplast Compartments

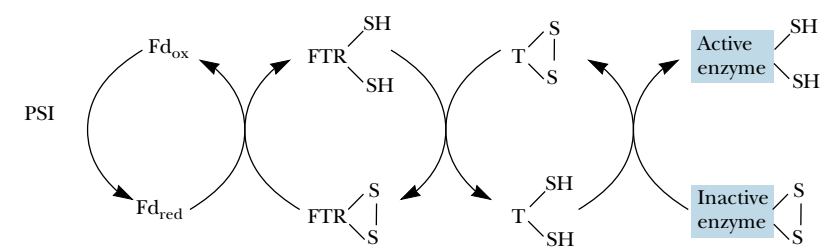
As discussed in Section 22.7, illumination of chloroplasts leads to light-driven pumping of protons into the thylakoid lumen, which causes pH changes in both the stroma and the thylakoid lumen (Figure 22.27). The stromal pH rises, typically to pH 8. Because rubisco and rubisco activase are more active at pH 8, CO₂ fixation is activated as stromal pH rises. *Fructose-1,6-bisphosphatase*, *ribulose-5-phosphate kinase*, and *glyceraldehyde-3-phosphate dehydrogenase* all have alkaline pH optima. Thus, their activities increase as a result of the light-induced pH increase in the stroma.

Light-Induced Generation of Reducing Power

Illumination of chloroplasts initiates photosynthetic electron transport, which generates reducing power in the form of reduced ferredoxin and NADPH. Several enzymes of CO₂ fixation, notably *fructose-1,6-bisphosphatase*, *sedoheptulose-1,7-bisphosphatase*, and *ribulose-5-phosphate kinase*, are activated upon reduction of specific Cys-Cys disulfide bonds to cysteine sulfhydryls. The reduced form of **thioredoxin** mediates this reaction. Thioredoxin is a small (12 kD) protein possessing in its reduced state a pair of sulfhydryls (—SH HS—), which upon oxidation form a disulfide bridge (—S—S—). Thioredoxin serves as the hydrogen carrier between NADPH or Fd_{red} and enzymes regulated by light (Figure 22.28).

Light-Induced Mg²⁺ Efflux from Thylakoid Vesicles

When light-driven proton pumping across the thylakoid membrane occurs, a concomitant efflux of Mg²⁺ ions from vesicles into the stroma is observed. This efflux of Mg²⁺ somewhat counteracts the charge accumulation due to H⁺



influx and is one reason why the membrane potential change in response to proton pumping is less in chloroplasts than in mitochondria (Eq. 22.5). Both *ribulose biphosphate carboxylase* and *fructose-1,6-biphosphatase* are Mg^{2+} -activated enzymes, and Mg^{2+} flux into the stroma as a result of light-driven proton pumping stimulates the CO_2 fixation pathway at these key steps. Activity measurements have indicated that fructose biphosphatase may be the rate-limiting step in the Calvin cycle. The recurring theme of fructose biphosphatase as the target of the light-induced changes in the chloroplasts implicates this enzyme as a key point of control in the Calvin cycle.

22.11 • The Ribulose Biphosphate Oxygenase Reaction: Photorespiration

As indicated, ribulose biphosphate carboxylase/oxygenase catalyzes an alternative reaction in which O_2 replaces CO_2 as the substrate added to RuBP (Figure 22.29a). The *ribulose-1,5-biphosphate oxygenase* reaction diminishes plant

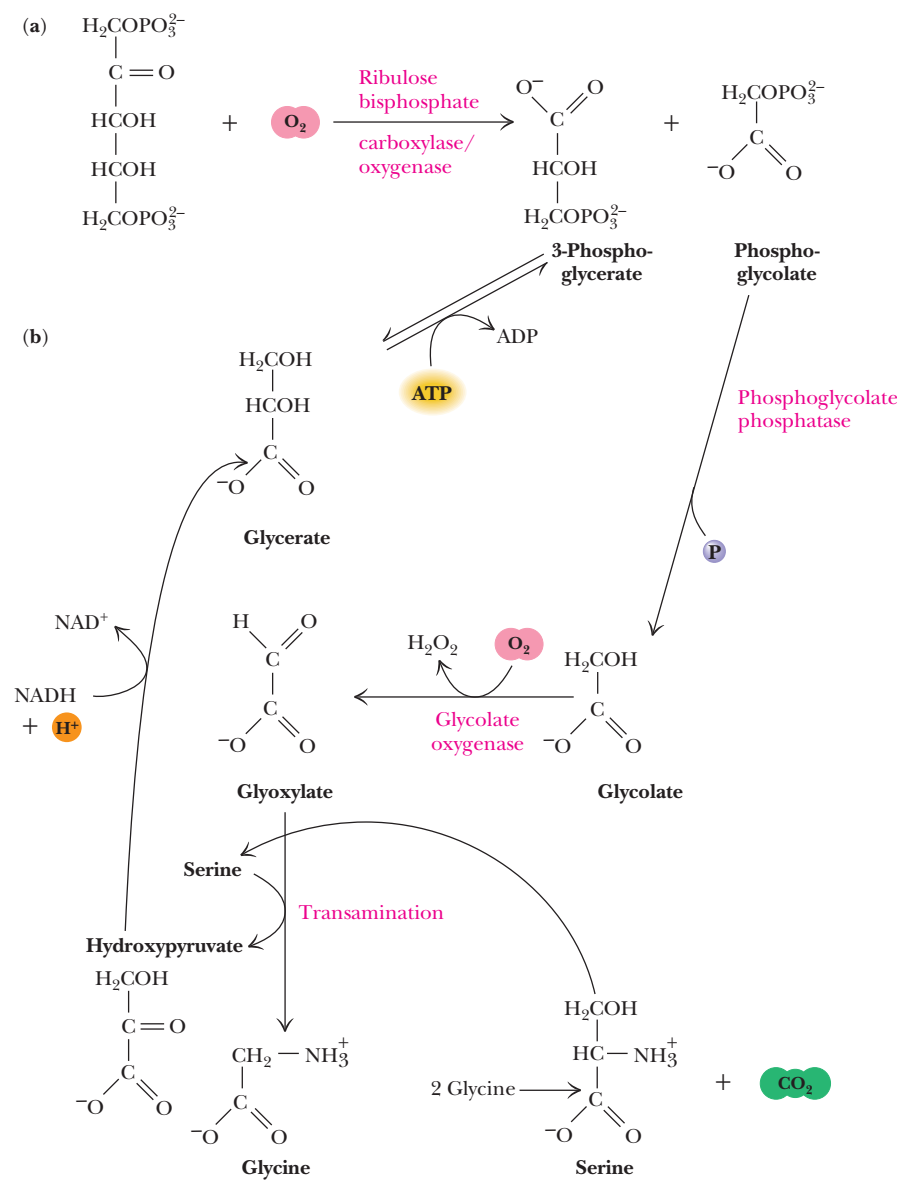


FIGURE 22.29 • The oxygenase reaction of rubisco. (a) The reaction of ribulose biphosphate carboxylase with O_2 in the presence of ribulose biphosphate leads to wasteful cleavage of RuBP to yield 3-phosphoglycerate and phosphoglycolate. (b) Conversion of phosphoglycolate to glycine. In mitochondria, two glycines from photorespiration are converted into one serine plus CO_2 . This step is the source of the CO_2 evolved in photorespiration. Transamination of glyoxylate to glycine by the product serine yields hydroxypyruvate; reduction of hydroxypyruvate yields glycerate, which can be phosphorylated to 3-phosphoglycerate. 3-Phosphoglycerate can fuel resynthesis of ribulose biphosphate by the Calvin cycle (Figure 22.25).

productivity because it leads to loss of RuBP, the essential CO₂ acceptor. The K_m for O₂ in this oxygenase reaction is about 200 μ M. Given the relative abundance of CO₂ and O₂ in the atmosphere and their relative K_m values in these rubisco-mediated reactions, the ratio of carboxylase to oxygenase activity *in vivo* is about 3 or 4 to 1.

The products of ribulose biphosphate oxygenase activity are *3-phosphoglycerate* and *phosphoglycolate*. Dephosphorylation and oxidation convert phosphoglycolate to **glyoxylate**, the α -keto acid of glycine (Figure 22.29b). Transamination yields glycine. Other fates of phosphoglycolate are also possible, including oxidation to CO₂, with the released energy being dissipated as heat. Obviously, agricultural productivity is dramatically lowered by this phenomenon, which, because it is a light-related uptake of O₂ and release of CO₂, is termed **photorespiration**. As we shall see, certain plants, particularly tropical grasses, have evolved means to circumvent photorespiration. These plants are more efficient users of light for carbohydrate synthesis.

22.12 • The C-4 Pathway of CO₂ Fixation

Tropical grasses are less susceptible to the effects of photorespiration, as noted earlier. Studies employing ¹⁴CO₂ as a tracer indicated that the first organic intermediate labeled in these plants was not a three-carbon compound but a four-carbon compound. Hatch and Slack, two Australian biochemists, first discovered this C-4 product of CO₂ fixation, and the C-4 pathway of CO₂ incorporation is named the *Hatch–Slack pathway* after them. The C-4 pathway is not an alternative to the Calvin cycle series of reactions or even a net CO₂ fixation scheme. Instead, it functions as a *CO₂ delivery system*, carrying carbon dioxide from the relatively oxygen-rich surface of the leaf to interior cells where oxygen is lower in concentration and hence less effective in competing with CO₂ in the rubisco reaction. Thus, the C-4 pathway is a means of avoiding photorespiration by sheltering the rubisco reaction in a cellular compartment away from high [O₂]. The C-4 compounds serving as CO₂ transporters are *malate* or *aspartate*.

Compartmentation of these reactions to prevent photorespiration involves the interaction of two cell types, *mesophyll cells* and *bundle sheath cells*. The mesophyll cells take up CO₂ at the leaf surface, where O₂ is abundant, and use it to carboxylate phosphoenolpyruvate to yield OAA in a reaction catalyzed by **PEP carboxylase** (Figure 22.30). This four-carbon dicarboxylic acid is then either reduced to malate by an **NADPH-specific malate dehydrogenase** or transaminated to give *aspartate* in the mesophyll cells.⁴ The 4-C CO₂ carrier (malate or aspartate) then is transported to the bundle sheath cells, where it is decarboxylated to yield CO₂ and a 3-C product. The CO₂ is then fixed into organic carbon by the Calvin cycle localized within the bundle sheath cells, and the 3-C product is returned to the mesophyll cells, where it is reconverted to PEP in preparation to accept another CO₂ (Figure 22.30). Plants that use the C-4 pathway are termed **C4 plants**, in contrast to those plants with the conventional pathway of CO₂ uptake (**C3 plants**).

⁴A number of different biochemical subtypes of C4 plants are known. They differ in whether OAA or malate is the CO₂ carrier to the bundle sheath cell and in the nature of the reaction by which the CO₂ carrier is decarboxylated to regenerate a 3-C product. In all cases, the 3-C product is returned to the mesophyll cell and reconverted to PEP.

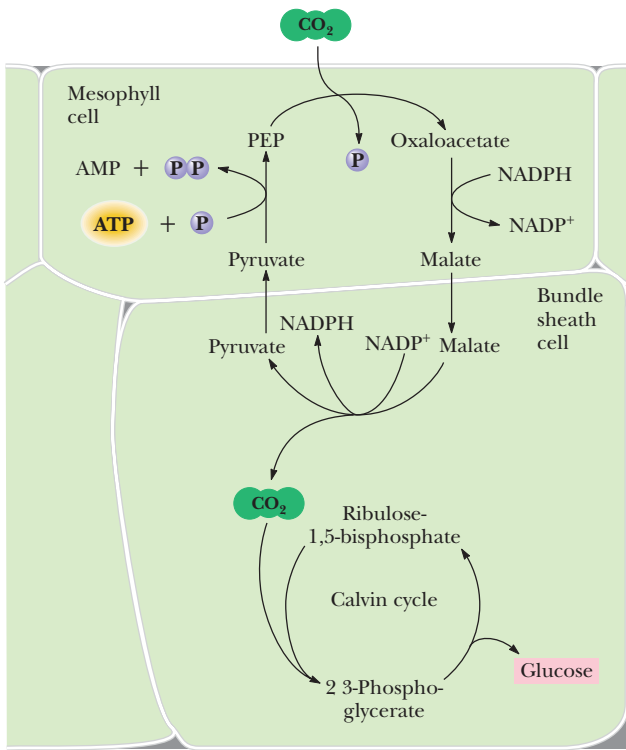
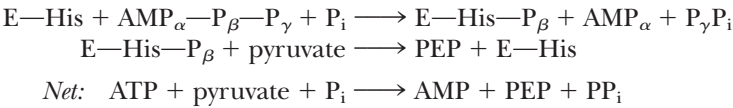


FIGURE 22.30 • Essential features of the compartmentation and biochemistry of the Hatch–Slack pathway of carbon dioxide uptake in C4 plants. Carbon dioxide is fixed into organic linkage by PEP carboxylase of mesophyll cells, forming OAA. Either malate (the reduced form of OAA) or aspartate (the aminated form) serves as the carrier transporting CO₂ to the bundle sheath cells. Within the bundle sheath cells, CO₂ is liberated by decarboxylation of malate or aspartate; the C-3 product is returned to the mesophyll cell. Formation of PEP by pyruvate:P_i dikinase reinitiates the cycle. The CO₂ liberated in the bundle sheath cell is used to synthesize hexose by the conventional rubisco–Calvin cycle series of reactions.

Intercellular Transport of Each CO₂ via a C-4 Intermediate Costs 2 ATP

The transport of each CO₂ requires the expenditure of two high-energy phosphate bonds. The energy of these bonds is expended in the phosphorylation of pyruvate to PEP (phosphoenolpyruvate) by the plant enzyme **pyruvate-P_i dikinase**; the products are PEP, AMP, and pyrophosphate (PP_i). This represents a unique phosphotransferase reaction in that both the β- and γ-phosphates of a single ATP are used to phosphorylate the two substrates, pyruvate and P_i. The reaction mechanism involves an enzyme phosphohistidine intermediate. The γ-phosphate of ATP is transferred to P_i, whereas formation of E-His-P occurs by addition of the β-phosphate from ATP:



Pyruvate-P_i dikinase is regulated by reversible phosphorylation of a threonine residue, the nonphosphorylated form being active. Interestingly, ADP is the phosphate donor in this interconvertible regulation. Despite the added metabolic expense of two phosphodiester bonds for each equivalent of carbon dioxide taken up, CO₂ fixation is more efficient in C4 plants, provided that light intensities and temperatures are both high. (As temperature rises, photorespiration in C3 plants rises and efficiency of CO₂ fixation falls.) Tropical grasses that are C4 plants include sugarcane, maize, and crabgrass. In terms of photosynthetic efficiency, cultivated fields of sugarcane represent the pinnacle of light-harvesting efficiency. Approximately 8% of the incident light energy on a sugarcane field appears as chemical energy in the form of CO₂ fixed into carbohydrate. This efficiency compares dramatically with the estimated pho-

tosynthetic efficiency of 0.2% for uncultivated plant areas. Research on photorespiration is actively pursued in hopes of enhancing the efficiency of agriculture by controlling this wasteful process. Only 1% of the 230,000 different plant species known are C4 plants; most are in hot climates.

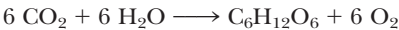
22.13 • Crassulacean Acid Metabolism

In contrast to C4 plants, which have separated CO₂ uptake and fixation into distinct cells in order to minimize photorespiration, succulent plants native to semiarid and tropical environments separate CO₂ uptake and fixation in time. Carbon dioxide (as well as O₂) enters the leaf through microscopic pores known as **stomata**, and water vapor escapes from plants via these same openings. In nonsucculent plants, the stomata are open during the day, when light can drive photosynthetic CO₂ fixation, and closed at night. Succulent plants, such as the *Cactaceae* (cacti) and *Crassulaceae*, cannot open their stomata during the heat of day because any loss of precious H₂O in their arid habitats would doom them. Instead, these plants open their stomata to take up CO₂ only at night, when temperatures are lower and water loss is less likely. This carbon dioxide is immediately incorporated into PEP to form OAA by PEP carboxylase; OAA is then reduced to malate by malate dehydrogenase and stored within vacuoles until morning. During the day, the malate is released from the vacuoles and decarboxylated to yield CO₂ and a 3-C product. The CO₂ is then fixed into organic carbon by rubisco and the reactions of the Calvin cycle. Because this process involves the accumulation of organic acids (OAA, malate) and is common to succulents of the *Crassulaceae* family, it is referred to as *crassulacean acid metabolism*, and plants capable of it are called *CAM plants*.

PROBLEMS

1. In photosystem I, P700 in its ground state has an $\mathcal{E}_o' = +0.4\text{ V}$. Excitation of P700 by a photon of 700-nm light alters the $\mathcal{E}_o'$ of P700* to -0.6 V . What is the efficiency of energy capture in this light reaction of P700?
2. What is the $\mathcal{E}_o'$ for the light-generated primary oxidant of photosystem II if the light-induced oxidation of water (which leads to O₂ evolution) proceeds with a $\Delta G^\circ'$ of -25 kJ/mol ?
3. Assuming that the concentrations of ATP, ADP, and P_i in chloroplasts are 3 mM, 0.1 mM, and 10 mM, respectively, what is the ΔG for ATP synthesis under these conditions? Photosynthetic electron transport establishes the proton-motive force driving photophosphorylation. What redox potential difference is necessary to achieve ATP synthesis under the foregoing conditions, assuming an electron pair is transferred per molecule of ATP generated?
4. ¹⁴C-labeled carbon dioxide is administered to a green plant, and shortly thereafter the following compounds are isolated from the plant: 3-phosphoglycerate, glucose, erythrose-4-phosphate, sedoheptulose-1,7-bisphosphate, ribose-5-phosphate. In which carbon atoms will radioactivity be found?
5. Write a balanced equation for the synthesis of a glucose molecule from ribulose-1,5-bisphosphate and CO₂ that involves the first three reactions of the Calvin cycle and subsequent conversion of the two glyceraldehyde-3-P molecules into glucose.

6. If noncyclic photosynthetic electron transport leads to the translocation of 3 H⁺/*e*[−] and cyclic photosynthetic electron transport leads to the translocation of 2 H⁺/*e*[−], what is the relative photosynthetic efficiency of ATP synthesis (expressed as the number of photons absorbed per ATP synthesized) for noncyclic *versus* cyclic photophosphorylation? (Assume that the CF₁CF₀ ATP synthase yields 1 ATP/3 H⁺.)
7. The photosynthetic CO₂ fixation pathway is regulated in response to specific effects induced in chloroplasts by light. What is the nature of these effects, and how do they regulate this metabolic pathway?
8. The overall equation for photosynthetic CO₂ fixation is



All the O atoms evolved as O₂ come from water; *none* comes from carbon dioxide. But 12 O atoms are evolved as 6 O₂, and only 6 O atoms appear as 6 H₂O in the equation. Also, 6 CO₂ have 12 O atoms, yet there are only 6 O atoms in C₆H₁₂O₆. How can you account for these discrepancies? (*Hint*: Consider the partial reactions of photosynthesis: ATP synthesis, NADP⁺ reduction, photolysis of water, and the overall reaction for hexose synthesis in the Calvin–Benson cycle.)

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Con pan y vino se anda el camino.
(With bread and wine you can walk
your road.)

Spanish proverb

OUTLINE

- 23.1 • Gluconeogenesis
- 23.2 • Regulation of Gluconeogenesis
- 23.3 • Glycogen Catabolism
- 23.4 • Glycogen Synthesis
- 23.5 • Control of Glycogen Metabolism
- 23.6 • The Pentose Phosphate Pathway

Chapter 23

Gluconeogenesis, Glycogen Metabolism, and the Pentose Phosphate Pathway



Bread and pasteries on a rack at a French bakery, Paris. Carbohydrates such as these provide a significant portion of human caloric intake. (© Steven Rothfeld/Tony Stone Images)

As shown in Chapters 18 and 19, the metabolism of sugars is an important source of energy for cells. Animals, including humans, typically obtain significant amounts of glucose and other sugars from the breakdown of starch and glycogen in their diets. Glucose can also be supplied via breakdown of cellular reserves of glycogen (in animals) or starch (in plants). Significantly, glucose also can be synthesized from noncarbohydrate precursors by a process known as *gluconeogenesis*. Each of these important pathways, as well as the synthesis of glycogen from glucose, will be examined in this chapter.

Another pathway of glucose catabolism, the *pentose phosphate pathway*, is the primary source of NADPH, the reduced coenzyme essential to most reductive biosynthetic processes. For example, NADPH is crucial to the biosynthesis of

fatty acids (Chapter 25) and amino acids (Chapter 26). The pentose phosphate pathway also results in the production of ribose-5-phosphate, an essential component of ATP, NAD⁺, FAD, coenzyme A, and particularly DNA and RNA. This important pathway will also be considered in this chapter.

23.1 • Gluconeogenesis

The ability to synthesize glucose from common metabolites is very important to most organisms. Human metabolism, for example, consumes about 160 ± 20 grams of glucose per day, about 75% of this in the brain. Body fluids carry only about 20 grams of free glucose, and glycogen stores normally can provide only about 180 to 200 grams of free glucose. Thus, the body carries only a little more than a one-day supply of glucose. If glucose is not obtained in the diet, the body must produce new glucose from noncarbohydrate precursors. The term for this activity is **gluconeogenesis**, which means the generation (*genesis*) of new (*neo*) glucose.

Further, muscles consume large amounts of glucose via glycolysis, producing large amounts of pyruvate. In vigorous exercise, muscle cells become anaerobic and pyruvate is converted to lactate. Gluconeogenesis salvages this pyruvate and lactate and reconverts it to glucose.

The Substrates of Gluconeogenesis

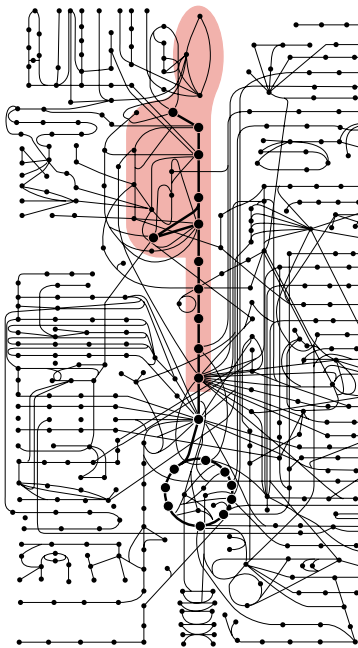
In addition to pyruvate and lactate, other noncarbohydrate precursors can be used as substrates for gluconeogenesis in animals. These include most of the amino acids, as well as glycerol and all the TCA cycle intermediates. On the other hand, fatty acids are not substrates for gluconeogenesis in animals, because most fatty acids yield only acetyl-CoA upon degradation, and animals cannot carry out net synthesis of sugars from acetyl-CoA. Lysine and leucine are the only amino acids that are not substrates for gluconeogenesis. These amino acids produce only acetyl-CoA upon degradation. Note also that acetyl-CoA is a substrate for gluconeogenesis when the glyoxylate cycle is operating (Chapter 20).

Nearly All Gluconeogenesis Occurs in the Liver and Kidneys in Animals

Interestingly, the mammalian organs that consume the most glucose, namely, brain and muscle, carry out very little glucose synthesis. The major sites of gluconeogenesis are the liver and kidneys, which account for about 90% and 10% of the body's gluconeogenic activity, respectively. Glucose produced by gluconeogenesis in the liver and kidney is released into the blood and is subsequently absorbed by brain, heart, muscle, and red blood cells to meet their metabolic needs. In turn, pyruvate and lactate produced in these tissues are returned to the liver and kidney to be used as gluconeogenic substrates.

Gluconeogenesis Is Not Merely the Reverse of Glycolysis

In some ways, gluconeogenesis is the reverse, or antithesis, of glycolysis. Glucose is synthesized, not catabolized; ATP is consumed, not produced; and NADH is oxidized to NAD⁺, rather than the other way around. However, gluconeogenesis cannot be *merely* the reversal of glycolysis, for two reasons. First, glycolysis is exergonic, with a Δ*G*°' of approximately -74 kJ/mol. If gluconeogenesis were merely the reverse, it would be a strongly endergonic process and could not occur spontaneously. Somehow the energetics of the process must be aug-



Gluconeogenesis, Glycogen Metabolism, and the Pentose Phosphate Pathway



mented so that gluconeogenesis can proceed spontaneously. Second, the processes of glycolysis and gluconeogenesis must be regulated in a reciprocal fashion so that when glycolysis is active, gluconeogenesis is inhibited, and when gluconeogenesis is proceeding, glycolysis is turned off. Both of these limitations are overcome by having unique reactions within the routes of glycolysis and gluconeogenesis, rather than a completely shared pathway.

Gluconeogenesis—Something Borrowed, Something New

The complete route of gluconeogenesis is shown in Figure 23.1, side by side with the glycolytic pathway. Gluconeogenesis employs three different reactions, catalyzed by three different enzymes, for the three steps of glycolysis that are

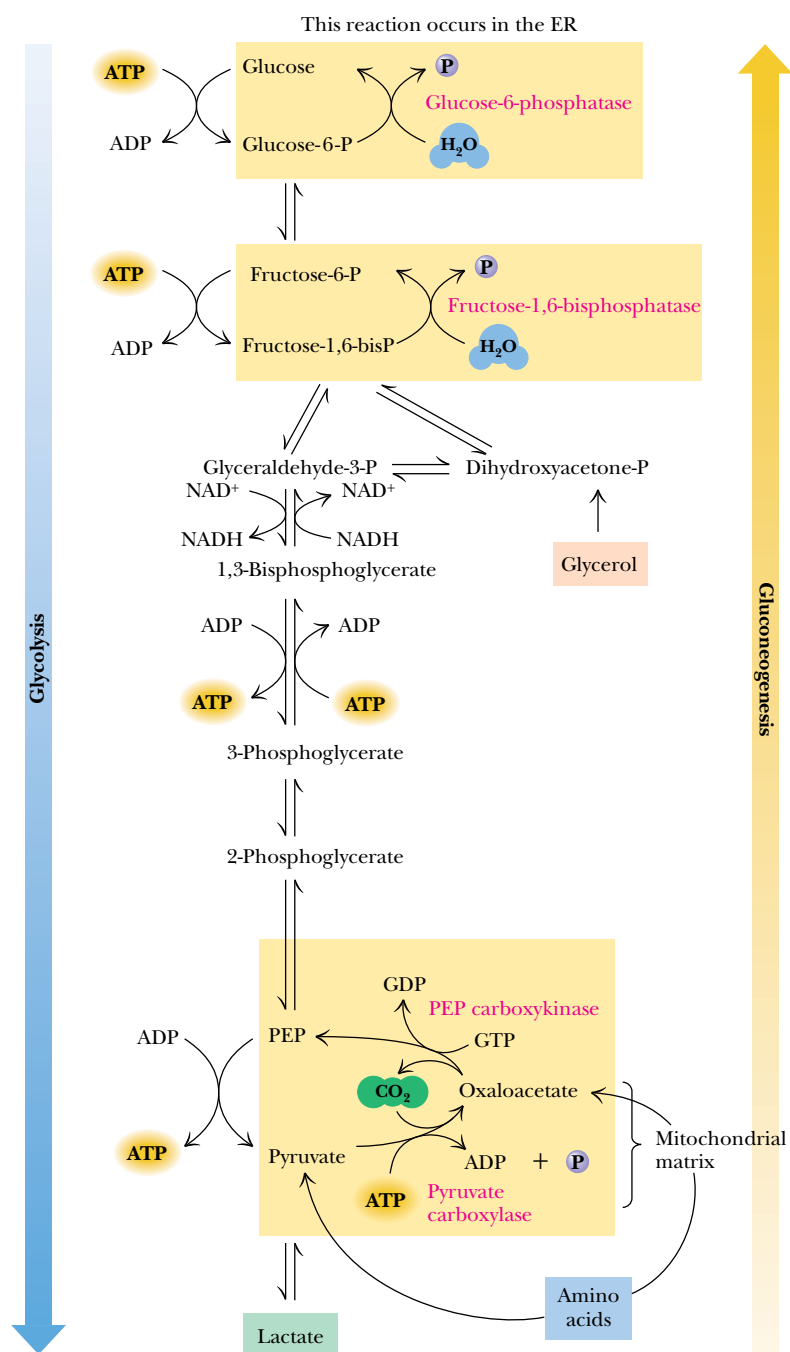


FIGURE 23.1 • The pathways of gluconeogenesis and glycolysis. Species in blue, green, and peach-colored shaded boxes indicate other entry points for gluconeogenesis (in addition to pyruvate).



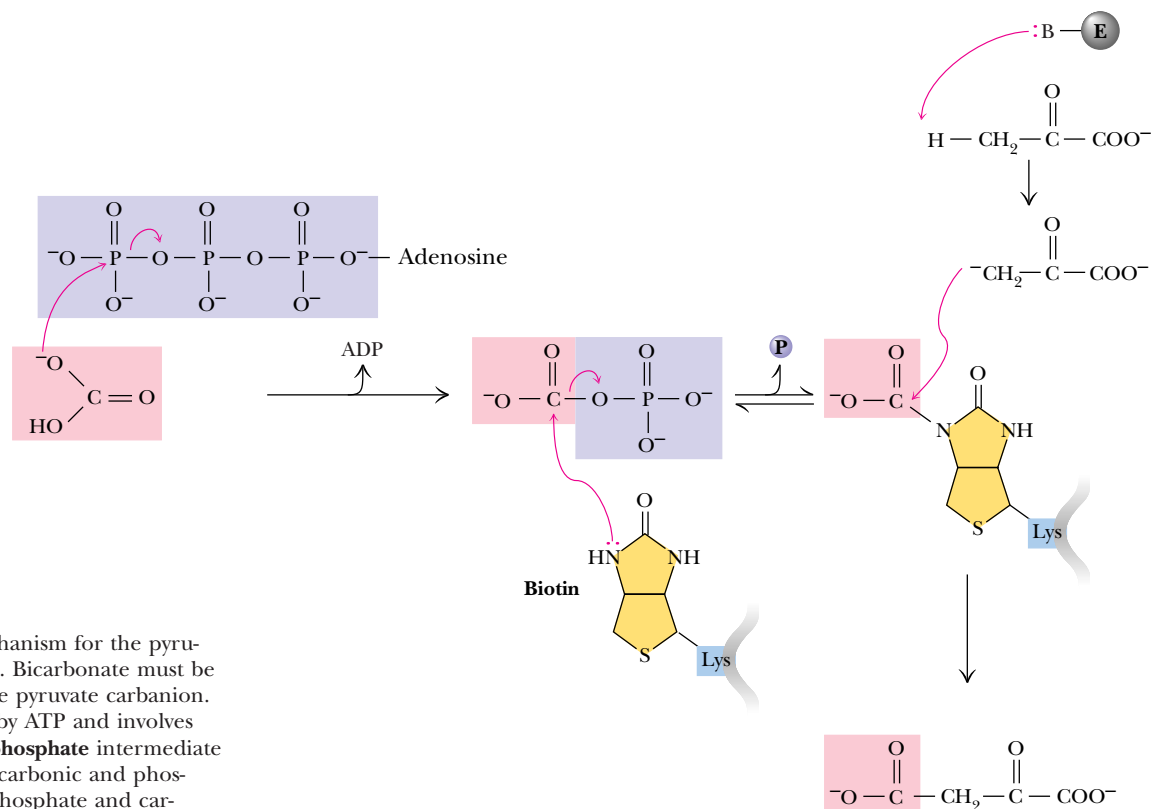


FIGURE 23.4 • A mechanism for the pyruvate carboxylase reaction. Bicarbonate must be activated for attack by the pyruvate carbanion. This activation is driven by ATP and involves formation of a **carbonylphosphate** intermediate—a mixed anhydride of carbonic and phosphoric acids. (Carbonylphosphate and carboxyphosphate are synonyms.)

Activation of pyruvate carboxylase by acetyl-CoA provides an important physiological regulation. Acetyl-CoA is the primary substrate for the TCA cycle, and oxaloacetate (formed by pyruvate carboxylase) is an important intermediate in both the TCA cycle and the gluconeogenesis pathway. If levels of ATP and/or acetyl-CoA (or other acyl-CoAs) are low, pyruvate is directed primarily into the TCA cycle, which eventually promotes the synthesis of ATP. If ATP and acetyl-CoA levels are high, pyruvate is converted to oxaloacetate and consumed in gluconeogenesis. Clearly, high levels of ATP and CoA derivatives are signs that energy is abundant and that metabolites will be converted to glucose (and perhaps even glycogen). If the energy status of the cell is low (in terms of ATP and CoA derivatives), pyruvate is consumed in the TCA cycle. Also, as noted in Chapter 20, pyruvate carboxylase is an important anaplerotic enzyme. Its activation by acetyl-CoA leads to oxaloacetate formation, replenishing the level of TCA cycle intermediates.

COMPARTMENTALIZED PYRUVATE CARBOXYLASE DEPENDS ON METABOLITE CONVERSION AND TRANSPORT The second interesting feature of pyruvate carboxylase is that it is found only in the *matrix* of the mitochondria. By contrast, the next enzyme in the gluconeogenic pathway, PEP carboxykinase, may be localized in the cytosol or in the mitochondria or both. For example, rabbit liver PEP carboxykinase is predominantly mitochondrial, whereas the rat liver enzyme is strictly cytosolic. In human liver, PEP carboxykinase is found both in the cytosol and in the mitochondria. Pyruvate is transported into the mitochondrial matrix, where it can be converted to acetyl-CoA (for use in the TCA cycle) and then to citrate (for fatty acid synthesis; see Figure 25.1). Alternatively, it may be converted directly to OAA by pyruvate carboxylase and used in glu-

coneogenesis. In tissues where PEP carboxykinase is found only in the mitochondria, oxaloacetate is converted to PEP, which is then transported to the cytosol for gluconeogenesis (Figure 23.6). However, in tissues that must convert some oxaloacetate to PEP in the cytosol, a problem arises. Oxaloacetate cannot be transported directly across the mitochondrial membrane. Instead, it must first be transformed into malate or aspartate for transport across the mitochondrial inner membrane (Figure 23.5). Cytosolic malate and aspartate must be reconverted to oxaloacetate before continuing along the gluconeogenic route.

(2) PEP Carboxykinase

The second reaction in the gluconeogenic pyruvate-PEP bypass is the conversion of oxaloacetate to PEP. Production of a high-energy metabolite such as PEP requires energy. The energetic requirements are handled in two ways here. First, the CO₂ added to pyruvate in the pyruvate carboxylase step is removed in the PEP carboxykinase reaction. Decarboxylation is a favorable process and helps to drive the formation of the very high-energy enol phosphate in PEP. This decarboxylation drives a reaction that would otherwise be highly endergonic. Note the inherent metabolic logic in this pair of reactions: pyruvate carboxylase consumed an ATP to drive a carboxylation, so that the PEP carboxykinase could use the decarboxylation to facilitate formation of PEP. Second, as shown in Figure 23.6, another high-energy phosphate is consumed by the carboxykinase. Mammals and several other species use GTP in this reaction, rather than ATP. The use of GTP here is equivalent to the consumption of an ATP, due to the activity of the **nucleoside diphosphate kinase** (see Figure 20.4). The substantial free energy of hydrolysis of GTP is crucial to the synthesis of PEP in this step. The overall Δ*G* for the pyruvate carboxylase and PEP carboxykinase reactions under physiological conditions in the liver is −22.6 kJ/mol. Once PEP is formed in this way, the phosphoglycerate mutase, phosphoglycerate kinase, glyceraldehyde-3-P dehydrogenase, aldolase, and triose phosphate isomerase reactions act to eventually form fructose-1,6-bisphosphate, as in Figure 23.1.

(3) Fructose-1,6-Bisphosphatase

The hydrolysis of fructose-1,6-bisphosphate to fructose-6-phosphate (Figure 23.7), like all phosphate ester hydrolyses, is a thermodynamically favorable (exergonic) reaction under standard-state conditions (Δ*G*^o = −16.7 kJ/mol). Under physiological conditions in the liver, the reaction is also exergonic (Δ*G* = −8.6 kJ/mol). **Fructose-1,6-bisphosphatase** is an allosterically regulated enzyme. Citrate stimulates bisphosphatase activity, but *fructose-2,6-bisphosphate* is a potent allosteric inhibitor. AMP also inhibits the bisphosphatase; the inhibition by AMP is enhanced by fructose-2,6-bisphosphate.

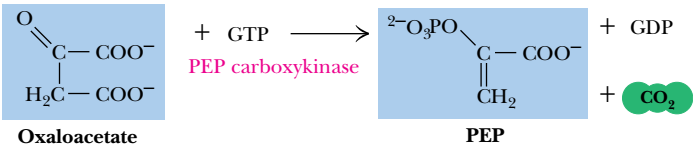


FIGURE 23.6 • The PEP carboxykinase reaction. GTP formed in this reaction can be converted to ATP by nucleoside diphosphate kinase, although liver cells in some species may not contain this enzyme.

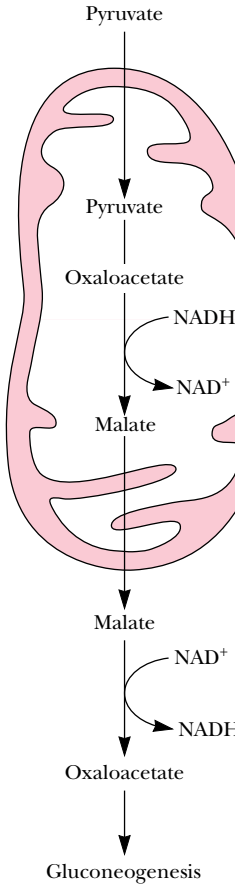


FIGURE 23.5 • Pyruvate carboxylase compartmentalized reaction. Pyruvate is converted to oxaloacetate in the mitochondrion. Because oxaloacetate cannot be transported directly across the mitochondrial membrane, it is reduced to malate, transported to the cytosol, and then oxidized back to oxaloacetate. Gluconeogenesis can continue.

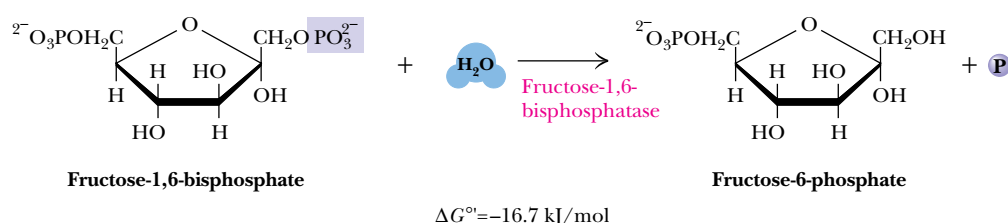
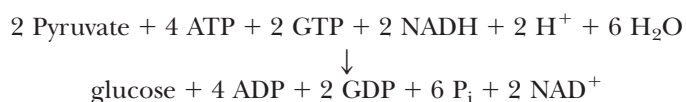


FIGURE 23.7 • The fructose-1,6-bisphosphatase reaction.

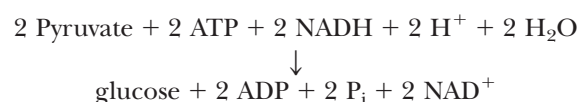
(4) *Glucose-6-Phosphatase*

The final step in the gluconeogenesis pathway is the conversion of glucose-6-phosphate to glucose by the action of **glucose-6-phosphatase**. This enzyme is present in the membranes of the endoplasmic reticulum of liver and kidney cells, but is absent in muscle and brain. For this reason, gluconeogenesis is not carried out in muscle and brain. Its membrane association is important to its function because (Figure 23.8) the substrate is hydrolyzed as it passes into the endoplasmic reticulum itself. Vesicles form from the endoplasmic reticulum membrane and diffuse to the plasma membrane and fuse with it, releasing their glucose contents into the bloodstream. The glucose-6-phosphatase reaction involves a phosphorylated enzyme intermediate, which may be a phosphohistidine (Figure 23.9). The ΔG for the glucose-6-phosphatase reaction in liver is -5.1 kJ/mol.

COUPLING WITH HYDROLYSIS OF ATP AND GTP DRIVES GLUCONEOGENESIS The net reaction for the conversion of pyruvate to glucose in gluconeogenesis is

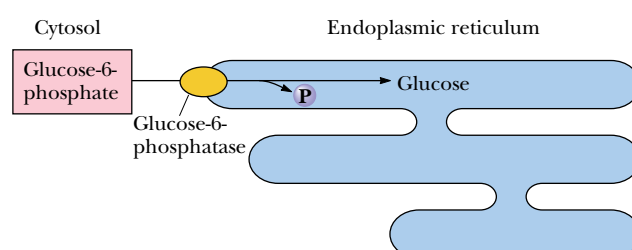


The net free energy change, $\Delta G'$, for this conversion is -37.7 kJ/mol. The consumption of a total of six nucleoside triphosphates drives this process forward. If glycolysis were merely reversed to achieve the net synthesis of glucose from pyruvate, the net reaction would be



and the overall ΔG° would be about +74 kJ/mol. Such a process would be highly endergonic, and therefore thermodynamically unfeasible. Hydrolysis of four additional high-energy phosphate bonds makes gluconeogenesis thermodynamically favorable. Under physiological conditions, however, gluconeogenesis is somewhat less favorable than at standard state, with an overall ΔG of -15.6 kJ/mol for the conversion of pyruvate to glucose.

FIGURE 23.8 • Glucose-6-phosphatase is localized in the endoplasmic reticulum membrane. Conversion of glucose-6-phosphate to glucose occurs during transport into the ER.



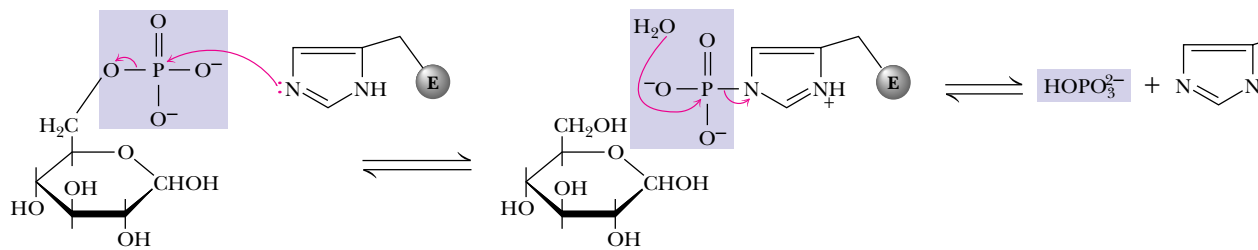


FIGURE 23.9 • The glucose-6-phosphate reaction involves formation of a pyridoxal phosphate intermediate.

LACTATE FORMED IN MUSCLES IS RECYCLED TO GLUCOSE IN THE LIVER A final point on the redistribution of lactate and glucose in the body serves to emphasize the metabolic interactions between organs. Vigorous exercise can lead to oxygen shortage (anaerobic conditions), and energy requirements must be met by increased levels of glycolysis. Under such conditions, glycolysis converts NAD^+ to NADH , yet O_2 is unavailable for regeneration of NAD^+ via cellular respiration. Instead, large amounts of NADH are reoxidized by the reduction of pyruvate to lactate. The lactate thus produced can be transported from muscle to the liver, where it is reoxidized by liver lactate dehydrogenase to yield pyruvate, which is converted eventually to glucose. In this way, the liver shares in the metabolic stress created by vigorous exercise. It exports glucose to muscle, which produces lactate, which can be processed by the liver into new glucose. This is referred to as the **Cori cycle** (Figure 23.10). Liver, with a typically high NAD^+/NADH ratio (about 700), readily produces more glucose than it can use. Muscle that is vigorously exercising will enter anaerobiosis and show a decreasing NAD^+/NADH ratio, which favors reduction of pyruvate to lactate.

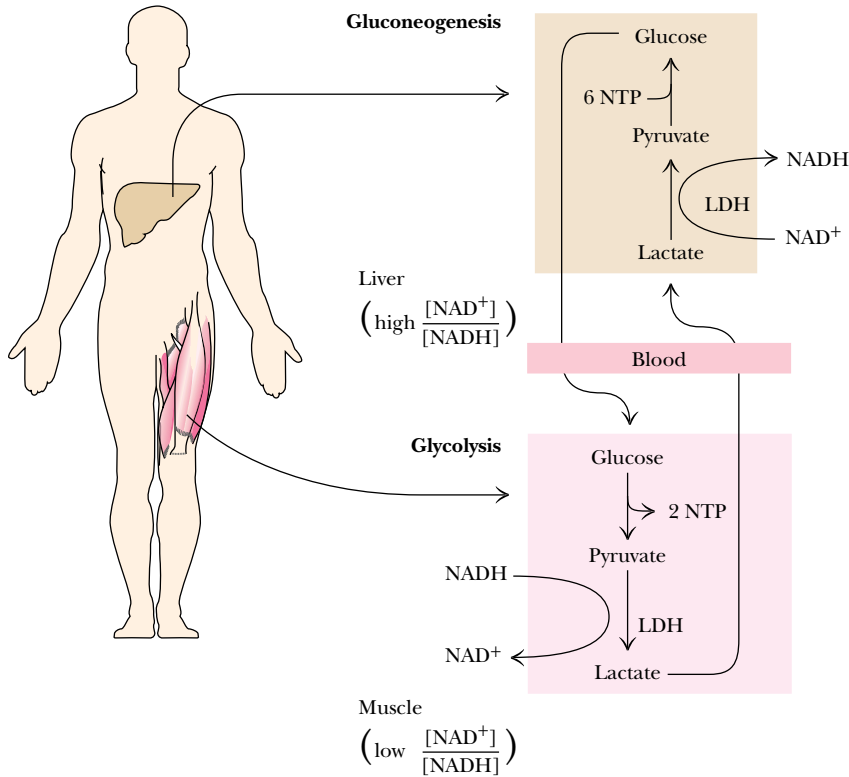
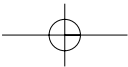


FIGURE 23.10 • The Cori cycle.



CRITICAL DEVELOPMENTS IN BIOCHEMISTRY

The Pioneering Studies of Carl and Gerty Cori

The Cori cycle is named for Carl and Gerty Cori, who received the Nobel Prize in physiology or medicine in 1947 for their studies of glycogen metabolism and blood glucose regulation. Carl Ferdinand Cori and Gerty Theresa Radnitz were both born in Prague (then in Austria). They earned medical degrees from the German University of Prague in 1920 and were married later that year. They joined the faculty of the Washington University School of Medicine in St. Louis in 1931. Their remarkable collaboration resulted in many fundamental advances in carbohydrate and glycogen metabolism. They were credited with the discovery of glucose-1-phosphate, also known at the time as the “Cori ester.”

They also showed that glucose-6-phosphate was produced from glucose-1-P by the action of phosphoglucomutase. They isolated and crystallized glycogen phosphorylase and elucidated the pathway of glycogen breakdown. In 1952, they showed that absence of glucose-6-phosphatase in the liver was the enzymatic defect in *von Gierke’s disease*, an inherited glycogen-storage disease. Six eventual Nobel laureates received training in their laboratory. Gerty Cori was the first American woman to receive a Nobel prize. Carl Cori said of their remarkable collaboration: “Our efforts have been largely complementary and one without the other would not have gone so far. . . .”

23.2 • Regulation of Gluconeogenesis

Nearly all of the reactions of glycolysis and gluconeogenesis take place in the cytosol. If metabolic control were not exerted over these reactions, glycolytic degradation of glucose and gluconeogenic synthesis of glucose could operate simultaneously, with no net benefit to the cell and with considerable consumption of ATP. This is prevented by a sophisticated system of **reciprocal control**, so that glycolysis is inhibited when gluconeogenesis is active, and vice versa. Reciprocal regulation of these two pathways depends largely on the energy status of the cell. When the energy status of the cell is low, glucose is rapidly degraded to produce needed energy. When the energy status is high, pyruvate and other metabolites are utilized for synthesis (and storage) of glucose.

In glycolysis, the three regulated enzymes are those catalyzing the strongly exergonic reactions: hexokinase (glucokinase), phosphofructokinase, and pyruvate kinase. As noted, the gluconeogenic pathway replaces these three reactions with corresponding reactions that are exergonic in the direction of glucose synthesis: glucose-6-phosphatase, fructose-1,6-bisphosphatase, and the pyruvate carboxylase–PEP carboxykinase pair, respectively. These are the three most appropriate sites of regulation in gluconeogenesis.

Gluconeogenesis Is Regulated by Allosteric and Substrate-Level Control Mechanisms

The mechanisms of regulation of gluconeogenesis are shown in Figure 23.11. Control is exerted at all of the predicted sites, but in different ways. Glucose-6-phosphatase is not under allosteric control. However, the K_m for the substrate, glucose-6-phosphate, is considerably higher than the normal range of substrate concentrations. As a result, glucose-6-phosphatase displays a near-linear dependence of activity on substrate concentrations and is thus said to be under **substrate-level control** by glucose-6-phosphate.

Acetyl-CoA is a potent allosteric effector of glycolysis and gluconeogenesis. It allosterically inhibits pyruvate kinase (as noted in Chapter 19) and activates pyruvate carboxylase. Because it also allosterically inhibits pyruvate dehydrogenase (the enzymatic link between glycolysis and the TCA cycle), the cellular fate of pyruvate is strongly dependent on acetyl-CoA levels. A rise in

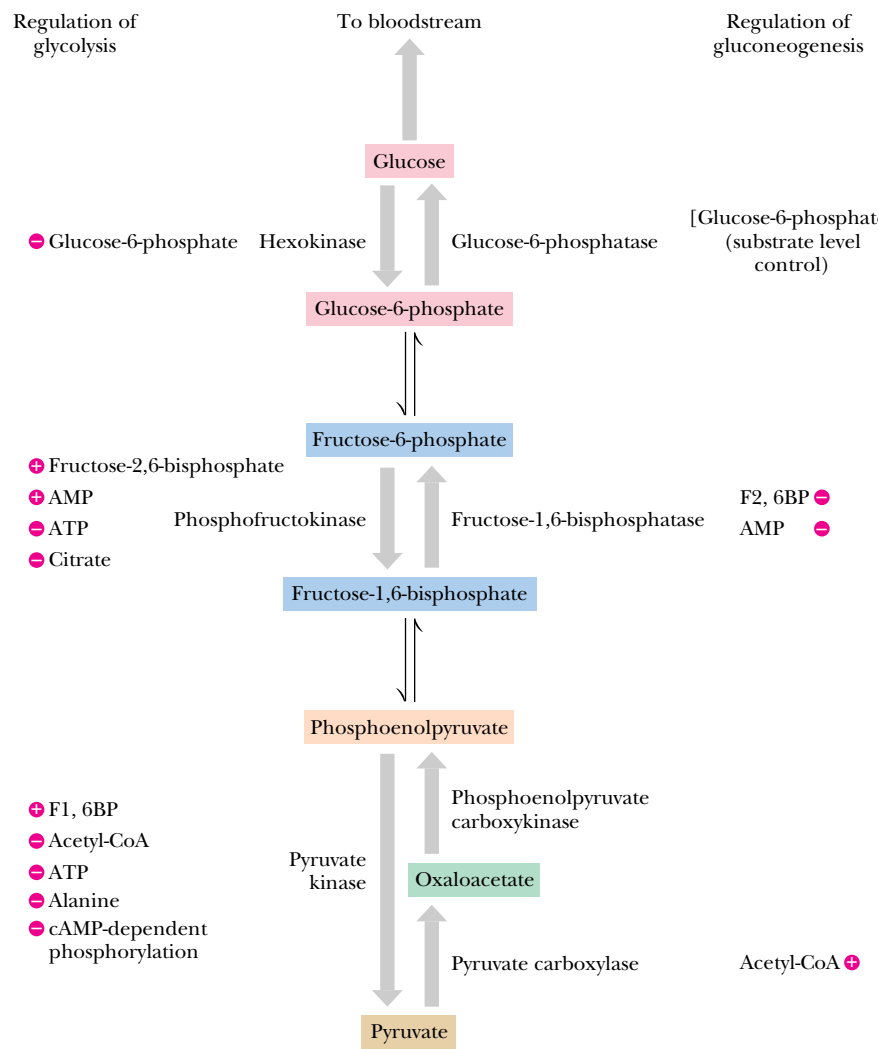


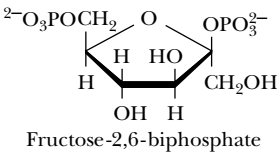
FIGURE 23.11 • The principal regulatory mechanisms in glycolysis and gluconeogenesis. Activators are indicated by plus signs; inhibitors by minus signs.

[acetyl-CoA] indicates that cellular energy levels are high and that carbon metabolites can be directed to glucose synthesis and storage. When acetyl-CoA levels drop, the activities of pyruvate kinase and pyruvate dehydrogenase increase and flux through the TCA cycle increases, providing needed energy for the cell.

Fructose-1,6-bisphosphatase is another important site of gluconeogenic regulation. This enzyme is inhibited by AMP and activated by citrate. These effects by AMP and citrate are the opposites of those exerted on phosphofructokinase in glycolysis, providing another example of reciprocal regulatory effects. When AMP levels increase, gluconeogenic activity is diminished and glycolysis is stimulated. An increase in citrate concentration signals that TCA cycle activity can be curtailed and that pyruvate should be directed to sugar synthesis instead.

Fructose-2,6-Bisphosphate—Allosteric Regulator of Gluconeogenesis

As described in Chapter 19, Emile Van Schaftingen and Henri-Géry Hers demonstrated in 1980 that fructose-2,6-bisphosphate is a potent stimulator of phosphofructokinase. Cognizant of the reciprocal nature of regulation in glycolysis and gluconeogenesis, Van Schaftingen and Hers also considered the

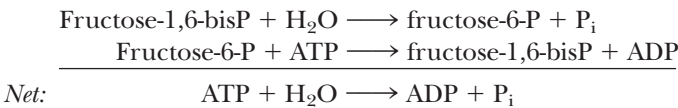


possibility of an opposite effect—inhibition—for fructose-1,6-bisphosphatase. In 1981 they reported that fructose-2,6-bisphosphate was indeed a powerful inhibitor of fructose-1,6-bisphosphatase (Figure 23.12). Inhibition occurs in either the presence or absence of AMP, and the effects of AMP and fructose-2,6-bisphosphate are synergistic.

Cellular levels of fructose-2,6-bisphosphate are controlled by **phosphofructokinase-2 (PFK-2)**, an enzyme distinct from the phosphofructokinase of the glycolytic pathway, and by **fructose-2,6-bisphosphatase (F-2,6-BPase)**. Remarkably, these two enzymatic activities are both found in the same protein molecule, which is an example of a **bifunctional**, or **tandem, enzyme** (Figure 23.13). The opposing activities of this bifunctional enzyme are themselves regulated in two ways. First, fructose-6-phosphate, the substrate of phosphofructokinase and the product of fructose-1,6-bisphosphatase, allosterically activates PFK-2 and inhibits F-2,6-BPase. Second, the phosphorylation by **cAMP-dependent protein kinase** of a single Ser residue on the 49-kD subunit of this dimeric enzyme exerts reciprocal control of the PFK-2 and F-2,6-BPase activities. Phosphorylation then inhibits PFK-2 activity (by increasing the K_m for fructose-6-phosphate) and stimulates F-2,6-BPase activity.

Substrate Cycles Provide Metabolic Control Mechanisms

If fructose-1,6-bisphosphatase and phosphofructokinase acted simultaneously, they would constitute a **substrate cycle** in which fructose-1,6-bisphosphate and fructose-6-phosphate became interconverted with net consumption of ATP:



Because substrate cycles such as this appear to operate with no net benefit to the cell, they were once regarded as metabolic quirks and were referred to as *futile cycles*. More recently, substrate cycles have been recognized as important devices for controlling metabolite concentrations.

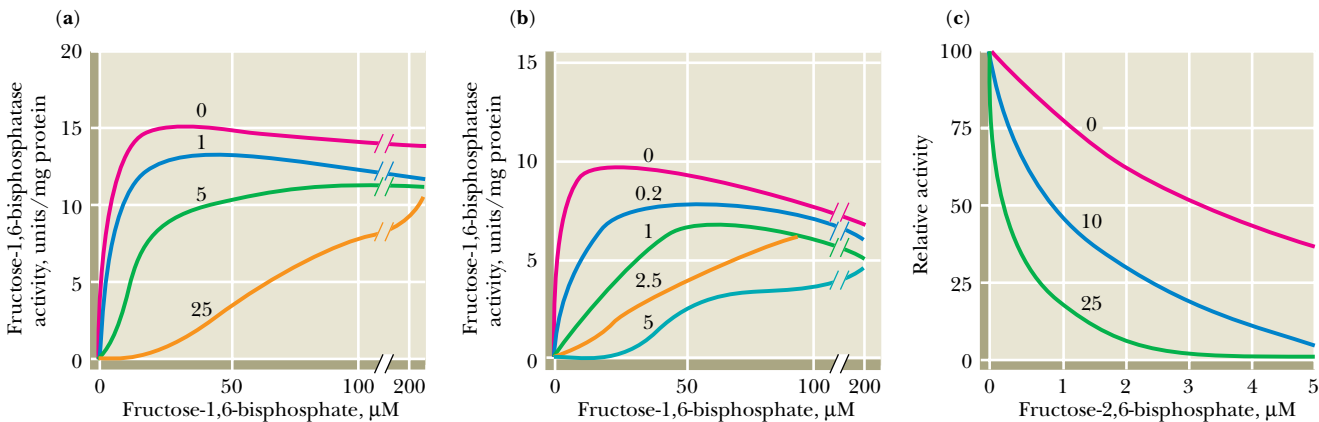


FIGURE 23.12 • Inhibition of fructose-1,6-bisphosphatase by fructose-2,6-bisphosphate in the (a) absence and (b) presence of 25 μM AMP. In (a) and (b), enzyme activity is plotted against substrate (fructose-1,6-bisphosphate) concentration. Concentrations of fructose-2,6-bisphosphate (in μM) are indicated above each curve. (c) The effect of AMP (0, 10, and 25 μM) on the inhibition of fructose-1,6-bisphosphatase by fructose-2,6-bisphosphate. Activity was measured in the presence of 10 μM fructose-1,6-bisphosphate. (Adapted from Van Schaftingen, E., and Hers, H.-G., 1981. Inhibition of fructose-1,6-bisphosphatase by fructose-2,6-bisphosphate. *Proceedings of the National Academy of Science, USA* 78:2861–2863.)

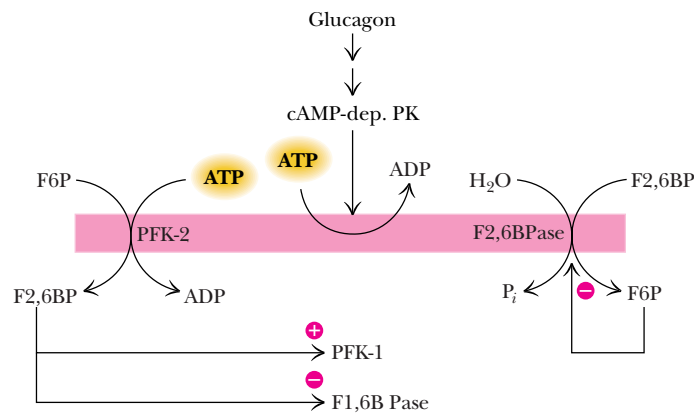


FIGURE 23.13 • Synthesis and degradation of fructose-2,6-bisphosphate are catalyzed by the same bifunctional enzyme.

The three steps in glycolysis and gluconeogenesis that differ constitute three such substrate cycles, each with its own particular metabolic *raison d'être*. Consider, for example, the regulation of the fructose-1,6-bisP–fructose-6-P cycle by fructose-2,6-bisphosphate. As already noted, fructose-1,6-bisphosphatase is subject to allosteric inhibition by fructose-2,6-bisphosphate, whereas phosphofructokinase is allosterically activated by fructose-2,6-bisP. The combination of these effects should permit *either* phosphofructokinase *or* fructose-1,6-bisphosphatase (but not both) to operate at any one time and should thus prevent futile cycling. For instance, in the **fasting state**, when food (i.e., glucose) intake is zero, phosphofructokinase (and therefore glycolysis) is inactive due to the low concentration of fructose-2,6-bisphosphate. In the liver, gluconeogenesis operates to provide glucose for the brain. However, in the fed state, up to 30% of fructose-1,6-bisphosphate formed from phosphofructokinase is recycled back to fructose-6-P (and then to glucose). Because the dependence of fructose-1,6-bisphosphatase activity on fructose-1,6-bisphosphate is sigmoidal in the presence of fructose-2,6-bisphosphate (Figure 23.12), substrate cycling occurs only at relatively high levels of fructose-1,6-bisphosphate. Substrate cycling in this case prevents the accumulation of excessively high levels of fructose-1,6-bisphosphate.

23.3 • Glycogen Catabolism

Dietary Glycogen and Starch Breakdown

As noted earlier, well-fed adult human beings normally metabolize about 160 g of carbohydrates each day. A balanced diet easily provides this amount, mostly in the form of starch, with smaller amounts of glycogen. If too little carbohydrate is supplied by the diet, glycogen reserves in liver and muscle tissue can also be mobilized. The reactions by which ingested starch and glycogen are digested are shown in Figure 23.14. The enzyme known as ***α*-amylase** is an important component of saliva and pancreatic juice. (***β*-Amylase** is found in plants. The *α*- and *β*-designations for these enzymes serve only to distinguish the two, and do not refer to glycosidic linkage nomenclature.) *α*-Amylase is an *endoglycosidase* that hydrolyzes *α*-(1 → 4) linkages of amylopectin and glycogen at random positions, eventually producing a mixture of maltose, maltotriose [with three *α*-(1 → 4)-linked glucose residues], and other small oligosaccharides. *α*-Amylase can cleave on either side of a glycogen or amylopectin branch point, but activity is reduced in highly branched regions of the polysaccharide and stops four residues from any branch point.

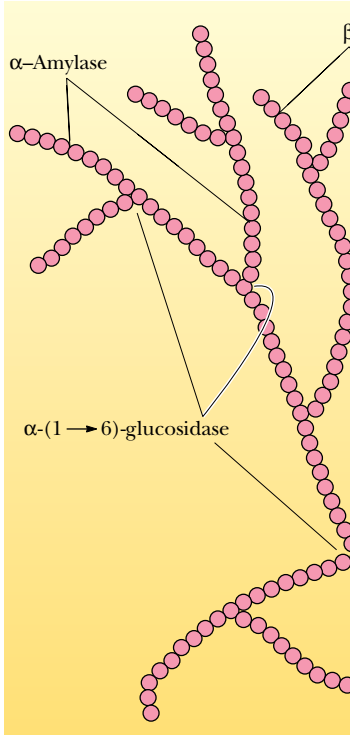


FIGURE 23.14 • Hydrolysis of glycogen and starch by *α*-amylase and *β*-amylase.

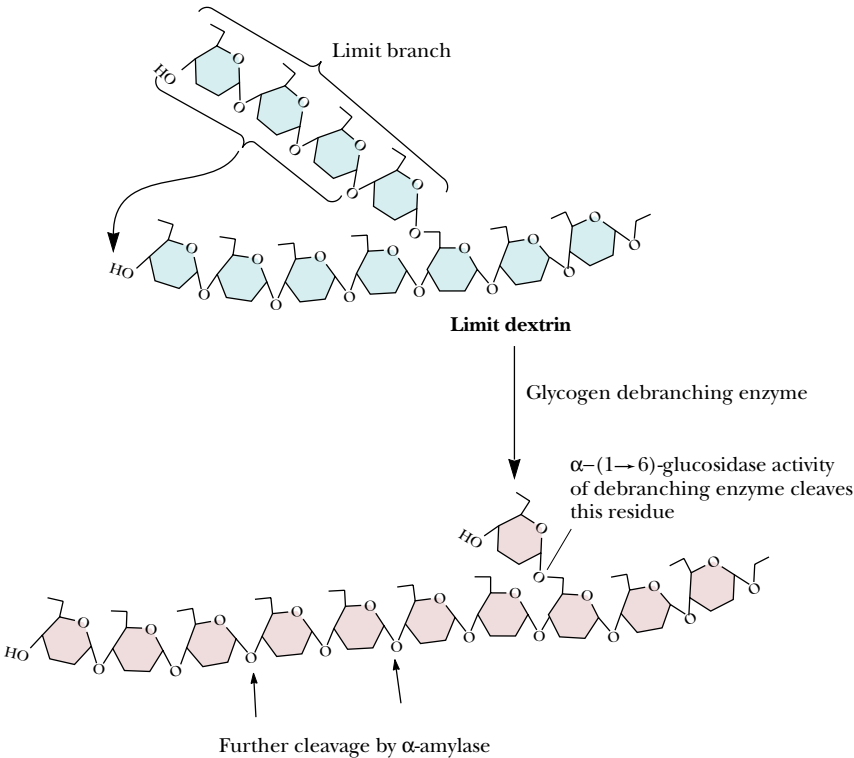
The highly branched polysaccharides that are left after extensive exposure to α -amylase are called **limit dextrins**. These structures can be further degraded by the action of a **debranching enzyme**, which carries out two distinct reactions. The first of these, known as **oligo(α 1,4 $\rightarrow$ α 1,4) glucantransferase** activity, removes a trisaccharide unit and transfers this group to the end of another, nearby branch (Figure 23.15). This leaves a single glucose residue in α -(1 $\rightarrow$ 6) linkage to the main chain. The **α -(1 $\rightarrow$ 6) glucosidase** activity of the debranching enzyme then cleaves this residue from the chain, leaving a polysaccharide chain with one branch fewer. Repetition of this sequence of events leads to complete degradation of the polysaccharide.

β -Amylase is an *exoglycosidase* that cleaves maltose units from the free, nonreducing ends of amylopectin branches, as in Figure 23.14. Like α -amylase, however, β -amylase does not cleave either the α -(1 $\rightarrow$ 6) bonds at the branch points or the α -(1 $\rightarrow$ 4) linkages near the branch points.

Metabolism of Tissue Glycogen

Digestion itself is a highly efficient process in which almost 100% of ingested food is absorbed and metabolized. Digestive breakdown of starch and glycogen is an unregulated process. On the other hand, tissue glycogen represents an important reservoir of potential energy, and it should be no surprise that the reactions involved in its degradation and synthesis are carefully controlled and regulated. Glycogen reserves in liver and muscle tissue are stored in the cytosol as granules exhibiting a molecular weight range from 6×10^6 to 1600×10^6 . These granular aggregates contain the enzymes required to synthesize and catabolize the glycogen, as well as all the enzymes of glycolysis.

FIGURE 23.15 • The reactions of glycogen debranching enzyme. Transfer of a group of three α -(1 $\rightarrow$ 4)-linked glucose residues from a limit branch to another branch is followed by cleavage of the α -(1 $\rightarrow$ 6) bond of the residue that remains at the branch point.



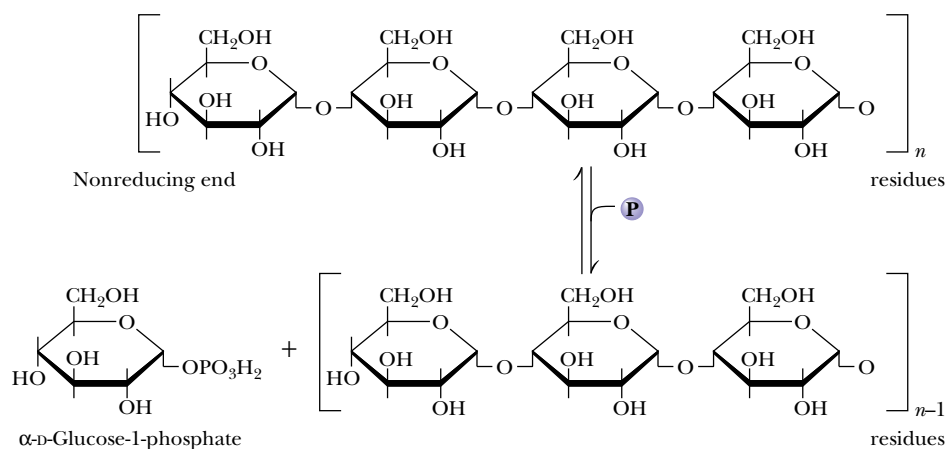


FIGURE 23.16 • The glycogen phosphorylase reaction.

The principal enzyme of glycogen catabolism is **glycogen phosphorylase**, a highly regulated enzyme that was discussed extensively in Chapter 15. The glycogen phosphorylase reaction (Figure 23.16) involves phosphorolysis at a nonreducing end of a glycogen polymer. The standard-state free energy change for this reaction is +3.1 kJ/mol, but the intracellular ratio of [P_i] to [glucose-1-P] approaches 100, and thus the actual ΔG *in vivo* is approximately −6 kJ/mol. There is an energetic advantage to the cell in this phosphorolysis reaction. If glycogen breakdown were hydrolytic and yielded glucose as a product, it would be necessary to phosphorylate the product glucose (with the expenditure of a molecule of ATP) to initiate its glycolytic degradation.

The glycogen phosphorylase reaction degrades glycogen to produce limit dextrins, which are further degraded by debranching enzyme, as already described.

23.4 • Glycogen Synthesis

Animals synthesize and store glycogen when glucose levels are high, but the synthetic pathway is not merely a reversal of the glycogen phosphorylase reaction. High levels of phosphate in the cell favor glycogen breakdown and prevent the phosphorylase reaction from synthesizing glycogen *in vivo*, in spite of the fact that $\Delta G'$ for the phosphorylase reaction actually favors glycogen synthesis. Hence, another reaction pathway must be employed in the cell for the net synthesis of glycogen. In essence, this pathway must activate glucose units for transfer to glycogen chains.

Glucose Units Are Activated for Transfer by Formation of Sugar Nucleotides

We are familiar with several examples of chemical activation as a strategy for group transfer reactions. Acetyl-CoA is an activated form of acetate, biotin and tetrahydrofolate activate one-carbon groups for transfer, and ATP is an activated form of phosphate. Luis Leloir, a biochemist in Argentina, showed in the 1950s that glycogen synthesis depended upon **sugar nucleotides**, which may be

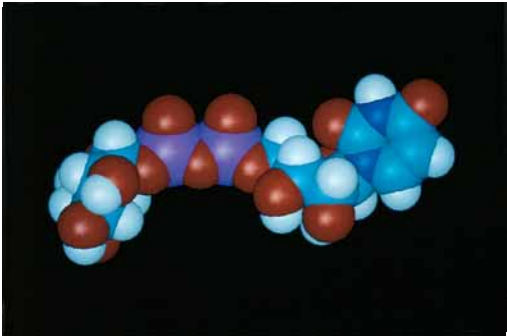
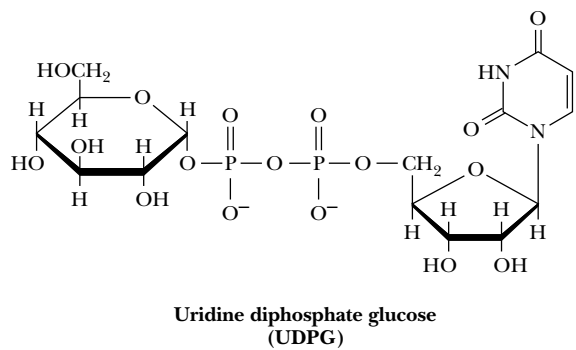


FIGURE 23.17 • The structure of UDP-glucose, a sugar nucleotide.

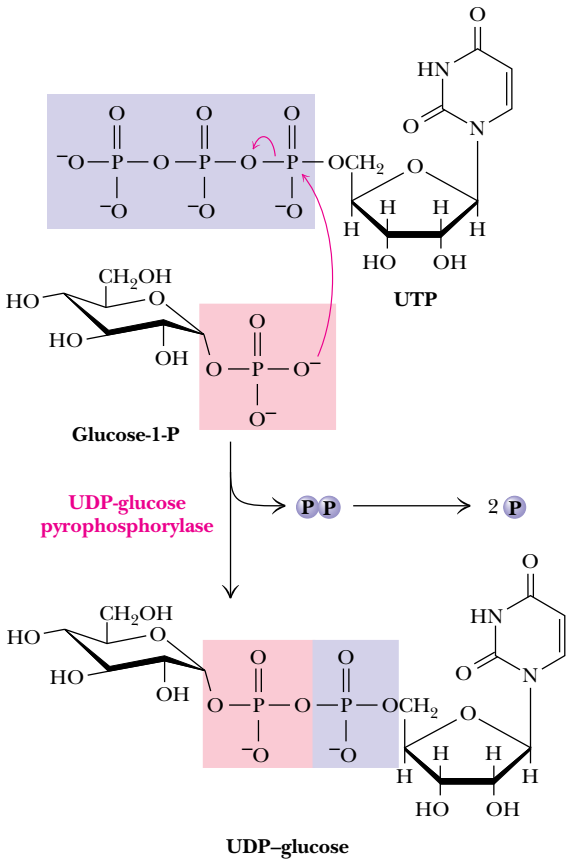
thought of as activated forms of sugar units (Figure 23.17). For example, formation of an ester linkage between the C-1 hydroxyl group and the β -phosphate of UDP activates the glucose moiety of **UDP-glucose**.

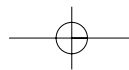
UDP-Glucose Synthesis Is Driven by Pyrophosphate Hydrolysis

Sugar nucleotides are formed from sugar-1-phosphates and nucleoside triphosphates by specific **pyrophosphorylase** enzymes (Figure 23.18). For example, **UDP-glucose pyrophosphorylase** catalyzes the formation of UDP-glucose from glucose-1-phosphate and uridine 5'-triphosphate:

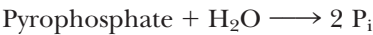


FIGURE 23.18 • The UDP-glucose pyrophosphorylase reaction is a phosphoanhydride exchange, with a phosphoryl oxygen of glucose-1-P attacking the α -phosphorus of UTP to form UDP-glucose and pyrophosphate.

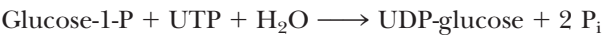




The reaction proceeds via attack by a phosphate oxygen of glucose-1-phosphate on the α -phosphorus of UTP, with departure of the pyrophosphate anion. The reaction is a reversible one, but—as is the case for many biosynthetic reactions—it is driven forward by subsequent hydrolysis of pyrophosphate:



The net reaction for sugar nucleotide formation (combining the preceding two equations) is thus



Sugar nucleotides of this type act as donors of sugar units in the biosynthesis of oligo- and polysaccharides. In animals, UDP-glucose is the donor of glucose units for glycogen synthesis, but ADP-glucose is the glucose source for starch synthesis in plants.

Glycogen Synthase Catalyzes Formation of α -(1 $\rightarrow$ 4) Glycosidic Bonds in Glycogen

The very large glycogen polymer is built around a tiny protein core. The first glucose residue is covalently joined to the protein **glycogenin** via an acetal linkage to a tyrosine–OH group on the protein. Sugar units are added to the glycogen polymer by the action of **glycogen synthase**. The reaction involves transfer of a glucosyl unit from UDP-glucose to the C-4 hydroxyl group at a nonreducing end of a glycogen strand. The mechanism proceeds by cleavage of the C–O bond between the glucose moiety and the β -phosphate of UDP-glucose, leaving an oxonium ion intermediate, which is rapidly attacked by the C-4 hydroxyl oxygen of a terminal glucose unit on glycogen (Figure 23.19). The reaction is exergonic and has a ΔG° of -13.3 kJ/mol .

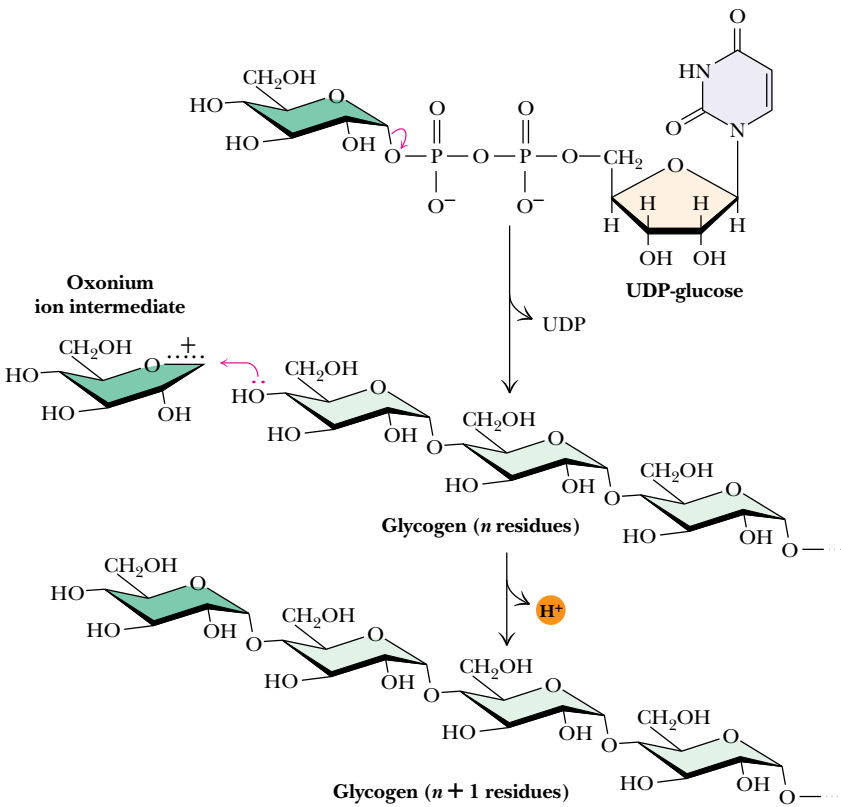
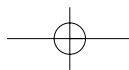


FIGURE 23.19 • The glycogen synthesis. Cleavage of the C–O bond of UDP-glucose yields an oxonium intermediate. The hydroxyl oxygen of the terminal glucose unit of a glycogen molecule completes the



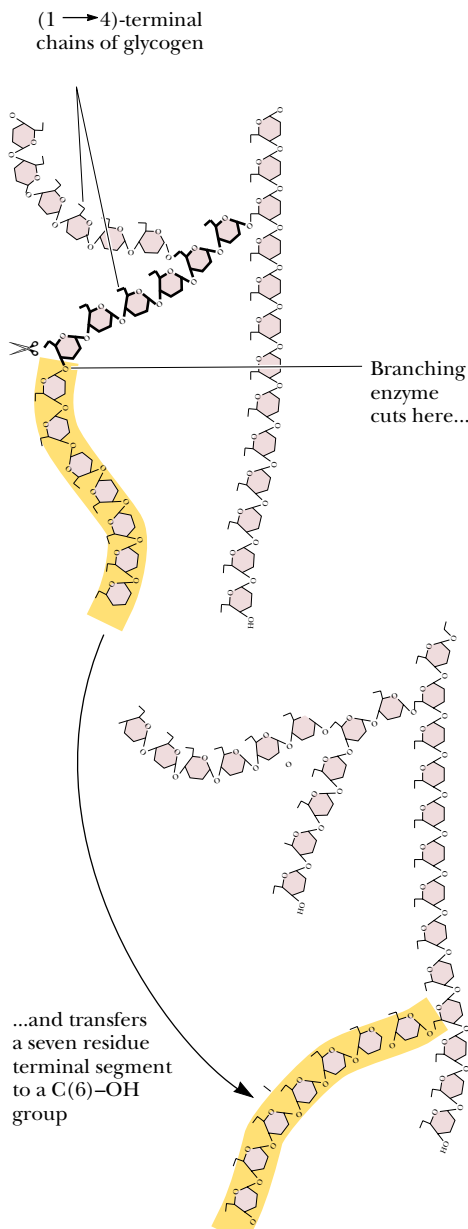


FIGURE 23.20 • Formation of glycogen branches by the branching enzyme. Six- or seven-residue segments of a growing glycogen chain are transferred to the C-6 hydroxyl group of a glucose residue on the same or a nearby chain.

Glycogen Branching Occurs by Transfer of Terminal Chain Segments

Glycogen is a branched polymer of glucose units. The branches arise from α -(1 → 6) linkages which occur every 8 to 12 residues. As noted in Chapter 7, the branches provide multiple sites for rapid degradation or elongation of the polymer and also increase its solubility. Glycogen branches are formed by **amylo-(1,4 → 1,6)-transglycosylase**, also known as *branching enzyme*. The reaction involves the transfer of a six- or seven-residue segment from the nonreducing end of a linear chain at least 11 residues in length to the C-6 hydroxyl of a glucose residue of the same chain or another chain (Figure 23.20). For each branching reaction, the resulting polymer has gained a new terminus at which growth can occur.

23.5 • Control of Glycogen Metabolism

Glycogen Metabolism Is Highly Regulated

Synthesis and degradation of glycogen must be carefully controlled so that this important energy reservoir can properly serve the metabolic needs of the organism. Glucose is the principal metabolic fuel for the brain, and the concentration of glucose in circulating blood must be maintained at about 5 mM for this purpose. Glucose derived from glycogen breakdown is also a primary energy source for muscle contraction. Control of glycogen metabolism is effected via reciprocal regulation of glycogen phosphorylase and glycogen synthase. Thus, activation of glycogen phosphorylase is tightly linked to inhibition of glycogen synthase, and vice versa. Regulation involves both allosteric control and covalent modification, with the latter being under hormonal control. The regulation of glycogen phosphorylase is discussed in detail in Chapter 15.

Regulation of Glycogen Synthase by Covalent Modification

Glycogen synthase also exists in two distinct forms which can be interconverted by the action of specific enzymes: active, dephosphorylated **glycogen synthase I** (glucose-6-P-independent) and less active phosphorylated **glycogen synthase D** (glucose-6-P-dependent). The nature of phosphorylation is more complex with glycogen synthase. As many as nine serine residues on the enzyme appear to be subject to phosphorylation, each site's phosphorylation having some effect on enzyme activity.

Dephosphorylation of both glycogen phosphorylase and glycogen synthase is carried out by **phosphoprotein phosphatase 1**. The action of phosphoprotein phosphatase 1 inactivates glycogen phosphorylase and activates glycogen synthase.

Hormones Regulate Glycogen Synthesis and Degradation

Storage and utilization of tissue glycogen, maintenance of blood glucose concentration, and other aspects of carbohydrate metabolism are meticulously regulated by hormones, including *insulin*, *glucagon*, *epinephrine*, and the *glucocorticoids*.

Insulin Is a Response to Increased Blood Glucose

The primary hormone responsible for conversion of glucose to glycogen is **insulin** (Figure 6.36). Insulin is secreted by special cells in the pancreas called the **islets of Langerhans**. *Secretion of insulin is a response to increased glucose in the*

DEEPER LOOK

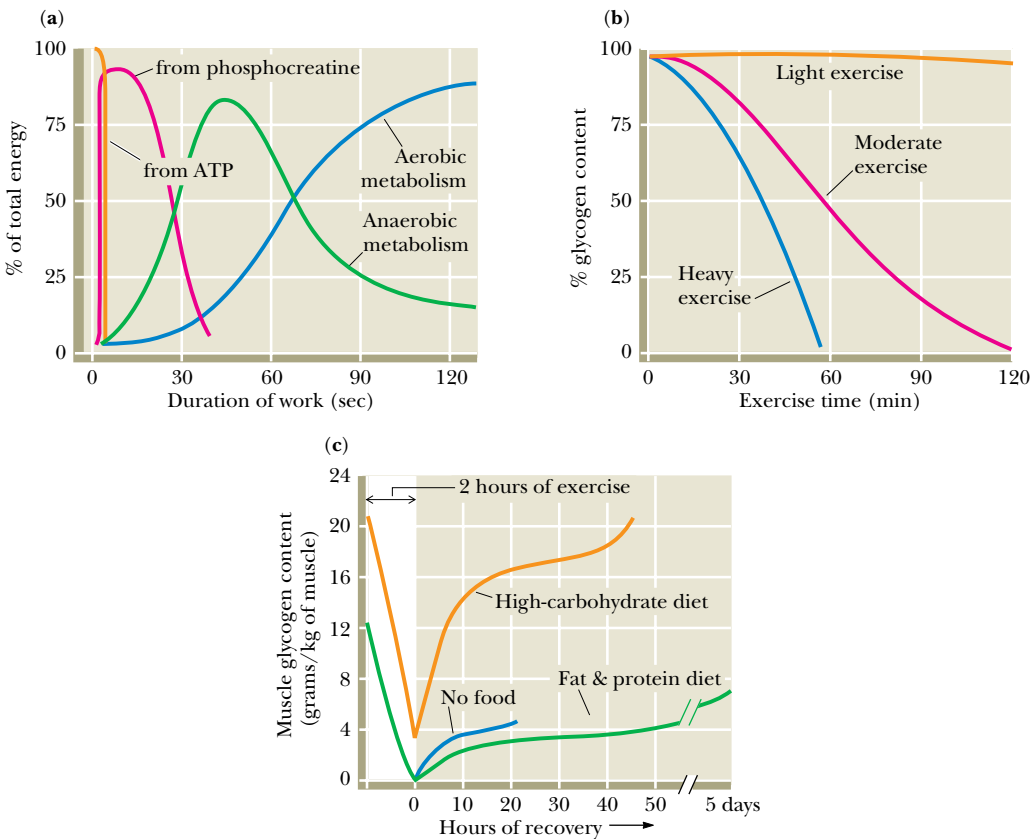
Carbohydrate Utilization in Exercise

Animals have a remarkable ability to “shift gears” metabolically during periods of strenuous exercise or activity. Metabolic adaptations allow the body to draw on different sources of energy (all of which produce ATP) for different types of activity. During periods of short-term, high-intensity exercise (e.g., a 100-m dash), most of the required energy is supplied directly by existing stores of ATP and creatine phosphate (Figure, part a). Long-term, low-intensity exercise (a 10-km run or a 42.2-km marathon) is fueled almost entirely by aerobic metabolism. Between these extremes is a variety of activities (an 800-m run, for example) that rely on anaerobic glycolysis—conversion of glucose to lactate in the muscles and utilization of the Cori cycle.

For all these activities, breakdown of muscle glycogen provides much of the needed glucose. The rate of glycogen con-

sumption depends upon the intensity of the exercise (Figure, part b). By contrast, glucose derived from gluconeogenesis makes only small contributions to total glucose consumed during exercise. During prolonged mild exercise, gluconeogenesis accounts for only about 8% of the total glucose consumed. During more intense exercise, this percentage becomes even lower.

Choice of diet has a dramatic effect on glycogen replenishment following exhaustive exercise. A diet consisting mainly of carbohydrates and fat results in very little recovery of muscle glycogen after 5 days (Figure, part c). On the other hand, a high-carbohydrate diet provides faster restoration of muscle glycogen. In this case, however, complete recovery of glycogen stores takes about 2 days.



(a) Contributions of the various energy sources to muscle activity during mild exercise. (b) Consumption of glycogen stores in fast-twitch muscles during light, moderate, and heavy exercise. (c) Rate of glycogen replenishment following exhaustive exercise. (a and c adapted from Rhodes and Pfланzer, 1992. Human Physiology. Philadelphia: Saunders College Publishing; b adapted from Horton and Terjung, 1988. Exercise, Nutrition and Energy Metabolism. New York: Macmillan.)

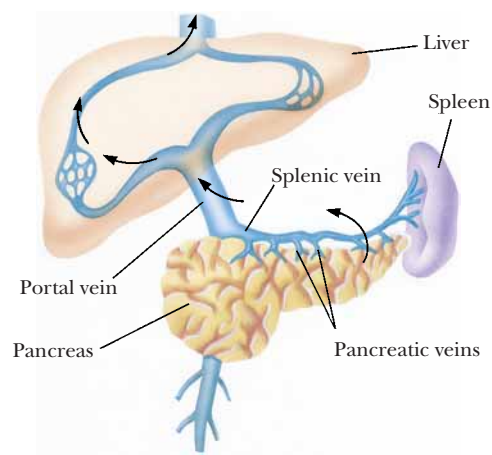


FIGURE 23.21 • The portal vein system carries pancreatic secretions such as insulin and glucagon to the liver and then into the rest of the circulatory system.

blood. When blood glucose levels rise (after a meal, for example), insulin is secreted from the pancreas into the *pancreatic vein*, which empties into the **portal vein system** (Figure 23.21), so that insulin traverses the liver before it enters the systemic blood supply. Insulin acts to rapidly lower blood glucose concentration in several ways. Insulin stimulates glycogen synthesis and inhibits glycogen breakdown in liver and muscle.

Several other physiological effects of insulin also serve to lower blood and tissue glucose levels (Figure 23.22). Insulin stimulates the active transport of glucose (and amino acids) across the plasma membranes of muscle and adipose tissue. Insulin also increases cellular utilization of glucose by inducing the synthesis of several important glycolytic enzymes, namely, glucokinase, phosphofructokinase, and pyruvate kinase. In addition, insulin acts to inhibit several enzymes of gluconeogenesis. These various actions enable the organism to respond quickly to increases in blood glucose levels.

Glucagon and Epinephrine Stimulate Glycogen Breakdown

Catabolism of tissue glycogen is triggered by the actions of the hormones **epinephrine** and **glucagon** (Figure 23.23). *In response to decreased blood glucose*, glucagon is released from the α cells in pancreatic islets of Langerhans. This peptide hormone travels through the blood to specific receptors on liver cell membranes. (Glucagon is active in liver and adipose tissue, but not in other tissues.) Similarly, signals from the central nervous system cause release of *epinephrine* (Figure 23.24)—also known as adrenaline—from the adrenal glands into the bloodstream. Epinephrine acts on liver and muscles. When either hormone binds to its receptor on the outside surface of the cell membrane, a cascade is initiated that activates glycogen phosphorylase and inhibits glycogen synthase. The result of these actions is *tightly coordinated stimulation of glycogen breakdown and inhibition of glycogen synthesis*.

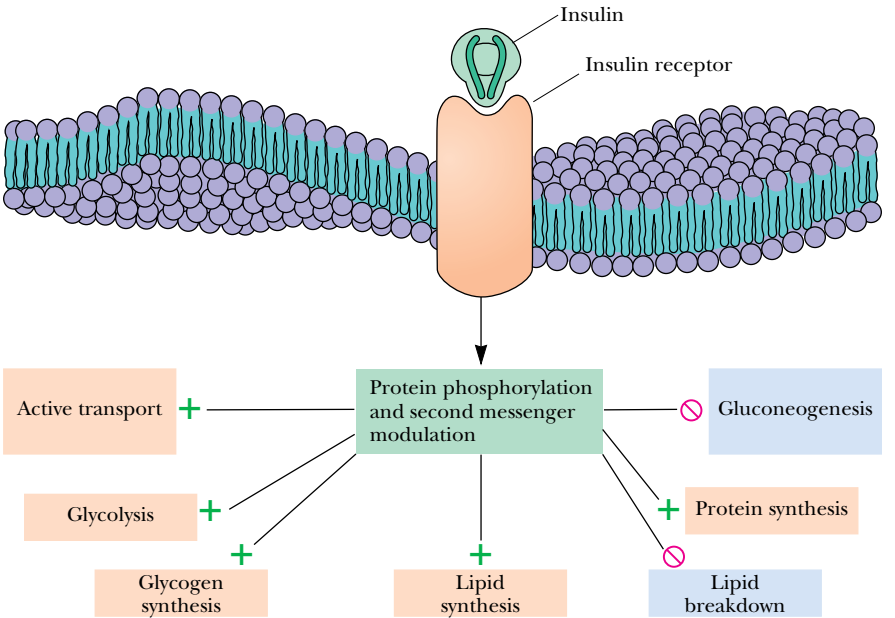


FIGURE 23.22 • The metabolic effects of insulin. As described in Chapter 34, binding of insulin to membrane receptors stimulates the protein kinase activity of the receptor. Subsequent phosphorylation of target proteins modulates the effects indicated.

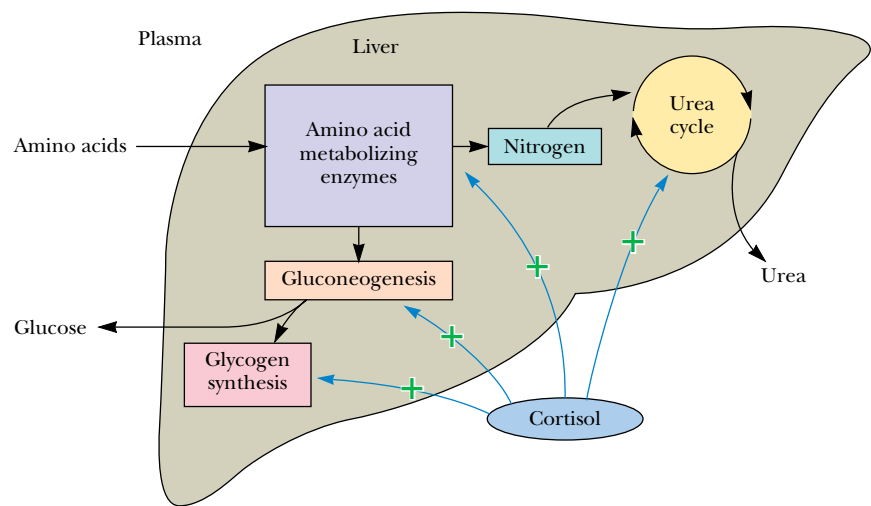


FIGURE 23.25 • The effects of cortisol on carbohydrate and protein metabolism in the liver.

23.6 • The Pentose Phosphate Pathway

Cells require a constant supply of NADPH for reductive reactions vital to biosynthetic purposes. Much of this requirement is met by a glucose-based metabolic sequence variously called the **pentose phosphate pathway**, the **hexose mono-phosphate shunt**, or the **phosphogluconate pathway**. In addition to providing NADPH for biosynthetic processes, this pathway produces *ribose-5-phosphate*, which is essential for nucleic acid synthesis. Several metabolites of the pentose phosphate pathway can also be shuttled into glycolysis.

An Overview of the Pathway

The pentose phosphate pathway begins with glucose-6-phosphate, a six-carbon sugar, and produces three-, four-, five-, six-, and seven-carbon sugars (Figure 23.26). As we will see, two successive oxidations lead to the reduction of NADP^+ to NADPH and the release of CO_2 . Five subsequent nonoxidative steps produce a variety of carbohydrates, some of which may enter the glycolytic pathway. The enzymes of the pentose phosphate pathway are particularly abundant in the cytoplasm of liver and adipose cells. These enzymes are largely absent in muscle, where glucose-6-phosphate is utilized primarily for energy production via glycolysis and the TCA cycle. These pentose phosphate pathway enzymes are located in the cytosol, which is the site of fatty acid synthesis, a pathway heavily dependent on NADPH for reductive reactions.

The Oxidative Steps of the Pentose Phosphate Pathway

(1) Glucose-6-Phosphate Dehydrogenase

The pentose phosphate pathway begins with the oxidation of glucose-6-phosphate. The products of the reaction are a cyclic ester (the lactone of phosphogluconic acid) and NADPH (Figure 23.27). **Glucose-6-phosphate dehydrogenase**, which catalyzes this reaction, is highly specific for NADP^+ . As the first step of a major pathway, the reaction is irreversible and highly regulated. Glucose-6-phosphate dehydrogenase is strongly inhibited by the product coenzyme, NADPH, and also by fatty acid esters of coenzyme A (which are

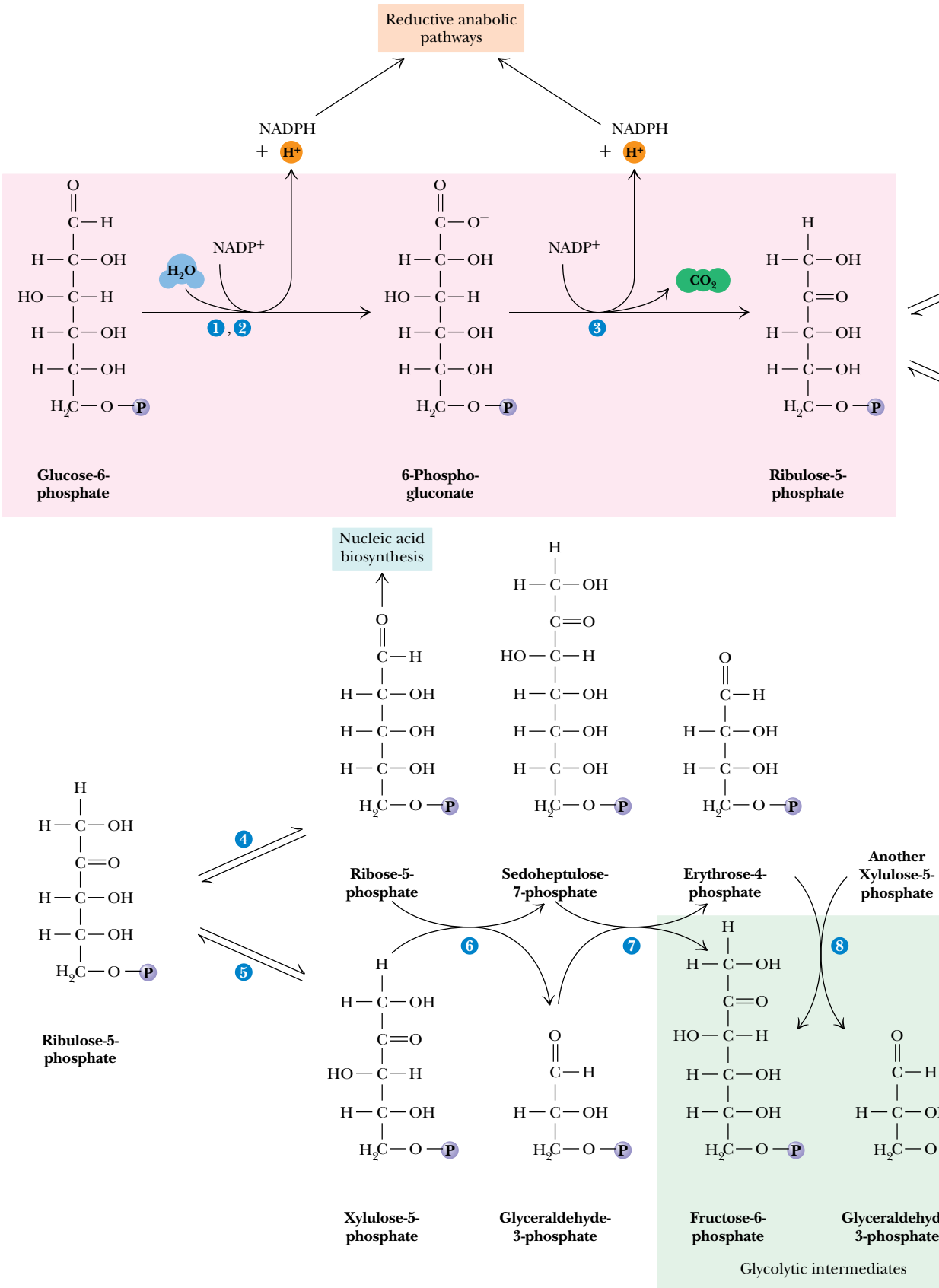


FIGURE 23.26 • The pentose phosphate pathway. The numerals in the blue circles indicate the steps discussed in the text.

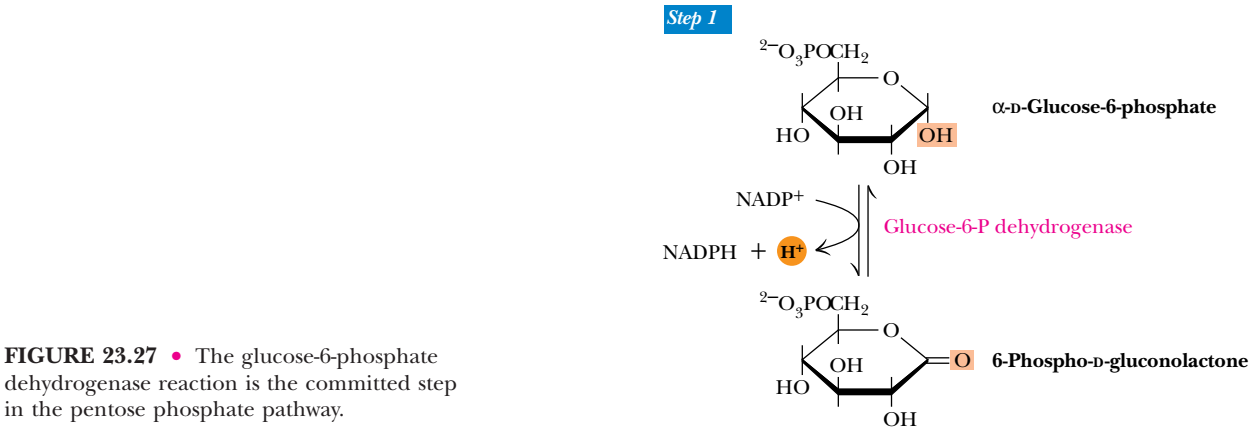


FIGURE 23.27 • The glucose-6-phosphate dehydrogenase reaction is the committed step in the pentose phosphate pathway.



intermediates of fatty acid biosynthesis). Inhibition due to NADPH depends upon the cytosolic NADP⁺/NADPH ratio, which in the liver is about 0.015 (compared to about 725 for the NAD⁺/NADH ratio in the cytosol).

(2) *Gluconolactonase*

The gluconolactone produced in step 1 is hydrolytically unstable and readily undergoes a spontaneous ring-opening hydrolysis, although an enzyme, **gluconolactonase**, accelerates this reaction (Figure 23.28). The linear product, the sugar acid 6-phospho-D-gluconate, is further oxidized in step 3.

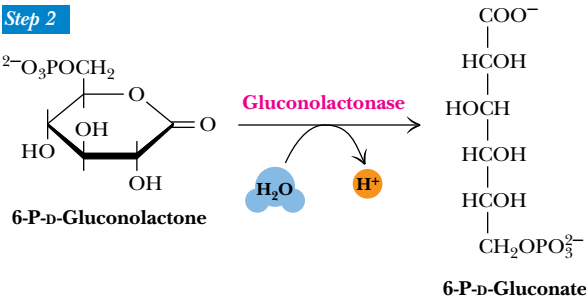
(3) *6-Phosphogluconate Dehydrogenase*

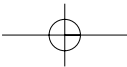
The oxidative decarboxylation of 6-phosphogluconate by **6-phosphogluconate dehydrogenase** yields D-ribulose-5-phosphate and another equivalent of NADPH. There are two distinct steps in this reaction (Figure 23.29): the initial NADP⁺-dependent dehydrogenation yields a β -keto acid, 3-keto-6-phosphogluconate, which is very susceptible to decarboxylation (the second step). The resulting product, D-ribulose-5-P, is the substrate for the nonoxidative reactions composing the rest of this pathway.

The Nonoxidative Steps of the Pentose Phosphate Pathway

This portion of the pathway begins with an isomerization and an epimerization, and it leads to the formation of either D-ribose-5-phosphate or D-xylulose-5-phosphate. These intermediates can then be converted into glycolytic intermediates or directed to biosynthetic processes.

FIGURE 23.28 • The gluconolactonase reaction.





Step 3

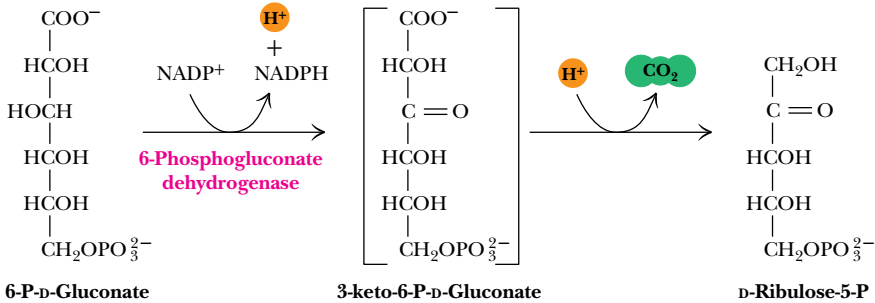


FIGURE 23.29 • The 6-phosphogluconate dehydrogenase reaction.

(4) Phosphopentose Isomerase

This enzyme interconverts ribulose-5-P and ribose-5-P via an enediol intermediate (Figure 23.30). The reaction (and mechanism) is quite similar to the phosphoglucose isomerase reaction of glycolysis, which interconverts glucose-6-P and fructose-6-P. The ribose-5-P produced in this reaction is utilized in the biosynthesis of coenzymes (including NADH, NADPH, FAD, and B₁₂), nucleotides, and nucleic acids (DNA and RNA). The net reaction for the first four steps of the pentose phosphate pathway is



(5) Phosphopentose Epimerase

This reaction converts ribulose-5-P to another ketose, namely, xylulose-5-P. This reaction also proceeds by an enediol intermediate, but involves an inversion at C-3 (Figure 23.31). In the reaction, an acidic proton located α- to a carbonyl carbon is removed to generate the enediolate, but the proton is added back to the same carbon from the opposite side. Note the distinction in nomenclature here. Interchange of groups on a single carbon is an epimerization, and interchange of groups between carbons is referred to as an isomerization.

To this point, the pathway has generated a pool of pentose phosphates. The ΔG' for each of the last two reactions is small, and the three pentose-5-phosphates coexist at equilibrium. The pathway has also produced two molecules of NADPH for each glucose-6-P converted to pentose-5-phosphate. The next three steps rearrange the five-carbon skeletons of the pentoses to produce three-, four-, six-, and seven-carbon units, which can be used for various metabolic purposes. Why should the cell do this? Very often, the cellular need for



Step 4

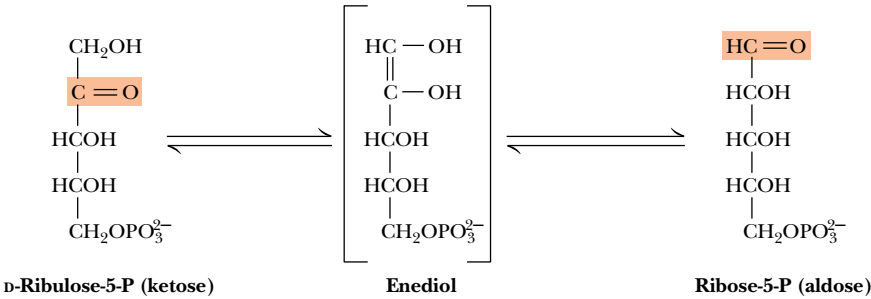
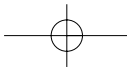


FIGURE 23.30 • The phosphopentose isomerase reaction involves an enediol intermediate.



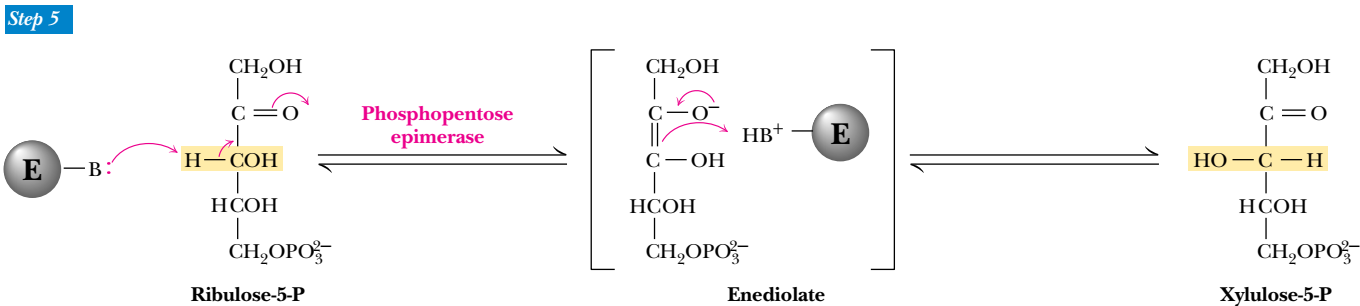


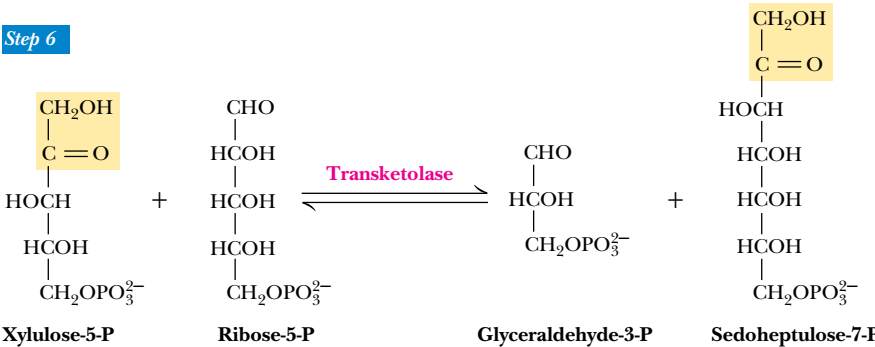
FIGURE 23.31 • The phosphopentose epimerase reaction interconverts ribulose-5-P and xylulose-5-phosphate. The mechanism involves an enediol intermediate and occurs with inversion at C-3.

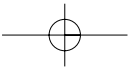
NADPH is considerably greater than the need for ribose-5-phosphate. The next three steps thus return some of the five-carbon units to glyceraldehyde-3-phosphate and fructose-6-phosphate, which can enter the glycolytic pathway. The advantage of this is that the cell has met its needs for NADPH and ribose-5-phosphate in a single pathway, yet at the same time it can return the excess carbon metabolites to glycolysis.

(6) and (8) Transketolase

The transketolase enzyme acts at both steps 6 and 8 of the pentose phosphate pathway. In both cases, the enzyme catalyzes the transfer of two-carbon units. In these reactions (and also in step 7, the transaldolase reaction, which transfers three-carbon units), the donor molecule is a ketose and the recipient is an aldose. In step 6, xylulose-5-phosphate transfers a two-carbon unit to ribose-5-phosphate to form glyceraldehyde-3-phosphate and sedoheptulose-7-phosphate (Figure 23.32). Step 8 involves a two-carbon transfer from xylulose-5-phosphate to erythrose 4-phosphate to produce another glyceraldehyde-3-phosphate and a fructose-6-phosphate (Figure 23.33). Three of these products enter directly into the glycolytic pathway. (The sedoheptulose-7-phosphate is taken care of in step 7, as we shall see.) Transketolase is a thiamine pyrophosphate–dependent enzyme, and the mechanism (Figure 23.34) involves abstraction of the acidic thiazole proton of TPP, attack by the resulting carbanion at the carbonyl carbon of the ketose phosphate substrate, expulsion of the glyceraldehyde-3-phosphate product, and transfer of the two-carbon unit. Transketolase can process a variety of 2-keto sugar phosphates in a similar manner. It is specific for ketose substrates with the configuration shown, but can accept a variety of aldose phosphate substrates.

FIGURE 23.32 • The transketolase reaction of step 6 in the pentose phosphate pathway.





Step 8

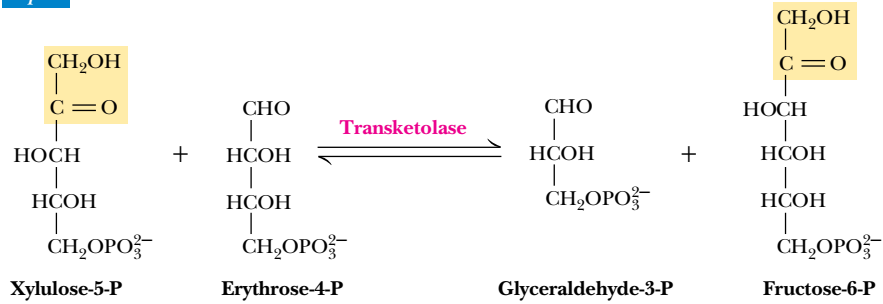


FIGURE 23.33 • The transketolase reaction of step 8 in the pentose phosphate pathway.

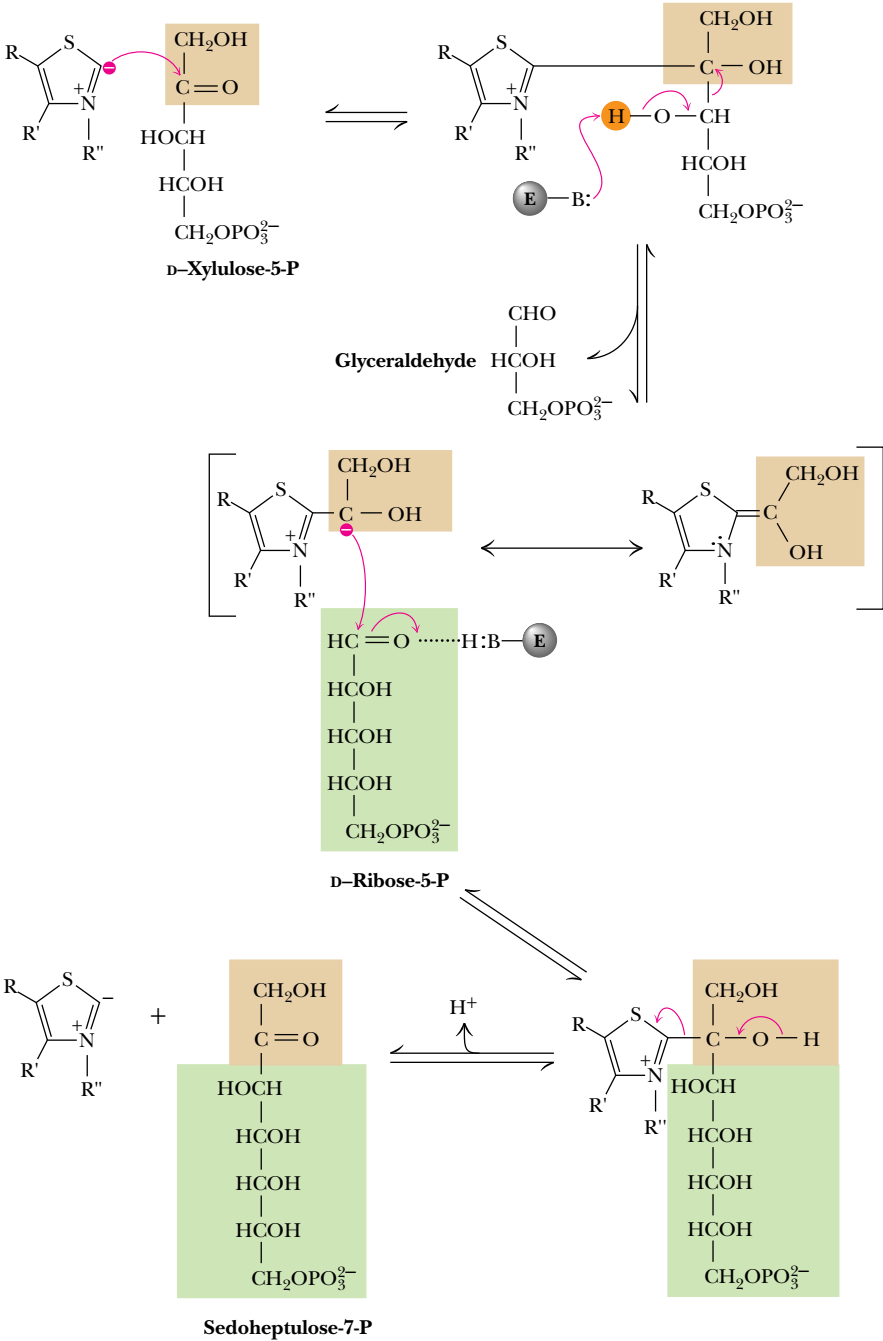


FIGURE 23.34 • The mechanism of the transketolase reaction. In the transketolase reaction, the group transferred in the transketolase reaction might best be described as an aldol, whereas the transferred group in the transaldolase reaction is actually a ketol. Despite this irony, these names persist for historical reasons.

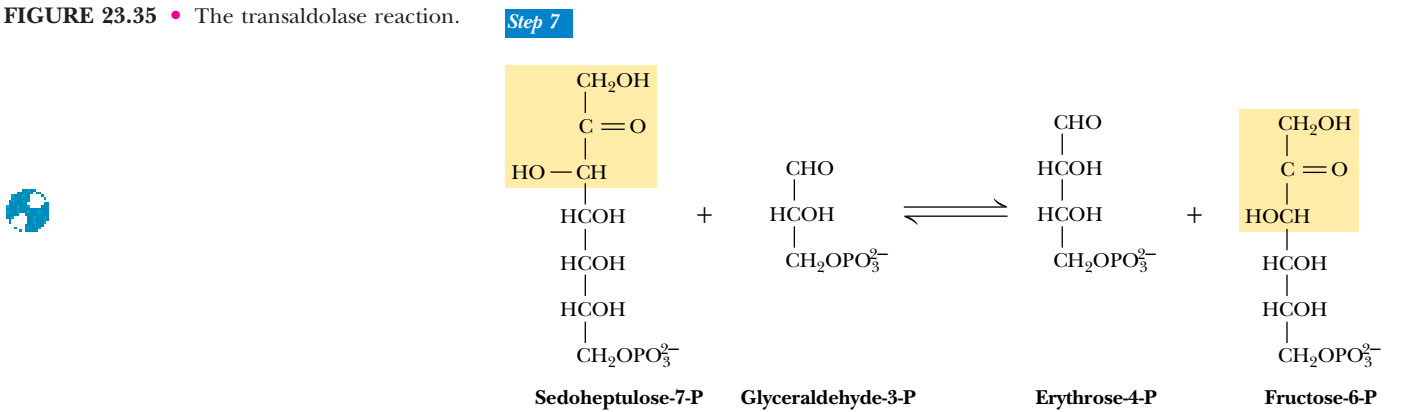
(7) *Transaldolase*

The transaldolase functions primarily to make a useful glycolytic substrate from the sedoheptulose-7-phosphate produced by the first transketolase reaction. This reaction (Figure 23.35) is quite similar to the aldolase reaction of glycolysis, involving formation of a Schiff base intermediate between the sedoheptulose-7-phosphate and an active-site lysine residue (Figure 23.36). Elimination of the erythrose-4-phosphate product leaves an enamine of dihydroxyacetone, which remains stable at the active site (without imine hydrolysis) until the other substrate comes into position. Attack of the enamine carbanion at the carbonyl carbon of glyceraldehyde-3-phosphate is followed by hydrolysis of the Schiff base (imine) to yield the product fructose-6-phosphate.

Utilization of Glucose-6-P Depends on the Cell’s Need for ATP, NADPH, and Ribose-5-P

It is clear that glucose-6-phosphate can be used as a substrate either for glycolysis or for the pentose phosphate pathway. The cell makes this choice on the basis of its relative needs for biosynthesis and for energy from metabolism. ATP can be produced in abundance if glucose-6-phosphate is channeled into glycolysis. On the other hand, if NADPH or ribose-5-phosphate is needed, glucose-6-phosphate can be directed to the pentose phosphate pathway. The molecular basis for this regulatory decision depends on the enzymes that metabolize glucose-6-phosphate in glycolysis and the pentose phosphate pathway. In glycolysis, phosphoglucosomerase converts glucose-6-phosphate to fructose-6-phosphate, which is utilized by phosphofructokinase (a highly regulated enzyme) to produce fructose-1,6-bisphosphate. In the pentose phosphate pathway, glucose-6-phosphate dehydrogenase (also highly regulated) produces gluconolactone from glucose-6-phosphate. Thus, the fate of glucose-6-phosphate is determined to a large extent by the relative activities of phosphofructokinase and glucose-6-P dehydrogenase. Recall (Chapter 19) that PFK is inhibited when the ATP/AMP ratio increases, and that it is inhibited by citrate but activated by fructose-2,6-bisphosphate. Thus, when the energy charge is high, glycolytic flux decreases. Glucose-6-P dehydrogenase, on the other hand, is inhibited by high levels of NADPH and also by the intermediates of fatty acid biosynthesis. Both of these are indicators that biosynthetic demands have been satisfied. If that is the case, glucose-6-phosphate dehydrogenase and the pen-

FIGURE 23.35 • The transaldolase reaction.



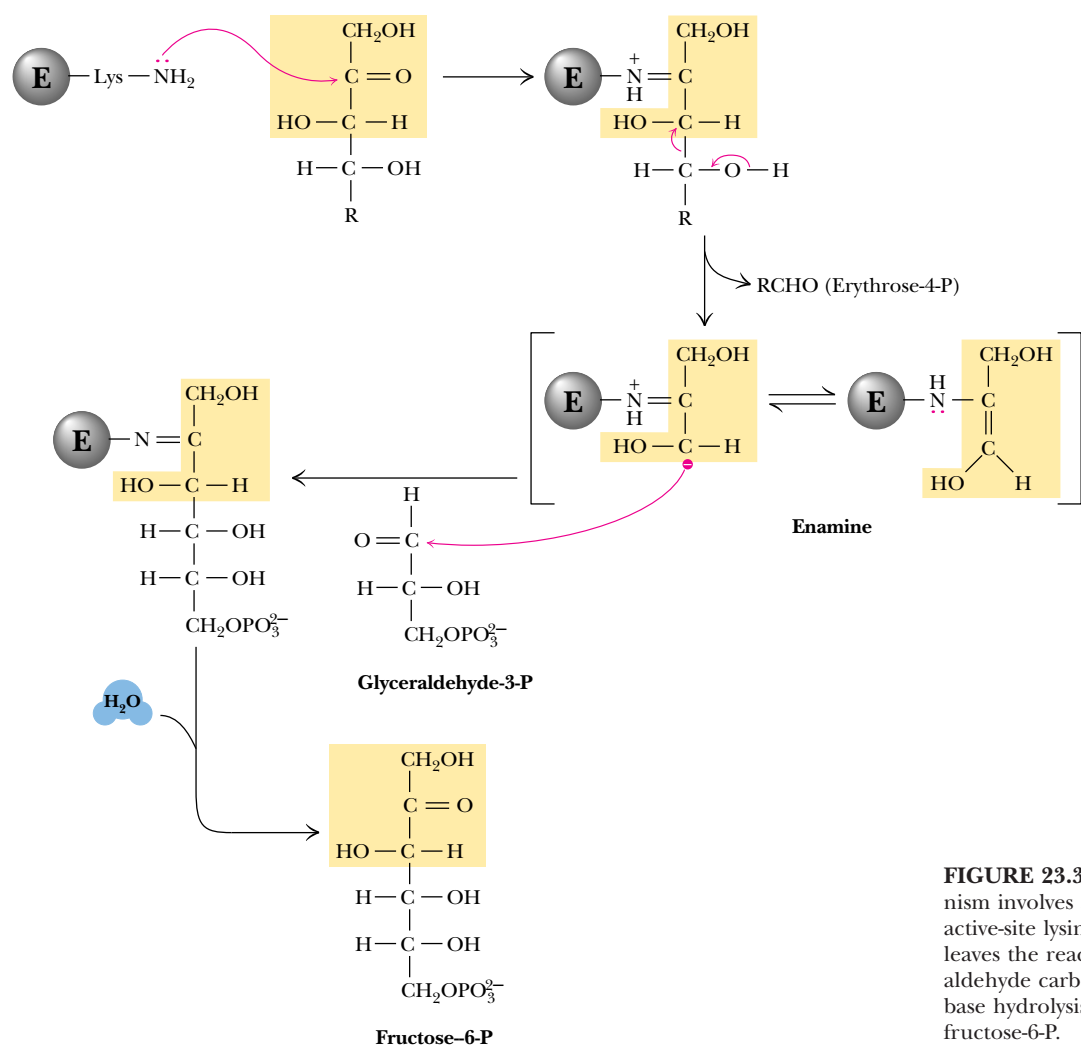
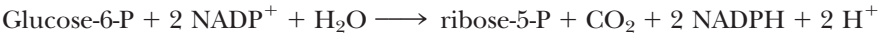


FIGURE 23.36 • The transaldolase reaction involves attack on the substrate by the active-site lysine. Departure of erythrose-4-P leaves the reactive enamine, which then reacts with the aldehyde carbon of glyceraldehyde-3-phosphate. Base hydrolysis yields the second product, fructose-6-P.

tose phosphate pathway are inhibited. If NADPH levels drop, the pentose phosphate pathway turns on, and NADPH and ribose-5-phosphate are made for biosynthetic purposes.

Even when the latter choice has been made, however, the cell must still be “cognizant” of the relative needs for ribose-5-phosphate and NADPH (as well as ATP). Depending on these relative needs, the reactions of glycolysis and the pentose phosphate pathway can be combined in novel ways to emphasize the synthesis of needed metabolites. There are four principal possibilities.

(1) BOTH RIBOSE-5-P AND NADPH ARE NEEDED BY THE CELL In this case, the first four reactions of the pentose phosphate pathway predominate (Figure 23.37). NADPH is produced by the oxidative reactions of the pathway, and ribose-5-P is the principal product of carbon metabolism. As stated earlier, the net reaction for these processes is



(2) MORE RIBOSE-5-P THAN NADPH IS NEEDED BY THE CELL Synthesis of ribose-5-P can be accomplished without production of NADPH if the oxidative steps of the pentose phosphate pathway are bypassed. The key to this route is the extrac-

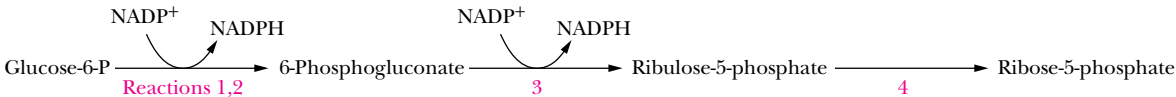
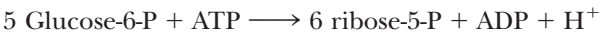


FIGURE 23.37 • When biosynthetic demands dictate, the first four reactions of the pentose phosphate pathway predominate and the principal products are ribose-5-P and NADPH.

tion of fructose-6-P and glyceraldehyde-3-P, but not glucose-6-P, from glycolysis (Figure 23.38). The action of transketolase and transaldolase on fructose-6-P and glyceraldehyde-3-P produces three molecules of ribose-5-P from two molecules of fructose-6-P and one of glyceraldehyde-3-P. In this route, as in case 1, no carbon metabolites are returned to glycolysis. The net reaction for this route is



(3) MORE NADPH THAN RIBOSE-5-P IS NEEDED BY THE CELL Large amounts of NADPH can be supplied for biosynthesis without concomitant production of ribose-5-P, if ribose-5-P produced in the pentose phosphate pathway is recycled to produce glycolytic intermediates. As shown in Figure 23.39, this alternative involves a complex interplay between the transketolase and transaldolase reac-

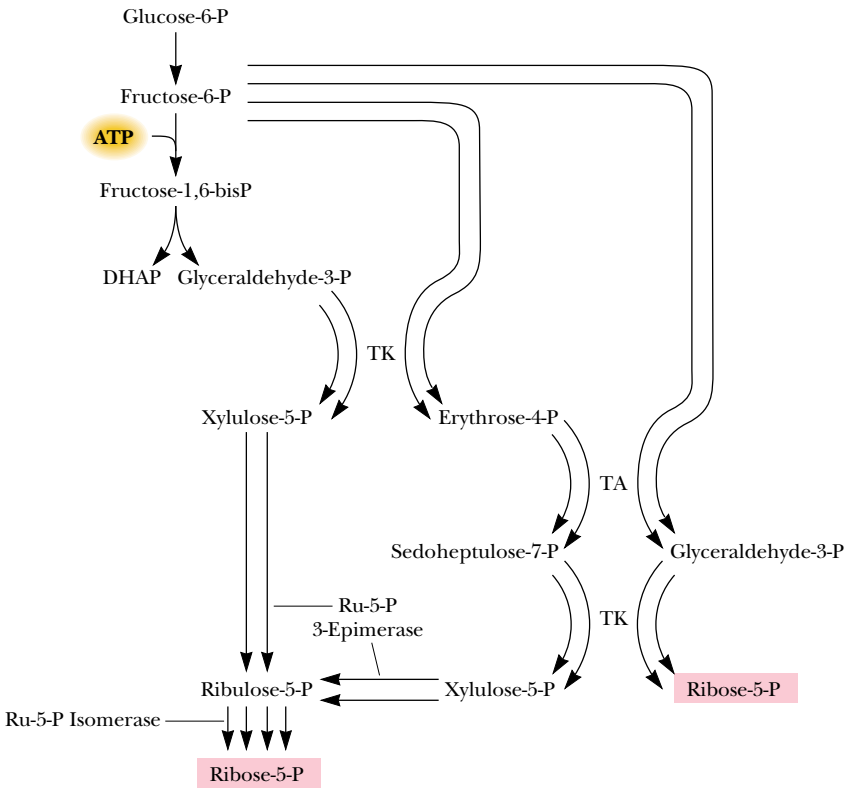


FIGURE 23.38 • The oxidative steps of the pentose phosphate pathway can be bypassed if the primary need is for ribose-5-P.

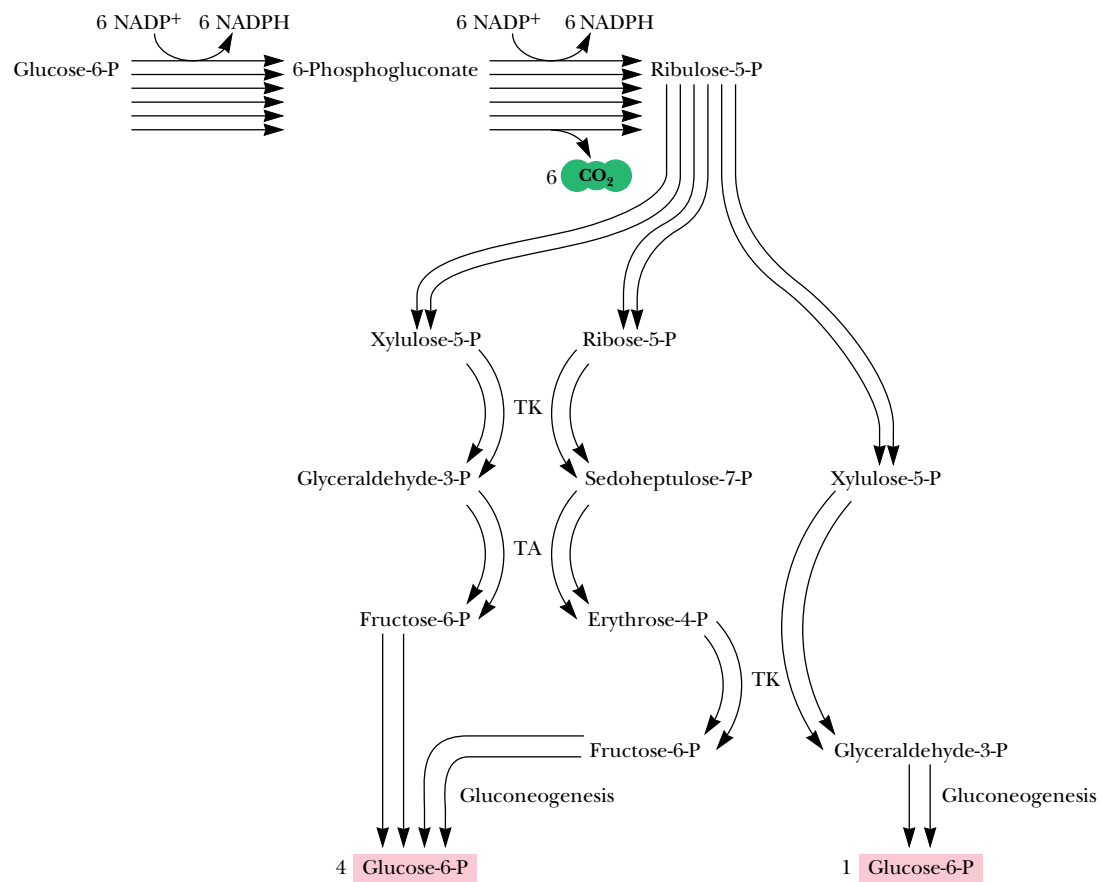
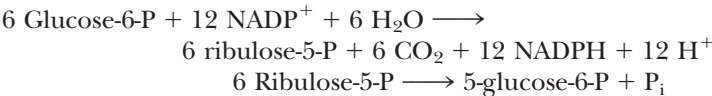


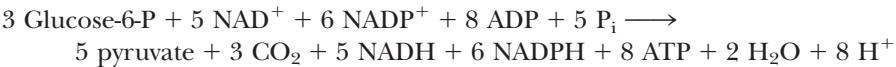
FIGURE 23.39 • Large amounts of NADPH can be produced by the pentose phosphate pathway without significant net production of ribose-5-P. In this version of the pathway, ribose-5-P is recycled to produce glycolytic intermediates.

tions to convert ribulose-5-P to fructose-6-P and glyceraldehyde-3-P, which can be recycled to glucose-6-P via gluconeogenesis. The net reaction for this process is



Note that in this scheme, the six hexose sugars have been converted to six pentose sugars with release of six molecules of CO₂, and the six pentoses are reconverted to five glucose molecules.

(4) BOTH NADPH AND ATP ARE NEEDED BY THE CELL, BUT RIBOSE-5-P IS NOT Under some conditions, both NADPH and ATP must be provided in the cell. This can be accomplished in a series of reactions similar to case 3, if the fructose-6-P and glyceraldehyde-3-P produced in this way proceed through glycolysis to produce ATP and pyruvate, which itself can yield even more ATP by continuing on to the TCA cycle (Figure 23.40). The net reaction for this alternative is



Note that, except for the three molecules of CO₂, all the other carbon from glucose-6-P is recovered in pyruvate.

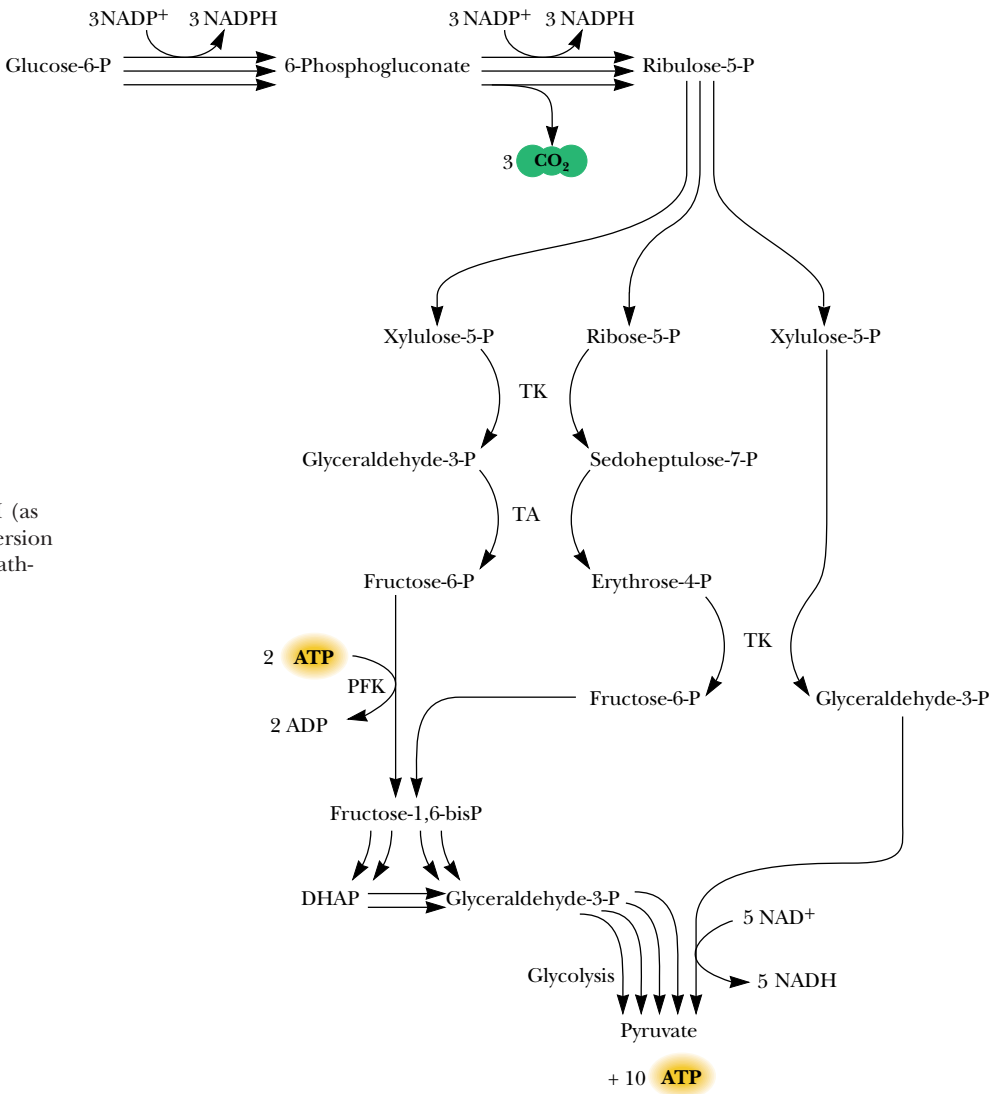
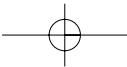
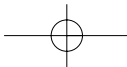


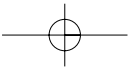
FIGURE 23.40 • Both ATP and NADPH (as well as NADH) can be produced by this version of the pentose phosphate and glycolytic pathways.

PROBLEMS

1. Consider the balanced equation for gluconeogenesis in Section 23.1. Account for each of the components of this equation and the indicated stoichiometry.
2. Calculate $\Delta G^\circ'$ and ΔG for gluconeogenesis in the erythrocyte, using data in Table 19.2 (assume $\text{NAD}^+/\text{NADH} = 20$, $[\text{GTP}] = [\text{ATP}]$, and $[\text{GDP}] = [\text{ADP}]$). See how closely your values match those in Section 23.1.
3. Use the data of Figure 23.12 to calculate the percent inhibition of fructose-1,6-bisphosphatase by $25\ \mu\text{M}$ fructose-2,6-bisphosphate when fructose-1,6-bisphosphate is (a) $25\ \mu\text{M}$ and (b) $100\ \mu\text{M}$.
4. Suggest an explanation for the exergonic nature of the glycogen synthase reaction ($\Delta G^\circ' = -13.3\ \text{kJ/mol}$). Consult Chapter 3 to review the energetics of high-energy phosphate compounds if necessary.
5. Using the values in Table 24.1 for body glycogen content and the data in part b of the illustration for A Deeper Look (page 759), calculate the rate of energy consumption by muscles in heavy exercise (in J/sec). Use the data for fast-twitch muscle.
6. What would be the distribution of carbon from positions 1, 3, and 6 of glucose after one pass through the pentose phosphate

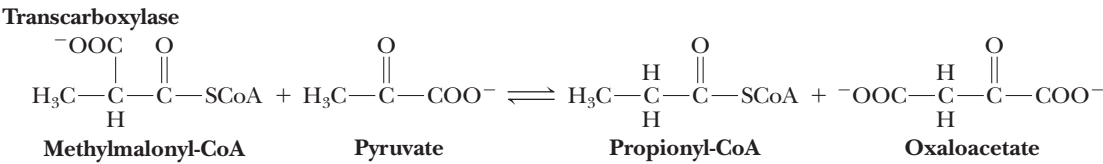
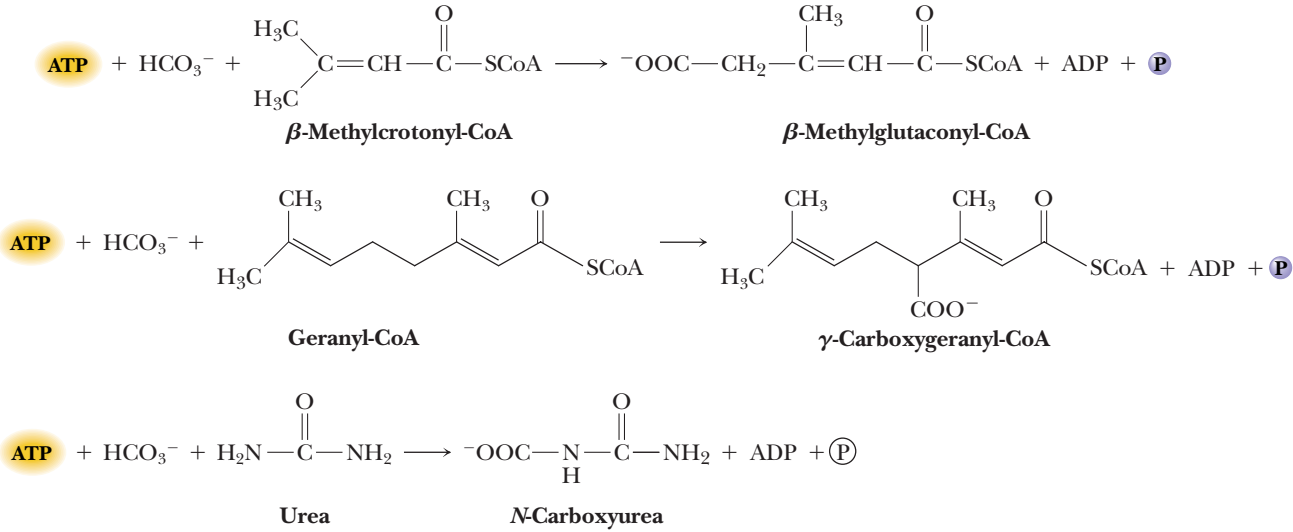
- pathway if the primary need of the organism is for ribose-5-P and the oxidative steps are bypassed (Figure 23.38)?
7. What is the fate of carbon from positions 2 and 4 of glucose-6-P after one pass through the scheme shown in Figure 23.40?
 8. Which reactions of the pentose phosphate pathway would be inhibited by NaBH_4 ? Why?
 9. Imagine a glycogen molecule with 8000 glucose residues. If branches occur every eight residues, how many reducing ends does the molecule have? If branches occur every 12 residues, how many reducing ends does it have? How many nonreducing ends does it have in each of these cases?
 10. Explain the effects of each of the following on the rates of gluconeogenesis and glycogen metabolism:
 - a. Increasing the concentration of tissue fructose-1,6-bisphosphate
 - b. Increasing the concentration of blood glucose
 - c. Increasing the concentration of blood insulin
 - d. Increasing the amount of blood glucagon
 - e. Decreasing levels of tissue ATP
 - f. Increasing the concentration of tissue AMP
 - g. Decreasing the concentration of fructose-6-phosphate



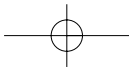
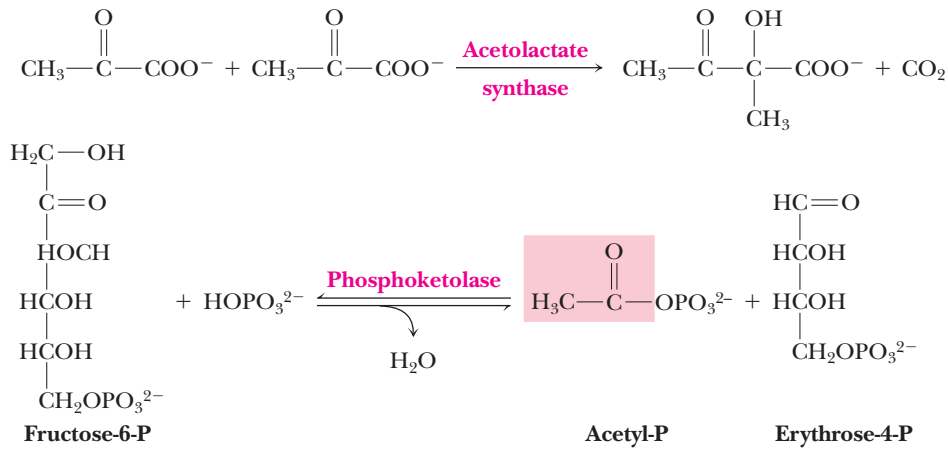


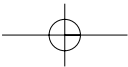
11. The free-energy change of the glycogen phosphorylase reaction is $\Delta G' = +3.1$ kJ/mol. If $[P_i] = 1$ mM, what is the concentration of glucose-1-P when this reaction is at equilibrium?

12. Based on the mechanism for pyruvate carboxylase (Figure 23.4), write reasonable mechanisms for the reactions shown below:



13. The mechanistic chemistry of the acetolactate synthase and phosphoketolase reactions (shown below) is similar to that of the transketolase reaction (Figure 23.34). Write suitable mechanisms for these reactions.





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Chapter 24

Fatty Acid Catabolism



The hummingbird's tremendous capacity to store and use fatty acids enables it to make migratory journeys of remarkable distances. (Two Hummingbirds Lithograph; The Academy of Natural Sciences of Philadelphia/Corbis Images)

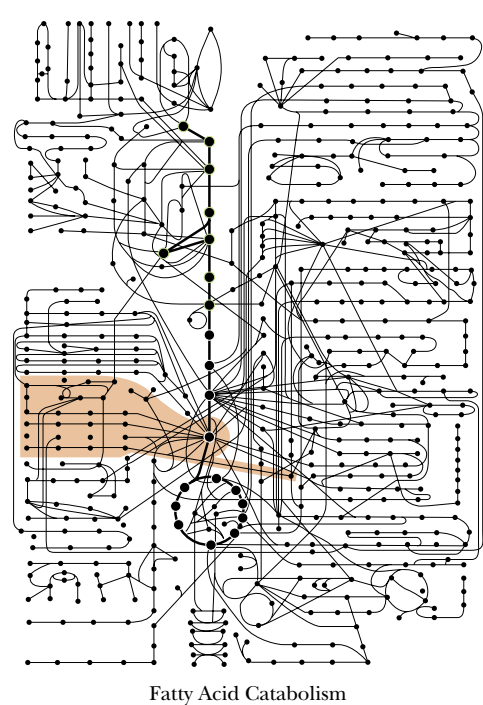
Fatty acids represent the principal form of stored energy for many organisms. There are two important advantages to storing energy in the form of fatty acids. (1) The carbon in fatty acids (mostly $\text{—CH}_2\text{—}$ groups) is almost completely reduced compared to the carbon in other simple biomolecules (sugars, amino acids). Therefore, oxidation of fatty acids will yield more energy (in the form of ATP) than any other form of carbon. (2) Fatty acids are not generally hydrated as mono- and polysaccharides are, and thus can pack more closely in storage tissues. This chapter will be devoted to several important aspects of fatty acid catabolism. Lipid biosynthetic processes will be considered in Chapter 25.

The fat is in the fire.

Proverbs, JOHN HEYWOOD (1497–1580)

OUTLINE

- 24.1 • Mobilization of Fats from Dietary Intake and Adipose Tissue
- 24.2 • β -Oxidation of Fatty Acids
- 24.3 • β -Oxidation of Odd-Carbon Fatty Acids
- 24.4 • β -Oxidation of Unsaturated Fatty Acids
- 24.5 • Other Aspects of Fatty Acid Oxidation
- 24.6 • Ketone Bodies



24.1 • Mobilization of Fats from Dietary Intake and Adipose Tissue

Modern Diets Are Often High in Fat

Fatty acids are acquired readily in the diet and can also be made from carbohydrates and the carbon skeletons of amino acids. Fatty acids provide 30% to 60% of the calories in the diets of most Americans. For our caveman and cave-woman ancestors, the figure was probably closer to 20%. Dairy products were apparently not part of their diet, and the meat they consumed (from fast-moving animals) was low in fat. In contrast, modern domesticated cows and pigs are actually bred for high fat content (and better taste). However, woolly mammoth burgers and saber-toothed tiger steaks are hard to find these days—even in the gourmet sections of grocery stores—and so, by default, we consume (and metabolize) large quantities of fatty acids.

Triacylglycerols Are a Major Form of Stored Energy in Animals

Although some of the fat in our diets is in the form of phospholipids, **triacylglycerols** are a major source of fatty acids. Triacylglycerols are also our principal stored energy reserve. As shown in Table 24.1, the energy available in stores of fat in the average person far exceeds the energy available from protein, glycogen, and glucose. Overall, fat accounts for approximately 83% of available energy, partly because more fat is stored than protein and carbohydrate, and partly because of the substantially higher energy yield per gram for fat compared with protein and carbohydrate. Complete combustion of fat yields about 37 kJ/g, compared with about 16 to 17 kJ/g for sugars, glycogen, and amino acids. In animals, fat is stored mainly as triacylglycerols in specialized cells called **adipocytes** or **adipose cells**. As shown in Figure 24.1, triacylglycerols, aggregated to form large globules, occupy most of the volume of adipose cells. Much smaller amounts of triacylglycerols are stored as small, aggregated globules in muscle tissue.

Hormones Signal the Release of Fatty Acids from Adipose Tissue

The pathways for liberation of fatty acids from triacylglycerols, either from adipose cells or from the diet, are shown in Figures 24.2 and 24.3. Fatty acids are mobilized from adipocytes in response to hormone messengers such as adren-



FIGURE 24.1 • Scanning electron micrograph of an adipose cell (fat cell). Globules of triacylglycerols occupy most of the volume of such cells. (Prof. P. Motta, Dept. of Anatomy, University “La Sapienza,” Rome/Science Photo Library/Photo Researchers, Inc.)

Table 24.1
Stored Metabolic Fuel in a 70-kg Person

Constituent	Energy (kJ/g dry weight)	Dry Weight (g)	Available Energy (kJ)
Fat (adipose tissue)	37	15,000	555,000
Protein (muscle)	17	6,000	102,000
Glycogen (muscle)	16	120	1,920
Glycogen (liver)	16	70	1,120
Glucose (extracellular fluid)	16	20	320
Total			660,360

Sources: Owen, O. E., and Reichard, G. A., Jr., 1971. *Progress in Biochemistry and Pharmacology* 6:177; Newsholme, E. A., and Leech, A. R., 1983. *Biochemistry for the Medical Sciences*. New York: Wiley.

aline, glucagon, and ACTH (adrenocorticotrophic hormone). These signal molecules bind to receptors on the plasma membrane of adipose cells and lead to the activation of adenylyl cyclase, which forms cyclic AMP from ATP. (Second messengers and hormonal signaling are discussed in Chapter 34.) In adipose cells, cAMP activates protein kinase A, which phosphorylates and activates a **triacylglycerol lipase** (also termed **hormone-sensitive lipase**) that hydrolyzes a fatty acid from C-1 or C-3 of triacylglycerols. Subsequent actions of **diacylglycerol lipase** and **monoacylglycerol lipase** yield fatty acids and glycerol. The cell then releases the fatty acids into the blood, where they are carried (in complexes with *serum albumin*) to sites of utilization.

Degradation of Dietary Fatty Acids Occurs Primarily in the Duodenum

Dietary triacylglycerols are degraded to a small extent (via fatty acid release) by lipases in the low-pH environment of the stomach, but mostly pass untouched into the duodenum. Alkaline pancreatic juice secreted into the

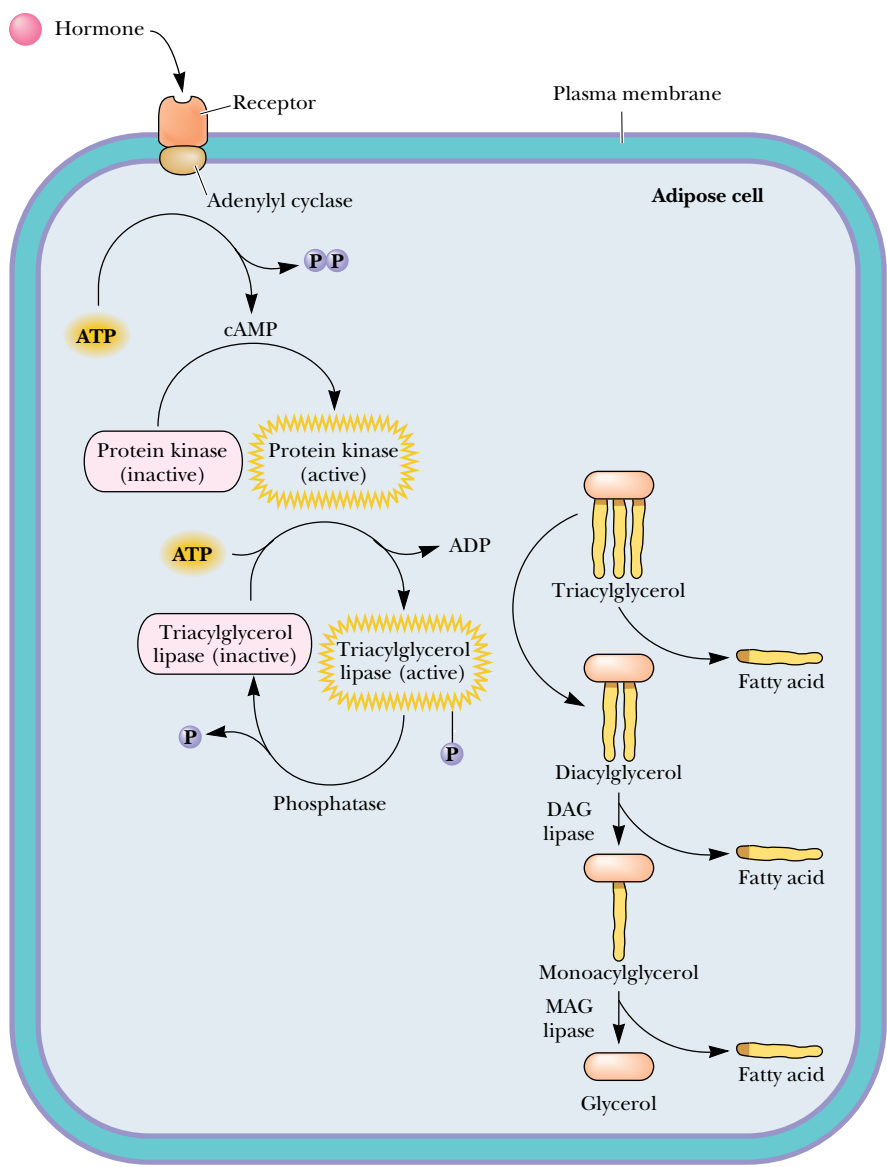


FIGURE 24.2 • Liberation of fatty acids from triacylglycerols in adipose tissue is hormone-dependent.

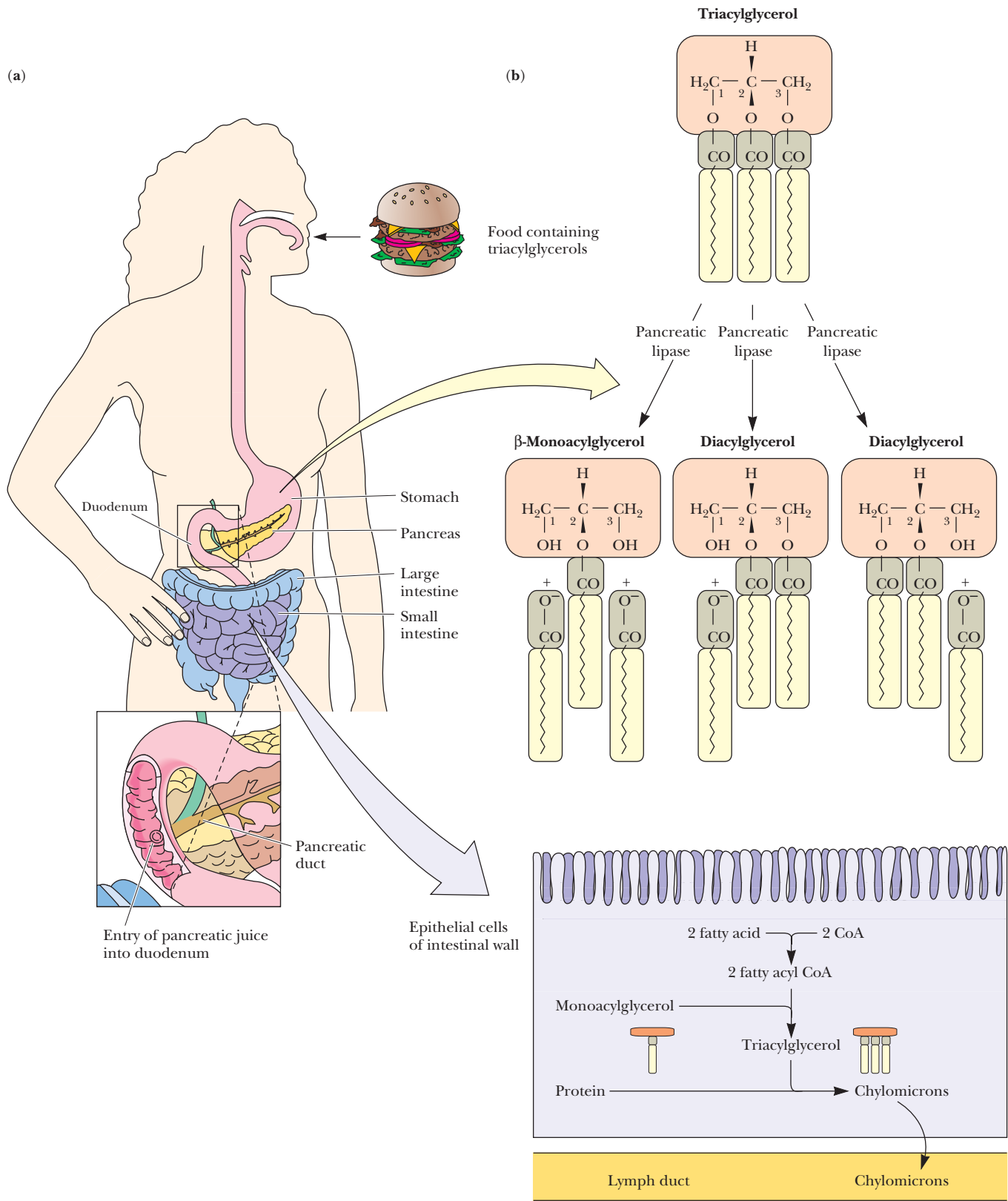
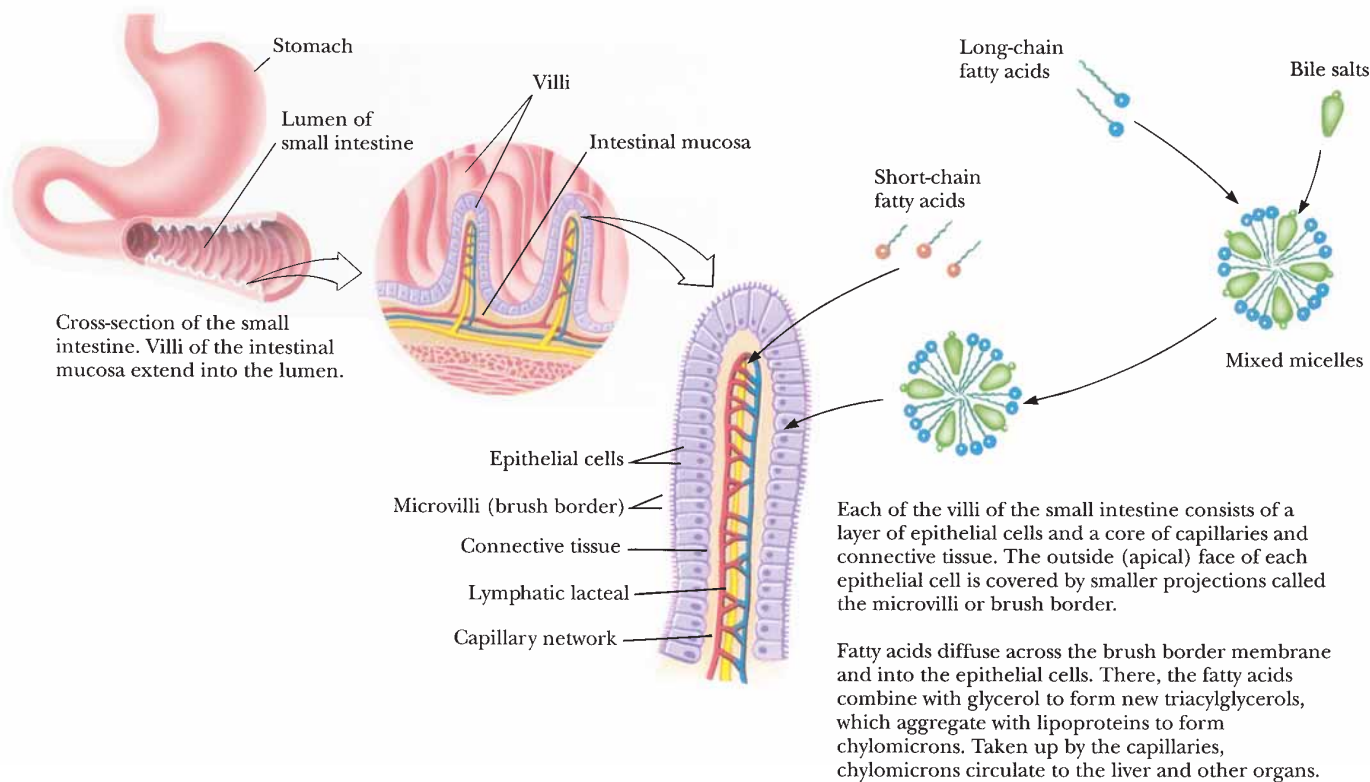


FIGURE 24.3 • (a) A duct at the junction of the pancreas and duodenum secretes pancreatic juice into the duodenum, the first portion of the small intestine. (b) Hydrolysis of triacylglycerols by pancreatic and intestinal lipases. Pancreatic lipases cleave fatty acids at the C-1 and C-3 positions. Resulting monoacylglycerols with fatty acids at C-2 are hydrolyzed by intestinal lipases. Fatty acids and monoacylglycerols are absorbed through the intestinal wall and assembled into lipoprotein aggregates termed *chylomicrons* (discussed in Chapter 25).



duodenum (Figure 24.3a) raises the pH of the digestive mixture, allowing hydrolysis of the triacylglycerols by pancreatic lipase and by nonspecific esterases, which hydrolyze the fatty acid ester linkages. Pancreatic lipase cleaves fatty acids from the C-1 and C-3 positions of triacylglycerols, and other lipases and esterases attack the C-2 position (Figure 24.3b). These processes depend upon the presence of **bile salts**, a family of carboxylic acid salts with steroid backbones (see also Chapter 25). These agents act as detergents to emulsify the triacylglycerols and facilitate the hydrolytic activity of the lipases and esterases. Short-chain fatty acids (10 carbons or less) released in this way are absorbed directly into the villi of the intestinal mucosa, whereas long-chain fatty acids, which are less soluble, form mixed micelles with bile salts, and are carried in this fashion to the surfaces of the epithelial cells that cover the villi (Figure 24.4). The fatty acids pass into the epithelial cells, where they are condensed with glycerol to form new triacylglycerols. These triacylglycerols aggregate with lipoproteins to form particles called **chylomicrons**, which are then transported into the lymphatic system and on to the bloodstream, where they circulate to the liver, lungs, heart, muscles, and other organs (Chapter 25). At these sites, the triacylglycerols are hydrolyzed to release fatty acids, which can then be oxidized in a highly exergonic metabolic pathway known as β -oxidation.

FIGURE 24.4 • In the small intestine, fatty acids combine with bile salts in mixed micelles, which deliver fatty acids to epithelial cells that cover the intestinal villi. Triacylglycerols are formed within the epithelial cells.

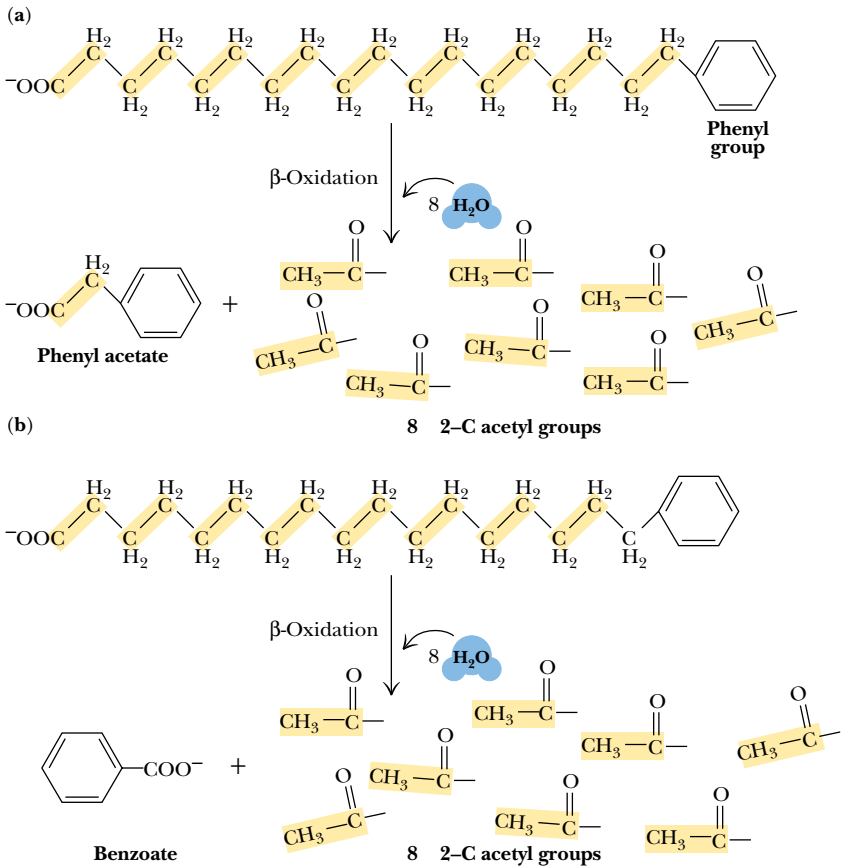
24.2 • β -Oxidation of Fatty Acids

Franz Knoop and the Discovery of β -Oxidation

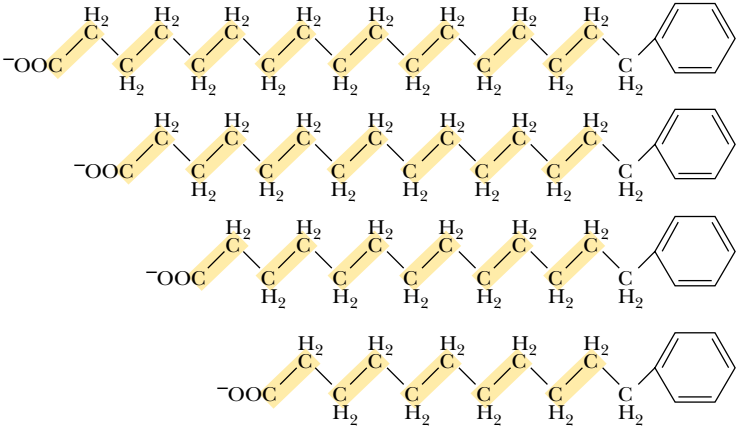
The earliest clue to the secret of fatty acid oxidation and breakdown came in the early 1900s, when Franz Knoop carried out experiments in which he fed dogs fatty acids in which the terminal methyl group had been replaced with a

phenyl ring (Figure 24.5). Knoop discovered that fatty acids containing an even number of carbon atoms were broken down to yield phenyl acetate as the final product, whereas fatty acids with an odd number of carbon atoms yielded benzoate as the final product (Figure 24.5). From these experiments, Knoop concluded that the fatty acids must be degraded by *oxidation at the β-carbon* (Figure 24.6), followed by cleavage of the C_α—C_β bond. Repetition of this process yielded 2-carbon units, which Knoop assumed must be acetate. Much later, Albert Lehninger showed that this degradative process took place in the mitochondria, and F. Lynen and E. Reichart showed that the 2-carbon unit released

FIGURE 24.5 • The oxidative breakdown of phenyl fatty acids observed by Franz Knoop. He observed that fatty acid analogs with even numbers of carbon atoms yielded phenyl acetate, whereas compounds with odd numbers of carbon atoms produced only benzoate.



Conclusion: Phenyl products shown can only result if carbons are removed in pairs



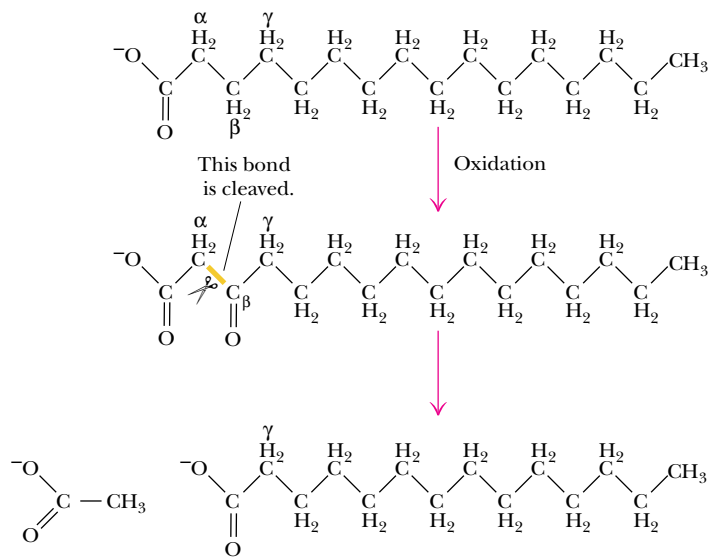


FIGURE 24.6 • Fatty acids are degraded by repeated cycles of oxidation at the β-carbon and cleavage of the C_α–C_β bond to yield acetate units.

is *acetyl-CoA*, not free acetate. Because the entire process begins with oxidation of the carbon that is “β” to the carboxyl carbon, the process has come to be known as **β-oxidation**.

Coenzyme A Activates Fatty Acids for Degradation

The process of β-oxidation begins with the formation of a thiol ester bond between the fatty acid and the thiol group of coenzyme A. This reaction, shown in Figure 24.7, is catalyzed by **acyl-CoA synthetase**, which is also called **acyl-CoA ligase** or **fatty acid thiokinase**. This condensation with CoA activates the fatty acid for reaction in the β-oxidation pathway. For long-chain fatty acids, this reaction normally occurs at the outer mitochondrial membrane, prior to entry of the fatty acid into the mitochondrion, but it may also occur at the surface of the endoplasmic reticulum. Short- and medium-length fatty acids undergo this activating reaction in the mitochondria. In all cases, the reaction is accompanied by the hydrolysis of ATP to form AMP and pyrophosphate. As shown in Figure 24.7, the two combined reactions have a net Δ*G*^{o'} of about –0.8 kJ/mol, so that the reaction is favorable but easily reversible. However, there is more to the story. As we have seen in several similar cases, the pyrophosphate produced in this reaction is rapidly hydrolyzed by inorganic pyrophosphatase to two molecules of phosphate, with a net Δ*G*^{o'} of about –33.6 kJ/mol.

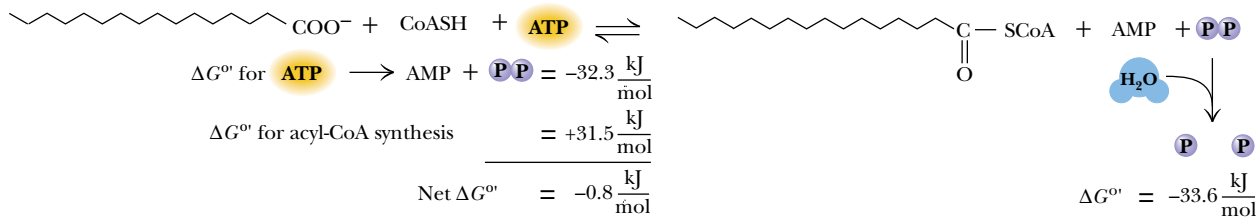


FIGURE 24.7 • The acyl-CoA synthetase reaction activates fatty acids for β-oxidation. The reaction is driven by hydrolysis of ATP to AMP and pyrophosphate and by the subsequent hydrolysis of pyrophosphate.

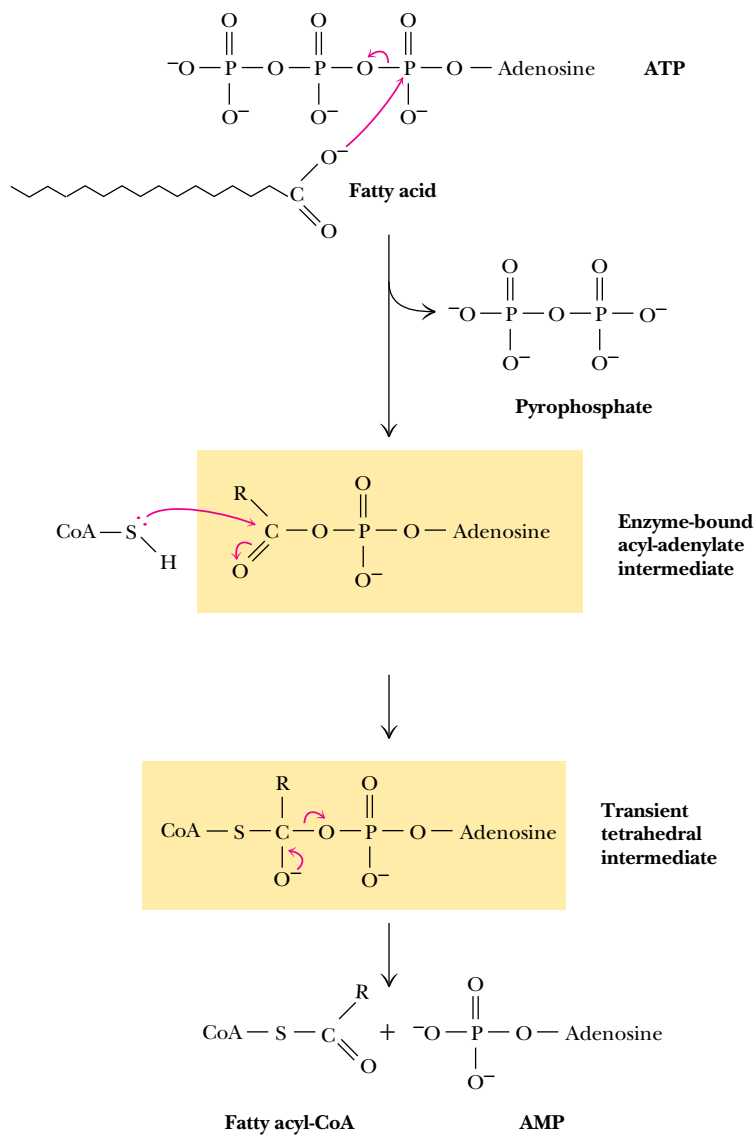


FIGURE 24.8 • The mechanism of the acyl-CoA synthetase reaction involves fatty acid carboxylate attack on ATP to form an acyl-adenylate intermediate. The fatty acyl CoA thioester product is formed by CoA attack on this intermediate.

Thus, pyrophosphate is maintained at a low concentration in the cell (usually less than 1 mM) and the synthetase reaction is strongly promoted. The mechanism of the acyl-CoA synthetase reaction is shown in Figure 24.8 and involves attack of the fatty acid carboxylate on ATP to form an *acyl-adenylate intermediate*, which is subsequently attacked by CoA, forming a fatty acyl-CoA thioester.

Carnitine Carries Fatty Acyl Groups Across the Inner Mitochondrial Membrane

All of the other enzymes of the β -oxidation pathway are located in the mitochondrial matrix. Short-chain fatty acids, as already mentioned, are transported into the matrix as free acids and form the acyl-CoA derivatives there. However, long-chain fatty acyl-CoA derivatives cannot be transported into the matrix directly. These long-chain derivatives must first be converted to *acylcarnitine* derivatives, as shown in Figure 24.9. **Carnitine acyltransferase I**, located on the outer side of the inner mitochondrial membrane, catalyzes the formation of

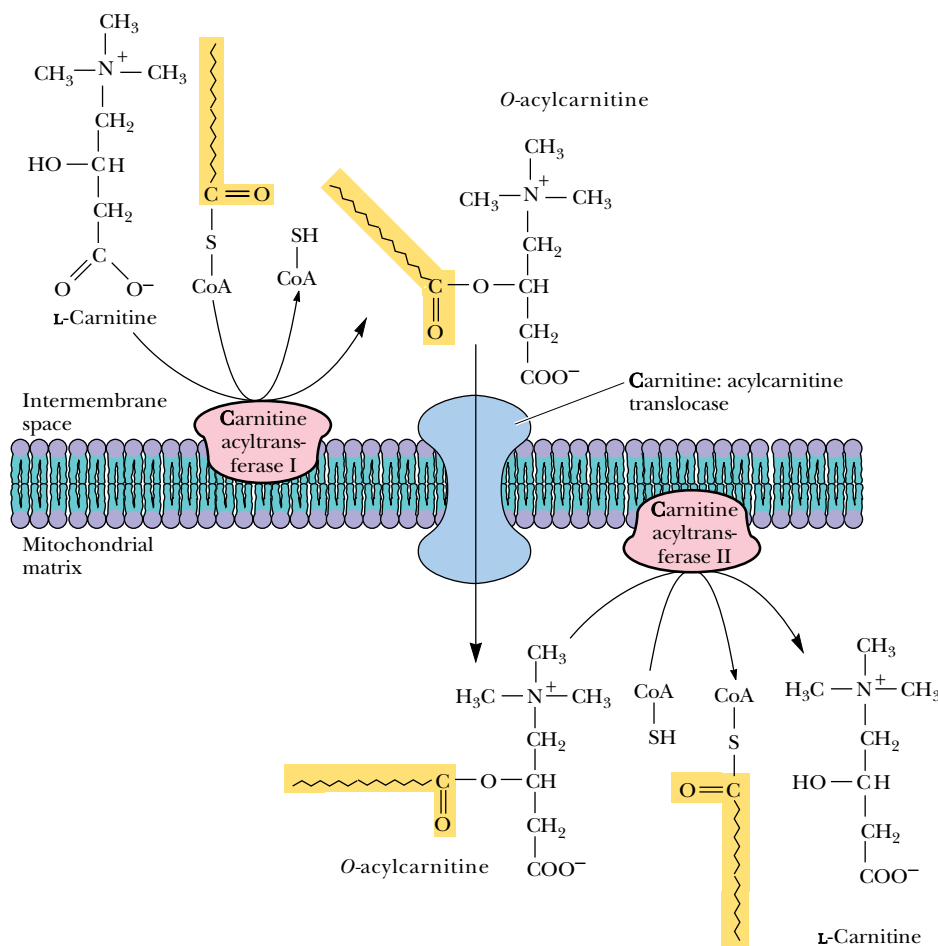


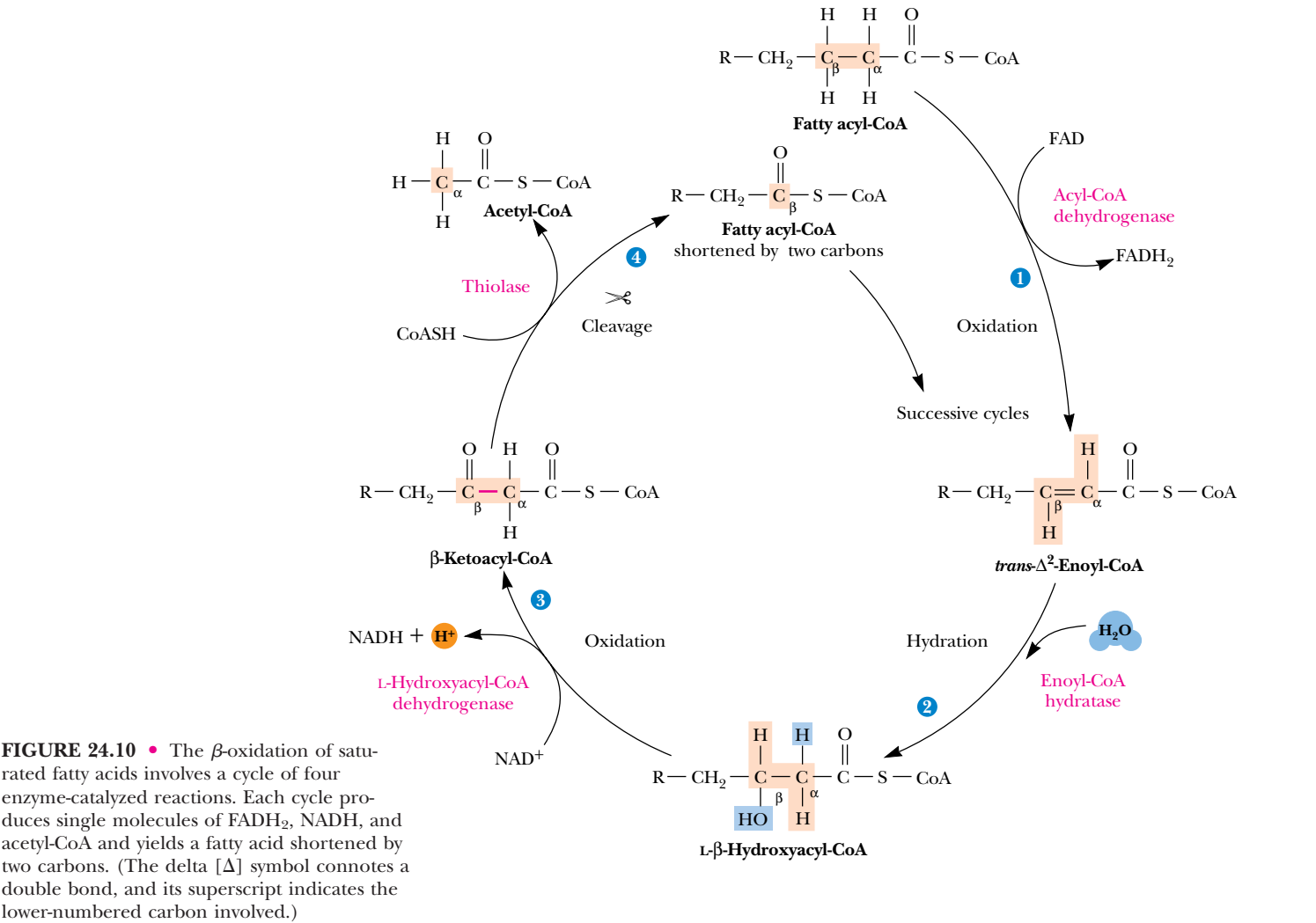
FIGURE 24.9 • The formation of acylcarnitines and their transport across the inner mitochondrial membrane. The process involves the coordinated actions of carnitine acyltransferases on both sides of the membrane and of a translocase that shuttles O-acylcarnitines across the membrane.

the O-acylcarnitine, which is then transported across the inner membrane by a **translocase**. At this point, the acylcarnitine is passed to **carnitine acyltransferase II** on the matrix side of the inner membrane, which transfers the fatty acyl group back to CoA to re-form the fatty acyl-CoA, leaving free carnitine, which can return across the membrane via the translocase.

Several additional points should be made. First, although oxygen esters usually have lower group-transfer potentials than thiol esters, the O—acyl bonds in acylcarnitines have high group-transfer potentials, and the transesterification reactions mediated by the acyltransferases have equilibrium constants close to 1. Second, note that eukaryotic cells maintain separate pools of CoA in the mitochondria and in the cytosol. The cytosolic pool is utilized principally in fatty acid biosynthesis (Chapter 25), and the mitochondrial pool is important in the oxidation of fatty acids and pyruvate, as well as some amino acids.

β -Oxidation Involves a Repeated Sequence of Four Reactions

For saturated fatty acids, the process of β -oxidation involves a recurring cycle of four steps, as shown in Figure 24.10. The overall strategy in the first three steps is to create a carbonyl group on the β -carbon by oxidizing the C_α — C_β bond to form an olefin, with subsequent hydration and oxidation. In essence, this cycle is directly analogous to the sequence of reactions converting succi-



nate to oxaloacetate in the TCA cycle. The fourth reaction of the cycle cleaves the β-keto ester in a reverse Claisen condensation, producing an acetate unit and leaving a fatty acid chain that is two carbons shorter than it began. (Recall from Chapter 20 that Claisen condensations involve attack by a nucleophilic agent on a carbonyl carbon to yield a β-keto acid.)

Acyl-CoA Dehydrogenase—The First Reaction of β-Oxidation

The first reaction, the oxidation of the C_α—C_β bond, is catalyzed by **acyl-CoA dehydrogenases**, a family of three soluble matrix enzymes (with molecular weights of 170 to 180 kD), which differ in their specificity for either long-, medium-, or short-chain acyl-CoAs. They carry noncovalently (but tightly) bound FAD, which is reduced during the oxidation of the fatty acid. As shown in Figure 24.11, FADH₂ transfers its electrons to an **electron transfer flavoprotein** (ETF). Reduced ETF is reoxidized by a specific oxidoreductase (an iron–sulfur protein), which in turn sends the electrons on to the electron transport chain at the level of coenzyme Q. As always, mitochondrial oxidation of FAD in this way eventually results in the net formation of about 1.5 ATP. The mechanism of the acyl-CoA dehydrogenase (Figure 24.12) involves deprotonation of the fatty acid chain at the α-carbon, followed by hydride transfer from the β-carbon to FAD. The structure of the medium-chain dehydrogenase from pig liver places an FAD molecule in an extended conformation between a bundle of α-helices and a distorted β-barrel (Figure 24.13).

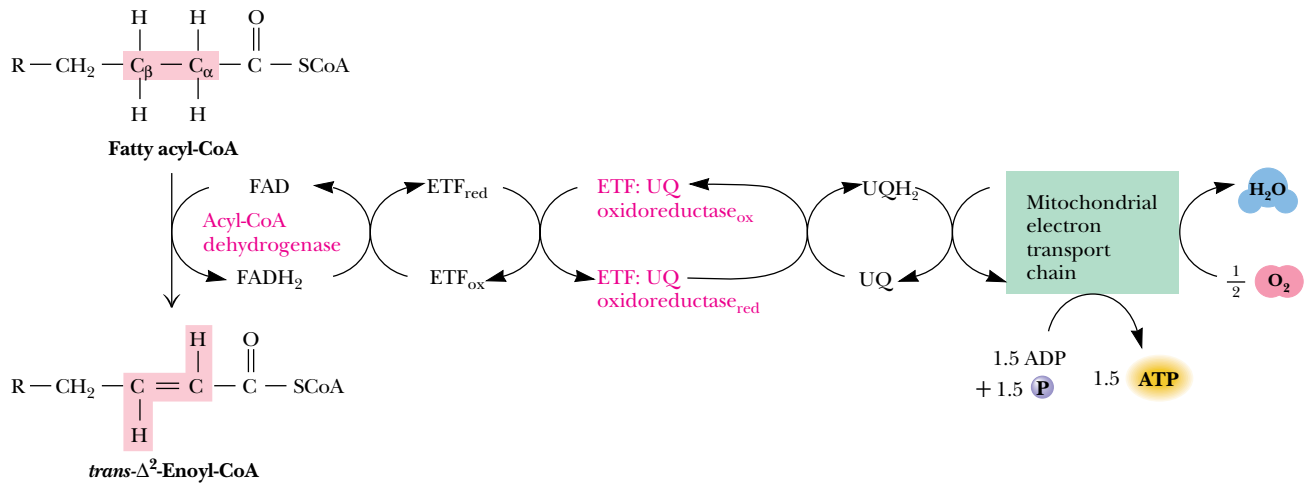


FIGURE 24.11 • The acyl-CoA dehydrogenase reaction. The two electrons removed in this oxidation reaction are delivered to the electron transport chain in the form of reduced coenzyme Q (UQH_2).

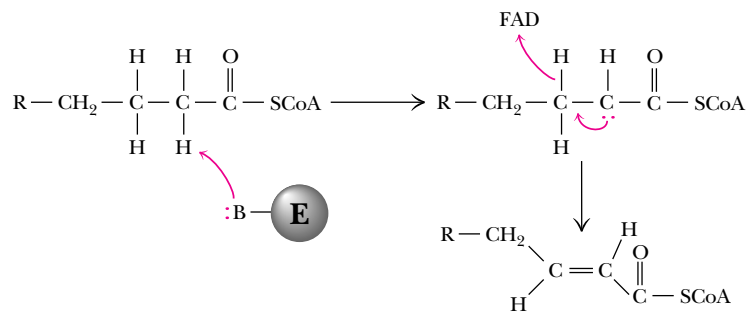


FIGURE 24.12 • The mechanism of acyl-CoA dehydrogenase. Removal of a proton from the α -C is followed by hydride transfer from the β -carbon to FAD.

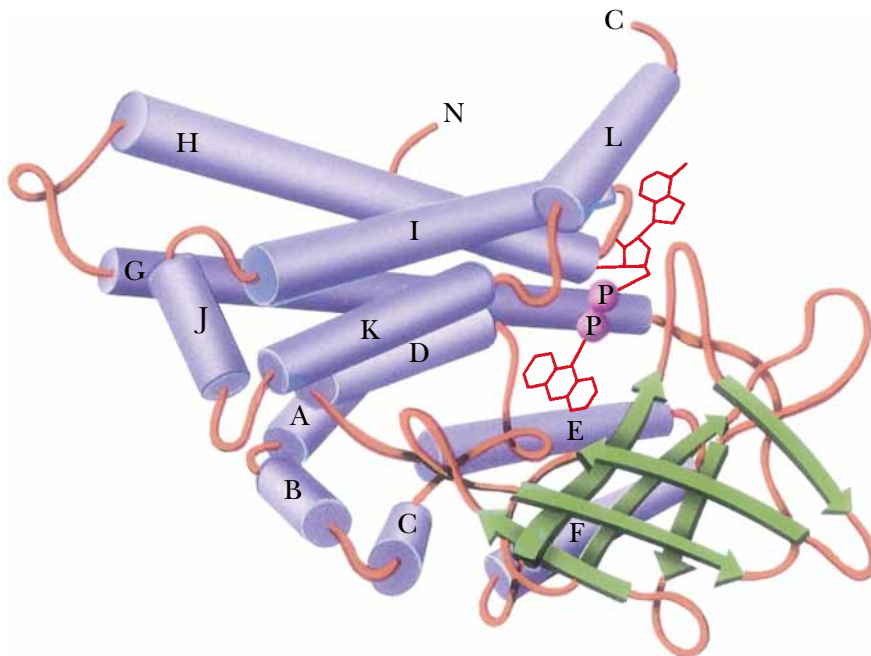


FIGURE 24.13 • The subunit structure of medium chain acyl-CoA dehydrogenase from pig liver mitochondria. Note the location of the bound FAD (red). (*Adapted from Kim, J.-T., and Wu, J., 1988. Structure of the medium-chain acyl-CoA dehydrogenase from pig liver mitochondria at 3-Å resolution. Proceedings of the National Academy of Sciences, USA 85:6671–6681.*)

A
D E E P E R L O O K

The Akee Tree

The akee (also spelled *ackee*) tree is native to West Africa and was brought to the Caribbean by African slaves. It was introduced to science by William Bligh, captain of the infamous sailing ship the *Bounty*, and its botanical name is (appropriately) *Blighia sapida* (the latter name from the Latin *sapidus* meaning “tasty”). A popular dish in the Caribbean consists of akee and salt fish.

*“Akee, rice, salt fish are nice,
And the rum is fine any time of year.”
—From the song Jamaica Farewell*

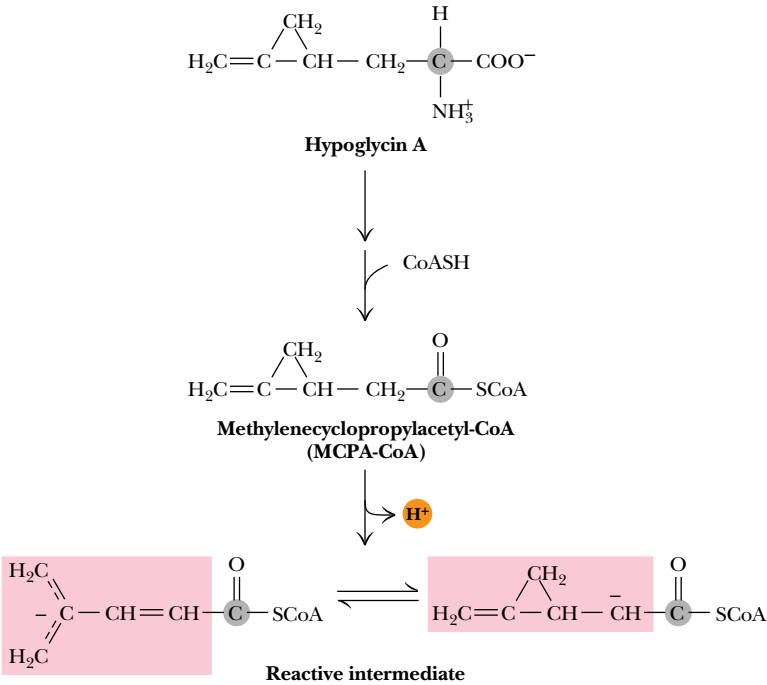


(R.R. Head/Earth Scenes/Animals, Animals)

A Metabolite of Hypoglycin from Akee Fruit
Inhibits Acyl-CoA Dehydrogenase

The unripened fruit of the **akee tree** contains **hypoglycin**, a rare amino acid (Figure 24.14). Metabolism of hypoglycin yields *methylenecyclopropylacetyl-CoA* (MCPA-CoA). Acyl-CoA dehydrogenase will accept MCPA-CoA as a substrate,

FIGURE 24.14 • The conversion of hypoglycin from akee fruit to a form that inhibits acyl-CoA dehydrogenase.



removing a proton from the α -carbon to yield an intermediate that irreversibly inactivates acyl-CoA dehydrogenase by reacting covalently with FAD on the enzyme. For this reason, consumption of unripened akee fruit can lead to vomiting and, in severe cases, convulsions, coma, and death. The condition is most severe in individuals with low levels of acyl-CoA dehydrogenase.

Enoyl-CoA Hydratase Adds Water Across the Double Bond

The next step in β -oxidation is the addition of the elements of H_2O across the new double bond in a stereospecific manner, yielding the corresponding hydroxyacyl-CoA (Figure 24.15). The reaction is catalyzed by **enoyl-CoA hydratase**. At least three different enoyl-CoA hydratase activities have been detected in various tissues. Also called **crotonases**, these enzymes specifically convert *trans*-enoyl-CoA derivatives to L- β -hydroxyacyl-CoA. As shown in Figure 24.15, these enzymes will also metabolize *cis*-enoyl-CoA (at slower rates) to give specifically D- β -hydroxyacyl-CoA. Recently, a novel enoyl-CoA hydratase was discovered, which converts *trans*-enoyl-CoA to D- β -hydroxyacyl-CoA, as shown in Figure 24.15.

L-Hydroxyacyl-CoA Dehydrogenase Oxidizes the β -Hydroxyl Group

The third reaction of this cycle is the oxidation of the hydroxyl group at the β -position to produce a β -ketoacyl-CoA derivative. This second oxidation reaction is catalyzed by **L-hydroxyacyl-CoA dehydrogenase**, an enzyme that requires NAD^+ as a coenzyme. NADH produced in this reaction represents metabolic energy. Each NADH produced in mitochondria by this reaction drives the synthesis of 2.5 molecules of ATP in the electron transport pathway. L-Hydroxyacyl-

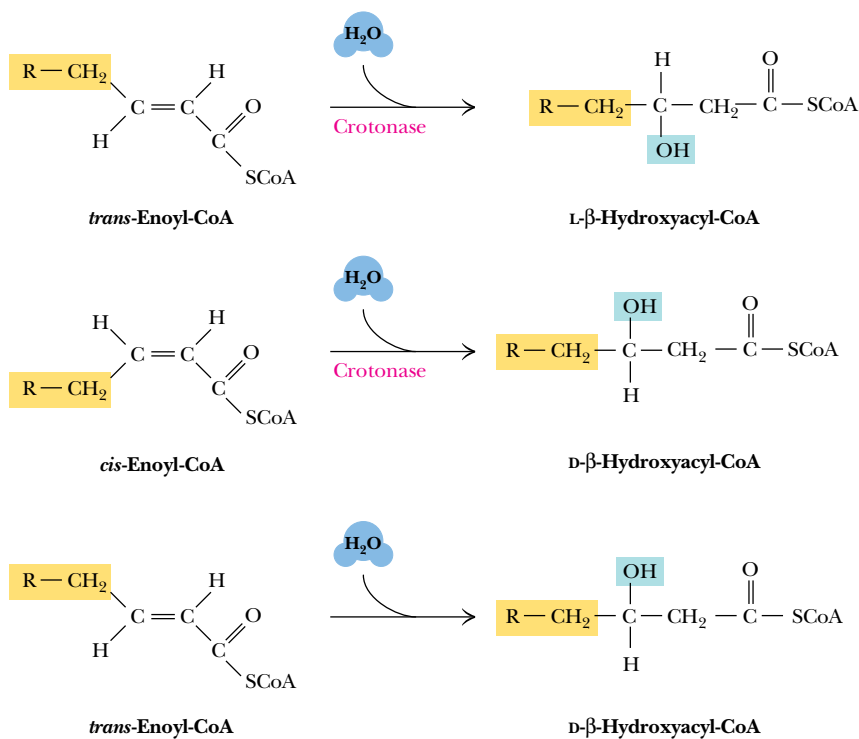


FIGURE 24.15 • The conversion of *trans*- and *cis*-enoyl CoA derivatives to L- and D- β -hydroxyacyl CoA, respectively. These reactions are catalyzed by enoyl-CoA hydratases (also called crotonases), enzymes that vary in their acyl-chain length specificity. A recently discovered enzyme converts *trans*-enoyl-CoA directly to D- β -hydroxyacyl-CoA.

FIGURE 24.16 • The L-β-hydroxyacyl-CoA dehydrogenase reaction.

CoA dehydrogenase shows absolute specificity for the L-hydroxyacyl isomer of the substrate (Figure 24.16). (D-Hydroxyacyl isomers, which arise mainly from oxidation of unsaturated fatty acids, are handled differently.)

β-Ketoacyl-CoA Intermediates Are Cleaved in the Thiolase Reaction

The final step in the β-oxidation cycle is the cleavage of the β-ketoacyl-CoA. This reaction, catalyzed by **thiolase** (also known as **β-ketothiolase**), involves the attack of a cysteine thiolate from the enzyme on the β-carbonyl carbon, followed by cleavage to give the enolate of acetyl-CoA and an enzyme-thioester intermediate (Figure 24.17). Subsequent attack by the thiol group of a second CoA and departure of the cysteine thiolate yields a new (shorter) acyl-CoA. If the reaction in Figure 24.17 is read in reverse, it is easy to see that it is a Claisen condensation—an attack of the enolate anion of acetyl-CoA on a thioester. Despite the formation of a second thioester, this reaction has a very favorable K_{eq} , and it drives the three previous reactions of β-oxidation.

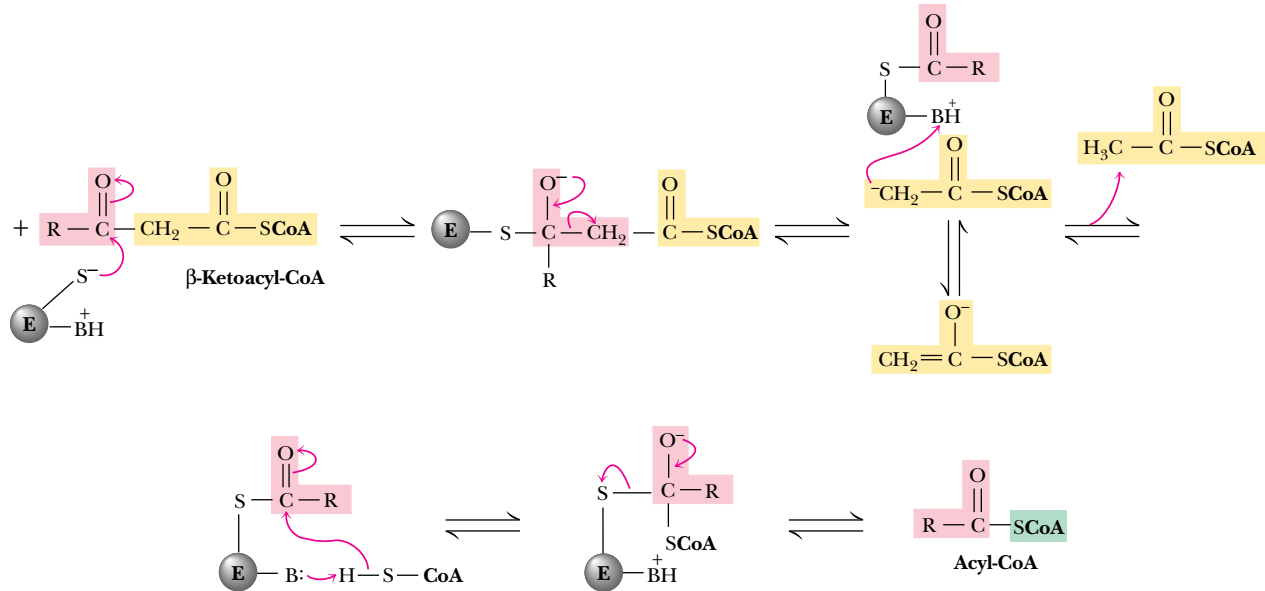


FIGURE 24.17 • The mechanism of the thiolase reaction. Attack by an enzyme cysteine thiolate group at the β-carbonyl carbon produces a tetrahedral intermediate, which decomposes with departure of acetyl-CoA, leaving an enzyme thioester intermediate. Attack by the thiol group of a second CoA yields a new (shortened) acyl-CoA.

Repetition of the β-Oxidation Cycle Yields a Succession of Acetate Units

In essence, this series of four reactions has yielded a fatty acid (as a CoA ester) that has been shortened by two carbons, and one molecule of acetyl-CoA. The shortened fatty acyl-CoA can now go through another β-oxidation cycle, as shown in Figure 24.10. Repetition of this cycle with a fatty acid with an even number of carbons eventually yields two molecules of acetyl-CoA in the final step. As noted in the first reaction in Table 24.2, complete β-oxidation of palmitic acid yields eight molecules of acetyl-CoA as well as seven molecules of FADH₂ and seven molecules of NADH. The acetyl-CoA can be further metabolized in the TCA cycle (as we have already seen). Alternatively, acetyl-CoA can also be used as a substrate in amino acid biosynthesis (Chapter 26). As noted in Chapter 23, however, acetyl-CoA cannot be used as a substrate for gluconeogenesis.

Complete β-Oxidation of One Palmitic Acid Yields 106 Molecules of ATP

If the acetyl-CoA is directed entirely to the TCA cycle in mitochondria, it can eventually generate approximately 10 high-energy phosphate bonds—that is, 10 molecules of ATP synthesized from ADP (Table 24.2). Including the ATP formed from FADH₂ and NADH, complete β-oxidation of a molecule of palmitoyl-CoA in mitochondria yields 108 molecules of ATP. Subtracting the two high-energy bonds needed to form palmitoyl-CoA, the substrate for β-oxidation, one concludes that β-oxidation of a molecule of palmitic acid yields 106 molecules of ATP. The Δ*G*°' for complete combustion of palmitate to CO₂ is −9790 kJ/mol. The hydrolytic energy embodied in 106 ATPs is 106 × 30.5 kJ/mol = 3233 kJ/mol, so the overall efficiency of β-oxidation under standard-state conditions is approximately 33%. The large energy yield from fatty acid oxidation is a reflection of the highly reduced state of the carbon in fatty acids. Sugars, in which the carbon is already partially oxidized, produce much less energy, carbon for carbon, than do fatty acids. The breakdown of fatty acids is regulated by a variety of metabolites and hormones. Details of this regulation are described in Chapter 25, following a discussion of fatty acid synthesis.

Table 24.2
Equations for the Complete Oxidation of Palmitoyl-CoA to CO₂ and H₂O

Equation	ATP Yield	Free Energy (kJ/mol) Yield
CH ₃ (CH ₂) ₁₄ CO—CoA + 7 [FAD] + 7 H ₂ O + 7 NAD ⁺ + 7 CoA → 8 CH ₃ CO-CoA + 7 [FADH ₂] + 7 NADH + 7 H ⁺ 7 [FADH ₂] + 10.5 P _i + 10.5 ADP + 3.5 O ₂ → 7 [FAD] + 17.5 H ₂ O + 10.5 ATP	10.5	320
7 NADH + 7 H ⁺ + 17.5 P _i + 17.5 ADP + 3.5 O ₂ → 7 NAD ⁺ + 24.5 H ₂ O + 17.5 ATP	17.5	534
8-Acetyl-CoA + 16 O ₂ + 80 ADP + 80 P _i → 8 CoA + 88 H ₂ O + 16 CO ₂ + 80 ATP	80	2440
CH ₃ (CH ₂) ₁₄ CO—CoA + 108 P _i + 108 ADP + 23 O ₂ → 108 ATP + 16 CO ₂ + 130 H ₂ O + CoA	108	3294
Energetic “cost” of forming palmitoyl-CoA from palmitate and CoA	<u>−2</u> 106	<u>−61</u> 3233

Migratory Birds Travel Long Distances on Energy from Fatty Acid Oxidation

Because they represent the most highly concentrated form of stored biological energy, fatty acids are the metabolic fuel of choice for sustaining the incredibly long flights of many migratory birds. Although some birds migrate over land masses and dine frequently, other species fly long distances without stopping to eat. The American golden plover flies directly from Alaska to Hawaii, a 3300-kilometer flight requiring 35 hours (at an average speed of nearly 60 miles/hr) and more than 250,000 wing beats! The ruby-throated hummingbird, which winters in Central America and nests in southern Canada, often flies nonstop across the Gulf of Mexico. These and similar birds accomplish these prodigious feats by storing large amounts of fatty acids (as triacylglycerols) in the days before their migratory flights. The percentage of dry-weight body fat in these birds may be as high as 70% when migration begins (compared with values of 30% and less for nonmigratory birds).



(a) Gerbil



(b) Ruby-throated hummingbird

Fatty Acid Oxidation Is an Important Source of Metabolic Water for Some Animals

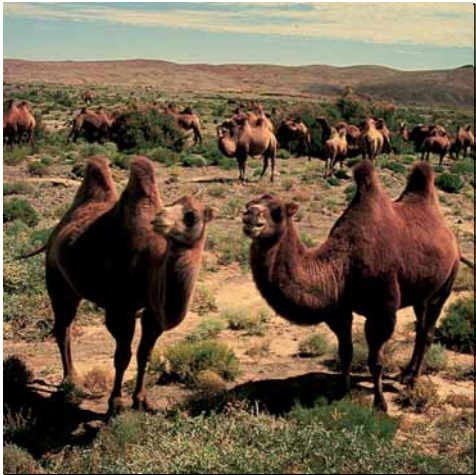
Large amounts of metabolic water are generated by β -oxidation (130 H₂O per palmitoyl-CoA). For certain animals—including desert animals, such as gerbils, and killer whales (which do not drink seawater)—the oxidation of fatty acids can be a significant source of dietary water. A striking example is the camel (Figure 24.18), whose hump is essentially a large deposit of fat. Metabolism of fatty acids from this store provides needed water (as well as metabolic energy) during periods when drinking water is not available. It might well be said that “the ship of the desert” sails on its own metabolic water!



(c) Golden plover



(d) Orca



(e) Camels

FIGURE 24.18 • Animals whose existence is strongly dependent on fatty acid oxidation: (a) gerbil, (b) ruby-throated hummingbird, (c) golden plover, (d) orca (killer whale), and (e) camels. (a, Photo Researchers, Inc.; b, Tom J. Ulrich/Visuals Unlimited; c, S. J. Krasemann/Photo Researchers, Inc.; d, © Francois Gohier/Photo Researchers, Inc.; e, © George Holton/Photo Researchers, Inc.)

24.3 • β-Oxidation of Odd-Carbon Fatty Acids

β-Oxidation of Odd-Carbon Fatty Acids Yields Propionyl-CoA

Fatty acids with odd numbers of carbon atoms are rare in mammals, but fairly common in plants and marine organisms. Humans and animals whose diets include these food sources metabolize odd-carbon fatty acids via the β-oxidation pathway. The final product of β-oxidation in this case is the 3-carbon propionyl-CoA instead of acetyl-CoA. Three specialized enzymes then carry out the reactions that convert propionyl-CoA to succinyl-CoA, a TCA cycle intermediate. (Because propionyl-CoA is a degradation product of methionine, valine, and isoleucine, this sequence of reactions is also important in amino acid catabolism, as we shall see in Chapter 26.) The pathway involves an initial carboxylation at the α-carbon of propionyl-CoA to produce D-methylmalonyl-CoA (Figure 24.19). The reaction is catalyzed by a biotin-dependent enzyme, **propionyl-CoA carboxylase**. The mechanism involves ATP-driven carboxylation of biotin at N₁, followed by nucleophilic attack by the α-carbanion of propionyl-CoA in a stereo-specific manner.

D-Methylmalonyl-CoA, the product of this reaction, is converted to the L-isomer by **methylmalonyl-CoA epimerase** (Figure 24.19). (This enzyme has often and incorrectly been called “methylmalonyl-CoA racemase.” It is not a racemase because the CoA moiety contains five other asymmetric centers.) The epimerase reaction also appears to involve a carbanion at the α-position (Figure 24.20). The reaction is readily reversible and involves a reversible dissociation of the acidic α-proton. The L-isomer is the substrate for methylmalonyl-CoA mutase. Methylmalonyl-CoA epimerase is an impressive catalyst. The pK_a for the proton that must dissociate to initiate this reaction is approximately 21! If binding of a proton to the α-anion is diffusion-limited, with k_{on} = 10⁹ M⁻¹ sec⁻¹, then the initial proton dissociation must be rate-limiting, and the rate constant must be

$k_{\text{off}} = K_a \cdot k_{\text{on}} = (10^{-21} \text{ M}) \cdot (10^9 \text{ M}^{-1} \text{ sec}^{-1}) = 10^{-12} \text{ sec}^{-1}$

The turnover number of methylmalonyl-CoA epimerase is 100 sec⁻¹, and thus the enzyme enhances the reaction rate by a factor of 10¹⁴.

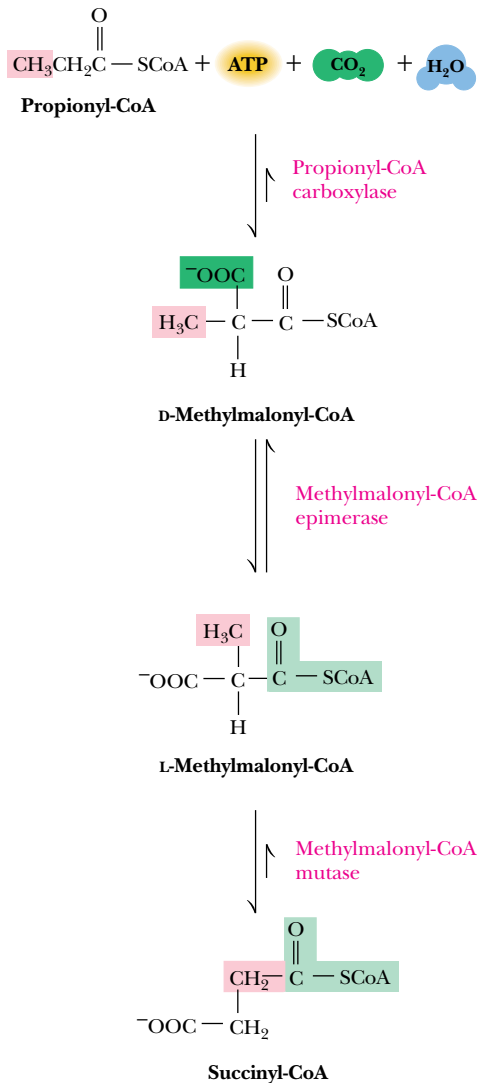


FIGURE 24.19 • The conversion of propionyl-CoA (formed from β-oxidation of odd-carbon fatty acids) to succinyl-CoA is carried out by a trio of enzymes as shown. Succinyl-CoA can enter the TCA cycle or be converted to acetyl-CoA.

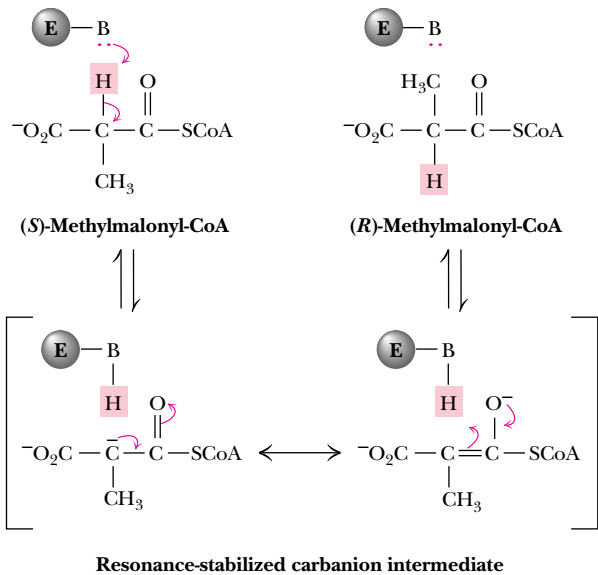


FIGURE 24.20 • The methylmalonyl-CoA epimerase mechanism involves a resonance-stabilized carbanion at the α-position.

A B₁₂-Catalyzed Rearrangement Yields Succinyl-CoA from L-Methylmalonyl-CoA

The third reaction, catalyzed by **methylmalonyl-CoA mutase**, is quite unusual because it involves a migration of the carbonyl-CoA group from one carbon to its neighbor (Figure 24.21). The mutase reaction is vitamin B₁₂-dependent and begins with **homolytic cleavage** of the Co³⁺—C bond in cobalamin, reducing the cobalt to Co²⁺. Transfer of a hydrogen atom from the substrate to the deoxyadenosyl group produces a methylmalonyl-CoA radical, which then can undergo a classic B₁₂-catalyzed rearrangement to yield a succinyl-CoA radical. Hydrogen transfer from the deoxyadenosyl group yields succinyl-CoA and regenerates the B₁₂ coenzyme.

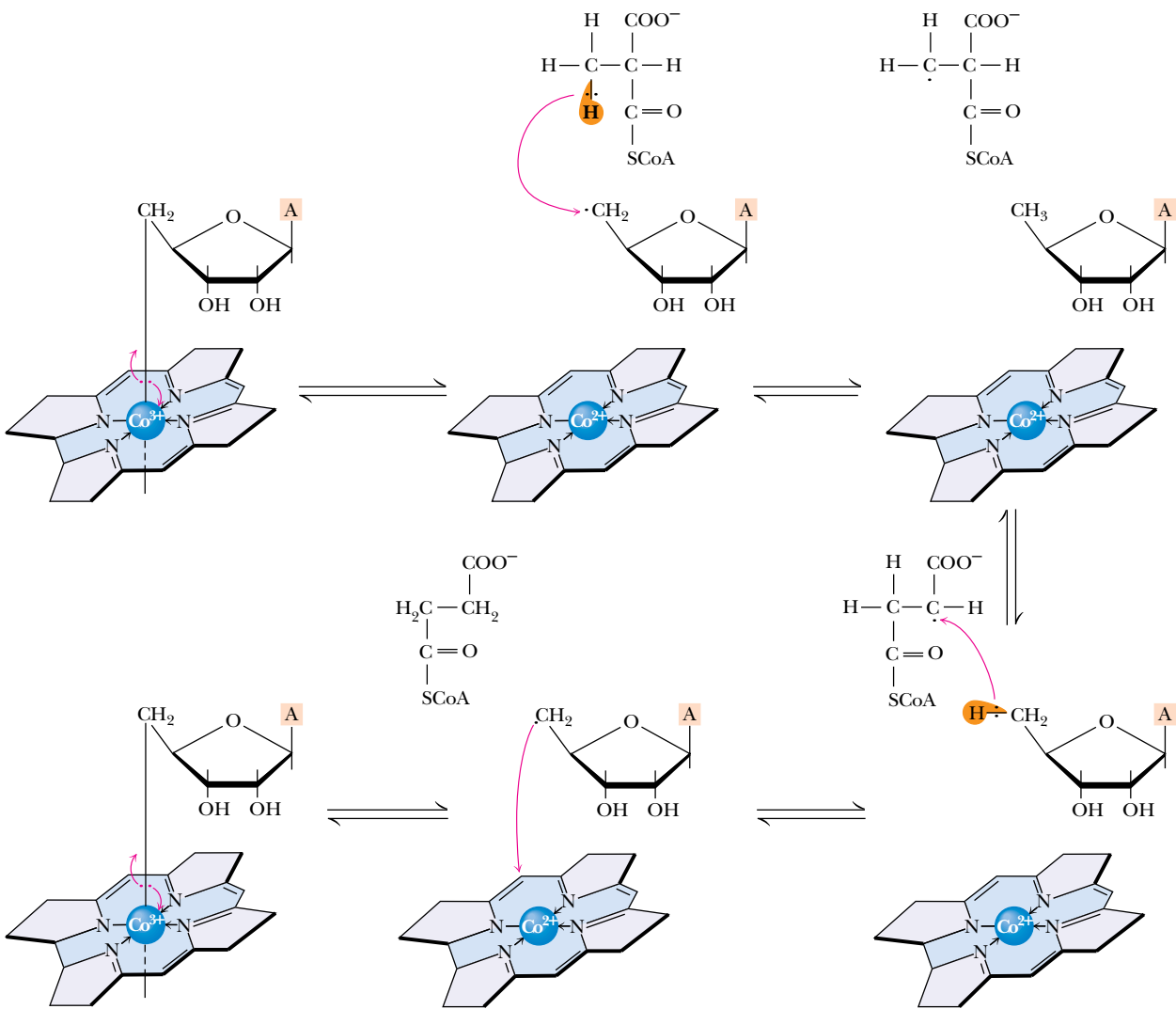


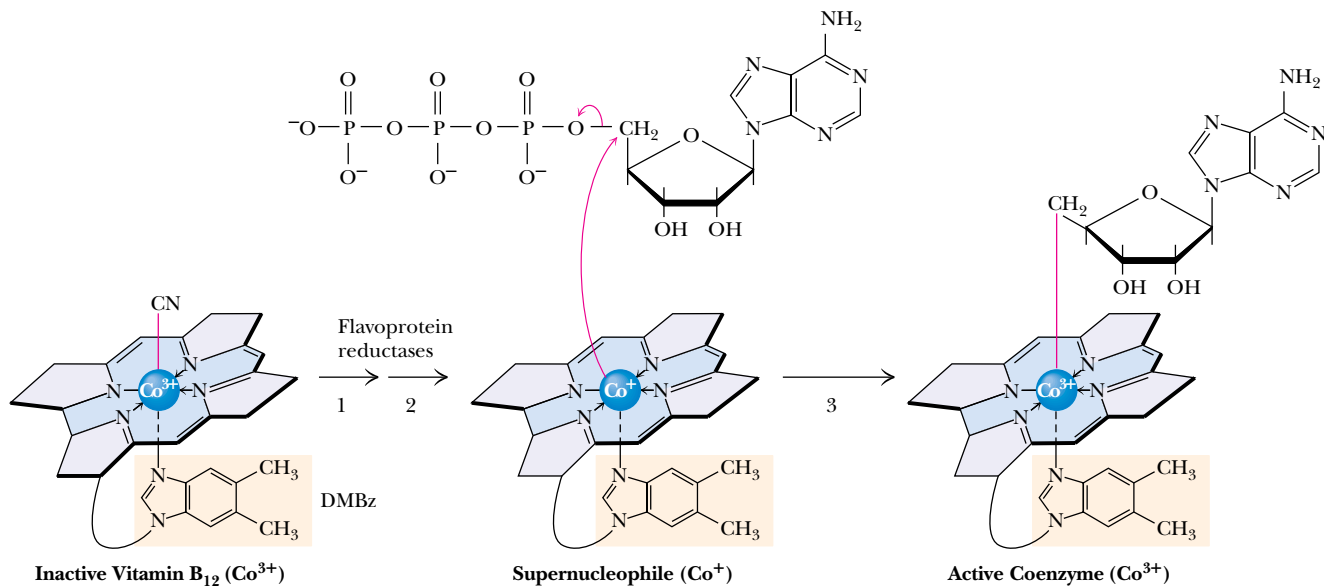
FIGURE 24.21 • A mechanism for the methylmalonyl-CoA mutase reaction. In the first step, Co³⁺ is reduced to Co²⁺ due to homolytic cleavage of the Co³⁺—C bond in cobalamin. Hydrogen atom transfer from methylmalonyl-CoA yields a methylmalonyl-CoA radical that can undergo rearrangement to form a succinyl-CoA radical. Transfer of an H atom regenerates the coenzyme and yields succinyl-CoA.

DEEPER LOOK

The Activation of Vitamin B₁₂

Conversion of inactive vitamin B₁₂ to active 5'-deoxyadenosylcobalamin is thought to involve three steps (see figure). Two *flavoprotein reductases* sequentially convert Co³⁺ in cyanocobalamin to the Co²⁺ state and then to the Co⁺ state. Co⁺ is an extremely powerful nucleophile. It attacks the C-5' carbon of ATP as shown, expelling the triphosphate anion to form 5'-deoxyadenosyl-

cobalamin. Because two electrons from Co⁺ are donated to the Co—carbon bond, the oxidation state of cobalt reverts to Co³⁺ in the active coenzyme. This is one of only two known adenosyl transfers (that is, nucleophilic attack on the ribose 5'-carbon of ATP) in biological systems. (The other is the formation of S-adenosylmethionine—see Chapter 26.)



Formation of the active coenzyme 5'-deoxyadenosylcobalamin from inactive vitamin B₁₂ is initiated by the action of flavoprotein reductases. The resulting Co⁺ species, dubbed a supernucleophile, attacks the 5'-carbon of ATP in an unusual adenosyl transfer.

Net Oxidation of Succinyl-CoA Requires Conversion to Acetyl-CoA

Succinyl-CoA derived from propionyl-CoA can enter the TCA cycle. Oxidation of succinate to oxaloacetate provides a substrate for glucose synthesis. Thus, although the acetate units produced in β -oxidation cannot be utilized in gluconeogenesis by animals, the occasional propionate produced from oxidation of odd-carbon fatty acids *can* be used for sugar synthesis. Alternatively, succinate introduced to the TCA cycle from odd-carbon fatty acid oxidation may be oxidized to CO₂. However, all of the 4-carbon intermediates in the TCA cycle are regenerated in the cycle and thus should be viewed as catalytic species. Net consumption of succinyl-CoA thus does not occur directly in the TCA cycle. Rather, the succinyl-CoA generated from β -oxidation of odd-carbon fatty acids must be converted to pyruvate and then to acetyl-CoA (which is completely oxidized in the TCA cycle). To follow this latter route, succinyl-CoA entering the TCA cycle must be first converted to malate in the usual way, and then transported from the mitochondrial matrix to the cytosol, where it is oxida-

FIGURE 24.22 • The malic enzyme reaction proceeds by oxidation of malate to oxaloacetate, followed by decarboxylation to yield pyruvate.

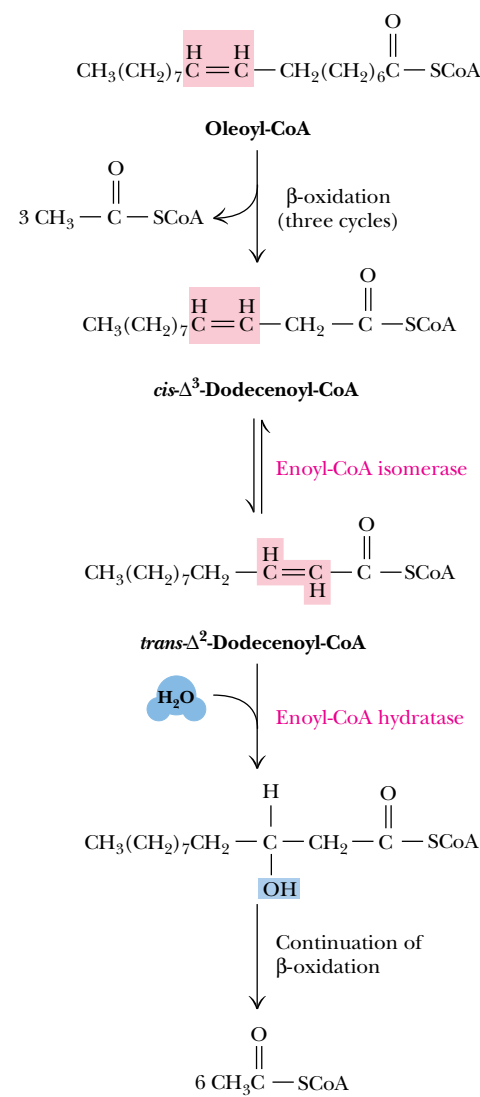
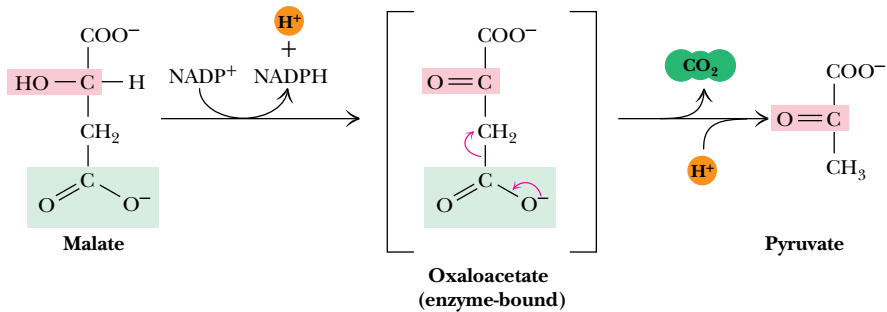


FIGURE 24.23 • β -Oxidation of unsaturated fatty acids. In the case of oleoyl-CoA, three β -oxidation cycles produce three molecules of acetyl-CoA and leave $cis\text{-}\Delta^3$ -dodecenoyl-CoA. Rearrangement of enoyl-CoA isomerase gives the $trans\text{-}\Delta^2$ species, which then proceeds normally through the β -oxidation pathway.

tively decarboxylated to pyruvate and CO_2 by **malic enzyme**, as shown in Figure 24.22. Pyruvate can then be transported back to the mitochondrial matrix, where it enters the TCA cycle via pyruvate dehydrogenase. Note that malic enzyme plays an important role in fatty acid synthesis (see Figure 25.1).

24.4 • β -Oxidation of Unsaturated Fatty Acids

An Isomerase and a Reductase Facilitate the β -Oxidation of Unsaturated Fatty Acids

Unsaturated fatty acids are also catabolized by β -oxidation, but two additional mitochondrial enzymes—an isomerase and a novel reductase—are required to handle the *cis*-double bonds of naturally occurring fatty acids. As an example, consider the breakdown of oleic acid, an 18-carbon chain with a double bond at the 9,10-position. The reactions of β -oxidation proceed normally through three cycles, producing three molecules of acetyl-CoA and leaving the degradation product $cis\text{-}\Delta^3$ -dodecenoyl-CoA, shown in Figure 24.23. This intermediate is not a substrate for acyl-CoA dehydrogenase. With a double bond at the 3,4-position, it is not possible to form another double bond at the 2,3- (or β) position. As shown in Figure 24.23, this problem is solved by **enoyl-CoA isomerase**, an enzyme that rearranges this $cis\text{-}\Delta^3$ double bond to a $trans\text{-}\Delta^2$ double bond. This latter species can proceed through the normal route of β -oxidation.

Degradation of Polyunsaturated Fatty Acids Requires 2,4-Dienoyl-CoA Reductase

Polyunsaturated fatty acids pose a slightly more complicated situation for the cell. Consider, for example, the case of linoleic acid shown in Figure 24.24. As with oleic acid, β -oxidation proceeds through three cycles, and enoyl-CoA isomerase converts the $cis\text{-}\Delta^3$ double bond to a $trans\text{-}\Delta^2$ double bond to permit one more round of β -oxidation. What results this time, however, is a $cis\text{-}\Delta^4$ enoyl-CoA, which is converted normally by acyl-CoA dehydrogenase to a $trans\text{-}\Delta^2$, $cis\text{-}\Delta^4$ species. This, however, is a poor substrate for the enoyl-CoA hydratase. This problem is solved by **2,4-dienoyl-CoA reductase**, the product of which depends on the organism. The mammalian form of this enzyme produces a $trans\text{-}\Delta^3$ enoyl product, as shown in Figure 24.24, which can be converted by an enoyl-CoA isomerase to the $trans\text{-}\Delta^2$ enoyl-CoA, which can then proceed normally through the β -oxidation pathway. *Escherichia coli* possesses a 2,4-dienoyl-CoA reductase that reduces the double bond at the 4,5-position to yield the $trans\text{-}\Delta^2$ enoyl-CoA product in a single step.

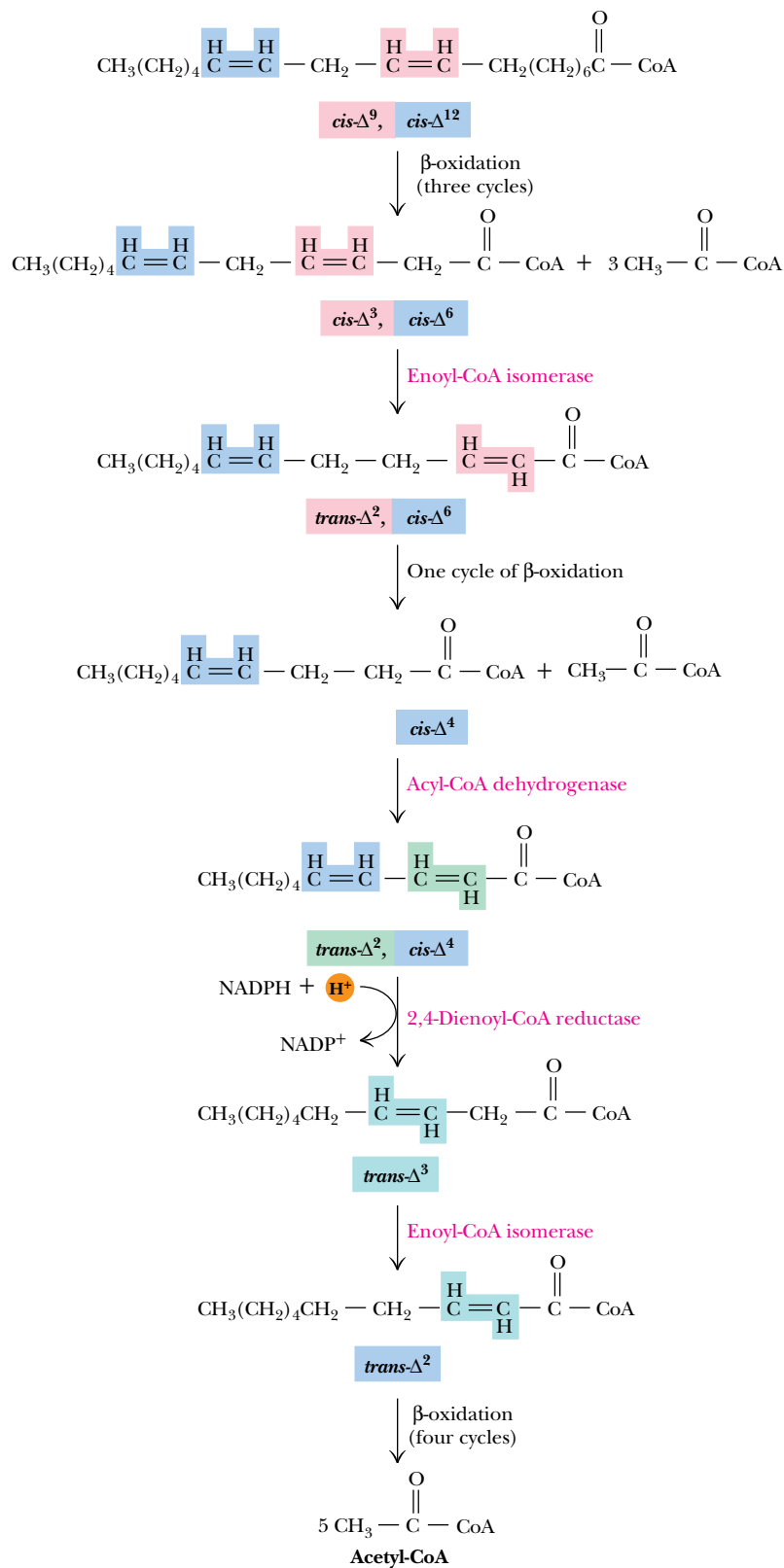


FIGURE 24.24 • The oxidation pathway for polyunsaturated fatty acids, illustrated for linoleic acid. Three cycles of β -oxidation on linoleoyl-CoA yield the $\text{cis-}\Delta^3$, $\text{cis-}\Delta^6$ intermediate, which is converted to a $\text{trans-}\Delta^2$, $\text{cis-}\Delta^6$ intermediate. An additional round of β -oxidation gives $\text{cis-}\Delta^4$ enoyl-CoA, which is oxidized to the $\text{trans-}\Delta^2$, $\text{cis-}\Delta^4$ species by acyl-CoA dehydrogenase. The subsequent action of 2,4-dienoyl-CoA reductase yields the $\text{trans-}\Delta^3$ product, which is converted by enoyl-CoA isomerase to the $\text{trans-}\Delta^2$ form. Normal β -oxidation then produces five molecules of acetyl-CoA.

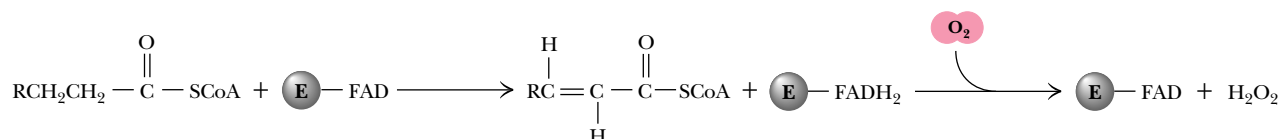


FIGURE 24.25 • The acyl-CoA oxidase reaction in peroxisomes.

24.5 • Other Aspects of Fatty Acid Oxidation

Peroxisomal β -Oxidation Requires FAD-Dependent Acyl-CoA Oxidase

Although β -oxidation in mitochondria¹ is the principal pathway of fatty acid catabolism, several other minor pathways play important roles in fat catabolism. For example, organelles other than mitochondria carry out β -oxidation processes, including *peroxisomes* and *glyoxysomes*. **Peroxisomes** are so named because they carry out a variety of flavin-dependent oxidation reactions, regenerating oxidized flavins by reaction with oxygen to produce hydrogen peroxide, H_2O_2 . Peroxisomal β -oxidation is similar to mitochondrial β -oxidation, except that the initial double bond formation is catalyzed by an FAD-dependent **acyl-CoA oxidase** (Figure 24.25). The action of this enzyme in the peroxisomes transfers the liberated electrons directly to oxygen instead of the electron transport chain. As a result, each 2-carbon unit oxidized in peroxisomes produces fewer ATPs. The enzymes responsible for fatty acid oxidation in peroxisomes are inactive with carbon chains of eight or fewer. Such short-chain products must be transferred to the mitochondria for further breakdown. Similar β -oxidation enzymes are also found in **glyoxysomes**—peroxisomes in plants that also carry out the reactions of the glyoxylate pathway.

Branched-Chain Fatty Acids and α -Oxidation

Although β -oxidation is universally important, there are some instances in which it cannot operate effectively. For example, branched-chain fatty acids with alkyl branches at odd-numbered carbons are not effective substrates for β -oxidation. For such species, **α -oxidation** is a useful alternative. Consider **phytol**, a breakdown product of chlorophyll that occurs in the fat of ruminant animals such as sheep and cows and also in dairy products. Ruminants oxidize phytol to phytanic acid, and digestion of phytanic acid in dairy products is thus an important dietary consideration for humans. The methyl group at C-3 will block β -oxidation, but, as shown in Figure 24.26, **phytanic acid α -hydroxylase** places an —OH group at the α -carbon, and **phytanic acid α -oxidase** decarboxylates it to yield *pristanic acid*. The CoA ester of this metabolite can undergo β -oxidation in the normal manner. The terminal product, isobutyryl-CoA, can be sent into the TCA cycle by conversion to succinyl-CoA.

Refsum's Disease Is a Result of Defects in α -Oxidation

The α -oxidation pathway is defective in **Refsum's disease**, an inherited metabolic disorder that results in defective night vision, tremors, and other neurologic abnormalities. These symptoms are caused by accumulation of phytanic acid in the body. Treatment of Refsum's disease requires a diet free of chloro-

¹Beta oxidation does not occur significantly in plant mitochondria.

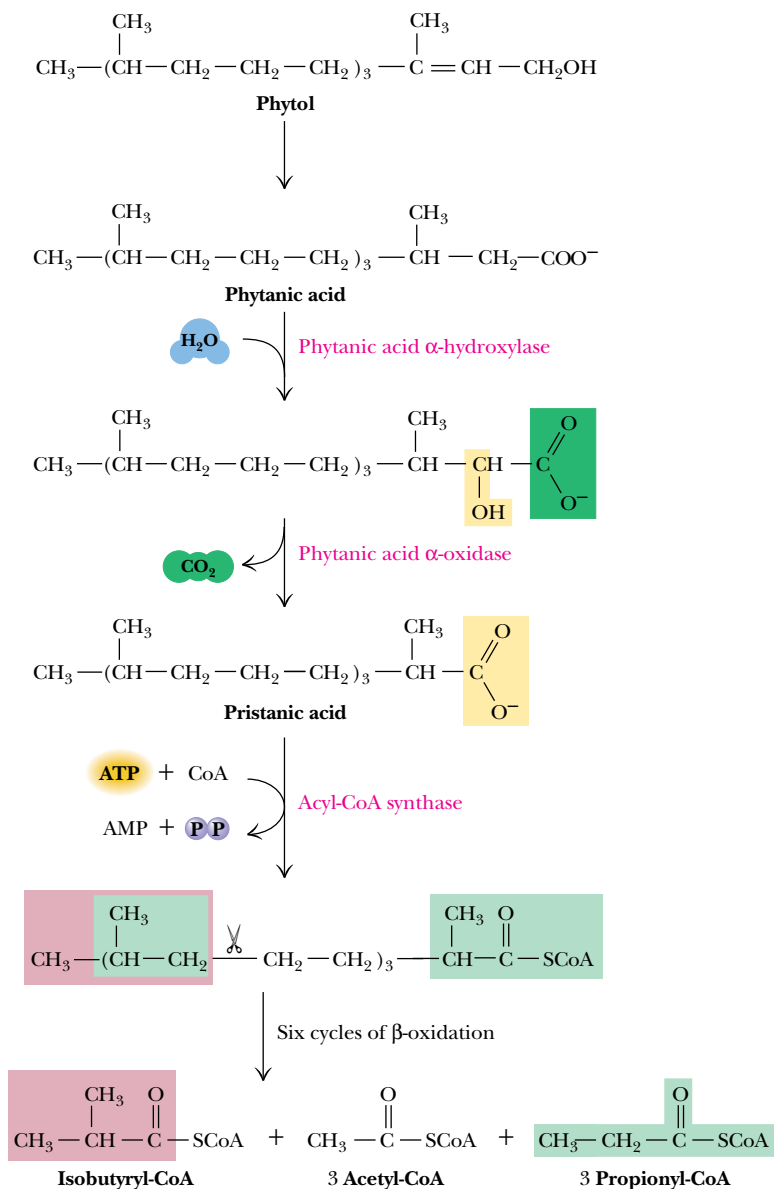


FIGURE 24.26 • Branched-chain fatty acids are oxidized by α -oxidation, as shown for phytanic acid. The product of the phytanic acid oxidase, pristanic acid, is a suitable substrate for normal β -oxidation. Isobutyryl-CoA and propionyl-CoA can both be converted to succinyl-CoA, which can enter the TCA cycle.

phyll, the precursor of phytanic acid. This regimen is difficult to implement because all green vegetables and even meat from plant-eating animals, such as cows, pigs, and poultry, must be excluded from the diet.

ω -Oxidation of Fatty Acids Yields Small Amounts of Dicarboxylic Acids

In the endoplasmic reticulum of eukaryotic cells, the oxidation of the terminal carbon of a normal fatty acid—a process termed ω -oxidation—can lead to the synthesis of small amounts of dicarboxylic acids (Figure 24.27). **Cytochrome P-450**, a monooxygenase enzyme that requires NADPH as a coenzyme and uses O_2 as a substrate, places a hydroxyl group at the terminal carbon. Subsequent oxidation to a carboxyl group produces a dicarboxylic acid. Either end can form an ester linkage to CoA and be subjected to β -oxidation, producing a

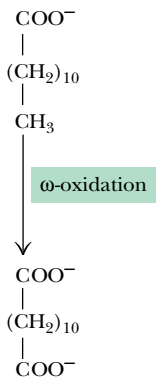


FIGURE 24.27 • Dicarboxylic acids can be formed by oxidation of the methyl group of fatty acids in a cytochrome P-450–dependent reaction.

variety of smaller dicarboxylic acids. (Cytochrome P-450–dependent monooxygenases also play an important role as agents of **detoxication**, the degradation and metabolism of toxic hydrocarbon agents.)

24.6 • Ketone Bodies

Ketone Bodies Are a Significant Source of Fuel and Energy for Certain Tissues

Most of the acetyl-CoA produced by the oxidation of fatty acids in liver mitochondria undergoes further oxidation in the TCA cycle, as stated earlier. However, some of this acetyl-CoA is converted to three important metabolites: acetone, acetoacetate, and β -hydroxybutyrate. The process is known as **keto-genesis**, and these three metabolites are traditionally known as **ketone bodies**, in spite of the fact that β -hydroxybutyrate does not contain a ketone function. These three metabolites are synthesized primarily in the liver but are important sources of fuel and energy for many peripheral tissues, including brain, heart, and skeletal muscle. The brain, for example, normally uses glucose as its source of metabolic energy. However, during periods of starvation, ketone bodies may be the major energy source for the brain. Acetoacetate and β -hydroxybutyrate are the preferred and normal substrates for kidney cortex and for heart muscle.

Ketone body synthesis occurs only in the mitochondrial matrix. The reactions responsible for the formation of ketone bodies are shown in Figure 24.28. The first reaction—the condensation of two molecules of acetyl-CoA to form acetoacetyl-CoA—is catalyzed by **thiolase**, which is also known as **acetoacetyl-CoA thiolase** or **acetyl-CoA acetyltransferase**. This is the same enzyme that carries out the thiolase reaction in β -oxidation, but here it runs in reverse. The second reaction adds another molecule of acetyl-CoA to give β -hydroxy- β -methylglutaryl-CoA, commonly abbreviated HMG-CoA. These two mitochondrial matrix reactions are analogous to the first two steps in cholesterol biosynthesis, a cytosolic process, as we shall see in Chapter 25. HMG-CoA is converted to acetoacetate and acetyl-CoA by the action of **HMG-CoA lyase** in a mixed aldol-Claisen ester cleavage reaction. This reaction is mechanistically similar to the reverse of the citrate synthase reaction in the TCA cycle. A membrane-bound enzyme, **β -hydroxybutyrate dehydrogenase**, then can reduce acetoacetate to β -hydroxybutyrate.

Acetoacetate and β -hydroxybutyrate are transported through the blood from liver to target organs and tissues, where they are converted to acetyl-CoA (Figure 24.29). *Ketone bodies are easily transportable forms of fatty acids that move through the circulatory system without the need for complexation with serum albumin and other fatty acid-binding proteins.*

Ketone Bodies and Diabetes Mellitus

Diabetes mellitus is the most common endocrine disease and the third leading cause of death in the United States, with approximately 6 million diagnosed cases and an estimated 4 million more borderline but undiagnosed cases. Diabetes is characterized by an abnormally high level of glucose in the blood. In **type I diabetes** (representing 10% or fewer of all cases), elevated blood glucose results from inadequate secretion of insulin by the islets of Langerhans in the pancreas. **Type II diabetes** (at least 90% of all cases) results from an insensitivity to insulin. Type II diabetics produce normal or even elevated levels of insulin, but owing to a shortage of insulin receptors (Chapter 34), their cells

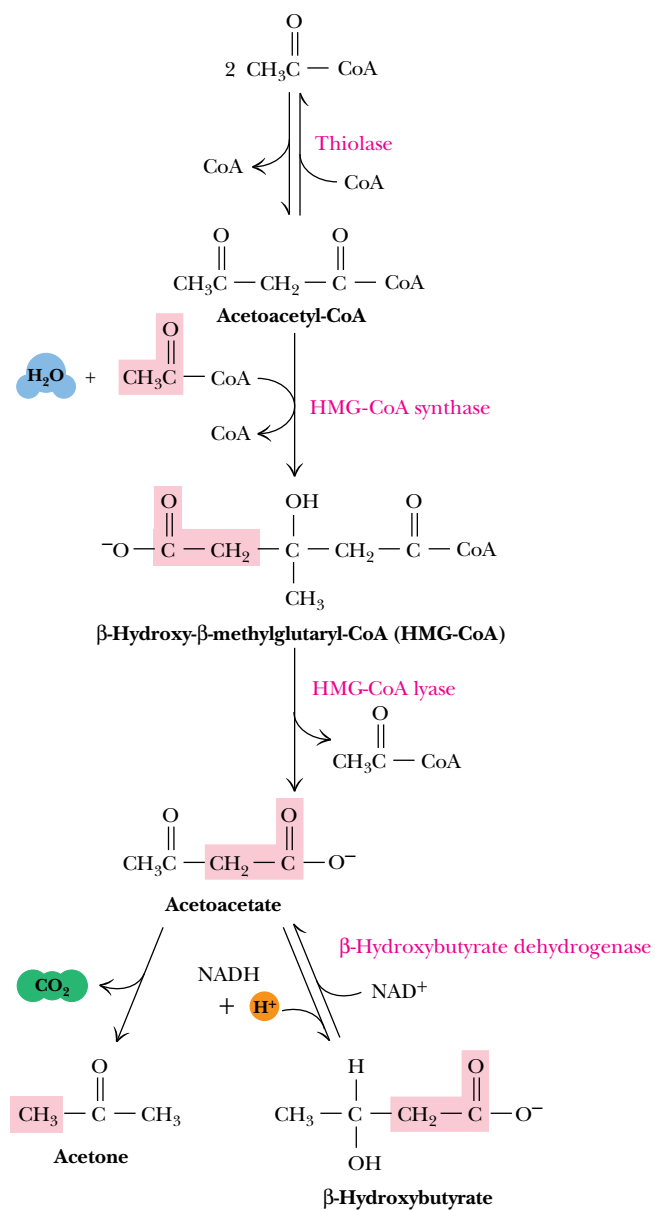


FIGURE 24.28 • The formation of ketone bodies, synthesized primarily in liver mitochondria.

are not responsive to insulin. In both cases, transport of glucose into muscle, liver, and adipose tissue is significantly reduced, and, despite abundant glucose in the blood, the cells are metabolically starved. They respond by turning to increased gluconeogenesis and catabolism of fat and protein. In type I diabetes, increased gluconeogenesis consumes most of the available oxaloacetate, but breakdown of fat (and, to a lesser extent, protein) produces large amounts of acetyl-CoA. This increased acetyl-CoA would normally be directed into the TCA cycle, but, with oxaloacetate in short supply, it is used instead for production of unusually large amounts of ketone bodies. Acetone can often be detected on the breath of type I diabetics, an indication of high plasma levels of ketone bodies.

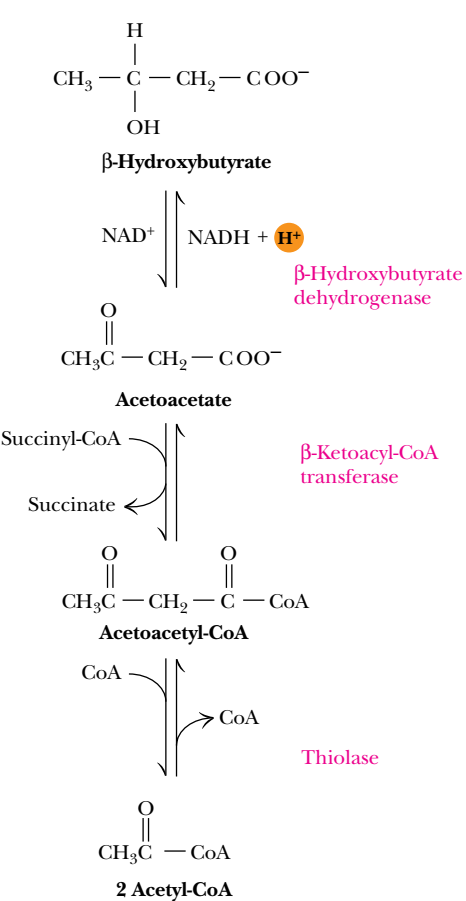


FIGURE 24.29 • Reconversion of ketone bodies to acetyl-CoA in the mitochondria of many tissues (other than liver) provides significant metabolic energy.

PROBLEMS

1. Calculate the volume of metabolic water available to a camel through fatty acid oxidation if it carries 30 lb of triacylglycerol in its hump.

2. Calculate the approximate number of ATP molecules that can be obtained from the oxidation of *cis*-11-heptadecenoic acid to CO₂ and water.

3. Phytanic acid, the product of chlorophyll that causes problems for individuals with Refsum's disease, is 3,7,11,15-tetramethyl hexadecanoic acid. Suggest a route for its oxidation that is consistent with what you have learned in this chapter. (*Hint*: The methyl group at C-3 effectively blocks hydroxylation and normal β -oxidation. You may wish to initiate breakdown in some other way.)

4. Even though acetate units, such as those obtained from fatty acid oxidation, cannot be used for *net* synthesis of carbohydrate in animals, labeled carbon from ¹⁴C-labeled acetate *can* be found in newly synthesized glucose (for example, in liver glycogen) in animal tracer studies. Explain how this can be. Which carbons of glucose would you expect to be the first to be labeled by ¹⁴C-labeled acetate?

5. What would you expect to be the systemic metabolic effects of consuming unripened akee fruit?

6. Overweight individuals who diet to lose weight often view fat in negative ways because adipose tissue is the repository of excess caloric intake. However, the "weighty" consequences might be even worse if excess calories were stored in other forms. Consider a person who is 10 lb "overweight," and estimate how much more he or she would weigh if excess energy were stored in the form of carbohydrate instead of fat.

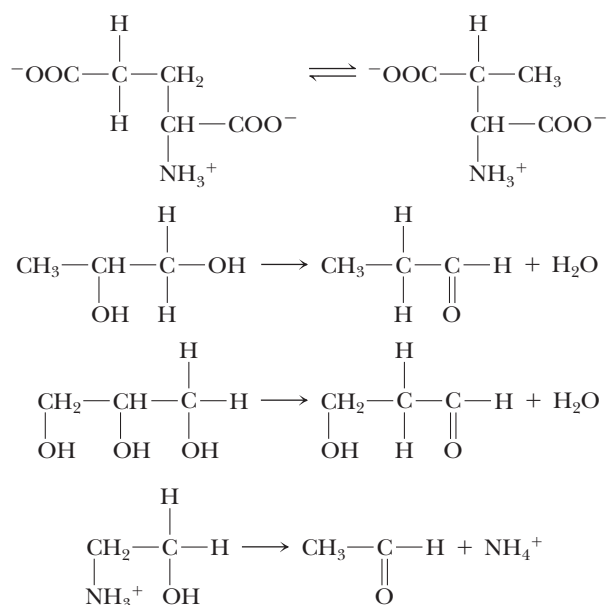
7. What would be the consequences of a deficiency in vitamin B₁₂ for fatty acid oxidation? What metabolic intermediates might accumulate?

8. Write properly balanced chemical equations for the oxidation to CO₂ and water of (a) myristic acid, (b) stearic acid, (c) α -linolenic acid, and (d) arachidonic acid.

9. How many tritium atoms are incorporated into acetate if a molecule of palmitic acid is oxidized in 100% tritiated water?

10. What would be the consequences of a carnitine deficiency for fatty acid oxidation?

11. Based on the mechanism for the methylmalonyl-CoA mutase (Figure 24.21), write reasonable mechanisms for the reactions shown below.



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To every thing there is a season, and a time for every purpose under heaven . . . A time to break down, a time to build up.
ECCLESIASTES 3:1–3

OUTLINE

- 25.1 • The Fatty Acid Biosynthesis and Degradation Pathways Are Different
- 25.2 • Biosynthesis of Complex Lipids
- 25.3 • Eicosanoid Biosynthesis and Function
- 25.4 • Cholesterol Biosynthesis
- 25.5 • Transport of Many Lipids Occurs via Lipoprotein Complexes
- 25.6 • Biosynthesis of Bile Acids
- 25.7 • Synthesis and Metabolism of Steroid Hormones

Chapter 25

Lipid Biosynthesis



Southern elephant seal, *Mirounga leonina*. (Gerald Lacz/Peter Arnold, Inc.)

We turn now to the biosynthesis of lipid structures. We begin with a discussion of the biosynthesis of fatty acids, stressing the basic pathways, additional means of elongation, mechanisms for the introduction of double bonds, and regulation of fatty acid synthesis. Sections then follow on the biosynthesis of glycerophospholipids, sphingolipids, eicosanoids, and cholesterol. The transport of lipids through the body in lipoprotein complexes is described, and the chapter closes with discussions of the biosynthesis of bile salts and steroid hormones.

25.1 • The Fatty Acid Biosynthesis and Degradation Pathways Are Different

We have already seen several cases in which the *synthesis* of a class of biomolecules is conducted differently from degradation (glycolysis versus gluconeogenesis and glycogen or starch breakdown versus polysaccharide synthesis, for

example). Likewise, the synthesis of fatty acids and other lipid components is different from their degradation. Fatty acid synthesis involves a set of reactions that follow a strategy different in several ways from the corresponding degradative process:

1. Intermediates in fatty acid synthesis are linked covalently to the sulfhydryl groups of special proteins, the **acyl carrier proteins**. In contrast, fatty acid breakdown intermediates are bound to the —SH group of coenzyme A.
2. Fatty acid synthesis occurs in the cytosol, whereas fatty acid degradation takes place in mitochondria.
3. In animals, the enzymes of fatty acid synthesis are components of one long polypeptide chain, the **fatty acid synthase**, whereas no similar association exists for the degradative enzymes. (Plants and bacteria employ separate enzymes to carry out the biosynthetic reactions.)
4. The coenzyme for the oxidation–reduction reactions of fatty acid synthesis is $\text{NADP}^+/\text{NADPH}$, whereas degradation involves the NAD^+/NADH couple.

Formation of Malonyl-CoA Activates Acetate Units for Fatty Acid Synthesis

The design strategy for fatty acid synthesis is this:

- a. Fatty acid chains are constructed by the addition of two-carbon units derived from *acetyl-CoA*.
- b. The acetate units are activated by formation of *malonyl-CoA* (at the expense of ATP).
- c. The addition of two-carbon units to the growing chain is driven by decarboxylation of malonyl-CoA.
- d. The elongation reactions are repeated until the growing chain reaches 16 carbons in length (palmitic acid).
- e. Other enzymes then add double bonds and additional carbon units to the chain.

Fatty Acid Biosynthesis Depends on the Reductive Power of NADPH

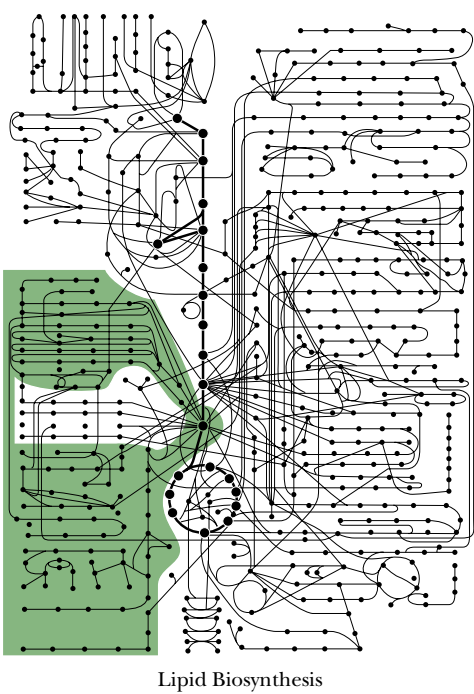
The net reaction for the formation of palmitate from acetyl-CoA is



(Levels of free fatty acids are very low in the typical cell. The palmitate made in this process is rapidly converted to CoA esters in preparation for the formation of triacylglycerols and phospholipids.)

Providing Cytosolic Acetyl-CoA and Reducing Power for Fatty Acid Synthesis

Eukaryotic cells face a dilemma in providing suitable amounts of substrate for fatty acid synthesis. Sufficient quantities of acetyl-CoA, malonyl-CoA, and NADPH must be generated *in the cytosol* for fatty acid synthesis. Malonyl-CoA is made by carboxylation of acetyl-CoA, so the problem reduces to generating sufficient acetyl-CoA and NADPH.



There are three principal sources of acetyl-CoA (Figure 25.1):

- 1. Amino acid degradation produces cytosolic acetyl-CoA.
- 2. Fatty acid oxidation produces mitochondrial acetyl-CoA.
- 3. Glycolysis yields cytosolic pyruvate, which (after transport into the mitochondria) is converted to acetyl-CoA by pyruvate dehydrogenase.

The acetyl-CoA derived from amino acid degradation is normally insufficient for fatty acid biosynthesis, and the acetyl-CoA produced by pyruvate dehydrogenase and by fatty acid oxidation cannot cross the mitochondrial membrane to participate directly in fatty acid synthesis. Instead, acetyl-CoA is linked with oxaloacetate to form citrate, which is transported from the mitochondrial matrix to the cytosol (Figure 25.1). Here it can be converted back into acetyl-CoA and oxaloacetate by **ATP-citrate lyase**. In this manner, mitochondrial acetyl-CoA becomes the substrate for cytosolic fatty acid synthesis. (Oxaloacetate returns to the mitochondria in the form of either pyruvate or malate, which is then reconverted to acetyl-CoA and oxaloacetate, respectively.)

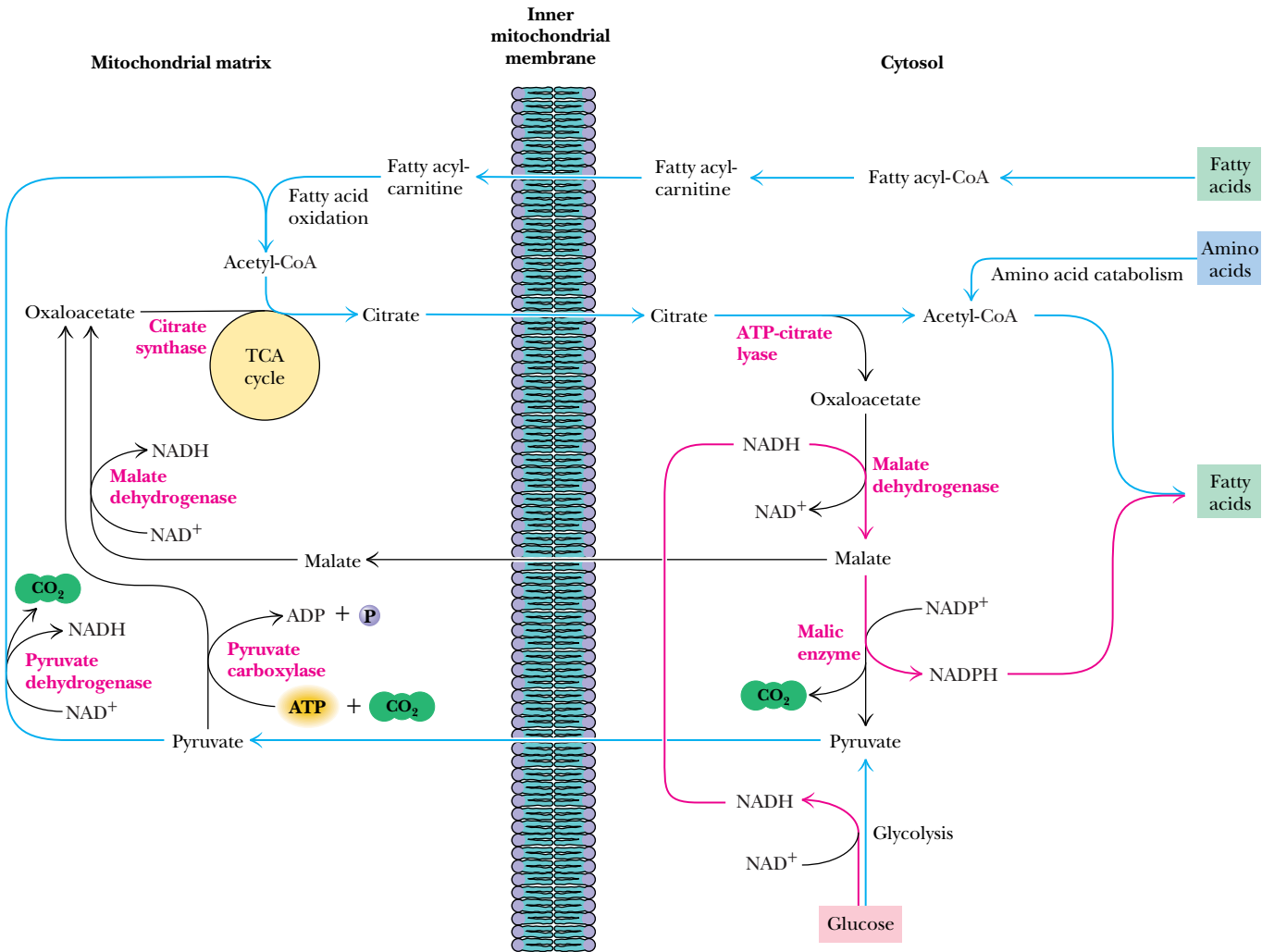
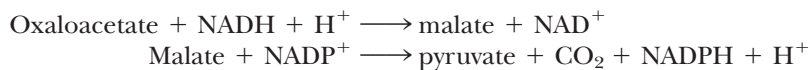


FIGURE 25.1 • The citrate-malate-pyruvate shuttle provides cytosolic acetate units and reducing equivalents (electrons) for fatty acid synthesis. The shuttle collects carbon substrates, primarily from glycolysis but also from fatty acid oxidation and amino acid catabolism. Most of the reducing equivalents are glycolytic in origin. Pathways that provide carbon for fatty acid synthesis are shown in blue; pathways that supply electrons for fatty acid synthesis are shown in red.

NADPH can be produced in the pentose phosphate pathway as well as by malic enzyme (Figure 25.1). Reducing equivalents (electrons) derived from glycolysis in the form of NADH can be transformed into NADPH by the combined action of malate dehydrogenase and malic enzyme:



How many of the 14 NADPH needed to form one palmitate (Eq. 25.1) can be made in this way? The answer depends on the status of malate. Every citrate entering the cytosol produces one acetyl-CoA and one malate (Figure 25.1). Every malate oxidized by malic enzyme produces one NADPH, at the expense of a decarboxylation to pyruvate. Thus, when malate is oxidized, one NADPH is produced for every acetyl-CoA. Conversion of 8 acetyl-CoA units to one palmitate would then be accompanied by production of 8 NADPH. (The other 6 NADPH required [Eq. 25.1] would be provided by the pentose phosphate pathway.) On the other hand, for every malate returned to the mitochondria, one NADPH fewer is produced.

Acetate Units Are Committed to Fatty Acid Synthesis by Formation of Malonyl-CoA

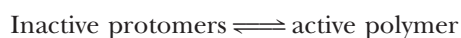
Rittenberg and Bloch showed in the late 1940s that acetate units are the building blocks of fatty acids. Their work, together with the discovery by Salih Wakil that bicarbonate is required for fatty acid biosynthesis, eventually made clear that this pathway involves synthesis of *malonyl-CoA*. The carboxylation of acetyl-CoA to form malonyl-CoA is essentially irreversible and is the **committed step** in the synthesis of fatty acids (Figure 25.2). The reaction is catalyzed by **acetyl-CoA carboxylase**, which contains a biotin prosthetic group. This carboxylase is the only enzyme of fatty acid synthesis in animals that is not part of the multi-enzyme complex called fatty acid synthase.

Acetyl-CoA Carboxylase Is Biotin-Dependent and Displays Ping-Pong Kinetics

The biotin prosthetic group of acetyl-CoA carboxylase is covalently linked to the ϵ -amino group of an active-site lysine in a manner similar to pyruvate carboxylase (Figure 23.3). The reaction mechanism is also analogous to that of pyruvate carboxylase (Figure 23.4): ATP-driven carboxylation of biotin is followed by transfer of the activated CO_2 to acetyl-CoA to form malonyl-CoA. The enzyme from *Escherichia coli* has three subunits: (1) a **biotin carboxyl carrier protein** (a dimer of 22.5-kD subunits); (2) **biotin carboxylase** (a dimer of 51-kD subunits), which adds CO_2 to the prosthetic group; and (3) **transcarboxylase** (an $\alpha_2\beta_2$ tetramer with 30-kD and 35-kD subunits), which transfers the activated CO_2 unit to acetyl-CoA. The long, flexible biotin–lysine chain (biocytin) enables the activated carboxyl group to be carried between the biotin carboxylase and the transcarboxylase (Figure 25.3).

Acetyl-CoA Carboxylase in Animals Is a Multifunctional Protein

In animals, acetyl-CoA carboxylase (ACC) is a filamentous polymer (4 to 8×10^6 D) composed of 230-kD protomers. Each of these subunits contains the biotin carboxyl carrier moiety, biotin carboxylase, and transcarboxylase activities, as well as allosteric regulatory sites. Animal ACC is thus a multifunctional protein. The polymeric form is active, but the 230-kD protomers are inactive. The activity of ACC is thus dependent upon the position of the equilibrium between these two forms:



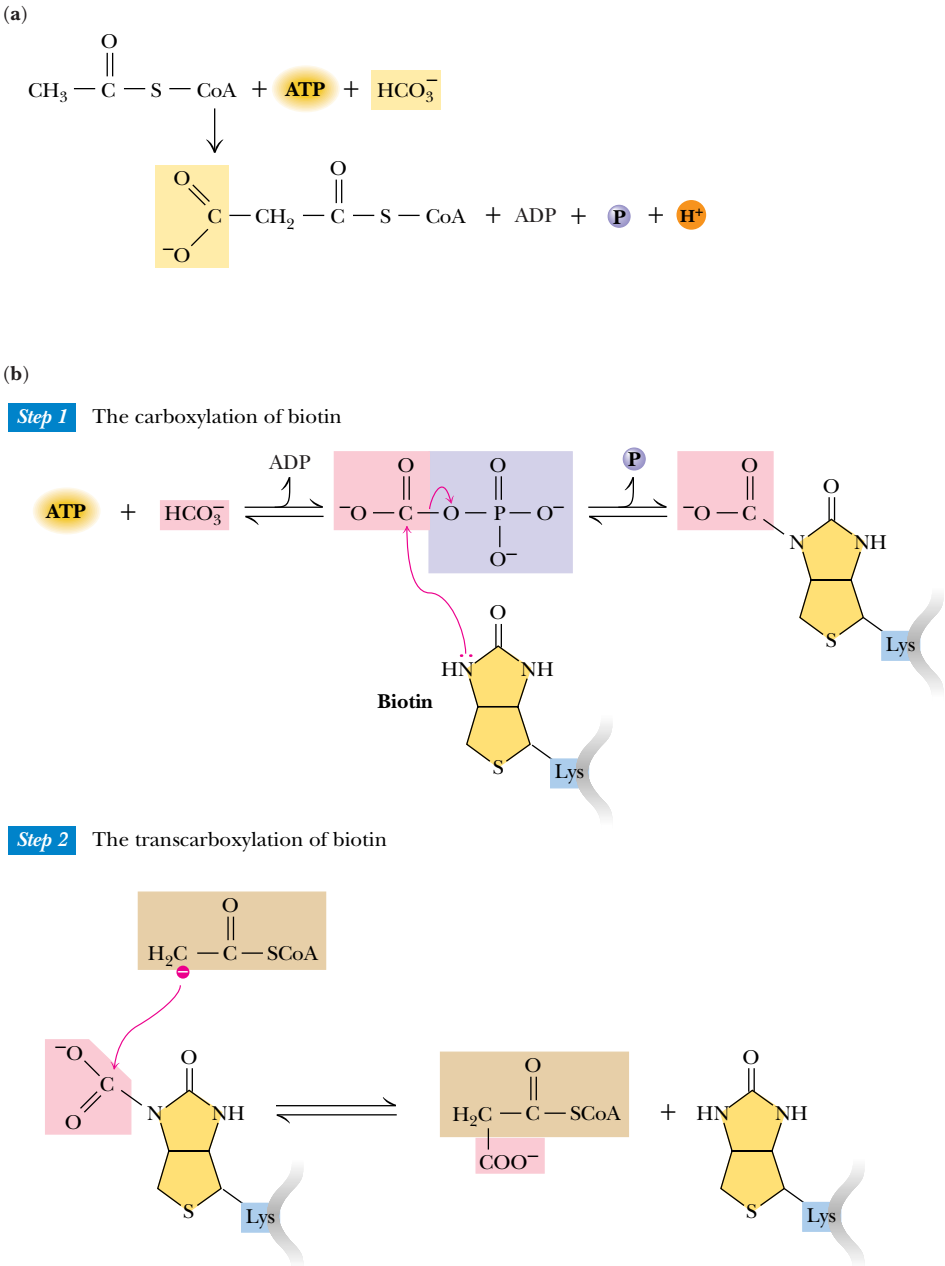


FIGURE 25.2 • (a) The acetyl-CoA carboxylase reaction produces malonyl-CoA for fatty acid synthesis. (b) A mechanism for the acetyl-CoA carboxylase reaction. Bicarbonate is activated for carboxylation reactions by formation of N-carboxybiotin. ATP drives the reaction forward, with transient formation of a carbonylphosphate intermediate (Step 1). In a typical biotin-dependent reaction, nucleophilic attack by the acetyl-CoA carbanion on the carboxyl carbon of N-carboxybiotin—a transcarboxylation—yields the carboxylated product (Step 2).



Because this enzyme catalyzes the committed step in fatty acid biosynthesis, it is carefully regulated. *Palmitoyl-CoA*, the final product of fatty acid biosynthesis, shifts the equilibrium toward the inactive protomers, whereas *citrate*, an important allosteric activator of this enzyme, shifts the equilibrium toward the active polymeric form of the enzyme. Acetyl-CoA carboxylase shows the kinetic behavior of a Monod–Wyman–Changeux V-system allosteric enzyme (Chapter 15).

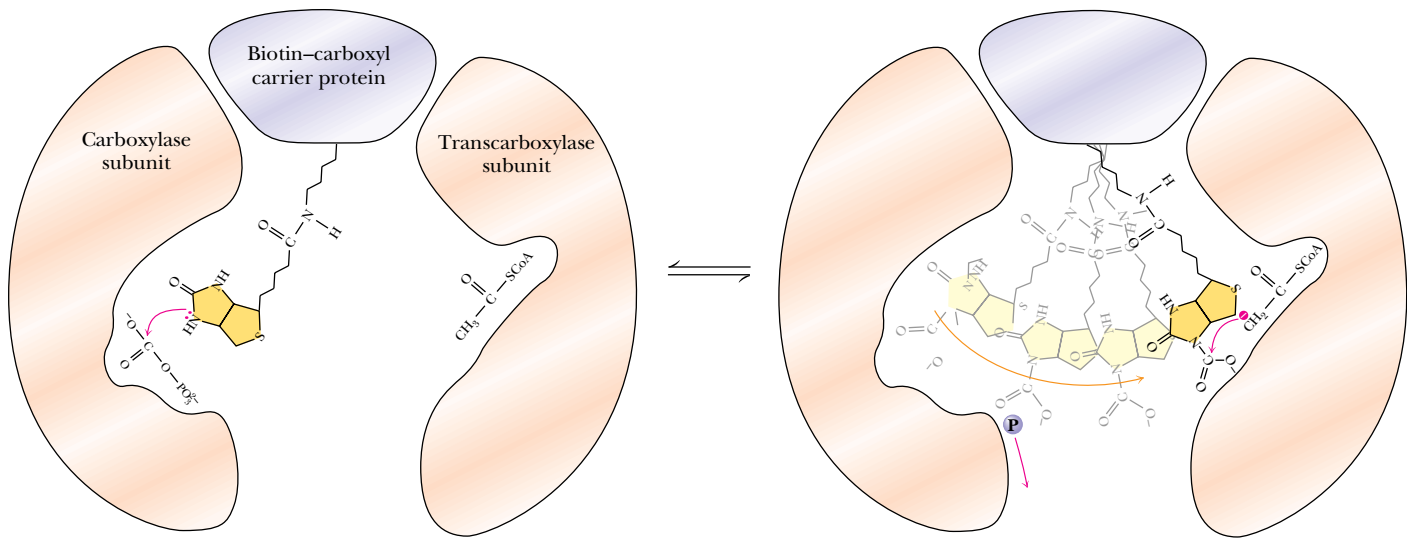


FIGURE 25.3 • In the acetyl-CoA carboxylase reaction, the biotin ring, on its flexible tether, acquires carboxyl groups from carbonylphosphate on the carboxylase subunit and transfers them to acyl-CoA molecules on the transcarboxylase subunits.

Phosphorylation of ACC Modulates Activation by Citrate and Inhibition by Palmitoyl-CoA

The regulatory effects of citrate and palmitoyl-CoA are dependent on the phosphorylation state of acetyl-CoA carboxylase. The animal enzyme is phosphorylated at 8 to 10 sites on each enzyme subunit (Figure 25.4). Some of these sites are reg-

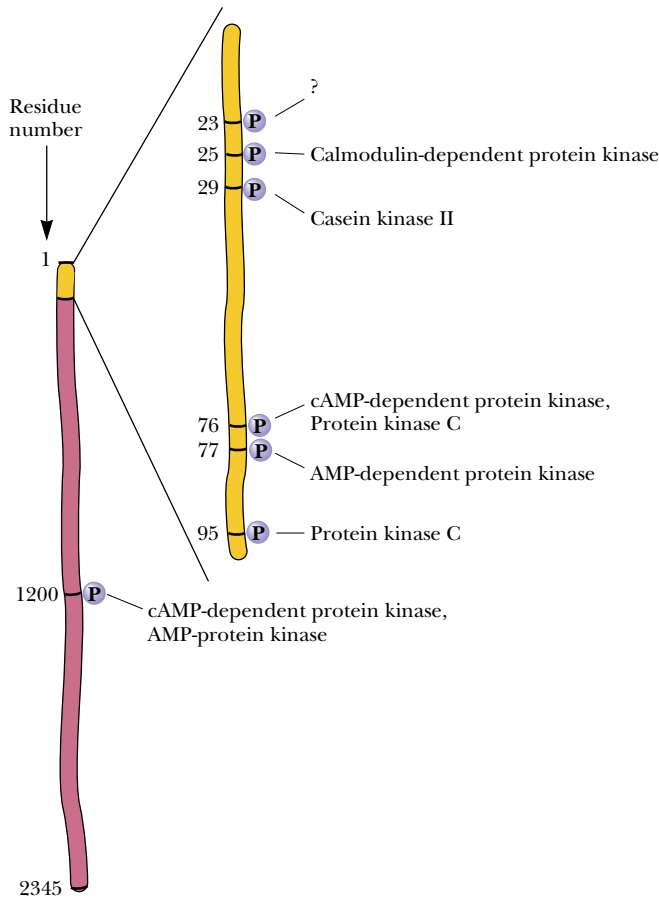


FIGURE 25.4 • Models of the acetyl-CoA carboxylase polypeptide, with phosphorylation sites indicated, along with the protein kinases responsible. Phosphorylation at Ser¹²⁰⁰ is primarily responsible for decreasing the affinity for citrate.

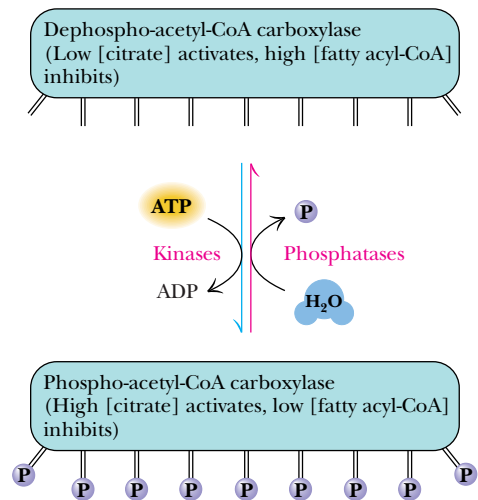


FIGURE 25.5 • The activity of acetyl-CoA carboxylase is modulated by phosphorylation and dephosphorylation. The dephospho form of the enzyme is activated by low [citrate] and inhibited only by high levels of fatty acyl-CoA. In contrast, the phosphorylated form of the enzyme is activated only by high levels of citrate, but is very sensitive to inhibition by fatty acyl-CoA.

ulatory, whereas others are “silent” and have no effect on enzyme activity. Unphosphorylated acetyl-CoA carboxylase binds citrate with high affinity and thus is active at very low citrate concentrations (Figure 25.5). Phosphorylation of the regulatory sites decreases the affinity of the enzyme for citrate, and in this case high levels of citrate are required to activate the carboxylase. The inhibition by fatty acyl-CoAs operates in a similar but opposite manner. Thus, low levels of fatty acyl-CoA inhibit the phosphorylated carboxylase, but the dephosphoenzyme is inhibited only by high levels of fatty acyl-CoA. Specific phosphatases act to dephosphorylate ACC, thereby increasing the sensitivity to citrate.

Acyl Carrier Proteins Carry the Intermediates in Fatty Acid Synthesis

The basic building blocks of fatty acid synthesis are acetyl and malonyl groups, but they are not transferred directly from CoA to the growing fatty acid chain. Rather, they are first passed to **acyl carrier protein** (or simply ACP), discovered by P. Roy Vagelos. This protein consists (in *E. coli*) of a single polypeptide chain of 77 residues to which is attached (on a serine residue) a **phosphopantetheine group**, the same group that forms the “business end” of coenzyme A. Thus, acyl carrier protein is a somewhat larger version of coenzyme A, specialized for use in fatty acid biosynthesis (Figure 25.6).

The enzymes that catalyze formation of acetyl-ACP and malonyl-ACP and the subsequent reactions of fatty acid synthesis are organized quite differently in different organisms. We first discuss fatty acid biosynthesis in bacteria and plants, where the various reactions are catalyzed by separate, independent proteins. Then we discuss the animal version of fatty acid biosynthesis, which involves a single multienzyme complex called **fatty acid synthase**.

Fatty Acid Synthesis in Bacteria and Plants

The individual steps in the elongation of the fatty acid chain are quite similar in bacteria, fungi, plants, and animals. The ease of purification of the separate enzymes from bacteria and plants made it possible in the beginning to sort out each step in the pathway, and then by extension to see the pattern of biosynthesis in animals. The reactions are summarized in Figure 25.7. The elongation reactions begin with the formation of acetyl-ACP and malonyl-ACP, which

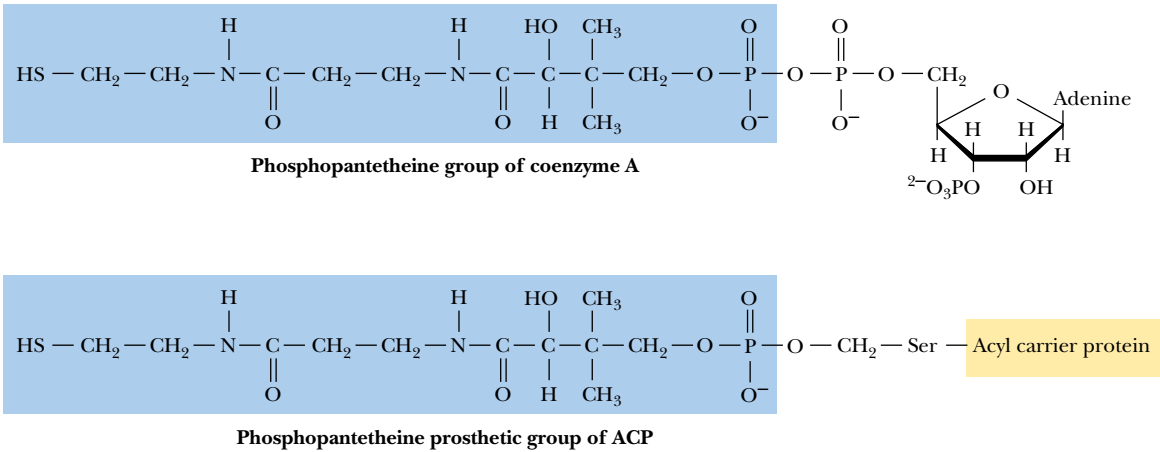


FIGURE 25.6 • Fatty acids are conjugated both to coenzyme A and to acyl carrier protein through the sulfhydryl of phosphopantetheine prosthetic groups.

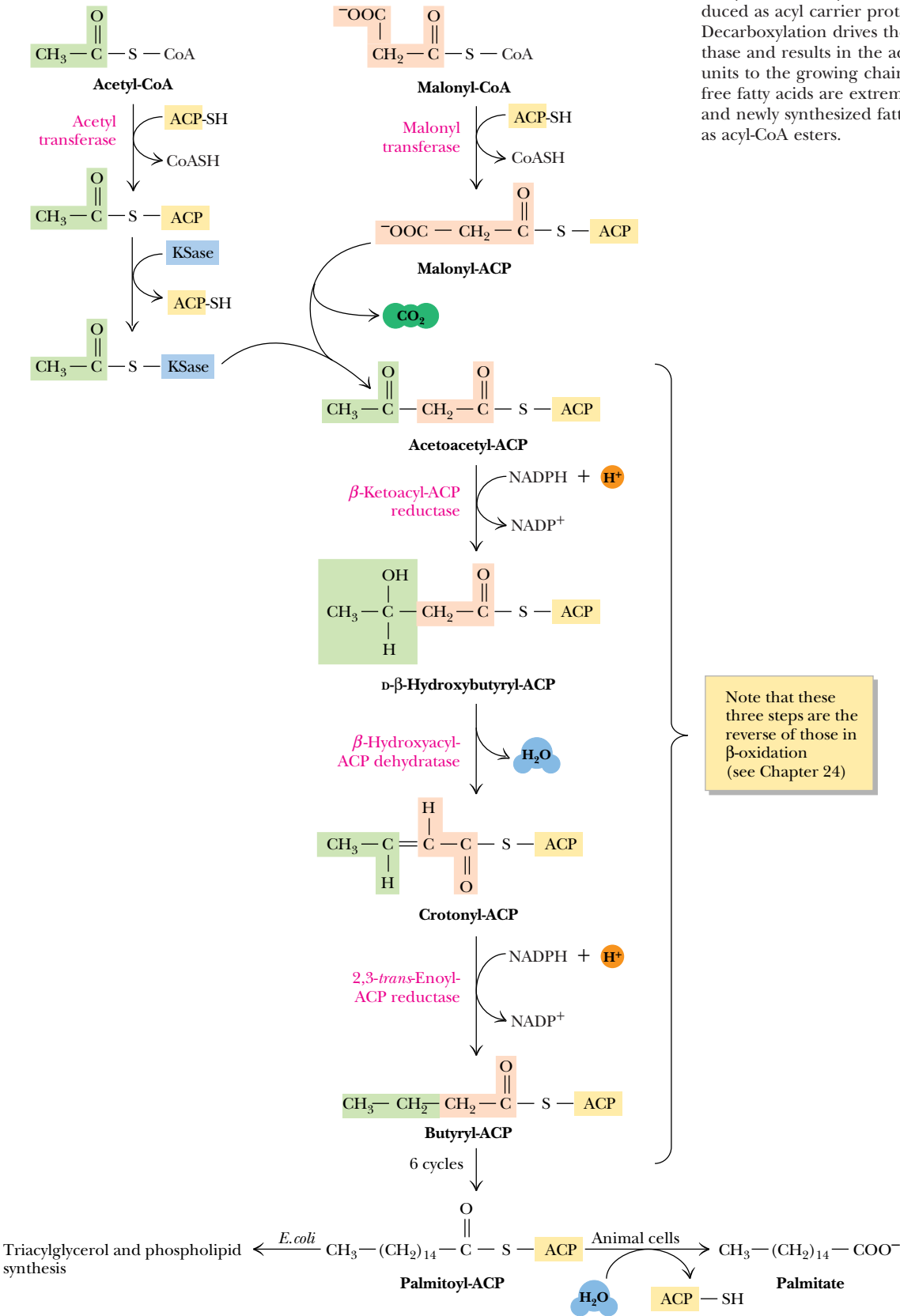


FIGURE 25.7 • The pathway of palmitate synthesis from acetyl-CoA and malonyl-CoA. Acetyl and malonyl building blocks are introduced as acyl carrier protein conjugates. Decarboxylation drives the β -ketoacyl-ACP synthase and results in the addition of two-carbon units to the growing chain. Concentrations of free fatty acids are extremely low in most cells, and newly synthesized fatty acids exist primarily as acyl-CoA esters.

A D E E P E R L O O K

Choosing the Best Organism for the Experiment

The selection of a suitable and relevant organism is an important part of any biochemical investigation. The studies that revealed the secrets of fatty acid synthesis are a good case in point.

The paradigm for fatty acid synthesis in plants has been the avocado, which has one of the highest fatty acid contents in the plant kingdom. Early animal studies centered primarily on

pigeons, which are easily bred and handled and which possess high levels of fats in their tissues. Other animals, richer in fatty tissues, might be even more attractive but more challenging to maintain. Grizzly bears, for example, carry very large fat reserves but are difficult to work with in the lab!

are formed by **acetyl transacylase (acetyl transferase)** and **malonyl transacylase (malonyl transferase)**, respectively. The acetyl transacylase enzyme is not highly specific—it can transfer other acyl groups, such as the propionyl group, but at much lower rates. (Fatty acids with odd numbers of carbons are made beginning with a propionyl group transfer by this enzyme.) Malonyl transacylase, on the other hand, is highly specific.

Decarboxylation Drives the Condensation of Acetyl-CoA and Malonyl-CoA

Another transacylase reaction transfers the acetyl group from ACP to **β -ketoacyl-ACP synthase (KSase)**, also known as **acyl-malonyl-ACP condensing enzyme**. The first actual elongation reaction involves the condensation of acetyl-ACP and malonyl-ACP by the β -ketoacyl-ACP synthase to form acetoacetyl-ACP (Figure 25.7). *One might ask at this point: Why is the three-carbon malonyl group used here as a two-carbon donor?* The answer is that this is yet another example of a decarboxylation driving a desired but otherwise thermodynamically unfavorable reaction. The decarboxylation that accompanies the reaction with malonyl-ACP drives the synthesis of acetoacetyl-ACP. Note that hydrolysis of ATP drove the carboxylation of acetyl-CoA to form malonyl-ACP, so, indirectly, ATP is responsible for the condensation reaction to form acetoacetyl-ACP. Malonyl-CoA can be viewed as a form of stored energy for driving fatty acid synthesis.

It is also worth noting that the carbon of the carboxyl group that was added to drive this reaction is the one removed by the condensing enzyme. Thus, all the carbons of acetoacetyl-ACP (*and* of the fatty acids to be made) are derived from acetate units of acetyl-CoA.

Reduction of the β -Carbonyl Group Follows a Now-Familiar Route

The next three steps—reduction of the β -carbonyl group to form a β -alcohol, followed by dehydration and reduction to saturate the chain (Figure 25.7)—look very similar to the fatty acid degradation pathway in reverse. However, there are two crucial differences between fatty acid biosynthesis and fatty acid oxidation (besides the fact that different enzymes are involved): First, the alcohol formed in the first step has the **D** configuration rather than the **L** form seen in catabolism, and, second, the reducing coenzyme is NADPH, although NAD^+ and FAD are the oxidants in the catabolic pathway.

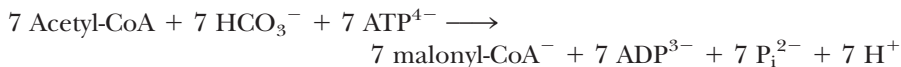
The net result of this biosynthetic cycle is the synthesis of a four-carbon unit, a butyryl group, from two smaller building blocks. In the next cycle of the process, this butyryl-ACP condenses with another malonyl-ACP to make a

six-carbon β -ketoacyl-ACP and CO_2 . Subsequent reduction to a β -alcohol, dehydration, and another reduction yield a six-carbon saturated acyl-ACP. This cycle continues with the net addition of a two-carbon unit in each turn until the chain is 16 carbons long (Figure 25.7). The β -ketoacyl-ACP synthase cannot accommodate larger substrates, so the reaction cycle ends with a 16-carbon chain. Hydrolysis of the C_{16} -acyl-ACP yields a palmitic acid and the free ACP.

In the end, seven malonyl-CoA molecules and one acetyl-CoA yield a palmitate (shown here as palmitoyl-CoA):



The formation of seven malonyl-CoA molecules requires



Thus, the overall reaction of acetyl-CoA to yield palmitic acid is



Note: These equations are stoichiometric and are charge balanced. See Problem 1 at the end of the chapter for practice in balancing these equations.

Fatty Acid Synthesis in Eukaryotes Occurs on a Multienzyme Complex

In contrast to bacterial and plant systems, the reactions of fatty acid synthesis beyond the acetyl-CoA carboxylase in animal systems are carried out by a special multienzyme complex called **fatty acid synthase (FAS)**. In yeast, this 2.4×10^6 D complex contains two different peptide chains, an α subunit of 213 kD and a β subunit of 203 kD, arranged in an $\alpha_6\beta_6$ dodecamer. The separate enzyme activities associated with each chain are shown in Figure 25.8. In animal systems, FAS is a dimer of identical 250-kD *multifunctional polypeptides*. Studies of the action of proteolytic enzymes on this polypeptide have led to a model involving three separate domains joined by flexible connecting sequences (Figure 25.9). The first domain is responsible for the binding of acetyl and malonyl building blocks and for the condensation of these units. This domain includes the acetyl transferase, the malonyl transferase, and the acyl-malonyl-ACP condensing enzyme (the β -ketoacyl synthase). The second domain is primarily responsible for the reduction of the intermediate synthesized in domain 1, and contains the acyl carrier protein, the β -ketoacyl reductase, the dehydratase, and the enoyl-ACP reductase. The third domain contains the thioesterase that liberates the product palmitate when the growing acyl chain reaches its limit length of 16 carbons. The close association of activities in this complex permits efficient exposure of intermediates to one active site and then the next. The presence of all these activities on a single polypeptide ensures that the cell will simultaneously synthesize all the enzymes needed for fatty acid synthesis.

The Mechanism of Fatty Acid Synthase

The first domain of one subunit of the fatty acid synthase interacts with the second and third domains of the other subunit; that is, the subunits are arranged in a head-to-tail fashion (Figure 25.9). The first step in the fatty acid synthase reaction is the formation of an acetyl-O-enzyme intermediate between the acetyl group of an acetyl-CoA and an active-site serine of the acetyl trans-

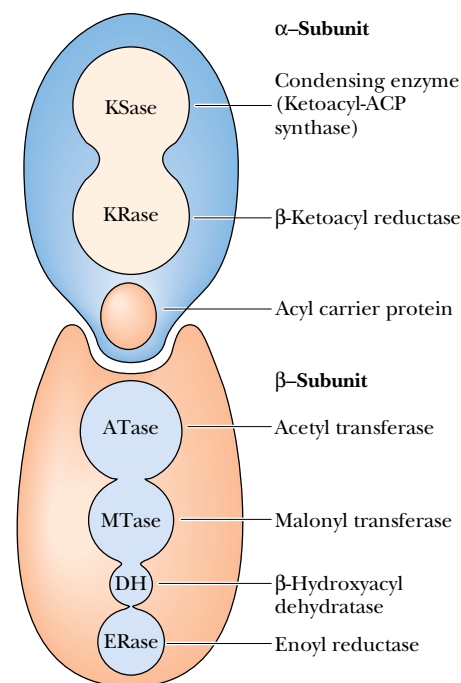
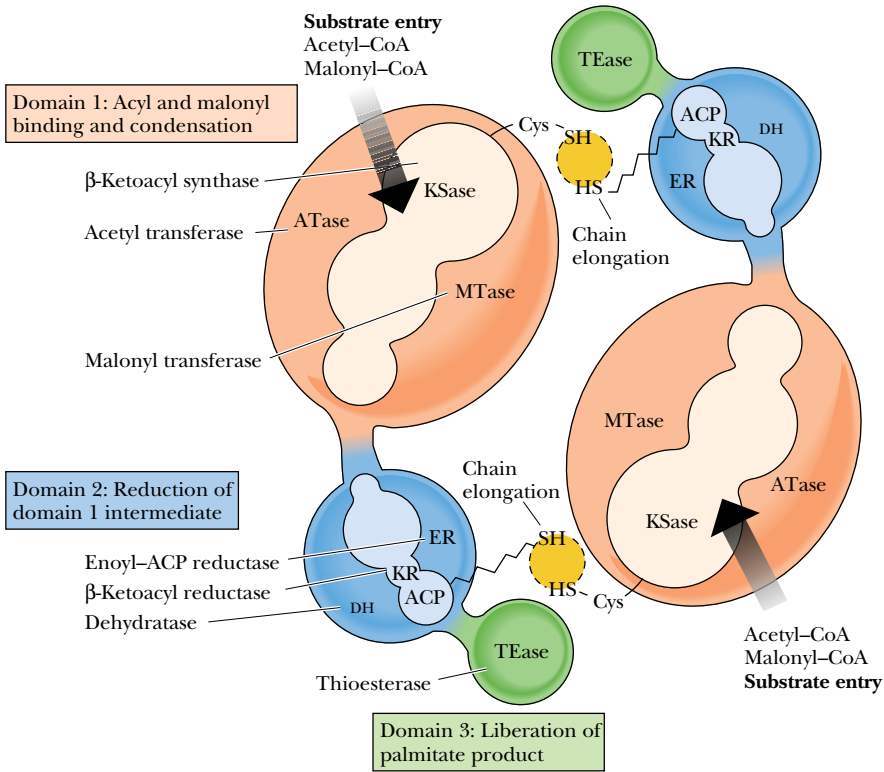


FIGURE 25.8 • In yeast, the functional groups and enzyme activities required for fatty acid synthesis are distributed between α and β subunits.

FIGURE 25.9 • Fatty acid synthase in animals contains all the functional groups and enzyme activities on a single multifunctional subunit. The active enzyme is a head-to-tail dimer of identical subunits. (Adapted from Wakil, S. J., Stoops, J. K., and Joshi, V. C., 1983. Annual Review of Biochemistry 52:556.)



ferase (Figure 25.10). In a similar manner, a malonyl-O-enzyme intermediate is formed between malonyl-CoA and a serine residue of the malonyl transferase. The acetyl group on the acetyl transferase is then transferred to the —SH group of the acyl carrier protein, as shown in Figure 25.11. The next step is the transfer of the acetyl group to the β -ketoacyl-ACP synthase, or condensing enzyme. This frees the acyl carrier protein to acquire the malonyl group from the malonyl transferase. The next step is the condensation reaction, in which decarboxylation facilitates the concerted attack of the remaining two-carbon unit of the acyl carrier protein at the carbonyl carbon of the acetate group on the condensing enzyme. Note that decarboxylation forms a transient, highly nucleophilic carbanion which can attack the acetate group.

The next three steps—reduction of the carbonyl to an alcohol, dehydration to yield a *trans*- α,β double bond, and reduction to yield a saturated chain—are identical to those occurring in bacteria and plants (Figure 25.7) and

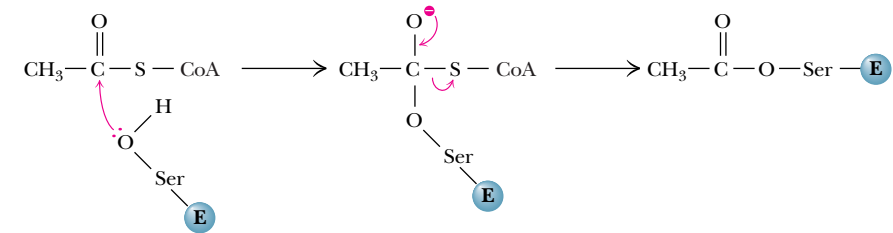


FIGURE 25.10 • Acetyl units are covalently linked to a serine residue at the active site of the acetyl transferase in eukaryotes. A similar reaction links malonyl units to the malonyl transferase.

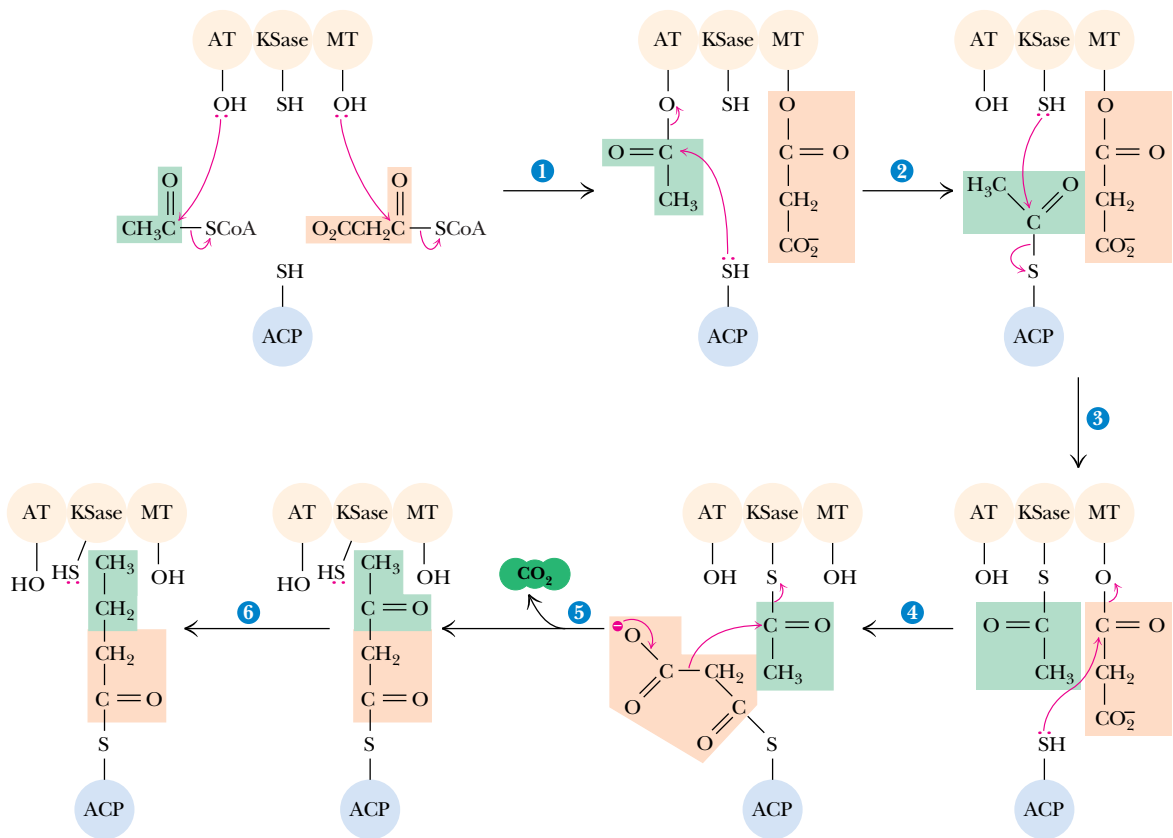


FIGURE 25.11 • The mechanism of the fatty acyl synthase reaction in eukaryotes. (1) Acetyl and malonyl groups are loaded onto acetyl transferase and malonyl transferase, respectively. (2) The acetate unit that forms the base of the nascent chain is transferred first to the acyl carrier protein domain and (3) then to the β-ketoacyl synthase. (4) Attack by ACP on the carbonyl carbon of a malonyl unit on malonyl transferase forms malonyl-ACP. (5) Decarboxylation leaves a reactive, transient carbanion that can attack the carbonyl carbon of the acetyl group on the β-ketoacyl synthase. (6) Reduction of the keto group, dehydration, and saturation of the resulting double bond follow, leaving an acyl group on ACP, and steps 3 through 6 repeat to lengthen the nascent chain.

resemble the reverse of the reactions of fatty acid oxidation (and the conversion of succinate to oxaloacetate in the TCA cycle). This synthetic cycle now repeats until the growing chain is 16 carbons long. It is then released by the thioesterase domain on the synthase. The amino acid sequence of the thioesterase domain is homologous with serine proteases; the enzyme has an active-site serine that carries out nucleophilic attack on the carbonyl carbon of the fatty acyl thioester to be cleaved.

Further Processing of C₁₆ Fatty Acids

Additional Elongation

As seen already, palmitate is the primary product of the fatty acid synthase. Cells synthesize many other fatty acids. Shorter chains are easily made if the chain is released before reaching 16 carbons in length. Longer chains are made through special elongation reactions, which occur both in the mitochondria and at the surface of the endoplasmic reticulum. The ER reactions are actually quite similar to those we have just discussed: addition of two-carbon units

at the carboxyl end of the chain by means of oxidative decarboxylations involving malonyl-CoA. As was the case for the fatty acid synthase, this decarboxylation provides the thermodynamic driving force for the condensation reaction. The mitochondrial reactions involve addition (and subsequent reduction) of acetyl units. These reactions (Figure 25.12) are essentially a reversal of fatty acid oxidation, with the exception that NADPH is utilized in the saturation of the double bond, instead of FADH₂.

Introduction of a Single *cis* Double Bond

Both prokaryotes and eukaryotes are capable of introducing a single *cis* double bond in a newly synthesized fatty acid. Bacteria such as *E. coli* carry out this process in an O₂-independent pathway, whereas eukaryotes have adopted an O₂-dependent pathway. There is a fundamental chemical difference between the two. The O₂-dependent reaction can occur anywhere in the fatty acid chain,

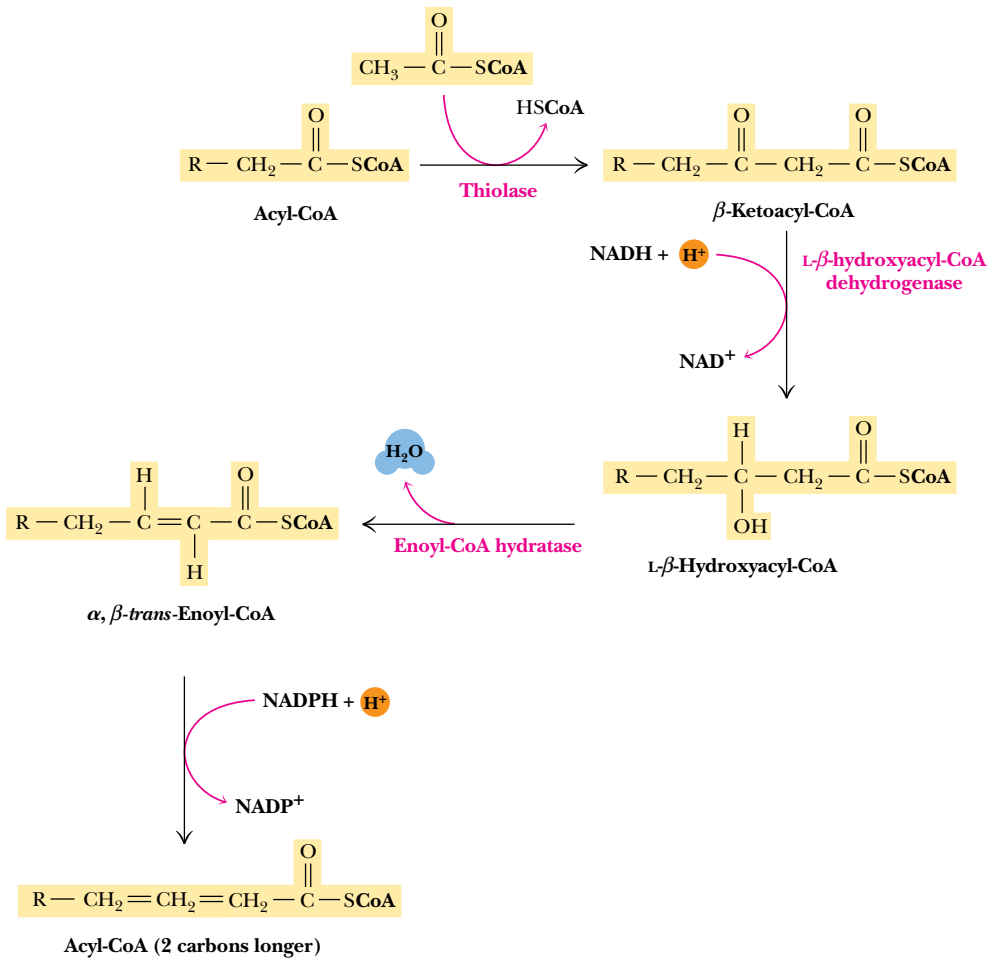


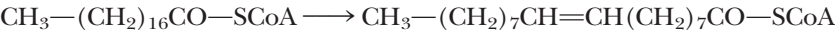
FIGURE 25.12 • Elongation of fatty acids in mitochondria is initiated by the thiolase reaction. The β-ketoacyl intermediate thus formed undergoes the same three reactions (in reverse order) that are the basis of β-oxidation of fatty acids. Reduction of the β-keto group is followed by dehydration to form a double bond. Reduction of the double bond yields a fatty acyl-CoA that is elongated by two carbons. Note that the reducing coenzyme for the second step is NADH, whereas the reductant for the fourth step is NADPH.

with no (additional) need to activate the desired bond toward dehydrogenation. However, in the absence of O₂, some other means must be found to activate the bond in question. Thus, in the bacterial reaction, dehydrogenation occurs while the bond of interest is still near the β-carbonyl or β-hydroxy group and the thioester group at the end of the chain.

In *E. coli*, the biosynthesis of a monounsaturated fatty acid begins with four normal cycles of elongation to form a 10-carbon intermediate, β-hydroxydecanoyl-ACP (Figure 25.13). At this point, **β-hydroxydecanoyl thioester dehydrase** forms a double bond β,γ to the thioester and in the *cis* configuration. This is followed by three rounds of the normal elongation reactions to form *palmitoleoyl*-ACP. Elongation may terminate at this point or may be followed by additional biosynthetic events. The principal unsaturated fatty acid in *E. coli*, *cis-vaccenic acid*, is formed by an additional elongation step, using *palmitoleoyl*-ACP as a substrate.

Unsaturations Reactions Occur in Eukaryotes in the Middle of an Aliphatic Chain

The addition of double bonds to fatty acids in eukaryotes does not occur until the fatty acyl chain has reached its full length (usually 16 to 18 carbons). Dehydrogenation of stearoyl-CoA occurs in the middle of the chain despite the absence of any useful functional group on the chain to facilitate activation:



This impressive reaction is catalyzed by **stearoyl-CoA desaturase**, a 53-kD enzyme containing a nonheme iron center. NADH and oxygen (O₂) are required, as are two other proteins: **cytochrome b₅ reductase** (a 43-kD flavoprotein) and **cytochrome b₅** (16.7 kD). All three proteins are associated with the endoplasmic reticulum membrane. Cytochrome b₅ reductase transfers a pair of electrons from NADH through FAD to cytochrome b₅ (Figure 25.14). Oxidation of reduced cytochrome b₅ is coupled to reduction of nonheme Fe³⁺ to Fe²⁺ in the desaturase. The Fe³⁺ accepts a pair of electrons (one at a time in a cycle) from cytochrome b₅ and creates a *cis* double bond at the 9,10-position of the stearoyl-CoA substrate. O₂ is the terminal electron acceptor in this fatty acyl desaturation cycle. Note that two water molecules are made, which means that four electrons are transferred overall. Two of these come through the reaction sequence from NADH, and two come from the fatty acyl substrate that is being dehydrogenated.

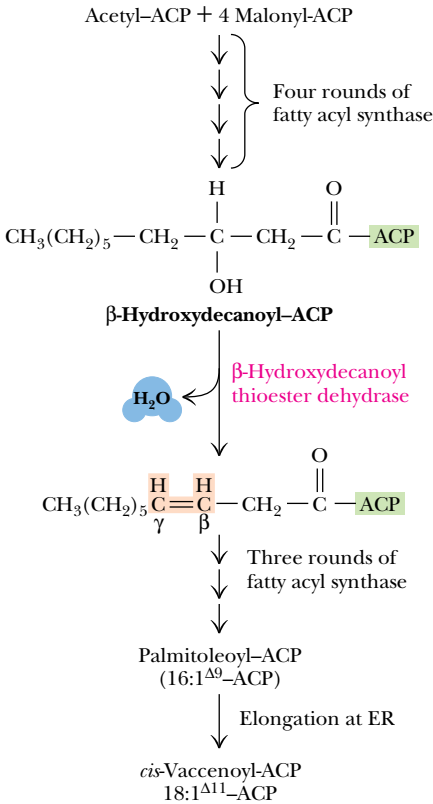


FIGURE 25.13 • Double bonds are introduced into the growing fatty acid chain in *E. coli* by specific dehydrases. Palmitoleoyl-ACP is synthesized by a sequence of reactions involving four rounds of chain elongation, followed by double bond insertion by β-hydroxydecanoyl thioester dehydrase and three additional elongation steps. Another elongation cycle produces *cis*-vaccenic acid.

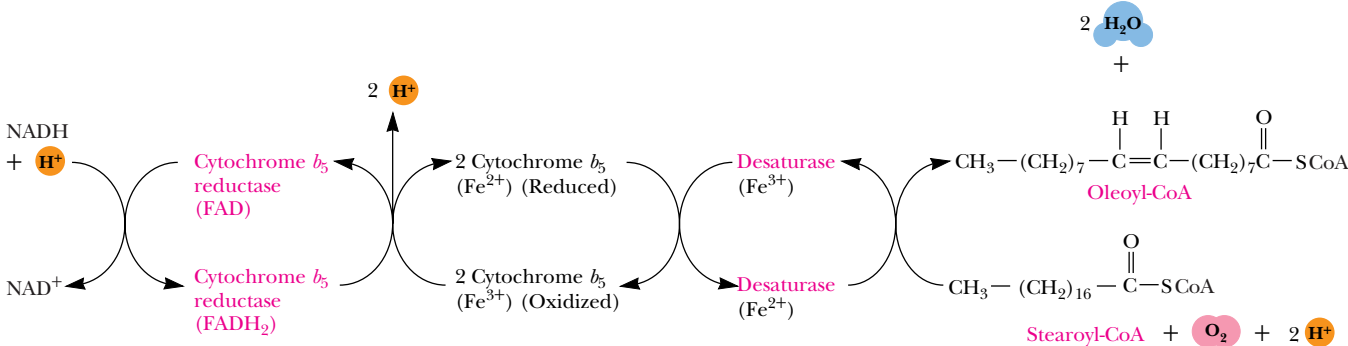


FIGURE 25.14 • The conversion of stearoyl-CoA to oleoyl-CoA in eukaryotes is catalyzed by stearoyl-CoA desaturase in a reaction sequence that also involves cytochrome b₅ and cytochrome b₅ reductase. Two electrons are passed from NADH through the chain of reactions as shown, and two electrons are also derived from the fatty acyl substrate.

The Unsaturation Reaction May Be Followed by Chain Elongation

Additional chain elongation can occur following this single desaturation reaction. The oleoyl-CoA produced can be elongated by two carbons to form a 20:1 *cis*- Δ^{11} fatty acyl-CoA. If the starting fatty acid is palmitate, reactions similar to the preceding scheme yield palmitoleoyl-CoA (16:1 *cis*- Δ^9), which subsequently can be elongated to yield *cis*-vaccenic acid (18:1 *cis*- Δ^{11}). Similarly, C₁₆ and C₁₈ fatty acids can be elongated to yield C₂₂ and C₂₄ fatty acids, such as are often found in sphingolipids.

Biosynthesis of Polyunsaturated Fatty Acids

Organisms differ with respect to formation, processing, and utilization of polyunsaturated fatty acids. *E. coli*, for example, does not have any polyunsaturated fatty acids. Eukaryotes *do* synthesize a variety of polyunsaturated fatty acids, certain organisms more than others. For example, plants manufacture double bonds between the Δ^9 and the methyl end of the chain, but mammals cannot. Plants readily desaturate oleic acid at the 12-position (to give linoleic acid) or at both the 12- and 15-positions (producing linolenic acid). Mammals require polyunsaturated fatty acids, but must acquire them in their diet. As such, they are referred to as **essential fatty acids**. On the other hand, mammals *can* introduce double bonds between the double bond at the 8- or 9-position and the carboxyl group. Enzyme complexes in the endoplasmic reticulum desaturate the 5-position, provided a double bond exists at the 8-position, and form a double bond at the 6-position if one already exists at the 9-position. Thus, oleate can be unsaturated at the 6,7-position to give an 18:2 *cis*- Δ^6, Δ^9 fatty acid.

Arachidonic Acid Is Synthesized from Linoleic Acid by Mammals

Mammals can add additional double bonds to unsaturated fatty acids in their diets. Their ability to make arachidonic acid from linoleic acid is one example (Figure 25.15). This fatty acid is the precursor for prostaglandins and other biologically active derivatives such as leukotrienes. Synthesis involves formation of a linoleoyl ester of CoA from dietary linoleic acid, followed by introduction of a double bond at the 6-position. The triply unsaturated product is then elongated (by malonyl-CoA with a decarboxylation step) to yield a 20-carbon fatty acid with double bonds at the 8-, 11-, and 14-positions. A second desaturation reaction at the 5-position followed by an **acyl-CoA synthetase** reaction (Chapter 24) liberates the product, a 20-carbon fatty acid with double bonds at the 5-, 8-, 11-, and 14-positions.

Regulatory Control of Fatty Acid Metabolism—An Interplay of Allosteric Modifiers and Phosphorylation–Dephosphorylation Cycles

The control and regulation of fatty acid synthesis is intimately related to regulation of fatty acid breakdown, glycolysis, and the TCA cycle. Acetyl-CoA is an important intermediate metabolite in all these processes. In these terms, it is easy to appreciate the interlocking relationships in Figure 25.16. Malonyl-CoA can act to prevent fatty acyl-CoA derivatives from entering the mitochondria by inhibiting the carnitine acyltransferase that is responsible for this transport. In this way, when fatty acid synthesis is turned on (as signaled by higher levels of malonyl-CoA), β -oxidation is inhibited. As we pointed out earlier, citrate is an important allosteric activator of acetyl-CoA carboxylase, and fatty acyl-CoAs

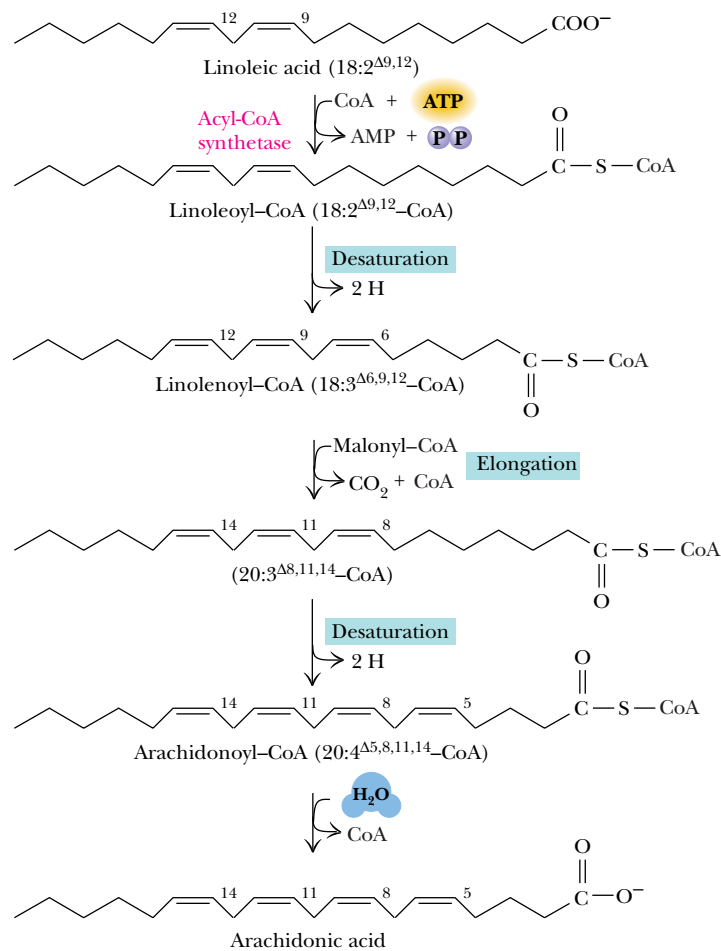


FIGURE 25.15 • Arachidonic acid is synthesized from linoleic acid in eukaryotes. This is the only means by which animals can synthesize fatty acids with double bonds at positions beyond C-9.

are inhibitors. The degree of inhibition is proportional to the chain length of the fatty acyl-CoA; longer chains show a higher affinity for the allosteric inhibition site on acetyl-CoA carboxylase. Palmitoyl-CoA, stearoyl-CoA, and arachidyl-CoA are the most potent inhibitors of the carboxylase.

Hormonal Signals Regulate ACC and Fatty Acid Biosynthesis

As described earlier, citrate activation and palmitoyl-CoA inhibition of acetyl-CoA carboxylase are strongly dependent on the phosphorylation state of the enzyme. This provides a crucial connection to hormonal regulation. Many of the enzymes that act to phosphorylate acetyl-CoA carboxylase (Figure 25.4) are controlled by hormonal signals. Glucagon is a good example (Figure 25.17). As noted in Chapter 23, glucagon binding to membrane receptors activates an intracellular cascade involving activation of adenylyl cyclase. Cyclic AMP produced by the cyclase activates a protein kinase, which then phosphorylates acetyl-CoA carboxylase. Unless citrate levels are high, phosphorylation causes inhibition of fatty acid biosynthesis. The carboxylase (and fatty acid synthesis)

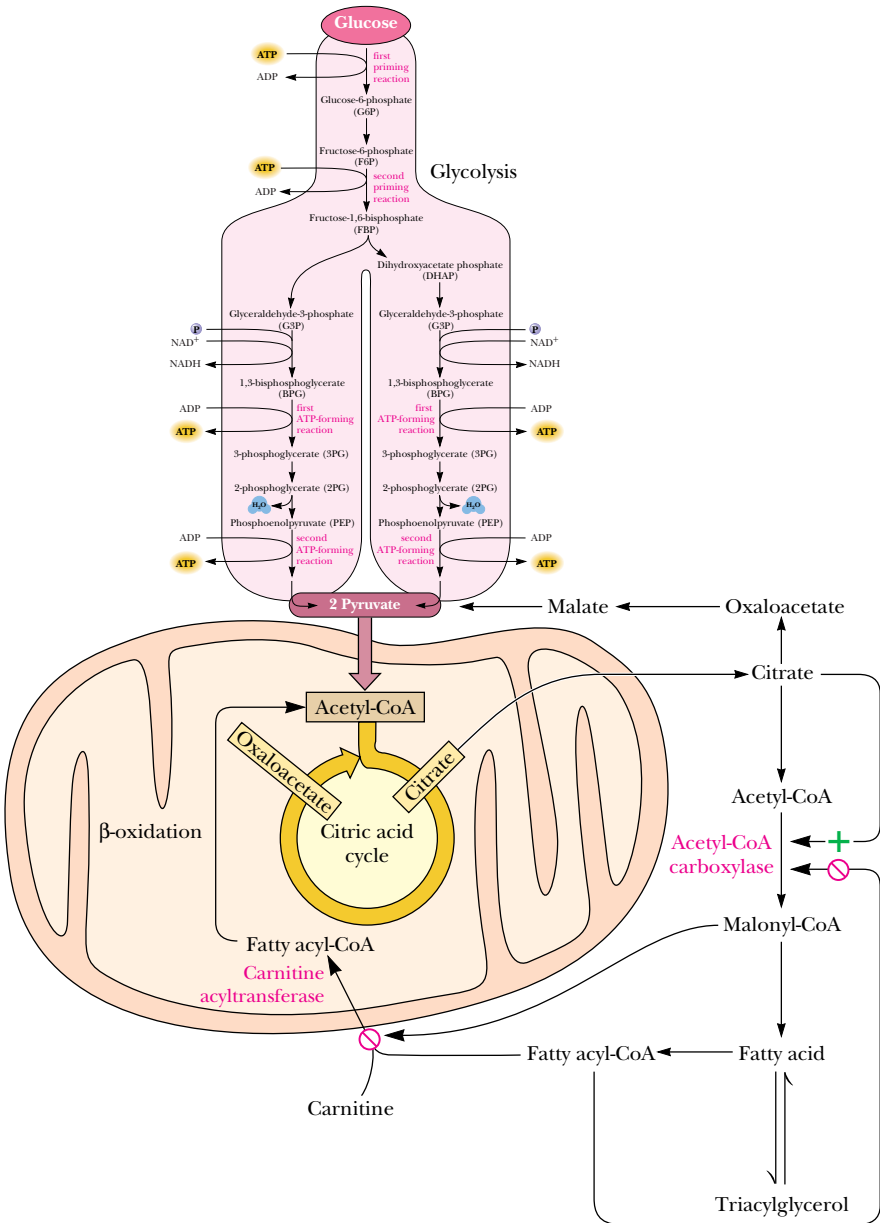


FIGURE 25.16 • Regulation of fatty acid synthesis and fatty acid oxidation are coupled as shown. Malonyl-CoA, produced during fatty acid synthesis, inhibits the uptake of fatty acylcarnitine (and thus fatty acid oxidation) by mitochondria. When fatty acyl CoA levels rise, fatty acid synthesis is inhibited and fatty acid oxidation activity increases. Rising citrate levels (which reflect an abundance of acetyl-CoA) similarly signal the initiation of fatty acid synthesis.

can be reactivated by a specific phosphatase, which dephosphorylates the carboxylase. Also indicated in Figure 25.17 is the simultaneous activation by glucagon of triacylglycerol lipases, which hydrolyze triacylglycerols, releasing fatty acids for β -oxidation. Both the inactivation of acetyl-CoA carboxylase and the activation of triacylglycerol lipase are counteracted by insulin, whose receptor acts to stimulate a phosphodiesterase that converts cAMP to AMP.

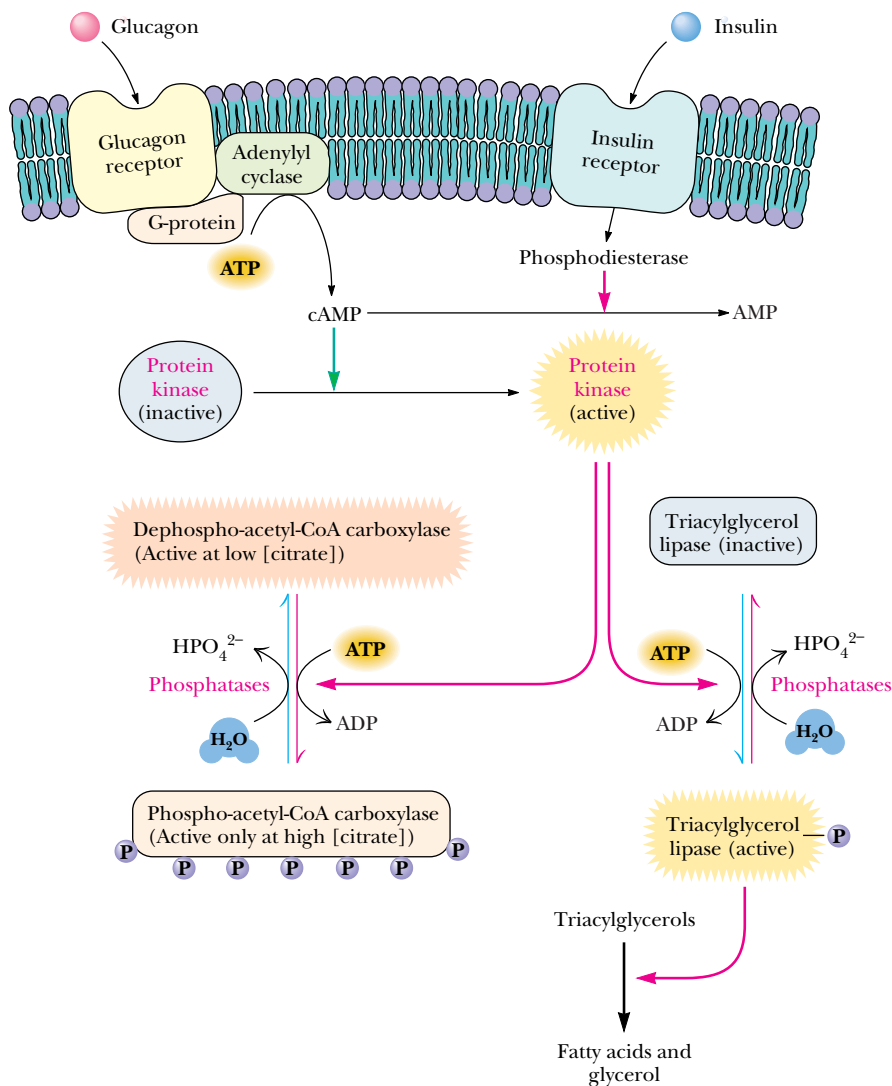


FIGURE 25.17 • Hormonal signals regulate fatty acid synthesis, primarily through actions on acetyl-CoA carboxylase. Availability of fatty acids also depends upon hormonal activation of triacylglycerol lipase.

25.2 • Biosynthesis of Complex Lipids

Complex lipids consist of backbone structures to which fatty acids are covalently bound. Principal classes include the **glycerolipids**, for which glycerol is the backbone, and **sphingolipids**, which are built on a sphingosine backbone. The two major classes of glycerolipids are **glycerophospholipids** and **triacylglycerols**. The **phospholipids**, which include both glycerophospholipids and sphingomyelins, are crucial components of membrane structure. They are also precursors of hormones such as the *eicosanoids* (e.g., *prostaglandins*) and signal molecules, such as the breakdown products of *phosphatidylinositol*.

Different organisms possess greatly different complements of lipids and therefore invoke somewhat different lipid biosynthetic pathways. For example,

sphingolipids and triacylglycerols are produced only in eukaryotes. In contrast, bacteria usually have rather simple lipid compositions. Phosphatidylethanolamine accounts for at least 75% of the phospholipids in *E. coli*, with phosphatidylglycerol and cardiolipin accounting for most of the rest. *E. coli* membranes possess no phosphatidylcholine, phosphatidylinositol, sphingolipids, or cholesterol. On the other hand, some bacteria (such as *Pseudomonas*) can synthesize phosphatidylcholine, for example. In this section and the one following, we consider some of the pathways for the synthesis of glycerolipids, sphingolipids, and the eicosanoids, which are derived from phospholipids.

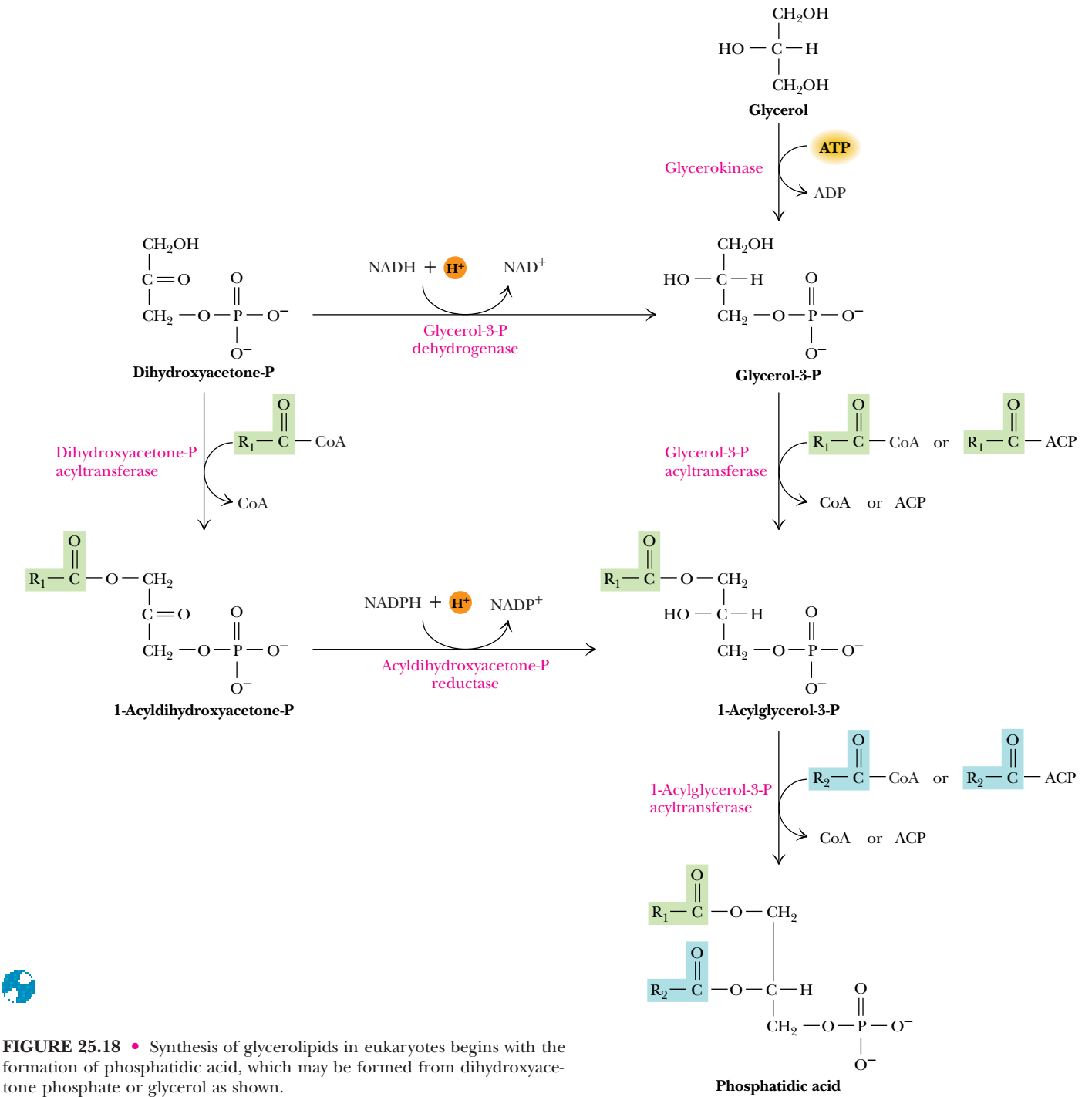


FIGURE 25.18 • Synthesis of glycerolipids in eukaryotes begins with the formation of phosphatidic acid, which may be formed from dihydroxyacetone phosphate or glycerol as shown.

Glycerolipid Biosynthesis

A common pathway operates in nearly all organisms for the synthesis of **phosphatidic acid**, the precursor to other glycerolipids. **Glycerokinase** catalyzes the phosphorylation of glycerol to form glycerol-3-phosphate, which is then acylated at both the 1- and 2-positions to yield phosphatidic acid (Figure 25.18). The first acylation, at position 1, is catalyzed by **glycerol-3-phosphate acyltransferase**, an enzyme that in most organisms is specific for saturated fatty acyl groups. Eukaryotic systems can also utilize **dihydroxyacetone phosphate** as a starting point for synthesis of phosphatidic acid (Figure 25.18). Again a specific acyltransferase adds the first acyl chain, followed by reduction of the backbone keto group by **acyldihydroxyacetone phosphate reductase**, using NADPH as the reductant. Alternatively, dihydroxyacetone phosphate can be reduced to glycerol-3-phosphate by **glycerol-3-phosphate dehydrogenase**.

Eukaryotes Synthesize Glycerolipids from CDP-Diacylglycerol or Diacylglycerol

In eukaryotes, phosphatidic acid is converted directly either to diacylglycerol or to *cytidine diphosphodiacylglycerol* (or simply *CDP-diacylglycerol*; Figure 25.19). From these two precursors, all other glycerophospholipids in eukaryotes are derived. Diacylglycerol is a precursor for synthesis of triacylglycerol, phosphatidylethanolamine, and phosphatidylcholine. Triacylglycerol is synthesized mainly in adipose tissue, liver, and intestines and serves as the principal energy storage molecule in eukaryotes. Triacylglycerol biosynthesis in liver and adipose tissue occurs via **diacylglycerol acyltransferase**, an enzyme bound to the cytoplasmic face of the endoplasmic reticulum. A different route is used, however, in intestines. Recall (Figure 24.3) that triacylglycerols from the diet are broken down to 2-monoacylglycerols by specific lipases. Acyltransferases then acylate 2-monoacylglycerol to produce new triacylglycerols (Figure 25.20).

Phosphatidylethanolamine Is Synthesized from Diacylglycerol and CDP-Ethanolamine

Phosphatidylethanolamine synthesis begins with phosphorylation of ethanolamine to form phosphoethanolamine (Figure 25.19). The next reaction involves transfer of a cytidyl group from CTP to form CDP-ethanolamine and pyrophosphate. As always, PP_i hydrolysis drives this reaction forward. A specific **phosphoethanolamine transferase** then links phosphoethanolamine to the diacylglycerol backbone. Biosynthesis of phosphatidylcholine is entirely analogous because animals synthesize it directly. All of the choline utilized in this pathway must be acquired from the diet. Yeast, certain bacteria, and animal livers, however, can convert phosphatidylethanolamine to phosphatidylcholine by methylation reactions involving S-adenosylmethionine (see Chapter 26).

Exchange of Ethanolamine for Serine Converts Phosphatidylethanolamine to Phosphatidylserine

Mammals synthesize phosphatidylserine (PS) in a calcium ion–dependent reaction involving aminoalcohol exchange (Figure 25.21). The enzyme catalyzing this reaction is associated with the endoplasmic reticulum and will accept phosphatidylethanolamine (PE) and other phospholipid substrates. A mitochondrial **PS decarboxylase** can subsequently convert PS to PE. No other pathway converting serine to ethanolamine has been found.

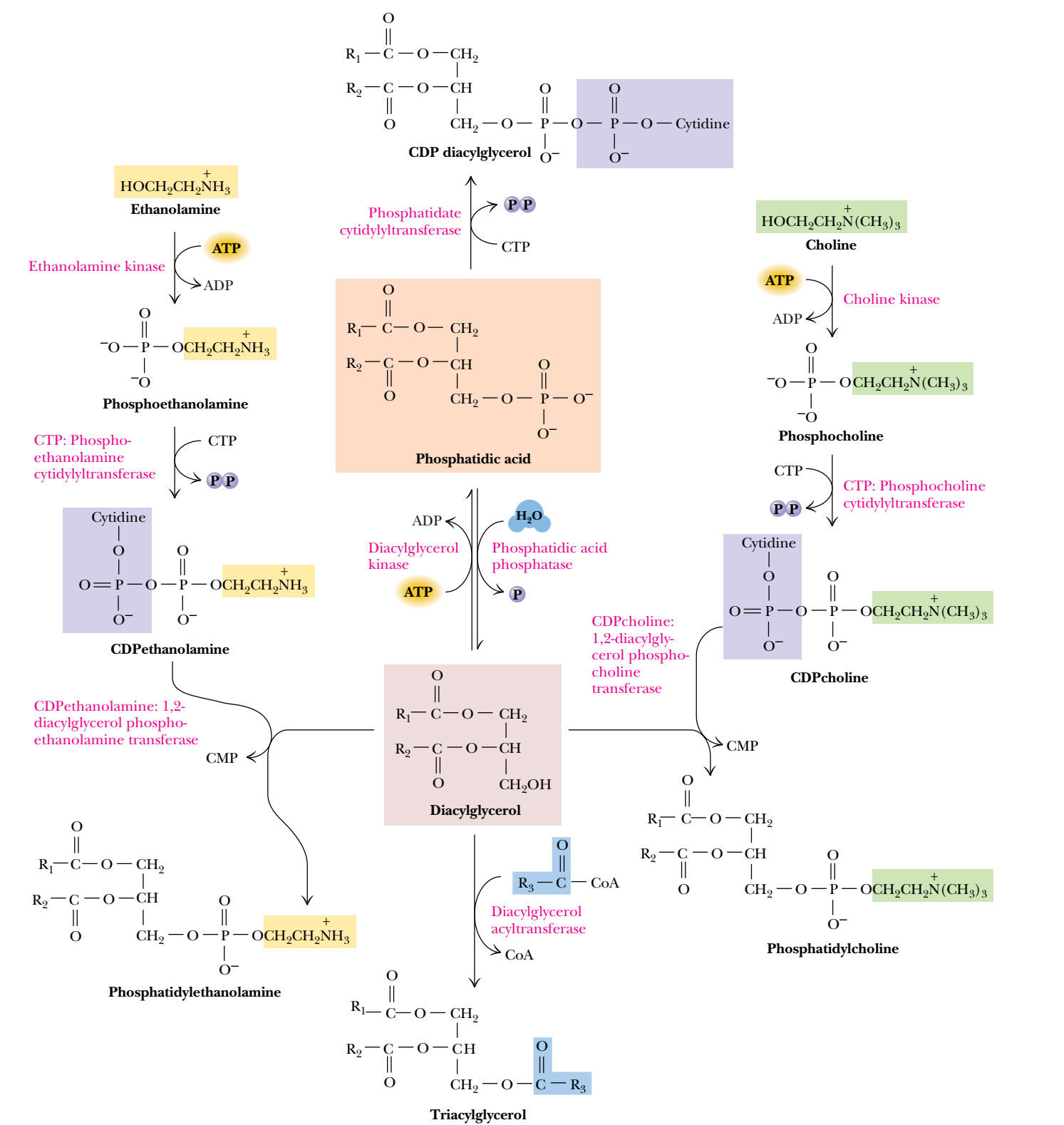


FIGURE 25.19 • Diacylglycerol and CDP-diacylglycerol are the principal precursors of glycerolipids in eukaryotes. Phosphatidylethanolamine and phosphatidylcholine are formed by reaction of diacylglycerol with CDP-ethanolamine or CDP-choline, respectively.

FIGURE 25.20 • Triacylglycerols are formed primarily by the action of acyltransferases on mono- and diacylglycerol. Acyltransferase in *E. coli* is an integral membrane protein (83 kD) and can utilize either fatty acyl-CoAs or acylated acyl carrier proteins as substrates. It shows a particular preference for palmitoyl groups. Eukaryotic acyltransferases use only fatty acyl-CoA molecules as substrates.

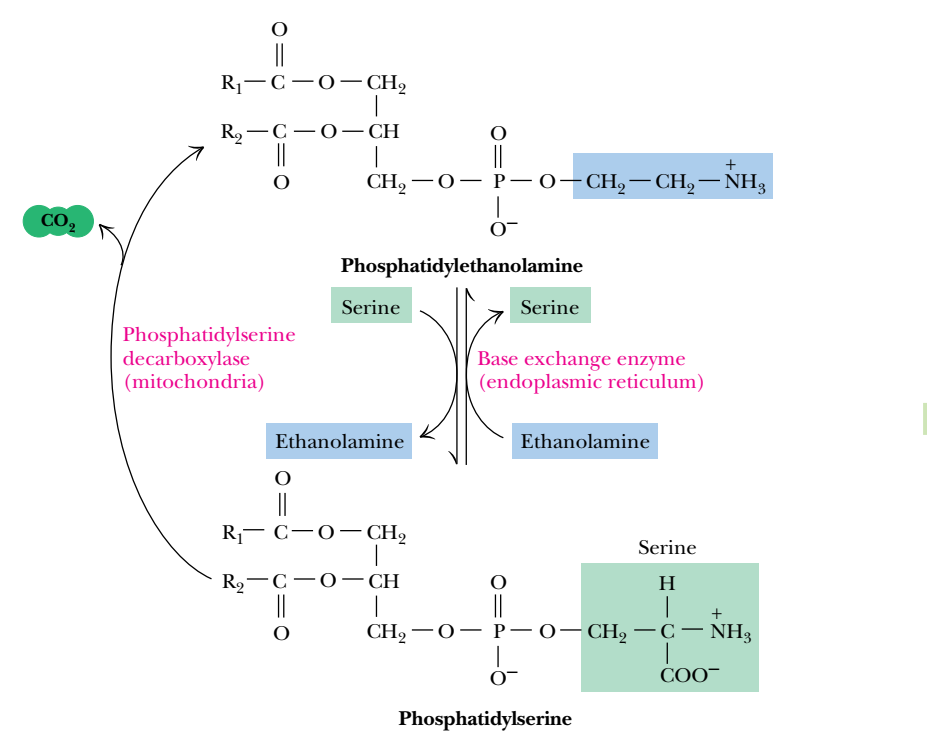


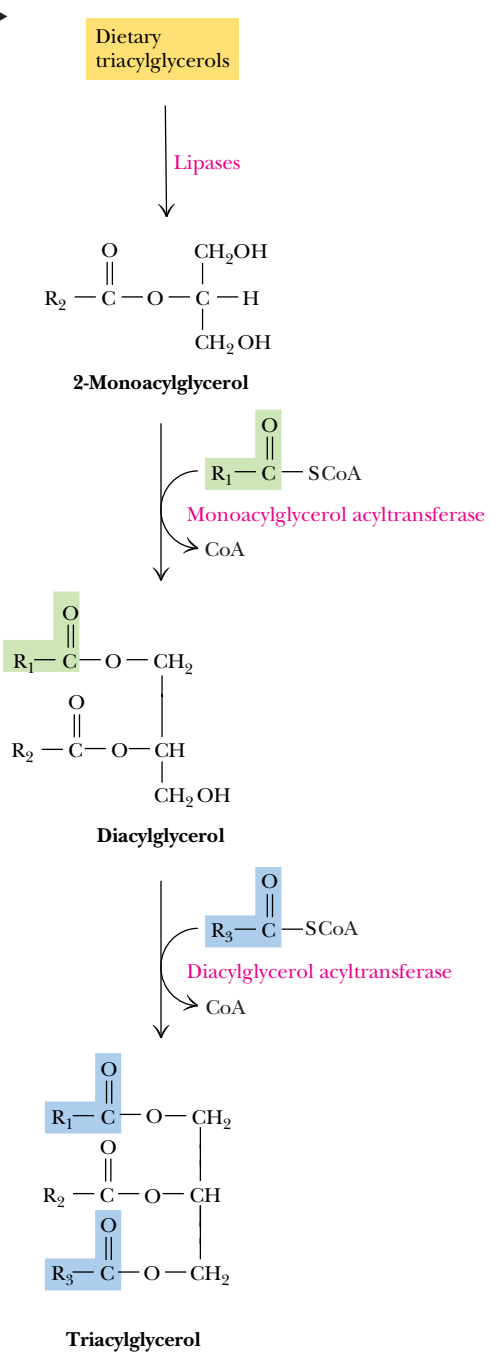
FIGURE 25.21 • The interconversion of phosphatidylethanolamine and phosphatidylserine in mammals.

Eukaryotes Synthesize Other Phospholipids via CDP-Diacylglycerol

Eukaryotes also use CDP-diacylglycerol, derived from phosphatidic acid, as a precursor for several other important phospholipids, including phosphatidylinositol (PI), phosphatidylglycerol (PG), and cardiolipin (Figure 25.22). PI accounts for only about 2 to 8% of the lipids in most animal membranes, but breakdown products of PI, including inositol-1,4,5-trisphosphate and diacylglycerol, are second messengers in a vast array of cellular signaling processes.

Dihydroxyacetone Phosphate Is a Precursor to the Plasmalogens

Certain glycerophospholipids possess alkyl or alkenyl ether groups at the 1-position in place of an acyl ester group. These glycerether phospholipids are synthesized from dihydroxyacetone phosphate (Figure 25.23). Acylation of dihydroxyacetone phosphate (DHAP) is followed by an exchange reaction, in which the acyl group is removed as a carboxylic acid and a long-chain alcohol



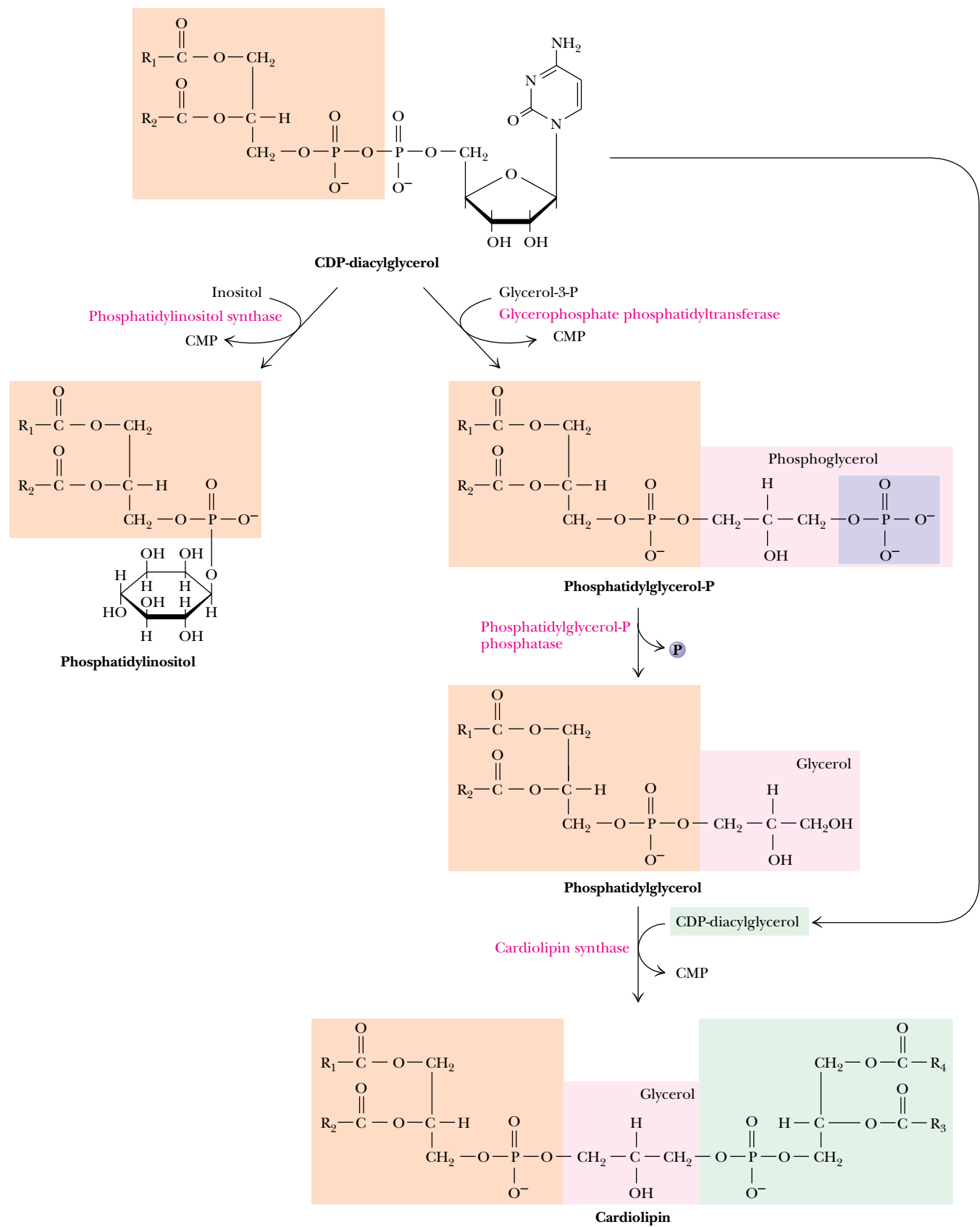
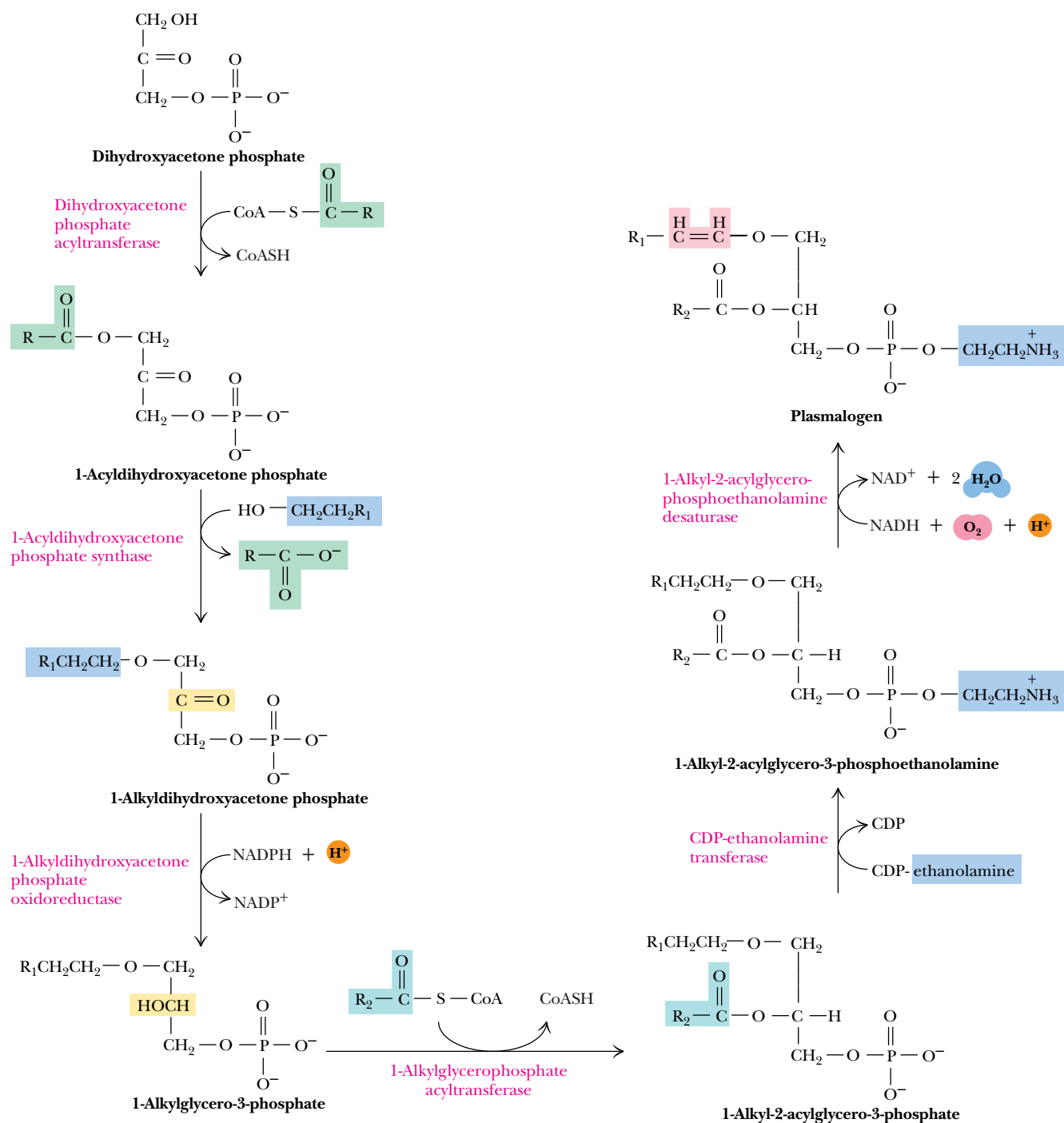


FIGURE 25.22 • CDP-diacylglycerol is a precursor of phosphatidylinositol, phosphatidylglycerol, and cardiolipin in eukaryotes.



adds to the 1-position. This long-chain alcohol is derived from the corresponding acyl-CoA by means of an **acyl-CoA reductase** reaction involving oxidation of two molecules of NADH. The 2-*keto* group of the DHAP backbone is then reduced to an alcohol, followed by acylation. The resulting 1-alkyl-2-acylglycero-3-phosphate can react in a manner similar to phosphatidic acid to produce ether analogs of phosphatidylcholine, phosphatidylethanolamine, and so forth (Figure 25.23). In addition, specific **desaturase** enzymes associated with the endoplasmic reticulum can desaturate the alkyl ether chains of these lipids

FIGURE 25.23 • Biosynthesis of plasmalogens in animals. Acylation at C-1 is followed by exchange of the acyl group for a long-chain alcohol. Reduction of the keto group at C-2 is followed by transferase reactions, which add an acyl group at C-2 and a polar head-group moiety, and a desaturase reaction that forms a double bond in the alkyl chain. The first two enzymes are of cytoplasmic origin, and the last transferase is located at the endoplasmic reticulum.

as shown. The products, which contain α,β -unsaturated ether-linked chains at the C-1 position, are **plasmalogens**; they are abundant in cardiac tissue and in the central nervous system. The desaturases catalyzing these reactions are distinct from but similar to those that introduce unsaturations in fatty acyl-CoAs. These enzymes use cytochrome b_5 as a cofactor, NADH as a reductant, and O_2 as a terminal electron acceptor.

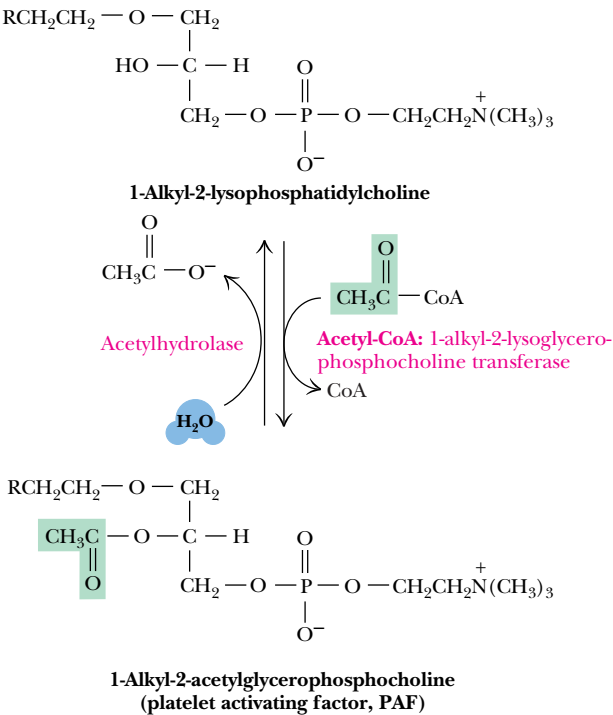
Platelet Activating Factor

A particularly interesting ether phospholipid with unusual physiological properties has recently been characterized. As shown in Figure 25.24, **1-alkyl-2-acetyl-glycerophosphocholine**, also known as **platelet activating factor**, possesses an alkyl ether at C-1 and an acetyl group at C-2. The very short chain at C-2 makes this molecule much more water-soluble than typical glycerolipids. Platelet activating factor displays a dramatic ability to dilate blood vessels (and thus reduce blood pressure in hypertensive animals) and to aggregate platelets.

Sphingolipid Biosynthesis

Sphingolipids, ubiquitous components of eukaryotic cell membranes, are present at high levels in neural tissues. The myelin sheath that insulates nerve axons is particularly rich in sphingomyelin and other related lipids. Prokaryotic organisms normally do not contain sphingolipids. Sphingolipids are built upon sphingosine backbones rather than glycerol. The initial reaction, which involves condensation of serine and palmitoyl-CoA with release of bicarbonate, is catalyzed by **3-ketosphinganine synthase**, a PLP-dependent enzyme (Figure 25.25). Reduction of the ketone product to form *sphinganine* is catalyzed by **3-keto-sphinganine reductase**, with NADPH as a reactant. In the next step, sphinganine is acylated to form N-acyl sphinganine, which is *then* desaturated to form ceramide. Sphingosine itself does not appear to be an intermediate in this pathway in mammals.

FIGURE 25.24 • Platelet activating factor, formed from 1-alkyl-2-lysophosphatidylcholine by acetylation at C-2, is degraded by the action of acetylhydrolase.



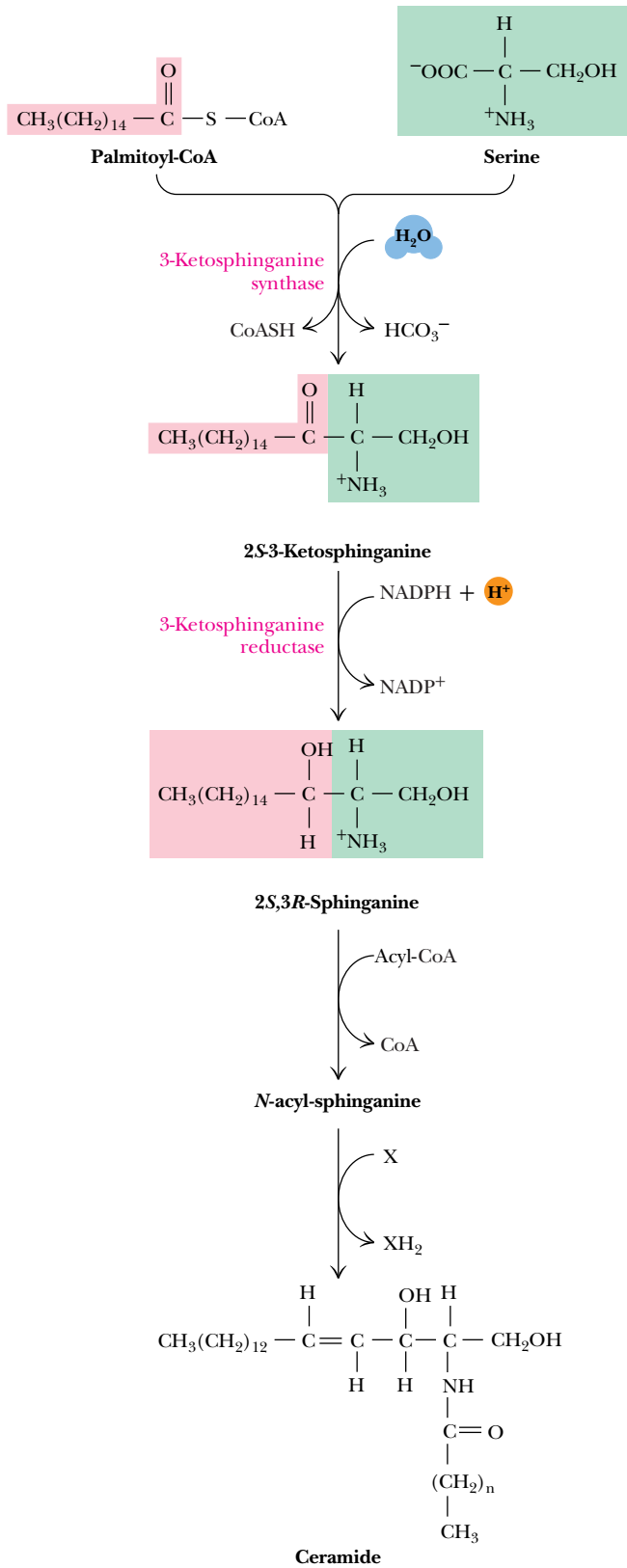


FIGURE 25.25 • Biosynthesis of sphingolipids in animals begins with the 3-ketosphinganine synthase reaction, a PLP-dependent condensation of palmitoyl-CoA and serine. Subsequent reduction of the keto group, acylation, and desaturation (via reduction of an electron acceptor, X) form ceramide, the precursor of other sphingolipids.



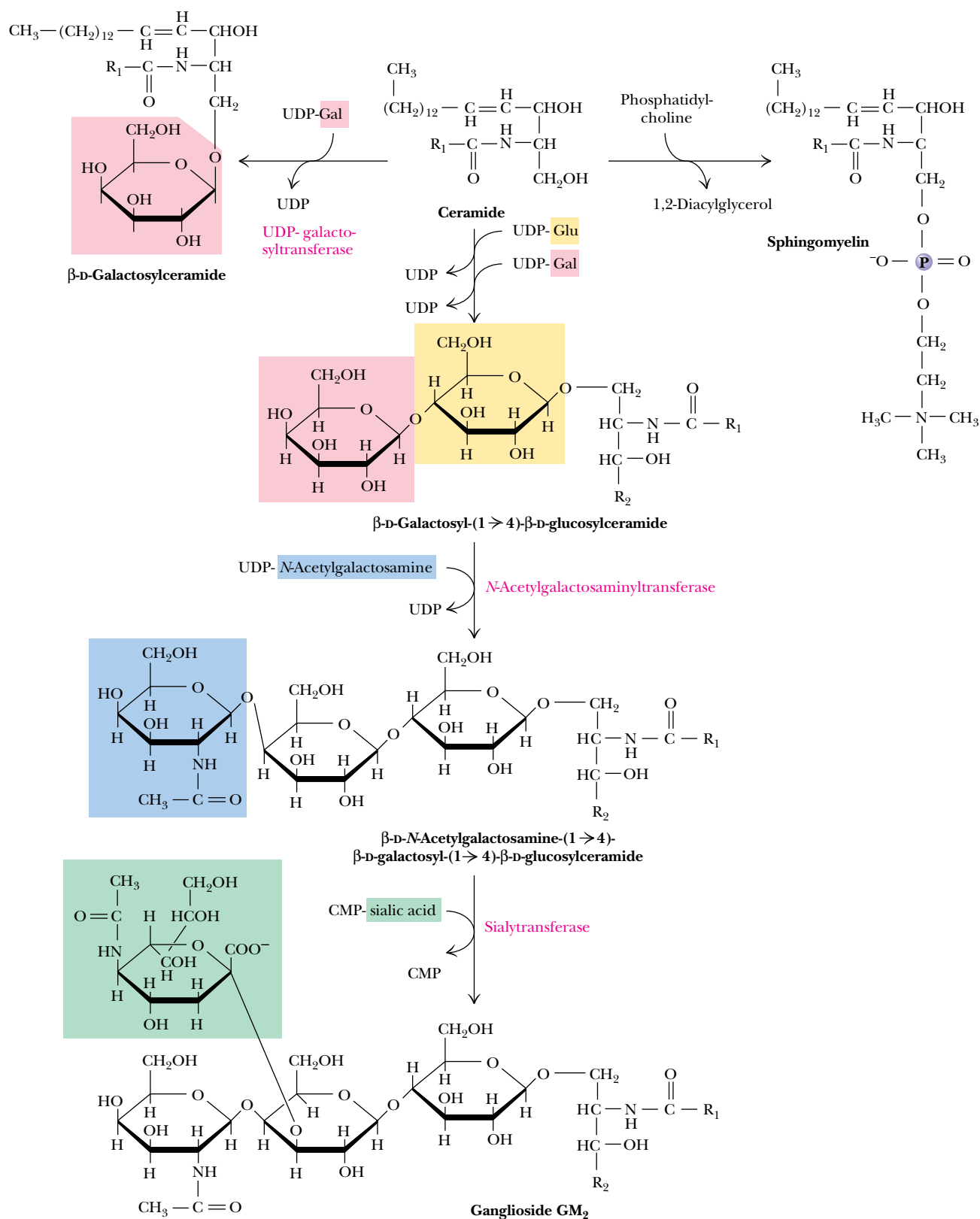


FIGURE 25.26 • Glycosylceramides (such as galactosylceramide), gangliosides, and sphingomyelins are synthesized from ceramide in animals.

Ceramide Is the Precursor for Other Sphingolipids and Cerebrosides

Ceramide is the building block for all other sphingolipids. Sphingomyelin, for example, is produced by transfer of phosphocholine from phosphatidylcholine (Figure 25.26). Glycosylation of ceramide by sugar nucleotides yields **cerebrosides**, such as galactosylceramide, which makes up about 15% of the lipids of myelin sheath structures. Cerebrosides that contain one or more sialic acid (N-acetylneuraminic acid) moieties are called **gangliosides**. Several dozen gangliosides have been characterized, and the general form of the biosynthetic pathway is illustrated for the case of ganglioside GM₂ (Figure 25.26). Sugar units are added to the developing ganglioside from nucleotide derivatives, including UDP-N-acetylglucosamine, UDP-galactose, and UDP-glucose.

25.3 • Eicosanoid Biosynthesis and Function

Eicosanoids, so named because they are all derived from 20-carbon fatty acids, are ubiquitous breakdown products of phospholipids. In response to appropriate stimuli, cells activate the breakdown of selected phospholipids (Figure 25.27). Phospholipase A₂ (Chapter 8) selectively cleaves fatty acids from the C-2 position of phospholipids. Often these are unsaturated fatty acids, among which is arachidonic acid. Arachidonic acid may also be released from phospholipids by the combined actions of phospholipase C (which yields diacylglycerols) and diacylglycerol lipase (which releases fatty acids).

Eicosanoids Are Local Hormones

Animal cells can modify arachidonic acid and other polyunsaturated fatty acids, in processes often involving cyclization and oxygenation, to produce so-called local hormones that (1) exert their effects at very low concentrations and (2) usually act near their sites of synthesis. These substances include the **prostaglandins** (PG) (Figure 25.27) as well as **thromboxanes** (Tx), **leukotrienes**, and other **hydroxyeicosanoic acids**. Thromboxanes, discovered in blood platelets (*thrombocytes*), are cyclic ethers (TxB₂ is actually a hemiacetal; see Figure 25.27) with a hydroxyl group at C-15.

Prostaglandins Are Formed from Arachidonate by Oxidation and Cyclization

All prostaglandins are cyclopentanoic acids derived from arachidonic acid. The biosynthesis of prostaglandins is initiated by an enzyme associated with the endoplasmic reticulum, called **prostaglandin endoperoxide synthase**, also known as **cyclooxygenase**. The enzyme catalyzes simultaneous oxidation and cyclization of arachidonic acid. The enzyme is viewed as having two distinct activities, cyclooxygenase and peroxidase, as shown in Figure 25.28.

A Variety of Stimuli Trigger Arachidonate Release and Eicosanoid Synthesis

The release of arachidonate and the synthesis or interconversion of eicosanoids can be initiated by a variety of stimuli, including histamine, hormones such as epinephrine and bradykinin, proteases such as thrombin, and even serum albumin. An important mechanism of arachidonate release and eicosanoid syn-

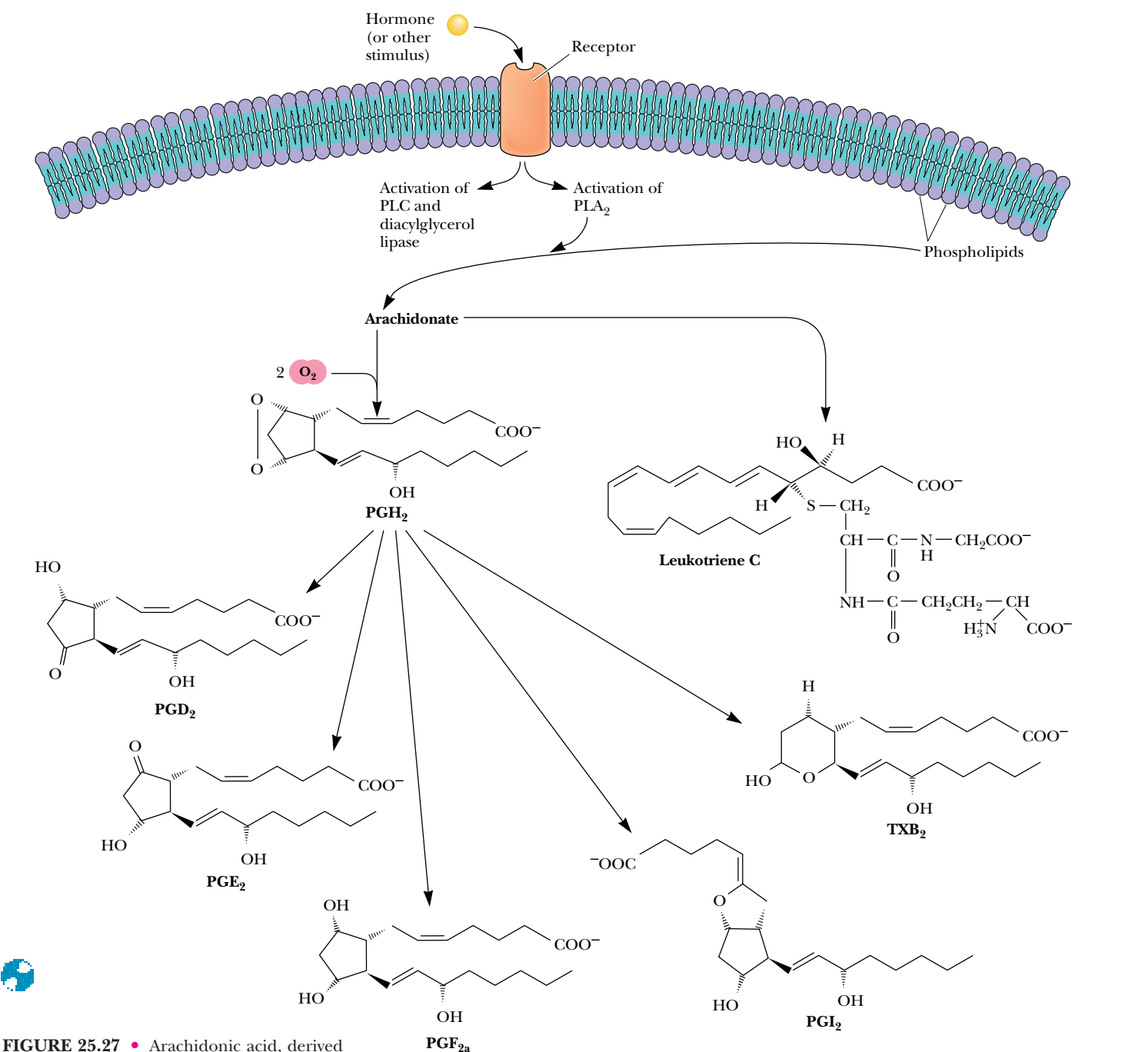


FIGURE 25.27 • Arachidonic acid, derived from breakdown of phospholipids (PL), is the precursor of prostaglandins, thromboxanes, and leukotrienes. The letters used to name the prostaglandins are assigned on the basis of similarities in structure and physical properties. The class denoted PGE, for example, consists of β -hydroxyketones that are soluble in ether, whereas PGF denotes 1,3-diols that are soluble in phosphate buffer. PGA denotes prostaglandins possessing α,β -unsaturated ketones. The number following the letters refers to the number of carbon-carbon double bonds. Thus, PGE₂ contains two double bonds.

thesis involves tissue injury and inflammation. When tissue damage or injury occurs, special inflammatory cells, **monocytes** and **neutrophils**, invade the injured tissue and interact with the resident cells (e.g., smooth muscle cells and fibroblasts). *This interaction typically leads to arachidonate release and eicosanoid synthesis.* Examples of tissue injury in which eicosanoid synthesis has been characterized include heart attack (myocardial infarction), rheumatoid arthritis, and ulcerative colitis.

A DEEPER LOOK

The Discovery of Prostaglandins

The name *prostaglandin* was given to this class of compounds by Ulf von Euler, their discoverer, in Sweden in the 1930s. He extracted fluids containing these components from human semen. Because he thought they originated in the prostate gland, he named them prostaglandins. Actually, they were synthesized in the seminal vesicles, and it is now known that similar substances are synthesized in most animal tissues (both male and female). Von Euler observed that injection of these substances into animals caused smooth muscle contraction and dramatic lowering of blood pressure.

Von Euler (and others) soon found that it is difficult to analyze and characterize these obviously interesting compounds because they are present at extremely low levels. Prostaglandin

$E_{2\alpha}$, or $PGE_{2\alpha}$, is present in human serum at a level of less than 10^{-14} M! In addition, they often have half-lives of only 30 seconds to a few minutes, not lasting long enough to be easily identified. Moreover, most animal tissues upon dissection and homogenization rapidly synthesize and degrade a variety of these substances, so the amounts obtained in isolation procedures are extremely sensitive to the methods used and highly variable even when procedures are carefully controlled.

Sune Bergstrom and his colleagues described the first structural determinations of prostaglandins in the late 1950s. In the early 1960s, dramatic advances in laboratory techniques such as NMR spectroscopy and mass spectrometry made further characterization possible.

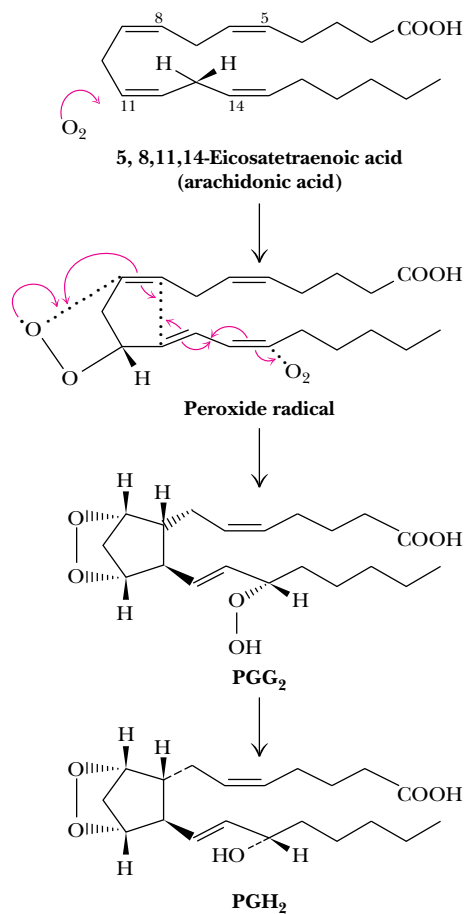


FIGURE 25.28 • Prostaglandin endoperoxide synthase, the enzyme that converts arachidonic acid to prostaglandin PGH_2 , possesses two distinct activities: cyclooxygenase (steps 1 and 2) and glutathione (GSSG)–dependent hydroperoxidase (step 3). Cyclooxygenase is the site of action of aspirin and many other analgesic agents.

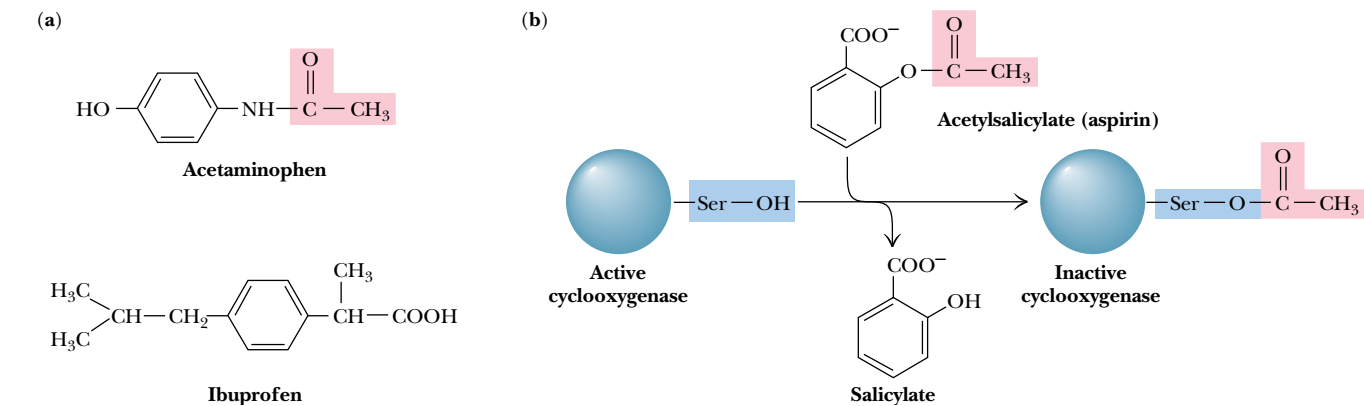


FIGURE 25.29 • (a) The structures of several common analgesic agents. Acetaminophen is marketed under the tradename Tylenol[®]. Ibuprofen is sold as Motrin[®], Nuprin[®], and Advil[®]. (b) Acetylsalicylate (aspirin) inhibits the cyclooxygenase activity of endoperoxide synthase via acetylation (covalent modification) of Ser⁵³⁰.

“Take Two Aspirin and . . .” Inhibit Your Prostaglandin Synthesis

In 1971, biochemist John Vane was the first to show that **aspirin** (acetylsalicylate; Figure 25.29) exerts most of its effects by inhibiting the biosynthesis of prostaglandins. Its site of action is prostaglandin endoperoxide synthase. Cyclooxygenase activity is destroyed when aspirin O-acetylates Ser⁵³⁰ on the enzyme. From this you may begin to infer something about how prostaglandins (and aspirin) function. Prostaglandins are known to enhance inflammation in animal tissues. Aspirin exerts its powerful anti-inflammatory effect by inhibiting this first step in their synthesis. Aspirin does not have any measurable effect on the peroxidase activity of the synthase. Other nonsteroidal anti-inflammatory agents, such as **ibuprofen** (Figure 25.29) and phenylbutazone, inhibit the cyclooxygenase by competing at the active site with arachidonate or with the peroxyacid intermediate (PGG₂, Figure 25.28). See A Deeper Look, page 834.

25.4 • Cholesterol Biosynthesis

The most prevalent steroid in animal cells is **cholesterol** (Figure 25.30). Plants contain no cholesterol, but they *do* contain other steroids very similar to cholesterol in structure (see page 256). Cholesterol serves as a crucial component of cell membranes and as a precursor to bile acids (e.g., cholate, glycocholate,

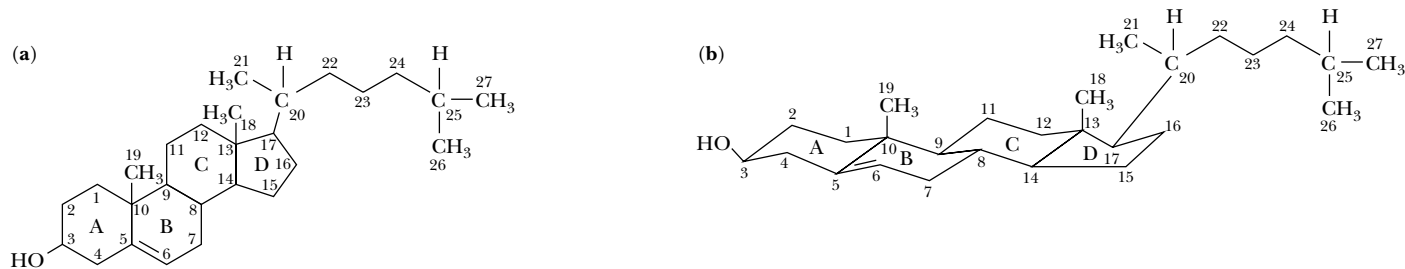


FIGURE 25.30 • The structure of cholesterol, drawn (a) in the traditional planar motif and (b) in a form that more accurately describes the conformation of the ring system.

taurocholate) and steroid hormones (e.g., testosterone, estradiol, progesterone). Also, vitamin D₃ is derived from *7-dehydrocholesterol*, the immediate precursor of cholesterol. Liver is the primary site of cholesterol biosynthesis.

Mevalonate Is Synthesized from Acetyl-CoA via HMG-CoA Synthase

The cholesterol biosynthetic pathway begins in the cytosol with the synthesis of mevalonate from acetyl-CoA (Figure 25.31). The first step is the **β-ketothiolase**-catalyzed Claisen condensation of two molecules of acetyl-CoA to form acetoacetyl-CoA. In the next reaction, acetyl-CoA and acetoacetyl-CoA join to form *3-hydroxy-3-methylglutaryl-CoA*, which is abbreviated *HMG-CoA*. The reaction—a second Claisen condensation—is catalyzed by **HMG-CoA synthase**. The third step in the pathway is the rate-limiting step in cholesterol biosynthesis. Here, HMG-CoA undergoes two NADPH-dependent reductions to produce *3R-mevalonate* (Figure 25.32). The reaction is catalyzed by **HMG-CoA reductase**, a 97-kD glycoprotein that traverses the endoplasmic reticulum membrane with its active site facing the cytosol. As the rate-limiting step, HMG-CoA reductase is the principal site of regulation in cholesterol synthesis.

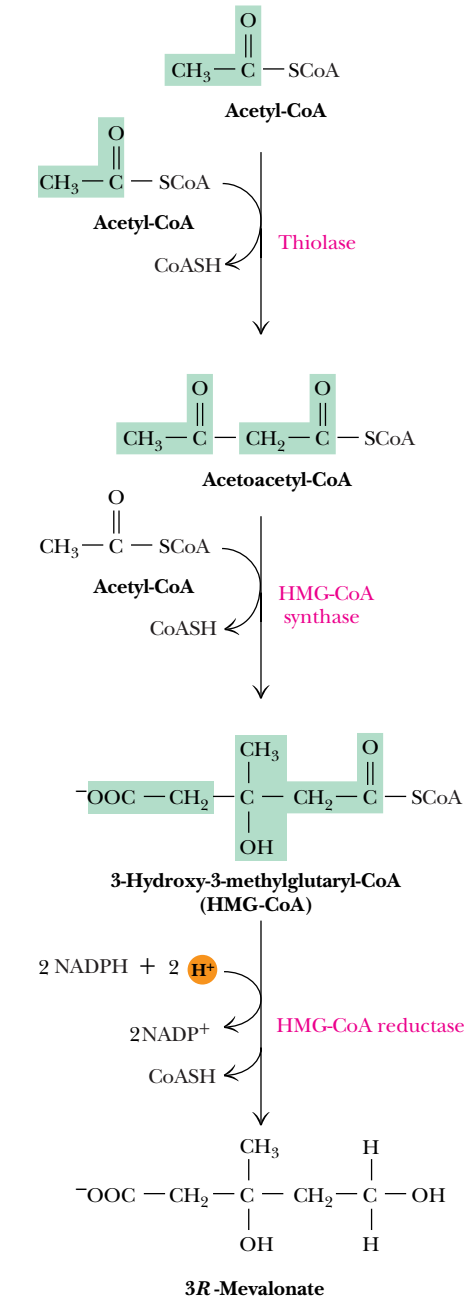
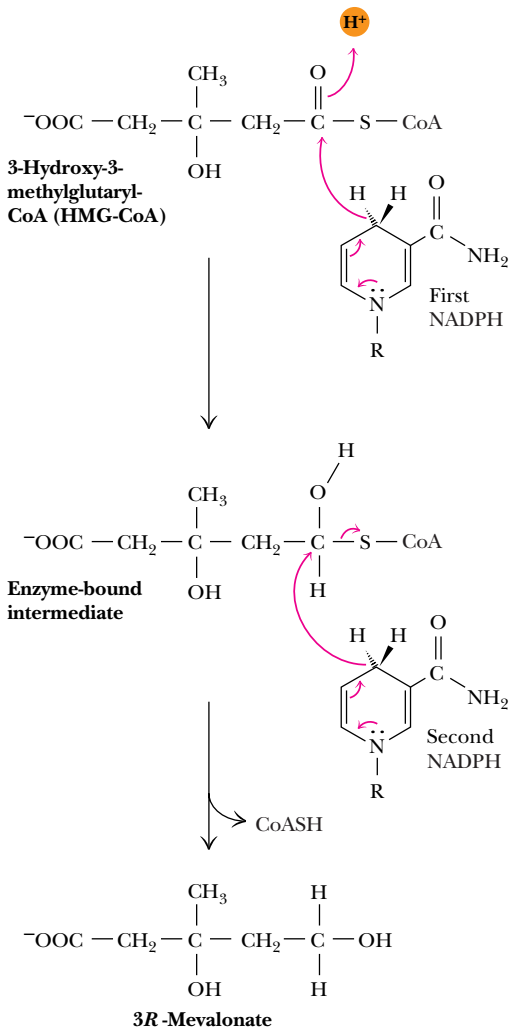


FIGURE 25.31 • The biosynthesis of 3R-mevalonate from acetyl-CoA.



FIGURE 25.32 • A reaction mechanism for HMG-CoA reductase. Two successive NADPH-dependent reductions convert the thioester, HMG-CoA, to a primary alcohol.

A D E E P E R L O O K

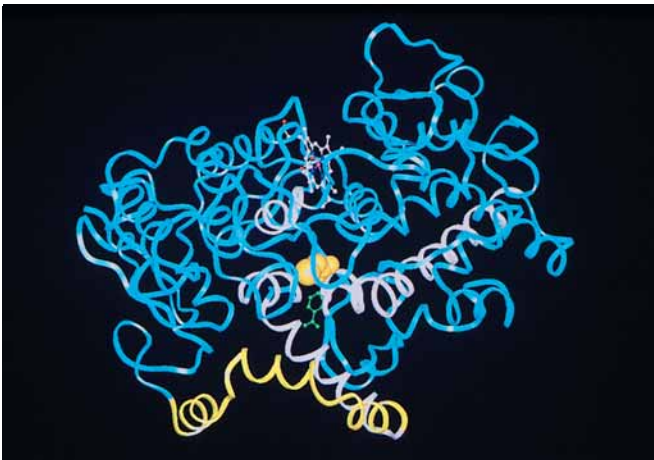
The Molecular Basis for the Action of Nonsteroidal Anti-inflammatory Drugs

Prostaglandins are potent mediators of inflammation. The first and committed step in the production of prostaglandins from arachidonic acid is the bis-oxygenation of arachidonate to prostaglandin PGG₂. This is followed by reduction to PGH₂ in a peroxidase reaction. Both these reactions are catalyzed by prostaglandin endoperoxide synthase, also known as PGH₂ synthase or cyclooxygenase, thus abbreviated COX. This enzyme is inhibited by the family of drugs known as nonsteroidal anti-inflammatory drugs, or NSAIDs. Aspirin, ibuprofen, flurbiprofen, and acetaminophen (trade name Tylenol®) are all NSAIDs.

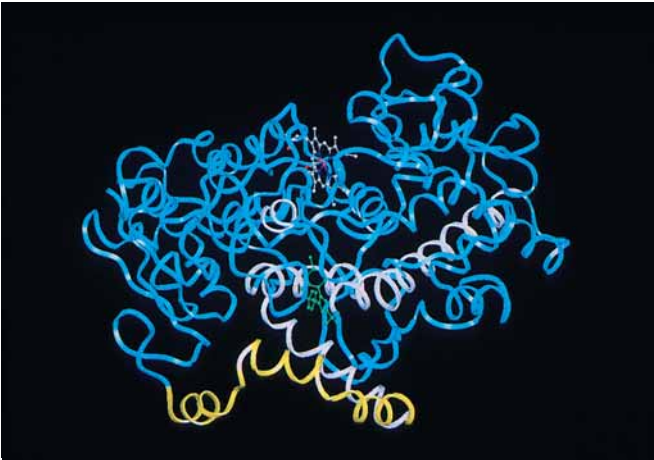
There are two isoforms of COX in animals: COX-1 (figure a), which carries out normal, physiological production of prostaglandins, and COX-2 (figure b), which is induced by cytokines, mitogens, and endotoxins in inflammatory cells and is responsible for the production of prostaglandins in inflammation.

The enzyme structure shown here is that of residues 33 to 583 of COX-1 from sheep, inactivated by bromoaspirin. These 551 residues comprise three distinct domains. The first of these, residues 33 to 72 (purple), form a small compact module that is similar to epidermal growth factor. The second domain, composed of residues 73 to 116 (yellow), forms a right-handed spiral of four α -helical segments along the base of the protein. These α -helical segments form a membrane-binding motif. The helical segments are amphipathic, with most of the hydrophobic residues (shown in green) facing away from the protein, where they can interact with a lipid bilayer. The third domain of the COX enzyme, the catalytic domain (in blue), is a globular structure that contains both the cyclooxygenase and peroxidase active sites.

The cyclooxygenase active site lies at the end of a long, narrow, hydrophobic tunnel or channel. Three of the α -helices of the membrane-binding domain lie at the entrance to this tunnel. The



(a)



(b)



Three different regulatory mechanisms are involved:

1. Phosphorylation by cAMP-dependent protein kinases inactivates the reductase. This inactivation can be reversed by two specific phosphatases (Figure 25.33).
2. Degradation of HMG-CoA reductase. This enzyme has a half-life of only three hours, and the half-life itself depends on cholesterol levels: high [cholesterol] means a short half-life for HMG-CoA reductase.
3. Gene expression—cholesterol levels control the amount of mRNA. If [cholesterol] is high, levels of mRNA coding for the reductase are reduced. If [cholesterol] is low, more mRNA is made. (Regulation of gene expression is discussed in Chapter 31.)

walls of the tunnel are defined by four α -helices, formed by residues 106 to 123, 325 to 353, 379 to 384, and 520 to 535 (shown in orange).

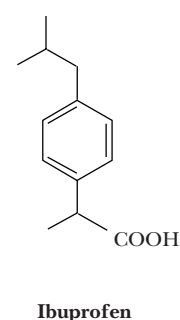
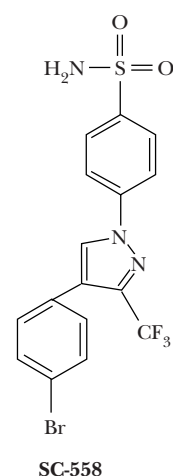
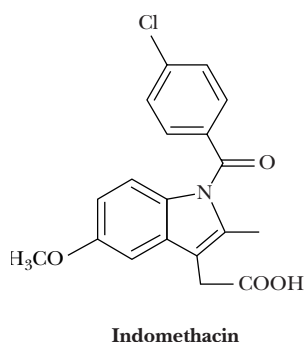
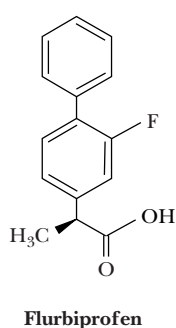
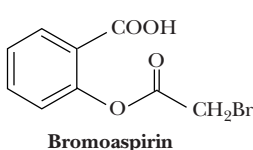
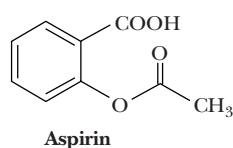
In this bromoaspirin-inactivated structure, Ser⁵³⁰, which lies along the wall of the tunnel, is bromoacetylated, and a molecule of salicylate is also bound in the tunnel. Deep in the tunnel, at the far end, lies Tyr³⁸⁵, a catalytically important residue. Heme-dependent peroxidase activity is implicated in the formation of a proposed Tyr³⁸⁵ radical, which is required for cyclooxygenase activity. Aspirin and other NSAIDs block the synthesis of prostaglandins by filling and blocking the tunnel, preventing the migration of arachidonic acid to Tyr³⁸⁵ in the active site at the back of the tunnel.

There are thought to be at least four different mechanisms of action for NSAIDs. Aspirin (and also bromoaspirin) covalently

modifies a residue in the tunnel, thus irreversibly inactivating both COX-1 and COX-2. Ibuprofen acts instead by competing in a reversible fashion for the substrate-binding site in the tunnel.

Flurbiprofen and indomethacin, which comprise the third class of inhibitors, cause a slow, time-dependent inhibition of COX-1 and COX-2, apparently via formation of a salt bridge between a carboxylate on the drug and Arg¹²⁰, which lies in the tunnel.

The drug SC-558 acts by a fourth mechanism, specifically inhibiting COX-2. It is a weak competitive inhibitor of COX-1 but inhibits COX-2 in a slow, time-dependent process. Specific COX-2 inhibitors will likely be the drugs of the future because they selectively block the inflammation mediated by COX-2, without the potential for stomach lesions and renal toxicity that arise from COX-1 inhibition.



A Thiolase Brainteaser

If acetate units can be condensed by the thiolase reaction to yield acetoacetate in the first step of cholesterol synthesis, why couldn't this same reaction also be used in fatty acid synthesis, avoiding all the complexity of the fatty acyl synthase? The answer is that the thiolase reaction is more or less reversible but slightly favors the cleavage reaction. In the cholesterol synthesis pathway, subsequent reactions, including HMG-CoA reductase and the following kinase reactions, pull the thiolase-catalyzed condensation forward. However, in the case of fatty acid synthesis, a succession of eight thiolase condensations would be distinctly unfavorable from an energetic perspective. Given the necessity of repeated reactions in fatty acid synthesis, it makes better energetic sense to use a reaction that is favorable in the desired direction.

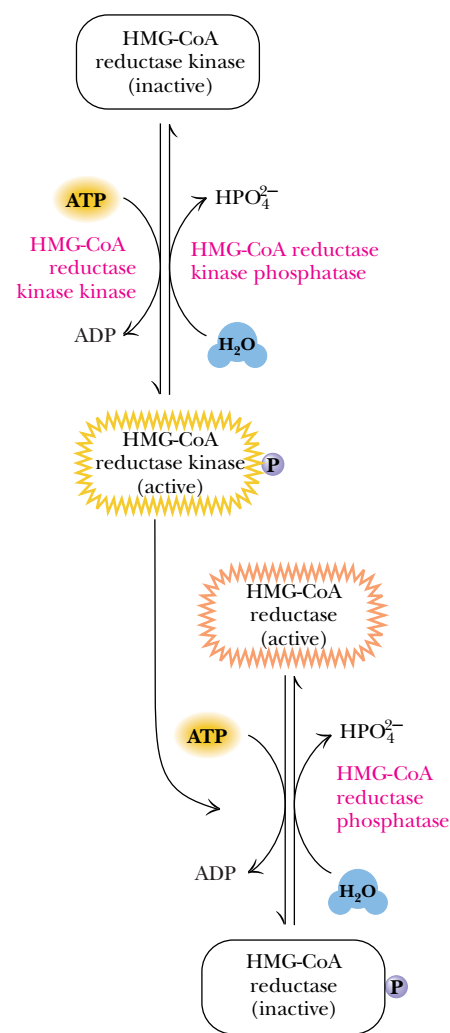


FIGURE 25.33 • HMG-CoA reductase activity is modulated by a cycle of phosphorylation and dephosphorylation.

Squalene Is Synthesized from Mevalonate

The biosynthesis of squalene involves conversion of mevalonate to two key 5-carbon intermediates, *isopentenyl pyrophosphate* and *dimethylallyl pyrophosphate*, which join to yield *farnesyl pyrophosphate* and then squalene. A series of four reactions converts mevalonate to isopentenyl pyrophosphate and then to dimethylallyl pyrophosphate (Figure 25.34). The first three steps each consume an ATP, two for the purpose of forming a pyrophosphate at the 5-position and the third to drive the decarboxylation and double bond formation in the third step. **Pyrophosphomevalonate decarboxylase** phosphorylates the 3-hydroxyl group, and this is followed by *trans* elimination of the phosphate and carboxyl groups to form the double bond in isopentenyl pyrophosphate. Isomerization of the double bond yields the dimethylallyl pyrophosphate. Condensation of these two 5-carbon intermediates produces *geranyl pyrophosphate*; addition of another 5-carbon isopentenyl group gives farnesyl pyrophosphate. Both steps in the production of farnesyl pyrophosphate occur with release of pyrophosphate, hydrolysis of which drives these reactions forward. Note too that the linkage of isoprene units to form farnesyl pyrophosphate occurs in a head-to-

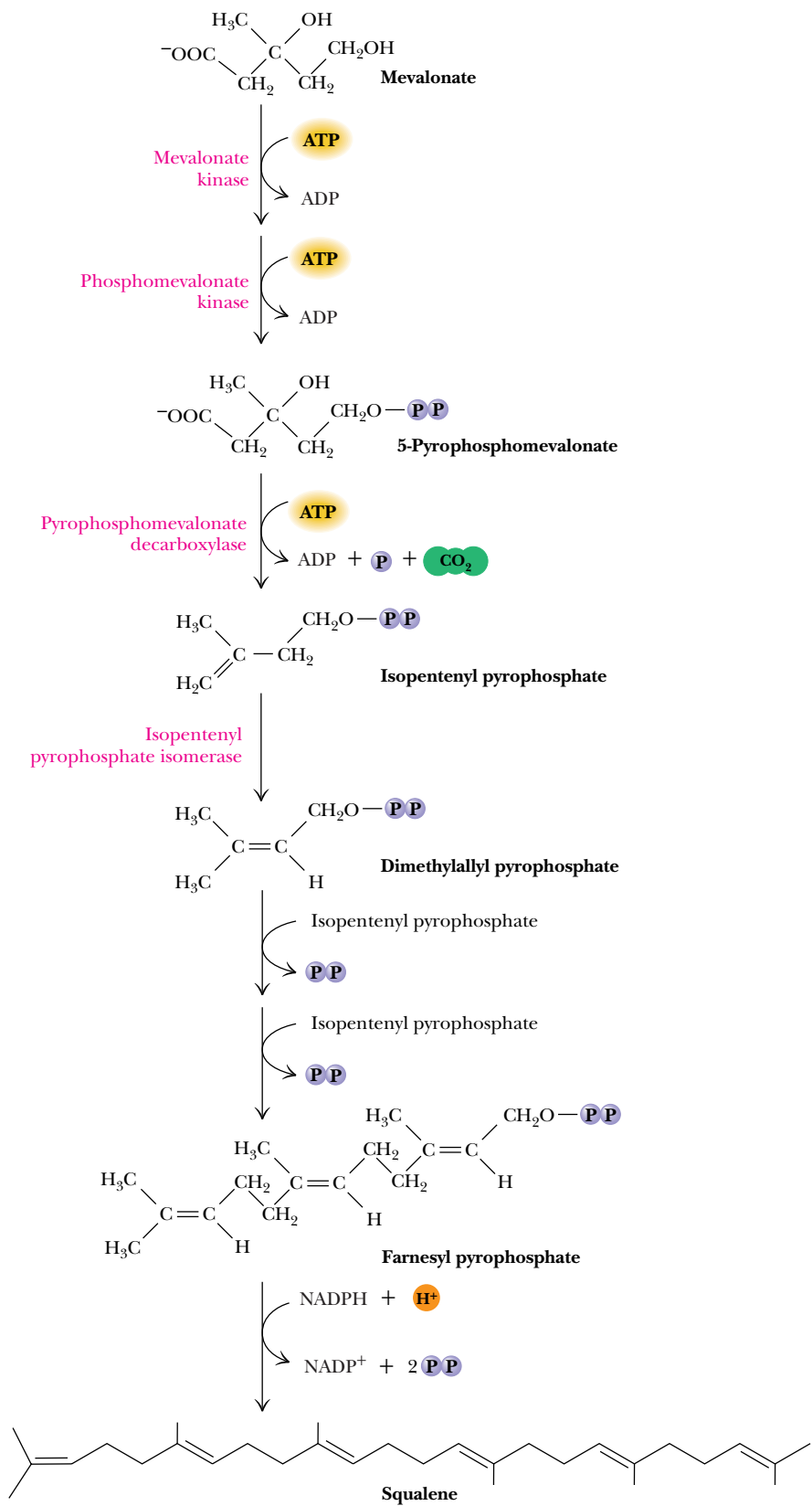


FIGURE 25.34 • The conversion of mevalonate to squalene.



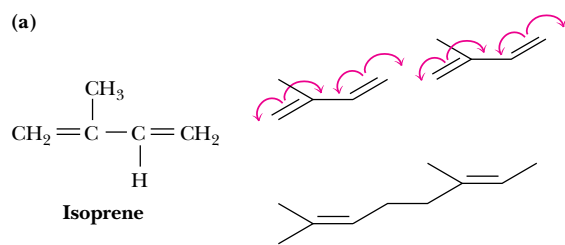
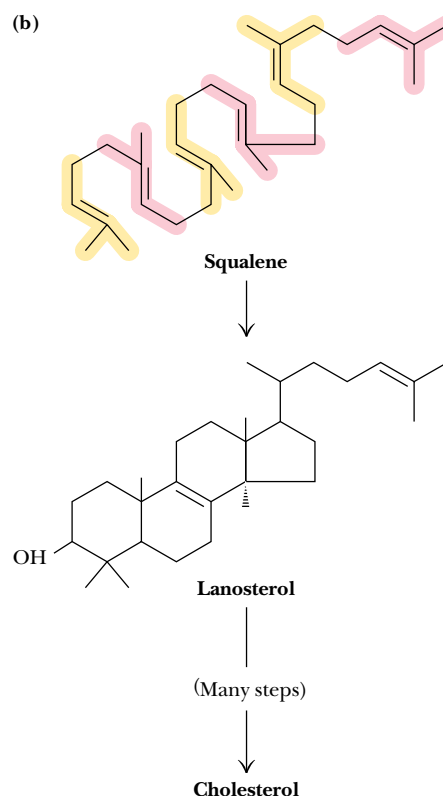
The Long Search for the Route of Cholesterol Biosynthesis

Heilbron, Kamm, and Owens suggested as early as 1926 that squalene is a precursor of cholesterol. That same year, H. J. Channon demonstrated that animals fed squalene from shark oil produced more cholesterol in their tissues. Bloch and Rittenberg showed in the 1940s that a significant amount of the carbon in the tetracyclic moiety and in the aliphatic side chain of cholesterol was derived from acetate. In 1934, Sir Robert Robinson suggested a scheme for the cyclization of squalene to form cholesterol before the biosynthetic link between acetate and squalene was understood. Squalene is actually a polymer of isoprene units, and Bonner and Arreguin suggested in 1949 that three acetate units could join to form 5-carbon *isoprene* units (see figure, part a).

In 1952, Konrad Bloch and Robert Langdon showed conclusively that labeled squalene is synthesized rapidly from labeled acetate and also that cholesterol is derived from squalene. Langdon, a graduate student of Bloch's, performed the critical experiments in Bloch's laboratory at the University of Chicago, while Bloch spent the summer in Bermuda attempting to demonstrate that radioactively labeled squalene would be converted to cholesterol in shark livers. As Bloch himself admitted, "All I was able to learn was that sharks of manageable length are very difficult to catch and their oily livers impossible to slice" (Bloch, 1987).

In 1953, Bloch, together with the eminent organic chemist R. B. Woodward, proposed a new scheme (see figure, part b) for the cyclization of squalene. (Together with Fyodor Lynen, Bloch received the Nobel Prize in medicine or physiology in 1964 for his work.) The picture was nearly complete, but one crucial question remained: How could isoprene be the intermediate in the

transformation of acetate into squalene? In 1956, Karl Folkers and his colleagues at Merck, Sharpe and Dohme isolated mevalonic acid and also showed that mevalonate was the precursor of isoprene units. The search for the remaining details (described in the text) made the biosynthesis of cholesterol one of the most enduring and challenging bioorganic problems of the forties, fifties, and sixties. Even today, several of the enzyme mechanisms remain poorly understood.



(a) An isoprene unit and a scheme for head-to-tail linking of isoprene units. (b) The cyclization of squalene to form lanosterol, as proposed by Bloch and Woodward.

tail fashion. This is the general rule in biosynthesis of molecules involving isoprene linkages. The next step—the joining of two farnesyl pyrophosphates to produce squalene—is a “tail-to-tail” condensation and represents an important exception to the general rule.

Squalene monooxygenase, an enzyme bound to the endoplasmic reticulum, converts squalene to *squalene-2,3-epoxide* (Figure 25.35). This reaction employs FAD and NADPH as coenzymes and requires O₂ as well as a cytosolic protein called **soluble protein activator**. A second ER membrane enzyme, **2,3-oxidosqualene lanosterol cyclase**, catalyzes the second reaction, which involves a succession of 1,2 shifts of hydride ions and methyl groups.

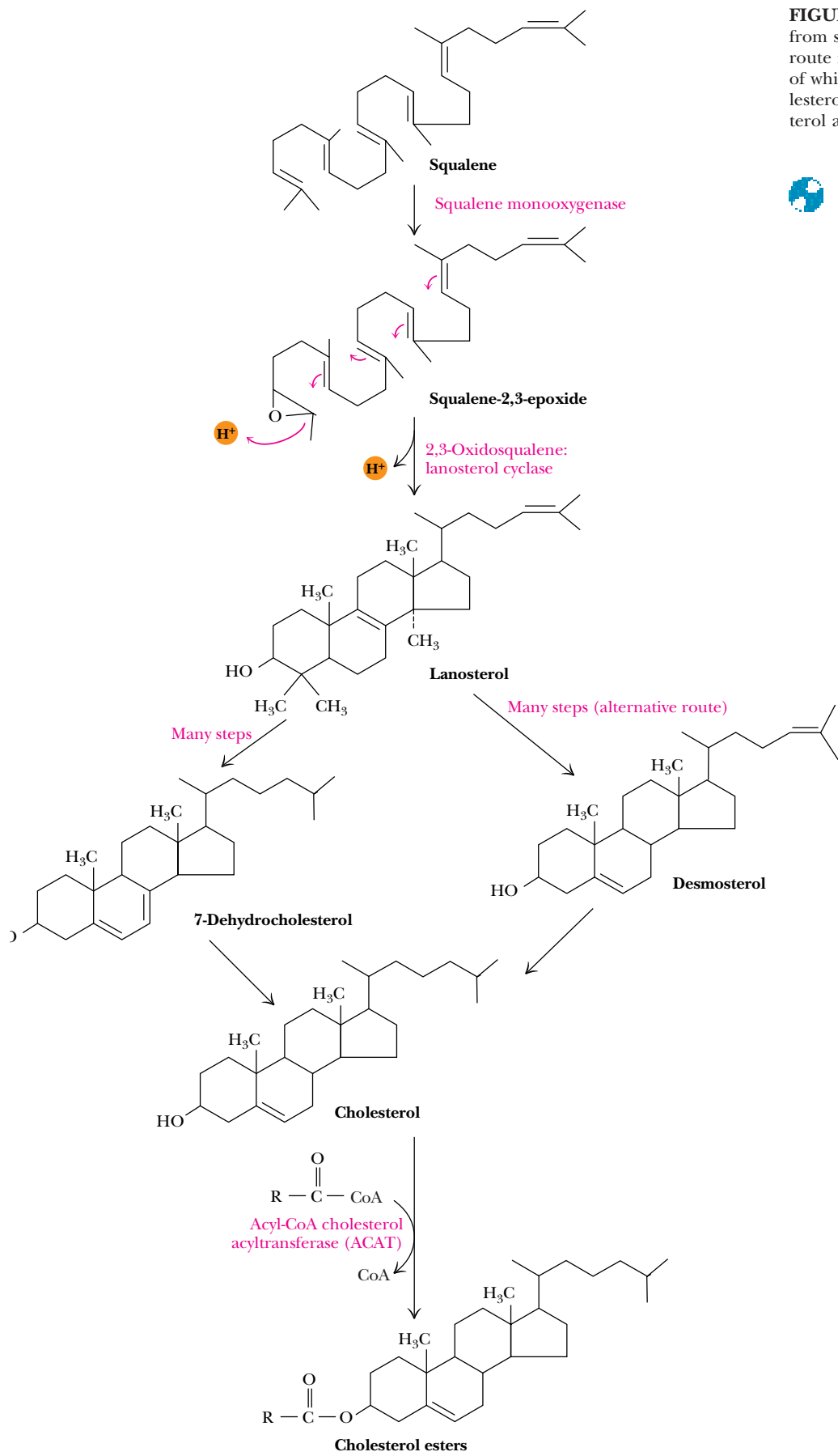


FIGURE 25.35 • Cholesterol is synthesized from squalene via lanosterol. The primary route from lanosterol involves 20 steps, the last of which converts 7-dehydrocholesterol to cholesterol. An alternative route produces desmosterol as the penultimate intermediate.



Table 25.1

Composition and Properties of Human Lipoproteins

Lipoprotein Class	Density (g/mL)	Diameter (nm)	Composition (% dry weight)			
			Protein	Cholesterol	Phospholipid	Triacylglycerol
HDL	1.063–1.21	5–15	33	30	29	8
LDL	1.019–1.063	18–28	25	50	21	4
IDL	1.006–1.019	25–50	18	29	22	31
VLDL	0.95–1.006	30–80	10	22	18	50
Chylomicrons	<0.95	100–500	1–2	8	7	84

Adapted from Brown, M., and Goldstein, J., 1987. In Braunwald, E., et al., eds., *Harrison’s Principles of Internal Medicine*, 11th ed. New York: McGraw-Hill; and Vance, D., and Vance, J., eds., 1985. *Biochemistry of Lipids and Membranes*. Menlo Park, CA: Benjamin/Cummings.

related to the relative amounts of lipid and protein in the complexes. Because most proteins have densities of about 1.3 to 1.4 g/mL, and lipid aggregates usually possess densities of about 0.8 g/mL, the more protein and the less lipid in a complex, the denser the lipoprotein. Thus, there are **high-density lipoproteins** (HDL), **low-density lipoproteins** (LDL), **intermediate-density lipoproteins** (IDL), **very low density lipoproteins** (VLDL), and also **chylomicrons**. Chylomicrons have the lowest protein-to-lipid ratio and thus are the lowest-density lipoproteins. They are also the largest.

The Structure and Synthesis of the Lipoproteins

HDL and VLDL are assembled primarily in the endoplasmic reticulum of the liver (with smaller amounts produced in the intestine), whereas chylomicrons form in the intestine. LDL is not synthesized directly, but is made from VLDL. LDL appears to be the major circulatory complex for cholesterol and cholesterol esters. The primary task of chylomicrons is to transport triacylglycerols. Despite all this, it is extremely important to note that each of these lipoprotein classes contains some of each type of lipid. The relative amounts of HDL and LDL are important in the disposition of cholesterol in the body and in the development of arterial plaques (Figure 25.36). The structures of the various

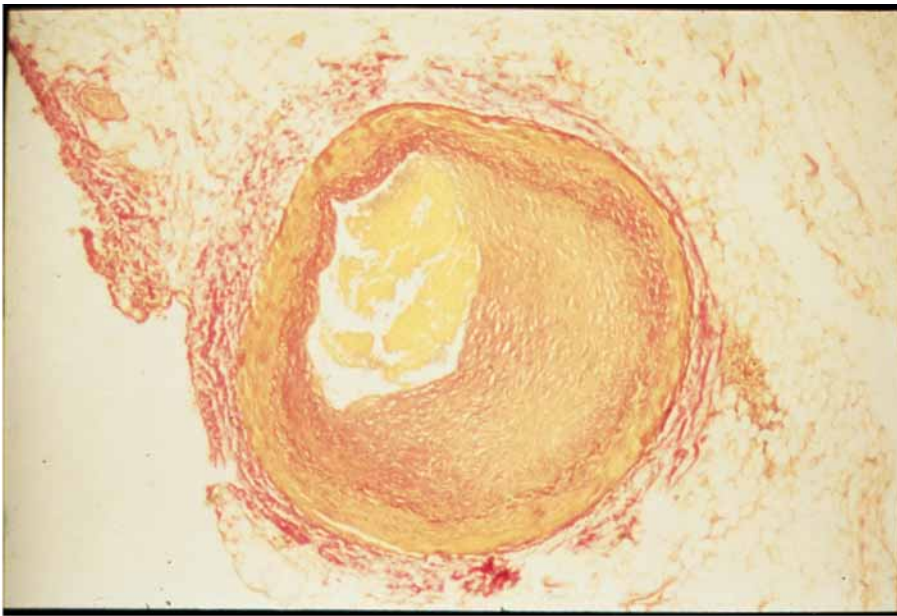


FIGURE 25.36 • Photograph of an arterial plaque. (Science Photo Library/Photo Researchers, Inc.)

lipoproteins are approximately similar, and they consist of a core of mobile triacylglycerols or cholesterol esters surrounded by a single layer of phospholipid, into which is inserted a mixture of cholesterol and proteins (Figure 25.37). Note that the phospholipids are oriented with their polar head groups facing outward to interact with solvent water, and that the phospholipids thus shield the hydrophobic lipids inside from the solvent water outside. The proteins also function as recognition sites for the various lipoprotein receptors throughout the body. A number of different apoproteins have been identified in lipoproteins (Table 25.2), and others may exist as well. The apoproteins are abundant in hydrophobic amino acid residues, as is appropriate for interactions with lipids. A **cholesterol ester transfer protein** also associates with lipoproteins.

Lipoproteins in Circulation Are Progressively Degraded by Lipoprotein Lipase

The livers and intestines of animals are the primary sources of circulating lipids. Chylomicrons carry triacylglycerol and cholesterol esters from the intestines to other tissues, and VLDLs carry lipid from liver, as shown in Figure 25.38. At

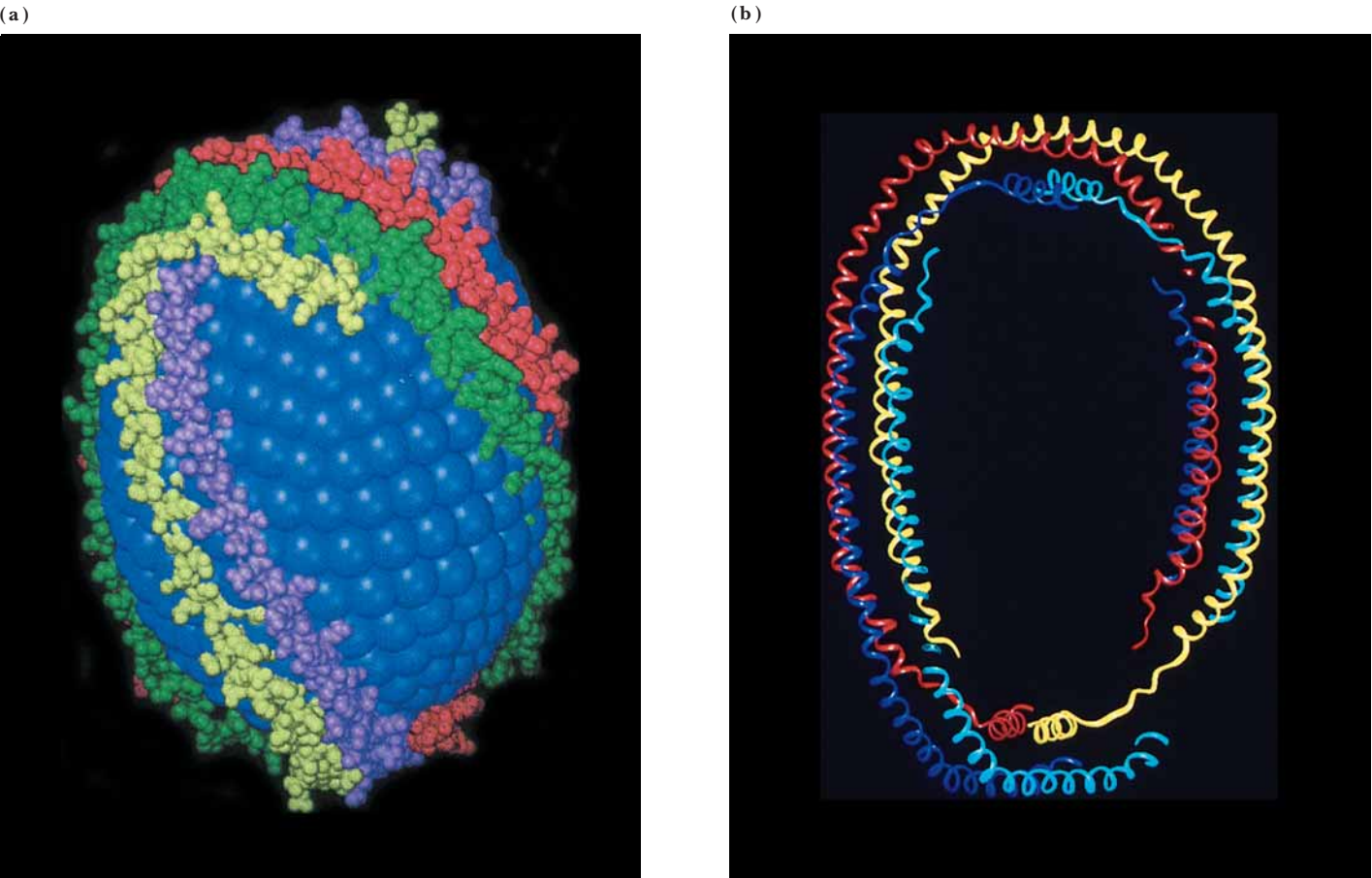


FIGURE 25.37 • A model for the structure of a typical lipoprotein. (a) A core of cholesterol and cholesteryl esters is surrounded by a phospholipid (monolayer) membrane. Apolipoprotein A-I is modeled here as a long amphipathic α -helix, with the nonpolar face of the helix embedded in the hydrophobic core of the lipid particle and the polar face of the helix exposed to solvent. (b) A ribbon diagram of apolipoprotein A-I. (Adapted from Borhani, D. W., Rogers, D. P., Engler, J. A., and Brouillette, C. G., 1997. Crystal structure of truncated human apolipoprotein A-I suggests a lipid-bound conformation. *Proceedings of the National Academy of Sciences* **94**:12291–12296.)

Table 25.2

Apoproteins of Human Lipoproteins

Apoprotein	M _r	Concentration in Plasma (mg/100 mL)	Distribution
A-1	28,300	90–120	Principal protein in HDL
A-2	8,700	30–50	Occurs as dimer mainly in HDL
B-48	240,000	<5	Found only in chylomicrons
B100	500,000	80–100	Principal protein in LDL
C-1	7,000	4–7	Found in chylomicrons, VLDL, HDL
C-2	8,800	3–8	Found in chylomicrons, VLDL, HDL
C-3	8,800	8–15	Found in chylomicrons, VLDL, IDL, HDL
D	32,500	8–10	Found in HDL
E	34,100	3–6	Found in chylomicrons, VLDL, IDL, HDL

Adapted from Brown, M., and Goldstein, J., 1987. In Braunwald, E., et al., eds., *Harrison's Principles of Internal Medicine*, 11th ed. New York: McGraw-Hill; and Vance, D., and Vance, J., eds., 1985. *Biochemistry of Lipids and Membranes*, Menlo Park, CA: Benjamin/Cummings.

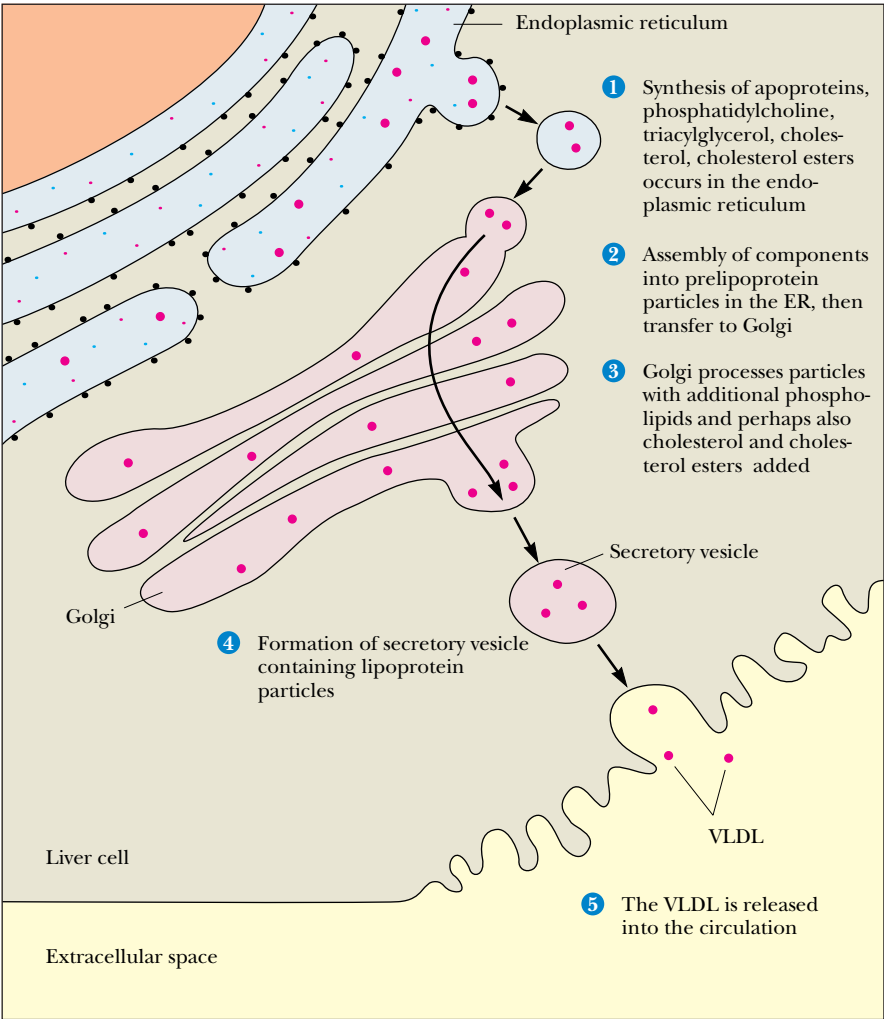


FIGURE 25.38 • Lipoprotein components are synthesized predominantly in the ER of liver cells. Following assembly of lipoprotein particles (red dots) in the ER and processing in the Golgi, lipoproteins are packaged in secretory vesicles for export from the cell (via exocytosis) and released into the circulatory system.

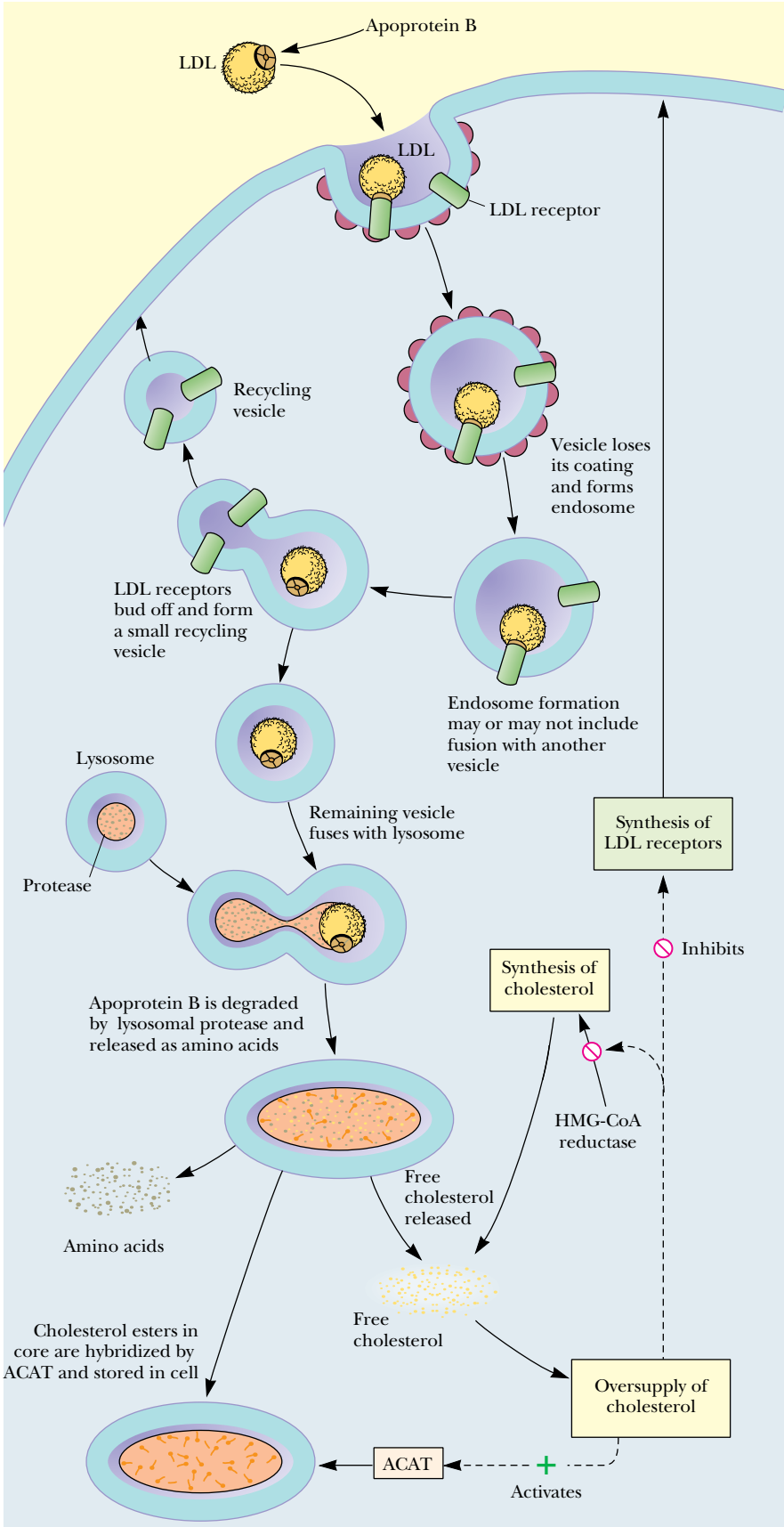


FIGURE 25.39 • Endocytosis and degradation of lipoprotein particles. (ACAT is acyl-CoA cholesterol acyltransferase.)

various target sites, particularly in the capillaries of muscle and adipose cells, these particles are degraded by **lipoprotein lipase**, which hydrolyzes triacylglycerols. Lipase action causes progressive loss of triacylglycerol (and apoprotein) and makes the lipoproteins smaller. This process gradually converts VLDL particles to IDL and then LDL particles, which are either returned to the liver for reprocessing or redirected to adipose tissues and adrenal glands. Every 24 hours, nearly half of all circulating LDL is removed from circulation in this way. The LDL binds to specific LDL receptors, which cluster in domains of the plasma membrane known as **coated pits** (discussed in subsequent paragraphs). These domains eventually invaginate to form **coated vesicles** (Figure 25.39). Within the cell, these vesicles fuse with lysosomes, and the LDLs are degraded by **lysosomal acid lipases**.

High-density lipoproteins (HDL) have much longer life spans in the body (5 to 6 days) than other lipoproteins. Newly formed HDL contains virtually no cholesterol ester. However, over time, cholesterol esters are accumulated through the action of **lecithin:cholesterol acyltransferase** (LCAT), a 59-kD glycoprotein associated with HDLs. Another associated protein, **cholesterol ester transfer protein**, transfers some of these esters to VLDL and LDL. Alternatively, HDLs function to return cholesterol and cholesterol esters to the liver. This latter process apparently explains the correlation between high HDL levels and reduced risk of cardiovascular disease. (High LDL levels, on the other hand, are correlated with an *increased* risk of coronary artery and cardiovascular disease.)

Structure of the LDL Receptor

The LDL receptor in plasma membranes (Figure 25.40) consists of 839 amino acid residues and is composed of five domains. These domains include an LDL-binding domain of 292 residues, a segment of about 350 to 400 residues containing N-linked oligosaccharides, a 58-residue segment of O-linked oligosaccharides, a 22-residue membrane-spanning segment, and a 50-residue segment extending into the cytosol. The clustering of receptors prior to the formation of coated vesicles requires the presence of this cytosolic segment.

Defects in Lipoprotein Metabolism Can Lead to Elevated Serum Cholesterol

The mechanism of LDL metabolism and the various defects that can occur therein have been studied extensively by Michael Brown and Joseph Goldstein, who received the Nobel Prize in medicine or physiology in 1985. **Familial hypercholesterolemia** is the term given to a variety of inherited metabolic defects that lead to greatly elevated levels of serum cholesterol—much of it in the form of LDL particles. The general genetic defect responsible for familial hypercholesterolemia is the absence or dysfunction of LDL receptors in the body. Only about half the normal level of LDL receptors is found in heterozygous individuals (persons carrying one normal gene and one defective gene). Homozygotes (with two copies of the defective gene) have few if any functional LDL receptors. In such cases, LDLs (and cholesterol) cannot be absorbed, and plasma levels of LDL (and cholesterol) are very high. Typical heterozygotes display serum cholesterol levels of 300 to 400 mg/dL, but homozygotes carry serum cholesterol levels of 600 to 800 mg/dL or even higher. There are two possible causes of an absence of LDL receptors. Either receptor synthesis does not occur at all, or the newly synthesized protein does not successfully reach the plasma membrane, due to faulty processing in the Golgi or faulty transport

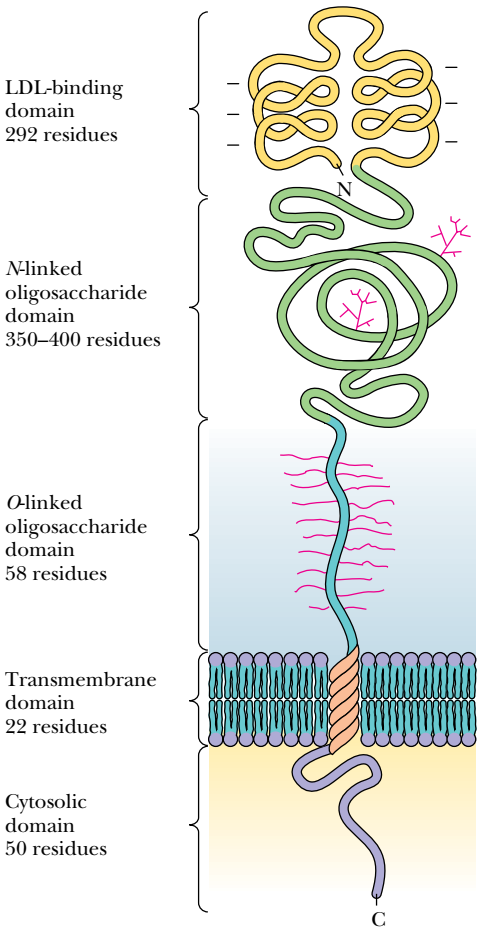


FIGURE 25.40 • The structure of the LDL receptor. The amino-terminal binding domain is responsible for recognition and binding of LDL apoprotein. The O-linked oligosaccharide-rich domain may act as a molecular spacer, raising the binding domain above the glycocalyx. The cytosolic domain is required for aggregation of LDL receptors during endocytosis.

to the plasma membrane. Even when LDL receptors are made and reach the plasma membrane, they may fail to function for two reasons. They may be unable to form clusters competent in coated pit formation because of folding or sequence anomalies in the carboxy-terminal domain, or they may be unable to bind LDL because of sequence or folding anomalies in the LDL-binding domain.

25.6 • Biosynthesis of Bile Acids

Bile acids, which exist mainly as **bile salts**, are polar carboxylic acid derivatives of cholesterol that are important in the digestion of food, especially the solubilization of ingested fats. The Na^+ and K^+ salts of *glycocholic acid* and *taurocholic acid* are the principal bile salts (Figure 25.41). Glycocholate and taurocholate are conjugates of *cholic acid* with glycine and taurine, respectively.

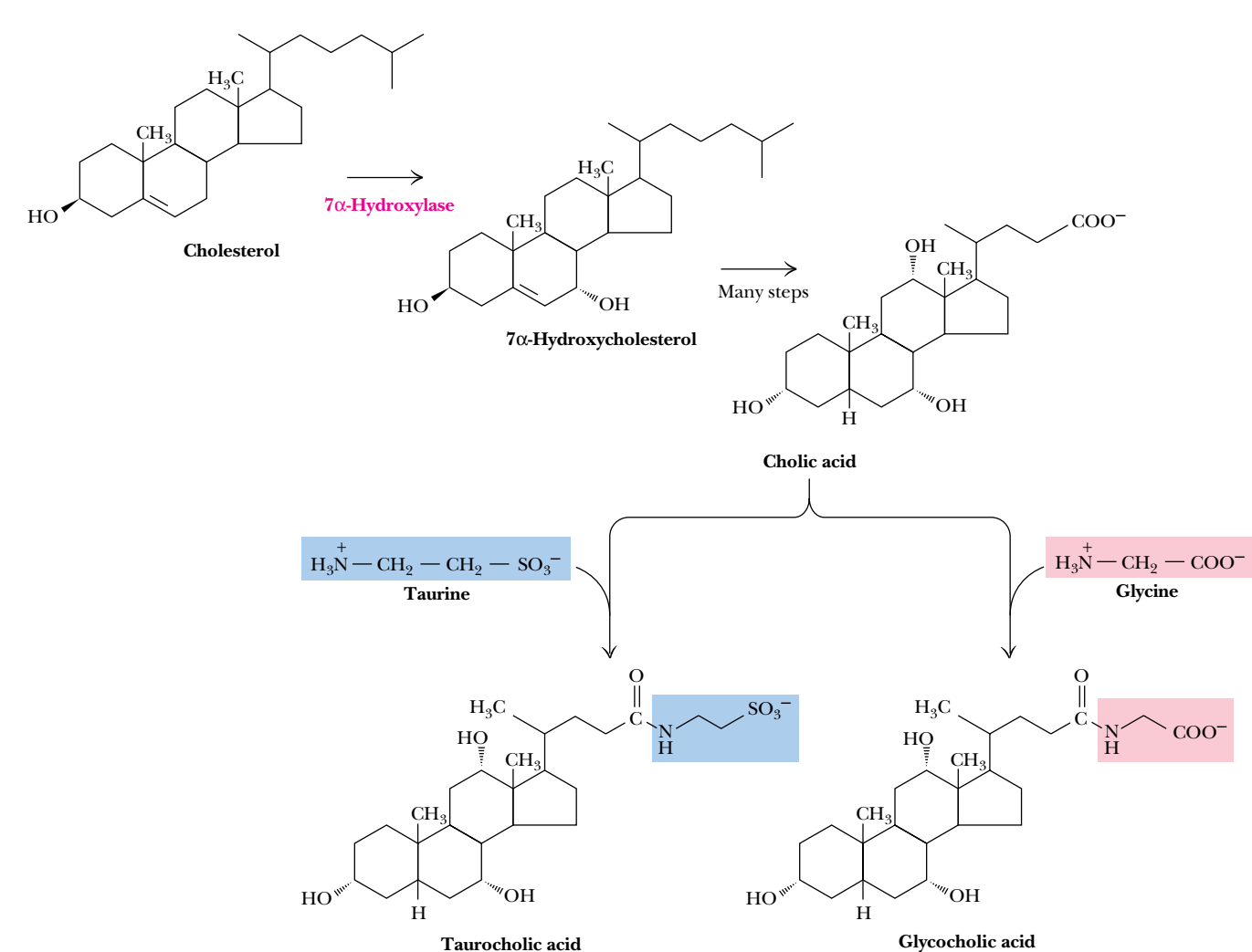


FIGURE 25.41 • Cholic acid, a bile salt, is synthesized from cholesterol via 7 α -hydroxycholesterol. Conjugation with taurine or glycine produces taurocholic acid and glycocholic acid, respectively. Taurocholate and glycocholate are freely water-soluble and are highly effective detergents.

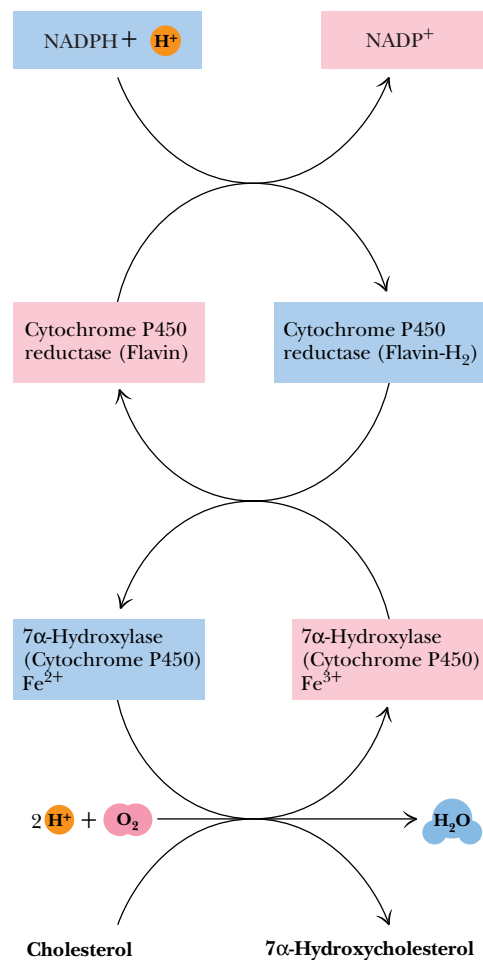


FIGURE 25.42 • The mixed-function oxidase activity of 7α-hydroxylase.

Because they contain both nonpolar and polar domains, these bile salt conjugates are highly effective as detergents. These substances are made in the liver, stored in the gallbladder, and secreted as needed into the intestines.

The formation of bile salts represents the major pathway for cholesterol degradation. The first step involves hydroxylation at C-7 (Figure 25.41). **7α-Hydroxylase**, which catalyzes the reaction, is a mixed-function oxidase involving *cytochrome P450*. **Mixed-function oxidases** use O₂ as substrate. One oxygen atom goes to hydroxylate the substrate, while the other is reduced to water (Figure 25.42). The function of cytochrome P450 is to activate O₂ for the hydroxylation reaction. Such hydroxylations are quite common in the synthetic routes for cholesterol, bile acids, and steroid hormones and also in detoxification pathways for aromatic compounds. Several of these are considered in the next section. 7α-Hydroxycholesterol is the precursor for cholic acid.

25.7 • Synthesis and Metabolism of Steroid Hormones

Steroid hormones are crucial signal molecules in mammals. (The details of their physiological effects are described in Chapter 34.) Their biosynthesis begins with the **desmolase** reaction, which converts cholesterol to preg-

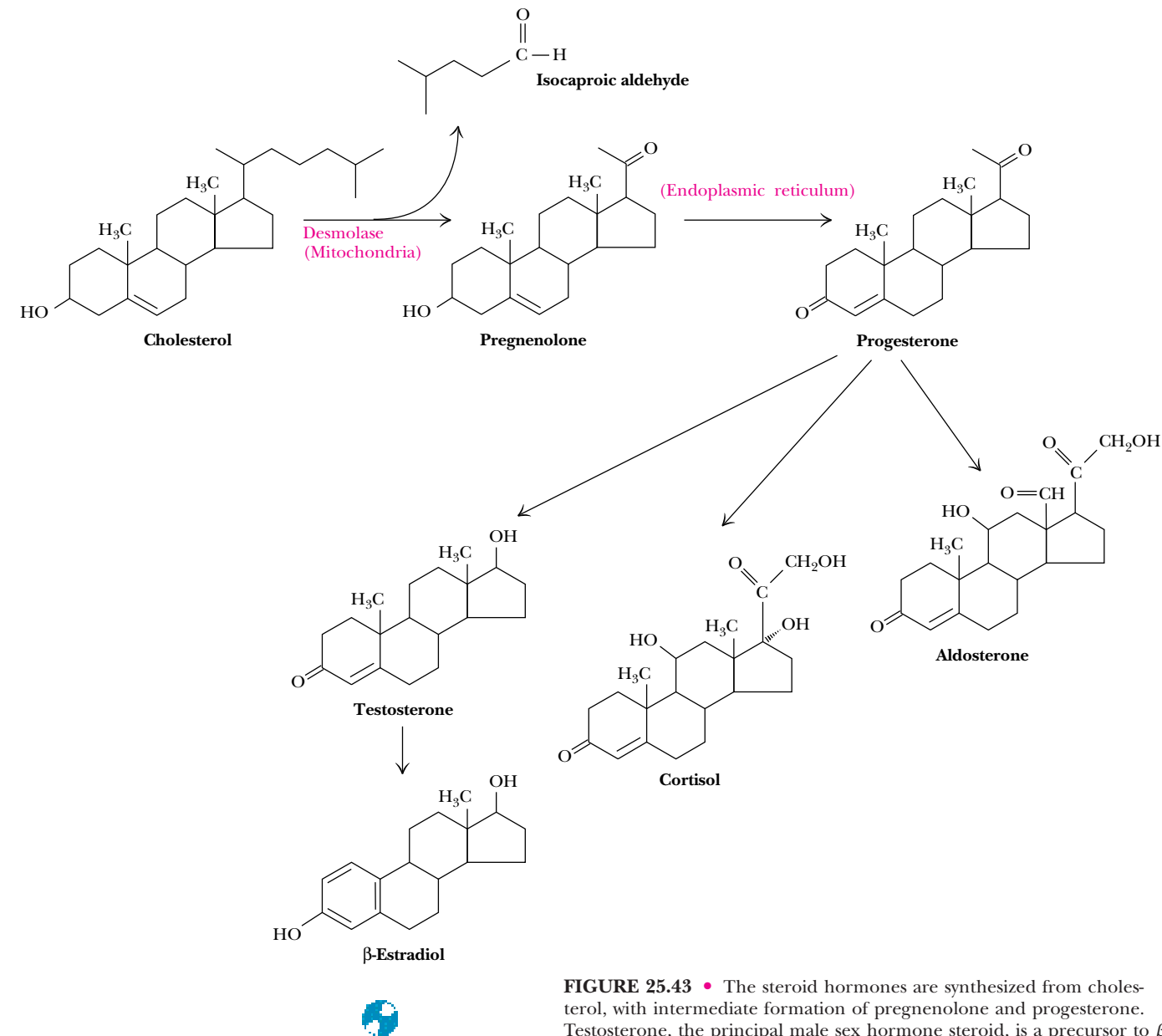


FIGURE 25.43 • The steroid hormones are synthesized from cholesterol, with intermediate formation of pregnenolone and progesterone. Testosterone, the principal male sex hormone steroid, is a precursor to β -estradiol. Cortisol, a glucocorticoid, and aldosterone, a mineralocorticoid, are also derived from progesterone.

nenolone (Figure 25.43). Desmolase is found in the mitochondria of tissues that synthesize steroids (mainly the adrenal glands and gonads). Desmolase activity includes two hydroxylases and utilizes cytochrome P450.

Pregnenolone and Progesterone Are the Precursors of All Other Steroid Hormones

Pregnenolone is transported from the mitochondria to the ER, where a hydroxyl oxidation and migration of the double bond yield progesterone. Pregnenolone synthesis in the adrenal cortex is activated by **adrenocorticotrophic hormone** (ACTH), a peptide of 39 amino acid residues secreted by the anterior pituitary gland.

Progesterone is secreted from the corpus luteum during the latter half of the menstrual cycle and prepares the lining of the uterus for attachment of a fertilized ovum. If an ovum attaches, progesterone secretion continues to ensure the successful maintenance of a pregnancy. Progesterone is also the precursor for synthesis of the **sex hormone steroids** and the **corticosteroids**. Male sex hormone steroids are called **androgens**, and female hormones, **estrogens**. Testosterone is an androgen synthesized in males primarily in the testes (and in much smaller amounts in the adrenal cortex). Androgens are necessary for sperm maturation. Even nonreproductive tissue (liver, brain, and skeletal muscle) is susceptible to the effects of androgens.

Testosterone is also produced primarily in the ovaries (and in much smaller amounts in the adrenal glands) of females as a precursor for the estrogens. β -Estradiol is the most important estrogen (Figure 25.43).

Steroid Hormones Modulate Transcription in the Nucleus

Steroid hormones act in a different manner from most hormones we have considered. In many cases, they do not bind to plasma membrane receptors, but rather pass easily across the plasma membrane. Steroids may bind directly to receptors in the nucleus or may bind to cytosolic steroid hormone receptors, which then enter the nucleus. In the nucleus, the hormone-receptor complex binds directly to specific nucleotide sequences in DNA, increasing transcription of DNA to RNA (Chapters 31 and 34).

Cortisol and Other Corticosteroids Regulate a Variety of Body Processes

Corticosteroids, including the *glucocorticoids* and *mineralocorticoids*, are made by the cortex of the adrenal glands on top of the kidneys. **Cortisol** (Figure 25.43) is representative of the **glucocorticoids**, a class of compounds that (1) stimulate gluconeogenesis and glycogen synthesis in liver (by signaling the synthesis of PEP carboxykinase, fructose-1,6-bisphosphatase, glucose-6-phosphatase, and glycogen synthase); (2) inhibit protein synthesis and stimulate protein degradation in peripheral tissues such as muscle; (3) inhibit allergic and inflammatory responses; (4) exert an immunosuppressive effect, inhibiting DNA replication and mitosis and repressing the formation of antibodies and lymphocytes; and (5) inhibit formation of fibroblasts involved in healing wounds and slow the healing of broken bones.

Aldosterone, the most potent of the **mineralocorticoids** (Figure 25.43), is involved in the regulation of sodium and potassium balances in tissues. Aldosterone increases the kidney's capacity to absorb Na^+ , Cl^- , and H_2O from the glomerular filtrate in the kidney tubules.

Anabolic Steroids Have Been Used Illegally To Enhance Athletic Performance

The dramatic effects of androgens on protein biosynthesis have led many athletes to the use of *synthetic androgens*, which go by the blanket term **anabolic steroids**. Despite numerous warnings from the medical community about side effects, which include kidney and liver disorders, sterility, and heart disease, abuse of such substances is epidemic. **Stanozolol** (Figure 25.44) was one of the agents found in the blood and urine of Ben Johnson following his record-setting performance in the 100-meter dash in the 1988 Olympic Games. Because use of such substances is disallowed, Johnson lost his gold medal, and Carl Lewis was declared the official winner.

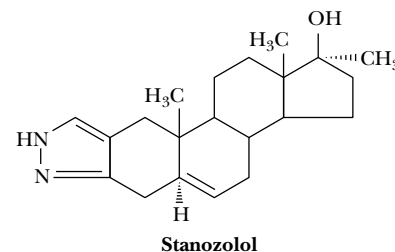


FIGURE 25.44 • The structure of stanozolol, an anabolic steroid.

PROBLEMS

- Carefully count and account for each of the atoms and charges in the equations for the synthesis of palmitoyl-CoA, the synthesis of malonyl-CoA, and the overall reaction for the synthesis of palmitoyl-CoA from acetyl-CoA.
- Use the relationships shown in Figure 25.1 to determine which carbons of glucose will be incorporated into palmitic acid. Consider the cases of both citrate that is immediately exported to the cytosol following its synthesis and citrate that enters the TCA cycle.
- Based on the information presented in the text and in Figures 25.4 and 25.5, suggest a model for the regulation of acetyl-CoA carboxylase. Consider the possible roles of subunit interactions, phosphorylation, and conformation changes in your model.
- Consider the role of the pantothenic acid groups in animal fatty acyl synthase and the size of the pantothenic acid group itself, and estimate a maximal separation between the malonyl transferase and the ketoacyl-ACP synthase active sites.
- Carefully study the reaction mechanism for the stearoyl-CoA desaturase in Figure 25.14, and account for all of the electrons flowing through the reactions shown. Also account for all of the hydrogen and oxygen atoms involved in this reaction, and convince yourself that the stoichiometry is correct as shown.

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- Write a balanced, stoichiometric reaction for the synthesis of phosphatidylethanolamine from glycerol, fatty acyl-CoA, and ethanolamine. Make an estimate of the $\Delta G^{\circ'}$ for the overall process.
- Write a balanced, stoichiometric reaction for the synthesis of cholesterol from acetyl-CoA.
- Trace each of the carbon atoms of mevalonate through the synthesis of cholesterol, and determine the source (i.e., the position in the mevalonate structure) of each carbon in the final structure.
- Suggest a structural or functional role for the O-linked saccharide domain in the LDL receptor (Figure 25.40).
- Identify the lipid synthetic pathways that would be affected by abnormally low levels of CTP.
- Determine the number of ATP equivalents needed to form palmitic acid from acetyl-CoA. (Assume for this calculation that each NADPH is worth 3.5 ATP.)
- Write a reasonable mechanism for the 3-ketosphinganine synthase reaction, remembering that it is a pyridoxal phosphate-dependent reaction.

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